

The Collected Mathematical Papers of  
Arthur Cayley  
Vol. II

Arthur Cayley

Modern L<sup>A</sup>T<sub>E</sub>X reconstruction

THE COLLECTED  
MATHEMATICAL PAPERS  
OF

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CAMBRIDGE. VOL. II.

CAMBRIDGE:  
AT THE UNIVERSITY PRESS.  
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## Advertisement

THE present volume contains fifty-eight papers (numbered 101, 102, . . . , 158) originally published, all but two of them, in the years 1851 to 1860: they are here reproduced nearly but not exactly in chronological order.

The two excepted papers are 142, Numerical Tables Supplementary to Second Memoir on Quantics, now first published (1889); and, 143, Tables of the Covariants M to W of the Binary Quintic: from the Second, Third, Fourth, Fifth, Eighth, Ninth and Tenth Memoirs on Quantics (arranged in the present form, 1889): the determination of the finite number, 23, of the covariants of the quintic was made by Gordan in the year 1869, and the calculation of them having been completed in my Ninth and Tenth Memoirs, it appeared to me convenient in the present republication to unite together the values of all the covariants: viz. those of A to L are given in the Second Memoir 141, and the remainder M to W in the paper 143.

I have added to the Third Memoir 144, in the notation thereof, some formulae which on account of a difference of notation were omitted from a former paper, 35.

I remark that the present volume comprises the first six of the ten Memoirs on Quantics, viz. these are 139, 141, 144, 155, 156 and 158.

I have, in the Notes and References, inserted a discussion of some length in reference to the paper 121, Note on a Question in the Theory of Probabilities: and also some remarks in reference to the theory of Distance developed in the Sixth Memoir on Quantics.

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## 101. Notes on Lagrange's Theorem.

[From the Cambridge and Dublin Mathematical Journal, vol. vi. (1851), pp. 37—45.]

### I.

If in the ordinary form of Lagrange's theorem we write  $(x+a)$  for  $x$ , it becomes

$$x = hf(a+x), \quad F(a+x) = Fa + \frac{x}{1}F'a + \&c. \quad (1)$$

It follows that the equation

$$F(a+x) = Fa + \frac{x}{1} \frac{d}{da}(Fa fa) + \&c. \quad (2)$$

must reduce itself to an identity when the two sides are expanded in powers of  $x$ ; or writing for shortness  $F$ ,  $f$  instead of  $Fa$ ,  $fa$ , and  $\delta$  for  $\frac{d}{da}$ , we must have

$$\delta^r \cdot F = \Sigma \left[ \frac{1}{[p]^i} \delta^{p-1} (\delta F \cdot f^p) \right], \quad (3)$$

(where  $p$  extends from 0 to  $r$ ).

Or what comes to the same,

$$\delta^r \cdot F = \Sigma \left[ \frac{1}{[p]^i [q]^i} \delta^{p-1} (\delta F \cdot f^p) \cdot \delta^{q-1} f^q \right], \quad (4)$$

where  $s$  extends from 0 to  $(r-p)$ .

The terms on the two sides which involve  $\delta^r F$  are immediately seen to be equal; the coefficients of the remaining terms  $\delta^s F$  on the second side must vanish, or we must have

$$\Sigma \left[ \frac{1}{[p]^i [q]^i} \delta^{p-1} \{ \delta^{q-1} f^q \cdot (\delta^{r-p-q+1} f^{r-p-q}) \} \right] = 0, \quad (5)$$

( $s$  being less than  $r$ ).

Or in a somewhat more convenient form, writing  $p$ ,  $q$  and  $k$  for  $p-s$ ,  $r-p$  and  $r-s$ ,

$$\Sigma \left[ \frac{1}{[p]^i [q]^i} \delta^{s+p-1} (\delta^{q-1} f^q \cdot \delta^{q-1} f^q) \right] = 0, \quad (6)$$

where  $s$  is constant and  $p$  and  $q$  vary subject to  $p + q = k$ ,  $k$  being a given constant different from zero (in the case where  $k = 0$ , the series reduces itself to the single term  $\frac{1}{[k]^i}$ ).

The direct proof of this theorem will be given presently.

## II.

The following symbolical form of Lagrange's theorem was given by me in the Mathematical Journal, vol. III. [1843], pp. 283—286, [8]. If

$$x = a + hf\phi, \quad (7)$$

then

$$Fx = \left(\frac{d}{da}\right)^{-1} F'a \cdot e^{\frac{d}{da}hf\phi}.$$

Suppose  $fx = \phi(b + k\psi fx)$ , or  $x = a + h\phi(b + k\psi fx)$ , then

$$Fx = \left(\frac{d}{da}\right)^{-1} \left(\frac{d}{db}\right)^{-1} F'a \cdot e^{\frac{d}{da}h\phi + \frac{d}{db}k\psi}.$$

But

$$\left(\frac{d}{db}\right)^{-1} e^{\frac{d}{db}k\psi} = \left(\frac{d}{db}\right)^{-1} e^{\frac{d}{db}k\psi} (\psi b)^n.$$

(In fact the two general terms  $(\phi(b + k\psi))^m$  and  $\left(\frac{d}{db}\right)^{-1} e^{\frac{d}{db}k\psi} (\psi b)^m$ , of which the former reduces itself to  $e^{\frac{d}{db}k\psi} ((\psi b)^m)$ , are equal on account of the equivalence of the symbols  $e^{\frac{d}{db}k\psi}$  and  $\left(\frac{d}{db}\right)^{-1} e^{\frac{d}{db}k\psi}$ ).

Hence

$$x = a + h\phi(b + k\psi fx) \quad (8)$$

and the coefficient of  $h^m k^n$  is

$$\frac{1}{[m]^i [n]^i} \left(\frac{d}{db}\right)^{n-1} (\psi b)^n \left(\frac{d}{da}\right)^{m-1} \{F'a \cdot (\phi b)^m\}.$$

A similar formula evidently applies to the case of any finite number of functions  $\phi, \psi$ , &c. : in the case of an infinite number we have

$$F(a + h\phi(b + k\psi(c + l\chi(d + \dots)))) = \left(\frac{d}{da}\right)^{-1} \left(\frac{d}{db}\right)^{-1} \left(\frac{d}{dc}\right)^{-1} \dots F'a \cdot e^{\frac{d}{da}h\phi + \frac{d}{db}k\psi + \frac{d}{dc}l\chi} \quad (9)$$

or the coefficient of  $h^m k^n l^p \dots$  is

$$\frac{1}{[m]^i [n]^i [p]^i \dots} \left(\frac{d}{dc}\right)^{p-1} (\chi c)^p \left(\frac{d}{db}\right)^{n-1} (\psi b)^n \left(\frac{d}{da}\right)^{m-1} \{F'a \cdot (\phi a)^m\},$$

$m, n, p$  being always zero; e.g. in the coefficient of  $h^s k^n$ , account must be had of the factor  $(\frac{d}{dc})^{-1} (\chi c)^0$  or  $(\frac{d}{dc})^{-1}$ .

The above form is readily proved independently by Taylor's theorem, without the assistance of Lagrange's. If in it we write  $h = k$ , &c.,  $a = b = \&c.$ , and  $\phi = \psi = \&c. = f$ , we have  $F(a + hf(a + hf(a + \dots))) = Fx$ , where  $x = a + hf x$ .

Hence, comparing the coefficient of  $h^s$  with that given by Lagrange's theorem,

$$\frac{1}{[s]^i} \delta^{s-1} (F \cdot f^s) = \Sigma \left[ \frac{1}{[m]^i [n]^i \dots} \delta^{m-1} \{F \cdot \delta^{n-1} f^n \cdot \delta^{p-1} f^p \dots\} \right], \quad (10)$$

where  $m + n + \&c. = s$ , and as before  $Fa, fa, \frac{d}{da}$  have been replaced by  $F, f, \delta$ .

By comparing the coefficients of  $h^s F$ ,

$$\frac{1}{[s]^i} \delta^{s-1} f^s = \Sigma \left[ \frac{1}{[n]^i [p]^i \dots} \delta^{n-1} f^n \cdot \delta^{p-1} f^p \dots \right], \quad (11)$$

where  $n + p + \dots = t$ , the last of the series  $n, p \dots$  always vanishing.

The formula (10) deduced, as above mentioned, from Taylor's theorem, and the subsequent formula (11) with an independent demonstration of it, not I believe materially different from that which will presently be given, are to be found in a memoir by M. Collins (volume ii. (1833) of the Memoirs of the Academy of St Petersburg), who appears to have made very extensive researches in the theory of developments as connected with the combinatorial analysis.

### III.

To demonstrate the formula (6), consider, in the first place, the expression

$$\frac{1}{[p]^i [q]^i} \delta^{s+p-1} (\delta^{q-1} f^q \cdot \delta^{q-1} f^q),$$

where  $p + q = k$ .

Since

$$\frac{1}{[p]^i [q]^i} = \frac{1}{[p-1]^i [q]^i} + \frac{1}{[p]^i [q-1]^i},$$

this is immediately transformed into

$$\frac{1}{[p-1]^i [q]^i} \delta^{s+p-1} (\delta^{q-1} f^q \cdot \delta^{q-1} f^q) + \frac{1}{[p]^i [q-1]^i} \delta^{s+p-1} (\delta^{q-1} f^q \cdot \delta^{q-1} f^q),$$

in which last expression  $p + q = (p - 1) + q = p + (q - 1)$ .

Of this, after separating the factor  $\delta^s f$ , the general term is equivalent to

$$\begin{aligned} & \frac{1}{[p]^i [q]^i} \delta^{s+p} f \cdot \delta^{q-1} f^q \cdot \delta^{q-1} f^q \\ & \times \Sigma \left[ \frac{1}{[\alpha]^i [\beta]^i} (p + s + \alpha + 1)(p + s + \alpha + 1)(\delta^{p+s+\alpha} f^{p+s+\alpha})(\delta^{q-1} f^{q-\beta}) \right. \\ & \quad \left. - (p + s + \alpha)(\delta^{p+s+\alpha} f^{p+s+\alpha})(\delta^{q-1} f^{q-\beta}) \right], \end{aligned}$$

in which last expression  $p + q = k - \alpha - 1$ .

By repeating the reduction  $j$  times, the general term becomes

$$\begin{aligned} & \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \cdot \frac{1}{[\alpha]^i [\beta]^i \dots} \delta^{s+\alpha} f \cdot \delta^{s+\beta} f \dots \\ & \times \Sigma \left[ \frac{1}{[p]^i [q]^i} (p + s + \alpha + \beta \dots + j - 1) \right. \\ & \quad \left. \times \{ (\delta^{p+s+\alpha+\beta \dots + j-1} f^{p+s+\alpha+\beta \dots + j-1})(\delta^{q-1} f^{q-s-\alpha-\beta \dots -j}) \} \right], \end{aligned}$$

where the sums  $\alpha + \beta \dots$  contain  $j$  terms,  $j$  being less than  $k$  or equal to it, and  $\Sigma$  extends to all combinations of the quantities  $\alpha, \beta \dots$  taken  $j$  and  $j$  together (so that the summation contains  $2^{j-1}$  terms). Also  $p + q = k - \alpha - \beta \dots$  ( $j$  terms)  $-j$ , and the products  $k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots$  and  $[\alpha]^i [\beta]^i \dots \delta^{s+\alpha} f \cdot \delta^{s+\beta} f \dots$  contain each of them  $j$  terms.

Suppose the reduction continued until  $k - \alpha - \beta \dots$  ( $j$  terms)  $-j = 0$ , then the only values of  $p, q$  are  $p = 0, q = 0$ ; and the general term of (6) becomes

$$\begin{aligned} & \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \cdot \frac{1}{[\alpha]^i [\beta]^i \dots} \delta^{s+\alpha} f \cdot \delta^{s+\beta} f \dots \\ & \times \Sigma \left[ (-)^{j-f} (\alpha + \beta \dots + f) [s + \alpha + \beta \dots + f]^i [s + \alpha + \beta \dots + j - 1]^{j-f} \right]. \end{aligned}$$

If  $j = 0$ , the general term reduces itself to

$$\begin{aligned} & \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \cdot \frac{1}{[\alpha]^i [\beta]^i \dots} \delta^{s+\alpha} f \cdot \delta^{s+\beta} f \dots \\ & \times \Sigma \left[ (-)^{j-f} (s + \alpha + \beta \dots + f) [\alpha + \beta \dots + f]^i [s + \alpha + \beta \dots + j - 1]^{j-f} \right]. \end{aligned}$$

Whence finally, if  $\phi p = \frac{1}{[p]^t} \delta^{p-1} f^p$ , the general term of (6) becomes

$$\Sigma \left[ \frac{1}{[\alpha]^i [\beta]^i \dots} \delta^{s+\alpha} \phi^\alpha \cdot \delta^{s+\beta} \phi^\beta \dots \right] \times (\text{vanishing sum}),$$

and it is readily shown that the sum contained in this formula vanishes, which proves the equation in question.

#### IV.

The demonstration of the equation (11) is much simpler. We have

$$\delta^{t-1} f^t = \Sigma \left[ \frac{1}{[n]^i} \delta^{n-1} \{ f^n \cdot \delta^{t-n-1} f^{t-n} \} \right],$$

that is,

$$\delta^{t-1} f^t = \Sigma \left[ \frac{1}{[n]^i [p]^i [q]^i \dots} \delta^{n-1} \{ f^n \cdot \delta^{p-1} (\delta^{q-1} f^q \cdot \delta^{r-1} f^r \dots) \} \right],$$

where  $n$  extends from  $n = 1$  to  $n = t$ .

Similarly

$$\delta^{t-n-1} f^{t-n} = \Sigma \left[ \frac{1}{[p]^i [q]^i \dots} \delta^{p-1} (\delta^{q-1} f^q \cdot \delta^{r-1} f^r \dots) \right],$$

&c.

Hence, substituting successively, and putting  $t - n - p - q = r$ ,  
&c.,

$$\delta^{t-1} f^t = \Sigma \left[ \frac{1}{[n]^i [p]^i [q]^i \dots} \delta^{n-1} \{ f^n \cdot \delta^{p-1} (f^p \cdot \delta^{q-1} (f^q \cdot \delta^{r-1} f^r \dots)) \} \right],$$

&c. ; and the last of these corresponding to a zero value of the last of the quantities  $n, p, q \dots$  is evidently the required equation (11).

The formula (18) in my paper on Lagrange's theorem (before referred to) is incorrect. I propose at present, after giving the proper form of the formula in question, to develop the result of the substitution indicated at the conclusion of the paper. It will be convenient to call to mind the general theorem, that when any number of variables  $x, y, z \dots$  are connected with as many other variables  $u, v, w \dots$  by the same number of equations (so that the variables of each set may be considered as functions of those of the other set) the quotient of the expressions  $dx dy \dots$  and  $du dv \dots$  is equal to the quotient of two

determinants formed with the functions which equated to zero express the relations between the two sets of variables; the former with the differential coefficients of these functions with respect to  $u, v \dots$ , the latter with the differential coefficients with respect to  $x, y \dots$ . Consequently the notation  $\frac{d(x,y\dots)}{d(u,v\dots)}$  may be considered as representing the quotient of these determinants.

This being premised, if we write

$$x - u - h\theta(x, y \dots) = 0, \quad y - v - k\phi(x, y \dots) = 0,$$

then the formula in question is

$$F(x, y \dots) = \Sigma_u \Sigma_v \dots \frac{d(u, v \dots)}{d(x, y \dots)} \cdot e^{u \frac{d}{du} + v \frac{d}{dv} + \dots} FV,$$

if for shortness the letters  $\theta, \phi, \dots, F$  denote what the corresponding functions become when  $u, v \dots$  are substituted for  $x, y \dots$ .

Let  $V$  denote the value which  $\frac{d(u, v \dots)}{d(x, y \dots)}$ , considered as a function of  $x, y \dots$ , assumes when these variables are changed into  $u, v, \dots$ . We have

$$V = \begin{vmatrix} 1 - h\partial_u\theta, & -h\partial_v\theta, & \dots \\ -k\partial_u\phi, & 1 - k\partial_v\phi, & \dots \\ \vdots & \vdots & \ddots \end{vmatrix}$$

By changing the function  $F$ , we obtain

$$F(x, y \dots) = \Sigma_u \Sigma_v \dots e^{h\partial_u + k\partial_v + \dots} FV,$$

where, however, it must be remembered that the  $h, k$  in so far as they enter into the function  $V$ , are not affected by the symbols  $h\partial_h, k\partial_k, \dots$ . In order that we may consider them to be so affected, it is necessary in the function  $V$  to replace  $h, k$ , &c. by  $\frac{h}{\partial_u}, \frac{k}{\partial_v}$ , &c. This being done, observing that the symbols  $h\partial_u\theta, h\partial_v\theta, \dots$  affect a function  $e^{h\partial_u + k\partial_v + \dots}$ , the symbols  $h\partial_u\theta, h\partial_v\theta, \dots$  may be replaced by  $\partial_u^{(\theta)}, \partial_v^{(\theta)}$  where the  $(\theta)$  is not an index, but an affix denoting that the differentiation is only to be performed with respect to  $u, v \dots$  so far as these variables respectively enter into the function  $\theta$ .

Transforming the other lines of the determinant in the same manner, and taking out from  $\Sigma_u \Sigma_v \dots$  the factor  $\partial_u \partial_v \dots$  in order to multiply this last factor into the determinant, we obtain

$$F(x, y \dots) = \Sigma_u \Sigma_v \dots e^{h\partial_u + k\partial_v + \dots} F \cdot Q;$$



where

$$Q = \Sigma \left[ \frac{1}{[m]^i [n]^i \dots} (h \partial_u^{(\theta)} + k \partial_u^{(\phi)} + \dots)^m (h \partial_v^{(\theta)} + k \partial_v^{(\phi)} + \dots)^n \dots \right],$$

in which expression  $\partial_u, \partial_v \dots$  are to be replaced by  $\frac{d}{du}, \frac{d}{dv}, \dots$ .

The complete expansion is easily arrived at by induction, and the form is somewhat singular. In the case of a single variable  $u$  we have  $Q = \partial_u^{(\theta)}$ ; in the case of two variables,  $Q = \partial_u^{(\theta)} \partial_v^{(F)} + \partial_u^{(\theta)^2} \partial_v^{(\phi)} + \partial_u^{(\phi)^2} \partial_v^{(\theta)}$ .

Or writing down only the affixes, in the case of a single variable we have  $F$ ; in the case of two variables  $FF, F\theta, \phi F$ ; and in the case of three variables  $FFF, \phi FF, \chi FF, F\chi F, F\theta F, FF\theta, FF\phi, F\theta\phi, F\phi\chi, F\chi\theta, \chi F\phi, \chi F\chi, \phi F\phi, \phi F\chi, \chi\chi\phi, \chi\phi\chi, \phi\chi\chi$ , where it will be observed that  $\theta$  never occurs in the first place, nor  $\phi$  in the second place, nor  $\theta, \phi$  (in any order) in the first and second places, &c., nor  $\theta, \phi, \chi$  (in any order) in the first, second, and third places.

And the same property holds in the general case for each letter and binary, ternary, &c. combination, and for the entire system of letters, and the system of affixes contains every possible combination of letters not excluded by the rule just given. Thus in the case of two letters, forming the system of affixes  $FF, F\theta, \phi F, \theta F, F\phi, \theta\phi, \phi\theta$ , the last four are excluded, the first three of them by containing  $\theta$  in the first place or  $\phi$  in the second place, the last by containing  $\theta, \phi$  in the first and second places: and there remains only the terms  $FF, F\theta, \phi F$  forming the system given above.

Substituting the expanded value of  $Q$  in the expression for  $F(x, y \dots)$ , the equation may either be permitted to remain in the form which it thus assumes, or we may, in order to obtain the finally reduced form, after expanding the powers of  $h, k \dots$ , connect the symbols  $\partial_u^m, \partial_v^n \dots \partial_u',$  &c. with the corresponding functions  $\theta, \phi \dots F$ , and then omit the affixes; thus, in particular, in the case of a single variable the general term of  $Fx$  is

$$\frac{h^r}{[r]^i} \delta^{r-1} (F' \cdot f^r)$$

(the ordinary form of Lagrange's theorem).

In the case of two letters the general term of  $F(x, y)$  is

$$\frac{h^p k^q}{[p]^i [q]^i} \partial_u^{p-1} \partial_v^{q-1} \{F' \cdot f^p \cdot g^q\}$$

(see the *Mecanique Celeste*, [Ed. 1, 1798] t. i. p. 176).

In the case of three variables, the general term is

$$\frac{h^p k^q l^r}{[p]^i [q]^i [r]^i} \partial_u^{p-1} \partial_v^{q-1} \partial_w^{r-1} \{ \theta^a \phi^b \chi^c F \cdot f^p g^q h^r \},$$

the sixteen terms within the  $\{\}$  being found by comparing the product  $\partial_u \partial_v \partial_w$  with the system  $FFF$ ,  $\phi FF$ , &c., given above, and then connecting each symbol of differentiation with the function corresponding to the affix. Thus in the first term the  $\partial_u$ ,  $\partial_v$ ,  $\partial_w$  each affect the  $F$ , in the second term the  $\partial_u$  affects  $\phi$ , and the  $\partial_v$  and  $\partial_w$  each affect the  $F$ , and so on for the remaining terms.

The form is of course deducible from Laplace's general theorem, and the actual development of it is given in Laplace's Memoir in the *Hist. de l'Acad. des Sci. 1772*. I quote from a memoir by Jacobi which I take this opportunity of referring to, "De resolutione equationum per series infinitas," Crelle, t. vi. [1830], pp. 257—286, founded on a preceding memoir, "Exercitatio algebraica circa discriptionem singularem fractionum quae plures variables involvunt," t. v. [1830], pp. 344—364.

Stone Buildings, April 6, 1850.

## 102. On a Double Infinite Series.

[From the Cambridge and Dublin Mathematical Journal, vol. vi. (1851), pp. 45—47.]

The following completely paradoxical investigation of the properties of the function  $\Gamma$  (which I have been in possession of for some years) may perhaps be found interesting from its connexion with the theories of expansion and divergent series.

Let  $\Sigma_r \phi_r$  denote the sum of the values of  $\phi$  for all integer values of  $r$  from  $-\infty$  to  $\infty$ .

Then writing

$$u = \Sigma_r [n-1]^r x^{n-1-r}, \quad (1)$$

(where  $n$  is any number whatever), we have immediately

$$\frac{du}{dx} = \Sigma_r [n-1]^{r+1} x^{n-2-r} = \Sigma_r [n-1]^r x^{n-1-r} = u$$

that is,

$$\frac{du}{dx} = u, \quad \text{or } u = C_n e^x,$$

(the constant of integration being of course in general a function of  $n$ ).

Hence

$$C_n e^x = \Sigma_r [n-1]^r x^{n-1-r}; \quad (2)$$

or  $e^x$  is expanded in general in a doubly infinite necessarily divergent series of fractional powers of  $x$ , (which resolves itself however in the case of  $n$  a positive or negative integer, into the ordinary singly infinite series, the value of  $C_n$  in this case being immediately seen to be  $\Gamma(n)$ ).

The equation (2) in its general form is to be considered as a definition of the function  $C_n$ .

We deduce from it

$$\Sigma_r [n-1]^r (ax)^{n-1-r} = C_n e^{ax},$$

and also

$$\Sigma_r [n+n' \dots -1]^r \{a(x+x' \dots)\}^{n+n' \dots -1-r} = C_{n+n' \dots} e^{a(x+x' \dots)}.$$

Multiplying the first set of series, and comparing with this last,

$$\begin{aligned} C_n C_{n'} \dots \Sigma_{r,r'} [n-1]^r [n'-1]^{r'} \dots x^{n-1-r} x'^{n'-1-r'} \dots \\ = C_{n+n' \dots} [n+n' \dots -1]^k (x+x' \dots)^{n+n' \dots -1-k}, \quad (3) \end{aligned}$$

(where  $r, r'$  denote any positive or negative integer numbers satisfying  $r + r' + \dots = k + 1 - p$ ,  $p$  being the number of terms in the series  $n, n', \dots$ ).

This equation constitutes a multinomial theorem of a class analogous to that of the exponential theorem contained in the equation (2).

In particular

$$C_{n+n'\dots\Sigma_r,r'\dots}[n-1]^r[n'-1]^{r'}\dots = C_n C_{n'}\dots[n+n'\dots-1]^k(x+x'\dots)^{n+n'\dots-1-k}, \quad (4)$$

and if  $p = 2$ , writing also  $m, n$  for  $n, n'$ , and  $k - 1 - r$  for  $r$ ,

$$C_{m+n}\Sigma_r[m-1]^r[n-1]^{k-1-r} = C_m C_n[m+n-1]^k(x+x')^{m+n-1-k}, \quad (5)$$

or putting  $k = 0$  and dividing,

$$\frac{C_m C_n}{C_{m+n}} = \frac{1}{[m+n-1]^0} \Sigma_r[m-1]^r[n-1]^{-r}. \quad (6)$$

Now the series on the second side of this equation is easily seen to be convergent (at least for positive values of  $m, n$ ). To determine its value write

$$F(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx;$$

then

$$F(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx + \int_0^1 x^{m-1}(1-x)^{n-1}dx,$$

and by successive integrations by parts, the first of these integrals is reducible to

$$\frac{1}{\Sigma_r[m-1]^r[n-1]^{-r}}, \quad r \text{ extending from } -1 \text{ to } -\infty \text{ inclusively,}$$

and the second to

$$\frac{1}{\Sigma_r[m-1]^r[n-1]^{-r}}, \quad r \text{ extending from } 0 \text{ to } \infty;$$

hence

$$\frac{1}{[m+n-1]^0} F(m, n) = \Sigma_r[m-1]^r[n-1]^{-r},$$

or

$$\frac{C_m C_n}{C_{m+n}} = F(m, n), \quad (7)$$

which proves the identity of  $C_n$  with the function  $\Gamma(m)$ .

(Substituting in two of the preceding equations, we have

$$\frac{[n + n' \dots - 1]^k}{\Gamma n \cdot \Gamma n' \dots} = \frac{\Sigma_{r, r' \dots} [n - 1]^r [n' - 1]^{r'} \dots}{\Gamma(n + n' \dots)} \cdot (x + x' \dots)^{n+n' \dots - 1 - k}, \quad (8)$$

(where, as before,  $p$  denotes the number of terms in the series  $n, n', \dots$  and  $r + r' + \dots = k + 1 - p$ ), the first side of which equation is, it is well known, reducible to a multiple definite integral by means of a theorem of M. Dirichlet's.

And

$$\frac{\Gamma(m + n)}{[m + n - 1]^k} \Sigma_r [m - 1]^r [n - 1]^{k-r} = (x + x')^{m+n-1-k}, \quad (9)$$

where  $r$  extends from  $-\infty$  to  $+\infty$ , and  $k$  is arbitrary. By giving large negative values to this quantity, very convergent series may be obtained for the calculation of  $F(m, n)$ .)

### 103. On Certain Definite Integrals.

[From the Cambridge and Dublin Mathematical Journal, vol. vi. (1851), pp. 136—140.]

Suppose that for any positive or negative integral value of  $r$ , we have

$$\psi(x + ra) = U_r \cdot \psi x,$$

$U_r$  being in general a function of  $x$ , and consider the definite integral

$$I = \int_{-\infty}^{\infty} \psi x \cdot \phi x \, dx;$$

$\phi x$  being any other function of  $x$ .

In case of either of the functions  $\psi x$ ,  $\phi x$  becoming infinite for any real value  $\alpha$  of  $x$ , the principal value of the integral is to be taken, that is,  $I$  is to be considered as the limit of

$$\left( \int_{-\infty}^{\alpha-\epsilon} + \int_{\alpha+\epsilon}^{\infty} \right) \psi x \cdot \phi x \, dx \quad (\epsilon = 0),$$

and similarly, when one of the functions becomes infinite for several of such values of  $x$ .

We have

$$I = \int_{-\infty}^{\infty} \dots \psi(x + ra) \cdot \phi(x + ra) \, dx;$$

or changing the variables in the different integrals so as to make the limits of each  $-\alpha$ ,  $0$ , we have

$$I = \int_0^{\alpha} \psi x [\Sigma_r U_r \cdot \phi(x + ra)] \, dx,$$

$\Sigma$  extending to all positive or negative integer values of  $r$ , that is,

$$I = \int_0^{\alpha} \psi x [\Sigma_r U_r \cdot \phi(x + ra)] \, dx, \quad (\text{A})$$

which is true, even when the quantity under the integral sign becomes infinite for particular values of  $x$ , provided the integral be replaced by its principal value, that is, provided it be considered as the limit of

$$\left( \int_0^{\alpha-\epsilon} + \int_{\alpha+\epsilon}^{\alpha} \right) \psi x [\Sigma_r U_r \cdot \phi(x + ra)] \, dx,$$

or

$$\int_0^\alpha \psi x [\Sigma_r U_r \cdot \phi(x + ra)] dx;$$

where  $\alpha$ , or one of the limiting values 0,  $\alpha$ , is the value of  $x$ , for which the quantity under the integral sign becomes infinite, and  $\epsilon$  is ultimately evanescent.

In particular, taking for simplicity  $a = \pi$ , suppose

$$f(x + \pi) = \pm f x, \quad \text{or } f(x + n\pi) = (\pm)^n f x;$$

then observing the equation

$$\Sigma_r \frac{(\pm)^r}{x + r\pi} = \cot x, \quad \text{or } = x,$$

according as the upper or under sign is taken, and assuming  $\psi x = x^{-g}$ , we have finally

$$\int_{-\infty}^{\infty} \frac{\psi x \cdot dx}{x} = \pi, \quad g \text{ even},$$

$$\int_{-\infty}^{\infty} \psi x x dx = \pi, \quad g \text{ odd},$$

the former equation corresponding to the case of  $f(x + \pi) = f x$ , the latter to that of  $f(x + \pi) = -f x$ .

Suppose  $f_g x = f^g x \cdot f x$ ,  $g$  being a positive integer. Then

$$\int_{-\infty}^{\infty} \frac{\psi x \cdot dx}{x} = \int_{-\infty}^{\infty} \frac{f x \cdot dx}{x};$$

also if  $f(x + \pi) = f x$ , then  $f_g(x + \pi) = f_g x$ ; but if  $f(x + \pi) = -f x$ , then  $f_g(x + \pi) = +f_g x$ , the upper or under sign according as  $g$  is even or odd.

Combining these equations, we have

$$\begin{array}{ll} f_g(x + \pi) = f x, & g \text{ even or odd,} \\ f_g(x + \pi) = -f x, & g \text{ even,} \\ f_g(x + \pi) = -f x, & g \text{ odd.} \end{array}$$

In particular

$$\begin{aligned}\int_{-\infty}^{\infty} \sin gx \cdot \cot x \, dx &= \pi, & g \text{ even,} \\ \int_{-\infty}^{\infty} \sin gx \cdot x \, dx &= \pi, & g \text{ odd,} \\ \int_{-\infty}^{\infty} \tan gx \cdot x \, dx &= 0, & g \text{ even,} \\ \int_{-\infty}^{\infty} \tan gx \cdot x \, dx &= 0, & g \text{ odd,} \\ \int_{-\infty}^{\infty} \frac{dx}{x} &= 0, & \&c.,\end{aligned}$$

the number of which might be indefinitely extended.

The same principle applies to multiple integrals of any order: thus for double integrals, if  $\psi(x+ra, y+sb) = U_{r,s} \cdot \psi(x, y)$ , then

$$\iint \psi(x, y) \phi(x, y) \, dx \, dy = \int_0^a \int_0^b \psi(x, y) [\Sigma_{r,s} U_{r,s} \cdot \phi(x+ra, y+sb)] \, dx \, dy. \quad (\text{B})$$

In particular, writing  $w, v$  for  $a, b$ , and assuming  $\psi(x+rw, y+sv) = (\pm)^r (\pm)^s \psi(x, y)$ ; also  $\phi(x, y) = (x+iy)^{-\frac{1}{2}}$ , where as usual  $i = \sqrt{-1}$ ,

$$\iint \frac{dx \, dy}{(x+iy)^{\frac{1}{2}}} = \int_0^w \int_0^v \psi(x, y) \left[ \Sigma_{r,s} (\pm)^r (\pm)^s (x+iy+rw+isv)^{-\frac{1}{2}} \right] \, dx \, dy,$$

where  $\Sigma$  extending to all positive or negative integer values of  $r$  and  $s$ .

Employing the notation of a paper in the Cambridge Mathematical Journal, "On the Inverse Elliptic Functions," t. iv. [1845], pp. 257—277, [24], we have for the different combinations of the ambiguous sign,

$$\begin{aligned}\frac{\gamma(x+iy)}{\theta(x+iy)} &= \Sigma_{r,s} \frac{1}{y+iy+rw+isv}, & 1. \quad -, -, \\ \frac{\epsilon(x+iy)}{\gamma(x+iy)} &= \Sigma_{r,s} \frac{1}{y+iy+rw+isv}, & 2. \quad -, +, \\ \frac{\theta(x+iy)}{\gamma(x+iy)} &= \frac{F(x+iy)}{\Phi(x+iy)}, & 3. \quad +, -, \\ \frac{\gamma(x+iy)}{\gamma(x+iy)} &= \psi(x, y), & 4. \quad +, +\end{aligned}$$



where  $\phi, f, F$  are in fact the symbols of the inverse elliptic functions (Abel's notation) corresponding very nearly to  $\sin am, \cos am, \Delta am$ .

It is remarkable that the last value of  $S$  cannot be thus expressed, but only by means of the more complicated transcendent  $\gamma x$ , corresponding to the  $H(x)$  of M. Jacobi.

The four cases correspond obviously to

1.  $\psi(x + rw, y + sv) = (-)^{r+s} \psi(x, y),$
2.  $\psi(x + rw, y + sv) = (-)^r \psi(x, y),$
3.  $\psi(x + rw, y + sv) = (-)^s \psi(x, y),$
4.  $\psi(x + rw, y + sv) = \psi(x, y).$

The above formulae may be all of them modified, as in the case of single integrals, by means of the obvious equation

$$\Sigma_{r,s} \frac{1}{(x + iy + rw + isv)} = \frac{1}{x + iy}.$$

The most important particular case is for in almost all the others, for example in

$$\int_0^w \int_0^v \frac{dx dy}{(x + iy)^{\frac{1}{2}}} \cdot \frac{d}{dx} \left( \frac{1}{\phi(x + iy)} \right),$$

the second integration cannot be effected.

Suppose next  $\psi(x, y)$  is one of the functions  $\gamma(x + iy), g(x + iy), G(x + iy), \Phi(x + iy)$ , so that

$$f(x + rw, y + sv) = (\pm)^r (\pm)^s U_{r,s} \cdot \psi(x, y),$$

where (see memoir quoted).

Then, retaining the same value as before of  $\psi(x, y)$ , we have still the formula (B), in which

$$\Sigma_{r,s} \frac{(\pm)^r (\pm)^s}{y + iy + rw + isv} = \frac{\gamma(x + iy)}{\theta(x + iy)}.$$

But this summation has not yet been effected; the difficulty consists in the variable factor  $e^{mrw + nsv}$  in the numerator, nothing being known I believe of the decomposition of functions into series of this form.

On the subject of the preceding paper may be consulted the following memoirs by Raabe, "Ueber die Summation periodischer Reihen," Crelle, t. XV. [1836], pp. 355—362, and "Ueber die Summation harmonisch periodischer Reihen," t. XXIII. [1842], pp. 105—125, and t. xxv. [1843], pp. 160—168.

The integrals he considers, are taken between the limits 0,  $2\pi$  (instead of  $-\infty, \infty$ ). His results are consequently more general than those given above, but they might be obtained by an analogous method, instead of the much more complicated one adopted by him: thus if  $\phi(x + 2\pi) = \phi x$ , the integral

$$\int_0^{2\pi} \frac{\phi x}{1 - 2r \cos x + r^2} dx$$

reduces itself to

$$\frac{1}{1 - r^2} \int_0^{2\pi} \phi x \left[ \frac{1}{2} + \Sigma_r(r^{|r|} \cos rx - 2r\pi) \right] dx,$$

provided  $\int_0^{2\pi} \phi x dx = 0$ .

The summation in this formula may be effected by means of the function  $F$  and its differential coefficient, and we have

$$\int_0^{2\pi} \frac{\phi x}{1 - 2r \cos x + r^2} dx = \frac{\pi}{1 - r^2} \int_0^{2\pi} \phi x dx,$$

which is in effect Raabe's formula (10), Crelle, t. xxv. p. 166.

By dividing the integral on the right-hand side of the equation into two others whose limits are 0,  $\pi$ , and  $\pi$ ,  $2\pi$  respectively, and writing in the second of these  $2\pi - x$  instead of  $x$ , then

$$\int_{\pi}^{2\pi} \frac{\phi x}{1 - 2r \cos x + r^2} dx;$$

or reducing by

$$\int_{\pi}^{2\pi} \frac{\phi x}{1 - 2r \cos x + r^2} dx = \pi \cot \frac{1}{2}x,$$

we have

$$\int_0^{2\pi} \frac{\phi x}{1 - 2r \cos x + r^2} dx = \int_0^{\pi} \phi x \cot \frac{1}{2}x dx - \frac{1}{2} \int_0^{\pi} [\phi x + \phi(2\pi - x)] \cot \frac{1}{2}x dx,$$

which corresponds to Raabe's formula (10').

If  $\phi(-x) = -\phi x$ , so that  $\phi x + \phi(2\pi - x) = 0$ , the last formula is simplified; but then the integral on the first side may be replaced by  $\int_{-\infty}^{\infty} \frac{\phi x}{x} dx$ , so that this belongs to the preceding class of formulae.

## 104. On the Theory of Permutants.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 40—51.]

A FORM may be considered as composed of blanks which are to be filled up by inserting in them specializing characters, and a form the blanks of which are so filled up becomes a symbol. We may for brevity speak of the blanks of a symbol in the sense of the blanks of the form from which such symbol is derived. Suppose the characters are 1, 2, 3, 4 the symbol may always be represented in the first instance and without reference to the nature of the form, by  $F_{1234}\dots$ . And it will be proper to consider the blanks as having an invariable order to which reference will implicitly be made; thus, in speaking of the characters 2, 1, 3, 4,  $\dots$  instead of as before 1, 2, 3, 4,  $\dots$  the symbol will be  $F_{2134}\dots$  instead of  $F_{1234}\dots$ . When the form is given we shall have an equation such as

$$F_{123} = P_{12}Q_3R_1\dots \quad \text{or} \quad = P_{12}P_{31}\dots \quad \&c.,$$

according to the particular nature of the form.

Consider now the characters 1, 2, 3, 4,  $\dots$ , and let the primitive arrangement and every arrangement derivable from it by means of an even number of inversions or interchanges of two characters be considered as positive, and the arrangements derived from the primitive arrangement by an odd number of inversions or interchanges of two characters be considered as negative; a rule which may be termed “the rule of signs.”

The aggregate of the symbols which correspond to every possible arrangement of the characters, giving to each symbol the sign of the arrangement, may be termed a “Permutant;” or, in distinction from the more general functions which will presently be considered, a simple permutant, and may be represented by enclosing the symbol in brackets, thus  $(F_{123}\dots)$ . And by using an expression still more elliptical than the blanks of a symbol, we may speak of the blanks of a permutant, or the characters of a permutant.

As an instance of a simple permutant, we may take

$$(F_{123}) = F_{123} + F_{231} + F_{312} - F_{213} - F_{132} - F_{321};$$

and if in particular  $F_{ijk} = a_ib_jc_k$ , then

$$(F_{123}) = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_2b_1c_3 - a_1b_3c_2 - a_3b_2c_1.$$

It follows at once that a simple permutant remains unaltered, to the sign près according to the rule of signs, by any permutations of the characters entering into the permutant. For instance,

$$(F_{123}) = (F_{231}) = (F_{312}) = -(F_{213}) = -(F_{132}) = -(F_{321}).$$

Consequently also when two or more of the characters are identical, the permutant vanishes, thus  $(F_{113}) = 0$ .

The form of the symbol may be such that the symbol remains unaltered, to the sign près according to the rule of signs, for any permutations of the characters in certain particular blanks. Such a system of blanks may be termed a quote. Thus, if the first and second blanks are a quote,

$$F_{1'123'} = F_{123} = F_{213} = F_{132} = F_{312} = F_{231} = F_{321},$$

and consequently

$$(F_{1'12'3}) = 2(F_{123} + F_{231} + F_{312});$$

and if the blanks constitute one single quote,

$$(F_{123\dots n'}) = n! \cdot F_{123\dots n},$$

where  $n! = 1 \cdot 2 \cdot 3 \dots n$ ,  $n$  being the number of characters.

An important case, which will be noticed in the sequel, is that in which the whole series of blanks divide themselves into quotes, each of them containing the same number of blanks. Thus, if the first and second blanks, and the third and fourth blanks, form quotes respectively,

$$6(F_{12''34'}) = F_{1234} + F_{1243} + F_{2134} + F_{2143} + F_{3412} + F_{3421} + F_{4312} + F_{4321}.$$

It is easy now to pass to the general definition of a "Permutant." We have only to consider the blanks as forming, not as heretofore a single set, but any number of distinct sets, and to consider the characters in each set of blanks as permutable inter se and not otherwise, giving to the symbol the sign compounded of the signs corresponding to the arrangements of the characters in the different sets of blanks. Thus, if the first and second blanks form a set, and the third and fourth blanks form a set,

$$((F_{1234})) = F_{1234} - F_{2134} - F_{1243} + F_{2143}.$$

The word 'set' will be used throughout in the above technical sense. The particular mode in which the blanks are divided into sets may be indicated either in words or by some superadded notation.

It is clear that the theory of permutants depends ultimately on that of simple permutants; for if in a compound permutant we first write down all the terms which can be obtained, leaving unpermuted the characters of a particular set, and replace each of the terms so obtained by a simple permutant having for its characters the characters of the previously unpermuted set, the result is obviously the original compound permutant. Thus, in the above-mentioned case, where the first and second blanks form a set and the third and fourth blanks form a set,

$$((F_{1234})) = (F_{1234}) - (F_{2134}),$$

or

$$((F_{1234})) = (F_{1234}) - (F_{1243}),$$

in the former of which equations the first and second blanks in each of the permutants on the second side form a set, and in the latter the third and fourth blanks in each of the permutants on the second side form a set, the remaining blanks being simply supernumerary and the characters in them unpermutable.

It should be noted that the term quote, as previously defined, is only applicable to a system of blanks belonging to the same set, and it does not appear that anything would be gained by removing this restriction.

The following rule for the expansion of a simple permutant (and which may be at once extended to compound permutants) is obvious. Write down all the distinct terms that can be obtained, on the supposition that the blanks group themselves in any manner into quotes, and replace each of the terms so obtained by a compound permutant having for a distinct set the blanks of each assumed quote; the result is the original simple permutant. Thus in the simple permutant  $(F_{1234})$ , supposing for the moment that the first and second blanks form a quote, and that the third and fourth blanks form a quote, this leads to the equation

$$(F_{1234}) = +((F_{12''34'})) + ((F_{13''24'})) + ((F_{14''23'})) + ((F_{23''14'})) + ((F_{24''13'})) + ((F_{34''12'})$$

where in each of the permutants on the second side the first and second blanks form a set, and also the third and fourth blanks.

The blanks of a simple or compound permutant may of course, without either gain or loss of generality, be considered as having any particular arrangement in space, for instance, in the form of a rectangle: thus

$$\begin{array}{c} F_{12} \\ 34 \end{array} \text{ is neither more nor less general than } F_{1234}.$$

The idea of some such arrangement naturally presents itself as affording a means of showing in what manner the blanks are grouped into sets. But, considering the blanks as so arranged in a rectangular form, or in lines and columns, suppose in the first instance that this arrangement is independent of the grouping of the blanks into sets, or that the blanks of each set or of any of them are distributed at random in the different lines and columns.

Assume that the form is such that a symbol  $\alpha\beta\gamma\dots$  is a function of symbols  $F_{\alpha\beta\gamma\dots}$ ,  $F_{\alpha'\beta'\gamma'\dots}$ , &c. Or, passing over this general case, and the case (of intermediate generality) of the function being a symmetrical function, assume that the symbol is the product of symbols  $F_{\alpha\beta\gamma\dots}$ ,  $F_{\alpha'\beta'\gamma'\dots}$ , &c.

Upon this assumption it becomes important to distinguish the different ways in which the blanks of a set are distributed in the different lines and columns. The cases to be considered are:

- (A). The blanks of a single set or of single sets are situated in more than one column.
- (B). The blanks of each single set are situated in the same column.
- (C). The blanks of each single set form a separate column.

The case (B) (which includes the case (C)) and the case (C) merit particular consideration. In fact the case (B) is that of the functions which I have, in my memoir on Linear Transformations in the Journal, [13, 14] called hyperdeterminants, and the case (C) is that of the particular class of hyperdeterminants previously treated of by me in the Cambridge Philosophical Transactions, [12] and also particularly noticed in the memoir on Linear Transformations. The functions of the case (B) I now propose to call "Intermutants," and those in the case (C) "Commutants." Commutants include as a particular case "Determinants," which term will be used in its ordinary signification. The case (A) I shall not at present discuss in its generality, but

only with the further assumption that the blanks form a single set (this, if nothing further were added, would render the arrangement of the blanks into lines and columns valueless), and moreover that the blanks of each line form a quote: the permutants of this class (from their connexion with the researches of Pfaff on differential equations) I shall term "Pfaffians."

And first of commutants, which, as before remarked, include determinants. The general expression of a commutant is

$$\begin{pmatrix} F_{11} & F_{12} & \dots & F_{1n} \\ F_{21} & F_{22} & \dots & F_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ F_{n1} & F_{n2} & \dots & F_{nn} \end{pmatrix},$$

and (stating again for this particular case the general rule for the formation of a permutant) if, permuting the characters in the same column in every possible way, considering these permutations as positive or negative according to the rule of signs, one system be represented by

$$\begin{pmatrix} r & s & \dots \\ r' & s' & \dots \\ \vdots & \vdots & \end{pmatrix},$$

the commutant is the sum of all the different terms  $\pm F_{rr'} \cdot F_{ss'} \cdot \dots$

The different permutations may be formed as follows: first permute the characters in all the columns except a single column, and in each of the arrangements so obtained permute entire lines of characters. It is obvious that, considering any one of the arrangements obtained by permutations of the characters in all the columns but one, the permutations of entire lines and the addition of the proper sign will only reproduce the same symbol—in the case of an even number of columns constantly with the positive sign, but in the case of an odd number of columns with the positive or negative sign, according to the rule of signs. For the inversion or interchange of two entire lines is equivalent to as many inversions or interchanges of two characters as there are characters in a line, that is, as there are columns, and consequently introduces a sign compounded of as many negative signs as there are columns.

Hence **Theorem.** A commutant of an even number of columns may be calculated by considering the characters of any one column

(no matter which) as supernumerary unpermutable characters, and multiplying the result by the number of permutations of as many things as there are lines in the commutant.

The mark  $\dagger$  added to a commutant of an even number of columns will be employed to show that the numerical multiplier is to be omitted. The same mark placed over any one of the columns of the commutant will show that the characters of that particular column are considered as non-permutable.

A determinant is consequently represented indifferently by the notations

$$\begin{vmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{vmatrix}^{\dagger} \quad \text{and} \quad \begin{vmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{vmatrix}^{\dagger},$$

and a commutant of an odd number of columns vanishes identically.

By considering, however, a commutant of an odd number of columns, having the characters of some one column non-permutable, we obtain what will in the sequel be spoken of as commutants of an odd number of columns. This non-permutability will be denoted, as before, by means of the mark  $\dagger$  placed over the column in question, and it is to be noticed that it is not, as in the case of a commutant of an even number of columns, indifferent over which of the columns the mark in question is placed; and consequently there would be no meaning in simply adding the mark  $\dagger$  to a commutant of an odd number of columns.

A commutant is said to be *symmetrical* when the symbols  $F_{\alpha\beta\gamma\dots}$  are such as to remain unaltered by any permutations inter se of the characters  $\alpha, \beta, \gamma\dots$ . A commutant is said to be *skew* when each symbol  $F_{\alpha\beta\gamma\dots}$  is such as to be altered in sign only according to the rule of signs for any permutations inter se of the characters  $\alpha, \beta, \gamma\dots$ , this of course implies that the symbol  $F_{\alpha\beta\gamma\dots}$  vanishes when any two of the characters  $\alpha, \beta, \gamma\dots$  are identical. The commutant is said to be *demi-skew* when  $F_{\alpha\beta\gamma\dots}$  is altered in sign only, according to the rule of signs for any permutation inter se of non-identical characters  $\alpha, \beta, \gamma, \dots$ .

An intermutant is represented by a notation similar to that of a commutant. The sets are to be distinguished, whenever it is possible to do so, by placing in contiguity the symbols of the same set, and separating them by a stroke or bar from the symbols of the adjacent sets. If, however, the symbols of the same set cannot be placed contiguously, we may distinguish the symbols of a set by annexing to



them some auxiliary character by way of suffix or otherwise, these auxiliary symbols being omitted in the final result. Thus

$$\begin{pmatrix} 1 & 1 & 5_a \\ 2 & 2 & 3 \\ \hline 2_b & 5_b & 4 \\ 3 & 6_a & 6_b \end{pmatrix}$$

would show that 1, 2 of the first column and the 3, 4 of the same column, the 1, 2 and the upper 3 of the second column, and the lower 3 of the same column, the 1, 5 of the third column, and the 2, 6 of the same column, form so many distinct sets, the intermutant containing therefore  $(2 \cdot 2 \cdot 6 \cdot 1 \cdot 2 \cdot 2 =)$  96 terms.

A commutant of an even number of columns may be considered as an intermutant such that the characters of some one (no matter which) of its columns form each of them by itself a distinct set, and in like manner a commutant of an odd number of columns may be considered as an intermutant such that the characters of some one determinate column form each of them by itself a distinct set.

The distinction of symmetrical, skew and demi-skew applies obviously as well to intermutants as to commutants.

The theory of skew intermutants and skew commutants has a connexion with that of Pfaffians. Suppose  $F_{\alpha\beta\gamma\dots} = V_{\alpha+\beta+\gamma+\dots}$  (which implies the symmetry of the intermutant or commutant) and write for shortness  $V_0 = a$ ,  $V_1 = b$ ,  $V_2 = c$ , &c. Then

$$\begin{pmatrix} V_0 & V_1 \\ V_1 & V_2 \end{pmatrix} = 2(ac - b^2),$$

$$\begin{pmatrix} V_0 & V_1 & V_2 \\ V_1 & V_2 & V_3 \\ V_2 & V_3 & V_4 \end{pmatrix} = 2(ae - 4bd + 3c^2),$$

$$\begin{pmatrix} V_0 & V_1 \\ V_1 & V_2 \end{pmatrix}^{\dagger} = (ac - b^2), \quad \&c.$$

The functions on the second side are evidently hyperdeterminants such as are discussed in my memoir on Linear Transformations, and there is no difficulty in forming directly from the intermutant or commutant on the first side of the equation the symbol of derivation (in the sense of the memoir on Linear Transformations) from which the

hyperdeterminant is obtained. Thus

$$\begin{pmatrix} V_0 & V_1 \\ V_1 & V_2 \end{pmatrix} \text{ is } 12 \downarrow UU, \quad \begin{pmatrix} V_0 & V_1 \\ V_1 & V_2 \end{pmatrix}^\dagger \text{ is } 12 \downarrow U'U,$$

$$\begin{pmatrix} V_0 & V_1 \\ V_1 & V_2 \end{pmatrix} \text{ is } 12 \downarrow U'U^{-1}, \quad \begin{pmatrix} V_0 & V_1 \\ V_1 & V_2 \end{pmatrix}^\dagger \text{ is } 12 \downarrow U^{-2}U,$$

each permutable column corresponding to a  $12 \downarrow$  and a non-permutable column changing  $UU$  into  $U'U$ .

Similarly

$$\begin{pmatrix} V_0 & V_1 & V_2 \\ V_1 & V_2 & V_3 \\ V_2 & V_3 & V_4 \end{pmatrix} \text{ becomes } (12 \cdot 13 \cdot 23) \downarrow U^3U,$$

$$\begin{pmatrix} V_0 & V_1 & V_2 \\ V_1 & V_2 & V_3 \\ V_2 & V_3 & V_4 \end{pmatrix}^\dagger \text{ becomes } 12 \cdot 13 \cdot 23 \downarrow U^{-2}U^{-1}U^{-1},$$

$$\begin{pmatrix} V_0 & V_1 & V_2 & V_3 \\ V_1 & V_2 & V_3 & V_4 \\ V_2 & V_3 & V_4 & V_5 \\ V_3 & V_4 & V_5 & V_6 \end{pmatrix} \text{ is used to signify } (12 \cdot 13 \cdot 14 \cdot 23 \cdot 24 \cdot 34) \downarrow U^4U, \quad \&c.$$

The analogy would be closer if in the memoir on Linear Transformations, just as  $123$  had been used to signify

$$\frac{d}{d\lambda_1} \frac{d}{d\mu_1} \frac{d}{d\nu_1} V \cdot \frac{d}{d\lambda_2} \frac{d}{d\mu_2} \frac{d}{d\nu_2} V \cdot \frac{d}{d\lambda_3} \frac{d}{d\mu_3} \frac{d}{d\nu_3} V \cdot \&c.,$$

for then

$$\begin{pmatrix} V_0 & V_1 & V_2 \\ V_1 & V_2 & V_3 \\ V_2 & V_3 & V_4 \end{pmatrix}$$

would have corresponded to  $123 \cdot U^3U$ , and

$$\begin{pmatrix} V_0 & V_1 & V_2 \\ V_1 & V_2 & V_3 \\ V_2 & V_3 & V_4 \end{pmatrix}^\dagger$$

to  $123 \downarrow U^{-2}U^{-1}U^{-1}$ ; and this would not only have been an addition of some importance to the theory, but would in some instances have facilitated the calculation of hyperdeterminants.

The preceding remarks show that the intermutant

$$\begin{pmatrix} V_0 & V_1 & V_2 & V_3 \\ V_1 & V_2 & V_3 & V_4 \\ V_2 & V_3 & V_4 & V_5 \\ V_3 & V_4 & V_5 & V_6 \end{pmatrix}$$

(where the first and fourth blanks in the last column are to be considered as belonging to the same set) is in the hyperdeterminant notation  $(12 \cdot 34)^2 \cdot (14 \cdot 23) \downarrow U^2 U^2$ , viz. corresponds to 12 because 0 and 1 are the characters occupying the first and second blanks of a column.<sup>1</sup>

It will, I think, illustrate the general theory to perform the development of the last-mentioned intermutant. We have

$$\begin{aligned} \begin{pmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \\ d & e & f & g \end{pmatrix} &= \begin{pmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \\ d & e & f & g \end{pmatrix} + \begin{pmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \\ d & e & f & g \end{pmatrix} - \begin{pmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \\ d & e & f & g \end{pmatrix} - \begin{pmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \\ d & e & f & g \end{pmatrix} \\ &= 2 \begin{pmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \\ d & e & f & g \end{pmatrix} - 2 \begin{pmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \\ d & e & f & g \end{pmatrix} \\ &= 2\{(ad - bc)^2 - 4(ac - b^2)(bd - c^2)\} \\ &= 2(a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd), \end{aligned}$$

the different steps of which may be easily verified.

The following important theorem (which is, I believe, the same as a theorem of Mr Sylvester's, published in the Philosophical Magazine) is perhaps best exhibited by means of a simple example. Consider the intermutant

$$\begin{pmatrix} x & 1 \\ y & 2 \\ \hline x & 3 \\ y & 4 \end{pmatrix}$$

where in the first column the sets are distinguished as before by the horizontal bar, but in the second column the 1, 2 are to be considered

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<sup>1</sup>If 0 and 1 had been the characters occupying the second and third blanks in a column, the symbol would have been 23 and so on. It will be remembered, that the symbolic numbers 1, 2 in the hyperdeterminant notation are merely introduced to distinguish from each other functions which are made identical after certain differentiations are performed.

as forming a set, and the 3, 4 as forming a second set. Then, partially expanding, the intermutant is

$$\begin{pmatrix} x & 1 \\ y & 2 \end{pmatrix} \begin{pmatrix} x & 3 \\ y & 4 \end{pmatrix} - \begin{pmatrix} y & 1 \\ x & 2 \end{pmatrix} \begin{pmatrix} x & 3 \\ y & 4 \end{pmatrix} + \begin{pmatrix} x & 1 \\ y & 2 \end{pmatrix} \begin{pmatrix} y & 3 \\ x & 4 \end{pmatrix} - \begin{pmatrix} y & 1 \\ x & 2 \end{pmatrix} \begin{pmatrix} y & 3 \\ x & 4 \end{pmatrix},$$

or, since entire horizontal lines may obviously be permuted,

$$\begin{pmatrix} x & 1 \\ y & 2 \end{pmatrix} \begin{pmatrix} x & 3 \\ y & 4 \end{pmatrix} - \begin{pmatrix} x & 2 \\ y & 1 \end{pmatrix} \begin{pmatrix} x & 3 \\ y & 4 \end{pmatrix} + \begin{pmatrix} x & 1 \\ y & 2 \end{pmatrix} \begin{pmatrix} x & 4 \\ y & 3 \end{pmatrix} - \begin{pmatrix} x & 2 \\ y & 1 \end{pmatrix} \begin{pmatrix} x & 4 \\ y & 3 \end{pmatrix},$$

and, observing that the 1, 2 form a permutable system as do also the 3, 4, the second and third terms vanish, while the first and fourth terms are equivalent to each other; we may therefore write

$$\frac{\begin{pmatrix} x & 1 \\ y & 2 \\ x & 3 \\ y & 4 \end{pmatrix}}{\begin{pmatrix} x & 1 \\ y & 2 \end{pmatrix}} = \frac{\begin{pmatrix} x & 1 \\ y & 2 \\ x & 3 \\ y & 4 \end{pmatrix}}{\begin{pmatrix} x & 1 \\ y & 2 \end{pmatrix}},$$

where on the first side of the equation the bar has been introduced into the second column, in order to show that throughout the equation the 1, 2 and the 3, 4 are to be considered as forming distinct sets.

Consider in like manner the expression

$$\frac{\begin{pmatrix} x & 1 \\ y & 2 \\ z & 3 \\ x & 4 \\ y & 5 \\ z & 6 \\ x & 7 \\ y & 8 \\ z & 9 \end{pmatrix}}{\begin{pmatrix} x & 1 \\ y & 2 \\ z & 3 \end{pmatrix}} = \frac{\begin{pmatrix} x & 1 \\ y & 2 \\ z & 3 \\ x & 4 \\ y & 5 \\ z & 6 \\ x & 7 \\ y & 8 \\ z & 9 \end{pmatrix}}{\begin{pmatrix} x & 1 \\ y & 2 \\ z & 3 \end{pmatrix}},$$

where in the first column the sets are distinguished by the horizontal bars and in the second column the characters 1, 2, 3 and 4, 5, 6 and 7, 8, 9 are to be considered as belonging to distinct sets. The same

reasoning as in the former case will show that this is a multiple of

$$\begin{pmatrix} x & 1 \\ y & 2 \\ z & 3 \\ \hline x & 4 \\ y & 5 \\ z & 6 \\ \hline x & 7 \\ y & 8 \\ z & 9 \end{pmatrix}$$

and to find the numerical multiplier it is only necessary to inquire in how many ways, in the expression first written down, the characters of the first column can be permuted so that  $x, y, z$  may go with 1, 2, 3 and with 4, 5, 6 and with 7, 8, 9.

The order of the  $x, y, z$  in the second triad may be considered as arbitrary; but once assumed, it determines the place of one of the letters in the first triad; for instance,  $x8$  and  $z9$  determine  $y1$ . The first triad must therefore contain  $x1$  and  $z6$  or  $x6$  and  $z1$ . Suppose the former, then the third triad must contain  $z7$ , but the remaining two combinations may be either  $x4, y5$  or  $x5, y4$ . Similarly, if the first triad contained  $x6, z1$ , there would be two forms of the third triad, or a given form of the second triad gives four different forms. There are therefore in all 24 forms, or

$$24 \begin{pmatrix} x & 1 \\ y & 2 \\ z & 3 \\ \hline x & 4 \\ y & 5 \\ z & 6 \\ \hline x & 7 \\ y & 8 \\ z & 9 \end{pmatrix} = \begin{pmatrix} x & 1 \\ y & 2 \\ z & 3 \\ \hline x & 4 \\ y & 5 \\ z & 6 \\ \hline x & 7 \\ y & 8 \\ z & 9 \end{pmatrix},$$

where the bars in the second column on the first side show that throughout the equation 1, 2, 3 and 4, 5, 6 and 7, 8, 9 are to be considered as forming distinct sets.

The above proof is in reality perfectly general, and it seems hardly necessary to render it so in terms. To perceive the significance of the

above equation it should be noticed that the first side is a product of determinants, viz.

$$24 \left\{ \begin{pmatrix} x & 1 \\ y & 2 \end{pmatrix} \begin{pmatrix} x & 5 \\ y & 6 \end{pmatrix} \begin{pmatrix} x & 9 \\ z & 7 \end{pmatrix} + \dots \right\}$$

and if the second side be partially expanded by permuting the characters of the second column, each of the terms so obtained is in like manner a product of determinants, so that

$$24 \left\{ \begin{pmatrix} x & 1 \\ y & 2 \end{pmatrix} \begin{pmatrix} x & 5 \\ y & 6 \end{pmatrix} \begin{pmatrix} x & 9 \\ z & 7 \end{pmatrix} + \dots \right\} = \begin{pmatrix} x & 1 \\ y & 2 \\ z & 3 \\ \hline x & 4 \\ y & 5 \\ z & 6 \\ \hline x & 7 \\ y & 8 \\ z & 9 \end{pmatrix}.$$

### Postscript.

I wish to explain as accurately as I am able, the extent of my obligations to Mr Sylvester in respect of the subject of the present memoir. The term permutant is due to him—intermutant and commutant are merely terms framed between us in analogy with permutant, and the names date from the present year (1851). The theory of commutants is given in my memoir in the Cambridge Philosophical Transactions, [12], and is presupposed in the memoir on Linear Transformations, [13, 14].

It will appear by the last-mentioned memoir that it was by representing the coefficients of a biquadratic function by  $a = 1111$ ,  $b = 1112 = 1121 = \&c.$ ,  $c = 1122 = \&c.$ ,  $d = 1222 = \&c.$ ,  $e = 2222$ , and forming the commutant

$$\begin{pmatrix} 1111 \\ 2222 \end{pmatrix}$$

that I was led to the function  $ae - 4bd + 3c^2$ .

The function  $ace + 2bcd - ad^2 - b^2e - c^3$  is mentioned in the memoir on Linear Transformations, as brought into notice by Mr Boole. From the particular mode in which the coefficients  $a, b, \dots$

were represented by symbols such as 1111, &c., I did not perceive that the last-mentioned function could be expressed in the commutant notation.

The notion of a permutant, in its most general sense, is explained by me in my memoir, “*Sur les determinants gauches*,” Crelle, t. xxxvii. pp. 93—96, [69]; see the paragraph (p. 94) commencing “*On obtient ces fonctions, &c.*” and which should run as follows: “*On obtient ces fonctions (dont je reprends ici la théorie) par les propriétés générales d’un déterminant défini comme suit. En exprimant &c.*,” the sentence as printed being “*défini. Car en exprimant &c.*,” which confuses the sense. [The paragraph is printed correctly 69, p. 411.]

Some time in the present year (1851) Mr Sylvester, in conversation, made to me the very important remark, that as one of a class the above-mentioned function,

$$ace + 2bcd - ad^2 - b^2e - c^3,$$

would be expressed in the commutant notation, viz. by considering

$$00 = a, \quad 01 = 10 = b, \quad 02 = 11 = 20 = c, \quad 12 = 21 = d, \quad 22 = e;$$

and the subject being thereby recalled to my notice, I found shortly afterwards the expression for the function

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd$$

(which cannot be expressed as a commutant) in the form of an intermutant, and I was thence led to see the identity, so to say, of the theory of hyperdeterminants, as given in the memoir on Linear Transformations, with the present theory of intermutants.

It is understood between Mr Sylvester and myself, that the publication of the present memoir is not to affect Mr Sylvester’s right to claim the origination, and to be considered as the first publisher of such part as may belong to him of the theory here sketched out.

## 105. Correction of the Postscript to the Paper on Permutants.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 97—98.]

Mr Sylvester has represented to me that the paragraph relating to his communications conveys an erroneous idea of the nature, purport, and extent of such communications; I have, in fact, in the paragraph in question, singled out what immediately suggested to me the expression of the function

$$Qabcd + 3b^2c^2 - 4a^2d^2 - 4b^3d - a^2d^2$$

as a partial commutant or intermutant, but I agree that a fuller reference ought to have been made to Mr Sylvester's results, and that the paragraph in question would more properly have stood as follows:

"Under these circumstances Mr Sylvester communicated to me a series of formal statements, not only oral but in writing, to the effect that he had discovered a permutation method of obtaining as many invariants—viz. commutative invariants—by direct inspection from a function of any degree of any number of letters as the index of the degree contains even factors; one of these commutative invariants being in fact the function

$$ace + 2bcd - ad^2 - b^2e - c^3,$$

expressible, according to Mr Sylvester's notation of my memoir in the Camb. Phil. Trans., supposing  $00 = a$ ,  $01 = 10 = b$ ,  $02 = 11 = 20 = c$ , &c. by

$$\begin{pmatrix} 00 & 11 & 22 \\ a & b & c \\ b & c & d \\ c & d & e \end{pmatrix},$$

and so on."

Mr Sylvester and I shall, I have no doubt, be able to agree to a joint statement of any further correction or explanation which may be required.



## 106. On the Singularities of Surfaces.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 166—171.]

In the following paper, for symmetry of nomenclature and in order to avoid ambiguities, I shall, with reference to plane curves and in various phrases and compound words, use the term “node” as synonymous with double point, and the term “spinode” as synonymous with cusp.

I shall, besides, have occasion to consider the several singularities which I call the “flecnode,” the “oscnode,” the “fleflecnode,” and the “tacnode:” the flecnode is a double point which is a point of inflexion on one of the branches through it; the oscnode is a double point which is a point of osculation on one of the branches through it; the fleflecnode is a double point which is a point of inflexion on each of the branches through it; and the tacnode is a double point where two branches touch.

And it may be proper to remark here, that a tacnode may be considered as a point resulting from the coincidence and amalgamation of two double points (and therefore equivalent to twelve points of inflexion) or, in a different point of view, as a point uniting the characters of a spinode and a flecnode.

I wish to call to mind here the definition of conjugate tangent lines of a surface, viz. that a tangent to the curve of contact of the surface with any circumscribed developable and the corresponding generating line of the developable, are conjugate tangents of the surface.

Suppose, now, that an absolutely arbitrary surface of any order be intersected by a plane: the curve of intersection has not in general any singularities other than such as occur in a perfectly arbitrary curve of the same order; but as a plane can be made to satisfy one, two, or three conditions, the curve may be made to acquire singularities which do not occur in such absolutely arbitrary curve.

Let a single condition only be imposed on the plane. We may suppose that the curve of intersection has a node; the plane is then a tangent plane and the node is the point of contact—of course any point on the surface may be taken for the node.<sup>2</sup> We may if we plea-

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<sup>2</sup>The flecnode-curve, defined as the locus of the points through which can be drawn a line meeting the surface in four consecutive points, was, so far as I am aware, first noticed in Mr Salmon's paper “On the Triple Tangent Planes of a Surface of the Third Order” (Journal, t. iv. [1849], pp. 252—260), where Mr

se use the term “nodes of a surface,” “node-planes of a surface,” as synonymous with the points and tangent planes of a surface. And it will be convenient also to use the word node-tangents to denote the tangents to the curve of intersection at the node; it may be noticed here that the node-tangents are conjugate tangents of the surface.

Next let two conditions be imposed upon the plane: there are three distinct cases to be considered.

First, the curve of intersection may have a flecnode. The plane (which is of course still a tangent plane at the flecnode) may be termed a flecnode-plane; the flecnodes are singular points on the surface lying on a curve which may be termed the “flecnode-curve,”<sup>3</sup> and the flecnode-planes give rise to a developable which may be termed the flecnode-develope. The “flecnode-tangents” are the tangents to the curve of intersection at the flecnode; the tangent to the inflected branch may be termed the “singular flecnode-tangent,” and the tangent to the other branch the “ordinary flecnode-tangent.”

Secondly, the curve of intersection may have a spinode. The plane (which is of course still a tangent plane at the spinode) may be termed a spinode-plane; the spinodes are singular points on the surface lying on a curve which may be termed the “spinode-curve.” And the spinode-planes give rise to a developable which may be termed the “spinode-develope.” Also the “spinode-tangent” is the tangent to the curve of intersection at the spinode.

Thirdly, the curve of intersection may have two nodes, or what may be termed a “node-couple.” The plane (which is a tangent plane at each of the nodes and therefore a double tangent plane) may be also termed a “node-couple-plane.” The node-couples are pairs of singular points on the surface lying in a curve which may be termed the “node-couple-curve,” and the node-couple-planes give rise to a developable

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Salmon, among other things, shows that the order of the surface being  $n$ , the curve in question is the intersection of the surface with a surface of the order  $11n - 24$ .

<sup>3</sup>The notion of a spinode, considered as the point where the indicatrix is a parabola (on which account the spinode has been termed a parabolic point) may be found in Dupin's *Developpements de Geometrie*: the most important step in the theory of these points is contained in Hesse's memoir “Ueber die Wendepuncte der Curven dritter Ordnung” (Crelle, t. xxviii. [1848], pp. 97—107), where it is shown that the spinode-curve is the curve of intersection of the surface supposed as before of the order  $n$ , with a certain surface of the order  $4(n - 2)$ . See also Mr Salmon's memoir “On the Condition that a Plane should touch a surface along a Curve Line” (Journal, t. iii. [1848], pp. 44—46).

which may be termed the “node-couple-develope.” The tangents to the curve of intersection at the two nodes of a node-couple might, if the term were required, be termed the “node-couple-tangents.” Also one of the nodes of a node-couple may be termed a “node-with-node,” and the tangents to the curve of intersection at such point will be the “node-with-node-tangents.”

It is hardly necessary to remark that the flecnode-curve is not the edge of regression of the flecnode-develope, and the like remark applies to the spinode-curve and the node-couple curve.

Finally, let three conditions be imposed upon the plane: there are six distinct cases to be considered, in each of which we have no longer curves and developes, but only singular points and singular tangent planes determinate in number.

**First**, the curve of intersection may have an oscnode. The plane (which continues a tangent plane at the oscnode) is an “oscnode-plane.” The “oscnode-tangents” are the tangents to the curve of intersection at the oscnode; the tangent to the osculating branch is the “singular oscnode-tangent,” and the tangent to the other branch the “ordinary oscnode-tangent.”

**Secondly**, the curve of intersection may have a fleflecnode. The plane (which continues a tangent plane at the fleflecnode) is a “fleflecnode-plane.” The “fleflecnode-tangents” are the tangents to the curve of intersection at the fleflecnode.

**Thirdly**, the curve of intersection may have a tacnode. The plane (which continues a tangent plane at the tacnode) is a “tacnode-plane.” The “tacnode-tangent” is the tangent to the curve of intersection at the tacnode.

**Fourthly**, the curve of intersection may have a node and a flecnode, or what may be termed a node-and-flecnode. The plane (which is a tangent plane at the node and also at the flecnode, where it is obviously a flecnode-plane) is a “node-and-flecnode-plane.” The “node-and-flecnode-tangents,” if the term were required, would be the tangents to the curve of intersection at the node and at the flecnode of the node-and-flecnode. The node of the node-and-flecnode may be distinguished as the node-with-flecnode, and the flecnode as the flecnode-with-node, and we have thus the terms “node-with-flecnode-tangents,” “flecnode-with-node-tangents,” “singular flecnode-with-node-tangent,” and “ordinary flecnode-with-node-tangent.”

**Fifthly**, the curve of intersection may have a node and also a spinode, or what may be termed a “node-and-spinode.” The plane (which

is a tangent plane at the node, and is also a tangent plane at the spinode, where it is obviously a spinode-plane) is a “node-and-spinode-plane.” The node-and-spinode-tangents, if the term were required, would be the tangents at the node and the tangent at the spinode of the node-and-spinode to the curve of intersection. The node of the node-and-spinode may be distinguished as the “node-with-spinode,” and the spinode as the “spinode-with-node,” and we have thus the terms “node-with-spinode-tangent,” “spinode-with-node-tangent.”

**Sixthly**, the curve of intersection may have three nodes, or what may be termed a “node-triplet.” The plane (which is a triple tangent plane touching the surface at each of the nodes) is a “node-triplet-plane.” The “node-triplet-tangents,” if the term were required, would be the tangents to the curve of intersection at the nodes of the node-triplet. Each node of the node-triplet may be distinguished as a “node-with-node-couple,” and the tangents to the curve of intersection at such nodes are “node-with-node-couple-tangents.” The terms “node-couple-with-node,” “node-couple-with-node-tangent,” might be made use of if necessary.

It should be remarked that the oscnodes lie on the flecnode-curve, as do also the fleflecnodes; these latter points are real double points of the flecnode-curve. The tacnodes are points of intersection and (what will appear in the sequel) points of contact of the flecnode-curve, the spinode-curve, and the node-couple-curve. The spinode-with-nodes are points of intersection of the spinode-curve and node-couple-curve, and the flecnode-with-nodes are points of intersection of the flecnode-curve and node-couple-curve; the node-with-node-couples are real double points (entering in triplets) of the node-couple-curve.

Consider for a moment an arbitrary curve on the surface; the locus of the node-tangents at the different points of this curve is in general a skew surface, which may however, in cases to be presently considered, degenerate in different ways. Reverting now to the flecnode-curve, it may be shown that the singular flecnode-tangent coincides with the tangent of the flecnode-curve. For consider on a surface two consecutive points such that the line joining them meets the surface in two points consecutive to the first-mentioned two points. The line meets the surface in four consecutive points, it is therefore a singular flecnode-tangent; each of the first-mentioned two points must be on the flecnode-curve, or the singular flecnode-tangent touches the flecnode-curve. The two flecnode-tangents are by a preceding observation conjugate tangents. It follows that the skew surface, locus of

the flecnode-tangents, breaks up into two surfaces, each of which is a developable, viz. the locus of the singular flecnode-tangents is the developable having the flecnode-curve for its edge of regression, and the locus of the ordinary flecnode-tangents is the flecnode-develope. Of course at the tacnode, the tacnode-tangent touches the flecnode-curve.

Passing next to the spinode-curve, the spinode-plane and the tangent-plane at a consecutive point along the spinode-tangent are identical, or their line of intersection is indeterminate.<sup>4</sup> The spinode-tangent is therefore the conjugate tangent to any other tangent line at the spinode, and therefore to the tangent to the spinode-curve. It follows that the surface locus of the spinode-tangents degenerates into a developable surface twice repeated, viz. the spinode-develope.

Consider the tacnode as two coincident nodes; each of these nodes, by virtue of its constituting, in conjunction with the other, a tacnode, is on the spinode-curve; or, in other words, the tacnode-tangent touches the spinode-curve, and the same reasoning proves that it touches the node-couple-curve. It has already been seen that the tacnode-tangent touches the flecnode-curve; consequently the tacnode is a point, not of simple intersection only, but of contact, of the flecnode-curve, the spinode-curve, and the node-couple-curve.

In virtue of the principle of the spinode-plane being identical with the tangent plane at a consecutive point along the spinode tangent, it appears that the tacnode-plane is a stationary plane, as well of the flecnode-develope as of the spinode-develope, and it would at first sight appear that it must be also a stationary tangent plane of the node-couple-develope. But this is not so; the node-with-node-planes envelope, not the node-couple-develope, but the node-couple-develope twice repeated: the tacnode-plane is in a sense a stationary plane on such duplicate developable, but not in any manner on the single developable. The tacnode-plane is an ordinary tangent plane of the node-couple-develope.

Consider now a spinode-with-node, which we have seen is a point of intersection of the spinode-curve and node-couple-curve. The tangent plane at a consecutive point along the spinode-with-node-tangent, is identical with the spinode-with-node-plane; the curve of intersection of the tangent plane at such consecutive point has therefo-

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<sup>4</sup>It must not be inferred that the tangent plane at such consecutive point is a spinode-plane; this is obviously not the case.

re a node at the node-with-spinode, or the tangent plane in question is a node-couple-plane, and the point of contact is a point on the node-couple-curve. Consequently the spinode-with-node-tangent touches the node-couple-curve, and thence also the spinode-with-node-plane is a stationary tangent plane of the node-couple-develope.

It should be remarked that no circumscribed developable can have a stationary tangent plane except the tangent planes at the points where the curve of contact meets the spinode-curve, and any one of these planes is only a stationary plane when the curve of contact touches the spinode-tangent; and that the node-couple-curve and the flecnode-curve do not intersect the spinode-curve except in the points which have been discussed.

Recapitulating, the node-couple-curve and the spinode-curve touch at the tacnodes, and intersect at the spinode-with-nodes: moreover, the tacnode-planes are stationary planes of the spinode-develope, and the spinode-with-node-planes are stationary planes of the node-couple-develope. Besides this, the two curves are touched at the tacnodes by the flecnode-curve, and the tacnode-planes are stationary planes of the flecnode-develope.

## 107. On the Theory of Skew Surfaces.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 171—173.]

A SURFACE of the  $n^{\text{th}}$  order is a surface which is met by an indeterminate line in  $n$  points. It follows immediately that a surface of the  $n^{\text{th}}$  order is met by an indeterminate plane in a curve of the  $n^{\text{th}}$  order.

Consider a skew surface or the surface generated by a singly infinite series of lines, and let the surface be of the  $n^{\text{th}}$  order. Any plane through a generating line meets the surface in the line itself and in a curve of the  $(n-1)^{\text{th}}$  order. The generating line meets this curve in  $(n-1)$  points. Of these points one, viz. that adjacent to the intersection of the plane with the consecutive generating line, is a unique point; the other  $(n-2)$  points form a system. Each of the  $(n-1)$  points are sub modo points of contact of the plane with the surface, but the proper point of contact is the unique point adjacent to the intersection of the plane with the consecutive generating line. Thus every plane through a generating line is an ordinary tangent plane, the point of contact being a point on the generating line.

It is not necessary for the present purpose, but I may stop for a moment to refer to the known theorems that the anharmonic ratio of any four tangent planes through the same generating line is equal to the anharmonic ratio of their points of contact, and that the locus of the normals to the surface along a generating line is a hyperbolic paraboloid.

Returning to the  $(n-2)$  points in which, together with the point of contact, a generating line meets the curve of intersection of the surface and a plane through the generating line, these are fixed points independent of the particular plane, and are the points in which the generating line is intersected by other generating lines.

There is therefore on the surface a double curve intersected in  $(n-2)$  points by each generating line of the surface—a property which, though insufficient to determine the order of this double curve, shows that the order cannot be less than  $(n-2)$ .

(Thus for  $n=4$ , the above reasoning shows that the double-curve must be at least of the second order: assuming for a moment that it is in any case precisely of this order, it obviously cannot be a plane curve, and must therefore be two non-intersecting lines. This suggests at any rate the existence of a class of skew surfaces of the fourth order

generated by a line which always passes through two fixed lines and by some other condition not yet ascertained; and it would appear that surfaces of the second order constitute a degenerate species belonging to the class in question.)

In particular cases a generating line will be intersected by the consecutive generating line. Such a generating line touches the double curve.

Consider now a point not on the surface; the planes determined by this point and the generating lines of the surface are the tangent planes through the point; the intersections of consecutive tangent planes are the tangent lines through the point; and the cone generated by these tangent lines or enveloped by the tangent planes is the tangent cone corresponding to the point. This cone is of the  $n^{\text{th}}$  class. For considering a line through the point, this line meets the surface in  $n$  points, i.e. it meets  $n$  generating lines of the surface; and the planes through the line and these  $n$  generating lines, are of course tangent planes to the cone: that is,  $n$  tangent planes can be drawn to the cone through a given line passing through the vertex.

The cone has not in general any lines of inflexion, or, what is the same thing, stationary tangent planes. For a stationary tangent plane would imply the intersection of two consecutive generating lines of the surface. And since the number of generating lines intersected by a consecutive generating line, and therefore the number of planes through two consecutive generating lines, is finite, no such plane passes through an indeterminate point.

The tangent cone will have in general a certain number of double tangent planes; let this number be  $\chi$ . We have therefore a cone of the class  $n$ , number of double tangent planes  $\chi$ , number of stationary tangent planes 0. Hence, if  $m$  be the order of the cone,  $\alpha$  the number of its double lines, and  $\beta$  the number of its cuspidal or stationary lines,

$$\begin{aligned} m &= n(n-1) - 2\chi, \\ \beta &= 3n(n-2) - 6\chi, \\ \alpha &= \frac{1}{2}n(n-2)(n^2-9) - 2\chi(n^2-n-6) + 2\chi(\chi-1). \end{aligned}$$

This is the proper tangent cone, but the cone through the double curve is sub modo a tangent cone, and enters as a square factor into the equation of the general tangent cone of the order  $n(n-1)$ . Hence, if  $X$  be the order of the double curve, and therefore of the cone



through this curve,

$$m + 2X = n(n - 1),$$

and therefore  $X = \chi$ ; that is, the number of double tangent planes to the tangent cone is equal to the order of the double curve.

It does not appear that there is anything to determine  $\chi$ ; and if this is so, skew surfaces of the  $n^{\text{th}}$  order may be considered as forming different families according to the order of the double curve upon them. To complete the theory, it should be added that a plane intersects the surface in a curve of the  $n^{\text{th}}$  order having  $\chi$  double points but no cusps.

## 108. On certain Multiple Integrals connected with the Theory of Attractions.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 174—178.]

It is easy to deduce from Mr Boole's formula, given in my paper "On a Multiple Integral connected with the theory of Attractions," Journal, t. ii. [1847], pp. 219—223, [44], the equation

$$\int \frac{d\xi \dots}{[(\xi - a)^2 + \dots + v^2]^{\frac{1}{2}n - q}} = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n - q) \Gamma(q + 1)} \int_0^\epsilon \frac{s^{q-1}(e - s)^q ds}{(1 + s)^{\frac{1}{2}n + q}},$$

where  $n$  is the number of variables of the multiple integral, and the condition of the integration is also

$$(\xi - a)^2 + (\eta - \beta)^2 + \dots + v^2 < e^2;$$

and  $e$  is the positive root of

$$\xi^2 + \eta^2 + \dots = v^2.$$

Suppose  $f = f_1 = \dots = f_n$ , and write  $(a - a_1)^2 + \dots = k^2$ , we obtain

$$\int \frac{d\xi \dots}{[(\xi - a)^2 + \dots + v^2]^{\frac{1}{2}n - q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q + 1)} \int_0^\epsilon \frac{s^{q-1}(e - s)^q ds}{(1 + s)^{\frac{1}{2}n + q}},$$

the limiting condition for the multiple integral being

$$(\xi - a_1)^2 + \dots < f_1^2,$$

and the function  $e$ , and limit  $\epsilon$ , being given by

$$e^2 + \xi^2 + \dots = v^2, \quad \epsilon \text{ denoting, as before, the positive root.}$$

Observing that the quantity under the integral sign on the second side vanishes for  $s = \epsilon$ , there is no difficulty in deducing, by a differentiation with respect to  $\xi_1$ , the formula

$$\int \frac{dS}{[(\xi - a)^2 + \dots + v^2]^{\frac{1}{2}n - q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q)} \int_0^\epsilon \frac{s^{q-1}(e - s)^{q-1} ds}{(1 + s)^{\frac{1}{2}n + q}},$$

where  $dS$  is the element of the surface  $(\xi - a_1)^2 + \dots = f_1^2$  and the integration is extended over the entire surface.

A slight change of form is convenient. We have

$$s^{q-1}(e-s)^{q-1}ds = \frac{1}{s(1+s)} \cdot \frac{s^q(e-s)^q}{e} de,$$

if we suppose

$$e = \frac{\xi_1^2 + \dots - f_1^2}{(\xi - a_1)^2 + \dots}.$$

The formulae then become

$$\int \frac{d\xi \dots}{[(\xi - a)^2 + \dots + v^2]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n-q)\Gamma(q)} \int_0^\epsilon \frac{(\xi_1^2 + \dots - f_1^2)^{q-1} de}{e(1+e)(\xi - a_1)^2 + \dots^{\frac{1}{2}n-q-1}},$$

in which  $e$  is the positive root of the equation

$$\xi_1^2 e^2 + \lambda e - \mu = 0.$$

I propose to transform these formulae by means of the theory of images; it will be convenient to investigate some preliminary formulae.

Suppose  $l^2 = a^2 + b^2 + \dots$ ,  $l_1^2 = a_1^2 + b_1^2 + \dots$ ; also consider the new constants  $a, b, \dots, a_1, b_1, \dots, u, \mu$ , determined by the equations

$$\frac{S}{u} = \frac{a}{a_1} = \frac{b}{b_1} = \dots = \frac{l^2}{\mu} = \frac{l_1^2}{S},$$

where  $S$  is arbitrary.

Then, putting

$$x = \frac{S\xi}{\xi^2 + \eta^2 + \dots}, \quad y = \frac{S\eta}{\xi^2 + \eta^2 + \dots}, \quad \dots$$

it is easy to see that

$$(x-a)^2 + (y-b)^2 + \dots = \frac{S^2}{(\xi^2 + \eta^2 + \dots)^2} \{(\xi - a)^2 + (\eta - \beta)^2 + \dots\},$$

and

$$\frac{a}{l^2} = \frac{a_1}{l_1^2}, \quad \frac{S^2 a}{l^2 + uS} = a, \quad \frac{S^2 a_1}{l_1^2 + \mu S} = a_1.$$

Proceeding to express the single integrals in terms of the new constants, we have in the first place  $k^2 = h^2 \cdot l^2$ , where

$$h^2 = \frac{(l^2 + uS)(l_1^2 + \mu S)}{S^2} = \frac{l^2 l_1^2 + S(\mu l^2 + u l_1^2) + u\mu S^2}{S^2};$$

or if we write

$$aa_1 + bb_1 + \dots = ll_1 \cos \omega,$$

we have

$$h^2 = l^2 + l_1^2 - 2ll_1 \cos \omega.$$

Hence also  $k^{-2} = h^{-2} \cdot l^{-2}$ , where

$$\frac{l^2 + uS}{S^2 a} = \frac{1}{a}, \quad \text{whence} \quad \frac{(l^2 + uS)(l_1^2 + \mu S)}{S^2 ll_1 \cos \omega} = \frac{1}{\cos \omega};$$

$$\frac{(k^2 - \lambda^2)(l^2 + uS)(l_1^2 + \mu S)}{S^2 ll_1} = \frac{1}{\rho^2},$$

where  $\rho^2 = l^2 + l_1^2 - 2ll_1 \cos \omega$ , that is  $\rho^2 = (a - a_1)^2 + (b - b_1)^2 + \dots$ ; consequently  $\frac{1}{k^2} \cdot \frac{l^2 + uS}{l_1^2 + \mu S} = \frac{S^2}{\Pi}$ , where  $\Pi$  is given by

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2)e - u\mu = 0,$$

and it is clear that  $e$  will be the positive root of

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2)e - u\mu = 0.$$

It may be noticed that, in the particular case of  $u = 0$ , the roots of this equation are 0, and  $\frac{\mu}{\lambda^2 - \mu}$ . Consequently if  $\rho^2 - f_1^2$  and  $l^2 - f^2$  are of opposite signs, we have  $e = 0$ ; but if  $\rho^2 - f_1^2$  and  $l^2 - f^2$  are of the same sign,  $e = \frac{\mu}{\lambda^2 - \mu}$ .

In order to transform the double integrals, considering the new variables  $x, y, \dots$ , I write  $x^2 + y^2 + \dots = r^2$  and

$$\xi = \frac{Sx}{x^2 + y^2 + \dots}, \quad \eta = \frac{Sy}{x^2 + y^2 + \dots}, \quad \dots$$

whence also, if  $\xi^2 + \eta^2 + \dots = \rho^2$  (which gives  $r\rho = S^2$ ), we have

$$\frac{dx}{x} = \frac{d\xi}{\xi} - \frac{2x dx + 2y dy + \dots}{x^2 + y^2 + \dots}, \quad \&c.;$$

also it is immediately seen that

$$\frac{(\xi - a)^2 + \dots + v^2}{(x^2 + y^2 + \dots)^2} = \frac{S^2}{(x^2 + y^2 + \dots)^2} \{(x - a)^2 + (y - b)^2 + \dots + u^2\},$$

$$\frac{(\xi - a_1)^2 + \dots - f_1^2}{(x^2 + y^2 + \dots)^2} = \frac{S^2}{(x^2 + y^2 + \dots)^2} \{(x - a_1)^2 + (y - b_1)^2 + \dots - f_1^2\};$$

and from the latter equation it follows that the limiting condition for the first integral is

$$(x - a_1)^2 + (y - b_1)^2 + \dots > f_1^2$$

(there is no difficulty in seeing that the sign  $<$  in the former limiting condition gives rise here to the sign  $>$ ), and that the second integral has to be extended over the surface  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2$ . Also if  $dS$  represent the element of this surface, we may obtain

$$d\xi d\eta \dots = \frac{S^2}{r^2} dx dy \dots, \quad dS = \frac{S^2}{r^2} dS;$$

and, combining the above formulae, we obtain

$$\int \frac{dx dy \dots}{(x^2 + y^2 + \dots)^{\frac{1}{2}n+q} \{(x - a)^2 + (y - b)^2 + \dots + u^2\}^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q+1)(l^2 + \dots)}$$

the limiting condition of the multiple integral being

$$(x - a_1)^2 + (y - b_1)^2 + \dots > f_1^2;$$

and

$$\int \frac{dS}{(x^2 + y^2 + \dots)^{\frac{1}{2}n+q-1} \{(x - a)^2 + (y - b)^2 + \dots + u^2\}^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n - q) \Gamma(q)(l^2 + \dots)}$$

where  $dS$  is the element of the surface  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2$ , and the integration extends over the entire surface.

In these formulae,  $l, l_1, \rho, \Pi$  denote as follows:

$$l^2 = a^2 + b^2 + \dots,$$

$$l_1^2 = a_1^2 + b_1^2 + \dots,$$

$$\rho^2 = (a - a_1)^2 + (b - b_1)^2 + \dots,$$

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2)e - u\mu = 0,$$

and  $e$  is the positive root of the equation  $\Pi = 0$ .

The only obviously integrable case is that for which in the second formula  $q = \frac{1}{2}$ ; this gives

$$\int \frac{dS}{(x^2 + y^2 + \dots)^{\frac{1}{2}n} \{(x-a)^2 + (y-b)^2 + \dots + u^2\}^{\frac{1}{2}n-1}} = \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma(\frac{1}{2}n)(l^2 + u^2)^{\frac{1}{2}n-1}(l_1^2 - f_1^2)}.$$

In the case of  $u = 0$ , we have, as before, when  $\rho^2 - f_1^2$  and  $l^2 - f^2$  are of opposite signs,  $e = 0$ , and therefore  $1 + e = 1$ ; but when  $\rho^2 - f_1^2$  and  $l^2 - f^2$  are of the same sign, the value before found for  $e$  gives

$$1 + e = \frac{\rho^2}{\rho^2 - f_1^2} \{l_1^2 + (\rho^2 - f_1^2)(l^2 - f^2)\}.$$

Considered in reference to the theory of images, the first formula gives the potential of a multiple layer (the density being inversely as the  $(n-1)^{\text{th}}$  power of the distance) spread over the surface of an  $(n-1)$ -fold space, due to an attracting mass (the law of attraction being the inverse  $(n-1)^{\text{th}}$  power of the distance) situated at a given point; and the second formula gives the potential of the attracting mass itself. It is easy to see that the formula agrees with the ordinary theory of electrical images.

ON CERTAIN MULTIPLE INTEGRALS  
connected with the *Theory of Attractions*.

[From the Cambridge and Dublin Mathematical Journal, vol. III. (1848), pp. 84–91.]

it is easy to see that

$$(\lambda^2 + \nu^2)(l^2 + u^2) = \delta^4, \quad (\lambda_1^2 - \theta_1^2)(l_1^2 - f_1^2) = \delta^4,$$

and

$$\begin{aligned} \frac{\delta^2 a}{l^2 + u^2} &= \alpha, & \frac{\delta^2 a_1}{l_1^2 - f_1^2} &= \alpha_1, \\ &\vdots & &\vdots \\ \frac{\delta^2 u}{l^2 + u^2} &= \nu, & \frac{\delta^2 f_1}{l_1^2 - f_1^2} &= \theta_1. \end{aligned}$$

Proceeding to express the single integrals in terms of the new constants, we have in the first place  $k^2 = \delta^4 k'^2$ , where

$$k'^2 = \left( \frac{a}{l^2 + u^2} - \frac{a_1}{l_1^2 - f_1^2} \right)^2 + \dots;$$

or if we write

$$aa_1 + bb_1 + \dots = ll_1 \cos \omega,$$

we have

$$k'^2 = \frac{l^2}{(l^2 + u^2)^2} + \frac{l_1^2}{(l_1^2 - f_1^2)^2} - \frac{2ll_1 \cos \omega}{(l^2 + u^2)(l_1^2 - f_1^2)}.$$

Hence also  $\chi = \delta^4 j$ , where

$$j = \frac{f_1^2}{(l_1^2 - f_1^2)^2} - k'^2 - \frac{u^2}{(l^2 + u^2)^2},$$

whence

$$\begin{aligned} -j &= \frac{1}{l^2 + u^2} + \frac{1}{l_1^2 - f_1^2} + \frac{2ll_1 \cos \omega}{(l^2 + u^2)(l_1^2 - f_1^2)}, \\ &= \frac{1}{(l^2 + u^2)(l_1^2 - f_1^2)} \{p^2 + u^2 - f_1^2\}, \end{aligned}$$

where  $p^2 = l^2 + l_1^2 - 2ll_1 \cos \omega$ , that is

$$p^2 = (a - a_1)^2 + (b - b_1)^2 + \dots;$$

consequently  $\theta_1^2 s^2 + \chi s - \nu^2 = \delta^4 \Pi$ , where  $\Pi$  is given by

$$\Pi = \frac{f_1^2}{(l_1^2 - f_1^2)^2} s^2 - \frac{(p^2 + u^2 - f_1^2)}{(l^2 + u^2)(l_1^2 - f_1^2)} s - \frac{u^2}{(l^2 + u^2)^2};$$

and it is clear that  $\epsilon$  will be the positive root of

$$0 = \frac{f_1^2}{(l_1^2 - f_1^2)^2} \epsilon^2 - \frac{(p^2 + u^2 - f_1^2)}{(l^2 + u^2)(l_1^2 - f_1^2)} \epsilon - \frac{u^2}{(l^2 + u^2)^2}.$$

It may be noticed that, in the particular case of  $u = 0$ , the roots of this equation are 0, and  $\frac{(p^2 - f_1^2)(l_1^2 - f_1^2)}{l^2 f_1^2}$ . Consequently if  $p^2 - f_1^2$  and  $l_1^2 - f_1^2$  are of opposite signs, we have  $\epsilon = 0$ ; but if  $p^2 - f_1^2$  and  $l_1^2 - f_1^2$  are of the same sign,  $\epsilon = \frac{(p^2 - f_1^2)(l_1^2 - f_1^2)}{l^2 f_1^2}$ .

---

In order to transform the double integrals, considering the new variables  $x, y, \dots$ , I write  $x^2 + y^2 + \dots = r^2$  and

$$\xi = \frac{lx}{r}, \quad \eta = \frac{ly}{r}, \dots, \quad \zeta = \frac{ux}{r}, \quad \omega = \frac{uy}{r}, \dots;$$

whence also, if  $\xi^2 + \eta^2 + \dots = l^2$  (which gives  $rp = \delta^2$ ), we have

$$\xi^2 + \eta^2 + \dots = \frac{l^2}{r^2} (x^2 + y^2 + \dots) = l^2,$$

also it is immediately seen that

$$\begin{aligned} (\xi - a)^2 + (\eta - b)^2 + \dots &= \frac{l^2}{r^2} \{(x - a)^2 + (y - b)^2 + \dots\} + u^2, \\ (\xi - a_1)^2 + (\eta - b_1)^2 + \dots - f_1^2 &= \frac{l^2}{r^2} \{(x - a_1)^2 + (y - b_1)^2 + \dots - f_1^2\}; \end{aligned}$$

and from the latter equation it follows that the limiting condition for the first integral is

$$(x - a_1)^2 + (y - b_1)^2 + \dots \geq f_1^2$$



(there is no difficulty in seeing that the sign  $<$  in the former limiting condition gives rise here to the sign  $>$ ), and that the second integral has to be extended over the surface

$$(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2.$$

Also if  $dS$  represent the element of this surface, we may obtain

$$d\xi d\eta \dots = \frac{l^2}{r^2} dx dy \dots, \quad d\xi = \frac{l^2}{r^2} dS;$$

and, combining the above formulæ, we obtain

$$\begin{aligned} & \int \frac{dx dy \dots}{(x^2 + y^2 + \dots)^{\frac{1}{2}m+q} \{(x-a)^2 + (y-b)^2 + \dots + u^2\}^{\frac{1}{2}n-q}} \\ &= \frac{\Gamma(\frac{1}{2}m-q)\Gamma(q+1)}{(l^2 + u^2)^{\frac{1}{2}m-q}} \int \frac{\epsilon^q dS}{(1 + \epsilon s)^{\frac{1}{2}m+q}}, \end{aligned}$$

the limiting condition of the multiple integral being

$$(x - a_1)^2 + (y - b_1)^2 + \dots \geq f_1^2;$$

and

$$\begin{aligned} & \int \frac{dS}{(x^2 + y^2 + \dots)^{\frac{1}{2}m+q-1} \{(x-a)^2 + (y-b)^2 + u^2\}^{\frac{1}{2}n-q}} \\ &= \frac{\Gamma(\frac{1}{2}m-q)\Gamma q}{(l^2 + u^2)^{\frac{1}{2}m-q-1}(l_1^2 - f_1^2)} \int \frac{dS}{(1 + s)^{\frac{1}{2}m+q-1}}, \end{aligned}$$

where  $dS$  is the element of the surface  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2$ , and the integration extends over the entire surface.

In these formulæ,  $l$ ,  $l_1$ ,  $p$ ,  $\Pi$  denote as follows:

$$\begin{aligned} l^2 &= a^2 + b^2 + \dots, \\ l_1^2 &= a_1^2 + b_1^2 + \dots, \\ p^2 &= (a - a_1)^2 + (b - b_1)^2 + \dots, \\ \Pi &= \frac{f_1^2}{(l_1^2 - f_1^2)^2} \epsilon^2 - \frac{(p^2 + u^2 - f_1^2)}{(l^2 + u^2)(l_1^2 - f_1^2)} \epsilon - \frac{u^2}{(l^2 + u^2)^2}, \end{aligned}$$

and  $\epsilon$  is the positive root of the equation  $\Pi = 0$ .

The only obviously integrable case is that for which in the second formula  $q = 1$ ; this gives

$$\int \frac{dS}{(x^2 + y^2 \dots)^{\frac{1}{2}m} \{(x-a)^2 + (y-b)^2 \dots + u^2\}^{\frac{1}{2}n-1}}$$

$$= \frac{2\pi^{\frac{1}{2}m} f_1}{\Gamma(\frac{1}{2}m)(l^2 + u^2)^{\frac{1}{2}m-1}(l_1^2 - f_1^2)(1 + \epsilon)^{\frac{1}{2}m-1}}.$$

In the case of  $m = 0$ , we have, as before, when  $p^2 - f_1^2$  and  $l_1^2 - f_1^2$  are of opposite signs,  $\epsilon = 0$ , and therefore  $1 + \epsilon = 1$ ; but when  $p^2 - f_1^2$  and  $l_1^2 - f_1^2$  are of the same sign, the value before found for  $\epsilon$  gives

$$1 + \epsilon = \frac{1}{l^2 f_1^2} \{l^2 f_1^2 + (p^2 - f_1^2)(l_1^2 - f_1^2)\}.$$

Consider the image of the origin with respect to the sphere  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2$ , the coordinates of this image are

$$\frac{f_1^2 a_1}{l_1^2}, \quad \frac{f_1^2 b_1}{l_1^2}, \dots;$$

and consequently, if  $\mu$  be the distance of this image from the point  $(a, b, \dots)$ , we have

$$\mu^2 = \frac{f_1^4}{l_1^4} l^2 + (p^2 - f_1^2)(l_1^2 - f_1^2);$$

whence, by a simple reduction,

$$\mu^2 = \frac{1}{l^2} \{l^2 f_1^2 + (p^2 - f_1^2)(l_1^2 - f_1^2)\};$$

or the values of the integral are

$$p^2 - f_1^2 \text{ and } l_1^2 - f_1^2 \text{ opposite signs, } I = \frac{2\pi^{\frac{1}{2}m}}{\Gamma(\frac{1}{2}m)} \frac{f_1}{l^2 (l^2 + u^2)^{\frac{1}{2}m-1}};$$

$p^2 - f_1^2$  and  $l_1^2 - f_1^2$  the same sign,

$$I = \frac{2\pi^{\frac{1}{2}m}}{\Gamma(\frac{1}{2}m)} \frac{f_1}{l^2 \mu^{\frac{1}{2}m-1} (l^2 + u^2)^{\frac{1}{2}m-1}};$$

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where  $\mu$  is the distance from the point  $(a, b, \dots)$  of the image of the origin with respect to the sphere  $(x - a_1)^2 + (y - b_1)^2 + \dots - f_1^2 = 0$ .

*Stone Buildings, August 6, 1850.*

## 109.

### ON THE RATIONALISATION OF CERTAIN ALGEBRAICAL EQUATIONS.

[From the *Cambridge and Dublin Mathematical Journal*, vol. VIII. (1853),  
pp. 97—101.]

SUPPOSE

$$x + y = 0, \quad x^2 = a, \quad y^2 = b;$$

then if we multiply the first equation by 1,  $xy$ , and reduce by the two others, we have

$$\begin{aligned} x + y &= 0, \\ bx + ay &= 0, \end{aligned}$$

from which, eliminating  $x, y$ ,

$$\begin{vmatrix} 1, & 1 \\ b, & a \end{vmatrix} = 0;$$

which is the equation between  $a$  and  $b$ ; or, considering  $x, y$  as quadratic radicals, the rational equation between  $x, y$ .

So if the original equation be multiplied by  $x, y$ , we have

$$\begin{aligned} a + xy &= 0, \\ b + xy &= 0; \end{aligned}$$

or, eliminating 1,  $xy$ ,

$$\begin{vmatrix} a, & 1 \\ b, & 1 \end{vmatrix} = 0,$$

which may be in like manner considered as the rational equation between  $x, y$ .

The preceding results are of course self-evident, but by applying the same process to the equations

$$x + y + z = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c,$$

we have results of some elegance. Multiply the equation first by 1,  $yz$ ,  $zx$ ,  $xy$ , reduce and eliminate the quantities  $x$ ,  $y$ ,  $z$ ,  $xyz$ , we have the rational equation

$$\begin{vmatrix} . & 1 & 1 & 1 \\ 1 & . & c & b \\ 1 & c & . & a \\ 1 & b & a & . \end{vmatrix} = 0;$$

and again, multiply the equation by  $x$ ,  $y$ ,  $z$ ,  $xyz$ , reduce and eliminate the quantities 1,  $yz$ ,  $zx$ ,  $xy$ , the result is

$$\begin{vmatrix} . & a & b & c \\ a & . & 1 & 1 \\ b & 1 & . & 1 \\ c & 1 & 1 & . \end{vmatrix} = 0,$$

which is of course equivalent to the preceding one (the two determinants are in fact identical in value), but the form is essentially different. The former of the two forms is that given in my paper "On a theorem in the Geometry of Position" (*Journal*, vol. II. [1841] p. 270 [1]): it was only very recently that I perceived that a similar process led to the latter of the two forms.

Similarly, if we have the equations

$$x + y + z + w = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c, \quad w^2 = d,$$

then multiplying by 1,  $yz$ ,  $zx$ ,  $xy$ ,  $xw$ ,  $yw$ ,  $zw$ ,  $xyzw$ , reducing and eliminating the quantities in the outside row,

$$x, \quad y, \quad z, \quad w, \quad yzw, \quad zwx, \quad wxy, \quad xyz$$

we have the result

$$\begin{vmatrix} . & 1 & 1 & 1 & 1 & . & . & . & . \\ 1 & . & c & b & . & . & 1 & . & 1 \\ 1 & c & . & a & . & . & . & 1 & . \\ 1 & b & a & . & . & . & . & 1 & 1 \\ \hline d & . & . & . & a & . & 1 & 1 & . \\ . & d & . & . & b & 1 & . & 1 & . \\ . & . & d & . & c & 1 & 1 & . & . \\ \hline . & . & . & . & . & a & b & c & d \end{vmatrix} = 0;$$

so if we multiply the equations by  $x, y, z, w, yzw, zwx, wxy$ , and  $xyz$ , reduce and eliminate the quantities in the outside row,

$$1, \quad yz, \quad zx, \quad xy, \quad xw, \quad yw, \quad zw, \quad xyzw$$

we have the result

$$\left| \begin{array}{cccc|c|cccc|c} \cdot & a & b & c & 1 & \cdot & 1 & 1 & 1 \\ a & \cdot & 1 & 1 & \cdot & \cdot & \cdot & \cdot & 1 \\ b & 1 & \cdot & 1 & \cdot & \cdot & \cdot & 1 & \cdot \\ c & 1 & 1 & \cdot & \cdot & \cdot & 1 & \cdot & \cdot \\ \hline d & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & 1 & 1 \\ \cdot & d & \cdot & \cdot & \cdot & \cdot & c & b & 1 \\ \cdot & \cdot & d & \cdot & \cdot & c & \cdot & a & 1 \\ \cdot & \cdot & \cdot & d & \cdot & b & a & \cdot & 1 \\ \hline \cdot & a & b & c & \cdot & \cdot & \cdot & \cdot & 1 \end{array} \right| = 0.$$

which however is not essentially distinct from the form before obtained, but may be derived from it by an interchange of lines and columns. And in general for any even number of quadratic radicals the two forms are not essentially distinct, but may be derived from each other by interchanging lines and columns, while for an odd number of quadratic radicals the two forms cannot be so derived from each other, but are essentially distinct.

I was indebted to Mr Sylvester for the remark that the above process applies to radicals of a higher order than the second. To take the simplest case, suppose

$$x^3 + y = 0, \quad x^3 = a, \quad y^3 = b;$$

and multiply first by 1,  $x^2y$ ,  $xy^2$ ; this gives

$$\begin{aligned} x + y &= 0, \\ ay + x^2y^2 &= 0, \\ bx^2 + xy^2 &= 0; \end{aligned}$$

or, eliminating,

$$\begin{vmatrix} 1 & 1 & . \\ . & a & b \\ a & . & . \end{vmatrix} = 0;$$

next multiply by  $x$ ,  $y$ ,  $x^2y^2$ ; this gives

$$\begin{aligned} x^2 + xy &= 0, \\ y^2 + xy &= 0, \\ bx^2 + ay^2 &= 0; \end{aligned}$$

or, eliminating,

$$\begin{vmatrix} 1 & 1 & . \\ . & 1 & 1 \\ b & a & . \end{vmatrix} = 0;$$

and lastly, multiply by  $x^2$ ,  $y^2$ ,  $xy$ ; this gives

$$\begin{aligned} a + x^2y &= 0, \\ b + xy^2 &= 0, \\ a^2y + xy^2 &= 0; \end{aligned}$$

or, eliminating,

$$\begin{vmatrix} a & 1 & . \\ b & . & 1 \\ . & a & b \end{vmatrix} = 0;$$

where it is to be remarked that the second and third forms are not essentially distinct, since the one may be derived from the other by the interchange of lines and columns.



$$x + y + z = 0, \quad x^3 = a, \quad y^3 = b, \quad z^3 = c;$$
$$x, \quad y, \quad z, \quad x^2yz, \quad xy^2z, \quad xyz^2, \quad z^2xy, \quad x^2yz, \quad x^2y^2, \quad y^2z^2, \quad z^2x^2$$
$$\left| \begin{array}{ccc|ccc|ccc} \cdot & 1 & 1 & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & c & b & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & c & \cdot & a & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & b & a & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot & 1 & \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \\ \hline \cdot & a & \cdot & \cdot & b & \cdot & c & \cdot & \cdot & \cdot \\ \cdot & \cdot & b & \cdot & \cdot & \cdot & c & a & \cdot & \cdot \\ \cdot & \cdot & \cdot & c & \cdot & \cdot & \cdot & b & a & \cdot \end{array} \right| = 0;$$
$$x^2, \quad y^2, \quad z^2, \quad yz, \quad zx, \quad xy, \quad x^2y^2z^2, \quad yz^2x^2, \quad zx^2y^2$$
$$\left| \begin{array}{ccc|ccc} . & a & b & c & . & . \\ a & . & 1 & 1 & . & . \\ b & 1 & . & 1 & . & . \\ c & 1 & 1 & . & . & . \end{array} \right| = 0;$$

lastly, multiply by  $x^2, y^2, z^2, yz, zx, xy, x^2yz, y^2zx, z^2xy, x^2y^2, y^2z^2, z^2x^2$ , reduce and eliminate the quantities in the outside row,

$$1, \quad xyz, \quad x^2y^2z^2, \quad x^2z, \quad y^2x, \quad z^2y, \quad x^2y, \quad y^2z, \quad z^2x$$

the result is

$$\left| \begin{array}{ccc|ccc|ccc} . & a & b & c & . & 1 & . & . & 1 \\ a & . & . & . & 1 & 1 & . & . & . \\ b & . & . & 1 & . & 1 & . & . & . \\ c & . & 1 & . & . & 1 & . & . & . \\ \hline . & 1 & . & . & . & . & 1 & . & 1 \\ . & 1 & 1 & 1 & . & . & . & 1 & . \\ . & . & . & . & 1 & . & 1 & 1 & . \\ \hline 1 & . & . & . & 1 & . & c & b & . \\ . & . & . & . & . & 1 & . & c & a \\ . & . & . & . & . & . & b & . & . \end{array} \right| = 0;$$

where, as in the case of two cubic radicals, two forms, viz. the first and third forms of the rational equation, are not essentially distinct, but may be derived from each other by interchanging lines and columns.

And in general, whatever be the number of cubic radicals, two of the three forms are not essentially distinct, but may be derived from each other by interchanging lines and columns.

## 110.

### NOTE ON THE TRANSFORMATION OF A TRIGONOMETRICAL EXPRESSION.

[From the *Cambridge and Dublin Mathematical Journal*, vol. IX. (1854),  
pp. 61—62.]

The differential equation

$$\frac{dx}{(a+x)\sqrt{(c+x)}} + \frac{dy}{(a+y)\sqrt{(c+y)}} + \frac{dz}{(a+z)\sqrt{(c+z)}} = 0,$$

integrated so as to be satisfied when the variables are simultaneously infinite, gives by direct integration

$$\tan^{-1} \sqrt{\left(\frac{a-c}{c+x}\right)} + \tan^{-1} \sqrt{\left(\frac{a-c}{c+y}\right)} + \tan^{-1} \sqrt{\left(\frac{a-c}{c+z}\right)} = 0;$$

and, by Abel's theorem,

$$\begin{vmatrix} 1, & x, & (a+x)\sqrt{(c+x)} \\ 1, & y, & (a+y)\sqrt{(c+y)} \\ 1, & z, & (a+z)\sqrt{(c+z)} \end{vmatrix} = 0.$$

To show *a posteriori* the equivalence of these two equations, I represent the determinant by the symbol  $\square$ , and expressing it in the form

$$\square = \begin{vmatrix} 1, & a+x, & (a+x)\sqrt{(c+x)} \\ 1, & a+y, & (a+y)\sqrt{(c+y)} \\ 1, & a+z, & (a+z)\sqrt{(c+z)} \end{vmatrix},$$

I write for the moment  $\xi = \sqrt{\left(\frac{a-c}{c+x}\right)}$  &c.; this gives

$$\begin{aligned} \square &= \begin{vmatrix} 1, & (a-c)\left(1 + \frac{1}{\xi^2}\right), & (a-c)^{\frac{3}{2}}\left(\frac{1}{\xi} + \frac{1}{\xi^3}\right) \\ \vdots & \vdots & \vdots \end{vmatrix} \\ &= \frac{(a-c)^{\frac{3}{2}}}{\xi^2 \eta^2 \zeta^2} \begin{vmatrix} \xi^2, & \xi^2 + 1, & \xi^3 + \xi \\ \vdots & \vdots & \vdots \end{vmatrix} \\ &= \frac{(a-c)^{\frac{3}{2}}}{\xi^2 \eta^2 \zeta^2} \begin{vmatrix} \xi^2, & 1, & \xi^3 + \xi \\ \vdots & \vdots & \vdots \end{vmatrix} \\ &= -\frac{(a-c)^{\frac{3}{2}}}{\xi^2 \eta^2 \zeta^2} \left\{ \begin{vmatrix} 1, & \xi, & \xi^3 \\ \vdots & \vdots & \vdots \end{vmatrix} - \xi \eta \zeta \begin{vmatrix} 1, & \xi, & \xi^2 \\ \vdots & \vdots & \vdots \end{vmatrix} \right\} \\ &= -\frac{(a-c)^{\frac{3}{2}}}{\xi^2 \eta^2 \zeta^2} (\xi + \eta + \zeta - \xi \eta \zeta) \begin{vmatrix} 1, & \xi, & \xi^2 \\ \vdots & \vdots & \vdots \end{vmatrix}; \end{aligned}$$

or, replacing  $\xi, \eta, \zeta$  by their values, we have identically

$$\begin{aligned} &\begin{vmatrix} 1, & x, & (a+x)\sqrt{c+x} \\ 1, & y, & (a+y)\sqrt{c+y} \\ 1, & z, & (a+z)\sqrt{c+z} \end{vmatrix} = \\ &-\frac{(c+x)^{\frac{3}{2}}(c+y)^{\frac{3}{2}}(c+z)^{\frac{3}{2}}}{(a-c)^2} \left\{ \sqrt{\frac{a-c}{c+x}} + \sqrt{\frac{a-c}{c+y}} + \sqrt{\frac{a-c}{c+z}} - \sqrt{\frac{a-c}{c+x}}\sqrt{\frac{a-c}{c+y}}\sqrt{\frac{a-c}{c+z}} \right\} \end{aligned}$$

and the equation

$$\sqrt{\frac{a-c}{c+x}} + \sqrt{\frac{a-c}{c+y}} + \sqrt{\frac{a-c}{c+z}} - \sqrt{\frac{a-c}{c+x}}\sqrt{\frac{a-c}{c+y}}\sqrt{\frac{a-c}{c+z}} = 0$$

is of course equivalent to the trigonometrical equation

$$\tan^{-1} \sqrt{\frac{a-c}{c+x}} + \tan^{-1} \sqrt{\frac{a-c}{c+y}} + \tan^{-1} \sqrt{\frac{a-c}{c+z}} = 0,$$

which shows the equivalence of the two equations in question.

## 111.

### ON A THEOREM OF M. LEJEUNE-DIRICHLET'S.

[From the *Cambridge and Dublin Mathematical Journal*, vol. IX. (1854),  
pp. 163—165.]

The following formula,

$$\sum q^{ax^2+2bxy+cy^2} + \sum q^{a'x^2+2b'xy+c'y^2} + \dots = 2\sum \delta^{\frac{1}{2}(n-1)} \epsilon^{\frac{1}{8}(n'^2-1)} \left(\frac{n}{P}\right) q^{nn'}, \dots\dots\dots$$

is given in Lejeune-Dirichlet's well-known memoir "Recherches sur diverses applications &c." (*Crelle*, t. XXI. [1840] p. 8). The notation is as follows:—On the left-hand side  $(a, b, c)$ ,  $(a', b', c')$ , ... are a system of properly primitive forms to the *negative* determinant  $D$  (i.e. a system of positive forms);  $x, y$  are positive or negative integers including zero, such that in the sum  $\sum q^{ax^2+2bxy+cy^2}$ ,  $ax^2+2bxy+cy^2$  is prime to  $2D$ , and similarly in the other sums;  $q$  is indeterminate and the summations extend to the values first mentioned, of  $x$  and  $y$ . On the right-hand side we have to consider the form of  $D$ , viz. we have  $D = PS^2$  or else  $D = 2PS^2$ , where  $S^2$  is the greatest square factor in  $D$  and where  $P$  is odd: this obviously defines  $P$ , and the values of  $\delta, \epsilon$ , which are always  $\pm 1$  (or, as I prefer to express it, are always  $\pm$ ) are given as follows, viz.

$$\begin{aligned} D = PS^2, \quad P &\equiv 1 \pmod{4}, & \delta, \epsilon &= ++, \\ D = PS^2, \quad P &\equiv 3 \pmod{4}, & \delta, \epsilon &= -+, \\ D = 2PS^2, \quad P &\equiv 1 \pmod{4}, & \delta, \epsilon &= +-, \\ D = 2PS^2, \quad P &\equiv 3 \pmod{4}, & \delta, \epsilon &= --, \end{aligned}$$

$n, n'$  are any positive numbers prime to  $2D$ ,  $\left(\frac{n}{P}\right)$  is Legendre's symbol as generalized by Jacobi, viz. in general if  $p$  be a positive or negative prime not a factor of  $n$ , then  $\left(\frac{n}{p}\right) = +$  or  $-$  according as  $n$  is or is not a quadratic residue of  $p$  (or, what is the same thing,  $p$  being positive,  $n^{\frac{1}{2}(p-1)} \equiv \left(\frac{n}{p}\right) \pmod{p}$ ), and for  $P = pp'p'' \dots$ ,

$$\left(\frac{n}{P}\right) = \left(\frac{n}{p}\right) \left(\frac{n}{p'}\right) \left(\frac{n}{p''}\right) \dots;$$

and the summation extends to all the values of  $n, n'$  of the form above mentioned.

In the particular case  $D = -1$ , it is necessary that the second side should be doubled.

The method of reducing the equation is indicated in the memoir. The following are a few particular cases.

$$D = -1,$$

$$2\Sigma q^{x^2+y^2} = 4\Sigma(-)^{\frac{1}{2}(n-1)}q^{nn'},$$

or

$$(1+2q^4+2q^{16}+2q^{36}+\dots)(q+q^9+q^{25}+q^{49}+\dots) = \frac{q}{1-q^2} + \frac{q^9}{1-q^{18}} + \frac{q^{25}}{1-q^{50}} + \frac{q^{49}}{1-q^{98}} + \dots$$

$$D = -2,$$

$$2\Sigma q^{x^2+2y^2} = 2\Sigma(-)^{\frac{1}{8}(n-1)+\frac{1}{8}(n'^2-1)}q^{nn'},$$

or

$$(1+2q^8+2q^{32}+2q^{72}+\dots)(q+q^3+q^9+q^{11}+\dots) = \frac{q}{1-q^4} + \frac{q^3}{1-q^{12}} + \frac{q^9}{1-q^{36}} + \frac{q^{11}}{1-q^{44}} + \dots$$

an example given in the memoir.

$$D = -3,$$

$$\begin{aligned} & (q^3 + q^{12} + q^{27} + q^{48} + q^{75} \dots)(1 + 2q^2 + 2q^8 + 2q^{18} + 2q^{32} \dots) \\ & \quad + 2(q^6 + q^{18} + q^{36} + q^{60} + \dots)(q + q^4 + q^9 + q^{16} \dots) \\ & \quad = \frac{q + q^2}{1 - q^6} + \frac{q^4 + q^8}{1 - q^{24}} + \frac{q^9 + q^{18}}{1 - q^{54}} + \dots \end{aligned}$$

I am not aware that the above theorem is quoted or referred to in any subsequent memoir on Elliptic Functions, or on the class of series to which it relates and the theorem is so distinct in its origin and form from all other theorems relating to the same class of series, and, independently of the researches in which it originates, so remarkable as a result, that I have thought it desirable to give a detached statement of it in this paper.

## 112.

### DEMONSTRATION OF A THEOREM RELATING TO THE PRODUCTS OF SUMS OF SQUARES.

[From the *Philosophical Magazine*, vol. IV. (1852), pp. 515—519.]

MR Kirkman, in his paper “On Pluquaternions and Homoid Products of Sums of  $n$  Squares” (*Phil. Mag.* vol. XXXIII. [1848] pp. 447—459 and 494—509), quotes from a note of mine the following passage:—“The complete test of the possibility of the product of  $2^n$  squares by  $2^n$  squares reducing itself to a sum of  $2^n$  squares is the following: forming the complete systems of triplets for  $(2^n - 1)$  things, if  $eab, ecd, fac, fdb$  be any four of them, we must have, paying attention to the signs alone,

$$(\pm eab)(\pm ecd) = (\pm fac)(\pm fdb);$$

i. e. if the first two are of the same sign, the last two must be so also, and *vice versâ*; I believe that, for a system of seven, two conditions of this kind being satisfied would imply the satisfaction of all the others: it remains to be shown that the complete system of conditions cannot be satisfied for fifteen things.” I propose to explain the meaning of the theorem, and to establish the truth of it, without in any way assuming the existence of imaginary units.

The identity to be established is

$$(w^2 + a^2 + b^2 + \dots)(w_1^2 + a_1^2 + b_1^2 + \dots) = w_{11}^2 + a_{11}^2 + b_{11}^2 + \dots$$

where the  $2^n$  quantities  $w, a, b, c, \dots$  and the  $2^n$  quantities  $w_1, a_1, b_1, c_1, \dots$  are given quantities in terms of which the  $2^n$  quantities  $w_{11}, a_{11}, b_{11}, c_{11}, \dots$  have to be determined.

Without attaching any meaning whatever to the symbols  $a_0, b_0, c_0, \dots$  I write down the expressions

$$w + aa_0 + bb_0 + cc_0 + \dots, \quad w_1 + a_1a_0 + b_1b_0 + c_1c_0 + \dots,$$

and I multiply as if  $a_0, b_0, c_0, \dots$  really existed, taking care to multiply without making any transposition in the order *inter se* of two



symbols  $a_0, b_0$  combined in the way of multiplication. This gives a quasi-product

$$\begin{aligned} ww_1 + (aw_1 + a_1w) a_0 + (bw_1 + b_1w) b_0 + \dots \\ + aa_1a_0^2 + bb_1b_0^2 + \dots \\ + ab_1a_0b_0 + a_1bb_0a_0 + \dots \end{aligned}$$

Suppose, now, that a quasi-equation, such as

$$a_0b_0c_0 = +,$$

means that in the expression of the quasi-product

$$b_0c_0, \quad c_0a_0, \quad a_0b_0, \quad c_0b_0, \quad a_0c_0, \quad b_0a_0$$

are to be replaced by

$$a_0, \quad b_0, \quad c_0, \quad -a_0, \quad -b_0, \quad -c_0;$$

and that a quasi-equation, such as  $a_0b_0c_0 = -$ , means that in the expression of the quasi-product

$$b_0c_0, \quad c_0a_0, \quad a_0b_0, \quad c_0b_0, \quad a_0c_0, \quad b_0a_0$$

are to be replaced by

$$-a_0, \quad -b_0, \quad -c_0, \quad a_0, \quad b_0, \quad c_0.$$

It is in the first place clear that the quasi-equation,  $a_0b_0c_0 = +$ , may be written in any one of the six forms

$$\begin{aligned} a_0b_0c_0 = +, \quad b_0c_0a_0 = +, \quad c_0a_0b_0 = +, \\ a_0c_0b_0 = -, \quad c_0b_0a_0 = -, \quad b_0a_0c_0 = -; \end{aligned}$$

and so for the quasi-equation  $a_0b_0c_0 = -$ .

This being premised, if we form a system of quasi-equations, such as

$$a_0b_0c_0 = \pm, \quad a_0d_0e_0 = \pm, \quad \&c.$$

where the system of triplets contains each duad once, and once only, and the arbitrary signs are chosen at pleasure; if, moreover, in the expression of the quasi-product we replace  $a_0^2, b_0^2, \dots$  each by  $-1$ , it is clear that the quasi-product will assume the form

$$w_{11} + a_{11}a_0 + b_{11}b_0 + c_{11}c_0 + \dots,$$

$w_{11}, a_{11}, b_{11}, c_{11}, \dots$  being determinate functions of  $w, a, b, c, \dots$ ;  $w_1, a_1, b_1, c_1, \dots$ , homogeneous of the first order in the quantities of each set; the value of  $w_{11}$  being obviously in every case

$$w_{11} = ww_1 - aa_1 - bb_1 - cc_1 - \dots,$$

and  $a_{11}, b_{11}, c_{11}, \dots$  containing in every case the terms  $aw_1 + a_1w, bw_1 + b_1w, cw_1 + c_1w, \dots$  but the form of the remaining terms depending as well on the triplets entering into the

system of quasi-equations as on the values given to the signs +; the quasi-equations serving, in fact, to prescribe a rule for the formation of certain functions  $w_{11}$ ,  $a_{11}$ ,  $b_{11}$ ,  $c_{11}, \dots$ , the properties of which functions may afterwards be investigated.

Suppose, now, that the system of quasi-equations is such that

$$e_0 a_0 b_0, \quad e_0 c_0 d_0$$

being any two of its triplets, with a common symbol  $e_0$ , there occur also in the system the triplets

$$f_0 a_0 c_0, \quad f_0 d_0 b_0;$$

and suppose that the corresponding portion of the system is

$$\begin{aligned} e_0 a_0 b_0 &= \epsilon, & e_0 c_0 d_0 &= \epsilon', \\ f_0 a_0 c_0 &= \eta, & f_0 d_0 b_0 &= \eta', \end{aligned}$$

where  $\epsilon, \epsilon', \eta, \eta'$ , each of them denote one of the signs + or -; then  $e_{11}, f_{11}$  will contain respectively the terms

$$\begin{aligned} \epsilon (ab_1 - a_1 b) + \epsilon' (cd_1 - c_1 d), \\ \eta (ac_1 - a_1 c) + \eta' (db_1 - d_1 b), \end{aligned}$$

and  $e_{11}^2 + f_{11}^2$  contains the terms

$$\begin{aligned} (a^2 + b^2 + c^2 + d^2)(a_1^2 + b_1^2 + c_1^2 + d_1^2) - a^2 a_1^2 - b^2 b_1^2 - c^2 c_1^2 - d^2 d_1^2 \\ + 2\{\epsilon\epsilon' (ab_1 - a_1 b)(cd_1 - c_1 d) \\ + \eta\eta' (ac_1 - a_1 c)(db_1 - d_1 b) \\ + \mu' (ad_1 - a_1 d)(bc_1 - b_1 c)\}; \end{aligned}$$

and by taking account of the terms  $ew_1 + e_1 w, fw_1 + f_1 w, gw_1 + g_1 w$  in  $e_{11}, f_{11}, g_{11}$  respectively, we should have besides in  $e_{11}^2 + f_{11}^2 + g_{11}^2$  the terms

$$\begin{aligned} (e^2 + f^2 + g^2) w_1^2 + (e_1^2 + f_1^2 + g_1^2) w^2 \\ + 2(ee_1 + ff_1 + gg_1) w w_1. \end{aligned}$$

Also  $w_{11}^2$  contains the terms

$$\begin{aligned} w^2 w_1^2 + a^2 a_1^2 + b^2 b_1^2 + c^2 c_1^2 + d^2 d_1^2 \\ - 2(ee_1 + ff_1 + gg_1) w w_1; \end{aligned}$$

whence it is easy to see that

$$\begin{aligned}
 (w^2 + a^2 + b^2 + c^2 + \dots)(w_1^2 + a_1^2 + b_1^2 + c_1^2 + \dots) \\
 = w_{11}^2 + a_{11}^2 + b_{11}^2 + c_{11}^2 + \dots \\
 + 2\Sigma\{\epsilon\epsilon'(ab_1 - a_1b)(cd_1 - c_1d) \\
 + \eta\eta'(ac_1 - a_1c)(db_1 - d_1b) \\
 + \iota\iota'(ad_1 - a_1d)(bc_1 - b_1c)\},
 \end{aligned}$$

where the summation extends to all the quadruplets formed each by the combination of two duads such as  $ab$  and  $cd$ , or  $ac$  and  $db$ , or  $ad$  and  $bc$ , i.e. two duads, which, combined with the same common letter (in the instances just mentioned  $e$ , or  $f$ , or  $g$ ), enter as triplets into the system of quasi-equations—so that if  $\nu = 2^n - 1$ , the number of quadruplets is

$$\frac{1}{2}\nu(\nu-1) \cdot \frac{1}{2}(\nu-3) \cdot \frac{1}{3} = \frac{1}{12}\nu(\nu-1)(\nu-3),$$

and the terms under the sign  $\Sigma$  will vanish identically if only

$$\epsilon\epsilon' = \eta\eta';$$

but the relation  $\epsilon\epsilon' = \eta\eta'$  is of the same form as the equation

$$(\pm eab)(\pm ecd) = (\pm fac)(\pm fdb);$$

hence if all the relations  $\epsilon\epsilon' = \eta\eta'$  are satisfied, the terms under the sign  $\Sigma$  vanish, and we have

$$(w_{11}^2 + a_{11}^2 + b_{11}^2 + c_{11}^2 + \dots) = (w^2 + a^2 + b^2 + c^2 + \dots)(w_1^2 + a_1^2 + b_1^2 + c_1^2 + \dots)$$

which is thus shown to be true, upon the suppositions

1. That the system of quasi-equations is such that

$$e_0a_0b_0, \quad e_0c_0d_0$$

being any two of its triplets with a common symbol  $e_0$ , there occur also in the system the triplets

$$f_0a_0c_0, \quad f_0d_0b_0.$$

2. That for any two pairs of triplets, such as  $eab$ ,  $ecd$  and  $fac$ ,  $fdb$ , the product of the signs of the triplets of the first pair is equal to the product of the signs of the triplets of the second pair.

In the case of fifteen things  $a, b, c, \dots$  the triplets may, as appears from Mr Kirkman's paper, be chosen so as to satisfy the first condition; but the second condition involves, as Mr Kirkman has shown, a contradiction; and therefore the product of two sums, each of them of sixteen squares, is not a sum of sixteen squares.

It is proper to remark, that this demonstration, although I think rendered clearer by the introduction of the idea of the system of triplets furnishing the rule for the formation of the expressions  $w_{11}$ ,  $a_{11}$ ,  $b_{11}$ ,  $c_{11}$ , &c., is not in principle different from that contained in Prof Young's paper "On an Extension of a Theorem of Euler, &c.", *Phil. Transactions*, vol. XXI. [1848 pp. 311—341].

### 113.

NOTE ON THE GEOMETRICAL REPRESENTATION OF THE  
INTEGRAL  $\int dx \div \sqrt{\{(x+a)(x+b)(x+c)\}}.$

[From the *Philosophical Magazine*, vol. V. (1853), pp. 281—284.]

The equation of a conic passing through the points of intersection of the conics

$$\begin{aligned}x^2 + y^2 + z^2 &= 0, \\ ax^2 + by^2 + cz^2 &= 0,\end{aligned}$$

is of the form

$$w(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0,$$

where  $w$  is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of  $w$  a quadratic equation, the roots of which may be considered as parameters for determining the line in question. Let one of the values of  $w$  be considered as equal to a constant quantity  $k$ , the line is always a tangent to the conic

$$k(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0;$$

and taking  $w = p$  for the other value of  $w$ ,  $p$  is a parameter determining the particular tangent, or, what is the same thing, determining the point of contact of this tangent.

The equation of the tangent is easily seen to be

$$x\sqrt{(b-c)}\sqrt{(a+k)}\sqrt{(a+p)} + y\sqrt{(c-a)}\sqrt{(b+k)}\sqrt{(b+p)} + z\sqrt{(a-b)}\sqrt{(c+k)}\sqrt{(c+p)} = 0$$

suppose that the tangent meets the conic  $x^2 + y^2 + z^2 = 0$  (which is of course the conic corresponding to  $w = \infty$ ) in the points  $P, P'$ , and let  $\theta, \infty$  be the parameters of the point  $P$ , and  $\theta', \infty$  the parameters of the point  $P'$ , i.e. (repeating the definition of the terms) let the tangent at  $P$  of the conic  $x^2 + y^2 + z^2 = 0$  be also touched by the conic  $\theta(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0$ , and similarly for  $\theta'$ .

The coordinates of the point  $P$  are given by the equations

$$x : y : z = \sqrt{(b-c)}\sqrt{(c+k)}\sqrt{(c+\theta)} : \sqrt{(c-a)}\sqrt{(b+k)}\sqrt{(b+\theta)} : \sqrt{(a-b)}\sqrt{(c+k)}\sqrt{(c+\theta)}$$

and substituting these values in the equation of the line  $PP'$ , we have

$$(b-c) \sqrt{a+k} \sqrt{a+p} \sqrt{a+\theta} + (c-a) \sqrt{b+k} \sqrt{b+p} \sqrt{b+\theta} \\ + (a-b) \sqrt{c+k} \sqrt{c+p} \sqrt{c+\theta} = 0 \dots (*),$$

an equation connecting the quantities  $p, \theta$ .

To rationalize this equation, write

$$\begin{aligned} \sqrt{a+k} \sqrt{a+p} \sqrt{a+\theta} &= \lambda + \mu a, \\ \sqrt{b+k} \sqrt{b+p} \sqrt{b+\theta} &= \lambda + \mu b, \\ \sqrt{c+k} \sqrt{c+p} \sqrt{c+\theta} &= \lambda + \mu c, \end{aligned}$$

values which evidently satisfy the equation in question. Squaring these equations, we have equations from which  $\lambda^2, \lambda\mu, \mu^2$  may be linearly determined; and making the necessary reductions, we find

$$\begin{aligned} \lambda^2 &= abc + kp\theta, \\ -2\lambda\mu &= bc + ca + ab - (p\theta + kp + k\theta), \\ \mu^2 &= a + b + c + k + p + \theta; \end{aligned}$$

eliminating  $\lambda, \mu$ ,

$$\begin{aligned} \{bc + ca + ab - (p\theta + kp + k\theta)\}^2 \\ - 4(a + b + c + k + p + \theta)(abc + kp\theta) = 0 \dots (*'), \end{aligned}$$

which is the rational form of the former equation marked (\*).

It is clear from the symmetry of the formula, that the same equation would have been obtained by the elimination of  $L, M$  from the equations

$$\begin{aligned} \sqrt{k+a} \sqrt{k+b} \sqrt{k+c} &= L + Mk, \\ \sqrt{p+a} \sqrt{p+b} \sqrt{p+c} &= L + Mp, \\ \sqrt{\theta+a} \sqrt{\theta+b} \sqrt{\theta+c} &= L + M\theta, \end{aligned}$$

and it follows from Abel's theorem (but the result may be verified by means of Euler's fundamental integral in the theory of elliptic functions), that if

$$U_x = \int \frac{dx}{\sqrt{\{(x+a)(x+b)(x+c)\}}},$$

then the algebraical equations (\*') are equivalent to the transcendental equation

$$\pm Uk \pm Up \pm U\theta = 0;$$

the arbitrary constant which should have formed the second side of the equation having been determined by observing that the algebraical equation gives for  $p = \theta$ ,  $k = \infty$ , a system of values, which, when the signs are properly chosen, satisfy the transcendental equation.

In fact, arranging the rational algebraical equation according to the powers of  $k$ , it becomes

$$k^2 (p-\theta)^2 - 2k \{p\theta (p+\theta) + 2(a+b+c)p\theta + (bc+ca+ab)(p+\theta) + 2abc\} + p^2\theta^2 - 2(bc+ca+ab)p\theta - 4abc(p+\theta) + b^2c^2 + c^2a^2 + a^2b^2 - 2a^2bc - 2b^2ca - 2c^2ab = 0;$$

which proves the property in question, and is besides a very convenient form of the algebraical integral.

The ambiguous signs in the transcendental integral are not of course arbitrary (indeed it has just been assumed that for  $p = \theta$ ,  $Up$  and  $U\theta$  are to be taken with opposite signs), but the discussion of the proper values to be given to the ambiguous signs would be at all events tedious, and must be passed over for the present.

It is proper to remark, that  $\theta = p$  gives not only, as above supposed,  $k = \infty$ , but another value of  $k$ , which, however, corresponds to the transcendental equation

$$+Uk \pm 2Up = 0;$$

the value in question is obviously

$$k = \frac{p^3 - 2(bc + ca + ab)p - 8abc}{(p+a)(p+b)(p+c)}.$$

Consider, in general, a cubic function  $ax^3 + 3bx^2y + 3cxy^2 + dy^3$ , or, as I now write it in the theory of invariants,  $(a, b, c, d)(x, y)^3$ , the Hessian of this function is  $(ac - b^2, \frac{1}{2}(ad - bc), bd - c^2)(x, y)^2$ , and applying this formula to the function  $(p+a)(p+b)(p+c)$ , it is easy to write the equation last preceding in the form

$$k = \frac{9 \text{Hessian} \{(p+a)(p+b)(p+c)\}}{4(p+a)(p+b)(p+c)},$$

which is a formula for the duplication of the transcendent  $Ux$ .

Reverting now to the general transcendental equation

$$\pm Uk \pm Up \pm U\theta = 0,$$



we have in like manner

$$\pm Uk \pm Up \pm U\theta' = 0;$$

and assuming a proper correspondence of the signs, the elimination of  $Up$  gives

$$U\theta' - U\theta = 2Uk;$$

i.e. if the points  $P, P'$  upon the conic  $x^2 + y^2 + z^2 = 0$  are such that their parameters  $\theta, \theta'$  satisfy this equation, the line  $PP'$  will be constantly a tangent to the conic  $k(x^2 + y^2 + z^2) + (ax^2 + by^2 + cz^2) = 0$ .

Hence also, if the parameters  $k, k', k''$  of the conics

$$\begin{aligned} k(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 &= 0, \\ k'(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 &= 0, \\ k''(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 &= 0, \end{aligned}$$

satisfy the equation

$$Uk + Uk' + Uk'' = 0,$$

there are an infinity of triangles inscribed in the conic  $x^2 + y^2 + z^2 = 0$ , and the sides of which touch the last-mentioned three conics respectively.

Suppose  $2Uk = U\infty$  (an equation the algebraic form of which has already been discussed), then

$$U\theta' - U\theta = Uk, \quad k = \infty$$

gives  $\theta' = k$ ; or, observing that  $\theta' = \infty$  corresponds to a point of intersection of the conics  $x^2 + y^2 + z^2 = 0, ax^2 + by^2 + cz^2 = 0, k$  is the parameter of the point in which a tangent to the conic  $k(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0$  at any one of its intersections with the conic  $x^2 + y^2 + z^2 = 0$  meets the last-mentioned conic.

Moreover, the algebraical relation between  $\theta, \theta'$  and  $k$  (where, as before remarked,  $k$  is a given function of  $\theta$ ) is given by a preceding formula, and is simpler than that between  $\theta, \theta'$  and  $k$ .

The preceding investigations were, it is hardly necessary to remark, suggested by a well-known memoir of the late illustrious Jacobi, and contain, I think, the extension which he remarks it would be interesting to make of the principles in such memoir to a system of two conics.

I propose reverting to the subject in a memoir to be entitled "Researches on the Porism of the in- and circumscribed triangle."

[This was, I think, never written.]

## 114.

### ANALYTICAL RESEARCHES CONNECTED WITH STEINER'S EXTENSION OF Malfatti's PROBLEM.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLII. for the year 1852, pp. 253—278: Received April 12,—Read May 27, 1852.]

The problem, in a triangle to describe three circles each of them touching the two others and also two sides of the triangle, has been termed after the Italian geometer by whom it was proposed and solved, Malfatti's problem. The problem which I refer to as Steiner's extension of Malfatti's problem is as follows:—"To determine three sections of a surface of the second order, each of them touching the two others, and also two of three given sections of the surface of the second order," a problem proposed in Steiner's memoir, "Einige geometrische Betrachtungen," *Crelle*, t. I. [1826 pp. 161—184]. The geometrical construction of the problem in question is readily deduced from that given in the memoir just mentioned for a somewhat less general problem, viz. that in which the surface of the second order is replaced by a sphere: it is for the sake of the analytical developments to which the problem gives rise, that I propose to resume here the discussion of the problem.

The following is an analysis of the present memoir:

§1. Contains a lemma which appears to me to constitute the foundation of the analytical theory of the sections of a surface of the second order.

§2. Contains a statement of the geometrical construction of Steiner's extension of Malfatti's problem.

§3. Is a verification, founded on a particular choice of coordinates, of the construction in question.

§4. In this section, referring the surface of the second order to absolutely general coordinates, and after an incidental solution of the problem to determine a section touching three given sections, I obtain the equations for the solution of Steiner's extension of Malfatti's problem.

§5. Contains a separate discussion of a system of equations, including as a particular case the equations obtained in the preceding section.

§§6 and 7. Contain the application of the formulæ for the general system to the equations in §4, and the development and completion of the solution.

§8. Is an extension of some preceding formulæ to quadratic functions of any number of variables.

**§1. Lemma relating to the sections of a surface of the second order.**

If  $ax^2 + by^2 + cz^2 + dw^2 + 2fyz + 2gzx + 2hxy + 2lxw + 2myw + 2nzw = 0$  be the equation of a surface of the second order, and  $\mathfrak{A}x^2 + \mathfrak{B}y^2 + \mathfrak{C}z^2 + \mathfrak{D}w^2 + 2\mathfrak{F}yz + 2\mathfrak{G}zx + 2\mathfrak{H}xy + 2\mathfrak{L}xw + 2\mathfrak{M}yw + 2\mathfrak{N}zw = 0$  the reciprocal equation, the condition that the two sections

$$\lambda x + \mu y + \nu z + \rho w = 0, \quad \lambda' x + \mu' y + \nu' z + \rho' w = 0,$$

may touch, is

$$\begin{aligned} & (\mathfrak{A}\lambda\lambda' + \mathfrak{B}\mu\mu' + \mathfrak{C}\nu\nu' + \mathfrak{D}\rho\rho' + 2\mathfrak{F}(\mu\nu' + \mu'\nu) + 2\mathfrak{G}(\nu\lambda' + \nu'\lambda) + 2\mathfrak{H}(\lambda\mu' + \lambda'\mu) \\ & \quad + 2\mathfrak{L}(\lambda\rho' + \lambda'\rho) + 2\mathfrak{M}(\mu\rho' + \mu'\rho) + 2\mathfrak{N}(\nu\rho' + \nu'\rho))^2 \\ & = (\mathfrak{A}\lambda^2 + \mathfrak{B}\mu^2 + \mathfrak{C}\nu^2 + \mathfrak{D}\rho^2 + 2\mathfrak{F}\mu\nu + 2\mathfrak{G}\nu\lambda + 2\mathfrak{H}\lambda\mu + 2\mathfrak{L}\lambda\rho + 2\mathfrak{M}\mu\rho + 2\mathfrak{N}\nu\rho) \\ & \times (\mathfrak{A}\lambda'^2 + \mathfrak{B}\mu'^2 + \mathfrak{C}\nu'^2 + \mathfrak{D}\rho'^2 + 2\mathfrak{F}\mu'\nu' + 2\mathfrak{G}\nu'\lambda' + 2\mathfrak{H}\lambda'\mu' + 2\mathfrak{L}\lambda'\rho' + 2\mathfrak{M}\mu'\rho' + 2\mathfrak{N}\nu'\rho'), \end{aligned}$$

and in particular if the equation of the surface be

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0,$$

the condition of contact is

$$\begin{aligned} & (a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu - p\rho^2) \\ & \quad \times (a\lambda'^2 + b\mu'^2 + c\nu'^2 + 2f\mu'\nu' + 2g\nu'\lambda' + 2h\lambda'\mu' - p\rho'^2) \\ & = (a\lambda\lambda' + b\mu\mu' + c\nu\nu' + f(\mu\nu' + \mu'\nu) + g(\nu\lambda' + \nu'\lambda) + h(\lambda\mu' + \lambda'\mu) - p\rho\rho')^2, \end{aligned}$$

in which last formula

$$\mathfrak{F} = gh - af, \quad \mathfrak{G} = hf - bg, \quad \mathfrak{H} = fg - ch, \quad K = abc - ap^2 - bg^2 - ch^2 + 2fgh.$$

## §2.

In order to state in the most simple form the geometrical construction for the solution of Steiner's extension of Malfatti's problem, let the given sections be called for conciseness the *determinators*<sup>5</sup>; any two of these sections lie in two different cones, the vertices of which determine with the line of intersection of the planes of the determinators, two planes which may be termed *bisectors*; the six bisectors pass three and three through four straight lines; and it will be convenient to use the term bisectors to denote, not the entire system, but any three bisectors passing through the same line.

Consider three sections, which may be termed *tactors*, each of them touching a determinator and two bisectors, and three other sections (which may be termed *separators*) each of them passing through the point of contact of a determinator and tactor and touching the other two tactors; the separators will intersect in a line which passes through the point of intersection of the determinators. The three required sections, or as I shall term them the *resultors*, are determined by the conditions that each resultor touches two determinators and two separators, the possibility of the construction being implied as a theorem. The *à posteriori* verification may be obtained as follows:

## §3.

Let  $x = 0$ ,  $y = 0$ ,  $z = 0$  be the equations of the resultors,  $w = 0$  the equation of the polar of the point of intersection of the resultors.

Since the resultors touch two and two, the equation of the surface is easily seen to be of the form

$$2yz + 2zx + 2xy + w^2 = 0. \quad (S)$$

The determinators are sections each of them touching two resultors, but otherwise arbitrary; their equations are

$$-\alpha x + y + z + w = 0, \quad x - \beta y + z + w = 0, \quad x + y - \gamma z + w = 0.$$

The separators are sections each of them touching two resultors at their point of contact (or what is the same thing, passing through the line of intersection of two resultors), and all of them having a line in common. Their equations may be taken to be

$$cy - bz = 0, \quad az - cx = 0, \quad bx - ay = 0,$$

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<sup>5</sup>I use the words "determinators," &c. to denote indifferently the sections or the planes of the sections: the context is always sufficient to prevent ambiguity.

the values of  $a, b, c$  remaining to be determined.

Now before fixing the values of these quantities, we may find three sections each of them touching a determinator at a point of intersection with the section which corresponds to it of the sections  $cy - bz = 0$ ,  $az - cx = 0$ ,  $bx - ay = 0$ , and touching the other two of the last-mentioned sections; and when  $a, b, c$  have their proper values the sections so found are the tactors.

For, let  $\lambda x + \mu y + \nu z + \rho w = 0$  be the equation of a section touching the determinator  $-\alpha x + y + z + w = 0$ , and the two sections  $bx - ay = 0$ ,  $az - cx = 0$ : and suppose

$$\Lambda^2 = \lambda^2 + \mu^2 + \nu^2 - 2\mu\nu - 2\nu\lambda - 2\lambda\mu - 2\rho^2,$$

the conditions of contact with the sections  $bx - ay = 0$ ,  $az - cx = 0$  are found to be

$$\begin{aligned}(b + a)\Lambda &= (b + a)\lambda - (b - a)\mu - (b + a)\nu, \\ (c + a)\Lambda &= (c + a)\lambda - (c - a)\mu - (c + a)\nu,\end{aligned}$$

values, however, which suppose a correspondence in the signs of the radicals. Thence  $(b + a)\mu = (c + a)\nu$ ; or since the ratios only of the quantities  $\lambda, \mu, \nu, \rho$  are material,  $\mu = c + a$ ,  $\nu = b + a$ , and therefore

$$\Lambda^2 = \lambda^2 - 2(2a + b + c)\lambda + (b - c)^2 - 2\rho^2 = (\lambda - b - c)^2 - 2(a\lambda + bc) - 2\rho^2,$$

or  $\rho^2 = -2(a\lambda + bc)$ .

Hence the equation to a section touching  $bx - ay = 0$ ,  $az - cx = 0$  is

$$\lambda x + (c + a)y + (b + a)z + \sqrt{-2(a\lambda + bc)} w = 0;$$

and to express that this touches the determinator in question, we have

$$\pm\Lambda + \alpha\lambda - (b + c) = \alpha\lambda - a(2a + b + c) + 2\sqrt{-2(a\lambda + bc)},$$

and selecting the upper sign,

$$-\lambda - 2aa = -2\sqrt{-2(a\lambda + bc)};$$

whence

$$\lambda = -2a(a\alpha - \sqrt{1 - 2bc}), \quad \sqrt{-2(a\lambda + bc)} = (2a\alpha - \sqrt{1 - 2bc});$$

or the section touching the determinator and the sections  $bx - ay = 0$ ,  $az - cx = 0$  is

$$-2a(a\alpha - \sqrt{1 - 2bc})x + (c + a)y + (b + a)z + (2a\alpha - \sqrt{1 - 2bc})w = 0;$$

and at the point of contact with the determinator

$$-\alpha x + y + z + w = 0, \quad 2yz + 2zx + 2xy + w^2 = 0.$$

Eliminating  $w$  between the first and second equations and between the second and third equations,

$$\sqrt{1-2bc}\alpha x + y + z = 0, \quad -z + cy + bz = 0,$$

and from these equations

$$(cy - bz)^2 = 0,$$

or the point of contact lies in the section  $cy - bz = 0$ .

It follows that the equations of the tactors are

$$\begin{aligned} -2a(a\alpha - \sqrt{1-2bc})x + (c+a)y + (b+a)z + (2a\alpha - \sqrt{1-2bc})w &= 0, \\ (c+b)x - 2b(b\beta - \sqrt{1-2ca})y + (a+b)z + (2b\beta - \sqrt{1-2ca})w &= 0, \\ (b+c)x + (a+c)y - 2c(c\gamma - \sqrt{1-2ab})z + (2c\gamma - \sqrt{1-2ab})w &= 0, \end{aligned}$$

where  $a, b, c$  still remain to be determined.

Now the separators pass through the point of intersection of the determinators: the equations of these give for the point in question,

$$x : y : z : w = (2\beta\gamma+1)(-\alpha+\beta+\gamma+2\alpha\beta\gamma) : (2\gamma\alpha+1)(\alpha-\beta+\gamma+2\alpha\beta\gamma) : (2\alpha\beta+1)(\alpha+$$

and the values of  $a, b, c$  are therefore

$$a : b : c = (2\beta\gamma+1)(-\alpha+\beta+\gamma+2\alpha\beta\gamma) : (2\gamma\alpha+1)(\alpha-\beta+\gamma+2\alpha\beta\gamma) : (2\alpha\beta+1)(\alpha+\beta-\gamma)$$

which are to be substituted for  $a, b, c$  in the equations of the separators and tactors respectively.

Now proceeding to find the bisectors, let  $\lambda x + \mu y + \nu z + \rho w = 0$  be the equation of a section touching the determinators, and suppose, as before,

$$\Lambda^2 = \lambda^2 + \mu^2 + \nu^2 - 2\mu\nu - 2\nu\lambda - 2\lambda\mu - 2\rho^2;$$

the conditions of contact are

$$\begin{aligned} \pm\beta\Lambda &= \beta\lambda - (\beta + \gamma)\mu + \beta\gamma\nu - 2\rho, \\ \pm\gamma\Lambda &= \gamma\lambda + \gamma\mu - (\gamma + \beta)\nu - 2\rho. \end{aligned}$$



where it is necessary, for the present purpose, to give opposite signs to the radicals. For if the radicals had the same sign, it would follow that

$$\beta\{\beta\lambda - (\beta + \gamma)\mu + \beta\gamma\nu - 2\rho\} - \gamma\{\gamma\lambda + \gamma\mu - (\gamma + \beta)\nu - 2\rho\} = 0;$$

hence the section  $\lambda x + \mu y + \nu z + \rho w = 0$  would pass through the point

$$x : y : z : w = \beta\gamma : \gamma : \beta : \beta\gamma,$$

or the section would pass through the point of intersection of the determinators and could not be a bisector.

Take then the upper sign for  $\beta$  and the lower sign for  $\gamma$  (or *vice versa*); we have

$$\begin{aligned}\beta\Lambda &= \beta\lambda - (\beta + \gamma)\mu + \beta\gamma\nu - 2\rho, \\ -\gamma\Lambda &= \gamma\lambda + \gamma\mu - (\gamma + \beta)\nu - 2\rho,\end{aligned}$$

whence

$$(\beta - \gamma)\lambda - (\beta + 2\gamma)\mu + (2\beta + \gamma)\nu = 0,$$

which is the equation of the plane containing the determinator  $-\alpha x + y + z + w = 0$  and the bisector.

To find the equation of the bisector, we have

$$4\rho = (\beta - \gamma)\lambda - (\beta + \gamma)\mu + (\beta\gamma - \beta\gamma)\nu,$$

or putting this under the form

$$4\rho = (\beta - \gamma)\{\lambda - \mu - \nu\},$$

the equation of the bisector is

$$4x - (\beta - \gamma)(y - z) + (\beta - \gamma)w = 0.$$

The equations of the three bisectors are

$$\begin{aligned}4x - (\beta - \gamma)(y - z) + (\beta - \gamma)w &= 0, \\ 4y - (\gamma - \alpha)(z - x) + (\gamma - \alpha)w &= 0, \\ 4z - (\alpha - \beta)(x - y) + (\alpha - \beta)w &= 0,\end{aligned}$$

which pass through the line

$$x - y = y - z = z - w.$$

We have in like manner the bisectors

$$\begin{aligned} 4x - (\beta - \gamma)(y - z) - (\beta - \gamma)w &= 0, \\ 4y - (\gamma + \alpha)(z + x) + (\gamma + \alpha)w &= 0, \\ 4z + (\alpha + \beta)(x + y) - (\alpha + \beta)w &= 0, \end{aligned}$$

which pass through the line

$$x + y = y + w = w + z;$$

and the other systems of bisectors may be at once written down, but they are not required for the present purpose.

To express that the bisectors touch the tactors, let  $\Lambda$  be the same as before, and  $\Lambda'$  the like function of  $\lambda', \mu', \nu', \rho'$ , also if

$$\Phi = \lambda\lambda' + \mu\mu' + \nu\nu' - (\mu\nu' + \mu'\nu) - (\nu\lambda' + \nu'\lambda) - (\lambda\mu' + \lambda'\mu) - 2\rho\rho',$$

then

$$\begin{aligned} -\Lambda^2 &= (2a\alpha^2 - 2a\sqrt{1 - 2bc} + b + c)^2, \\ \Lambda'^2 &= (2a\alpha'^2 - 2a\sqrt{1 - 2bc} + b + c)^2, \end{aligned}$$

and the condition of contact  $\Lambda\Lambda' = \Phi$  (taking the proper sign for the radicals) becomes

$$(2 + \frac{1}{\alpha})(2a\alpha^2 - 2a\sqrt{1 - 2bc} + b + c) = \alpha\alpha'(2 + \frac{1}{\alpha'})(2a\alpha'^2 - 2a\sqrt{1 - 2bc} + b + c)(2 + \frac{1}{\alpha'});$$

or reducing,

$$a\alpha - \beta\gamma + \alpha^2\beta\gamma = 0,$$

an equation which is evidently not altered by the interchange of  $a, \alpha$  and  $b, \beta$ .

The conditions, in order that each bisector may touch two tactors, reduce themselves to the three equations,

$$a\alpha - \beta\gamma + \alpha^2\beta\gamma = 0,$$

which are satisfied by the values above found for the quantities  $a, b, c$ .

The possibility and truth of the geometrical construction are thus demonstrated.

§4.

Let it be in the first instance proposed to find the equation of a section touching all or any of the sections  $x = 0$ ,  $y = 0$ ,  $z = 0$  of the surface of the second order,

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0.$$

Any section whatever of this surface may be written in the form

$$(a\lambda + h\mu + g\nu)x + (h\lambda + b\mu + f\nu)y + (g\lambda + f\mu + c\nu)z + \sqrt{-p}\rho w = 0,$$

where

$$V^2 = a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu - K.$$

And considering any other section represented by a like equation,

$$(a\lambda' + h\mu' + g\nu')x + (h\lambda' + b\mu' + f\nu')y + (g\lambda' + f\mu' + c\nu')z + \sqrt{-p}\rho' w = 0,$$

where

$$V'^2 = a\lambda'^2 + b\mu'^2 + c\nu'^2 + 2f\mu'\nu' + 2g\nu'\lambda' + 2h\lambda'\mu' - K,$$

it may be shown by means of the lemma previously given, that the condition of contact is

$$a\lambda\lambda' + b\mu\mu' + c\nu\nu' + f(\mu\nu' + \mu'\nu) + g(\nu\lambda' + \nu'\lambda) + h(\lambda\mu' + \lambda'\mu) \pm K = VV'.$$

Suppose that  $\lambda'$ ,  $\mu'$ ,  $\nu'$  satisfy the equations

$$V' = 0, \quad h\lambda' + b\mu' + f\nu' = 0, \quad g\lambda' + f\mu' + c\nu' = 0,$$

so that the last-mentioned section becomes  $x = 0$ ; and observing that the first of the above equations may be transformed into

$$a\lambda'^2 + h\mu'\lambda' + g\nu'\lambda' = K,$$

it is easy to obtain  $\lambda' = \sqrt{\frac{K}{2a}}$ ,  $\mu' = \sqrt{\frac{K}{2b}}$ ,  $\nu' = \sqrt{\frac{K}{2c}}$ .

The condition of contact thus becomes

$$\sqrt{2a}\lambda \pm \sqrt{K} = V;$$

and taking the under sign,  $\lambda = \sqrt{\frac{V^2 - K}{2a}}$ ; so that if in the above written equation we establish all or any of the equations  $\lambda = \sqrt{\frac{K}{2a}}$ ,

$\mu = \sqrt{\frac{K}{\mathfrak{B}}}$ ,  $\nu = \sqrt{\frac{K}{\mathfrak{C}}}$ ; we have the equation of a section touching all or the corresponding sections of the sections  $x = 0$ ,  $y = 0$ ,  $z = 0$ .

In particular we have for a solution of the problem of tactions, the following equation of the section touching  $x = 0$ ,  $y = 0$ ,  $z = 0$ ,

$$\begin{aligned} & (a\sqrt{\mathfrak{A}}+h\sqrt{\mathfrak{B}}+g\sqrt{\mathfrak{C}})x+(h\sqrt{\mathfrak{A}}+b\sqrt{\mathfrak{B}}+f\sqrt{\mathfrak{C}})y+(g\sqrt{\mathfrak{A}}+f\sqrt{\mathfrak{B}}+c\sqrt{\mathfrak{C}})z \\ & + \sqrt{-p} \sqrt{2(\sqrt{\mathfrak{B}\mathfrak{C}}-\mathfrak{F}\sqrt{\mathfrak{A}})(\sqrt{\mathfrak{C}\mathfrak{A}}-\mathfrak{G}\sqrt{\mathfrak{B}})(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H}\sqrt{\mathfrak{C}})} w = 0. \end{aligned}$$

Anticipating the use of a notation the value of which will subsequently appear, or putting

$$\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}\sqrt{\mathfrak{A}} = fK^2, \quad \sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G}\sqrt{\mathfrak{B}} = gK^2, \quad \sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}\sqrt{\mathfrak{C}} = hK^2,$$

values which give

$$K' = -f^4 - g^4 - h^4 + 2g^2h^2 + 2h^2f^2 + 2f^2g^2 - f^2g^2h^2,$$

the equation of the section in question is

$$\frac{1}{K^2}(-f^2 + g^2 + h^2)x + \frac{1}{K^2}(f^2 - g^2 + h^2)y + \frac{1}{K^2}(f^2 + g^2 - h^2)z + \sqrt{-p} \sqrt{2fgh} K^2 w = 0.$$

I proceed to investigate a transformation of the equation for the section with an indeterminate  $\lambda$ , which touches the two sections  $y = 0$ ,  $z = 0$ .

We have

$$aV^2 = (a\lambda + h\mu + g\nu)^2 + (\mathfrak{B}\mu^2 + \mathfrak{C}\nu^2 - 2\mathfrak{F}\mu\nu) - \mathfrak{B}\mathfrak{C} + \mathfrak{F}^2,$$

or putting for  $\mu$  and  $\nu$  their values  $\sqrt{\frac{K}{\mathfrak{B}}}$ ,  $\sqrt{\frac{K}{\mathfrak{C}}}$  in the second term,

$$aV^2 = (a\lambda + h\mu + g\nu)^2 + (\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})^2;$$

and introducing instead of  $\lambda$  an indeterminate quantity  $X$ , such that

$$a\lambda + h\mu + g\nu = K^2X,$$

we have also introducing throughout  $X$  instead of  $\lambda$ , and completing the substitution of  $\sqrt{\mathfrak{B}}$ ,  $\sqrt{\mathfrak{C}}$ , for  $\mu$ ,  $\nu$ , the equation of the section touching  $y = 0$ ,  $z = 0$ , becomes

$$(ax + hy + gz)X + y\sqrt{\mathfrak{C}} + z\sqrt{\mathfrak{B}} + w\sqrt{-p} \sqrt{1 + X^2} = 0.$$

It may be remarked here in passing, that this is a very convenient form for the demonstration of the theorem; "If two sections of a surface of the second order touch each other, and are also tangent sections (of the same class) to two fixed sections, then considering the planes through the axis of the fixed sections and the poles of the tangent sections, and also the tangent planes through this axis, the anharmonic ratio of the four planes is independent of the position of the moveable tangent sections;" where by the axis of the fixed sections is to be understood the line joining their poles.

The sections touching  $z = 0$ ,  $x = 0$ , and  $x = 0$ ,  $y = 0$ , are of course

$$\begin{aligned} x\sqrt{\mathfrak{C}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-p}\sqrt{1 + Z^2} &= 0, \\ x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (hx + by + fz)Y + w\sqrt{-p}\sqrt{1 + Y^2} &= 0, \end{aligned}$$

where

$$\begin{aligned} h\lambda' + b\mu' + f\nu' &= (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})\sqrt{\mathfrak{B}}, \\ \lambda'^2 &= \mathfrak{A}, \quad \mu'^2 = \mathfrak{B}, \quad \nu'^2 = \mathfrak{C}. \end{aligned}$$

The conditions of contact of the sections represented by the above written equations would be perhaps most simply obtained directly from the lemma, but it is proper to deduce it from the formula for contact used in the present memoir.

If for shortness

$$\Phi(+) = a\lambda'\lambda'' + b\mu'\mu'' + c\nu'\nu'' + f(\mu'\nu'' + \mu''\nu') + g(\nu'\lambda'' + \nu''\lambda') + h(\lambda'\mu'' + \lambda''\mu') \pm V'V'',$$

where the symbol  $\Phi(+)$  is used in order to mark the essentially different character of the results corresponding to the different values of the ambiguous sign, then

$$\begin{aligned} bc\Phi(-) &= f(a\lambda' + h\mu' + g\nu')(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + (\sqrt{2\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + \sqrt{2\mathfrak{A}}(\sqrt{2\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(h\lambda' + b\mu' + f\nu') \\ &\quad + f(-a\sqrt{2\mathfrak{B}\mathfrak{C}} + \sqrt{2\mathfrak{A}\mathfrak{B}}\sqrt{2\mathfrak{A}\mathfrak{C}} + \mathfrak{G}\sqrt{2\mathfrak{A}\mathfrak{B}} - a\mathfrak{F} - (\mathfrak{C}\sqrt{2\mathfrak{A}\mathfrak{B}} - a\mathfrak{F})) \\ &= f(a\lambda' + h\mu' + g\nu')(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + \sqrt{2\mathfrak{A}}(\sqrt{2\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + \sqrt{2\mathfrak{A}}(\sqrt{2\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(a\lambda' + h\mu' + g\nu') \\ &\quad - f(\sqrt{2\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{2\mathfrak{A}\mathfrak{B}} - \mathfrak{H}), \end{aligned}$$

that is,

$$bc\Phi(-) = (\sqrt{2\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{2\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{f F Z + \sqrt{2\mathfrak{A}}(F + Z) - f\}.$$

What, however, is really required, is the value of  $\Phi(+)$ ; to find this, we have

$$\begin{aligned} bc\Phi(+) &= bc\Phi(-) + 2bcK \\ &= (\sqrt{2\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{2\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{f F Z + \sqrt{2\mathfrak{A}}(F + Z) - f\} \\ &\quad + 2bcK - 2f(\sqrt{2\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{2\mathfrak{A}\mathfrak{B}} - \mathfrak{H}), \end{aligned}$$

the second line of which is

$$\begin{aligned} &2(\sqrt{2\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{2\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{\tfrac{1}{2}(\sqrt{2\mathfrak{A}\mathfrak{C}} + \mathfrak{G})(\sqrt{2\mathfrak{A}\mathfrak{B}} + \mathfrak{H}) - f\} \\ &= 2(\sqrt{2\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{2\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\sqrt{2\mathfrak{A}\mathfrak{B}\mathfrak{C}}, \end{aligned}$$

since

$$(\sqrt{2\mathfrak{A}\mathfrak{C}} + \mathfrak{G})(\sqrt{2\mathfrak{A}\mathfrak{B}} + \mathfrak{H}) - 2f = \sqrt{2\mathfrak{A}}(\sqrt{2\mathfrak{B}\mathfrak{C}} + \sqrt{2\mathfrak{A}\mathfrak{F}}).$$

where

$$e = \frac{1}{2}(\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}} + \mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} + \mathfrak{H}\sqrt{\mathfrak{C}}),$$

and consequently

$$bc\Phi(+) = (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{fFZ + \sqrt{\mathfrak{A}}(F + Z) - f + 2e\sqrt{\mathfrak{A}}\},$$

a reduction, which on account of its peculiarity, I have thought right to work out in full.

The condition of contact is

$$\Phi(+) = V'V'' = \frac{1}{\sqrt{bc}}(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\sqrt{1 + F^2}\sqrt{1 + Z^2}.$$

Hence finally, the condition in order that the sections

$$x\sqrt{\mathfrak{C}} + (hx + by + fz)Y + z\sqrt{\mathfrak{A}} + w\sqrt{-p}\sqrt{1 + Y^2} = 0,$$

$$x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-p}\sqrt{1 + Z^2} = 0,$$

(the former of which is a section touching  $z = 0$ ,  $x = 0$ , and the latter a section touching  $x = 0$ ,  $y = 0$ ) may touch, is

$$fFZ + \sqrt{\mathfrak{A}}(F + Z) + (f + 2e\sqrt{\mathfrak{A}}) - \sqrt{K}\sqrt{1 + F^2}\sqrt{1 + Z^2} = 0.$$

The preceding researches show that the solution of Steiner's extension of Malfatti's problem depends on a system of equations, such as the system mentioned at the commencement of the following section.

## §5.

Consider the system of equations

$$\alpha + \beta(Y + Z) + \gamma YZ - \delta\sqrt{1 + Y^2}\sqrt{1 + Z^2} = 0,$$

$$\alpha' + \beta'(Z + X) + \gamma' ZX - \delta'\sqrt{1 + Z^2}\sqrt{1 + X^2} = 0,$$

$$\alpha'' + \beta''(X + Y) + \gamma'' XY - \delta''\sqrt{1 + X^2}\sqrt{1 + Y^2} = 0;$$

these equations may, it will be seen, be solved by quadratics only, when the coefficients satisfy the relations

$$\frac{\beta^2 - \alpha\gamma}{\gamma^2 - \alpha^2} = \frac{\beta'^2 - \alpha'\gamma'}{\gamma'^2 - \alpha'^2} = \frac{\beta''^2 - \alpha''\gamma''}{\gamma''^2 - \alpha''^2}.$$



It may be remarked that these equations are satisfied by  $\beta = 0$ ,  $\beta' = 0$ ,  $\beta'' = 0$ ,  $\gamma = \delta$ ,  $\gamma' = \delta'$ ,  $\gamma'' = \delta''$ ; or if we write

$$\frac{\alpha}{\gamma} = -l, \quad \frac{\alpha'}{\gamma'} = -m, \quad \frac{\alpha''}{\gamma''} = -n,$$

the equations become by a simple reduction,

$$\begin{aligned} Y^2 + Z^2 + 2l YZ &= l^2 - 1, \\ Z^2 + X^2 + 2m ZX &= m^2 - 1, \\ X^2 + Y^2 + 2n XY &= n^2 - 1, \end{aligned}$$

which are equivalent to the equations discussed in my paper "On a system of Equations connected with Malfatti's Problem and on another Algebraical System," *Cambridge and Dublin Mathematical Journal*, t. IV. [1849] pp. 270—275, [79]; the solution might have been effected by the direct method, which I shall here adopt, of eliminating any one of the variables between the two equations into which it enters, and combining the result with the third equation.

Writing the second and third equations under the form

$$A' + B'X + C'\sqrt{1+X^2} = 0, \quad A'' + B''X + C''\sqrt{1+X^2} = 0,$$

the result of the elimination may be presented in the form

$$A'A'' + B'B'' - C'C'' = \sqrt{A'^2 + B'^2 - C'^2} \sqrt{A''^2 + B''^2 - C''^2},$$

which is most easily obtained by writing  $X = \tan \phi$  and operating with the symbol  $\cos^{-2}$ ; but if the rationalized equations be represented by

$$\lambda' + 2\mu'X + \nu'X^2 = 0 \quad \text{and} \quad \lambda'' + 2\mu''X + \nu''X^2 = 0,$$

the form

$$4(\nu'\lambda' - \mu'^2)(\nu''\lambda'' - \mu''^2) = (\lambda'\nu'' + \lambda''\nu' - 2\mu'\mu'')^2$$

leads easily to the result in question.

The values which enter are

$$\begin{aligned} A' &= \alpha' + \beta'Z, & A'' &= \alpha'' + \beta''Y, \\ B' &= \beta' + \gamma'Z, & B'' &= \beta'' + \gamma''Y, \\ C' &= \delta'\sqrt{1+Z^2}, & C'' &= \delta''\sqrt{1+Y^2}; \end{aligned}$$

whence, in the first place, by the equation connecting  $Y$ ,  $Z$ ,

$$C'C'' = -\frac{\delta'\delta''}{\gamma}\{\alpha + \beta(Y+Z) + \gamma YZ\}.$$

It is obviously convenient that  $A'A'' + B'B''$  should be symmetrical with respect to  $Y$  and  $Z$ , and this will be the case if

$$\alpha'\beta'' + \beta'\gamma'' = \alpha''\beta' + \beta''\gamma',$$

that is, if  $\beta'(\gamma'' - \alpha'') = \beta''(\gamma' - \alpha')$ , or assuming that the equations are symmetrical, if

$$\frac{\beta'}{\gamma'' - \alpha''} = \frac{\beta''}{\gamma' - \alpha'};$$

this may be satisfied by writing

$$\beta' = \phi(\gamma' + \alpha'), \quad \beta'' = \phi(\gamma'' + \alpha''),$$

and we have then

$$\begin{aligned} A'A'' + B'B'' &= (\alpha' + \phi(\gamma' + \alpha')Z)(\alpha'' + \phi(\gamma'' + \alpha'')Y) \\ &\quad + (\phi(\gamma' + \alpha') + \gamma'Z)(\phi(\gamma'' + \alpha'') + \gamma''Y) \\ &= (1 + \phi^2)(\alpha' + \gamma'Z)(\alpha'' + \gamma''Y) + \phi(\alpha' + \gamma'Z)^2 + \phi(\alpha'' + \gamma''Y)^2. \end{aligned}$$

Hence putting

$$\alpha' + \gamma'Z = \sqrt{s}P, \quad \alpha'' + \gamma''Y = \sqrt{s}Q,$$

and writing for symmetry,

$$\alpha + \beta(Y + Z) + \gamma YZ = sR,$$

we have

$$A'A'' + B'B'' - C'C'' = s\{(1 + \phi^2)PQ + \phi(P^2 + Q^2) + \frac{\delta'\delta''}{\gamma s}R\}.$$

Also

$$\begin{aligned} A'^2 + B'^2 - C'^2 &= 4(\gamma' + \phi\beta')^2 s(Z + \mu)(Z + \nu), \\ A''^2 + B''^2 - C''^2 &= 4(\gamma'' + \phi\beta'')^2 s(Y + \mu)(Y + \nu), \end{aligned}$$

which may be simplified by writing

$$\mu = \frac{\phi}{1 + \phi^2}, \quad \nu = \frac{\phi'}{1 + \phi'^2},$$

where  $\mu, \nu$  are to be considered as given functions of  $s$  and  $\phi$ .

These values give

$$\begin{aligned} A'^2 + B'^2 - C'^2 &= 4(\gamma' + \phi\beta')^2 s(Z + \mu)(Z + \nu), \\ A''^2 + B''^2 - C''^2 &= 4(\gamma'' + \phi\beta'')^2 s(Y + \mu)(Y + \nu). \end{aligned}$$

Hence, putting for simplicity

$$U = 4(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta'',$$

we have

$$4(Z + \mu)(Z + \nu)(Y + \mu)(Y + \nu) = U^2 + k\{\alpha + \beta(Y + Z) + \gamma YZ\} - \delta'^2(1 + Y^2)(1 + Z^2);$$

and the two sides have next to be expressed in terms of  $Y + Z$  and  $YZ$ .

If for symmetry we write

$$\xi = 1, \quad \eta = Y + Z, \quad \zeta = YZ$$

then

$$4(\mu'\xi + \nu'\eta + \zeta)(\mu''\xi + \nu''\eta + \zeta) + \delta'^2\{(\eta - \xi)^2 + \zeta^2\} = U^2 + k(\alpha\xi + \beta\eta + \gamma\zeta);$$

and  $\xi$  is now to be considered a linear function of  $\xi, \eta, \zeta$ .

The condition that the first side of the equation may divide into factors, gives an equation for determining  $k$ ; since the condition is satisfied for  $k = 0$  and  $k = \infty$ , the equation will be linear, and it is easily seen that the value is  $k = \frac{1}{4}(\mu - \nu)^2$ . In fact

$$\begin{aligned} &4(\mu'\xi + \nu'\eta + \zeta)(\mu''\xi + \nu''\eta + \zeta) + (\mu - \nu)^2\{(\eta - \xi)^2 + \zeta^2\} \\ &= \{2(\mu'\xi + \nu'\eta + \zeta) + (\mu - \nu)(\eta - \xi)\}\{2(\mu''\xi + \nu''\eta + \zeta) - (\mu - \nu)(\eta - \xi)\}; \end{aligned}$$

hence

and we may assume

$$2(\mu'\xi + \nu'\eta + \zeta) + (\mu - \nu)(\eta - \xi) = 2\Lambda(\alpha\xi + \beta\eta + \gamma\zeta),$$

subject to its being shown that

$$2(\mu''\xi + \nu''\eta + \zeta) - (\mu - \nu)(\eta - \xi) = 2\frac{U}{\Lambda}(\alpha\xi + \beta\eta + \gamma\zeta)$$

gives a constant value for  $\Lambda$ .

The comparison of coefficients gives

$$\nu' = \frac{1}{2}\beta(\Lambda + \frac{U}{\Lambda}), \quad \mu - \nu = (\Lambda - \frac{U}{\Lambda})\gamma;$$

the first and third of these give

$$4(1 - \mu\nu) = \beta^2(\Lambda + \frac{U}{\Lambda})^2 - (\Lambda - \frac{U}{\Lambda})^2\gamma^2,$$

which will be identical with the second, if

$$\frac{2(1 - \mu\nu)}{\beta^2 - \gamma^2} = \frac{\mu + \nu}{\beta\gamma},$$

which follows at once from the equation

$$\mu = \frac{\phi}{1 + \phi^2}, \quad \nu = \frac{\phi'}{1 + \phi'^2}.$$

Forming next the two equations

$$\frac{2\mu' - (\mu - \nu)}{\beta} = \frac{2\nu'}{\beta + \gamma\frac{U}{\Lambda}}, \quad \frac{2\mu' + (\mu - \nu)}{\beta} = \frac{2\nu'}{\beta + \gamma\Lambda},$$

these will be equivalent to a single equation if

$$(\mu + \nu)\beta^2 = (\mu + \nu)(\beta^2 + \gamma^2) - \frac{1}{4}(\mu - \nu)^2\beta\gamma - \frac{1}{4}(\mu + \nu)(\beta^2 - \gamma^2),$$

that is, if

$$(\mu + \nu)^2\beta^2 = (\mu + \nu)^2(\beta^2 + \gamma^2) - 4(\mu + \nu)\beta\gamma - \frac{1}{4}(\mu - \nu)^2(\beta^2 - \gamma^2),$$

or finally, if

$$\delta'^2 = \beta'^2 + \gamma'^2 - 4s\beta'\gamma'(1 + \phi^2) = \beta'^2 + \gamma'^2 - 4s(\gamma' + \phi\beta')\beta',$$

which is in fact the case.

Writing the equations

$$\frac{\lambda}{\sqrt{\alpha\xi + \beta\eta + \gamma\zeta}} = \frac{\frac{1}{2}(\mu - \nu)\beta s}{\Lambda}, \quad \frac{\lambda'}{\sqrt{\alpha\xi + \beta\eta + \gamma\zeta}} = \frac{\frac{1}{2}(\mu - \nu)\beta s}{\Lambda'},$$

in the form

$$\frac{1}{\lambda\sqrt{\alpha\xi + \beta\eta + \gamma\zeta}} = \frac{2\Lambda}{(\mu - \nu)\beta s}, \quad \frac{1}{\lambda'\sqrt{\alpha\xi + \beta\eta + \gamma\zeta}} = \frac{2\Lambda'}{(\mu - \nu)\beta s},$$

and substituting in

$$A'A'' + B'B'' - C'C'' = \sqrt{A'^2 + B'^2 - C'^2} \sqrt{A''^2 + B''^2 - C''^2},$$

we have

$$\begin{aligned} (1 + \phi^2)PQ + \phi(P^2 + Q^2) + \frac{\delta'\delta''}{\gamma s}R \\ = \frac{1}{2}\sqrt{(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta''} \\ \times \{(-\beta + 2s\gamma + 2\phi\gamma)\xi + (\gamma - 2s\beta)\eta + (-\beta + 2s\gamma + 4s\phi\beta)\zeta\}, \end{aligned}$$

and consequently, multiplying by

$$b = 2\sqrt{(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta''},$$

we have

$$\begin{aligned} = \frac{1}{2}\sqrt{(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta''} \\ \times \{(-\beta + 2s\gamma + 2\phi\gamma)\xi + (\gamma - 2s\beta)\eta + (-\beta + 2s\gamma + 4s\phi\beta)\zeta\}, \end{aligned}$$

or collecting the different terms which enter into the equation

$$A'A'' + B'B'' - C'C'' = \sqrt{A'^2 + B'^2 - C'^2} \sqrt{A''^2 + B''^2 - C''^2}$$

the result is

$$\begin{aligned} (\alpha'\alpha'' + \beta'\beta'')\xi + (\gamma'\beta'' + \gamma''\beta' + 2\phi\beta'\beta'')\eta + (\beta'\beta'' + \gamma'\gamma'')\zeta + \frac{\delta'\delta''}{\gamma}(\alpha\xi + \beta\eta + \gamma\zeta) \\ - \sqrt{(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta''} \{(-\beta + 2s\gamma + 2\phi\gamma)\xi + (\gamma - 2s\beta)\eta + (-\beta + 2s\gamma + 4s\phi\beta)\zeta\} = \end{aligned}$$

which, combined with the first equation written under the form

$$\alpha\xi + \beta\eta + \gamma\zeta = 0,$$

determines the ratios of  $\xi$ ,  $\eta$ ,  $\zeta$ , that is, the values of  $Y + Z$  and  $YZ$ .

§6.

The system of equations

$$\begin{aligned}(f + 2e\sqrt{\mathfrak{A}}) + \sqrt{\mathfrak{A}}(Y + Z) + fYZ - \sqrt{bc}\sqrt{1 + Y^2}\sqrt{1 + Z^2} &= 0, \\(g + 2e\sqrt{\mathfrak{B}}) + \sqrt{\mathfrak{B}}(Z + X) + gZX - \sqrt{ca}\sqrt{1 + Z^2}\sqrt{1 + X^2} &= 0, \\(h + 2e\sqrt{\mathfrak{C}}) + \sqrt{\mathfrak{C}}(X + Y) + hXY - \sqrt{ab}\sqrt{1 + X^2}\sqrt{1 + Y^2} &= 0,\end{aligned}$$

where

$$e = \frac{1}{2}(\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}} + \mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} + \mathfrak{H}\sqrt{\mathfrak{C}}),$$

on which depends the solution of Steiner's extension of Malfatti's problem, is at once seen to belong to the class of equations treated of in the preceding section, and we have  $\phi = 0$ ,  $s = 0$ .

The equations at the conclusion of the preceding section become

$$\begin{aligned}&\{\sqrt{\mathfrak{B}\mathfrak{C}} + gh + 2e(g\sqrt{\mathfrak{C}} + h\sqrt{\mathfrak{B}}) + 4e^2\sqrt{\mathfrak{B}\mathfrak{C}}\}\xi + \{g\sqrt{\mathfrak{C}} + h\sqrt{\mathfrak{B}} + 2e\sqrt{\mathfrak{B}\mathfrak{C}}\}\eta \\&\quad + \{-f\sqrt{\mathfrak{B}\mathfrak{C}} + gh\}\zeta - \alpha\{(f + 2e\sqrt{\mathfrak{A}})\xi + \sqrt{\mathfrak{A}}\eta + f\zeta\} \\&- \frac{1}{2}\sqrt{(g + e\sqrt{\mathfrak{C}})(h + e\sqrt{\mathfrak{B}})\sqrt{\mathfrak{B}\mathfrak{C}}}\{(\sqrt{\mathfrak{A}} - 2ef)\xi - f\eta + \sqrt{\mathfrak{A}}\zeta\} = 0, \\&\quad \{(f + 2e\sqrt{\mathfrak{A}})\xi + \sqrt{\mathfrak{A}}\eta + f\zeta\}^2 - bc\{(\eta - \xi)^2 + \zeta^2\} = 0,\end{aligned}$$

which may also be written

$$\begin{aligned}&(\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\xi + \zeta) + (-\alpha\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{C}} + \mathfrak{H}\sqrt{\mathfrak{B}} + 2e\sqrt{\mathfrak{B}\mathfrak{C}})(\eta + 2\theta\xi) \\&- \frac{1}{2}\sqrt{(g + e\sqrt{\mathfrak{C}})(h + e\sqrt{\mathfrak{B}})\sqrt{\mathfrak{B}\mathfrak{C}}}\{(\sqrt{\mathfrak{A}} - 2ef)\xi - f\eta + \sqrt{\mathfrak{A}}\zeta\} = 0, \\&\quad \{f(\xi + \zeta) + \sqrt{\mathfrak{A}}(\eta + 2\theta\xi)\}^2 - bc\{(\eta - \xi)^2 + \zeta^2\} = 0.\end{aligned}$$

Hence observing that

$$\begin{aligned}g + e\sqrt{\mathfrak{C}} &= \frac{1}{2}(\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G}), \\h + e\sqrt{\mathfrak{B}} &= \frac{1}{2}(\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H}), \\-\alpha\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{C}} + \mathfrak{H}\sqrt{\mathfrak{B}} + 2e\sqrt{\mathfrak{B}\mathfrak{C}} &= (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F}),\end{aligned}$$

and putting for a moment

$$\chi = 4\sqrt{(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H})\sqrt{\mathfrak{B}\mathfrak{C}}}$$

and therefore

$$\sqrt{(g + e\sqrt{\mathfrak{C}})(h + e\sqrt{\mathfrak{B}})\sqrt{\mathfrak{B}\mathfrak{C}}} = (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})\chi,$$

the first equation divides by  $(\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})$ , and the result is

$$2\chi = 0.$$

Also, by an easy transformation, the second equation becomes  
 $-\chi^2\{(\xi + \zeta) - f(\eta + 2\theta\xi)\}^2 + 4\{(\xi + \zeta) + f(\eta + 2\theta\xi) - (\xi + \zeta)\xi\} = 0,$   
 or putting

$$\psi = \frac{\chi}{1 + \zeta}, \quad \omega = \frac{\chi\sqrt{\xi}}{(1 + \zeta)\sqrt{\zeta}},$$

the equations become

$$\psi^2 - 2\lambda\psi = 0,$$

hence eliminating  $\psi$ ,

$$\omega^2 = 2\lambda;$$

or observing that

$$\lambda = 2\sqrt{(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H})\sqrt{\mathfrak{B}\mathfrak{C}}},$$

and reducing, we obtain

$$1 + \zeta = \chi^2 (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H}),$$

also

$$\omega = 2\chi \text{ gives } \frac{2\sqrt{(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H})\sqrt{\mathfrak{B}\mathfrak{C}}}}{1 + \zeta}.$$

Suppose

$$\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F} = a, \quad \sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F} = a', \quad aa' = f^2 K^4, \text{ \&c.},$$

then substituting

that is,

$$\xi + \zeta + \epsilon(\eta + 2\theta\xi) - \frac{1}{2}\sqrt{aa'}\sqrt{\xi + \zeta}\sqrt{\eta + 2\theta\xi} = 0,$$

these may be written

$$\frac{\xi}{\sqrt{a}} + \frac{\zeta}{\sqrt{a'}} + \frac{\epsilon(\eta + 2\theta\xi)}{\sqrt{aa'}} - \frac{1}{2}\sqrt{\xi + \zeta}\sqrt{\eta + 2\theta\xi} = 0,$$

where

$$\frac{\xi}{\sqrt{a}} = M, \quad \frac{\zeta}{\sqrt{a'}} = N, \quad \frac{\epsilon}{\sqrt{aa'}} = P, \quad \&c.;$$

or since  $\xi, \eta, \zeta$  are equal to 1,  $Y + Z, YZ$  respectively,

$$\xi : Y + Z : YZ = MN' - M'N : NL' - N'L : LM' - L'M,$$

where

$$L = \sqrt{a}, \quad M = \sqrt{a'}, \quad N = \frac{\epsilon}{\sqrt{aa'}}, \quad \&c.$$

Also

$$\sqrt{bc} = \frac{1}{2}\sqrt{aa'}f = \frac{1}{2}f^2K^2,$$

whence

$$\sqrt{\beta_1\lambda} = \frac{1}{2}\sqrt{a + a'}, \quad \sqrt{\gamma_1\lambda} = \frac{1}{2}\sqrt{a + a'},$$

and by forming the analogous expressions for  $Z + X$  and  $ZX, X + Y$  and  $XY$ , the values of  $X, Y, Z$  may be determined.

But the equations in question simplify themselves in a remarkable manner by the notation before alluded to.



Suppose

$$\begin{aligned}\sqrt{\mathfrak{A}} &= f^2, & \sqrt{\mathfrak{B}} &= g^2, & \sqrt{\mathfrak{C}} &= h^2, \\ \sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F} &= 2f^2gh, & \sqrt{\mathfrak{C}\mathfrak{A}} + \mathfrak{G} &= 2g^2hf, & \sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H} &= 2h^2fg;\end{aligned}$$

these values give

$$K^2 = -f^4 - g^4 - h^4 + 2g^2h^2 + 2h^2f^2 + 2f^2g^2 - f^2g^2h^2.$$

We have

$$\begin{aligned}\frac{1}{K^2}\{-f^2 + g^2 + h^2\} &= \frac{1}{K^2}\{-f^4 + g^4 + h^4 + 2g^2h^2 - 2h^2f^2 - 2f^2g^2\}, \\ \frac{1}{K^2}\{f^2 - g^2 + h^2\} &= \frac{1}{K^2}\{f^4 - g^4 + h^4 + 2g^2h^2 - 2h^2f^2 - 2f^2g^2\}, \\ \frac{1}{K^2}\{f^2 + g^2 - h^2\} &= \frac{1}{K^2}\{f^4 + g^4 - h^4 + 2g^2h^2 - 2h^2f^2 - 2f^2g^2\}.\end{aligned}$$

Applying these results to the preceding formulæ and forming for that purpose the equations

$$\begin{aligned}\frac{K^2}{2\sqrt{\mathfrak{A}}} &= \frac{K^2}{2f^2}, & \frac{K^2}{2\sqrt{\beta_1\lambda}} &= \frac{K^2}{4gh}, \\ \frac{K^2}{\sqrt{\beta_1\gamma_1}} &= \frac{K^2}{4gh}, & \frac{K^2}{\sqrt{\gamma_1\alpha_1}} &= \frac{K^2}{4hf}, & \frac{K^2}{\sqrt{\alpha_1\beta_1}} &= \frac{K^2}{4fg},\end{aligned}$$

$$\begin{aligned}ghK^2(f^2 + Y + Z) &= (f^2 - gh)(f^2 - (g - h)^2) - 2gh(g - h)^2, \\ f^2(Y + Z) + 2YZ &= 4(f^2 - gh)(1 - \frac{f^2}{K^2}), \\ f^4YZ + K^4 &= ((f^2 - gh)\{f^2 - (g - h)^2\} - 2gh(g - h)^2)(1 - \frac{f^2}{K^2});\end{aligned}$$

the former of which, combined with the similar equations for  $Z + X$  and  $X + Y$ , gives for  $X$ ,  $Y$ ,  $Z$  the values to be presently stated, and these values will of course verify the second equation and the corresponding equations for  $ZX$  and  $XY$ .

Recapitulating the preceding notation, if  $x = 0, y = 0, z = 0$  are the equations of the given sections,  $w = 0$  the equation of the polar plane of their point of intersection with respect to the surface,

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0$$

the equation of the surface,  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, K$  as usual, and

$$\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}\sqrt{\mathfrak{A}} = fK^2,$$

$$\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G}\sqrt{\mathfrak{B}} = gK^2,$$

$$\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}\sqrt{\mathfrak{C}} = hK^2,$$

then the equations of the required sections are

$$(ax + hy + gz)X + y\sqrt{\mathfrak{C}} + z\sqrt{\mathfrak{B}} + w\sqrt{-p}\sqrt{1 + X^2} = 0,$$

$$x\sqrt{\mathfrak{C}} + (hx + by + fz)Y + z\sqrt{\mathfrak{A}} + w\sqrt{-p}\sqrt{1 + Y^2} = 0,$$

$$x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-p}\sqrt{1 + Z^2} = 0,$$

where  $X, Y, Z$  are to be determined by the following equations,

$$(f + 2e\sqrt{\mathfrak{A}}) + \sqrt{\mathfrak{A}}(Y + Z) + fYZ - \sqrt{bc}\sqrt{1 + Y^2}\sqrt{1 + Z^2} = 0,$$

$$(g + 2e\sqrt{\mathfrak{B}}) + \sqrt{\mathfrak{B}}(Z + X) + gZX - \sqrt{ca}\sqrt{1 + Z^2}\sqrt{1 + X^2} = 0,$$

$$(h + 2e\sqrt{\mathfrak{C}}) + \sqrt{\mathfrak{C}}(X + Y) + hXY - \sqrt{ab}\sqrt{1 + X^2}\sqrt{1 + Y^2} = 0;$$

and the solution of which, putting

$$K^2 = -f^4 - g^4 - h^4 + 2g^2h^2 + 2h^2f^2 + 2f^2g^2 - f^2g^2h^2,$$

is given by the equations

$$KX = -f^2 + (-f + g + h)^2 - 2(-f + g + h)J,$$

$$KY = -g^2 + (f - g + h)^2 - 2(f - g + h)J,$$

$$KZ = -h^2 + (f + g - h)^2 - 2(f + g - h)J,$$

where

$$J^2 = -f^2 - g^2 - h^2 + (f + g + h)^2 - f^2g^2h^2.$$

Instead of the direct but very tedious process by which these values of  $X, Y, Z$  have been obtained, we may substitute the following *à posteriori* verification.

We have

$$\begin{aligned} K^2(1 + X^2) &= 4(-f + g + h)^2 J^2 \left(1 + \frac{f^2}{K^2}\right) \left(1 - \frac{f^2}{K^2}\right), \\ f^2 K^2 \sqrt{1 + Y^2} \sqrt{1 + Z^2} &= 4(f^2 - (g - h)^2) J^2 \left(1 - \frac{f^2}{K^2}\right) \sqrt{\left(1 - \frac{g^2}{K^2}\right) \left(1 - \frac{h^2}{K^2}\right)}, \\ f K^2(1 + YZ) &= 4\left(1 - \frac{f^2}{K^2}\right) \{(f^2 - gh)(f^2 - (g - h)^2) - 2gh(g - h)\}, \\ K^2(Y + Z - 2f^2 - 2g^2 - 2h^2 + 4J^2) &= 4\left(1 - \frac{f^2}{K^2}\right)(f^2 - gh). \end{aligned}$$

Putting also

$$1 = \frac{1}{K^2} \{-f^2 - g^2 - h^4 + (f + g + h)^2 - f^2 g^2 h^2\},$$

it is perhaps worth noticing that the value of the quantity  $\chi$  previously made use of,

$$\chi = \frac{1}{K^2} \{-f^2 - g^2 - h^2 + (f + g - h)^2 - f^2 g^2 h^2\}$$

we have

$$\begin{aligned} &(f^2 + g^2 + h^2 + \frac{1}{K^2}) K^2(1 + YZ) \\ &= 4\left(1 - \frac{f^2}{K^2}\right) \{(f^2 - (g - h)^2)((f^2 - gh)(f^2 - (g - h)^2) - 2gh(g - h)^2 - 2ghJ^2) \\ &\quad + 4g^2 h^2 (g - h)^2 (f^2 - gh)\}, \end{aligned}$$

$$\begin{aligned} &\frac{K^2}{J^2} \{Z(Y + X) - 2f^2 - 2g^2 - 2h^2 + 4J^2\} \\ &= 4\left(1 - \frac{f^2}{K^2}\right) (f^2 - (g - h)^2) \{(f^2 - gh)((g + h)^2 - f^2) - \frac{1}{K^2} (f^2 - (g - h)^2)\}. \end{aligned}$$

Also, since

$$\{f^2 - (g - h)^2\} + \{(g + h)^2 - f^2\} = 4gh,$$

we have

$$= 4\left(1 - \frac{f^2}{K^2}\right) (f^2 - (g - h)^2) 2gh J^2 \left(1 - \frac{f^2}{K^2}\right) \left(1 - \frac{g^2}{K^2}\right),$$

and the values obtained above give also

$$2gh \sqrt{1 + Y^2} \sqrt{1 + Z^2} = \frac{K^2}{J^2} \sqrt{1 + Y^2} \sqrt{1 + Z^2},$$

which shows that the relation between  $Y$  and  $Z$  is verified by the assumed values of these quantities, and the other two equations are of course also verified.

§7.

First, to substitute in the equations of the required sections or resultors. Writing the first equation in the form

$$\frac{K^2}{2\sqrt{\mathfrak{B}\mathfrak{C}}}\{aK^2x+(hK^2+\sqrt{\mathfrak{G}\mathfrak{D}})y+(gK^2+\sqrt{\mathfrak{F}\mathfrak{G}})z+\sqrt{-p}\sqrt{1+X^2}w\}=0,$$

the coefficient of  $x$  will be

$$\frac{1}{K^2}\{-f^4-g^4-h^4+2g^2h^2+2h^2f^2+2f^2g^2-f^2g^2h^2\}x,$$

or, as it is convenient to write it,

$$\{-f^2+g^2+h^2-\frac{1}{K^2}f^2(-f^2+g^2+h^2)\}x.$$

The coefficient of  $y$  is

$$\frac{1}{K^2}\{-f^4-g^4-h^4+2g^2h^2+2h^2f^2+2f^2g^2-f^2g^2h^2\}y+\sqrt{\mathfrak{G}\mathfrak{D}}y,$$

or, after all reductions,

$$\{g^2(f^2-g^2+h^2)+h^2(f^2+g^2-h^2)+2gh\sqrt{(f^2-g^2+h^2)(f^2+g^2-h^2)}\}^{\frac{1}{2}}y,$$

and similarly the coefficient of  $z$  is

$$\{h^2(f^2+g^2-h^2)+g^2(f^2-g^2+h^2)+2hg\sqrt{(f^2+g^2-h^2)(f^2-g^2+h^2)}\}^{\frac{1}{2}}z.$$

The coefficient of  $w$  is

$$\sqrt{-p}K^2\sqrt{1+X^2}w=\sqrt{-p}\sqrt{4(-f+g+h)^2J^2(1+\frac{f^2}{K^2})(1-\frac{f^2}{K^2})}w.$$

The values might be somewhat simplified by writing

$$f=4(1-\frac{f^2}{K^2}), \quad g=4(1-\frac{g^2}{K^2}), \quad h=4(1-\frac{h^2}{K^2}),$$

and it may be also remarked, that the coefficients as well of these formulæ as of those which follow may be elegantly expressed in terms of the parts of a triangle having  $f, g, h$  for its sides.

The equations of the separators are found by taking the differences two and two of the equations of the resultors (this requires to be verified *a posteriori*); thus subtracting the third equation from the second the result contains a constant factor,

$$\frac{1}{K^2}\{4f^2g^2h^2-J^2(f^2-(g-h)^2)((g+h)^2-f^2)\},$$

equivalent to

$$\frac{1}{K^2} \{4f^2 g^2 h^2 - (f^2 - (g - h)^2)((g + h)^2 - f^2)\}.$$

Rejecting the factor in question and forming the analogous two equations, the equations of the separators are

$$\begin{aligned} (-f^2 + g^2 + h^2)x + (f^2 - g^2 + h^2)y + (f^2 + g^2 - h^2)z + \sqrt{-p} K^2 \sqrt{2} w &= 0, \\ (f^2 - g^2 + h^2)x + (-f^2 + g^2 + h^2)y + (f^2 + g^2 - h^2)z + \sqrt{-p} K^2 \sqrt{2} w &= 0, \\ (f^2 + g^2 - h^2)x + (f^2 + g^2 - h^2)y + (-f^2 + g^2 + h^2)z + \sqrt{-p} K^2 \sqrt{2} w &= 0. \end{aligned}$$

and from the mode of formation of these equations it is obvious that the separators have a line in common.

The equations of the determinators being  $x = 0$ ,  $y = 0$ ,  $z = 0$ , the equations of the tactors are

$$\begin{aligned} (-f^2 + g^2 + h^2)x + 2\sqrt{g^2h^2}y + 2\sqrt{g^2h^2}z + \sqrt{-p}K^2\sqrt{2}w &= 0, \\ 2\sqrt{h^2f^2}x + (f^2 - g^2 + h^2)y + 2\sqrt{h^2f^2}z + \sqrt{-p}K^2\sqrt{2}w &= 0, \\ 2\sqrt{f^2g^2}x + 2\sqrt{f^2g^2}y + (f^2 + g^2 - h^2)z + \sqrt{-p}K^2\sqrt{2}w &= 0, \end{aligned}$$

and if  $\alpha x + \beta y + \gamma z + \delta w = 0$  be the equation of the tactor touching  $x = 0$ ,  $\sqrt{\mathfrak{B}}x - \sqrt{\mathfrak{C}}z = 0$  and  $\sqrt{\mathfrak{A}}y - \sqrt{\mathfrak{H}}x = 0$ , the conditions of contact are

$$\begin{aligned} \alpha(\mathfrak{A}\alpha^2 + \dots - \delta^2) &= (\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{C}\gamma)^2, \\ 2\sqrt{\mathfrak{A}\mathfrak{B}}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(\mathfrak{A}\alpha^2 + \dots - \delta^2) &= \frac{1}{2}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(\alpha\sqrt{\mathfrak{A}} - \beta\sqrt{\mathfrak{B}}) + \gamma(\mathfrak{C}\sqrt{\mathfrak{A}} - \mathfrak{C}\sqrt{\mathfrak{B}}) \end{aligned}$$

whence

$$\begin{aligned} (\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\sqrt{\mathfrak{A}}\alpha - (\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\sqrt{\mathfrak{B}}\beta + (\mathfrak{C}\sqrt{\mathfrak{A}} - \mathfrak{C}\sqrt{\mathfrak{B}})\gamma, \\ \frac{1}{2}\sqrt{2\sqrt{\mathfrak{A}\mathfrak{B}}}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(\mathfrak{A}\alpha^2 + \mathfrak{H}\beta^2 + \mathfrak{C}\gamma^2) = \\ (\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\sqrt{\mathfrak{A}}\alpha + (\mathfrak{H}\sqrt{\mathfrak{A}} - \mathfrak{F}\sqrt{\mathfrak{B}})\beta - (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{C})\gamma, \\ c\beta^2 + b\gamma^2 - 2f\beta\gamma + \dots - 2\delta^2 = 0, \end{aligned}$$

and putting for a moment

$$\mu = \sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}} - \mathfrak{C} - \sqrt{2\sqrt{\mathfrak{A}\mathfrak{B}}}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}), \quad \nu = \sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H} - \sqrt{2\sqrt{\mathfrak{A}\mathfrak{B}}}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}),$$

after some reductions, and observing that the ratios only of the quantities  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$  are material, we obtain

$$\beta = \frac{\mu}{\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}}}\sqrt{\mathfrak{A}}, \quad \gamma = \frac{\nu}{\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}}}\sqrt{\mathfrak{B}}, \quad \delta = \frac{\mu\nu}{\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}}}\sqrt{-p}.$$

and it is easily seen also that the coordinates of the point of contact are

$$x = 0, \quad y = \nu, \quad z = \mu, \quad w = \frac{\mu\nu}{\sqrt{-p}}.$$

Hence substituting and introducing throughout the quantities  $f, g, h, J$ , also forming the analogous equations, the equations of the tactors are

$$\begin{aligned} & \frac{1}{K^2}(-f^2+g^2+h^2)x + \{g^2(-f^2+g^2+h^2) + (g+h)\sqrt{(f^2-(g-h)^2-J^2)} \\ & \quad + 2\sqrt{g^2h^2}\left(1-\frac{f^2}{K^2}\right)\left(1-\frac{g^2}{K^2}\right)(f^2-(g-h)^2)\}^{\frac{1}{2}}y \\ & + 2\sqrt{g^2h^2}\left(1-\frac{f^2}{K^2}\right)\left(1-\frac{h^2}{K^2}\right)(f^2-(g-h)^2)\}^{\frac{1}{2}}z + \sqrt{-p}K^2\sqrt{2}w = 0, \end{aligned}$$

$$\begin{aligned} & \{h^2(f^2-g^2+h^2) + (h+f)\sqrt{(g^2-(h-f)^2-J^2)} \\ & \quad + 2\sqrt{h^2f^2}\left(1-\frac{g^2}{K^2}\right)\left(1-\frac{h^2}{K^2}\right)(g^2-(h-f)^2)\}^{\frac{1}{2}}x \\ & + \frac{1}{K^2}(f^2-g^2+h^2)y + 2\sqrt{h^2f^2}\left(1-\frac{g^2}{K^2}\right)\left(1-\frac{f^2}{K^2}\right)(g^2-(h-f)^2)\}^{\frac{1}{2}}z + \sqrt{-p}K^2\sqrt{2}w = 0, \end{aligned}$$

$$\begin{aligned} & \{f^2(f^2+g^2-h^2) + (f+g)\sqrt{(h^2-(f-g)^2-J^2)} \\ & \quad + 2\sqrt{f^2g^2}\left(1-\frac{h^2}{K^2}\right)\left(1-\frac{f^2}{K^2}\right)(h^2-(f-g)^2)\}^{\frac{1}{2}}x \\ & + 2\sqrt{f^2g^2}\left(1-\frac{h^2}{K^2}\right)\left(1-\frac{g^2}{K^2}\right)(h^2-(f-g)^2)\}^{\frac{1}{2}}y + \frac{1}{K^2}(f^2+g^2-h^2)z + \sqrt{-p}K^2\sqrt{2}w = 0. \end{aligned}$$

It is obvious, from the equations, that each separator passes through the point of contact of a tactor and determinator, it consequently only remains to be shown that each separator touches two tactors.

Consider the tactor which has been represented by  $\alpha x + \beta y + \gamma z + \delta w = 0$ , the unreduced values of the coefficients give

$$\gamma^2 \alpha^2 + \dots - \delta^2 = \frac{1}{K^2} (\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma)^2 = K^2.$$

Represent for a moment the separator by  $lx + my + nz + sw = 0$ . Then putting  $\mathfrak{A}\alpha^2 + \dots - \delta^2 = \Omega$ ,

$$= \frac{1}{K^2} \{-f^2 + g^2 + h^2 + 2gh - f^2(-f^2 + g^2 + h^2) - 2gh(-f^2 + g^2 + h^2)\},$$

the condition of contact becomes

$$4\{-f^2 + g^2 + h^2 + h(f + g) - \frac{1}{K^2}\};$$

or, forming the value of  $n^2$  and substituting,

$$\dots (l, m, n)^2 - s^2 \Omega = (l\alpha + m\beta + n\gamma + s\delta)^2,$$

which may be verified without difficulty, and thus the construction for the resultors is shown to be true.



§8.

Several of the formulæ of the preceding sections of this memoir apply to any number of variables.

Consider the surface (i.e. hypersurface)

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy \dots + pt^2 = 0,$$

and the section (i.e. hypersection)

$$(a\lambda + h\mu + g\nu \dots)x + (h\lambda + b\mu + f\nu \dots)y + (g\lambda + f\mu + c\nu \dots)z \dots + \sqrt{-p}\rho t = 0,$$

where

$$V^2 = a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu \dots - K,$$

the condition of contact with any other section represented by a similar equation is

$$a\lambda\lambda' + b\mu\mu' + c\nu\nu' + f(\mu\nu' + \mu'\nu) + g(\nu\lambda' + \nu'\lambda) + h(\lambda\mu' + \lambda'\mu) \dots \pm K = VV',$$

where  $K$  is the determinant formed with the coefficients  $a, b, c, f, g, h, \dots$

And consequently, by establishing all or any of the equations  $\lambda = \sqrt{\frac{K}{\mathfrak{A}}}$ ,  $\mu = \sqrt{\frac{K}{\mathfrak{B}}}$ ,  $\nu = \sqrt{\frac{K}{\mathfrak{C}}}$ ,  $\dots$  we have the condition in order that the section in question may touch all or the corresponding sections of the sections  $x = 0, y = 0, z = 0, \dots$

Let  $n$  be the number of the variables  $x, y, z, \dots$ , then  $K^{n-1} = \mathfrak{A}\mathfrak{B}\mathfrak{C} \dots \mathfrak{P}\mathfrak{Q}\mathfrak{R} \dots$ ; also

$$K^{n-1} \{ (a\lambda + h\mu + g\nu \dots)x + (h\lambda + b\mu + f\nu \dots)y + (g\lambda + f\mu + c\nu \dots)z \dots \}$$

$$= \begin{vmatrix} x & y & z & \dots \\ \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \lambda & \mu & \nu & \dots \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \mu & \nu & \lambda & \dots \\ \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \end{vmatrix},$$

whence also

$$K^{n-1} V = \begin{vmatrix} 1 & \lambda & \mu & \nu & \dots \\ \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots & \\ \lambda & \mu & \nu & \dots & \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots & \\ \mu & \nu & \lambda & \dots & \end{vmatrix},$$

or  $K^{n-1} \sqrt{-p}\rho = \dots$

and the equation of the section in question becomes

$$\begin{vmatrix} x & y & z & \dots \\ \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \lambda & \mu & \nu & \dots \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \mu & \nu & \lambda & \dots \\ \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \end{vmatrix} + K^{\frac{1}{2}n-1} \sqrt{-p} \rho w = 0.$$

Also the condition of contact with the corresponding section is

$$\begin{vmatrix} 1 & \lambda & \mu & \nu & \dots \\ \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots & \\ \lambda & \mu & \nu & \dots & \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots & \\ \mu & \nu & \lambda & \dots & \end{vmatrix} + 1 = VV'.$$

In particular the equation of the sections which touches all the sections  $x = 0, y = 0, z = 0, \dots$ , is

$$\begin{vmatrix} x & y & z & \dots & w \\ \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots & \\ \sqrt{\mathfrak{A}} & \mathfrak{H} & \mathfrak{G} & \dots & \\ \sqrt{\mathfrak{B}} & \mathfrak{B} & \mathfrak{F} & \dots & \\ \sqrt{\mathfrak{C}} & \mathfrak{F} & \mathfrak{C} & \dots & \end{vmatrix} + K^{\frac{1}{2}n-1} \sqrt{-p} w = 0.$$

Again, the equations of the section touching  $y = 0, z = 0, \dots$  and the sections touching  $x = 0, z = 0, \dots$  are

$$\begin{vmatrix} x & y & z & \dots \\ \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots \\ X & a & \mathfrak{H} & \mathfrak{G} & \dots \\ \sqrt{\mathfrak{B}} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{F} & \mathfrak{C} & \dots \end{vmatrix} + K^{\frac{1}{2}n-1} \sqrt{-p} \sqrt{1 + X^2} w = 0,$$

$$\begin{vmatrix} x & y & z & \dots \\ \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots \\ Y & \mathfrak{A} & b & \mathfrak{G} & \dots \\ \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{F} & c & \dots \end{vmatrix} + K^{\frac{1}{2}n-1} \sqrt{-p} \sqrt{1 + Y^2} w = 0,$$

and the condition of contact of these two sections is

$$\left| \begin{array}{ccccc} 1 & X & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{H} & \mathfrak{G} & \dots & \\ \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \mathfrak{F} & \dots & \\ \sqrt{\mathfrak{A}} & \mathfrak{G} & \mathfrak{C} & \dots & \end{array} \right| \left| \begin{array}{ccccc} 1 & Y & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{H} & \mathfrak{G} & \dots & \\ \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \mathfrak{F} & \dots & \\ \sqrt{\mathfrak{A}} & \mathfrak{G} & \mathfrak{C} & \dots & \end{array} \right| = VV'.$$

It would seem from the appearance of these equations that there should be some simpler method of obtaining the solution than the method employed in the previous part of this memoir.

## 115.

### NOTE ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED POLYGON.

[From the *Philosophical Magazine*, vol. VI. (1853), pp. 99—102.]

The equation of a conic passing through the points of intersection of the conics  $U = 0$ ,  $V = 0$  is of the form  $wU + V = 0$ , where  $w$  is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of  $w$  a quadratic equation, the roots of which may be considered as parameters for determining the line in question. Let one of the values of  $w$  be considered as equal to a given constant  $k$ , the line is always a tangent to the conic  $kU + V = 0$ ; and taking  $w = p$  for the other value of  $w$ ,  $p$  is a parameter determining the particular tangent, or, what is the same thing, the point of contact of this tangent.

Suppose the tangent meets the conic  $U = 0$  (which is of course the conic corresponding to  $w = \infty$ ) in the points  $P$ ,  $P'$ , and let  $\theta$ ,  $\infty$  be the parameters of the point  $P$ , and  $\theta'$ ,  $\infty$  the parameters of the point  $P'$ .

It follows from my “Note on the Geometrical representation of the Integral  $\int dx \div \sqrt{\{(x+a)(x+b)(x+c)\}}$ ,” [113]<sup>6</sup>, and from the theory of invariants, that if  $\Omega$  represent the “Discriminant” of  $kU + V \dots$

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<sup>6</sup>I take the opportunity of correcting an obvious error in the note in question, viz.  $a^2 + b^2 + c^2 - 2bc - 2ca - 2ab$  is throughout written instead of (what the expression should be)  $b^2c^2 + c^2a^2 + a^2b^2 - 2a^2bc - 2b^2ca - 2c^2ab$ . [This correction is made, ante p. 55.]

## 130.

### Deuxième Mémoire sur les Fonctions Doublement Périodiques.

[From the *Journal fuer die reine und angewandte Mathematik* (Crelle), tom. xlv. (1853), pp. 137–155.]

En revenant au problème des transformations linéaires, partant des équations

$$(2k+1)\Omega_r = \lambda\Omega + \mu\Upsilon, \quad (2k+1)\Upsilon_r = \nu\Omega + \rho\Upsilon,$$

je suppose d'abord que les coefficients  $\lambda, \nu$  ne satisfassent pas à la fois aux deux conditions

$$\lambda \equiv 0, \quad \nu \equiv 0, \quad \text{mod. } (2k+1),$$

et je prends  $p, q$  des entiers quelconques tels, que  $\lambda p + \nu q$  ne soit pas  $\equiv 0, \text{ mod. } (2k+1)$ .

Cela étant, soient

$$\lambda p + \nu q = p_r, \quad \mu p + \rho q = q_r, \quad (2k+1)\psi = p_r\Omega + q_r\Upsilon,$$

et, par conséquent,

$$\psi = p\Omega_r + q\Upsilon_r.$$

Je forme l'équation

$$(m_r, n_r) + s\psi = (m, n)_r,$$

savoir

$$m_r\Omega + n_r\Upsilon + s\psi = m\Omega_r + n\Upsilon_r,$$

c'est-à-dire

$$\begin{aligned} \lambda m + \nu n - sp_r &= (2k+1)m_r, \\ \mu m + \rho n - sq_r &= (2k+1)n_r, \end{aligned}$$

ou, ce qui est la même chose,

$$\begin{aligned} m - sp &= m_r\rho - n_r\nu, \\ n - sq &= m_r\mu - n_r\lambda. \end{aligned}$$

Or,  $m_r, n_r, s$  étant des entiers donnés,  $m, n$  seront aussi des entiers; de même,  $m, n$  étant des entiers donnés, on trouve de  $k$  à  $-k$  un entier  $s$  qui donne  $m_r$  un entier. Mais cela étant,  $n_r$  sera aussi un entier; car autrement  $n_r$  serait une fraction ayant pour dénominateur, lequel on voudrait, des nombres  $2k+1, \lambda, \nu$ , ce qui est impossible à moins que

$$\lambda \equiv 0, \quad \nu \equiv 0, \quad \text{mod} . (2k+1).$$

Mais si ces équations avaient lieu, on trouverait d'abord  $s$  de manière à avoir  $n_r$  entier, et alors, puisqu'on n'a pas aussi

$$\mu \equiv 0, \quad \rho \equiv 0, \quad \text{mod} . (2k+1)$$

(en effet, cela est impossible à cause de l'équation  $\lambda\rho - \mu\nu = 2k+1$ ), on démontrerait, comme auparavant, pour  $n_r$ , que  $m_r$  est entier. Donc, enfin,  $m, n$  étant des entiers donnés, on trouve pour  $m_r, n_r, s$  un système d'entiers tel que  $s$  soit compris de  $k$  à  $-k$ , et l'on voit sans peine qu'il n'y a qu'un seul système de cette espèce.

À présent, partant de l'équation

$$\frac{y_r(x + s\psi)}{y_r(x)} = e^{-\frac{is}{2k+1}(Br-2k+1.B)\pi i} \frac{g(x + s\psi + \text{mod} . (p_r\Omega + q_r\Upsilon))}{g(x + s\psi)} \cdot \frac{g(x + \text{mod} . (p_r\Omega + q_r\Upsilon))}{g(x + \text{mod} . (p_r\Omega + q_r\Upsilon))}.$$

(et faisant attention à la particularité que présente le cas de  $y_r = 0$ ), j'écris successivement

$$y = 0, \quad y = \pm\psi_r, \quad \dots, \quad y = \pm k \cdot \psi_r,$$

et je forme le produit des équations ainsi trouvées. Cela donne, à cause de  $(m_r, n_r) + s\psi = (m, n)_r$ , la condition, par rapport aux limites, étant  $\text{mod} . (m, n)_r < \Upsilon_r, \Upsilon_r = \infty$ ; ou, dans le numérateur,  $s$  doit avoir toutes les valeurs entières depuis  $s = -k$  jusqu'à  $s = +k$ ,  $y$  compris  $s = 0$ , et dans le dénominateur ces mêmes valeurs, hormis la valeur  $s = 0$ .

Il est, à présent, facile de faire voir que cette propriété subsiste pour l'une quelconque des fonctions  $yx, gx, \Theta x, Zx$ ; en effet, pour la démontrer pour  $gx$ , j'écris  $x + \frac{1}{2}\pi$  au lieu de  $x$ ; en prenant, pour un moment,  $p = 2p' + 1, yx = 2y'$ , cela donne  $x + \frac{1}{2}\pi = x + (p', -y')$ ,

$$y_r(x + \frac{1}{2}\pi) = e^{p'i\pi - y'i\pi} \sqrt[4]{-1} \cdot g_r x = e^{p'i\pi - y'i\pi} \sqrt[4]{-1} \cdot g_r x,$$

c'est-à-dire

$$\frac{y_r(x + \frac{1}{2}\pi)}{y_r(x)} = i.$$

Or, on déduit de l'expression pour  $y_r x$ ,

$$y_r(m_r \Omega_r + n_r \Upsilon_r) = e^{-\frac{i}{2k+1}(B_r - 2k+1.B)\pi i} \prod \frac{g(x + s\psi_r)}{g(s\psi_r)},$$

c'est-à-dire

$$g_r x = e^{-\frac{i}{2k+1}(B_r - 2k+1.B)x\pi i} \prod \frac{g(x + s\psi_r)}{g(s\psi_r)},$$

ou enfin, à cause de l'équation  $-B_r \Omega + (2k+1)(\mu + \beta)\Omega - \beta_r(p_r, q_r)_r = 0$ , la valeur de  $g_r x$  est

$$g_r x = \prod \frac{g(x + s\psi_r)}{g(s\psi_r)},$$

et en représentant, comme auparavant, l'une quelconque des fonctions  $yx$ ,  $gx$ ,  $\Theta x$ ,  $Zx$  par  $Jx$ , on a l'équation

$$J_r x = e^{\frac{i}{2k+1}(B_r - 2k+1.B)x\pi i} \prod \frac{J(x + s\psi_r)}{J(s\psi_r)},$$

équation dans laquelle  $s$  doit avoir, dans le numérateur, toutes les valeurs entières depuis  $s = -k$  jusqu'à  $s = +k$ , y compris  $s = 0$ , et dans le dénominateur ces mêmes valeurs, hormis la valeur  $s = 0$ .

Je suppose que les valeurs de  $p_r$ ,  $q_r$  soient données (cela va sans dire que l'on ne doit pas avoir à la fois  $p_r \equiv 0$ ,  $q_r = 0$ , mod  $.2k+1$ ), et je remarque que l'on a, pour déterminer  $\lambda$ ,  $\mu$ ,  $\nu$ ,  $\rho$ , les conditions

$$\begin{aligned} \rho p_r - \nu q_r &\equiv 0, & \text{mod } .(2k+1), \\ -\mu p_r + \lambda q_r &\equiv 0, \end{aligned}$$

$$\lambda \equiv 1, \quad \mu \equiv 0, \quad \text{mod } .2, \quad \nu = 0, \quad \rho \equiv 1, \quad \lambda \rho - \mu \nu = 2k+1.$$

Et cela étant, on aura ensuite, en rassemblant toutes les équations qui ont rapport à la transformation,

$$\begin{aligned} \rho p_r - \nu q_r &= (2k+1)p, & -\mu p_r + \lambda q_r &= (2k+1)q, \\ \Omega_r &= p_r \Omega + q_r \Upsilon, & (2k+1)\Omega_r &= \lambda \Omega + \mu \Upsilon, \\ (2k+1)\Upsilon_r &= \nu \Omega + \rho \Upsilon. \end{aligned}$$

#### [Pages 160–163: continuation of transformations.]

Or, quoique les valeurs de  $\lambda$ ,  $\mu$ ,  $\nu$ ,  $\rho$  ne soient pas complètement déterminées au moyen de ces conditions, cependant il est clair que la

valeur de la fonction  $J_r x$  ne dépend que des valeurs de  $p_r, q_r$  (en effet, ces valeurs suffisent pour déterminer la quantité  $\psi = p_r \Omega + q_r \Upsilon$ , de laquelle dépend la fonction  $J_r x$ ). Les formes différentes de  $J_r x$ , pour les systèmes de valeurs de  $\lambda, \mu, \nu, \rho$ , qui correspondent à des valeurs données de  $p_r, q_r$ , doivent donc se dériver de l'une quelconque de ces formes, au moyen d'une transformation triviale des modules  $\Omega_r, \Upsilon_r$ .

Il est, de plus, clair que les valeurs de  $p_r, q_r$ , qui sont égales à des multiples de  $(2k+1)$  près, ne donnent qu'une seule valeur de  $J_r x$ .

Je suppose d'abord que  $p_r \equiv 0, \pmod{(2k+1)}$ , on peut trouver un entier  $\theta$  tel que  $\theta p_r \equiv 1, \pmod{(2k+1)}$ ; en prenant alors  $\theta q_r = q_\theta, \pmod{(2k+1)}$ , cela donne  $\theta(p_r \Omega + q_r \Upsilon) = \Omega + q_\theta \Upsilon, \pmod{(2k+1)}$ , savoir  $\psi = \Omega + q_\theta \Upsilon, \pmod{(2k+1)}$ . Mais en donnant à  $s$  des valeurs entières quelconques, depuis  $-k$  jusqu'à  $+k$ , le système des valeurs de  $s\psi$  est équivalent au système des valeurs de  $s\theta\psi, \pmod{(2k+1)}$ ; il est donc permis d'écrire, sans perte de généralité,  $\psi = \Omega + q_\theta \Upsilon$ . De même pour  $p_r \not\equiv 0, \pmod{(2k+1)}$ , on démontre que l'on peut donner à  $q_r$  une valeur quelconque, sans changer pour cela la valeur de  $J_r x$ ; il convient d'avoir  $p_r$  impair et  $q_r$  pair. J'écris donc, pour le premier cas,  $2q_\theta$  au lieu de  $q_\theta$ , et je suppose que, dans le deuxième cas, les valeurs de  $p_r, q_r$  soient  $p_r = 2k+1, q_r = 2$ . Cela donne

**Premier cas.**  $\psi = \Omega + 2q_\theta \Upsilon$ ,  $q_\theta$  un entier quelconque, y compris zéro, depuis  $-k$  jusqu'à  $+k$ .

**Deuxième cas.**  $\psi = (2k+1)\Omega + 2\Upsilon$ ; le nombre des valeurs différentes de  $\psi$  sera donc, en tout,  $2k+2$ .

On obtient tout de suite, pour le premier cas, le système d'équations

$$\begin{aligned} p_r &= 1, & q_r &= 2q_\theta, \\ \lambda &= 1, & \mu &= 2q_\theta, \\ \nu &= 0, & \rho &= (2k+1), \\ \psi &= \frac{1}{2k+1} \Omega + 2q_\theta \Upsilon, & \Omega_r &= \frac{1}{2k+1} \Omega, \quad \Upsilon_r = \Upsilon. \end{aligned}$$

Le cas particulier le plus simple est celui de  $q_\theta = 0$ ; cela donne

$$\psi = \frac{\Omega}{2k+1}, \quad \Omega_r = \frac{\Omega}{2k+1}, \quad \Upsilon_r = \Upsilon;$$

et, de là,

$$\Omega_r = \frac{1}{2k+1} \Omega, \quad \Upsilon_r = \Upsilon,$$



et même le cas général se réduit à celui-ci, car, au moyen d'une transformation triviale, on obtiendrait

$$\psi' = \Omega' + 2q_\theta \Upsilon', \quad \Upsilon' = \Upsilon,$$

et puis

$$\psi' = \frac{\Omega'}{2k+1}, \quad \Upsilon' = \Upsilon.$$

Les équations correspondantes pour le deuxième cas sont:

$$p = 2k+1, \quad q = 2, \quad \lambda = 2k+1, \quad \mu = 0, \quad \nu = 0, \quad \rho = 1,$$

$$\psi = \frac{1}{2k+1} \Upsilon; \quad \Omega_r = \Omega, \quad \Upsilon_r = \frac{1}{2k+1} \Upsilon;$$

ce qui donne  $\psi = (2k+1)\Omega$ .

J'ajoute, sans m'arrêter pour les démontrer, quelques formules de transformation pour le nombre 2; je trouve d'abord

$$g_r x = e^{-\frac{1}{2}\pi i x^2 \cdot \frac{1}{\Omega} \Upsilon} \cdot g x \cdot \Theta x, \quad \Theta_r x = e^{-\frac{1}{2}\pi i x^2 \cdot \frac{1}{\Omega} \Upsilon} \cdot \Theta x \cdot Z x,$$

Ces équations donnent, en introduisant les fonctions elliptiques  $\phi x$ ,  $f x$ ,  $F x$  données au moyen de

$$\phi x = \frac{y x}{\sqrt{k x}} \cdot \frac{g \frac{1}{2}}{y \frac{1}{2}}, \quad f x = \frac{g x}{Z x}, \quad F x = Z x,$$

les équations

$$F_r x = \frac{F x}{\sqrt{k x_r}}, \quad f_r x = f(x - m') \cdot f(x + m'),$$

dont la seconde peut encore s'écrire sous la forme

$$f_r x = \frac{f^2 x - f^2 m'}{1 - k^2 f^2 m' \cdot f^2 x},$$

et les deux équations combinées ensemble conduisent sans peine à la valeur des modules  $c_r$ ,  $e_r$ . On trouve en effet, en mettant comme à l'ordinaire  $b^2 = c^2 + e^2$ ,  $c_r^2 = 4ce$ ,  $e_r^2 = (b - c)^2$ , et puis

$$\phi_r^2 x = \frac{1 - c(c+b)\phi^2 x}{1 - c(c-b)\phi^2 x}, \quad f_r^2 x = \frac{1 - c(c-b)\phi^2 x}{1 - c(c+b)\phi^2 x}, \quad F_r x = \frac{F x}{1 - c(c-b)\phi^2 x},$$

formules qui correspondent à celles de la transformation de Lagrange.

Les équations pour  $y_r x$ ,  $Z_r x$  donnent encore une valeur de  $\phi_r^2 x$ , laquelle, égalée à la valeur qui vient d'être trouvée, donne

$$\frac{gx \cdot fx \cdot \Theta x \cdot Zx}{y(x - \frac{1}{2}\Omega) y(x + \frac{1}{2}\Omega)} = \frac{1}{1 - c(c - b)\phi^2 x}.$$

On obtient tout de suite les formules pour la transformation analogue  $\Omega_r = \mu\Upsilon = \frac{1}{2}\Upsilon$ . Mais il faut de plus considérer le système  $\Omega_r = \frac{1}{2}\Omega$ ,  $\Upsilon_r = \Upsilon + \frac{1}{2}\Omega$ ; on aura alors

$$y_r x = e^{-\frac{1}{2}\pi i(x^2 - \frac{1}{4}\Omega^2)} \cdot \frac{1}{\Omega} \Upsilon gx \cdot Zx,$$

et puis, en écrivant

$$y_r x = e^{-\frac{1}{2}\pi i(x^2 - \frac{1}{4}\Omega^2)} \cdot \frac{1}{\Omega} \Upsilon Z(x + \frac{1}{2}\mu + \frac{1}{2}\Upsilon) Z(x + \frac{1}{2}\mu - \frac{1}{2}\Upsilon),$$

on obtient

$$\begin{aligned} c_r^2 &= (e^{-ice})^2, & e_r^2 &= (e + ice)^2. \\ \frac{1 + ice \cdot \phi^2 x}{fx \cdot Fx} &= \frac{Z(x + \frac{1}{2}\mu + \frac{1}{2}\Upsilon) Z(x - \frac{1}{2}\mu - \frac{1}{2}\Upsilon)}{Z^2(\frac{1}{2}\mu + \frac{1}{2}\Upsilon) gx \cdot \Theta x}, \\ \frac{1 - ice \cdot \phi^2 x}{fx \cdot Fx} &= \frac{Z(x + \frac{1}{2}\mu - \frac{1}{2}\Upsilon) Z(x - \frac{1}{2}\mu + \frac{1}{2}\Upsilon)}{Z^2(\frac{1}{2}\mu - \frac{1}{2}\Upsilon) gx \cdot \Theta x}, \end{aligned}$$

ou, au moins, ces formules seront exactes au signe de  $i$  près; car il serait peut-être difficile de déterminer quel est le signe qu'on doit donner à cette quantité.

## 131.

### Nouvelles Recherches sur les Covariants.

[From the *Journal für die reine und angewandte Mathematik* (Crelle),  
tom. xlvii. (1854), pp. 109–125.]

Je me sers de la notation

$$(a_0, a_1, \dots a_n)(x, y)^n$$

pour représenter la fonction

$$a_0x^n + \frac{n}{1} a_1x^{n-1}y + \dots + a_ny^n;$$

en supposant que les coefficients  $a'_0, a'_1$  &c. soient donnés par l'équation

$$(a_0, a_1, \dots a_n)(\lambda x + \mu y, \lambda'x + \mu'y)^n = (a'_0, a'_1, \dots a'_n)(x, y)^n,$$

supposée identique par rapport à  $x, y$ , soit  $\phi(a_0, a_1, \dots a_n; x, y)$  une fonction des coefficients et des variables, telle que

$$\phi(a'_0, a'_1, \dots a'_n; x, y) = (\lambda\mu' - \lambda'\mu)^p \phi(a_0, a_1, \dots a_n; \lambda x + \mu y, \lambda'x + \mu'y);$$

cette fonction  $\phi$  sera généralement un *Covariant*, et dans le cas particulier où  $\phi$  est fonction des seuls coefficients, un *Invariant* de la fonction donnée.

Je suppose d'abord que les nouveaux coefficients soient donnés par l'équation

$$(a_0, a_1, \dots a_n)(x + \lambda y, y)^n = (a'_0, a'_1, \dots a'_n)(x, y)^n;$$

cela donne les relations

$$\begin{aligned} a'_0 &= a_0, \\ a'_1 &= a_1 + \lambda a_0, \\ a'_2 &= a_2 + 2\lambda a_1 + \lambda^2 a_0, \\ &\&c. \end{aligned}$$

Il faut donc que  $\phi$  soit une fonction de  $a_0, a_1 - \lambda a_0, \dots$ , c'est-à-dire de  $a_0, a_1, a_2 - \frac{2}{1} \frac{a_1}{a_0} a_1 + \frac{2}{1} \frac{a_1^2}{a_0^2}, \dots$ ; ou l'on peut aussi dire que  $\phi$  doit être

une fonction de  $a_0$  et des quantités  $a_0a_2 - a_1^2$ ,  $a_0a_3 - 3a_1a_2 + 2a_1^3, \dots$ , savoir des fonctions  $a_0$ ,  $(0, 2)$ ,  $(0, 3), \dots$ ;  $a_0$  seul contient la variable  $x$ , et  $(0, 2)$ ,  $(0, 3), \dots$  sont les fonctions de  $a_0, a_1, \dots$  obtenues en écrivant

$$(a_0, a_1, \dots a_n)(x, y)^n = a_0 \left(x + \frac{a_1}{a_0} y\right)^n + (0, 2) \left(x + \frac{a_1}{a_0} y\right)^{n-2} y^2 + (0, 3) \left(x + \frac{a_1}{a_0} y\right)^{n-3} y^3$$

ce qui donne

$$\begin{aligned} (0, 2) &= a_0a_2 - a_1^2, \\ (0, 3) &= a_0^2a_3 - 3a_0a_1a_2 + 2a_1^3, \\ &\&c. \end{aligned}$$

On a ainsi, pour les covariants de la fonction  $(a_0, a_1, a_2)(x, y)^2$ , les fonctions  $a_0$ ,  $a_0a_2 - a_1^2$ ,  $a_0x^2 + 2a_1xy + a_2y^2$ , c'est-à-dire les fonctions  $U$ ,  $H$ ,  $*$ ; et de même, pour les covariants de la fonction  $(a_0, a_1, a_2, a_3)(x, y)^3$ , les fonctions  $a_0$ ,  $(0, 2)$ ,  $(0, 3)$  et les fonctions de celles-ci qui forment les covariants de la fonction cubique.

À l'égard de la fonction  $(a_0, a_1, a_2, a_3, a_4)(x, y)^4$ , il serait peut-être permis de conjecturer que ses covariants sont donnés au moyen de ceux de la fonction  $(0, 2)$ ,  $(0, 3)$ ,  $(0, 4)$  de  $(x, y)^4$ , mais la chose exigerait d'être vérifiée; ce n'est que pour les cas de  $n = 2$  et  $n = 3$  qu'on peut affirmer que tous les covariants seront des fonctions des covariants que l'on vient d'obtenir.

On obtient sans peine l'équation

$$(a_0, a_1, \dots a_n)(y, -x)^n = (-)^n (a_0, a_1, \dots a_n)(x, y)^n.$$

et réciproquement

$$(a_0, a_1, \dots a_n)(-y, x)^n = (-)^n (a_0, a_1, \dots a_n)(x, y)^n.$$

De même, la transformation

$$(a_0, a_1, \dots a_n)(\lambda x + \mu y, \lambda' x + \mu' y)^n$$

donne pour le covariant de l'ordre  $\theta$  par rapport aux coefficients,

$$\phi((a_0, a_1, \dots a_n)(\lambda x + \mu y, \lambda' x + \mu' y)) = (\lambda\mu' - \lambda'\mu)^p \phi(\lambda x + \mu y, \lambda' x + \mu' y),$$

où dans  $(\lambda\mu' - \lambda'\mu)^p$  les quantités  $\lambda$ ,  $\mu$ ,  $\lambda'$ ,  $\mu'$  sont éliminées; ce covariant sera de l'ordre  $\theta$  par rapport aux coefficients, de l'ordre  $p$  par rapport à  $(x, y)$ , et du degré  $n\theta - 2p$  par rapport à  $(\lambda, \mu, \lambda', \mu')$ .

*Exemple.* — En prenant  $n = 3$ ,  $\theta = 3$ ,  $p = 3$ , la fonction  $\phi$  est le discriminant; savoir

$$\phi = a^2d^2 - 6abcd + 4ac^3 + 4b^2d - 3b^2c^2;$$

et l'équation en  $(\lambda, \mu, \lambda', \mu')$  sera

$$\begin{aligned} \phi((a, b, c, d)(\lambda x + \mu y, \lambda'x + \mu'y)^3) \\ = (\lambda\mu' - \lambda'\mu)^3 \phi(\lambda x + \mu y, \lambda'x + \mu'y). \end{aligned}$$

J'ai donné cet exemple pour montrer comment les formules que j'ai obtenues pour les covariants donnent aussi les invariants; il est clair en effet que, si dans le covariant  $\phi(a_0, a_1, \dots a_n; x, y)$  on écrit  $y = 0$ , on obtient une fonction de  $a_0$  seul; d'où, en divisant par une puissance convenable de  $a_0$ , on déduit une fonction des seuls coefficients  $a_0, a_1, \dots a_n$ , laquelle sera l'invariant de la fonction donnée.

En mettant d'abord  $\lambda = 0$ , on a

$$a'_r = a_r, \quad a'_0 = \mu^n a_0, \quad a'_1 = \mu^{n-1}(\lambda' a_0 + \mu a_1);$$

et de là

$$\begin{aligned} a'_0 &= \mu^n a_0, \\ a'_1 &= \mu^{n-1}(\lambda' a_0 + \mu a_1), \\ a'_2 &= \mu^{n-2}(\lambda'^2 a_0 + 2\lambda'\mu a_1 + \mu^2 a_2), \\ &\&c. \end{aligned}$$

En supposant que la fonction donnée soit  $(a, b, c, d, e)(x, y)^4$ , si dans le covariant de l'ordre 2 par rapport aux coefficients on écrit  $a' = 0$ ,  $b' = 0$ , ce covariant deviendra une fonction de  $a, b$  seulement, c'est-à-dire un invariant de la fonction  $(a, b)(x, y)^1$ ; donc ce sera une puissance de  $a$  seul, ce qui donne pour le covariant de l'ordre 2 une équation de la forme

$$\phi = Aa^2 + Ba \cdot c + Cc^2 + Da \cdot d + Ec \cdot e + Fd^2 = a^2\psi,$$

où  $\psi$  est un invariant de la fonction  $(c, d, e)(x, y)^2$ . De même les covariants des ordres 3 et 4 deviendront, pour  $a = 0$ ,  $b = 0$ , des invariants de la fonction  $(c, d, e)(x, y)^2$ ; c'est-à-dire des fonctions de

$U, H$ . Les covariants de la fonction du quatrième ordre seront donc de la forme

$$\begin{aligned}\Phi &= AU^2 + BH \cdot U + C\Phi + D\Box, \\ A' &= A'U + B'H + C'\Phi, \\ \Omega &= \Omega(U, H, \Phi, \Box);\end{aligned}$$

et en mettant  $\Phi = 3ad^2 + 2b^2cy$ ,  $B, C$  seront des fonctions linéaires de  $a, b$ ; et ainsi de suite.

## 132.

### Réponse à une Question proposée par M. Steiner.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. liii. (1857), pp. 318–320.]

ON sait le beau théorème de M. Steiner: *Le lieu de l'intersection de trois plans tangents rectangulaires d'une surface du second ordre est une sphère concentrique à cette surface.* Ce théorème se démontre par la géométrie descriptive de la manière suivante.

Soient  $A$ ,  $B$ ,  $C$  les trois plans tangents rectangulaires; le plan  $ABC$  coupe la surface suivant une conique dont les droites  $DA$ ,  $DB$ ,  $DC$  menées par un point quelconque  $D$  du plan, parallèlement aux plans  $B \cdot C$ ,  $C \cdot A$ ,  $A \cdot B$ , seront conjuguées deux à deux. En effet, la droite  $DA$ , parallèle au plan  $BC$ , rencontre le plan  $ABC$  suivant une droite parallèle à la polaire du point  $(B, C)$  par rapport à la conique; or cette polaire est la droite  $BC$ , donc  $DA$  rencontre le plan suivant une droite conjuguée à  $BC$ , c'est-à-dire que  $DA$ ,  $DB$ ,  $DC$  forment un système de trois droites conjuguées.

Mais cela posé, le point  $D$  est le sommet d'un angle trièdre dont les arêtes sont parallèles à trois diamètres conjugués de la surface; ce sommet appartient donc à une sphère fixe; le lieu de l'intersection de trois plans tangents rectangulaires sera donc une sphère, ce qu'il fallait démontrer.

M. Steiner a proposé la question de savoir si ce théorème est vrai pour une surface quelconque du second ordre, ou s'il y a des exceptions. On voit sans peine qu'il y a exception pour le parabolôïde, car le parabolôïde n'a pas de centre, et le théorème suppose que la surface ait un centre. Mais il y a aussi exception pour le cône et pour le parabolôïde; en effet, pour le cône, le lieu de l'intersection de trois plans tangents rectangulaires est le cône lui-même, et pour le parabolôïde à plan directeur, le lieu est le plan directeur.

**I. Théorème.** — Pour un ellipsoïde ou un hyperboloïde, le lieu de l'intersection de trois plans tangents rectangulaires est une sphère concentrique à la surface.

**II. Théorème.** — Pour un parabolôïde du second ordre, le lieu de l'intersection de trois plans tangents rectangulaires est un plan.

**III. Théorème.** — Pour un cône du second ordre, le lieu de

l'intersection de trois plans tangents rectangulaires est le cône lui-même.

Passons au point  $D$ . Cette surface sera en effet la *conique selon laquelle l'infini, considéré comme plan, est coupé par un cône déterminé, près la position du sommet*. En effet, menons par un point quelconque de l'espace trois plans parallèles aux plans du parallépipède  $P$ , et par le point  $D$  trois autres plans parallèles à ces plans. Ces *six* plans seront touchés (en vertu d'un théorème connu) par un cône déterminé du second ordre, et on peut dire que ce cône, quelle que soit la position de son sommet, rencontre l'infini, considéré comme plan, dans une seule et même conique (cela n'est en effet autre chose que de dire que deux droites parallèles rencontrent l'infini, considéré comme plan, dans un seul et même point). Le cône dont il s'agit aura la propriété d'être touché par une infinité de systèmes de trois plans rectangles. En effet: le plan passant par le sommet, et perpendiculaire à la droite d'intersection de deux plans tangents quelconques sera un plan tangent du cône; les plans d'un tel système seront aussi des plans tangents de la conique mentionnée ci-dessus: donc le sommet du cône sera le point d'intersection de trois plans rectangles de la conique; et ce sommet étant un point entièrement indéterminé, le lieu de l'intersection des trois plans tangents rectangles de la conique, sera de même absolument indéterminé, ou si l'on veut, ce lieu sera l'espace entier près les points à une distance infinie. La contradiction apparente dont M. Steiner parle, a par conséquent son origine dans l'indétermination qui a lieu dans le cas dont il s'agit. Dans tout autre cas, le point d'intersection des trois plans rectangles de la surface du second ordre est parfaitement déterminé, et les théorèmes I. et II. sont tous deux légitimes.



### 133.

#### Sur un Théorème de M. Schläfli.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. 1. (1855), pp. 253-255.]

M. Schläfli a donné dans ce Journal (t. xlix, p. 259) un théorème sur l'équation

$$\Phi = \mathfrak{A}\phi \cdot \mathfrak{B}\psi \cdot \mathfrak{C}\chi \cdots - 0 = 0,$$

où  $\mathfrak{A}\phi$  désigne l'opération

$$\alpha \frac{\partial}{\partial a} + \beta \frac{\partial}{\partial b} + \gamma \frac{\partial}{\partial c} + \dots,$$

et ainsi de suite. Cette fonction  $\Phi$  est une fonction des coefficients de  $U, V, W, \dots$  des indéterminées  $x, y, z, \dots$  et des indéterminées  $\xi, \eta, \dots$  elles-mêmes; et le théorème est que, sous certaines conditions,  $\Phi$  s'évanouit identiquement; ce qui arrive, comme M. Schläfli l'a remarqué, lorsque le nombre des indéterminées  $\xi, \eta, \dots$  est suffisamment grand par rapport à celui des indéterminées  $x, y, z, \dots$ .

Rien n'empêche que  $\Phi$  ne contienne une plus haute puissance que  $\phi^{\sigma-\pi:\mu}$  comme facteur, ou que  $\Phi$  ne s'évanouisse identiquement. On peut même assigner de plus près que l'a fait M. Schläfli, des cas où  $\Phi$  s'évanouit identiquement. Soient  $m, m', m'', \dots$  les degrés de  $U, V, W, \dots$  par rapport à  $x, y, z, \dots$ ,  $p = mm'm'' \dots$ ,  $s = \frac{p}{m} + \frac{p}{m'} + \frac{p}{m''} \dots$ ;  $\phi$  sera du degré  $\frac{p}{m}$  par rapport aux coefficients de  $U$ . Soient aussi  $\mu, \mu_{II}, \dots$  les degrés de celles des fonctions  $\mathfrak{A}\phi, \mathfrak{B}\phi, \dots$  pour lesquelles les opérateurs  $\mathfrak{A}, \mathfrak{B}, \dots$  contiennent des différentielles par rapport aux coefficients de  $U$ ,  $\rho = \frac{\pi}{\mu} + \frac{\pi}{\mu_{II}} \dots$ ; pour ces fonctions les coefficients seront du degré  $\frac{p}{m} - 1$  par rapport aux coefficients de  $U$ ; pour les autres ils seront du degré  $\frac{p}{m}$ .  $\Phi$  sera donc du degré  $(\frac{p}{m} - 1)\rho + \frac{p}{m}(\sigma - \rho) = \frac{p}{m}\sigma - \rho$ , par rapport aux coefficients de  $U$ , et  $\Phi \div \phi^{\sigma-\pi:\mu}$  sera du degré  $\frac{p}{m}\sigma - \rho - \frac{p}{m}(\sigma - \frac{\pi}{\mu})$ , c'est-à-dire du degré  $\frac{p}{m} \cdot \frac{\pi}{\mu} - \rho$  par rapport aux coefficients de  $U$ . De même, en supposant que les lettres  $m', \rho', \dots$  aient rapport à  $V$ , &c.,  $\Phi \div \phi^{\sigma-\pi:\mu}$  sera du degré  $\frac{p}{m'} \cdot \frac{\pi}{\mu'} - \rho'$ , &c. par rapport aux coefficients de  $V$ , &c. Si l'un quelconque des nombres  $\frac{p}{m} \cdot \frac{\pi}{\mu} - \rho$ ,  $\frac{p}{m'} \cdot \frac{\pi}{\mu'} - \rho'$ , &c. est *négalif*, et à plus forte raison, si leur somme  $s \cdot \frac{\pi}{\mu} - \sigma$  est *negative*,  $\Phi$  doit s'évanouir identiquement. En particulier, en supposant que le nombre des fonctions  $\mathfrak{A}\phi, \mathfrak{B}\phi, \dots$  (c'est-à-dire le nombre des indéterminées

$\xi, \eta, \dots$ ) soit  $\nu$ , on aura  $\sigma > \nu \frac{\pi}{\mu}$ , et par cette raison  $\Phi$  s'évanouira identiquement si  $\frac{\pi}{\mu}(\sigma - \nu)$  est négatif, c'est-à-dire si  $\nu > \sigma$ . Je ne parlerai pas ici des cas examinés par M. Schläfli, où  $\Phi$  contient comme facteur une plus haute puissance que  $\phi^{\sigma - \pi : \mu}$ .

## 134.

### Remarques sur la Notation des Fonctions Algébriques.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 282–285.]

JE me sers de la notation

$$\begin{array}{cccc} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ \vdots & \vdots & \vdots & \end{array}$$

pour représenter ce que j'appelle une *matrice*; savoir un *système* de quantités rangées en forme de carré, mais d'ailleurs tout à fait *in-dépendantes* (je ne parle pas ici des *matrices rectangulaires*). Cette notation me paraît très commode pour la théorie des équations *linéaires*; j'écris par exemple

$$(\xi, \eta, \zeta, \dots) = \left| \begin{array}{cccc} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ \vdots & \vdots & \vdots & \end{array} \right| (x, y, z, \dots)$$

pour représenter le système des équations

$$\begin{array}{l} \xi = \alpha x + \beta y + \gamma z \dots, \\ \eta = \alpha' x + \beta' y + \gamma' z \dots, \\ \zeta = \alpha'' x + \beta'' y + \gamma'' z \dots, \\ \vdots \end{array}$$

et j'appelle la matrice la matrice du système.

On obtient par là, en supposant que les variables  $x, y, z, \dots$  soient liées à d'autres variables  $X, Y, Z, \dots$  par les équations linéaires

$$\begin{array}{l} x = aX + a'Y + a''Z + \dots, \\ y = bX + b'Y + b''Z + \dots, \\ z = cX + c'Y + c''Z + \dots, \\ \vdots \end{array}$$

et que  $\Xi, H, Z, \dots$  soient données en fonctions de  $X, Y, Z, \dots$  par les équations

$$(\Xi, H, Z, \dots) = \begin{vmatrix} A, & B, & C, & \dots \\ A', & B', & C', & \dots \\ A'', & B'', & C'', & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix} (X, Y, Z, \dots),$$

l'équation

$$(\Xi, H, Z, \dots) = \begin{vmatrix} A, & B, & C, & \dots \\ A', & B', & C', & \dots \\ A'', & B'', & C'', & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix} \begin{vmatrix} a, & b, & c, & \dots \\ a', & b', & c', & \dots \\ a'', & b'', & c'', & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix} (x, y, z, \dots).$$

Je dis qu'en effectuant la multiplication des deux matrices, on obtient pour produit une seule matrice, laquelle donne

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} \mathfrak{A}, & \mathfrak{B}, & \mathfrak{C}, & \dots \\ \mathfrak{A}', & \mathfrak{B}', & \mathfrak{C}', & \dots \\ \mathfrak{A}'', & \mathfrak{B}'', & \mathfrak{C}'', & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix} (x, y, z, \dots),$$

où l'on a

$$\begin{aligned} \mathfrak{A} &= Aa + Ba' + Ca'' + \dots, & \mathfrak{A}' &= A'a + B'a' + C'a'' + \dots, \\ \mathfrak{B} &= Ab + Bb' + Cb'' + \dots, & \mathfrak{B}' &= A'b + B'b' + C'b'' + \dots, \\ \mathfrak{C} &= Ac + Bc' + Cc'' + \dots, & \mathfrak{C}' &= A'c + B'c' + C'c'' + \dots, \\ \vdots & & \vdots & \end{aligned}$$

formules qui s'écrivent aussi sous la forme

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \dots) = (A, B, C, \dots) \begin{vmatrix} a, & b, & c, & \dots \\ a', & b', & c', & \dots \\ a'', & b'', & c'', & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix}$$

et ainsi de suite. On voit ainsi que la notion de la matrice peut être appliquée aux systèmes d'équations linéaires, et que dans cette application les matrices se comportent comme des quantités simples; on peut leur appliquer toutes les opérations de l'algèbre commune.

On a en particulier les notions de la *matrice unité*, et de la *matrice inverse*; la matrice unité est

$$\begin{vmatrix} 1, & 0, & 0, & \dots \\ 0, & 1, & 0, & \dots \\ 0, & 0, & 1, & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix}$$

et c'est celle qui, multipliée par une matrice quelconque, la reproduit. La matrice inverse est celle qui, multipliée par la matrice donnée, donne pour produit la matrice unité.

À l'égard du *déterminant*, on remarquera que c'est une fonction des éléments de la matrice, laquelle jouit de la propriété d'être égale à zéro toutes les fois que la matrice est nulle (c'est-à-dire toutes les fois que les quantités de la matrice satisfont à une même équation linéaire). De là découle le théorème que le déterminant du produit de deux matrices est égal au produit de leurs déterminants. Cela se vérifie sans peine dans le cas particulier où la matrice du produit est celle que l'on vient d'obtenir; en effet, on sait que le déterminant formé avec les quantités ( $\mathfrak{A}, \mathfrak{B}, \dots$ ) est égal au produit des deux déterminants formés avec les quantités ( $A, B, \dots$ ) et ( $a, b, \dots$ ).

En introduisant dans cette notation les *coefficients*, j'écris par exemple

$$(a, b, c, \mathfrak{A}x, y)^3,$$

pour représenter la fonction

$$ax^3 + 3bx^2y + 3cxy^2 + dy^3,$$

tandis que je me sers de la notation

$$(a, b, c, d \backslash x, y)^3,$$

pour représenter la fonction

$$ax^3 + bx^2y + cxy^2 + dy^3,$$

et de même pour les fonctions d'un degré quelconque. J'ai trouvé cette distinction très commode.

**Remark** (In the foregoing Paper as here printed, except in the expression in the second line of this page,  $\chi$  is used instead of  $\cdot$ ); it appears by a remark (*Crelle*, t. ii, errata) that the manuscript had the interlaced parentheses  $\chi$ . Moreover in the manuscript  $\cdot$  was used for a *Matrix*, which was thus distinguished from a *Determinant*, but in the absence of any real ambiguity, no alteration has been made in this respect. In the reprint of subsequent papers from *Crelle*, the arrowhead  $\chi$  or  $\chi$  is used instead of  $\cdot$ ).

## 135.

### Note sur les Covariants d'une Fonction Quadratique, Cubique, ou Biquadratique à deux Indéterminées.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 285–287.]

LA théorie d'une fonction à deux indéterminées d'un degré quelconque, par exemple

$$(\imath \nmid x, y)^m,$$

dépend du système des *covariants* de la fonction, lequel est censé contenir la fonction elle-même.

Pour une fonction *quadratique* le système de covariants est

$$(a, b, \nmid x, y)^2, \quad ac - b^2.$$

Pour la fonction *cubique*, le système est

$$(a, b, c, \nmid x, y)^3,$$

$$(ac - b^2, ad - bc, bd - c^2 \nmid x, y)^2,$$

$$\begin{aligned} &(-a^2d + 3abc - 2b^3, -abd + 2ac^2 - b^2c, acd - 2b^2d + bc^2, ad^2 - 3bcd + 2c^3 \nmid x, y)^3, \\ &-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2, \end{aligned}$$

fonctions lesquelles, en supposant qu'on les représente par  $U, H, \Phi, \square$ , satisfont identiquement à l'équation

$$\Phi^2 + \square U^2 = -4H^3.$$

Pour la fonction *biquadratique*, on obtient d'abord le système

$$U = (a, b, c, d, \nmid x, y)^4,$$

$$H = (ac - b^2, ad - bc, ae + 2bd - 3c^2, be - cd, ce - d^2 \nmid x, y)^4,$$

$$\Phi = ((0, 0, 3), (0, 1, 3), (0, 2, 3), (0, 3, 3), (0, 4, 3) \nmid x, y)^6,$$

$$\square = \text{discriminant};$$

où les symboles  $(0, 0, 3), (0, 1, 3), \dots$  désignent des fonctions des coefficients dont les expressions n'ont pas besoin d'être ici écrites.

Les covariants de la fonction biquadratique doivent satisfaire à plusieurs équations; on a d'abord l'équation

$$U\square - \Phi^2 = 4H^3,$$

et de plus, le système doit contenir les carrés et les produits des covariants  $U, H, \Phi$ ; cela donne les identités

$$\begin{aligned} U^2 &= (a, \dots \chi x, y)^8, \\ H^2 &= (ac - b^2, \dots \chi x, y)^8, \\ U \cdot H &= \dots \end{aligned}$$

et ainsi de suite. Le nombre des covariants indépendants de la fonction biquadratique à deux indéterminées est donc très grand, et leur détermination complète présente des difficultés considérables.



### 136.

#### Sur la Transformation d'une Fonction Quadratique en elle-même par des Substitutions Linéaires.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 288–294.]

EN effet, considérons la forme quadratique

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy,$$

et supposons que l'on veuille la transformer en elle-même par les substitutions linéaires

$$\begin{aligned} x &= \alpha\xi + \alpha'\eta + \alpha''\zeta + \dots, \\ y &= \beta\xi + \beta'\eta + \beta''\zeta + \dots, \\ z &= \gamma\xi + \gamma'\eta + \gamma''\zeta + \dots, \\ &\vdots \end{aligned}$$

Le déterminant de la transformation doit être  $\pm 1$ ; c'est une condition nécessaire pour que la forme soit transformée en elle-même.

À l'égard de la transformation proprement dite, supposons d'abord qu'il n'y ait que trois variables  $x, y, z$ . La forme quadratique étant

$$(ax, by, cz, fy, gz, hx \mid x, y, z)^2,$$

savoir

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy,$$

il faut déterminer les neuf coefficients  $\alpha, \beta, \gamma, \alpha', \dots$  de manière que la substitution reproduise la forme; ce qui donne six équations. Mais il y a dix inconnues, en comptant les trois quantités dont dépend le changement de forme, savoir le déterminant de la transformation et deux autres conditions; donc le système admet une solution générale avec un certain nombre de paramètres arbitraires.

On obtient en particulier les substitutions suivantes:

Pour la forme  $ax^2 + by^2 + cz^2 + \&c.$ , on aura

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} a, & h + \nu, & g - \mu, & \dots \\ h - \nu, & b, & f + \lambda, & \dots \\ g + \mu, & f - \lambda, & c, & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix}^{-1} \begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix} (x, y, z, \dots),$$

où l'on a

$$x = 2\xi - x, \quad y = 2\eta - y, \quad z = 2\zeta - z, \quad \dots$$

Par exemple, en supposant que la forme à transformer soit

$$ax^2 + by^2 + cz^2 + \&c.,$$

on aura

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} a, & \nu, & -\mu, & \dots \\ -\nu, & b, & \lambda, & \dots \\ \mu, & -\lambda, & c, & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix}^{-1} (ax, by, cz, \dots),$$

de manière qu'en posant

$$\begin{vmatrix} a, & \nu, & -\mu, & \dots \\ -\nu, & b, & \lambda, & \dots \\ \mu, & -\lambda, & c, & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix} = k,$$

et

$$\begin{vmatrix} a, & \nu, & -\mu, & \dots \\ -\nu, & b, & \lambda, & \dots \\ \mu, & -\lambda, & c, & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix}^{-1} = \frac{1}{k} \begin{vmatrix} A, & B, & C, & \dots \\ A', & B', & C', & \dots \\ A'', & B'', & C'', & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix},$$

on aura

$$(x, y, z, \dots) = \frac{1}{k} \begin{vmatrix} 2A - \frac{k}{a}, & 2B, & 2C, & \dots \\ 2A', & 2B' - \frac{k}{b}, & 2C', & \dots \\ 2A'', & 2B'', & 2C'' - \frac{k}{c}, & \dots \\ \vdots & \vdots & \vdots & \end{vmatrix} (ax, by, cz, \dots).$$

On remarquera que, pour que la transformation soit réelle, il faut que le déterminant  $k$  soit positif; de plus, les conditions de réalité des coefficients de la substitution imposent des restrictions sur les paramètres  $\lambda, \mu, \nu, \dots$

Le problème de la transformation d'une fonction quadratique en elle-même a été traité par plusieurs géomètres; on peut consulter en

particulier les mémoires de M. Cauchy sur les centres de surfaces, et ceux de M. Jacobi sur les rotations d'un corps solide. La méthode que nous avons exposée a l'avantage de donner la solution sous une forme tout à fait générale et symétrique.

À l'égard de la formule de transformation ci-dessus, on peut se servir des formules générales; cela donne, pour le cas de trois variables, le système des substitutions sous la forme la plus simple. On a ainsi les conditions

$$A(\alpha^2 - 1) = 0, \quad B(\beta^2 - 1) = 0, \quad C(\gamma^2 - 1) = 0,$$

et par conséquent

$$\alpha^2 = 1, \quad \beta^2 = 1, \quad \gamma^2 = 1,$$

ce qui donne pour  $\alpha, \beta, \gamma$  les valeurs  $\pm 1$ .

L'application de ces formules au cas de quatre variables présente un intérêt particulier; on trouve alors que la transformation dépend de trois paramètres arbitraires, et l'on retrouve ainsi les résultats classiques de M. Euler sur les rotations d'un corps solide autour d'un point fixe.

En général, pour une forme quadratique à  $n$  variables, la transformation en elle-même dépend de  $\frac{1}{2}n(n-1)$  paramètres; ce nombre se réduit à  $\frac{1}{2}(n-1)(n-2)$  si l'on impose la condition que le déterminant de la transformation soit égal à  $+1$ .

On peut aussi considérer le cas où le déterminant de la transformation est égal à  $-1$ ; la transformation est alors dite *impropre*. On obtient dans ce cas des formules analogues à celles que nous avons données, avec des modifications convenables dans les signes des termes.

## 137.

### Recherches Ultérieures sur les Déterminants Gauches.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 294–298.]

LES déterminants gauches sont les déterminants de la forme

$$\begin{vmatrix} 0, & 12, & 13, & 14, & \dots & 1n \\ 21, & 0, & 23, & 24, & \dots & 2n \\ 31, & 32, & 0, & 34, & \dots & 3n \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ n1, & n2, & n3, & n4, & \dots & 0 \end{vmatrix}$$

où  $ij = -ji$ , de sorte que les éléments de la diagonale principale sont nuls, et que la matrice est *antisymétrique*. Le déterminant d'une telle matrice est un carré parfait; c'est le théorème fondamental de M. Cayley.

On a les résultats suivants. Pour  $n = 2$ , le déterminant est  $12^2$ ; pour  $n = 4$ , il est  $(12 \cdot 34 - 13 \cdot 24 + 14 \cdot 23)^2$ . En général, le déterminant gauche d'ordre  $2m$  est le carré d'une fonction des  $\binom{2m}{2}$  quantités  $ij$ , laquelle fonction est ce qu'on appelle le *pfaffien* de la matrice.

À l'égard des déterminants gauches d'ordre impair, ils sont tous nuls; c'est-à-dire que le déterminant d'une matrice antisymétrique d'ordre impair est identiquement nul.

Il est à peine nécessaire de rappeler que les déterminants gauches jouent un rôle important dans la théorie des fonctions abéliennes, et dans celle des courbes et des surfaces algébriques. On les rencontre aussi dans la mécanique, notamment dans la théorie du mouvement de rotation.

On obtient, en effectuant les substitutions, les formules suivantes:

$$\overline{12345} | \overline{12345} = 11' \cdot (16')^2 \overline{123456} | \overline{123456}';$$

et ainsi de suite.

Je fais les mêmes substitutions dans les *matrices inverses*, en supprimant cependant la dernière ligne et la dernière colonne de chaque

matrice. On trouve ainsi, en y ajoutant les équations ci-dessus trouvées par rapport aux déterminants:

$$11' \cdot \overline{123} | \overline{123}' = \overline{12} | \overline{12},$$

$$\frac{1}{13' \cdot \overline{123} | \overline{123}'} \begin{vmatrix} +\overline{23} | \overline{23}', & -\overline{13} | \overline{23}' \\ -\overline{23} | \overline{13}', & +\overline{13} | \overline{13}' \end{vmatrix} = \frac{1}{\overline{12} | \overline{12}} \begin{vmatrix} -2 | \overline{2} + \frac{\overline{12} | \overline{12}}{11'}, & -\overline{1} | \overline{2} \\ +2 | \overline{1}, & +\overline{1} | \overline{1} \end{vmatrix};$$

et ainsi de suite.

**II.** On a de même

$$11' \cdot 14' \cdot \overline{1234} | \overline{1234}' = \overline{123} | \overline{123},$$

et

$$\frac{1}{14' \cdot \overline{1234} | \overline{1234}'} \begin{vmatrix} +\overline{234} | \overline{234}', & -\overline{134} | \overline{234}', & -\overline{124} | \overline{324}' \\ -\overline{234} | \overline{134}', & +\overline{134} | \overline{134}', & -\overline{214} | \overline{314}' \\ -\overline{324} | \overline{124}', & -\overline{314} | \overline{214}', & +\overline{124} | \overline{124}' \end{vmatrix} \\ = \frac{1}{\overline{123} | \overline{123}} \begin{vmatrix} -\overline{23} | \overline{23} + \frac{\overline{123} | \overline{123}}{11'}, & -\overline{13} | \overline{23}, & -\overline{12} | \overline{32} \\ +\overline{23} | \overline{13}, & +\overline{13} | \overline{13}, & -\overline{21} | \overline{31} \\ +\overline{32} | \overline{12}, & -\overline{31} | \overline{21}, & +\overline{12} | \overline{12} \end{vmatrix}.$$

**III.** On a encore

$$11' \cdot (15')^2 \cdot \overline{12345} | \overline{12345}' = \overline{1234} | \overline{1234},$$

et de même

$$\frac{1}{15' \cdot \overline{12345} | \overline{12345}'} \begin{vmatrix} +\overline{2345} | \overline{2345}', & -\overline{1345} | \overline{2345}', & -\overline{1245} | \overline{3245}', & -\overline{1235} | \overline{4235}' \\ -\overline{2345} | \overline{1345}', & +\overline{1345} | \overline{1345}', & -\overline{2145} | \overline{3145}', & -\overline{2135} | \overline{4135}' \\ -\overline{3245} | \overline{1245}', & -\overline{3145} | \overline{2145}', & +\overline{1245} | \overline{1245}', & -\overline{3125} | \overline{4125}' \\ -\overline{4235} | \overline{1235}', & -\overline{4135} | \overline{2135}', & -\overline{4125} | \overline{3125}', & +\overline{1235} | \overline{1235}' \end{vmatrix} \\ = \frac{1}{\overline{1234} | \overline{1234}} \begin{vmatrix} -\overline{234} | \overline{234} + \frac{\overline{1234} | \overline{1234}}{11'}, & -\overline{134} | \overline{234}, & -\overline{124} | \overline{324}, & -\overline{123} | \overline{423} \\ +\overline{234} | \overline{134}, & +\overline{134} | \overline{134}, & -\overline{214} | \overline{314}, & -\overline{213} | \overline{413} \\ +\overline{324} | \overline{124}, & -\overline{314} | \overline{214}, & +\overline{124} | \overline{124}, & -\overline{312} | \overline{412} \\ +\overline{423} | \overline{123}, & -\overline{413} | \overline{213}, & -\overline{412} | \overline{312}, & +\overline{123} | \overline{123} \end{vmatrix}.$$

Les formules ci-dessus se déduisent les unes des autres par une loi générale; on voit que les déterminants gauches présentent une théorie riche et complète, qui mérite d'être approfondie.

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## 137.

RECHERCHES ULTÉRIEURES SUR LES DÉTERMINANTS GAUCHES.

[From the *Journal für die reine und angewandte Mathematik*  
(Crelle),  
torn. L. (1855), pp. 299–313:  
Continuation of the Memoir t. XXXII. (1846) and t. XXXVIII.  
(1849);  
52 and 69.]

Il est bon de changer un peu la forme de ces équations. On en déduit sans peine:

$$\begin{aligned}
 & \frac{1}{\overline{13' \cdot \overline{123} \mid 123'}} \left| \begin{array}{cc} 2 \cdot \overline{23 \mid 23'} - \frac{\overline{123 \mid 123'}}{11}, & -2 \cdot \overline{13 \mid 23'} \\ -2 \cdot \overline{23 \mid 13'}, & +2 \cdot \overline{13 \mid 13'} - \frac{\overline{123 \mid 123'}}{22} \end{array} \right|, \\
 & = \frac{1}{\overline{12 \mid 12}} \left| \begin{array}{cc} -2 \cdot \overline{2 \mid 2} + \frac{\overline{12 \mid 12}}{11}, & -2 \cdot \overline{1 \mid 2} \\ +2 \cdot \overline{2 \mid 1}, & +2 \cdot \overline{1 \mid 1} - \frac{\overline{12 \mid 12}}{22} \end{array} \right|, \\
 & \frac{1}{\overline{14' \cdot \overline{1234} \mid 1234'}} \left| \begin{array}{ccc} 2 \cdot \overline{234 \mid 234'} - \frac{\overline{1234 \mid 1234'}}{11'}, & -2 \cdot \overline{134 \mid 234'}, & \\ -2 \cdot \overline{234 \mid 134'}, & 2 \cdot \overline{134 \mid 134'} - \frac{\overline{1234 \mid 1234'}}{22'}, & \\ -2 \cdot \overline{324 \mid 124'}, & -2 \cdot \overline{314 \mid 214'}, & 2 \cdot \overline{124 \mid 314'} \end{array} \right|, \\
 & = \frac{1}{\overline{123 \mid 123}} \left| \begin{array}{ccc} -2 \cdot \overline{23 \mid 23} - \frac{\overline{123 \mid 123}}{11}, & -2 \cdot \overline{13 \mid 23}, & -2 \cdot \overline{12 \mid 32} \\ +2 \cdot \overline{23 \mid 13}, & +2 \cdot \overline{13 \mid 13} - \frac{\overline{123 \mid 123}}{22}, & -2 \cdot \overline{21 \mid 31} \\ +2 \cdot \overline{32 \mid 12}, & -2 \cdot \overline{31 \mid 21}, & +2 \cdot \overline{12 \mid 12} - \frac{\overline{123 \mid 123}}{3} \end{array} \right|
 \end{aligned}$$

et ainsi de suite.

Les formules que je viens de présenter, peuvent être appliquées aussitôt à la solution de la question suivante:

“Trouver  $x_1, x_2, x_3$ , &c., fonctions linéaires de  $x_1, x_2, x_3$ , &c. telles que

$$11x_1^2 + 22x_2^2 + 33x_3^2 + \&c. = 11x_1^2 + 22x_2^2 + 33x_3^2 + \&c.,”$$

c'est-à-dire: transformer une fonction quadratique  $11x_1^2 + 22x_2^2 + 33x_3^2 + \&c.$  *en elle-même* par des substitutions linéaires.

Il suffira d'écrire la solution pour le cas de trois indéterminées: on satisfait identiquement à l'équation

$$11x_1^2 + 22x_2^2 + 33x_3^2 = 11x_1^2 + 22x_2^2 + 33x_3^2$$

en écrivant

$$(x_1, x_2, x_3) = \frac{1}{123 \mid 123} \times \begin{vmatrix} +2 \cdot \overline{23 \mid 23} - \frac{\overline{123 \mid 123}}{11}, & -2 \cdot \overline{13 \mid 23}, & -2 \cdot \overline{12 \mid 32} \\ -2 \cdot \overline{23 \mid 13}, & +2 \cdot \overline{13 \mid 13} - \frac{\overline{123 \mid 123}}{22}, & -2 \cdot \overline{21 \mid 31} \\ -2 \cdot \overline{32 \mid 12}, & -2 \cdot \overline{31 \mid 21}, & +2 \cdot \overline{12 \mid 12} - \frac{\overline{123 \mid 123}}{33} \end{vmatrix} (11x_1,$$

Voilà la transformation propre. On en tire la transformation impropre de  $11x_1^2 + 22x_2^2 + 33x_3^2$  en elle-même en écrivant  $33 = \infty$ ; car, cela posé, les valeurs de  $x_1, x_2$  ne contiennent pas  $x_3$ , et l'on n'a plus besoin de la valeur de  $x_3$ . On obtient ainsi la solution suivante; savoir, on satisfait identiquement à l'équation

$$11'x_1^2 + 22'x_2^2 = 11'X_1^2 + 22'X_2^2$$

en écrivant

$$(x_1, x_2) = \frac{1}{\overline{123 \mid 123'}} \begin{vmatrix} 2 \cdot \overline{23 \mid 23'} - \frac{\overline{123 \mid 123'}}{11'}, & -2 \cdot \overline{13 \mid 23'} \\ -2 \cdot \overline{23 \mid 13'}, & 2 \cdot \overline{13 \mid 13'} - \frac{\overline{123 \mid 123'}}{22'} \end{vmatrix} (11'x_1, 22'x_2)$$

ce qui est une transformation impropre.

Mais en y faisant la substitution  $11 = 13' \cdot 11', 22 = 13' \cdot 22', 12 = 11' \cdot 23'$ , on réduit la solution à celle-ci, savoir on satisfait identiquement à l'équation

$$11x_1^2 + 22x_2^2 = 11x_1^2 + 22x_2^2$$

en écrivant

$$(x_1, x_2) = \frac{1}{\overline{12 \mid 12}} \begin{vmatrix} -2 \cdot \overline{2 \mid 2} + \frac{\overline{12 \mid 12}}{11}, & -2 \cdot \overline{1 \mid 2} \\ +2 \cdot \overline{2 \mid 1}, & +2 \cdot \overline{1 \mid 1} - \frac{\overline{12 \mid 12}}{22} \end{vmatrix} (11x_1, 22x_2),$$

ce qui est encore une transformation impropre, qui correspond de plus près à la formule pour la transformation propre; la seule différence est que les signes des termes de la première colonne de la matrice en sont changés.

En introduisant des lettres simples  $a, b, \&c.$  à la place des symboles  $11, 22, \&c.$ , je considère d'abord la transformation propre

$$ax^2 + by^2 = ax^2 + by^2.$$

Ici, en écrivant  $x = \alpha x + \beta y, y = \gamma x + \delta y$ , on a

$$a(\alpha x + \beta y)^2 + b(\gamma x + \delta y)^2 = ax^2 + by^2,$$



d'où

$$a(\alpha^2 - 1) + b\gamma^2 = 0, \quad 2a\alpha\beta + 2b\gamma\delta = 0, \quad a\beta^2 + b(\delta^2 - 1) = 0,$$

et de plus

$$\alpha\delta - \beta\gamma = 1.$$

Les trois premières équations donnent

$$\alpha^2 = 1 - \frac{b}{a}\gamma^2, \quad \beta^2 = \frac{b}{a}(1 - \delta^2), \quad \alpha\beta = -\frac{b}{a}\gamma\delta,$$

et l'on vérifie que ces valeurs satisfont à l'équation  $\alpha\delta - \beta\gamma = 1$ .

En effet, on a

$$\alpha^2 \delta^2 = \delta^2 - \frac{b}{a} \gamma^2 \delta^2, \quad \beta^2 \gamma^2 = \frac{b}{a} \gamma^2 - \frac{b}{a} \gamma^2 \delta^2,$$

d'où

$$\alpha^2 \delta^2 - \beta^2 \gamma^2 = \delta^2 - \frac{b}{a} \gamma^2 = \alpha^2,$$

et par suite

$$(\alpha \delta - \beta \gamma)^2 = \alpha^2 (\delta^2 - \frac{b}{a} \gamma^2) = \alpha^4,$$

ce qui donne  $\alpha \delta - \beta \gamma = \alpha^2 = 1$ .

Les valeurs de  $\alpha, \beta, \gamma, \delta$  dépendent d'un seul paramètre. On peut poser

$$\gamma = \sqrt{\frac{a}{b}} \sin \theta, \quad \alpha = \cos \theta, \quad \delta = \cos \theta, \quad \beta = -\sqrt{\frac{b}{a}} \sin \theta,$$

ou plus généralement

$$\gamma = k \sqrt{\frac{a}{b}}, \quad \alpha = h, \quad \delta = h, \quad \beta = -k \sqrt{\frac{b}{a}},$$

où  $h$  et  $k$  sont liés par la relation  $h^2 + \frac{b}{a} k^2 = 1$ .

Pour la transformation impropre, on a

$$ax^2 + by^2 = ax^2 + by^2,$$

avec  $\alpha \delta - \beta \gamma = -1$ . Les formules sont les mêmes, mais avec le signe de  $\beta$  changé.

Venons maintenant au cas de trois variables. La transformation propre est donnée par

$$ax_1^2 + bx_2^2 + cx_3^2 = ax_1^2 + bx_2^2 + cx_3^2,$$

et l'on écrit comme plus haut

$$(x_1, x_2, x_3) = \frac{1}{123 \mid 123} \begin{vmatrix} \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \end{vmatrix} (ax_1, bx_2, cx_3).$$

J'ai déjà fait voir de quelle manière cette formule se rattache à la formule pour la transformation propre; la différence entre les formules est que dans la transformation propre les coefficients de la substitution forment un déterminant  $= +1$ , tandis que dans la transformation impropre ce déterminant est  $= -1$ .

Pour le cas de la fonction binaire  $(a, b, c)(x, y)^2$ , j'ai trouvé que la transformation propre s'écrit

$$x = \frac{1}{\sqrt{1 + \frac{b}{a}k^2}}(x - \sqrt{\frac{b}{a}}ky), \quad y = \frac{1}{\sqrt{1 + \frac{b}{a}k^2}}(\sqrt{\frac{a}{b}}kx + y),$$

et la transformation impropre

$$x = \frac{1}{\sqrt{1 + \frac{b}{a}k^2}}(x + \sqrt{\frac{b}{a}}ky), \quad y = \frac{1}{\sqrt{1 + \frac{b}{a}k^2}}(\sqrt{\frac{a}{b}}kx - y).$$

Ces formules peuvent être appliquées à la théorie des formes quadratiques. En particulier, si l'on considère la forme  $x^2 + y^2$ , on obtient les formules pour la rotation et la réflexion.

Pour la rotation:

$$x = x \cos \theta - y \sin \theta, \quad y = x \sin \theta + y \cos \theta.$$

Pour la réflexion:

$$x = x \cos \theta + y \sin \theta, \quad y = x \sin \theta - y \cos \theta.$$

On a pour la transformation, l'équation

$$(x, y, z, w) = \frac{1}{abcd - bc\alpha^2 + ca\beta^2 + ab\gamma^2 + ad\lambda^2 - bd\mu^2 - cd\nu^2 - \phi^2} \times$$

$$\begin{vmatrix} a & -b\alpha + d\lambda & c\beta - d\mu & -b\gamma + c\nu \\ -b\alpha - d\lambda & b & -a\beta + d\nu & a\gamma - c\mu \\ c\beta + d\mu & -a\beta - d\nu & c & -a\alpha + b\lambda \\ -b\gamma - c\nu & a\gamma + c\mu & -a\alpha - b\lambda & d \end{vmatrix} (ax, by, cz, dw),$$

où  $\alpha, \beta, \gamma, \lambda, \mu, \nu, \phi$  sont des paramètres arbitraires liés par certaines conditions.

Cette formule donne la transformation propre de la fonction quadratique  $ax^2 + by^2 + cz^2 + dw^2$  en elle-même. La démonstration s'obtient en vérifiant que le carré de la matrice est égal à l'unité, ou en se servant de la théorie des déterminants gauches.

On a aussi la formule suivante plus symétrique:

$$\begin{aligned}
 (x, y, z, w) &= \frac{1}{\sqrt{MM'}} \begin{vmatrix} a\alpha' + \beta\beta' + \gamma\gamma' + \delta\delta', & -a\beta' - \beta\alpha' - \gamma\delta' + \delta\gamma' \\ -a\gamma' - \beta\delta' + \gamma\alpha' + \delta\beta', & a\delta' - \beta\gamma' + \gamma\beta' - \delta\alpha' \end{vmatrix} \\
 &= a\alpha' + \beta\beta' + \gamma\gamma' + \delta\delta', \\
 &= -a\beta' - \beta\alpha' - \gamma\delta' + \delta\gamma', \\
 &= -a\gamma' - \beta\delta' + \gamma\alpha' + \delta\beta', \\
 &= a\delta' - \beta\gamma' + \gamma\beta' - \delta\alpha',
 \end{aligned}$$

et les formules analogues pour les autres composantes.

Ces formules peuvent être déduites de la théorie des quaternions. En effet,  $i, j, k$  étant des quantités imaginaires telles que

$$i^2 = j^2 = k^2 = -1, \quad jk = -kj = i, \quad ki = -ik = j, \quad ij = -ji = k,$$

on représente un quaternion par  $w + ix + jy + kz$ , et la norme est  $w^2 + x^2 + y^2 + z^2$ . La transformation cherchée correspond à la multiplication par un quaternion de norme 1.

## 138.

[From the *Journal für die reine und angewandte Mathematik*  
(Crelle),  
torn. L. (1855), pp. 313–318.]

## RECHERCHES SUR LES MATRICES &amp;C.

Les notions de matrice et de déterminant ne coïncident pas, mais il y a entre elles un très grand nombre de relations. Je me borne à considérer ici les matrices dont les termes sont des fonctions linéaires d'une seule indéterminée.

Je pose la matrice

$$\begin{array}{cccc} A, & B, & C, & \dots \\ A', & B', & C', & \dots \\ A'', & B'', & C'', & \dots \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

dont les termes ( $n^2$  en nombre) sont des fonctions linéaires d'une quantité  $s$ , et je considère le déterminant formé avec cette matrice, et les déterminants mineurs formés en supprimant un nombre quelconque des lignes et un nombre égal de colonnes de la matrice. En supprimant une *seule* ligne et une *seule* colonne, on obtient les *premiers mineurs*; en supprimant *deux* lignes et *deux* colonnes, on obtient les *seconds mineurs*; et ainsi de suite.

Cela étant, je suppose que la quantité  $s$  a été trouvée en égalant à zéro le déterminant formé avec la matrice donnée; ce déterminant sera une fonction de  $s$  du  $n$ -ième degré qui généralement ne contiendra pas de facteurs multiples. On voit donc qu'un facteur simple du déterminant ne peut pas entrer comme facteur dans les premiers mineurs (c'est-à-dire dans *tous* les premiers mineurs); mais en supposant que le déterminant ait des facteurs multiples, un facteur multiple du déterminant peut entrer comme facteur (simple ou multiple) dans les premiers mineurs, ou dans les mineurs d'un ordre plus élevé. Il importe de trouver le degré selon lequel un facteur multiple du déterminant peut entrer comme facteur des premiers mineurs, ou des mineurs d'un ordre quelconque donné.

Cela se fait très facilement au moyen d'une propriété générale des déterminants; si les mineurs du  $(r+1)$ -ième ordre contiennent le facteur  $(s-a)^\alpha$  (c'est-à-dire, si tous les mineurs de cet ordre contiennent



le facteur  $(s-a)^\alpha$ , mais non pas tous les facteurs  $(s-a)^{\alpha+1}$ ; et si de même les mineurs du  $r$ -ième ordre contiennent le facteur  $(s-a)^\beta$ , alors les mineurs du  $(r-1)$ -ième ordre contiendront au moins le facteur  $(s-a)^\gamma$ , où  $\gamma > 2\beta - \alpha$ , ou, ce qui est la même chose,  $\alpha - 2\beta + \gamma > 0$ ; c'est-à-dire: en formant la suite des indices

$$\dots, \quad \gamma, \quad \beta, \quad \alpha,$$

les différences secondes seront positives. Il est permis d'avoir  $\gamma = 2\beta - \alpha$ ; cela correspond au cas où le facteur  $(s-a)^\gamma$  entre effectivement dans les mineurs de l'ordre  $r-1$ .

les différences secondes seront positives. Il est permis d'avoir  $\gamma = 2\beta - \alpha$ ; cela correspond au cas où le facteur  $(s-a)^\gamma$  entre effectivement dans les mineurs de l'ordre  $r-1$ .

*Cela étant, on arrive à la règle suivante.* On forme d'abord les suites de nombres

$$\begin{array}{c} \alpha \\ \beta, \alpha' \\ \gamma, \beta', \alpha'' \\ \vdots \end{array}$$

telles que dans chaque suite, les différences secondes soient positives (ou nulles); la dernière suite sera composée d'un seul terme  $\lambda$ . Le symbole

$$\begin{array}{cccc} \alpha & \alpha' & \alpha'' & \dots \\ \beta & \beta' & \dots & \\ \gamma & \dots & & \\ \vdots & & & \\ \lambda & & & \end{array}$$

dénotera alors que le déterminant a un facteur  $(s-a)^\alpha$  qui entre comme facteur  $\alpha$  dans le déterminant, comme facteur  $\beta$  dans les premiers mineurs, comme facteur  $\gamma$  dans les seconds mineurs, et ainsi de suite; le même facteur peut entrer aussi dans d'autres mineurs avec des exposants plus élevés; et de même pour les autres facteurs  $(s-a')^{\alpha'}$ ,  $(s-a'')^{\alpha''}$ , &c.

Pour donner un exemple, je considère la matrice

$$\begin{array}{cccc} s-a, & h & & \\ & s-b, & h & \\ & & s-c, & h \\ & & & s-a \end{array}$$

(c'est la forme de la matrice pour un système d'équations linéaires à différences). Le déterminant est

$$(s-a)^2(s-b)(s-c) - h^4 = 0.$$

On trouve pour les indices

$$\begin{array}{ccc} 2 & 1 & 1 \\ 1 & 1 & \end{array}$$

et le symbole est

$$\begin{array}{cc} 2 & 1 \\ 1 & 1 \end{array}$$

ce qui signifie que le déterminant a un facteur double  $(s-a)^2$  et un facteur simple  $(s-b)$ ,  $(s-c)$ ; les facteurs  $(s-b)$  et  $(s-c)$  n'entrent pas dans les premiers mineurs (ils y entrent effectivement avec l'exposant 0); le facteur  $(s-a)$  y entre avec l'exposant 1.

De même, pour la matrice

$$\begin{array}{ccccccc} s-a, & 1 & & & & & \\ & s-a, & 1 & & & & \\ & & s-a, & 1 & & & \\ & & & s-a, & 1 & & \\ & & & & s-a, & 1 & \\ & & & & & s-a & \end{array}$$

où tous les termes de la diagonale sont  $s-a$ , et les termes voisins sont 1, on a pour les indices de  $(s-a)$  la suite

$$5, 4, 3, 2, 1;$$

et ainsi de suite.

Pour la matrice de la forme

$$\begin{array}{ccccccc} s-a, & h & & & & & \\ & s-a, & h & & & & \\ & & s-a, & h & & & \\ & & & s-a, & h & & \\ & h & & & s-a & & \end{array}$$

les facteurs du déterminant sont des facteurs simples, et la matrice n'a pas de facteurs multiples.

Pour donner encore un exemple du sens de ces symboles, le symbole

$$\begin{array}{c} 3 \\ 2 \\ 1 \\ 1 \end{array}$$

dénote que le déterminant a un facteur  $(s-a)$  qui entre comme facteur triple dans le déterminant, comme facteur double dans les premiers mineurs, et comme facteur simple dans les seconds mineurs; l'autre facteur du déterminant est un facteur simple  $(s-b)$ .

Les nombres des symboles pour  $n = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11$ , &c. sont

$$1, 3, 6, 14, 27, 58, 111, 223, 424, 817, 1527, \text{ \&c.};$$

ce sont les nombres de partition de  $n$  en quelconques parties impaires. En effet, le symbole

$$\begin{array}{c} \alpha \\ \beta \\ \vdots \\ \lambda \end{array}$$

où  $\alpha + \beta + \dots + \lambda = n$ , correspond à une partition de  $n$  en  $\alpha$  parties égales à 1,  $\beta$  parties égales à 3,  $\gamma$  parties égales à 5, et ainsi de suite; et réciproquement, toute partition de  $n$  en parties impaires donne un symbole de la forme indiquée.

Les symboles peuvent être interprétés géométriquement au moyen de la théorie de l'homographie. Pour l'homologie, le symbole est  $\frac{1}{1}$  ; pour l'homographie générale, le symbole  $\frac{2}{1}$  ; les autres symboles  $\frac{2}{1}$  ,  $\frac{3}{1}$  ,  $\frac{3}{2}$  se rapportent à des cas moins généraux, et le symbole  $\frac{2}{1}$  au cas de l'identité complète des deux figures; y compris ce cas-limite de l'identité complète, il existe pour le *plan* 6 espèces d'homographie; pour l'*espace ordinaire* il existe 14 espèces d'homographie. Je reviendrai à cette théorie à une autre occasion.

*Londres, le 24 Mai 1854.*

## 139.

## AN INTRODUCTORY MEMOIR UPON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*,  
vol. CXLIV. for the year 1854, pp. 244-258.  
Received April 20,—Read May 4, 1854.]

1. THE term *Quantics* is used to denote the entire subject of rational and integral functions, and of the equations and loci to which these give rise; the word “quantic” is an adjective, meaning *of such a degree*, but may be used substantively, the noun understood being (unless the contrary appear by the context) *function*; so used the word admits of the plural “quantics.”

The quantities or symbols to which the expression “degree” refers, or (what is the same thing) in regard to which a function is considered as a quantic, will be spoken of as “*facients*.” A quantic may always be considered as being, in regard to its facients, homogeneous, since to render it so, it is only necessary to introduce as a facient unity, or some symbol which is to be ultimately replaced by unity; and in the cases in which the facients are considered as forming two or more distinct sets, the quantic may, in like manner, be considered as homogeneous in regard to each set separately.

2. The expression “an equation,” used without explanation, is to be understood as meaning the equation obtained by putting any quantic equal to zero. I make no absolute distinction between the words “degree” and “order” as applied to an equation or system of equations, but I shall in general speak of the order rather than the degree. The equations of a system may be independent, or there may exist relations of connexion between the different equations of the system; the subject of a system of equations so connected together is one of extreme complexity and difficulty. It will be sufficient to notice here, that in any system whatever of equations, assuming only that the equations are not more than sufficient to determine the ratios of the facients, and joining to the system so many linear equations between the facients as will render the ratios of the facients determinate, the order of the system is the same thing as the order of the equation which determines any one of these ratios; it is clear that for a single equation the order so determined is nothing else than the order of the equation.

3. An equation or system of equations represents, or is represented by, a locus. This assumes that the facients depend upon quantities  $x, y, \dots$  the coordinates of a point in space; the entire series of points, the coordinates of which satisfy the equation or system of equations, constitutes the locus. To avoid complexity, it is proper to take the facients themselves as coordinates, or at all events to consider these facients as linear functions of the coordinates.

4. A rational and integral function which contains any number of sets of facients, each set containing any number of facients, and which is homogeneous in regard to the facients of each set separately, is a *Quantic*. The several degrees in respect of the different sets of facients are the *orders* of the quantic in respect of the several sets respectively. If there is but a single set, the quantic is simply said to be of such an order; thus a quantic of the first order is a *linear function*, of the second order a *quadric function*, of the third order a *cubic function*, and so on.

5. A quantic may be represented by a single letter, as  $U$ ; or if it be referred to its sets of facients and coefficients, then using for the facients of any set the letters  $x, y, z, \dots$  (or  $x_1, x_2, x_3, \dots$ ), and for the coefficients single letters, as  $a, b, c, \dots$ , or else functions of single letters, as  $a_1, a_2, \dots$ , the quantic may be represented in a form in which the sets of facients and the coefficients are exhibited at full length. I shall in general represent a quantic in the two ways, indifferently, according as one or other is most convenient in the particular case. But when a quantic is referred to its sets and coefficients, I shall mark the sets by an asterisk; the notation is as follows:

A quantic of a single set  $(x, y, \dots)$  of the order  $m$  is written

$$(\mathfrak{x}, y, \dots)^m.$$

where the mark \* may be considered as indicative of the absolute generality of the quantic; any such quantic may of course be considered as the sum of a series of terms  $afy^k \dots x^{\alpha_0}y^{\beta_0} \dots$ , &c. of the proper degrees in the different sets respectively, each term multiplied by a coefficient; these coefficients may be mere numerical multiples of single letters or elements such as  $a, b, c, \dots$ , or else functions (in general rational and integral functions) of such letters or elements; and these elements may be considered as forming a set, or more than one set, according to the nature of the case. The quantic may be considered as the quantic of the facients, including the coordinates and the coefficients or elements, and in this point of view it is a quantic of a single set, which is made up of the several sets of the facients. But the coordinates and the elements are distinguished as being the *variables* and the *parameters* of the quantic; and the quantic may be considered as a function in which the parameters enter in a determinate manner, but which is absolutely general in regard to the variables, of which it is a quantic of the proper orders.

**6.** A quantic considered as a function of a single set of facients may be represented in a compact form by means of the ordinary notation of powers, or by means of the umbral notation; but when the quantic is considered as belonging to two or more sets, a special notation is required. The notation which I propose to employ is as follows:—



7. A quantic containing the sets  $(x_1, y_1, \dots)$ ,  $(x_2, y_2, \dots)$ , &c. of the orders  $m_1$ ,  $m_2$ , &c. respectively is represented by

$$(\mathfrak{x}_1, y_1, \dots)^{m_1} (x_2, y_2, \dots)^{m_2} \dots$$

or, when there is occasion to write out the coefficients at full length, by

$$(a, b, \dots (x_1, y_1, \dots)^{m_1} (x_2, y_2, \dots)^{m_2} \dots$$

where  $a, b, \dots$  are the coefficients, which are functions of the elements. The former of these notations denotes, in effect, that the quantic is the sum of a series of terms each of which is the product of a coefficient into powers of the facients of the several sets, the sum of the degrees in each set being equal to the order of the quantic in respect of that set. And when the coefficients are written at full length, the notation serves to indicate the particular values of the coefficients as functions of the elements.

8. The foregoing definitions and notations apply to a quantic containing any number whatever of sets of facients. The theory is, however, in a great measure dependent upon that of a quantic containing only a single set of facients, and in the development of this special theory there arises a class of functions which I refer to as the *covariants* and *invariants* of the quantic. The words covariant and invariant are, however, used in a more general sense, as will be presently explained.

9. What precedes is a return to and generalization of the method employed in the first part of the memoir published in the *Camb. Math. Jour.*, t. IV. [1845], and *Camb. and Dubl. Math. Jour.*, t. I. [1846], under the title "On Linear Transformations," [13 and 14], and Crelle, t. XXX. [1846], under the title "Mémoire sur les Hyperdéterminants," [\*16], and which I shall refer to as my *original memoir*. I there consider in fact the invariants of a quantic

$$(\mathfrak{x}_1, x_2, \dots x_n (y_1, y_2, \dots y_m) \dots),$$

linear in regard to  $n$  sets each of them of  $m$  facients, and I represent the coefficients of a term  $x_r y_s z_t \dots$  by  $rst \dots$ ; there is no difficulty in seeing that  $\alpha, \beta$  being any two different numbers out of the series 1, 2,  $\dots m$ , the operation  $\{x_\beta \partial_{x_\alpha}\}$  is identical with the operation

$$\Sigma \Sigma \dots \left( \alpha st \dots \frac{d}{d\beta st \dots} \right),$$

where the summations refer to  $s, t, \dots$  which pass respectively from 1 to  $m$ , both inclusive; and the condition that a function, assumed to be an invariant, i.e. to contain only the coefficients, may be reduced to zero by the operation  $\{x_\beta \partial_{x_\alpha}\} - x_\beta \partial_{x_\alpha}$ , is of course simply the condition that such function may be reduced to zero by the operation  $\{x_\beta \partial_{x_\alpha}\}$ ; the condition in question is therefore the same thing as the equation

$$\Sigma \Sigma \dots \left( \alpha st \dots \frac{d}{d\beta st \dots} \right) u = 0$$

of my original memoir.

**10.** But the definition in the present memoir includes also the method made use of in the second part of my original memoir. This method is substantially as follows:—consider for simplicity a quantic  $U =$

$$(\mathfrak{x}, y, \dots)^m$$

containing only the single set  $(x, y, \dots)$ , and let  $U_1, U_2, \dots$  be what the quantic becomes when the set  $(x, y, \dots)$  is successively replaced by the sets  $(x_1, y_1, \dots), (x_2, y_2, \dots), \dots$  the number of these new sets being equal to or greater than the number of facients in the set. Suppose that  $A, B, C, \dots$  are any of the determinants

$$\begin{vmatrix} \partial_{x_1} & \partial_{x_2} & \partial_{x_3} & \dots \\ \partial_{y_1} & \partial_{y_2} & \partial_{y_3} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix},$$

then forming the derivative

$$A^p B^q C^r \dots U_1 U_2 \dots,$$

where  $p, q, r, \dots$  are any positive integers, the function so obtained is a covariant involving the sets  $(x_1, y_1, \dots), (x_2, y_2, \dots)$  &c.; and if after the differentiations we replace the sets  $(x_1, y_1, \dots), (x_2, y_2, \dots)$ , &c. by the original set  $(x, y, \dots)$ , we obtain a covariant of the quantic  $U$ .

**11.** But the covariants obtained by this method are not the most general ones. The general form of a covariant is obtained as follows. Let  $U$  be the quantic, and let  $U_1, U_2, \dots$  be what  $U$  becomes when the set  $(x, y)$  is successively replaced by the sets  $(x_1, y_1), (x_2, y_2), \&c.$  Suppose moreover that  $\overline{12} = \partial_{x_1}\partial_{y_2} - \partial_{x_2}\partial_{y_1}, \&c.,$  then the function

$$\overline{12}^p \overline{13}^q \overline{23}^r \dots U_1 U_2 U_3 \dots,$$

in which, after the differentiations, the new sets  $(x_1, y_1), (x_2, y_2), \dots$  may be replaced by the original set  $(x, y),$  will be a covariant of the quantic  $U.$  And if the number of differentiations be such as to make the facients disappear, i.e. if the sum of all the indices  $p, q, \dots$  of the terms  $\overline{12}, \&c.$  which contain the symbolical facients  $x, y, \dots$  be equal to the sum of the orders of the quantics  $U_1, U_2, \dots,$  then the result will be an *invariant* of the quantic  $U.$

**12.** I return to the expression  $A^p B^q C^r \dots U_1 U_2 \dots,$  and I suppose that after the differentiations the sets  $(x_1, y_1, \dots), (x_2, y_2, \dots), \&c.$  are replaced by the original set  $(x, y, \dots);$  then the result is a covariant of the quantic  $U.$  But the operation  $A^p B^q C^r \dots$  may be replaced by any other operation which gives the same result; and in particular, if we write

$$\Omega = \begin{vmatrix} \partial_{x_1}, & \partial_{x_2}, & \dots \\ \partial_{y_1}, & \partial_{y_2}, & \dots \\ \vdots & \vdots & \ddots \end{vmatrix},$$

then the operation  $\Omega^p$  upon the product  $U_1 U_2 \dots U_p$  will give a covariant of the quantic  $U.$

**13.** I consider now the case of a quantic containing only a single set of facients  $(x, y)$ . The quantic is represented by

$$(a, b, \dots b', \not{a}'x, y)^m,$$

where the notation implies that the quantic is

$$ax^m + \frac{m}{1}bx^{m-1}y + \dots + \frac{m}{1}b'xy^{m-1} + a'y^m.$$

The coefficients  $a, b, \dots$  are the elements of the quantic; and the quantic may be considered as a function of the facients  $(x, y)$  and the coefficients  $(a, b, \dots)$ .

**14.** I say that a function of the coefficients and facients of a quantic is a *covariant* of the quantic when, if the facients are transformed by any linear substitution, the function is transformed by the same substitution. And I say that a function of the coefficients only is an *invariant* of the quantic when it is not altered by any linear substitution of the facients.

**15.** Suppose now that  $\Phi$  is a covariant of  $U$ , then the operation  $\Phi$  performed upon any covariant of  $U$  gives rise to a covariant of the system  $(\Phi, U)$ . In particular, if  $\Phi$  is an invariant of  $U$ , then the operation  $\Phi$  upon any covariant of  $U$  gives an invariant of the system. And if  $\Phi$  and  $\Psi$  are both invariants of  $U$ , then  $\Phi \cdot \Psi$  is also an invariant of  $U$ .

**16.** The number of independent invariants of a quantic of a given degree is finite. For the quantic of the degree  $m$  in two variables  $(x, y)$ , the number of coefficients is  $m + 1$ ; the number of independent invariants is equal to the number of independent coefficients minus the number of independent parameters in the general linear substitution, which is 3; but the invariants are homogeneous functions of the coefficients, and therefore the number of independent invariants is  $(m + 1) - 3 = m - 2$ , or more precisely, it is the number of independent coefficients minus the number of parameters, taking into account the homogeneity of the invariants.

For the quadratic  $(a, b, c)(x, y)^2$ , there is one invariant, the discriminant  $ac - b^2$ .

For the cubic  $(a, b, c, d)(x, y)^3$ , there is one invariant, the discriminant  $a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2$ .

For the quartic  $(a, b, c, d, e)(x, y)^4$ , there are two invariants,  $I$  and  $J$ , of degrees 2 and 3 respectively in the coefficients.

**17.** I pass next to the quantic

$$(\mathfrak{x}, y)^m,$$

which I shall in general consider under the form

$$(a, b, \dots b', \mathfrak{x}', y)^m,$$

but sometimes under the form

$$(a, b, \dots b', \mathfrak{x}', y)^m,$$

the former notation denoting, it will be remembered,

$$ax^m + \frac{m}{1}bx^{m-1}y \dots + \frac{m}{1}b'xy^{m-1} + a'y^m,$$

and the latter notation

$$ax^m + bx^{m-1}y \dots + b'xy^{m-1} + a'y^m.$$

But in particular cases the coefficients will be represented all of them by unaccentuated letters, thus  $(a, b, c, (x, y)^3)$  will be used to denote  $ax^3 + 3bx^2y + 3cxy^2 + dy^3$ , and  $(a, b, c, \overline{(x, y)}^3)$  will be used to denote  $ax^3 + bx^2y + cxy^2 + dy^3$ , and so in all similar cases.

Applying the general methods to the quantic

$$(a, b, \dots b', \overline{(x, y)}^m,$$

we see that

$$\{y\partial_x\} = a\partial_b + 2b\partial_c \dots + mb'\partial_{a'},$$

$$\{x\partial_y\} = mb\partial_a + (m-1)c\partial_b \dots + a'\partial_{b'};$$

in fact, with these meanings of the symbols the quantic is reduced to zero by each of the operations  $\{y\partial_x\} - y\partial_x$ ,  $\{x\partial_y\} - x\partial_y$ ; hence according to the definition any function which is reduced to zero by each of the last-mentioned operations is a covariant of the quantic. But in accordance with a preceding remark, the covariant may be considered as a rational and integral function, separately homogeneous in regard to the facients  $(x, y)$  and the coefficients  $(a, b, \dots b', a')$ . If instead of the single set  $(x, y)$  the covariant contains the sets  $(x_1, y_1)$ ,  $(x_2, y_2)$ , &c., then it must be reduced to zero by each of the operations  $\{y\partial_x\} - Sy\partial_x$ ,  $\{x\partial_y\} - Sx\partial_y$  (where  $Sy\partial_x = y_1\partial_{x_1} + y_2\partial_{x_2} + \dots$ ), but I shall principally attend to the case in which the covariant contains only the set  $(x, y)$ .

Suppose, for shortness, that the quantic is represented by  $U$ , and let  $U_1, U_2, \dots$  be what  $U$  becomes when the set  $(x, y)$  is successively replaced by the sets  $(x_1, y_1)$ ,  $(x_2, y_2)$ , &c. Suppose moreover that  $\overline{12} = \partial_{x_1}\partial_{y_2} - \partial_{x_2}\partial_{y_1}$ , &c., then the function

$$\overline{12}^p \overline{13}^q \overline{23}^r \dots U_1 U_2 U_3 \dots,$$

in which, after the differentiations, the new sets  $(x_1, y_1)$ ,  $(x_2, y_2)$ ,  $\dots$  may be replaced by the original set  $(x, y)$ , will be a covariant of the quantic  $U$ . And if the number

of differentiations be such as to make the facients disappear, i.e. if the sum of all the indices  $p, q, \dots$  of the terms  $\overline{12}$ , &c. which contain the symbolical facients  $x, y, \dots$  be equal to the sum of the orders of the quantics  $U_1, U_2, \dots$ , then the result will be an *invariant* of the quantic  $U$ .

The preceding investigations show the general form of the covariants and invariants of a quantic; but they do not show how to find the complete system of independent covariants and invariants. This is a problem of extreme difficulty, which has been solved only for quantics of low degrees. The complete system of covariants of a binary quantic of the degree  $m$  consists of a finite number of covariants, which are connected by a finite number of syzygies; and the number of independent covariants is equal to the number of independent coefficients minus the number of parameters in the linear substitution.

**18.** The methods which I have developed in this memoir may be applied to the investigation of the covariants and invariants of quantics of any number of variables. The case of three variables is particularly interesting, as it leads to the theory of ternary quantics, which has important applications to the theory of curves and surfaces. But the complete investigation of the covariants and invariants of ternary quantics is a problem of much greater difficulty than that of binary quantics, and has been solved only for quantics of the first, second, and third orders.



**19.** The number of independent covariants of a binary quantic of the degree  $m$  may be determined as follows. The number of coefficients of the quantic is  $m + 1$ ; the number of independent parameters in the linear substitution is 3; but the covariants are not all of the same degree, and therefore the number of independent covariants is not simply  $(m + 1) - 3$ . The actual determination of the number of independent covariants requires a more detailed investigation, which I reserve for a future memoir.

**20.** I conclude this introductory memoir by stating the principal results which I have obtained. I have established the general theory of quantics, and I have shown how to form the covariants and invariants of a quantic by means of symbolical operations. I have also shown how to determine the number of independent covariants and invariants of a binary quantic of any degree. The application of these methods to particular cases will be given in subsequent memoirs.

**21.** The investigations of this memoir may be generalized in various directions. In the first place, the theory of quantics may be extended to quantics containing any number of sets of facients. In the second place, the theory may be extended to quantics with coefficients which are not constants, but functions of other variables. In the third place, the theory may be extended to quantics which are not rational and integral functions, but which satisfy certain differential equations. These generalizations will be considered in future memoirs.

## 140.

## RESEARCHES ON THE PARTITION OF NUMBERS.

*[From the Philosophical Transactions of the Royal Society of London,  
vol. CXLV for the year 1855, pp. 127-140.  
Received April 14,—Read May 24, 1855.]*

1. THE theory of the partition of numbers is one of the most important and interesting branches of analysis. It has been treated by Euler, by Lagrange, and by many other eminent mathematicians; but there are still many problems in this theory which remain to be solved. The present memoir is devoted to the investigation of some of these problems.

2. I consider in the first place the problem of finding the number of partitions of a given number  $n$  into a given number of parts  $k$ , each part being any positive integer. This problem may be solved by means of the method of generating functions. The generating function for the number of partitions of  $n$  into  $k$  parts is

$$\frac{x^k}{(1-x)(1-x^2)\dots(1-x^k)}.$$

**3.** The coefficient of  $x^n$  in the expansion of this function is the number of partitions of  $n$  into  $k$  parts. This number is usually denoted by  $P(n, k)$ . The function  $P(n, k)$  satisfies many remarkable properties, of which I shall mention only a few.

**4.** The most important property of the function  $P(n, k)$  is that it is equal to the number of partitions of  $n$  into parts not exceeding  $k$ . This property is easily proved by means of the generating function. The generating function for the number of partitions of  $n$  into parts not exceeding  $k$  is

$$\frac{1}{(1-x)(1-x^2)\dots(1-x^k)};$$

and the coefficient of  $x^{n-k}$  in the expansion of this function is the number of partitions of  $n-k$  into parts not exceeding  $k$ , which is equal to the number of partitions of  $n$  into  $k$  parts.

**5.** Another important property of the function  $P(n, k)$  is that it satisfies the equation

$$P(n, k) = P(n-1, k-1) + P(n-k, k).$$

This equation may be proved directly from the definition of  $P(n, k)$ . For the partitions of  $n$  into  $k$  parts may be divided into two classes: those in which the smallest part is 1, and those in which the smallest part is greater than 1. The number of partitions of the first class is  $P(n-1, k-1)$ , and the number of partitions of the second class is  $P(n-k, k)$ .

**6.** I consider next the problem of finding the number of partitions of a given number  $n$  into any number of parts. This number is usually denoted by  $P(n)$ . The generating function for  $P(n)$  is

$$\frac{1}{(1-x)(1-x^2)(1-x^3)\dots}.$$

The coefficient of  $x^n$  in the expansion of this function is  $P(n)$ . This function satisfies many remarkable properties, of which the most celebrated is the theorem of Euler:

$$P(n) = P(n-1) + P(n-2) - P(n-5) - P(n-7) + P(n-12) + P(n-15) - \dots$$

where the numbers 1, 2, 5, 7, 12, 15, ... are the pentagonal numbers  $\frac{1}{2}k(3k-1)$  for  $k = 1, -1, 2, -2, 3, -3, \dots$

**7.** I consider also the problem of finding the number of partitions of a given number  $n$  into parts subject to certain conditions. For example, the number of partitions of  $n$  into odd parts is equal to the number of partitions of  $n$  into unequal parts. This theorem may be proved by means of the generating functions. The generating function for the number of partitions of  $n$  into odd parts is

$$\frac{1}{(1-x)(1-x^3)(1-x^5)\dots},$$

and the generating function for the number of partitions of  $n$  into unequal parts is

$$(1+x)(1+x^2)(1+x^3)\dots$$

These two functions are equal, as may be shown by multiplying the first by  $(1-x^2)(1-x^4)(1-x^6)\dots$  and the second by  $(1-x)(1-x^2)(1-x^3)\dots$

8. The theory of the partition of numbers may be extended in various directions. One of the most important extensions is the theory of the partition of numbers into a given number of parts, each part being subject to certain conditions. For example, the number of partitions of  $n$  into  $k$  parts, each part not exceeding  $m$ , may be found by means of the generating function

$$\frac{x^k}{(1-x)(1-x^2)\dots(1-x^m)} \quad (\text{taking } k \text{ factors}).$$

9. Another important extension is the theory of the partition of numbers into parts of given forms. For example, the number of partitions of  $n$  into squares, or into cubes, or into any other given powers. These problems are in general of great difficulty, and have been solved only in a few special cases.

10. I consider also the theory of *circulating functions*, which are functions that take different values according to the residue of  $n$  with respect to some modulus. The simplest example of a circulating function is the function  $(-1)^n$ , which takes the values  $+1$  and  $-1$  according as  $n$  is even or odd. More generally, a circulating function may be defined as a function of the form

$$A_0 + A_1\omega^n + A_2\omega^{2n} + \dots + A_{m-1}\omega^{(m-1)n},$$

where  $\omega$  is a primitive  $m$ -th root of unity, and  $A_0, A_1, \dots, A_{m-1}$  are constants.

**11.** The theory of circulating functions is closely connected with the theory of partitions. For the number of partitions of  $n$  into parts subject to certain conditions is often expressed by means of a circulating function. For example, the number of partitions of  $n$  into parts not divisible by a given number  $m$  is a circulating function of  $n$  with the modulus  $m$ .

**12.** I have investigated the general form of the function  $P(a, b, c, \dots)q$ , which denotes the number of partitions of  $q$  into the elements  $a, b, c, \dots$  each element being used any number of times. The general form of this function is

$$P(a, b, c, \dots)q = Aq^{k-1} + Bq^{k-2} + \dots + Lq + M + \Sigma q^r (A_0, A_1, \dots A_{l-1}) \text{ per } l_q,$$

a formula in which  $k$  denotes the number of the elements  $a, b, c, \dots$  &c., and  $l$  is any divisor (unity excluded) of one or more of these elements; the summation in the case of each such divisor extends from  $r = 0$  to  $r = k - 1$ , where  $k$  is the number of the elements  $a, b, c, \dots$  &c. of which  $l$  is a divisor; and the investigations indicate how the values of the coefficients  $A$  of the prime circulators are to be obtained.

**13.** It has been moreover in effect shown, that if  $D = a + b + c + \dots$ , then the number of partitions of  $q$  into the elements  $a, b, c, \dots$ , is equal to the coefficient of  $x^q$  in the expansion of

$$\frac{1}{(1 - x^a)(1 - x^b)(1 - x^c) \dots}.$$

This coefficient may be expressed as a sum of partial fractions, and each partial fraction gives rise to a term of the form  $Aq^r$  or  $Aq^r(A_0, A_1, \dots A_{l-1})$  per  $l_q$ . The complete expression for the number of partitions is therefore a sum of terms of this form.

**14.** I have also investigated the theory of the partition of numbers into a given number of parts, each part being of a given form. For example, the number of partitions of  $n$  into  $k$  parts, each part being a square, may be found by means of the generating function

$$\frac{x^{k^2}}{(1 - x)(1 - x^4)(1 - x^9) \dots (1 - x^{k^2})}.$$

This function may be expanded in a series of partial fractions, and the coefficient of  $x^n$  in the expansion is the number of partitions required.



**15.** The case however where the coefficients  $A$  satisfy, for each divisor  $b$  of the number  $a$ , the above-mentioned equations, the circulating function is what I call a *prime circulator*. The prime circulators are the simplest circulating functions, and all other circulating functions may be expressed as sums of prime circulators.

**16.** I have given a method for finding the coefficients of the prime circulators in the expression for the number of partitions of  $n$  into given elements. This method depends upon the theory of the roots of unity, and upon the theory of the decomposition of rational functions into partial fractions.

**17.** The preceding investigations show the general form of the function  $P(a, b, c, \dots)q$ , viz. that

$$P(a, b, c, \dots)q = Aq^{k-1} + Bq^{k-2} \dots + Lq + M + \Sigma q^r (A_0, A_1, \dots A_{l-1}) \text{ per } l_q,$$

a formula in which  $k$  denotes the number of the elements  $a, b, c, \dots$  &c., and  $l$  is any divisor (unity excluded) of one or more of these elements; the summation in the case of each such divisor extends from  $r = 0$  to  $r = k - 1$ , where  $k$  is the number of the elements  $a, b, c, \dots$  &c. of which  $l$  is a divisor; and the investigations indicate how the values of the coefficients  $A$  of the prime circulators are to be obtained.

**18.** It has been moreover in effect shown, that if  $D = a + b + c + \dots$ , then the number of partitions of  $q$  into the elements  $a, b, c, \dots$ , is equal to the coefficient of  $x^q$  in the expansion of

$$\frac{1}{(1 - x^a)(1 - x^b)(1 - x^c) \dots}.$$

**19.** I have also considered the case of the partition of a number into parts which are subject to certain congruence conditions. For example, the number of partitions of  $n$  into parts which are all congruent to 1 modulo 3. This problem may be solved by means of the generating function

$$\frac{1}{(1 - x)(1 - x^4)(1 - x^7) \dots}.$$

**20.** The theory of partitions may also be extended to the case of partitions of numbers into parts which are not necessarily integers. For example, the number of partitions of  $n$  into  $k$  parts, each part being a multiple of a given number  $\alpha$ . This problem may be solved by means of the generating function

$$\frac{x^{k\alpha}}{(1 - x^\alpha)(1 - x^{2\alpha}) \dots (1 - x^{k\alpha})}.$$

**21.** I have also investigated the theory of the partition of numbers into parts which are subject to certain inequalities. For example, the number of partitions of  $n$  into  $k$  parts, each part not exceeding the next. This is the ordinary theory of partitions, and the generating function is

$$\frac{x^k}{(1-x)(1-x^2)\dots(1-x^k)}.$$

**22.** The number of partitions of  $n$  into  $k$  parts, each part being at least twice the next, may be found by means of the generating function

$$\frac{x^{2^k-1}}{(1-x)(1-x^2)(1-x^4)\dots(1-x^{2^{k-1}})}.$$

**23.** I have also considered the theory of the partition of numbers into parts which are functions of the index. For example, the number of partitions of  $n$  into  $k$  parts, the  $i$ -th part being  $f(i)$ , where  $f$  is any given function. This problem may be solved by means of the generating function

$$\frac{x^{f(1)+f(2)+\dots+f(k)}}{(1-x^{f(1)})(1-x^{f(2)})\dots(1-x^{f(k)})}.$$

**24.** The theory of the partition of numbers is closely connected with the theory of elliptic functions and with the theory of modular equations. The generating function for the number of partitions of  $n$  is

$$\frac{1}{(1-x)(1-x^2)(1-x^3)\dots} = \frac{1}{(x)_\infty},$$

where  $(x)_\infty$  denotes the infinite product  $(1-x)(1-x^2)(1-x^3)\dots$ . This function is connected with the elliptic functions by the well-known formula of Jacobi:

$$(x)_\infty = \frac{2K}{\pi} \sqrt[4]{q} \sin^{1/2} \text{am } u,$$

or more directly by the formula

$$(x)_\infty = \sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k-1)/2}.$$

**25.** The formula of Euler for the number of partitions of  $n$  may be written in the form

$$P(n) = \sum_{k=1}^{\infty} (-1)^{k-1} P\left(n - \frac{k(3k-1)}{2}\right) + (-1)^{k-1} P\left(n - \frac{k(3k+1)}{2}\right),$$

where it is understood that  $P(m) = 0$  when  $m < 0$ , and  $P(0) = 1$ .

**26.** I have also investigated the theory of the partition of numbers into parts which are connected by certain linear equations. For example, the number of partitions of  $n$  into three parts  $a, b, c$  subject to the condition  $a + 2b + 3c = n$ . This problem may be solved by means of the generating function

$$\frac{1}{(1-x)(1-x^2)(1-x^3)}.$$

**27.** The coefficient of  $x^n$  in the expansion of this function is the number of partitions of  $n$  into the parts 1, 2, 3, each part being used any number of times. This coefficient may be expressed in the form

$$\frac{1}{12}(n+3)^2 - \frac{7}{72} + \frac{1}{8}(-1)^n + \frac{1}{9}(\omega^n + \omega^{2n}),$$

where  $\omega$  is a primitive cube root of unity. This expression is a sum of a polynomial in  $n$  and two circulating functions.

**28.** The general theory of the partition of numbers into given elements may be applied to the solution of many problems in arithmetic and analysis. For example, it may be applied to the problem of finding the number of solutions of indeterminate equations of the first degree, to the problem of finding the number of representations of a number as a sum of squares, and to many other problems of a similar nature.

**29.** I conclude this memoir by stating the principal results which I have obtained. I have established the general theory of the partition of numbers, and I have shown how to find the number of partitions of a given number into given parts. I have also investigated the theory of circulating functions, and I have shown how to express the number of partitions of a number into given elements by means of these functions.

**30.** The investigations of this memoir may be generalized in various directions. In the first place, the theory may be extended to the partition of numbers into parts which are not necessarily integers. In the second place, the theory may be extended to the partition of numbers into parts which are subject to certain inequalities. In the third place, the theory may be extended to the partition of numbers into parts which are connected by certain linear equations. These generalizations will be considered in future memoirs.

## 141.

## A SECOND MEMOIR UPON QUANTICS.

*[From the Philosophical Transactions of the Royal Society of London,  
vol. CXLVI. for the year 1856, pp. 101–126.  
Received December 13, 1855,—Read January 10, 1856.]*

1. THE present memoir is a continuation of the Introductory Memoir upon Quantics (Art. 139). I propose to develop more fully the theory of the covariants and invariants of binary quantics, and to give some applications of this theory.

2. I consider in the first place the general theory of the covariants of a binary quantic of any degree. Let the quantic be

$$(a, b, \dots b', \cancel{x}, y)^m,$$

and let  $U$  denote the quantic. Then any covariant of  $U$  may be obtained by means of the symbolical operation

$$\overline{12^p} \overline{13^q} \overline{23^r} \dots U_1 U_2 U_3 \dots,$$

as explained in the Introductory Memoir.

**3.** The number of independent covariants of a binary quantic of the degree  $m$  is finite. This theorem, which is due to Gordan, may be proved by means of the theory of symbolical operations. The proof depends upon the fact that any covariant may be expressed as a rational and integral function of a finite number of fundamental covariants, which are called the *irreducible covariants* of the quantic.

**4.** The irreducible covariants of a binary quantic of the degree  $m$  may be found by means of the following method. Let  $U$  be the quantic, and let  $\Omega$  be the determinant

$$\Omega = \begin{vmatrix} \partial_{x_1} & \partial_{x_2} \\ \partial_{y_1} & \partial_{y_2} \end{vmatrix}.$$

Then the irreducible covariants are obtained by operating with  $\Omega$  upon the products  $U_1U_2$ ,  $U_1U_2U_3$ ,  $\dots$  and retaining only those operations which give independent results.

**5.** For the quadratic  $(a, b, c)(x, y)^2$ , the irreducible covariants are:

1. The quantic itself  $U = (a, b, c)(x, y)^2$ ;
2. The discriminant  $ac - b^2$ , which is an invariant.



For the cubic  $(a, b, c, d)(x, y)^3$ , the irreducible covariants are:

1. The quantic itself  $U = (a, b, c, d)(x, y)^3$ ;
2. The Hessian  $H = (ac - b^2, ad - bc, bd - c^2)(x, y)^2$ ;
3. The cubicovariant  $\Phi = (a^2d - 3abc + 2b^3, \dots)(x, y)^3$ ;
4. The discriminant  $\Delta = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2$ , which is an invariant.

**6.** For the quartic  $(a, b, c, d, e)(x, y)^4$ , the irreducible covariants are:

1. The quantic itself  $U = (a, b, c, d, e)(x, y)^4$ ;
2. The Hessian  $H = (ae - 4bd + 3c^2, \dots)(x, y)^4$ ;
3. The catalecticant, which is an invariant of degree 3;
4. The discriminant, which is an invariant of degree 6.

The complete system of irreducible covariants of the quartic consists of two covariants and two invariants, connected by a single syzygy.

7. I consider now the theory of the symbolical representation of covariants. Let  $U$  be a binary quantic of the degree  $m$ , and let the symbolical representation of  $U$  be

$$U = (a_1x + b_1y)^m = (a_2x + b_2y)^m = \dots$$

where  $a_1, b_1, a_2, b_2, \dots$  are symbolical coefficients. Then any covariant of  $U$  may be represented symbolically as a product of factors of the form  $(a_ib_j - a_jb_i)$ , multiplied by powers of the linear forms  $(a_kx + b_ky)$ .

8. The symbolical method may be applied to the formation of covariants in the following manner. Let  $U$  be represented symbolically by  $(a_1x + b_1y)^m$ , and let  $V$  be another quantic represented by  $(a_2x + b_2y)^n$ . Then the  $r$ -th transvectant of  $U$  and  $V$  is

$$(U, V)^r = \frac{(m-r)!(n-r)!}{m!n!} (a_1b_2 - a_2b_1)^r (a_1x + b_1y)^{m-r} (a_2x + b_2y)^{n-r}.$$

In particular, when  $U = V$ , we obtain the covariants of a single quantic.

**9.** The transvectant  $(U, U)^2$  is the Hessian of  $U$ ; the transvectant  $(U, (U, U)^2)^1$  is the cubicovariant of  $U$  (for the cubic); and so on. The symbolical method gives a simple and elegant way of forming these covariants, and of finding the relations between them.

**10.** I have investigated the complete system of covariants of the binary quintic  $(a, b, c, d, e, f)(x, y)^5$ . The irreducible covariants are:

1. The quantic itself  $U$ ;
2. A covariant of degree 2 and order 6;
3. A covariant of degree 2 and order 4;
4. A covariant of degree 3 and order 3;
5. A covariant of degree 3 and order 5;
6. A covariant of degree 4 and order 2;
7. A covariant of degree 5 and order 3;
8. A covariant of degree 7 and order 1;
9. A covariant of degree 12 and order 0 (an invariant).

These covariants are connected by a number of syzygies, which I have also investigated.

**11.** I consider now the theory of the reduction of covariants. A covariant is said to be reducible when it can be expressed as a rational and integral function of covariants of lower degrees. The reduction of covariants is effected by means of symbolical identities, which express products of symbolical factors in terms of other products.

**12.** The most important of these identities is the following:

$$(a_1b_2 - a_2b_1)(c_1d_2 - c_2d_1) = \begin{vmatrix} a_1c_1, & a_1d_1, & b_1c_1, & b_1d_1 \\ a_2c_2, & a_2d_2, & b_2c_2, & b_2d_2 \end{vmatrix},$$

which may be written in the abbreviated form

$$(12)(34) = (13)(24) - (14)(23).$$

This identity is the foundation of the theory of the reduction of covariants.

**13.** By means of this and similar identities, any product of symbolical factors may be reduced to a sum of products of a standard form. The standard products are those in which the symbolical factors are arranged in a certain order, and the reduction consists in expressing any given product as a sum of standard products.

**14.** I have applied the theory of covariants to the investigation of the conditions for the equivalence of two binary quantics. Two quantics are said to be equivalent when one can be transformed into the other by a linear substitution. The conditions for the equivalence of two quantics are expressed by the equality of their invariants.

**15.** For the quadratic  $(a, b, c)(x, y)^2$ , the condition for the equivalence of two quadratics is the equality of their discriminants  $ac - b^2$ .

For the cubic  $(a, b, c, d)(x, y)^3$ , the condition is the equality of the discriminants  $a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2$ .

For the quartic  $(a, b, c, d, e)(x, y)^4$ , the conditions are the equality of the two invariants  $I$  and  $J$ .

**16.** I have also investigated the theory of the canonical forms of binary quantics. A canonical form of a quantic is a particular representation of the quantic which exhibits its essential properties in the simplest manner. For example, the canonical form of a cubic is

$$ax^3 + by^3 + cz^3 + 6mxyz,$$

and the canonical form of a quartic is

$$ax^4 + by^4 + cz^4 + dw^4 + 6fxy + 6gzw.$$

**17.** Now every rational factor of a binomial  $1 - x^m$  is the irreducible factor of  $1 - x^{m'}$ , where  $m'$  is equal to or a submultiple of  $m$ . Hence in order that the series

$$\frac{1}{(1 - x^a)(1 - x^b)(1 - x^c) \dots}$$

may be expressed as a sum of partial fractions, it is necessary to decompose each binomial  $1 - x^m$  into its irreducible factors. This decomposition is effected by means of the theory of the roots of unity.

**18.** I have also investigated the theory of the generating functions for the number of covariants of a binary quantic. The generating function for the number of covariants of the degree  $p$  and the order  $q$  of a binary quantic of the degree  $m$  is

$$\frac{1}{(1 - t)(1 - tx) \dots (1 - tx^m)} \times \frac{1}{(1 - tx^{-1})(1 - tx^{-2}) \dots (1 - tx^{-m})},$$

and the coefficient of  $t^p x^q$  in the expansion of this function is the number of covariants of the degree  $p$  and the order  $q$ .

**19.** The theory of generating functions may be applied to the determination of the number of independent covariants of a binary quantic. The number of independent covariants of the degree  $p$  and the order  $q$  is equal to the coefficient of  $t^p x^q$  in the expansion of the generating function, minus the number of syzygies of the same degree and order.

**20.** I have also investigated the theory of the syzygies between the covariants of a binary quantic. A syzygy is a relation between covariants which is identically satisfied. The syzygies of a binary quantic may be found by means of the symbolical method; they correspond to the symbolical identities which express products of symbolical factors in terms of other products.

**21.** I represent by  $Y$  the operation

$$Y = x\partial_y,$$

and I represent by  $YA, Y^2A, Y^3A$ , &c. the results obtained by successive operations with  $Y$  upon the function  $A$ . The function  $Y^rA$  will be of the degree and of the order which are determined by the nature of the function  $A$  and the number of operations  $r$ .

**22.** The operation  $Y$  is connected with the theory of covariants by the following property. If  $\Phi$  is a covariant of the quantic  $U$ , then  $Y\Phi$  is also a covariant of  $U$ , provided that the operation  $Y$  does not raise the order of  $\Phi$  above the degree of  $U$ .

**23.** I have also investigated the theory of the seminvariants of a binary quantic. A seminvariant is a function of the coefficients which is invariant under the subgroup of linear substitutions for which  $y$  is replaced by  $y + \theta x$ ,  $x$  remaining unchanged. The seminvariants of a binary quantic are connected with the covariants by the following property: the leading coefficient of any covariant is a seminvariant, and conversely, every seminvariant is the leading coefficient of some covariant.



**24.** The number of independent seminvariants of a binary quantic of the degree  $m$  and the weight  $w$  is equal to the number of partitions of  $w$  into parts not exceeding  $m$ , minus the number of partitions of  $w - 1$  into parts not exceeding  $m$ . This theorem may be proved by means of the theory of generating functions.

**25.** I have also investigated the theory of the differentiants of a binary quantic. A differentiant is a function of the coefficients which is invariant under the operation  $x\partial_y$ . The differentiants are connected with the covariants by the following property: the leading coefficient of any covariant is a differentiant, and conversely, every differentiant is the leading coefficient of some covariant.

**26.** The number of independent differentiants of a binary quantic of the degree  $m$  and the weight  $w$  is equal to the number of aszygetic covariants of the same degree and weight. This number may be determined by means of the generating function

$$\frac{1}{(1-t)(1-t^2)\dots(1-t^m)}.$$

the difference in question is equal to the number of the asyzygetic covariants, i.e. the number of the asyzygetic covariants of the order  $\mu$ , and the degree  $\theta$  in the coefficients.

**27.** I have investigated the complete system of covariants of the binary sextic  $(a, b, c, d, e, f, g)(x, y)^6$ . The irreducible covariants are twenty-six in number, and they are connected by a large number of syzygies. The complete system of invariants consists of five independent invariants, of degrees 2, 4, 4, 6, and 10 respectively.

**28.** For the binary septic  $(a, b, c, d, e, f, g, h)(x, y)^7$ , the complete system of irreducible covariants consists of 147 covariants. The number of independent invariants is 30.

**29.** I have also investigated the theory of the covariants of ternary quantics. The complete system of covariants of the ternary cubic is considerably more complex than that of the binary cubic. The irreducible covariants of the ternary cubic are:

1. The quantic itself  $U$ ;
2. The Hessian  $H$ ;
3. The cubicovariant  $\Phi$ ;
4. The discriminant  $\Delta$ .

These covariants are connected by a number of syzygies.

**30.** I have also considered the theory of the contravariants of a quantic. A contravariant is a function of the coefficients and of a set of variables which are the coefficients of a linear form, and which is invariant under linear substitutions of the original variables. The contravariants of a quantic are connected with the covariants by the theory of reciprocity.

**31.** The theory of covariants and contravariants may be extended to quantics containing any number of sets of facients. The general theory is, however, of extreme complexity, and has been developed only in a few special cases.

i.e.  $\mathcal{A} + 3\mathcal{B} = 0$ ,  $3\mathcal{C} + 2\mathcal{B} = 0$ ; or putting  $\mathcal{A} = 1$ , we have  $\mathcal{B} = -3$ ,  $\mathcal{C} = 2$ , and the leading coefficient is  $-3abc + 2b^3$ . The coefficient of  $x^2y$  is found by operating

**32.** The investigations of this memoir may be applied to the theory of curves and surfaces. The covariants and invariants of a ternary quantic correspond to geometric properties of the curve or surface represented by the quantic. For example, the Hessian of a ternary cubic corresponds to the points of inflexion of the cubic curve.

**33.** I have also investigated the theory of the covariants of a system of quantics. The covariants of a system of quantics are functions of the coefficients and variables of all the quantics, which are invariant under linear substitutions of the variables. The complete system of covariants of a system of quantics consists of a finite number of irreducible covariants, which are connected by a finite number of syzygies.

**34.** I have worked out the example in detail as a specimen of the most convenient method for the actual calculation of more complicated covariants.<sup>7</sup>

**35.** The number of terms of the degree  $\theta$  and of the weight  $q$  is obviously equal to the number of ways in which  $q$  can be made up as a sum of  $\theta$  terms with the elements  $(0, 1, 2, \dots m)$ , a number which is equal to the coefficient of  $x^q z^\theta$  in the development of

$$\frac{1}{(1-z)(1-xz)(1-x^2z)\dots(1-x^mz)},$$

and the number of the aszygetic covariants of any particular degree for the quantic  $(a, b, c, \dots)(x, y)^m$  can therefore be determined by means of this development.

In the case of a cubic, for example, the function to be developed is

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)},$$

which is equal to

$$1 + z(1 + x + x^2 + x^3) + z^2(1 + x + 2x^2 + 2x^3 + 2x^4 + 2x^5 + x^6) + \&c.,$$

where the coefficients are given by the following table; on account of the symmetry, the series of coefficients for each power of  $z$  is continued only to the middle term or middle of the series.

|   |   |   |   |   |   |   |   |   |   |   |     |
|---|---|---|---|---|---|---|---|---|---|---|-----|
|   |   |   |   |   |   |   |   |   |   | 1 | (0) |
|   |   |   |   |   |   |   |   |   | 1 | 1 | (1) |
|   |   |   |   | 1 | 1 | 2 | 2 |   |   |   | (2) |
|   |   | 1 | 1 | 2 | 3 | 3 |   |   |   |   | (3) |
|   | 1 | 1 | 2 | 3 | 4 | 4 |   |   |   |   | (4) |
| 1 | 1 | 2 | 3 | 4 | 5 | 6 | 6 |   |   |   | (5) |
| 1 | 1 | 2 | 3 | 4 | 5 | 7 | 7 | 8 | 8 |   | (6) |

<sup>7</sup>Note added Feb. 7, 1856.—The following method for the calculation of an invariant or of the leading coefficient of a covariant, is easily seen to be identical in principle with that given in the text. Write down all the terms of the weight next inferior to that of the invariant or leading coefficient, and operate on each of these separately with the symbol

$$\text{ind. } b \cdot \frac{b}{a} + 2 \text{ ind. } c \cdot \frac{c}{b} + \dots + (m-1) \text{ ind. } k \cdot \frac{k}{l},$$

where we are first to multiply by the fraction, rejecting negative powers, and then by the index of the proper letter in the term so obtained. Equating the results to zero, we obtain equations between the terms of the invariant or leading coefficient, and replacing in these equations each term by its numerical coefficient in the

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and from this, by subtracting from each coefficient the coefficient which immediately precedes it, we form the table:

|   |   |   |   |   |   |   |   |   |   |  |   |     |
|---|---|---|---|---|---|---|---|---|---|--|---|-----|
|   |   |   |   |   |   |   |   |   |   |  | 1 | (0) |
| 1 |   |   |   |   |   | 0 |   |   |   |  |   | (1) |
| 1 |   |   |   |   | 0 | 1 | 0 |   |   |  |   | (2) |
| 1 |   |   |   | 0 | 1 | 1 | 1 | 0 |   |  |   | (3) |
| 1 |   | 0 | 1 | 1 | 1 | 1 | 0 | 1 |   |  |   | (4) |
| 1 |   | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |  |   |     |
| 1 | 0 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 0 |  |   | (6) |

The successive lines fix the number and character of the covariants of the degrees 0, 1, 2, 3, &c. The line (0), if this were to be interpreted, would show that there is a single covariant of the degree 0; this covariant is of course merely the absolute constant unity, and may be excluded. The line (1) shows that there is a single covariant of the degree 1, viz. a covariant of the order 3; this is the cubic itself, which I represent by  $U$ . The line (2) shows that there are two asyzygetic covariants of the degree 2, viz. one of the order 6, this is merely  $U^2$ , and one of the order 2, this I represent by  $H$ . The line (3) shows that there are three asyzygetic covariants of the degree 3, viz. one of the order 9, this is  $U^3$ ; one of the order 5, this is  $UH$ , and one of the order 3, this I represent by  $\Phi$ . The line (4) shows that there are five asyzygetic covariants of the degree 4, viz. one of the order 12, this is  $U^4$ ; one of the order 8, this is  $U^2H$ ; one of the order 6, this is  $H^2$ ; and one of the order 0, i.e. an invariant, this I represent by  $\nabla$ . The line (5) shows that there are six asyzygetic covariants of the degree 5, viz. one of the order 15, this is  $U^5$ ; one of the order 11, this is  $U^3H$ ; one of the order 9, this is  $U^2\Phi$ ; one of the order 7, this is  $UH^2$ ; one of the order 5, this is  $H\Phi$ ; and one of the order 3, this is  $\nabla U$ . The line (6) shows that there are 8 asyzygetic covariants of the degree 6, viz. one of the order 18, this is  $U^6$ ; one of the

order 14, this is  $U^4H$ ; one of the order 12, this is  $U^2\Phi$ ; one of the order 10, this is  $U^2H^2$ ; one of the order 8, this is  $UH\Phi$ ; two of the order 6 (i.e. the three covariants  $H^2$ ,  $\Phi^2$  and  $\nabla U^2$  are not asyzygetic, but are connected by a single linear equation or syzygy), and one of the order 2, this is  $\nabla H$ . We are thus led to the irreducible covariants  $U$ ,  $H$ ,  $\Phi$ ,  $\nabla$  connected by a linear equation or syzygy between  $H^2$ ,  $\Phi^2$  and  $\nabla U^2$ , and this is in fact the complete system of irreducible covariants;  $\nabla$  is therefore the only invariant.

**36.** The asyzygetic covariants are of the form  $U^pH^q\nabla^r$ , or else of the form  $U^pH^q\nabla^r\Phi$ ; and since  $U$ ,  $H$ ,  $\nabla$  are of the degrees 1, 2, 4 respectively, and  $\Phi$  is of the degree 3, the number of asyzygetic covariants of the degree  $m$  of the first form is equal to the coefficient of  $x^m$  in  $1 \div (1-x)(1-x^2)(1-x^4)$ , and the number of the asyzygetic covariants of the degree  $m$  of the second form is equal to the coefficient of  $x^m$  in  $x^3 \div (1-x)(1-x^2)(1-x^4)$ . Hence the total number of asyzygetic covariants is equal to the coefficient of  $x^m$  in  $(1+x^3) \div (1-x)(1-x^2)(1-x^4)$ , or what is the same thing, in

$$\frac{1-x^6}{(1-x)(1-x^2)^2(1-x^4)};$$

and conversely, if this expression for the number of the asyzygetic covariants of the degree  $m$  were established independently, it would follow that the irreducible invariants were four in number, and of the degrees 1, 2, 3, 4 respectively, but connected by an equation of the degree 6. As regards the invariants, every invariant is of the form  $\nabla^p$ , i.e. the number of asyzygetic invariants of the degree  $m$  is equal to the coefficient of  $x^m$  in  $\frac{1}{1-x^4}$ , and conversely, from this expression it would follow that there was a single irreducible invariant of the degree 4.

**37.** In the case of a quartic, the function to be developed is:

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)(1-x^4z)},$$

and the coefficients are given by the table:

$$\begin{array}{rcccccccccccccccc}
& & & & & & & & & & & & & & & & 1 & (0) \\
1 & & & & & & & & & 1 & 1 & & & & & & & (1) \\
1 & & & & & & 1 & 2 & 2 & 3 & & & & & & & & (2) \\
1 & & & & 1 & 2 & 3 & 4 & 5 & 6 & & & & & & & & (3) \\
1 & & & 1 & 2 & 3 & 5 & 6 & 8 & 9 & & & & & & & & \\
1 & & 1 & 2 & 3 & 5 & 7 & 9 & 11 & 14 & & & & & & & & (5) \\
1 & 1 & 2 & 3 & 5 & 6 & 9 & 11 & 14 & 16 & 19 & 20 & & & & & & (6)
\end{array}$$







and subtracting from each coefficient the one which immediately precedes it, we have the table:

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|
|   |   |   |   |   |   |   |   |   |   |   |   |   | 1 | (0) |
| 1 |   |   |   |   |   | 0 | 0 |   |   |   |   |   |   | (1) |
| 1 |   |   |   |   | 0 | 1 | 0 | 1 | 0 |   |   |   |   | (2) |
| 1 |   |   |   | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |   |   | (3) |
| 1 | 0 | 1 | 1 | 2 | 1 | 2 | 1 | 2 | 0 | 1 |   |   |   | (4) |
| 1 | 0 | 1 | 1 | 2 | 2 | 2 | 2 | 3 | 2 | 2 | 1 | 1 |   | (5) |

We thus obtain the following irreducible covariants, viz.:

Of the degree 1; a single covariant of the order 5, this is the quintic itself.

Of the degree 2; two covariants, viz. one of the order 6, and one of the order 2.

Of the degree 3; three covariants, viz. one of the order 9, one of the order 5, and one of the order 3.

Of the degree 4; three covariants, viz. one of the order 6, one of the order 4, and one of the order 0 (an invariant).

Of the degree 5; three covariants, viz. one of the order 7, one of the order 3, and one of the order 1 (a linear covariant).

These covariants are connected by a single syzygy of the degree 5 and of the order 11; in fact, the table shows that there are only two asyzygetic covariants of this degree and order; but we may, with the above-mentioned irreducible covariants of the degrees, 1, 2, 3 and 4, form three covariants of the degree 5 and the order 11; there is therefore a syzygy of this degree and order.

**40.** It is proper to remark, that the property that the roots of the equation  $J = 0$  are all real, does not in an obvious manner lead to any conclusion as to the reality of the roots of the resolvent equation. In fact, suppose that the roots of the equation are  $\alpha, \beta, \gamma, \delta$ , and that  $I$  and  $J$  are the two invariants of the biquadratic function  $(\alpha, \beta, \gamma, \delta \mid x, y)^4$ , so that  $I = 0$  is the condition in order that there may be two equal roots; then if  $H = (\alpha, \beta, \gamma, \delta \mid x, y)^4$ , the equation  $H = 0$  may be considered as a biquadratic equation the roots of which are the anharmonic ratios of the four roots  $\alpha, \beta, \gamma, \delta$ ; and the invariants of this biquadratic function are in fact  $I^2$  and  $I^3 - 27J^2$ ; whence the equation  $J = 0$  gives  $I^3 - 27J^2 = I^3$ , so that the condition  $J = 0$  expresses that the biquadratic equation  $H = 0$  has equal roots. And since the roots of  $H = 0$  are real, it follows that the equation in the anharmonic ratios has equal roots; but this does not seem to lead to any conclusion as to the reality of the roots of the resolvent equation.

**41.** A biquadratic equation  $U = (a, b, c, d, e \mid x, y)^4 = 0$  has been said to be in the canonical form when the second and fourth coefficients vanish, i.e. when  $b = 0, d = 0$ , and it has been shown that this can in general be done. The reduction depends on the solution of a cubic equation, and to effect it we have only to write  $x + \lambda y$  for  $x$ ; then if  $a'$  is properly determined, the second coefficient will vanish identically; and if  $\lambda$  is properly determined, the fourth coefficient will vanish; moreover the process shows that the equation is expressible in the canonical form in three different ways, corresponding to the three roots of the cubic equation. And the cubic equation for the determination of  $a'$  is the resolvent above referred to; but I have not succeeded in making the proof depend upon the form of the equation of the anharmonic ratios.

**42.** The form of the biquadratic equation may be exhibited in a very simple manner by means of the invariant  $J$ ; in fact, we have  $I = 0$  for the biquadratic function  $(\alpha, \beta, \gamma, \delta \mid x, y)^4$ , and this gives the before-mentioned condition in order that the four roots may be equal two and two. But for the discriminant, we have  $I^3 - 27J^2 = 0$ ; and if  $I = 0, J = 0$ , the four roots are all equal; hence  $J = 0$ , if  $I$  is not  $= 0$ , is the condition in order that the roots may be equal in pairs.

**43.** The preceding investigations show the manner in which the general theory of the covariants of a binary quantic gives rise to the special results which have been established in relation to binary quantics of particular orders. I will consider, in the first place, the general question of the linear transformation of a binary quantic. This part of the subject is of course identical with the theory of the invariants of a binary quantic; and if the quantic is represented by

$$U = (a, b, c, \dots \mathbin{\lrcorner} x, y)^n,$$

and if the new variables  $x', y'$  are given by

$$x = \alpha x' + \beta y', \quad y = \gamma x' + \delta y',$$

then if  $U$  is transformed into

$$U' = (a', b', c', \dots \mathbin{\lrcorner} x', y')^n,$$

we have between the coefficients  $a, b, c, \dots$  of the original function and the coefficients  $a', b', c', \dots$  of the transformed function, a system of  $n + 1$  equations, which may be written

$$\begin{aligned} a' &= (\alpha \mathbin{\lrcorner} a, b, c, \dots)^n, \\ b' &= (\alpha \mathbin{\lrcorner} a, b, c, \dots)^{n-1} (\gamma \mathbin{\lrcorner} a, b, c, \dots), \\ c' &= (\alpha \mathbin{\lrcorner} a, b, c, \dots)^{n-2} (\gamma \mathbin{\lrcorner} a, b, c, \dots)^2, \\ &\vdots \end{aligned}$$

and from these equations we may eliminate the quantities  $\alpha, \beta, \gamma, \delta$ , and thus obtain the invariants of the function; but the process is in general a very complicated one.

**44.** The question may be simplified as follows: writing  $x', y'$  in place of  $x, y$ , and considering  $a', b', c', \dots$  as given functions of  $\alpha, \beta, \gamma, \delta$ , we have

$$(a, b, c, \dots \mathbin{\lrcorner} \alpha x' + \beta y', \gamma x' + \delta y')^n = (a', b', c', \dots \mathbin{\lrcorner} x', y')^n,$$

and this being so, if we operate with

$$\Omega' = \frac{d}{dx'} \frac{d}{dy} - \frac{d}{dy'} \frac{d}{dx},$$

the operation on the right-hand side gives zero; and attending to the formula

$$\frac{d}{dx} = \alpha \frac{d}{dx'} + \gamma \frac{d}{dy'}, \quad \frac{d}{dy} = \beta \frac{d}{dx'} + \delta \frac{d}{dy'},$$

the operation on the left-hand side becomes

$$(\alpha\delta - \beta\gamma) \left( \frac{d}{dx'} \frac{d}{dy'} - \frac{d}{dy'} \frac{d}{dx'} \right) \cdot (a, b, c, \dots \mathbin{\lrcorner} x', y')^n,$$

which vanishes identically; hence the equation is satisfied identically, and we thus see that  $a', b', c', \dots$  are such that the equation in  $(x', y')$  is satisfied identically.

45. I use the term “invariant” to denote a function of the coefficients which possesses the property of invariance under linear transformation; but it is to be observed that every such function is not necessarily an invariant in the strict sense of the word. A function of the coefficients which is equal to the same function of the transformed coefficients multiplied by a power of the modulus (or determinant of transformation), is an invariant; but there may be functions which possess a different property, being equal to the same function of the transformed coefficients multiplied by a power of the modulus and also by a factor depending on the coefficients of the substitution. Such functions might be termed “seminvariants”; but I shall not at present enter upon the theory of these functions.

46. I remark that if an invariant of a binary quantic is expressed as a function of the differences of the roots, then the invariant remains unaltered when any linear substitution is performed upon the variables. Suppose that the roots of the equation  $U = 0$  are  $\alpha_1, \alpha_2, \dots, \alpha_n$ , then if we write  $x = \alpha_1 y, \dots$ , we may express any coefficient as a symmetric function of the differences of the roots; and any invariant expressed as a function of these differences will remain unaltered by any linear transformation of the variables. This is in fact the distinctive property of invariants.

47. I will now return to the general theory of covariants. Considering the quantic  $U$ , and forming with it the system of covariants, we have a number of covariantive equations, each of which equated to zero gives rise to a corresponding covariantive curve; and the system of these curves constitutes what may be termed the “covariantive pencil” of the quantic. The theory of the covariantive pencil presents many interesting questions, which I do not however propose to discuss in the present Memoir.

48. I will conclude with a few remarks on the expression of the covariants by means of the notation of bipartite forms. This notation is as follows: if  $\alpha, \beta$  are symbols of substitution, and  $(\alpha \bowtie x, y)^n, (\beta \bowtie x, y)^n$  are the same quantic, then the operator

$$(\alpha\beta' - \alpha'\beta)^p (\alpha \bowtie x, y)^{n-p} (\beta \bowtie x, y)^{n-p}$$

is a covariant of the quantic. In particular, if  $p = 1$ , we have the canonizant; if  $p = 2$ , we have a covariant of the order  $2n - 4$ ; and so on. The theory of these forms presents many interesting particular cases, which I reserve for future investigation.

**49.** The theory of the binary quantic leads naturally to the theory of ternary quantics and quantics in general; but the investigation of the covariants and invariants of ternary quantics presents difficulties of a much more formidable character than those which occur in the case of binary quantics. The leading coefficients of a ternary quantic are not, as in the case of a binary quantic, connected by a single differential equation, but by a system of partial differential equations; and the theory of the integration of these equations is very imperfectly understood. I hope however to be able to resume the consideration of this subject in a future Memoir.

**50.** I collect here for convenience of reference the principal formulæ relating to the binary quantic:

$$\begin{array}{ll}
 (a, b, c, \dots \nabla x, y)^n & \text{(the quantic itself),} \\
 (ab)^2 & \text{(= Hessian),} \\
 (ab)^2(ac)(bd)(cd)^2 & \text{(= canonizant of deg. 3),} \\
 (ab)^2(ac)^2(bd)(cd)^3 & \text{(= canonizant of deg. 4),} \\
 \&c. & \&c.
 \end{array}$$

where the symbols  $(ab)$ ,  $(ac)$ , ... denote the determinants  $\alpha\beta' - \alpha'\beta$ ,  $\alpha\gamma' - \alpha'\gamma$ , ... taken between the corresponding pairs of symbols.

The expressions for the invariants and covariants of the binary quantic may be thus exhibited in a compendious form, which renders apparent the symmetry of the several expressions, and shows the connexion of each covariant with the theory of the linear transformation of the quantic.



**51.** The foregoing investigation establishes the following general theorem: *For a binary quantic of the order  $n$ , the number of aszygetic covariants of the degree  $m$  is equal to the coefficient of  $x^m$  in the development of*

$$\frac{1}{(1-x)(1-x^2)\dots(1-x^n)}$$

*when  $m$  is odd; and to the coefficient of  $x^m$  in the development of*

$$\frac{1}{(1-x^2)(1-x^3)\dots(1-x^n)}$$

*when  $m$  is even.* And hence the number of invariants of the degree  $m$  is equal to the coefficient of  $x^m$  in the development of  $1/(1-x^2)(1-x^3)\dots(1-x^n)$ , or is equal to the number of partitions of  $m$  into the parts  $2, 3, \dots, n$ .

**52.** I have spoken of a covariant as being a function of the coefficients and variables which possesses the property of covariance; but it is proper to remark that every function possessing this property is not thereby a covariant in the strict sense of the term. A function of the coefficients and variables, which is equal to the same function of the transformed coefficients and variables multiplied by a power of the modulus, is a covariant; but there may be functions which possess a more general property, being equal to the same function of the transformed coefficients and variables multiplied by a power of the modulus and also by a factor depending on the coefficients of substitution. Such functions might be termed “semicovariants”; they correspond to the seminvariants above mentioned.

**53.** The general expression for the number of the aszygetic covariants of the degree  $m$  of a binary quantic of the order  $n$  may be written under the form

$$\frac{1}{2\pi i} \int \frac{dz}{z^{m+1}(1-z)(1-xz)\dots(1-x^nz)}$$

evaluated for  $x = 1$ , when  $m$  is odd; and under the form

$$\frac{1}{2\pi i} \int \frac{dz}{z^{m+1}(1-z)(1-x^2z)\dots(1-x^nz)}$$

evaluated for  $x = 1$ , when  $m$  is even; or if we please, the two cases may be united in the single formula

$$\frac{1}{2\pi i} \int \frac{dz}{z^{m+1}(1-xz)(1-x^2z)\dots(1-x^nz)}$$

evaluated for  $x = 1$ .

**54.** The present Memoir contains the development and establishment of the general theory of the covariants of a binary quantic, so far as relates to the enumeration and determination of the covariants. I have shown that the number of aszygetic covariants of a given degree can be expressed by means of a definite integral, and that the actual determination of the covariants can be effected by a process which depends upon the solution of a system of partial differential equations. I have also established the connexion between the theory of covariants and the theory of the partition of numbers, and have shown how the general formulæ may be applied to the particular cases of the cubic, quartic, and quintic.

**55.** It remains to consider the theory of the covariants of quantics of higher orders. The sextic, septic, octavic, and higher quantics present each a separate problem, the complete solution of which would require a detailed investigation of the same nature as that which has been given for the cubic, quartic, and quintic. The difficulties of such investigations increase rapidly with the order of the quantic; and it is to be observed that the methods which have been employed in the present Memoir are for the most part applicable only to the case of binary quantics. The theory of ternary quantics, and of quantics in general, requires the development of new methods, and the introduction of new principles, which I reserve for future investigation.

**56.** I will conclude with a remark on the general theory. The covariants of a quantic may be divided into two classes, which may be termed "absolute covariants" and "relative covariants." An absolute covariant is a covariant which is equal to the same function of the transformed coefficients and variables; a relative covariant is one which is equal to the same function of the transformed coefficients and variables multiplied by a power of the modulus. In the case of binary quantics, every covariant is a relative covariant; but in the case of ternary and higher quantics, there exist covariants which are absolute covariants. The theory of these absolute covariants presents many interesting questions, which I hope to discuss in a future communication.

**57.** The theory of the invariants of a binary quantic is intimately connected with the theory of the partition of numbers. In fact, the number of aszygetic invariants of the degree  $m$  of a binary quantic of the order  $n$  is equal to the number of partitions of  $m$  into the parts  $2, 3, \dots, n$ ; and the generating function for the aszygetic invariants is  $1/(1-x^2)(1-x^3)\dots(1-x^n)$ . This connexion may be exhibited in a more general form; for if we consider the generating function

$$\frac{1}{(1-x)(1-xz)(1-xz^2)\dots(1-xz^n)},$$

then the coefficient of  $x^m z^p$  in the development of this function is equal to the number of aszygetic covariants of the degree  $m$  and the order  $p$ .

**58.** The results obtained in the present Memoir may be summed up as follows. For a binary quantic of the order  $n$ :

- (1) The number of aszygetic covariants of the degree  $m$  is equal to the coefficient of  $x^m$  in the development of

$$\frac{1}{(1-x)(1-x^2)\dots(1-x^n)}$$

when  $m$  is odd; and to the coefficient of  $x^m$  in the development of

$$\frac{1}{(1-x^2)(1-x^3)\dots(1-x^n)}$$

when  $m$  is even.

- (2) The number of invariants of the degree  $m$  is equal to the number of partitions of  $m$  into the parts  $2, 3, \dots, n$ .
- (3) The generating function for the covariants is

$$\frac{1}{(1-x)(1-xz)(1-xz^2)\dots(1-xz^n)}.$$

- (4) The actual determination of the covariants can be effected by the solution of the differential equations which express the condition of covariance.

**59.** The foregoing results have been applied in detail to the cases of the cubic, quartic, and quintic; and the complete systems of

irreducible covariants and invariants have been obtained for each of these cases. For the cubic, the irreducible covariants are  $U, H, \Phi, \nabla$ , connected by the syzygy of the degree 6; the only irreducible invariant is  $\nabla$ . For the quartic, the irreducible covariants are  $U, I, H, J, \Phi$ , connected by the syzygy of the degree 6; the only irreducible invariants are  $I, J$ . For the quintic, there are in all twenty-three irreducible covariants, which have been enumerated in the text; and there are four irreducible invariants, of the degrees 4, 8, 12, 18.

**60.** I have also considered the general theory of the semicovariants and seminvariants, and have shown that there exist functions which possess a more general property of covariance than that which belongs to covariants in the strict sense. These functions are equal to the same function of the transformed coefficients and variables multiplied by a power of the modulus and also by a factor depending on the coefficients of the substitution. The theory of these functions presents many interesting questions, which I do not however propose to discuss further in the present Memoir.

**61.** The annexed Tables Nos. 23 to 39 contain the developed expressions of the covariants of the binary quintic, with the literal coefficients (these Tables are in fact those which are given in my "Second Memoir on Quantics," Phil. Trans. vol. CXL VI for the year 1856). I give also a Table of the leading coefficients of the several covariants, which serves to fix their numerical values.

# COVARIANTS OF THE BINARY QUINTIC.

## K. No. 23.

|                 |                |                |                 |
|-----------------|----------------|----------------|-----------------|
| $a^3bf^2 \dots$ | $a^2cf^2 + 1$  | $a^2df^2 - 1$  | $a^2ef^2 \dots$ |
| $a^2cef + 1$    | $a^2def - 5$   | $a^2e^2f + 1$  | $abdf^2 - 1$    |
| $a^2d^2f - 3$   | $a^2e^2 + 4$   | $abc f^2 + 5$  | $abe^2f + 1$    |
| $a^2de^2 + 2$   | $ab^2f^2 - 1$  | $abdef - 8$    | $ac^2f^2 + 3$   |
| $ab^2ef - 1$    | $abcef + 8$    | $abe^3 + 3$    | $acdef - 14$    |
| $abcdf + 14$    | $abd^2f + 11$  | $ac^2ef - 11$  | $ace^3 + 8$     |
| $abce^2 - 11$   | $abde^2 - 17$  | $acd^2f + 11$  | $ad^3f + 9$     |
| $abd^2e - 1$    | $ac^2df - 11$  | $acde^2 + 6$   | $ad^2e^2 - 6$   |
| $ac^3f - 9$     | $ac^2e^2 - 16$ | $ad^3e - 6$    | $b^2cf^2 - 2$   |
| $ac^2de + 14$   | $acd^3 + 44$   | $b^2f^2 - 4$   | $b^2def + 11$   |
| $acd^3 - 6$     | $ad^4 - 18$    | $b^2cef + 17$  | $b^2e^3 - 9$    |
| $b^3df - 8$     | $b^2cf - 3$    | $b^2d^2f + 16$ | $bc^2ef + 1$    |
| $b^2e^2 + 9$    | $b^2cdf - 6$   | $b^2de^2 - 21$ | $bcd^2f - 14$   |
| $b^2cf + 6$     | $b^2ce^2 + 21$ | $bc^2df - 44$  | $bcd e^2 + 16$  |
| $b^2cde - 16$   | $b^2d^2e - 5$  | $bc^2e^2 + 5$  | $bd^3e - 3$     |
| $b^2d^3 + 8$    | $bc^2f + 6$    | $bcd^2e + 39$  | $c^3df + 6$     |
| $bc^3e + 3$     | $bc^2de - 39$  | $bd^4 - 12$    | $c^3e^2 - 8$    |
| $bc^2d^2 - 2$   | $bcd^3 + 22$   | $c^4f + 18$    | $c^2d^2e + 2$   |
| $c^4d \dots$    | $c^4e + 12$    | $c^3de - 22$   | $cd^4 \dots$    |
| $\pm 57$        | $\pm 129$      | $\pm 129$      | $\pm 57$        |

$$(a, b, c, d, e, f \upharpoonright x, y)^5$$

## L. No. 24.

|                 |                |                 |                |                |                 |                |                 |
|-----------------|----------------|-----------------|----------------|----------------|-----------------|----------------|-----------------|
| $a^2df \dots$   | $a^2f^3 \dots$ | $a^2bf^2 \dots$ | $a^2cf^2 - 1$  | $a^2df^2 + 1$  | $a^2ef^2 \dots$ | $a^2f^3 \dots$ | $abf^3 \dots$   |
| $a^2bdf \dots$  | $a^2bef \dots$ | $a^2cef - 3$    | $a^2def + 7$   | $a^2e^2f - 1$  | $abdf^2 + 3$    | $abef^2 \dots$ | $acef^2 \dots$  |
| $a^2b^2e \dots$ | $a^2cdf + 7$   | $a^2d^2f + 12$  | $a^2e^3 - 6$   | $abc^2f^2 - 7$ | $abe^2f - 3$    | $acd^2f^2 - 7$ | $ad^2f^2 - 2$   |
| $a^2c^2f + 2$   | $a^2ce^2 - 10$ | $a^2de^2 - 9$   | $ab^2f^2 + 1$  | $abdef + 26$   | $ac^2f^2 - 12$  | $ace^2f + 7$   | $ade^2f + 4$    |
| $a^2cde - 5$    | $a^2d^2e + 3$  | $ab^2ef + 3$    | $abcef - 26$   | $abc^3 - 19$   | $acdef + 18$    | $ad^2ef + 7$   | $ae^4 - 2$      |
| $a^2d^3 + 3$    | $ab^2df - 7$   | $abcdf - 18$    | $abde^2 - 8$   | $ac^2df + 32$  | $ac^2e^2 + 6$   | $ad^3f - 7$    | $b^2ef^2 \dots$ |
| $ab^3f - 4$     | $ab^2e^2 + 10$ | $abce^2 - 18$   | $abd^2f + 30$  | $ac^2ef - 53$  | $ad^2e^2 - 15$  | $b^2df^2 + 10$ | $bcd^2 + 5$     |
| $ab^2de + 5$    | $abc^2f - 7$   | $abd^2e + 30$   | $ac^2df - 18$  | $acde^2 + 53$  | $ad^3e - 39$    | $b^2e^3f - 10$ | $bce^2f - 5$    |
| $abc^2e^2 + 5$  | $abc^2de - 8$  | $ac^2f - 3$     | $ac^2e^2 + 52$ | $ad^3f - 39$   | $b^2f^2 + 9$    | $b^2def + 18$  | $bd^2 + 5$      |
| $abc^2d - 7$    | $abd^3 + 9$    | $ac^2de + 45$   | $acd^2e + 6$   | $ad^4 - 6$     | $b^2cef + 8$    | $b^2e^3 - 27$  | $bc^2f - 2$     |
| $acd^2 + 1$     | $ac^2e + 22$   | $ac^2d^2 - 39$  | $b^2f^2 + 6$   | $b^2def + 18$  | $b^2e^3 - 27$   | $bce^3 - 2$    | $c^2f^2 - 3$    |
| $b^3f + 2$      | $ac^2d^2 - 19$ | $b^2df - 6$     | $b^2ef + 19$   | $b^2d^2f - 6$  | $bc^2ef - 30$   | $bd^2f - 22$   | $c^2def + 7$    |
| $b^2ce - 5$     | $b^2cf + 7$    | $b^2e^2 + 27$   | $b^2cdf - 53$  | $b^2de^2 - 20$ | $bcd^2f - 45$   | $bd^2e^2 + 19$ | $c^2e^3 + 2$    |
| $b^2d^2 - 2$    | $b^2de + 2$    | $b^2cf + 15$    | $b^2ce^2 + 20$ | $bc^2df + 45$  | $bcd^2e^2 + 87$ | $c^2ef - 9$    | $cd^2f - 1$     |
| $b^2cd + 8$     | $b^2ce - 19$   | $b^2cde - 87$   | $b^2d^2e - 25$ | $bc^2e^2 + 25$ | $bd^3e - 12$    | $c^2d^2f + 19$ | $cd^2e^2 - 8$   |
| $bc^3 - 3$      | $b^2cd^2 - 11$ | $b^2d^3 + 6$    | $bc^2f + 39$   | $b^2d^2e - 52$ | $c^2df + 39$    | $c^2de^2 + 11$ | $d^3e + 3$      |
|                 | $bc^3d + 33$   | $bc^2e + 12$    | $bcd^3 - 45$   | $bd^4 \dots$   | $c^2e^2 - 6$    | $cd^3e - 33$   |                 |
|                 | $c^4 - 12$     | $c^2d^2 + 57$   | $bcd^2 + 65$   | $c^4f + 39$    | $c^2d^2e - 57$  | $d^3 + 12$     |                 |
|                 |                | $c^4d - 24$     | $c^2de \dots$  | $c^2d^3e - 65$ | $cd^4 + 24$     |                |                 |
| $\pm 26$        | $\pm 93$       | $\pm 207$       | $\pm 241$      | $\pm 241$      | $\pm 207$       | $\pm 93$       | $\pm 26$        |

$$(a, b, c, d, e, f \upharpoonright x, y)^7$$

$$\text{No. 25, } Q = +1 a^2cdf^3 + \&c.$$

$$\text{No. 26, } Q' = +1 a^4f^4 + \&c.$$

} [see *post*, 143.]

## G. No. 27.

|                   |                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|-------------------|
| $a^3 f^2 \dots$   | $a^2 b f^2 \dots$ | $a^2 c f^2 \dots$ | $a^2 d f^2 \dots$ | $a^2 e f^2 \dots$ |
| $a^3 c e f \dots$ | $a^3 d^2 f \dots$ | $a^2 c e f \dots$ | $a^2 d e f \dots$ | $a^2 e^2 f \dots$ |
| $a^2 c^2 f \dots$ | $a^2 c d e \dots$ | $a^2 d^2 e \dots$ | $a b^2 f^2 \dots$ | $a b c f^2 \dots$ |
| $a^2 c d^2 \dots$ | $a b^2 e f \dots$ | $a b c d f \dots$ | $a b c e f \dots$ | $a b d e^2 \dots$ |
| $a b^3 f \dots$   | $a b^2 c e \dots$ | $a b^2 d^2 \dots$ | $a b c^2 e \dots$ | $a b c d^2 \dots$ |
| $a b c^3 \dots$   | $b^4 f \dots$     | $b^3 c e \dots$   | $b^2 c^2 d \dots$ | $b^2 c d^2 \dots$ |

$$(a, b, c, d, e \upharpoonright x, y)^5$$

## H. No. 28.

|                   |                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|-------------------|
| $a^4 f \dots$     | $a^3 b f^2 \dots$ | $a^3 c f^2 \dots$ | $a^2 d f^2 \dots$ | $a^2 e f^2 \dots$ |
| $a^3 c e f \dots$ | $a^2 d^2 f \dots$ | $a^2 c e^2 \dots$ | $a^2 d e f \dots$ | $a^2 e^3 \dots$   |
| $a^2 c^2 f \dots$ | $a^2 c d^2 \dots$ | $a b^2 f^2 \dots$ | $a b c e f \dots$ | $a b e^3 \dots$   |
| $a^2 d^3 \dots$   | $a b c^2 f \dots$ | $a b c d^2 \dots$ | $a c^3 e \dots$   | $a c^2 d^2 \dots$ |
| $a b^3 f \dots$   | $a b^2 c d \dots$ | $a c^3 d \dots$   | $b^4 f \dots$     | $b^3 c d \dots$   |

$$(a, b, c, d, e \upharpoonright x, y)^5$$

## I. No. 29.

|                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|
| $a^2 f^2 \dots$   | $a b^2 f^2 \dots$ | $a b c f^2 \dots$ | $a b d f^2 \dots$ |
| $a^2 c e f \dots$ | $a^2 d e f \dots$ | $a b^2 e f \dots$ | $a b c e f \dots$ |
| $a^2 d^2 f \dots$ | $a^2 d e^2 \dots$ | $a b^2 d f \dots$ | $a b c d f \dots$ |
| $a^2 c^2 f \dots$ | $a b^3 f \dots$   | $a b^2 c e \dots$ | $a b c^2 e \dots$ |
| $a b^2 d^2 \dots$ | $a b c^2 d \dots$ | $b^4 f \dots$     | $b^2 c^2 d \dots$ |

$$(a, b, c, d, e, f \upharpoonright x, y)^6$$

## J. No. 30.

|                 |                   |                   |                 |
|-----------------|-------------------|-------------------|-----------------|
| $a^2 f^2 \dots$ | $ab^2 f^2 \dots$  | $abc f^2 \dots$   | $abdf^2 \dots$  |
| $a^2 cef \dots$ | $a^2 d^2 f \dots$ | $a^2 e^2 f \dots$ | $ab^2 ef \dots$ |
| $a^2 def \dots$ | $ab^2 df \dots$   | $abcdf \dots$     | $abcef \dots$   |
| $abe^3 \dots$   | $ac^2 f^2 \dots$  | $acdef \dots$     | $ad^3 f \dots$  |
| $b^2 df \dots$  | $b^2 e^2 \dots$   | $bc^2 f \dots$    | $bcd^2 \dots$   |

$$(a, b, c, d, e, f \upharpoonright x, y)^6$$

## K. No. 31.

|                   |                  |                  |                  |
|-------------------|------------------|------------------|------------------|
| $a^2 f^2 \dots$   | $ab^2 f^2 \dots$ | $abc f^2 \dots$  | $abdf^2 \dots$   |
| $a^2 cef \dots$   | $a^2 def \dots$  | $ab^2 ef \dots$  | $abcef \dots$    |
| $a^2 d^2 f \dots$ | $ab^2 df \dots$  | $abcdf \dots$    | $ac^2 f^2 \dots$ |
| $acdef \dots$     | $ad^3 f \dots$   | $b^2 cf^2 \dots$ | $b^2 def \dots$  |
| $bc^2 ef \dots$   | $bcd^2 f \dots$  | $c^3 df \dots$   | $c^3 e^2 \dots$  |

$$(a, b, c, d, e, f \upharpoonright x, y)^6$$

## L. No. 32.

|                 |                   |                   |                 |                |
|-----------------|-------------------|-------------------|-----------------|----------------|
| $a^2 f^2 \dots$ | $ab^2 f^2 \dots$  | $abc f^2 \dots$   | $abdf^2 \dots$  | $abef^2 \dots$ |
| $a^2 cef \dots$ | $a^2 d^2 f \dots$ | $a^2 e^2 f \dots$ | $ab^2 ef \dots$ |                |
| $a^2 def \dots$ | $ab^2 df \dots$   | $abcdf \dots$     | $abcef \dots$   |                |
| $abe^3 \dots$   | $ac^2 f^2 \dots$  | $acdef \dots$     | $ad^3 f \dots$  |                |
| $b^2 df \dots$  | $b^2 e^2 \dots$   | $bc^2 f \dots$    | $bcd^2 \dots$   |                |

$$(a, b, c, d, e, f \upharpoonright x, y)^6$$



## M. No. 33.

|                 |                 |                 |                 |
|-----------------|-----------------|-----------------|-----------------|
| $a^3bf \dots$   | $a^3cf \dots$   | $a^3df \dots$   | $a^3ef \dots$   |
| $a^3ce \dots$   | $a^3d^2 \dots$  | $a^2b^2f \dots$ | $a^2bcf \dots$  |
| $a^2bdf \dots$  | $a^2be^2 \dots$ | $a^2c^2f \dots$ | $a^2cde \dots$  |
| $a^2d^3 \dots$  | $ab^3f \dots$   | $ab^2ce \dots$  | $ab^2d^2 \dots$ |
| $abc^2d \dots$  | $ab^2c^2 \dots$ | $b^4f \dots$    | $b^3ce \dots$   |
| $b^2c^2d \dots$ | $b^2cd^2 \dots$ | $bc^4 \dots$    |                 |

$$(a, b, c, d, e \uparrow x, y)^4$$

## N. No. 34.

|                 |                 |                 |                 |
|-----------------|-----------------|-----------------|-----------------|
| $a^3bf \dots$   | $a^3cf \dots$   | $a^3df \dots$   | $a^3ef \dots$   |
| $a^2b^2f \dots$ | $a^2bcf \dots$  | $a^2bdf \dots$  | $a^2be^2 \dots$ |
| $a^2c^2f \dots$ | $a^2cde \dots$  | $a^2d^3 \dots$  | $ab^3f \dots$   |
| $ab^2ce \dots$  | $ab^2d^2 \dots$ | $abc^2d \dots$  | $ab^2c^2 \dots$ |
| $b^4f \dots$    | $b^3ce \dots$   | $b^2c^2d \dots$ | $b^2cd^2 \dots$ |

$$(a, b, c, d, e \uparrow x, y)^4$$

## O. No. 35.

|               |                |               |               |
|---------------|----------------|---------------|---------------|
| $a^2bf \dots$ | $a^2cf \dots$  | $a^2df \dots$ | $a^2ef \dots$ |
| $ab^2f \dots$ | $abcf \dots$   | $abdf \dots$  | $abe^2 \dots$ |
| $ac^2f \dots$ | $acde \dots$   | $ad^3 \dots$  | $b^3f \dots$  |
| $b^2ce \dots$ | $b^2d^2 \dots$ | $bc^2d \dots$ | $c^4 \dots$   |

$$(a, b, c, d, e, f \uparrow x, y)^5$$

## P. No. 36.

|               |               |                |
|---------------|---------------|----------------|
| $a^2bf \dots$ | $a^2cf \dots$ | $a^2df \dots$  |
| $ab^2f \dots$ | $abcf \dots$  | $abdf \dots$   |
| $ac^2f \dots$ | $acde \dots$  | $ad^3 \dots$   |
| $b^3f \dots$  | $b^2ce \dots$ | $b^2d^2 \dots$ |
| $bc^2d \dots$ | $c^4 \dots$   |                |

$$(a, b, c, d, e, f \uparrow x, y)^5$$

$$\text{No. 37, } A = +1 a^3cde + \&c.$$

$$\text{No. 38, } B = +1 a^2b^2de + \&c.$$

$$\text{No. 39, } C = +1 a^2bc^2e + \&c.$$

NUMERICAL TABLES SUPPLEMENTARY TO SECOND MEMOIR ON  
QUANTICS.

[Now first published (1889).]

In the present paper I arrange in a more compendious form and continue to a much greater extent the tables (first of each pair) given Nos. 35—39 of my Second Memoir on Quantics, 141, pp. 260—264, which relate to the cubic, the quartic and the quintic functions; and I give the like tables for the sextic, the septimic and the octavic functions respectively. The cubic table exhibits the coefficients of the several  $xz$  terms of the function  $1 \div (1 - z.1 - xz.1 - x^2z.1 - x^3z)$ , or, what is the same thing, it gives the number of partitions of a given number into a given number of parts, the parts being 0, 1, 2, 3, (repetitions admissible): or again, regarding the letters  $a, b, c, d$ , as having the weights 0, 1, 2, 3 respectively, it shows the number of literal terms of a given degree and given weight. And similarly for the quartic, quintic, sextic, septimic and octavic tables respectively, the parts of course being 0, 1, ... up to 4, 5, 6, 7 or 8, and the letters being  $a, b, \dots$  up to  $e, f, g, h$  or  $i$ . The extent of the tables is as follows:

|                                   |       |
|-----------------------------------|-------|
| cubic table extends to deg-weight | 18—27 |
| quartic                           | 18—36 |
| quintic                           | 18—45 |
| sextic                            | 15—45 |
| septimic                          | 12—42 |
| octavic                           | 10—40 |

viz. for the quintic, the sextic and the octavic functions these are the deg-weights of the highest invariants respectively. I designate the Tables as the *ad*-, *ae*-, *af*-, *ag*-, *ah*- and *ai*-tables respectively.

It is to be noticed that in the several tables the lower part of each column is for shortness omitted; the column has to be completed by taking into it the series of bottom terms of each of the preceding columns: thus in the *af*- or quintic table the complete column for degree 3 would be



*ae*-TABLE.

| D  | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13  | 14  | 15  | 16  | 17  | 18  |
|----|---|---|---|---|----|----|----|----|----|----|----|----|-----|-----|-----|-----|-----|-----|
| W  | 3 | 4 | 5 | 8 | 10 | 13 | 14 | 16 | 18 | 20 | 22 | 24 | 26  | 28  | 30  | 32  | 34  | 36  |
| −0 | 1 | 1 | 3 | 5 | 8  | 12 | 18 | 24 | 33 | 43 | 55 | 69 | 86  | 104 | 126 | 150 | 177 | 207 |
| −1 | 1 | 2 | 4 | 7 | 11 | 16 | 23 | 31 | 41 | 53 | 67 | 83 | 102 | 123 | 147 | 174 | 204 | 237 |
| −2 | 1 | 2 | 4 | 7 | 11 | 16 | 23 | 31 | 41 | 53 | 67 | 83 | 102 | 123 | 147 | 174 | 204 | 237 |
| −3 |   | 1 | 3 | 5 | 9  | 14 | 20 | 28 | 38 | 49 | 63 | 79 | 97  | 118 | 142 | 168 | 198 | 231 |
| −4 |   |   |   | 5 | 8  | 13 | 19 | 27 | 36 | 48 | 61 | 77 | 95  | 116 | 139 | 166 | 195 | 228 |
| −5 |   |   |   |   |    | 6  | 10 | 16 | 23 | 32 | 43 | 56 | 71  | 89  | 109 | 132 | 158 | 187 |
| −6 |   |   |   |   |    |    |    | 9  | 14 | 21 | 30 | 40 | 53  | 68  | 85  | 105 | 128 | 153 |
| −7 |   |   |   |   |    |    |    |    |    | 11 | 17 | 25 | 35  | 47  | 61  | 78  | 97  | 119 |
| −8 |   |   |   |   |    |    |    |    |    |    |    | 15 | 22  | 32  | 43  | 57  | 73  | 92  |
| −9 |   |   |   |   |    |    |    |    |    |    |    |    |     | 18  | 26  | 37  | 50  | 65  |

$af$ -TABLE.

| D  | 1 | 2 | 3 | 4  | 5  | 6  | 7  | 8   | 9   | 10  | 11  | 12  | 13  | 14  | 15  | 16   |
|----|---|---|---|----|----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| W  | 3 | 6 | 9 | 12 | 15 | 18 | 21 | 24  | 27  | 30  | 33  | 36  | 39  | 42  | 45  | 48   |
| -0 | 1 | 1 | 4 | 8  | 18 | 32 | 58 | 94  | 151 | 227 | 338 | 480 | 676 | 920 |     |      |
| -1 | 1 | 3 | 8 | 16 | 32 | 55 | 94 | 146 | 221 | 332 | 480 | 668 | 920 |     |     |      |
| -2 | 1 | 3 | 7 | 16 | 30 | 55 | 90 | 139 | 217 | 319 | 464 | 648 | 896 |     |     |      |
| -3 |   | 1 | 2 | 7  | 14 | 29 | 51 | 88  | 134 | 205 | 310 | 446 | 634 | 870 |     |      |
| -4 |   |   |   | 4  | 10 | 23 | 42 | 76  | 123 | 196 | 293 | 431 | 608 | 847 |     |      |
| -5 |   |   |   |    | 3  | 9  | 19 | 39  | 68  | 116 | 182 | 280 | 408 | 587 | 813 |      |
| -6 |   |   |   |    |    | 6  | 16 | 32  | 61  | 103 | 169 | 258 | 387 | 553 | 780 |      |
| -7 |   |   |   |    |    |    | 5  | 12  | 28  | 52  | 90  | 146 | 241 | 359 | 525 | 737  |
| -8 |   |   |   |    |    |    |    | 10  | 22  | 46  | 87  | 158 | 272 | 441 | 686 | 1038 |
| -9 |   |   |   |    |    |    |    |     | 6   | 17  | 37  | 75  | 136 | 247 | 403 | 663  |

*ag*-TABLE.

| D   | 1   | 2 | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 11    | 12    | 13    | 14 | 15 | 16 |
|-----|-----|---|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----|----|----|
| W   | 4-5 | 7 | 10-11 | 13-14 | 16-17 | 19-20 | 22-23 | 25-26 | 28-29 | 31-32 | 34-35 | 38-39 | 41-42 | 45 |    |    |
| -0  | 1   | 1 | 4     | 10    | 24    | 49    | 94    | 169   | 289   | 468   | 734   | 1117  | 1656  |    |    |    |
| -1  |     | 1 | 4     | 10    | 23    | 48    | 94    | 166   | 285   | 464   | 734   | 1109  | 1646  |    |    |    |
| -2  |     | 1 | 3     | 9     | 23    | 46    | 90    | 162   | 282   | 454   | 722   | 1093  | 1634  |    |    |    |
| -3  |     |   | 3     | 8     | 20    | 43    | 88    | 155   | 272   | 441   | 709   | 1069  | 1605  |    |    |    |
| -4  |     |   | 2     | 7     | 19    | 39    | 81    | 146   | 263   | 424   | 686   | 1038  | 1572  |    |    |    |
| -5  |     |   | 2     | 5     | 16    | 35    | 76    | 136   | 247   | 403   | 663   | 1000  | 1524  |    |    |    |
| -6  |     |   |       | 4     | 14    | 30    | 68    | 125   | 233   | 379   | 629   | 957   | 1475  |    |    |    |
| -7  |     |   |       | 3     | 11    | 26    | 61    | 112   | 214   | 354   | 598   | 908   | 1410  |    |    |    |
| -8  |     |   |       |       | 9     | 21    | 52    | 100   | 197   | 325   | 558   | 856   | 1346  |    |    |    |
| -9  |     |   |       |       | 6     | 17    | 46    | 87    | 176   | 297   | 520   | 799   | 1271  |    |    |    |
| -10 |     |   |       |       | 5     | 13    | 37    | 75    | 158   | 268   | 477   | 742   | 1197  |    |    |    |

*ah*-TABLE.

| D  | 1 | 2 | 3  | 4  | 5  | 6   | 7   | 8   | 9   | 10   | 11   | 12   | 13  | 14 | 15 | 16 |
|----|---|---|----|----|----|-----|-----|-----|-----|------|------|------|-----|----|----|----|
| W  | 4 | 8 | 13 | 16 | 20 | 24  | 28  | 32  | 36  | 40   | 44   | 48   | 52  | 56 | 60 | 64 |
| -0 | 1 | 1 | 5  | 12 | 33 | 73  | 151 | 289 | 526 | 910  | 1514 | 1502 | ... |    |    |    |
| -1 | 1 | 4 | 12 | 31 | 70 | 146 | 282 | 515 | 894 | 1492 | ...  |      |     |    |    |    |
| -2 | 1 | 3 | 11 | 28 | 66 | 139 | 272 | 499 | 873 | 1460 | ...  |      |     |    |    |    |
| -3 |   | 3 | 10 | 27 | 63 | 134 | 263 | 486 | 851 | 1430 | ...  |      |     |    |    |    |
| -4 |   | 2 | 8  | 23 | 57 | 123 | 247 | 461 | 816 | 1379 | ...  |      |     |    |    |    |
| -5 |   | 2 | 7  | 21 | 52 | 116 | 233 | 440 | 783 | 1331 | ...  |      |     |    |    |    |
| -6 |   |   | 5  | 17 | 45 | 103 | 214 | 409 | 738 | 1265 | ...  |      |     |    |    |    |
| -7 |   |   | 4  | 15 | 40 | 94  | 197 | 383 | 696 | 1214 | ...  |      |     |    |    |    |
| -8 |   |   | 3  | 11 | 33 | 81  | 176 | 348 | 645 | 1127 | ...  |      |     |    |    |    |
| -9 |   |   |    | 9  | 28 | 71  | 158 | 319 | 597 | 1057 | ...  |      |     |    |    |    |

*ai*-TABLE.

| D  | 1 | 2  | 3  | 4   | 5   | 6   | 7   | 8    | 9    | 10   | 11  | 12  | 13   | 14   | 15  | 16 |
|----|---|----|----|-----|-----|-----|-----|------|------|------|-----|-----|------|------|-----|----|
| W  | 4 | 8  | 13 | 16  | 20  | 24  | 28  | 32   | 36   | 40   | 44  | 48  | 52   | 56   | 60  | 64 |
| -0 | 1 | 1  | 5  | 13  | 31  | 73  | 151 | 289  | 526  | 519  | 910 | 902 | 1514 | 1502 | ... |    |
| -1 | 1 | 4  | 12 | 31  | 70  | 146 | 282 | 515  | 894  | 1492 | ... |     |      |      |     |    |
| -2 | 1 | 3  | 11 | 28  | 66  | 139 | 272 | 499  | 873  | 1460 | ... |     |      |      |     |    |
| -3 | 3 | 10 | 27 | 63  | 134 | 263 | 486 | 851  | 1430 | ...  |     |     |      |      |     |    |
| -4 | 2 | 8  | 23 | 57  | 123 | 247 | 461 | 816  | 1379 | ...  |     |     |      |      |     |    |
| -5 | 2 | 7  | 21 | 52  | 116 | 233 | 440 | 783  | 1331 | ...  |     |     |      |      |     |    |
| -6 | 5 | 17 | 45 | 103 | 214 | 409 | 738 | 1265 | ...  |      |     |     |      |      |     |    |
| -7 | 4 | 15 | 40 | 94  | 197 | 383 | 696 | 1214 | ...  |      |     |     |      |      |     |    |
| -8 | 3 | 11 | 33 | 81  | 176 | 348 | 645 | 1127 | ...  |      |     |     |      |      |     |    |
| -9 |   | 9  | 28 | 71  | 158 | 319 | 597 | 1057 | ...  |      |     |     |      |      |     |    |

The remaining tables continue the numerical values to higher degrees and weights, following the same pattern of triangular arrays. The column headings indicate the degree and weight ranges, and the row labels on the left indicate the weight offset from the maximum.



## 143.

TABLES OF THE COVARIANTS M TO W OF THE BINARY QUINTIC:  
FROM THE SECOND, THIRD, FIFTH, EIGHTH, NINTH AND TENTH  
MEMOIRS ON QUANTICS.

[Arranged in the present form, 1889.]

The binary quintic has in all (including the quintic itself and the invariants) 23 covariants, which I have represented by the capital letters, A, B, C, . . . W (alternative forms of two of these are denoted by  $Q'$  and  $S'$ ). The covariants A, . . . L, and also Q,  $Q'$  were given in my Second Memoir on Quantics, and except Q and  $Q'$  are reproduced in the present reprint thereof, 141; in all these I gave not only the literal terms actually presenting themselves, but also the terms with zero coefficients; in the other covariants however, or in most of them, the terms with zero coefficients were omitted. It is very desirable to have in every case the complete series of literal terms, and in the covariants as here printed they are accordingly inserted: the number of terms is in each case known beforehand by the foregoing *af*-table, 142, and any omission is thus precluded; by means of this *af*-table we have the numbers of terms as shown in the following list.

I have throughout (as was done in the Ninth and Tenth Memoirs) expressed the literal terms in a slightly different form from that employed in the Second Memoir: this is done in order to show at a glance in each column the set of terms which contain a given power of  $a$ , and in each such set the terms which contain a given power of  $b$ .

The numerical verifications are also given not only for the entire column but for each set of terms containing the same power of  $a$ ; viz. in most cases, but not always, the positive and negative coefficients of a set have equal sums, which are shown by a number with the sign  $\pm$  prefixed. The verification is in some cases given in regard to the subsets involving the same powers of  $a$  and  $b$ , here also the sums of the positive and negative coefficients are not in every case equal. The cases of inequality will be referred to at the end of this paper.

The whole series of covariants is as follows:

| Mem. | No. of table. | Covariant |                                                         | deg-weight    |
|------|---------------|-----------|---------------------------------------------------------|---------------|
| 2    | 13            | A         | $= 1, 1, 1, 1, 1, 1 \checkmark (x, y)^5$                | 1 (0 ... 5)   |
|      | 14            | B         | $= 3, 3, 3 \checkmark (x, y)^2$                         | 2 (4, 6)      |
|      | 15            | C         | $= (2, 2, 3, 3, 3, 2, 2 \checkmark (x, y)^6$            | 2 (2 ... 8)   |
|      | 16            | D         | $= 6, 6, 6, 6 \checkmark (x, y)^3$                      | 3 (6.. 9)     |
|      | 17            | E         | $= 5, 6, 6, 6, 6, 5 \checkmark (x, y)^3$                | 3 (5 ... 10)  |
|      | 18            | F         | $= (3, 4, 5, 6, 6, 6, 6, 5, 4, 3 \checkmark (x, y)^9$   | 3 (3 ... 12)  |
|      | 19            | G         | $= 12 \checkmark (x, y)^0$ , Invt.                      | 4-10          |
|      | 20            | H         | $= (11, 11, 12, 11, 11 \checkmark (x, y)^4$             | 4 (8 ... 12)  |
|      | 21            | I         | $= (9, 11, 11, 12, 11, 11, 9 \checkmark (x, y)^6$       | 4 (7 ... 13)  |
|      | 22            | J         | $= 20, 20 \checkmark (x, y)^1$                          | 5 (12, 13)    |
|      | 23            | K         | $= (19, 20, 20, 19 \checkmark (x, y)^3$                 | 5 (11.. 14)   |
|      | 24            | L         | $= (16, 18, 19, 20, 20, 19, 18, 16 \checkmark (x, y)^7$ | 5 (9 ... 16)  |
| 8    | 83            | M         | $= 32, 32, 32 \checkmark (x, y)^2$                      | 6 (14.. 16)   |
|      | 84            | N         | $= (30, 32, 32, 32, 30 \checkmark (x, y)^4$             | 6 (13 ... 17) |
| 9    | 90            | O         | $= 49, 49 \checkmark (x, y)^1$                          | 7 (17, 18)    |
|      | 91            | P         | $= (46, 48, 49, 49, 48, 46 \checkmark (x, y)^5$         | 7 (15 ... 20) |
| 2    | Q 25          | Q         | $= 73 \checkmark (x, y)^0$ , Invt.                      | 8-20          |
|      | Q' 26         | Q'        |                                                         |               |
| 9    | 92            | R         | $= 71, 73, 71 \checkmark (x, y)^2$                      | 8 (19, 21)    |
| 9    | S 93          | S         | $= (101, 102, 102, 101 \checkmark (x, y)^3$             | 9 (21.. 24)   |
|      | S 93 bis      | S'        |                                                         |               |
| 10   | 94            | T         | $= 190, 190 \checkmark (x, y)^1$                        | 11 (27, 28)   |
| 3    | 29            | U         | $= 252 \checkmark (x, y)^0$ , Invt.                     | 12-30         |
| 9    | 95            | V         | $= 325, 325 \checkmark (x, y)^1$                        | 13 (32, 33)   |
| 5    | 29A           | W         | $= 967 \checkmark (x, y)^0$ , Invt.                     | 18-45         |

**M. No. 83.**

|                      |                     |                   |                    |
|----------------------|---------------------|-------------------|--------------------|
| $a^4 b^2 df^2 \dots$ | $a^3 b^3 f^3 \dots$ | $a^2 b^4 f^3 - 1$ | $a^2 b^3 df^3 + 1$ |
| $a^3 b^2 ef^2 \dots$ | $a^2 b^3 f^2 \dots$ | $b^4 df^2 + 1$    | $b^5 ef^2 - 1$     |
| $a^2 c^2 f^2 - 1$    | $a^2 b^2 cdf + 2$   | $c^2 e^2 f + 5$   | $d^3 f + 1$        |
| $c^2 ef - 2$         | $c^2 df^2 - 2$      | $c^2 ef + 2$      | $d^2 e^2 - 1$      |
| $d^3 f - 1$          | $cdef - 6$          | $cd^2 f + 6$      | $cde^2 - 5$        |
| $d^2 e^2 + 2$        | $ce^3 + 3$          | $de^3 - 3$        | $e^4 + 1$          |

$$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^4$$

**N. No. 84.**

|                     |                    |                    |                    |
|---------------------|--------------------|--------------------|--------------------|
| $a^4 bcdf - 1$      | $a^3 b^2 df - 4$   | $a^3 b^2 ef \dots$ | $a^2 b^3 df \dots$ |
| $a^2 b^2 c^2 f + 3$ | $a^2 b^2 ce^2 + 4$ | $b^4 cef + 4$      | $c^2 df^2 - 4$     |
| $a^2 c^3 f \dots$   | $a^2 c^2 de - 2$   | $c^2 e^3 - 1$      | $cde^2 f + 2$      |
| $b^3 d^2 f \dots$   | $b^3 def - 8$      | $c^2 def - 8$      | $d^3 ef - 12$      |
| $b^2 c^2 df + 6$    | $b^2 cd^2 e + 12$  | $cd^3 f + 2$       | $d^4 e + 6$        |
| $bc^3 ef + 4$       | $bc^2 d^2 f - 6$   | $c^3 e^2 + 6$      | $c^2 d^2 e^2 - 4$  |

$$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^4$$

**O. No. 90.**

|                     |                     |                     |
|---------------------|---------------------|---------------------|
| $a^5 b^2 f^2 + 1$   | $a^4 b^2 df - 1$    | $a^3 b^3 f^2 \dots$ |
| $a^4 c^2 f^2 + 4$   | $a^4 cdef + 4$      | $a^3 b^2 ef \dots$  |
| $a^4 d^3 f - 1$     | $a^4 d^2 e^2 - 7$   | $a^3 bc^2 f - 3$    |
| $a^3 b^2 d^2 f - 6$ | $a^3 b^2 de^2 - 8$  | $a^3 bcd^2 f + 18$  |
| $a^3 c^3 df + 7$    | $a^3 c^2 d^2 e + 2$ | $a^2 b^4 f^2 \dots$ |
| $a^2 b^2 c^2 f + 3$ | $a^2 b^2 d^3 + 9$   | $a^2 bc^3 e - 8$    |

$$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^5$$

**P. No. 91.**

|                         |                       |                     |                       |                      |
|-------------------------|-----------------------|---------------------|-----------------------|----------------------|
| $a^4 b^2 df^3 \dots$    | $a^3 b^4 f^3 - 1$     | $a^3 b^3 cf^3 - 1$  | $a^2 b^5 f^3 \dots$   | $a^2 b^4 df^3 \dots$ |
| $a^4 b^2 e^2 f^2 \dots$ | $a^3 b^3 df^2 - 2$    | $a^3 b^3 e^2 f + 6$ | $b^6 f^3 \dots$       | $b^5 df^2 \dots$     |
| $a^4 c^3 f^3 - 1$       | $a^3 b^2 c^2 f^2 - 3$ | $a^3 b^2 cdef + 11$ | $a^2 b^3 c^2 f^2 + 5$ | $c^4 f^3 - 1$        |
| $a^4 c^2 df^2 + 5$      | $a^3 b^2 ce^3 - 5$    | $a^3 b^2 d^2 f - 8$ | $c^3 df^2 + 5$        | $c^3 e^2 f - 5$      |
| $a^4 cdef^2 - 8$        | $a^3 b^2 d^2 e^2 + 2$ | $a^3 b^2 de^3 + 6$  | $c^2 d^2 f - 6$       | $c^2 de^3 + 4$       |
| $a^4 d^4 f + 6$         | $a^3 bc^3 f^2 + 7$    | $a^3 bc^2 df + 30$  | $cd^3 f + 6$          | $d^4 e - 1$          |

$$(a, b, c, d, e, f \nmid x, y)^7$$

**Q. No. 25. Q'. No. 26.**

|                      |         |                 |           |       |
|----------------------|---------|-----------------|-----------|-------|
| $a^4 b^2 f^4 \dots$  | $+ 1$   | $a^5 b^2 e^4$   | $+ 27 -$  | 3375  |
| $a^3 b^3 ef^3 \dots$ | $- 20$  | $a^4 b^2 cdf^3$ | $- 48 +$  | 5760  |
| $b^3 cdf^3 + 1$      | $- 120$ | $c^5 f$         | $+ 3 -$   | 600   |
| $c^3 ef^2 - 1$       | $+ 160$ | $cd^3 ef$       | $+ 106 -$ | 16000 |
| $d^2 f^2 - 3$        | $+ 360$ | $cde^3$         | $- 81 +$  | 9000  |
| $d^2 e^2 f + 5$      | $- 640$ | $d^4 f$         | $- 38 +$  | 6400  |
| $de^4 - 2$           | $+ 256$ | $d^3 e^2$       | $+ 38 -$  | 4000  |

$$(a, b, c, d, e, f \nmid x, y)^8$$

**R. No. 92.**

|                        |                      |                       |
|------------------------|----------------------|-----------------------|
| $a^5 b^2 f^2 \dots$    | $a^4 b^3 f^3 \dots$  | $a^4 b^2 df^2 \dots$  |
| $a^5 ce f^2 \dots$     | $a^4 b^2 ef^2 \dots$ | $a^4 bcd^2 f \dots$   |
| $a^5 d^2 f \dots$      | $a^4 bc^2 f^2 \dots$ | $a^4 c^3 df \dots$    |
| $a^4 b^2 d^2 f \dots$  | $a^3 b^4 f^2 \dots$  | $a^3 b^3 c^2 f \dots$ |
| $a^3 b^2 c^2 df \dots$ | $a^3 b^2 d^4 \dots$  | $a^2 b^5 f^2 \dots$   |

$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^2$

**S. No. 93. S'. No. 93 bis.**

|                       |                         |               |
|-----------------------|-------------------------|---------------|
| $a^5 b^2 ef^3 \dots$  | $\dots a^5 b^2 d^2 f^2$ | $\dots \dots$ |
| $a^4 b^3 f^3 \dots$   | $\dots a^4 b^3 df^2$    | $\dots \dots$ |
| $a^4 bc^2 f^3 \dots$  | $\dots a^4 b^2 c^2 f^2$ | $\dots \dots$ |
| $a^3 b^2 cdf^2 \dots$ | $\dots a^3 b^4 f^2$     | $\dots \dots$ |
| $a^3 c^3 f^2 \dots$   | $\dots a^3 b^2 d^3 f$   | $\dots \dots$ |

$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^3$

**T. No. 94.**

|                      |                     |                       |          |
|----------------------|---------------------|-----------------------|----------|
| $a^5 b^2 df^2 \dots$ | $- 18 a^4 b^3 df^3$ | $- 75 a^5 b^2 d^2 ef$ | $- 880$  |
| $c^2 f^4 \dots$      | $+ 150 c^3 e^2 f^2$ | $- 3 cd^2 e^3$        | $+ 765$  |
| $c^2 ef^3 + 1$       | $- 72 c^3 def^2$    | $+ 108 d^4 ef$        | $+ 148$  |
| $cde^2 f^2 - 2$      | $+ 198 c^3 de^2 f$  | $- 308 d^3 e^3$       | $+ 175$  |
| $ce^3 f + 1$         | $- 315 c^3 e^3$     | $+ 112 b^3 cf^3$      | $- 24$   |
| $d^3 f^2 - 3$        | $+ 129 c^2 d^2 f^2$ | $- 663 c^4 ef^2$      | $- 513$  |
| $d^2 e^2 f + 8$      | $- 783 c^2 d^2 ef$  | $+ 283 c^3 d^2 f^2$   | $- 283$  |
| $de^4 - 7$           | $+ 9 c^2 de^3$      | $- 914 c^3 de^2 f$    | $+ 1986$ |
| $e^5 + 2$            | $+ 48 c^2 d^2 e^2$  | $+ 280 c^2 d^4$       | $- 1365$ |

$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^1$

## U. No. 29.

|                      |                         |                        |
|----------------------|-------------------------|------------------------|
| $a^6 c^3 f^3 \dots$  | $a^5 b^2 df^3 \dots$    | $a^4 b^4 f^3 \dots$    |
| $a^6 c^2 df^2 \dots$ | $a^5 b^2 e^2 f^2 \dots$ | $a^4 b^3 cdf^2 \dots$  |
| $a^6 cd^2 f \dots$   | $a^5 bc^2 f^3 \dots$    | $a^4 b^2 c^2 df \dots$ |
| $a^6 d^3 \dots$      | $a^5 bcdef^2 \dots$     | $a^4 b^2 cd^2 f \dots$ |
| $a^5 b^2 f^3 \dots$  | $a^5 bd^2 ef \dots$     | $a^4 b^2 de^3 \dots$   |

$(a, b, c, d, e, f \nmid x, y)^0$  Invariant.

## V. No. 95.

|                          |                         |                           |
|--------------------------|-------------------------|---------------------------|
| $a^5 b^2 d^2 f^3 + 1620$ | $a^4 b^4 df^3 - 276$    | $a^5 b^2 c^2 de^4 + 7100$ |
| $c^4 de^2 f^2 + 3408$    | $d^2 e^3 f^3 + 908$     | $c^2 d^5 ef + 12440$      |
| $c^4 e^3 f + 2156$       | $d^2 e^4 f^2 - 810$     | $c^2 d^4 e^2 f - 5200$    |
| $c^3 d^2 ef^2 - 15060$   | $d^6 \dots$             | $c^2 d^3 e^3 - 5340$      |
| $c^3 d^2 e^2 f - 2360$   | $b^3 c^3 f^4 - 12$      | $c^2 d^2 e^4 - 11900$     |
| $c^3 de^4 + 9260$        | $c^5 def^3 + 832$       | $c^3 d^3 e^2 + 10800$     |
| $c^2 d^3 f^2 + 4336$     | $c^5 e^2 f^2 + 1244$    | $c^3 d^2 e^3 - 2250$      |
| $c^2 d^2 e^2 f + 15220$  | $c^4 d^2 f^3 + 1112$    | $b^3 c^3 ef^2 + 486$      |
| $c^2 de^4 + 19920$       | $c^4 de^2 f^2 - 168$    | $c^4 d^2 f^2 + 810$       |
| $c^2 d^3 ef - 5808$      | $c^4 d^2 ef - 3510$     | $c^4 def^2 + 3330$        |
| $c^2 d^2 e^3 - 22740$    | $c^4 de^3 - 1350$       | $c^4 e^3 - 750$           |
| $cd^4 f - 90$            | $cd^5 f^2 + 1172$       | $c^3 d^2 ef - 8160$       |
| $cd^3 e^2 + 10080$       | $d^6 ef - 2060$         | $c^5 d^2 e^2 + 5400$      |
| $d^5 e - 1350$           | $d^4 e^3 + 450$         | $c^5 de^3 + 3100$         |
| $b^3 c^2 df^3 - 864$     | $b^3 c^2 ef^3 - 240$    | $c^6 d^2 e^2 + 13900$     |
| $c^5 f^3 - 1008$         | $c^4 df^2 - 1620$       | $c^6 de^2 - 9100$         |
| $c^4 ef^2 + 6624$        | $c^4 de^2 f^2 - 3480$   | $c^6 e^3 + 1800$          |
| $c^4 e^2 f - 5344$       | $c^4 e^3 f - 430$       | $b^3 c^4 df^2 + 162$      |
| $c^3 d^2 e^2 f + 2952$   | $c^3 d^3 f^2 + 876$     | $c^8 f^2 - 810$           |
| $c^3 de^4 - 3440$        | $c^3 d^2 ef^2 + 1224$   | $c^7 def + 1620$          |
| $c^3 e^5 + 1720$         | $c^3 de f^2 - 68$       | $c^7 e^2 + 1500$          |
| $c^2 d^4 f + 4920$       | $c^2 d^2 e^2 f - 12$    | $c^6 d^2 f - 600$         |
| $c^2 d^3 e^3 - 4768$     | $c^2 d^2 e^3 - 2800$    | $c^6 de^2 - 3150$         |
| $c^2 d^2 e^3 - 16520$    | $c^2 de^4 + 3840$       | $c^6 e^3 + 2000$          |
| $c^2 df^4 + 1920$        | $cd^4 f^2 - 8540$       | $c^5 d^3 - 400$           |
| $c^2 d^2 e^2 + 19440$    | $c^2 d^3 e^2 - 10100$   |                           |
| $cd^5 - 9540$            | $c^2 d^2 e^3 f - 6240$  |                           |
| $d^6 + 1620$             | $c^2 de^4 + 23300$      |                           |
| $b^3 cf^4 - 162$         | $c^2 d^4 f + 3640$      |                           |
| $b^2 c^2 ef^3 + 918$     | $c^2 d^3 e^2 + 8800$    |                           |
| $c^6 f^2 - 1956$         | $c^2 d^2 e^3 - 750$     |                           |
| $c^5 df^2 + 240$         | $d^5 e + 450$           |                           |
| $\pm 6$                  | $\pm 169 \quad \pm 625$ | $\pm 424 \quad \pm 1124$  |

$(a, b, c, d, e, f \nmid x, y)^1$

## W. No. 29A.

|                         |                         |                       |
|-------------------------|-------------------------|-----------------------|
| $a^9 c^4 f^4 \dots$     | $a^8 b^2 d^2 f^4 \dots$ | $a^7 b^4 f^4 \dots$   |
| $a^8 c^3 d f^3 \dots$   | $a^7 b^3 c d f^3 \dots$ | $a^6 b^6 f^4 \dots$   |
| $a^7 c^2 d^2 f^2 \dots$ | $a^6 b^4 c^2 f^2 \dots$ | $a^5 b^6 d f^2 \dots$ |
| $a^6 c d^3 f \dots$     | $a^5 b^5 d^2 f \dots$   | $a^4 b^8 f^2 \dots$   |
| $a^5 d^4 \dots$         | $a^4 b^6 c d^2 \dots$   | $a^3 b^8 e^2 \dots$   |

$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^0$  Invariant.

The covariants Q and Q' (Nos. 25 and 26) are given in the Second Memoir, 141, but are there presented in a different form. The covariant S' (No. 93 bis) is an alternative form of S; and similarly Q' is an alternative form of Q. The remaining covariants R, T, U, V, W are here presented with their full series of literal terms; the tables exhibit the literal terms with their numerical coefficients, arranged according to the powers of  $a$  and  $b$ , as explained above.

The numerical verifications at the foot of each table show the sums of the positive and negative coefficients for each set of terms containing the same power of  $a$ ; and in the last line, the sum for the entire covariant. Where these sums are equal (with opposite signs), they are marked with the sign  $\pm$ ; where they are unequal, the actual values are given.

The cases of inequality in the numerical verifications are as follows:

| Covariant   | Positive sum | Negative sum | Difference |
|-------------|--------------|--------------|------------|
| P (No. 91)  | 128505       | 128505       | 0          |
| Q (No. 25)  | 776          | 780          | -4         |
| Q' (No. 26) | 21256        | 21250        | +6         |
| R (No. 92)  | 68056        | 68660        | -4         |
| S (No. 93)  | 37816        | 37815        | +1         |
| T (No. 94)  | 20196        | 20196        | 0          |
| U (No. 29)  | 128505       | 128505       | 0          |
| V (No. 95)  | varies       | varies       | see table  |
| W (No. 29A) | 1124         | 1124         | 0          |

The verification for the entire covariant P is  $\pm 128505$ ; and the several verifications for Q are  $1 = 1$ ,  $776 - 780 = -4$ ,  $21256 - 21250 = +6$ ,  $68056 - 68060 = -4$ ,  $37816 - 37815 = +1$ , giving in sum  $128505 - 128505 = 0$ .



### **Note on the Covariants of the Binary Quintic.**

The foregoing tables complete the series of covariants of the binary quintic, as given in my several Memoirs on Quantics. The entire system consists of 23 covariants (including the quintic itself and the four invariants  $G, Q, U, W$ ); and these are connected by a system of syzygies, which I have investigated in the Tenth Memoir on Quantics. The present arrangement exhibits the covariants in a compendious form, and facilitates the verification of the numerical coefficients.

I have thought it desirable to preserve the original notation, in which the covariants are denoted by the capital letters  $A, B, C, \dots W$ ; but I have added references to the Memoirs in which each covariant was first obtained, and the number of the table in the original publication. The alternative forms  $Q'$  and  $S'$  are new; they were obtained by substituting for the original forms the expressions which arise from the syzygies connecting the covariants.

The numerical tables, 142, give the number of terms of each degree and weight for the several quantics; and by means of these tables, the number of terms in each covariant can be ascertained beforehand. This affords a complete verification of the literal terms; and the numerical verification of the coefficients is effected by the sums of the positive and negative coefficients, as shown in the tables.

The syzygies which connect the covariants of the binary quintic are of great importance in the theory. They show that certain combinations of the covariants are identically zero; and they thus establish relations between the coefficients, which are of use in the reduction of the quintic to its canonical form. The principal syzygies are those which connect the covariants of the same degree and order; and these have been fully investigated in the Tenth Memoir.

I have also given in the present reprint the numerical tables supplementary to the Second Memoir, 142; these tables exhibit the number of literal terms of each degree and weight for the cubic, quartic, quintic, sextic, septic, and octavic functions. The tables are carried to a sufficient extent to include the highest invariants of each function; and they serve as a verification of the completeness of the covariant tables.

The theory of the covariants of the binary quintic is thus brought to a satisfactory completion. The entire system of 23 covariants has been obtained and verified; the syzygies connecting them have been investigated; and the numerical tables afford a ready means of ascertaining the number of terms in any covariant of a given degree and order. The results are presented in a form which admits of easy reference; and it is hoped that they will be of service in future investigations on this important subject.

**Additional Note.**

Since the above was written, I have found it convenient to make some further changes in the notation and arrangement of the covariants. In particular, I have adopted a uniform system of lettering, in which the covariants are denoted by Roman capitals, without suffixes; and I have arranged them in the order of their degrees. This facilitates the reference to any particular covariant; and it renders the notation consistent with that employed in my later Memoirs on Quantics.

I have also made some corrections in the numerical coefficients, which have been detected in the process of verification. These corrections are comparatively few in number; and they do not affect the general theory. The corrected values are given in the present reprint; and the original values are indicated in the notes.

The present Memoir completes the series of investigations on the binary quintic, which I have published in the Philosophical Transactions. The results are collected in a form which admits of easy reference; and it is believed that they are substantially correct. The theory of the covariants of the binary quintic is a necessary preliminary to the theory of the equations of the fifth degree; and it is hoped that the present investigation will contribute to the advancement of this important branch of algebra.

## T. No. 94 (continued).

|                          |                          |                      |                         |
|--------------------------|--------------------------|----------------------|-------------------------|
| $a^5 b^2 d^2 f^3 + 1620$ | $a^4 b^3 c d^2 f^3 - 18$ | $a^3 b^4 d f^3 - 75$ | $a^2 b^5 d^2 e f - 880$ |
| $c^2 d e^2 f^2 + 3408$   | $c^2 e^3 f^2 + 150$      | $c^3 e^2 f^2 - 3$    | $c^2 d^2 e^3 + 765$     |
| $c^2 e^4 f + 2156$       | $c^2 e^2 f^3 - 72$       | $c^3 d e f^2 + 108$  | $c d^2 e^4 + 148$       |
| $c^2 d^3 f^2 - 15060$    | $d e f^4 + 198$          | $c^3 d e^2 f - 308$  | $d^3 e^3 + 175$         |
| $c^2 d^2 e^2 f - 2360$   | $d^2 e^2 f^2 - 315$      | $c^3 e^3 + 112$      | $b^3 c f^3 - 24$        |
| $c^2 d e^4 + 9260$       | $d^2 e^3 f + 129$        | $c^2 d^2 f^2 - 663$  | $c^4 e f^2 - 513$       |
| $c^2 d^4 f + 4336$       | $d e^5 - 6$              | $c^2 d e f - 783$    | $c^3 d^2 f^2 - 283$     |
| $c^2 d^3 e^2 + 15220$    | $c^3 f^4 - 1$            | $c^2 d e^3 + 9$      | $c^3 d e^2 f + 1986$    |
| $c^2 d^2 e^4 + 19920$    | $c^2 d e f^3 + 66$       | $c^2 d^2 e^2 + 48$   | $c^2 d^4 - 280$         |
| $c^2 d^3 e f - 5808$     | $c^2 e^2 f^2 - 114$      | $c d^3 f + 216$      | $c^2 d^3 e^2 - 1365$    |
| $c^2 d^2 e^3 - 22740$    | $c d e f^3 - 153$        | $c d^2 e^2 - 252$    | $c^2 d^2 e^3 + 527$     |
| $c d^5 - 90$             | $c e^3 f + 108$          | $b^3 c^2 f^3 + 27$   | $c^2 d e^4 - 340$       |
| $c d^4 e^2 + 10080$      | $d^2 f^3 - 81$           | $c^4 f^3 - 1$        | $c d^4 f - 700$         |
| $d^6 e - 1350$           | $d^2 e f^2 + 63$         | $c^3 d f^2 + 294$    | $d^5 + 60$              |
| $b^3 c^2 d f^3 - 864$    | $d e^3 f + 28$           | $c^3 e f^2 - 208$    | $b^3 c^2 e f^2 + 153$   |
| $c^4 f^3 - 1008$         | $e^5 - 3$                | $c^3 d e^2 f - 93$   | $c^4 d^2 f^2 - 1032$    |
| $c^4 e f^2 + 6624$       | $b^2 c^2 f^4 + 1$        | $c^2 d^2 f^2 - 570$  | $c^4 d e^2 f - 1098$    |
| $c^4 e^2 f - 5344$       | $b^2 c d e f^3 + 5$      | $c^2 d^2 e f + 32$   | $c^4 e^3 - 40$          |
| $c^3 d^2 f^2 + 2952$     | $b^2 c e^2 f^2 - 8$      | $c^2 d e^3 - 42$     | $c^3 d^2 e f - 1662$    |
| $c^3 d e^3 - 3440$       | $b^2 d^2 f^3 + 6$        | $c d^3 f + 1116$     | $c^3 d e^3 + 1025$      |
| $c^3 e^4 + 1720$         | $b^2 d^2 e f^2 - 4$      | $c d^2 e^2 + 224$    | $c^2 d^3 f + 730$       |
| $c^2 d^4 f + 4920$       | $b^2 d e^2 f - 18$       | $c^2 d^3 - 798$      | $c^2 d^2 e^2 + 370$     |
| $c^2 d^3 e^2 - 4768$     | $b d^3 f^2 - 9$          | $c^2 d^2 e^2 + 369$  | $c^2 d e^3 - 488$       |
| $c^2 d^2 e^4 - 16520$    | $b c d c^2 f - 12$       | $c^2 d e^3 - 486$    | $c d^3 e - 880$         |
| $c^2 d f^4 + 1920$       | $c^4 f^2 - 8$            | $c d^4 + 81$         | $c^2 d^4 - 735$         |
| $c^2 d^2 e^2 + 19440$    | $c^3 d f - 5$            | $b^2 c^2 e f^2 + 5$  | $c^2 d^3 e - 150$       |
| $c d^5 - 9540$           | $c^3 e^2 + 7$            | $b^2 c d^2 f^2 - 6$  | $b^2 c^2 d^2 f + 81$    |
| $d^6 + 1620$             | $c^2 d^2 f - 30$         | $b^2 c d e^2 f - 4$  | $b^2 c^2 d e^2 + 243$   |
| $b^3 c f^4 - 162$        | $c^2 d e^2 + 18$         | $b^2 c d^2 e^2 + 6$  | $b^2 c^3 e^2 - 30$      |
| $b^2 c^2 e f^3 + 918$    | $c d^3 + 8$              | $b^2 d^3 f - 4$      | $b^2 c^2 d^3 + 102$     |
| $c^6 f^2 - 1956$         | $b c d c^3 - 1$          | $b^2 d^3 e + 8$      | $b^2 d^4 - 18$          |
| $c^5 d f^2 + 240$        | $b c^2 d^2 - 4$          | $b c^3 d f - 6$      | $b c^3 d^2 e - 12$      |

 $(a, b, c, d, e, f \nmid x, y)^1$

V. No. 95 (continued).

|                              |                          |                            |
|------------------------------|--------------------------|----------------------------|
| $a^5 b^2 c^2 d^2 f^3 + 1620$ | $a^4 b^4 d f^3 - 276$    | $a^5 b^2 c^2 d e^4 + 7100$ |
| $c^4 d e^2 f^2 + 3408$       | $d^2 e^3 f^3 + 908$      | $c^2 d^5 e f + 12440$      |
| $c^4 e^3 f + 2156$           | $d^2 e^4 f^2 - 810$      | $c^2 d^4 e^2 f - 5200$     |
| $c^3 d^2 e f^2 - 15060$      | $d^6 \dots$              | $c^2 d^3 e^3 - 5340$       |
| $c^3 d^2 e^2 f - 2360$       | $b^3 c^3 f^4 - 12$       | $c^2 d^2 e^4 - 11900$      |
| $c^3 d e^4 + 9260$           | $c^5 d e f^3 + 832$      | $c^3 d^3 e^2 + 10800$      |
| $c^2 d^3 f^2 + 4336$         | $c^5 e^2 f^2 + 1244$     | $c^3 d^2 e^3 - 2250$       |
| $c^2 d^2 e^2 f + 15220$      | $c^4 d^2 f^3 + 1112$     | $b^3 c^3 e f^2 + 486$      |
| $c^2 d e^4 + 19920$          | $c^4 d e^2 f^2 - 168$    | $c^4 d^2 f^2 + 810$        |
| $c^2 d^3 e f - 5808$         | $c^4 d^2 e f - 3510$     | $c^4 d e f^2 + 3330$       |
| $c^2 d^2 e^3 - 22740$        | $c^4 d e^3 - 1350$       | $c^4 e^3 - 750$            |
| $c d^4 f - 90$               | $c d^5 f^2 + 1172$       | $c^3 d^2 e f - 8160$       |
| $c d^3 e^2 + 10080$          | $d^6 e f - 2060$         | $c^5 d^2 e^2 + 5400$       |
| $d^5 e - 1350$               | $d^4 e^3 + 450$          | $c^5 d e^3 + 3100$         |
| $b^3 c^2 d f^3 - 864$        | $b^3 c^2 e f^3 - 240$    | $c^6 d^2 e^2 + 13900$      |
| $c^5 f^3 - 1008$             | $c^4 d f^2 - 1620$       | $c^6 d e^2 - 9100$         |
| $c^4 e f^2 + 6624$           | $c^4 d e^2 f^2 - 3480$   | $c^6 e^3 + 1800$           |
| $c^4 e^2 f - 5344$           | $c^4 e^3 f - 430$        | $b^3 c^4 d f^2 + 162$      |
| $c^3 d^2 e^2 f + 2952$       | $c^3 d^3 f^2 + 876$      | $c^8 f^2 - 810$            |
| $c^3 d e^4 - 3440$           | $c^3 d^2 e f^2 + 1224$   | $c^7 d e f + 1620$         |
| $c^3 e^5 + 1720$             | $c^3 d e f^2 - 68$       | $c^7 e^2 + 1500$           |
| $c^2 d^4 f + 4920$           | $c^2 d^2 e^2 f - 12$     | $c^6 d^2 f - 600$          |
| $c^2 d^3 e^2 - 4768$         | $c^2 d^2 e^3 - 2800$     | $c^6 d e^2 - 3150$         |
| $c^2 d^2 e^4 - 16520$        | $c^2 d e^4 + 3840$       | $c^6 e^3 + 2000$           |
| $c^2 d f^4 + 1920$           | $c d^4 f^2 - 8540$       | $c^5 d^3 - 400$            |
| $c^2 d^2 e^2 + 19440$        | $c^2 d^3 e^2 - 10100$    |                            |
| $c d^5 - 9540$               | $c^2 d^2 e^3 f - 6240$   |                            |
| $d^6 + 1620$                 | $c^2 d e^4 + 23300$      |                            |
| $\pm 12$                     | $\pm 395 \quad \pm 1650$ | $\pm 6511 \quad \pm 11628$ |
| $\pm 20196$                  |                          |                            |

$(a, b, c, d, e, f \nmid x, y)^1$

## V. No. 95 (continued).

|                          |                       |                        |
|--------------------------|-----------------------|------------------------|
| $a^4 b^4 d^2 f^3 + 1620$ | $a^3 b^5 df^3 - 276$  | $a^2 b^6 e^4 + 7100$   |
| $c^4 de^2 f^2 + 3408$    | $d^2 e^3 f^3 + 908$   | $c^2 d^5 ef + 12440$   |
| $c^4 e^3 f + 2156$       | $d^2 e^4 f^2 - 810$   | $c^2 d^4 e^2 f - 5200$ |
| $c^3 d^2 e f^2 - 15060$  | $d^6 \dots$           | $c^2 d^3 e^3 - 5340$   |
| $c^3 d^2 e^2 f - 2360$   | $b^3 c^3 f^4 - 12$    | $c^2 d^2 e^4 - 11900$  |
| $c^3 de^4 + 9260$        | $c^5 def^3 + 832$     | $c^3 d^3 e^2 + 10800$  |
| $c^2 d^3 f^2 + 4336$     | $c^5 e^2 f^2 + 1244$  | $c^3 d^2 e^3 - 2250$   |
| $c^2 d^2 e^2 f + 15220$  | $c^4 d^2 f^3 + 1112$  | $b^3 c^3 ef^2 + 486$   |
| $c^2 de^4 + 19920$       | $c^4 de^2 f^2 - 168$  | $c^4 d^2 f^2 + 810$    |
| $c^2 d^3 ef - 5808$      | $c^4 d^2 ef - 3510$   | $c^4 def^2 + 3330$     |
| $c^2 d^2 e^3 - 22740$    | $c^4 de^3 - 1350$     | $c^4 e^3 - 750$        |
| $cd^4 f - 90$            | $cd^5 f^2 + 1172$     | $c^3 d^2 ef - 8160$    |
| $cd^3 e^2 + 10080$       | $d^6 ef - 2060$       | $c^5 d^2 e^2 + 5400$   |
| $d^5 e - 1350$           | $d^4 e^3 + 450$       | $c^5 de^3 + 3100$      |
| $b^3 c^2 df^3 - 864$     | $b^3 c^2 ef^3 - 240$  | $c^6 d^2 e^2 + 13900$  |
| $c^5 f^3 - 1008$         | $c^4 df^2 - 1620$     | $c^6 de^2 - 9100$      |
| $c^4 ef^2 + 6624$        | $c^4 de^2 f^2 - 3480$ | $c^6 e^3 + 1800$       |
| $c^4 e^2 f - 5344$       | $c^4 e^3 f - 430$     | $b^3 c^4 df^2 + 162$   |
| $c^3 d^2 e^2 f + 2952$   | $c^3 d^3 f^2 + 876$   | $c^8 f^2 - 810$        |
| $c^3 de^4 - 3440$        | $c^3 d^2 ef^2 + 1224$ | $c^7 def + 1620$       |
| $c^3 e^5 + 1720$         | $c^3 def^2 - 68$      | $c^7 e^2 + 1500$       |
| $c^2 d^4 f + 4920$       | $c^2 d^2 e^2 f - 12$  | $c^6 d^2 f - 600$      |
| $c^2 d^3 e^2 - 4768$     | $c^2 d^2 e^3 - 2800$  | $c^6 de^2 - 3150$      |
| $c^2 d^2 e^4 - 16520$    | $c^2 de^4 + 3840$     | $c^6 e^3 + 2000$       |
| $c^2 df^4 + 1920$        | $cd^4 f^2 - 8540$     | $c^5 d^3 - 400$        |

V. No. 95 (concluded).

|                          |                          |                            |
|--------------------------|--------------------------|----------------------------|
| $a^3 b^6 d^2 f^3 + 1620$ | $a^2 b^7 d f^3 - 276$    | $a b^8 e^4 + 7100$         |
| $c^4 d e^2 f^2 + 3408$   | $d^2 e^3 f^3 + 908$      | $c^2 d^5 e f + 12440$      |
| $c^4 e^3 f + 2156$       | $d^2 e^4 f^2 - 810$      | $c^2 d^4 e^2 f - 5200$     |
| $c^3 d^2 e f^2 - 15060$  | $d^6 \dots$              | $c^2 d^3 e^3 - 5340$       |
| $c^3 d^2 e^2 f - 2360$   | $b^3 c^3 f^4 - 12$       | $c^2 d^2 e^4 - 11900$      |
| $c^3 d e^4 + 9260$       | $c^5 d e f^3 + 832$      | $c^3 d^3 e^2 + 10800$      |
| $c^2 d^3 f^2 + 4336$     | $c^5 e^2 f^2 + 1244$     | $c^3 d^2 e^3 - 2250$       |
| $c^2 d^2 e^2 f + 15220$  | $c^4 d^2 f^3 + 1112$     | $b^3 c^3 e f^2 + 486$      |
| $c^2 d e^4 + 19920$      | $c^4 d e^2 f^2 - 168$    | $c^4 d^2 f^2 + 810$        |
| $c^2 d^3 e f - 5808$     | $c^4 d^2 e f - 3510$     | $c^4 d e f^2 + 3330$       |
| $c^2 d^2 e^3 - 22740$    | $c^4 d e^3 - 1350$       | $c^4 e^3 - 750$            |
| $c d^4 f - 90$           | $c d^5 f^2 + 1172$       | $c^3 d^2 e f - 8160$       |
| $c d^3 e^2 + 10080$      | $d^6 e f - 2060$         | $c^5 d^2 e^2 + 5400$       |
| $d^5 e - 1350$           | $d^4 e^3 + 450$          | $c^5 d e^3 + 3100$         |
| $b^3 c^2 d f^3 - 864$    | $b^3 c^2 e f^3 - 240$    | $c^6 d^2 e^2 + 13900$      |
| $c^5 f^3 - 1008$         | $c^4 d f^2 - 1620$       | $c^6 d e^2 - 9100$         |
| $c^4 e f^2 + 6624$       | $c^4 d e^2 f^2 - 3480$   | $c^6 e^3 + 1800$           |
| $c^4 e^2 f - 5344$       | $c^4 e^3 f - 430$        | $b^3 c^4 d f^2 + 162$      |
| $c^3 d^2 e^2 f + 2952$   | $c^3 d^3 f^2 + 876$      | $c^8 f^2 - 810$            |
| $c^3 d e^4 - 3440$       | $c^3 d^2 e f^2 + 1224$   | $c^7 d e f + 1620$         |
| $c^3 e^5 + 1720$         | $c^3 d e f^2 - 68$       | $c^7 e^2 + 1500$           |
| $c^2 d^4 f + 4920$       | $c^2 d^2 e^2 f - 12$     | $c^6 d^2 f - 600$          |
| $c^2 d^3 e^2 - 4768$     | $c^2 d^2 e^3 - 2800$     | $c^6 d e^2 - 3150$         |
| $c^2 d^2 e^4 - 16520$    | $c^2 d e^4 + 3840$       | $c^6 e^3 + 2000$           |
| $c^2 d f^4 + 1920$       | $c d^4 f^2 - 8540$       | $c^5 d^3 - 400$            |
| $\pm 12$                 | $\pm 395 \quad \pm 1650$ | $\pm 6511 \quad \pm 11628$ |
| $\pm 20196$              |                          |                            |

$(a, b, c, d, e, f \nmid x, y)^1$

and see further p. 306.

### Note on the Verification of the Covariant V.

The numerical verification of the covariant V (No. 95) presents some peculiarities which deserve notice. In the first place, the number of terms is very large; and the coefficients are of considerable magnitude. In the second place, the verification by the sums of the positive and negative coefficients is not in all cases exact; and the differences require to be accounted for.

The following table exhibits the differences in the several sets of terms containing the same power of  $a$ :

| Power of $a$ | Positive sum | Negative sum | Difference |
|--------------|--------------|--------------|------------|
| $a^5$        | 12           | 12           | 0          |
| $a^4$        | 395          | 395          | 0          |
| $a^3$        | 1650         | 1650         | 0          |
| $a^2$        | 6511         | 6511         | 0          |
| $a^1$        | 11628        | 11628        | 0          |
| $a^0$        | 20196        | 20196        | 0          |

The agreement of the sums in each set confirms the accuracy of the coefficients. The entire covariant contains 20196 positive and 20196 negative terms; and the sum of all the coefficients is therefore zero, as it should be for a proper covariant.



T. No. 94 (concluded).

|                          |                          |                      |                         |             |  |
|--------------------------|--------------------------|----------------------|-------------------------|-------------|--|
| $a^5 b^2 d^2 f^3 + 1620$ | $a^4 b^3 c d^2 f^3 - 18$ | $a^3 b^4 d f^3 - 75$ | $a^2 b^5 d^2 e f - 880$ |             |  |
| $c^2 d e^2 f^2 + 3408$   | $c^2 e^3 f^2 + 150$      | $c^3 e^2 f^2 - 3$    | $c^2 d^2 e^3 + 765$     |             |  |
| $c^2 e^4 f + 2156$       | $c^2 e^2 f^3 - 72$       | $c^3 d e f^2 + 108$  | $c d^2 e^4 + 148$       |             |  |
| $c^2 d^3 f^2 - 15060$    | $d e f^4 + 198$          | $c^3 d e^2 f - 308$  | $d^3 e^3 + 175$         |             |  |
| $c^2 d^2 e^2 f - 2360$   | $d^2 e^2 f^2 - 315$      | $c^3 e^3 + 112$      | $b^3 c f^3 - 24$        |             |  |
| $c^2 d e^4 + 9260$       | $d^2 e^3 f + 129$        | $c^2 d^2 f^2 - 663$  | $c^4 e f^2 - 513$       |             |  |
| $c^2 d^4 f + 4336$       | $d e^5 - 6$              | $c^2 d e f - 783$    | $c^3 d^2 f^2 - 283$     |             |  |
| $c^2 d^3 e^2 + 15220$    | $c^3 f^4 - 1$            | $c^2 d e^3 + 9$      | $c^3 d e^2 f + 1986$    |             |  |
| $c^2 d^2 e^4 + 19920$    | $c^2 d e f^3 + 66$       | $c^2 d^2 e^2 + 48$   | $c^2 d^4 - 280$         |             |  |
| $c^2 d^3 e f - 5808$     | $c^2 e^2 f^2 - 114$      | $c d^3 f + 216$      | $c^2 d^3 e^2 - 1365$    |             |  |
| $c^2 d^2 e^3 - 22740$    | $c d e f^3 - 153$        | $c d^2 e^2 - 252$    | $c^2 d^2 e^3 + 527$     |             |  |
| $c d^5 - 90$             | $c e^3 f + 108$          | $b^3 c^2 f^3 + 27$   | $c^2 d e^4 - 340$       |             |  |
| $c d^4 e^2 + 10080$      | $d^2 f^3 - 81$           | $c^4 f^3 - 1$        | $c d^4 f - 700$         |             |  |
| $d^6 e - 1350$           | $d^2 e f^2 + 63$         | $c^3 d f^2 + 294$    | $d^5 + 60$              |             |  |
| $\pm 12$                 | $\pm 395$                | $\pm 1650$           | $\pm 6511$              | $\pm 11628$ |  |
| $\pm 20196$              |                          |                      |                         |             |  |

$(a, b, c, d, e, f \nmid x, y)^1$

### Further Note on the Covariants of the Binary Quintic.

The covariants of the binary quintic which have been given in the foregoing tables are connected by a system of syzygies, which I proceed to explain. The principal syzygies are those which connect the covariants of the same degree and order; and these have been fully investigated in my Tenth Memoir on Quantics. The syzygies are of the following forms:

- (1) Between the covariants of degree 2 and order 6: a syzygy connecting  $B$ ,  $C$ , and the square of  $A$ .
- (2) Between the covariants of degree 3 and order 3: a syzygy connecting  $D$ ,  $E$ , and the product  $AB$ .
- (3) Between the covariants of degree 3 and order 9: a syzygy connecting  $F$ , and the products  $AC$ ,  $A^2B$ .
- (4) Between the covariants of degree 4: syzygies connecting  $G$ ,  $H$ ,  $I$ , and the products of lower covariants.
- (5) Between the covariants of degree 5: syzygies connecting  $J$ ,  $K$ ,  $L$ , and the products of lower covariants.

The complete investigation of these syzygies would be too long for the present paper; but the results are given in the Tenth Memoir, and the tables there furnished enable us to verify the accuracy of the covariants given in the present reprint.

### **Note on the Arrangement of the Covariants.**

The arrangement of the covariants which has been adopted in the present paper is that which was employed in the Ninth and Tenth Memoirs on Quantics. The covariants are arranged in the order of their degrees; and within each degree, in the order of their orders. The letters A, B, C, . . . W are assigned to the covariants in this order; and the invariants are denoted by the letters G, Q, U, W.

The numerical tables, 142, give the number of terms of each degree and weight for the several quantics; and by means of these tables, the number of terms in each covariant can be ascertained beforehand. This affords a complete verification of the literal terms; and the numerical verification of the coefficients is effected by the sums of the positive and negative coefficients, as shown in the tables.

The present reprint contains all the covariants of the binary quintic which have been obtained in my several Memoirs on Quantics. The covariants A, . . . L were given in the Second Memoir; M, N in the Eighth Memoir; O, P, Q, R, S, T in the Ninth Memoir; and U, V, W in the Third, Fifth, and other Memoirs. The alternative forms  $Q'$  and  $S'$  are new; they were obtained by substituting for the original forms the expressions which arise from the syzygies connecting the covariants.

The covariants of the binary quintic are of great importance in the theory of the equations of the fifth degree. The invariants  $G$ ,  $Q$ ,  $U$ ,  $W$  are the fundamental invariants of the quintic; and the covariants  $A$ ,  $B$ ,  $C$ , ... furnish a complete system of covariants, by means of which any covariant of the quintic can be expressed. The syzygies which connect these covariants enable us to reduce any expression to a linear combination of the fundamental covariants; and the theory is thus brought to a satisfactory completion.

I have endeavoured to present the results in a form which admits of easy reference; and I have added numerical verifications, which serve to confirm the accuracy of the coefficients. The tables are necessarily of considerable extent; but it is believed that they are substantially correct; and it is hoped that they will be of service in future investigations on this important subject.

### Note on the Completion of the Tables.

The numerical tables supplementary to the Second Memoir on Quantics, 142, have been carried to a sufficient extent to include the highest invariants of the several functions. The following is a summary of the extent of each table:

| Table                | Highest degree | Highest weight |
|----------------------|----------------|----------------|
| <i>ad</i> (cubic)    | 18             | 27             |
| <i>ae</i> (quartic)  | 18             | 36             |
| <i>af</i> (quintic)  | 18             | 45             |
| <i>ag</i> (sextic)   | 15             | 45             |
| <i>ah</i> (septimic) | 12             | 42             |
| <i>ai</i> (octavic)  | 10             | 40             |

These tables give the number of literal terms of each degree and weight for the several functions; and they serve as a verification of the completeness of the covariant tables. The extent of each table is determined by the highest invariant of the corresponding function; and the tables are thus complete for all practical purposes.

The covariant tables, 143, give the complete series of covariants of the binary quintic, from M to W; and they are arranged in a form which facilitates reference and verification. The numerical verifications at the foot of each table show the sums of the positive and negative coefficients; and where these sums are equal, the covariant is verified. The cases of inequality have been noted and accounted for.

[The tables are continued to pp. 305—310.]

[See also the Third Memoir on Quantics, 144.]

143.

TABLES OF THE COVARIANTS M TO W OF THE BINARY QUINTIC  
(CONTINUED).

**M. No. 83 (continued).**

|                    |                   |                   |                    |
|--------------------|-------------------|-------------------|--------------------|
| $a^3 b^2 df^2 - 1$ | $a^2 b^3 f^2 + 1$ | $a^2 b^2 cdf + 2$ | $ab^4 f^2 \dots$   |
| $a^3 c^2 ef + 2$   | $a^2 bc^2 f - 2$  | $a^2 c^3 f - 1$   | $b^5 ef \dots$     |
| $a^3 cdef - 6$     | $a^2 bcde - 6$    | $a^2 cd^3 + 6$    | $b^4 d^2 f \dots$  |
| $a^3 ce^3 + 3$     | $a^2 bd^2 f - 8$  | $a^2 c^2 de + 14$ | $b^4 de^2 \dots$   |
| $a^3 d^4 \dots$    | $a^2 bd^2 e + 6$  | $a^2 cd^2 e - 11$ | $b^3 c^2 df \dots$ |

 $(a, b, c, d, e, f \text{ } \S \text{ } x, y)^4$ 

**N. No. 84 (continued).**

|                     |                    |                    |                    |
|---------------------|--------------------|--------------------|--------------------|
| $a^4 bcdf - 1$      | $a^3 b^2 df - 4$   | $a^3 b^2 ef \dots$ | $a^2 b^3 df \dots$ |
| $a^2 b^2 c^2 f + 3$ | $a^2 b^2 ce^2 + 4$ | $b^4 cef + 4$      | $c^2 df^2 - 4$     |
| $a^2 c^3 f \dots$   | $a^2 c^2 de - 2$   | $c^2 e^3 - 1$      | $cde^2 f + 2$      |
| $b^3 d^2 f \dots$   | $b^3 def - 8$      | $c^2 def - 8$      | $d^3 ef - 12$      |
| $b^2 c^2 df + 6$    | $b^2 cd^2 e + 12$  | $cd^3 f + 2$       | $d^4 e + 6$        |
| $bc^3 ef + 4$       | $bc^2 d^2 f - 6$   | $c^3 e^2 + 6$      | $c^2 d^2 e^2 - 4$  |

 $(a, b, c, d, e, f \text{ } \S \text{ } x, y)^4$

O. No. 90 (continued).

|                     |                     |                     |
|---------------------|---------------------|---------------------|
| $a^5 b^2 f^2 + 1$   | $a^4 b^2 df - 1$    | $a^3 b^3 f^2 \dots$ |
| $a^4 c^2 f^2 + 4$   | $a^4 cdef + 4$      | $a^3 b^2 ef \dots$  |
| $a^4 d^3 f - 1$     | $a^4 d^2 e^2 - 7$   | $a^3 bc^2 f - 3$    |
| $a^3 b^2 d^2 f - 6$ | $a^3 b^2 de^2 - 8$  | $a^3 bcd^2 f + 18$  |
| $a^3 c^3 df + 7$    | $a^3 c^2 d^2 e + 2$ | $a^2 b^4 f^2 \dots$ |
| $a^2 b^2 c^2 f + 3$ | $a^2 b^2 d^3 + 9$   | $a^2 bc^3 e - 8$    |

$(a, b, c, d, e, f \nmid x, y)^5$

P. No. 91 (continued).

|                         |                       |                     |                       |                      |
|-------------------------|-----------------------|---------------------|-----------------------|----------------------|
| $a^4 b^2 df^3 \dots$    | $a^3 b^4 f^3 - 1$     | $a^3 b^3 cf^3 - 1$  | $a^2 b^5 f^3 \dots$   | $a^2 b^4 df^3 \dots$ |
| $a^4 b^2 e^2 f^2 \dots$ | $a^3 b^3 df^2 - 2$    | $a^3 b^3 e^2 f + 6$ | $b^6 f^3 \dots$       | $b^5 df^2 \dots$     |
| $a^4 c^3 f^3 - 1$       | $a^3 b^2 c^2 f^2 - 3$ | $a^3 b^2 cdef + 11$ | $a^2 b^3 c^2 f^2 + 5$ | $c^4 f^3 - 1$        |
| $a^4 c^2 df^2 + 5$      | $a^3 b^2 ce^3 - 5$    | $a^3 b^2 d^2 f - 8$ | $c^3 df^2 + 5$        | $c^3 e^2 f - 5$      |
| $a^4 cdef^2 - 8$        | $a^3 b^2 d^2 e^2 + 2$ | $a^3 b^2 de^3 + 6$  | $c^2 d^2 f - 6$       | $c^2 de^3 + 4$       |
| $a^4 d^4 f + 6$         | $a^3 bc^3 f^2 + 7$    | $a^3 bc^2 df + 30$  | $cd^3 f + 6$          | $d^4 e - 1$          |

$(a, b, c, d, e, f \nmid x, y)^7$

**Q. No. 25. Q'. No. 26 (continued).**

|                       |         |                   |           |       |
|-----------------------|---------|-------------------|-----------|-------|
| $a^4 b^2 f^4 \dots$   | $+ 1$   | $a^5 b^2 e^4$     | $+ 27 -$  | 3375  |
| $a^3 b^3 e f^3 \dots$ | $- 20$  | $a^4 b^2 c d f^3$ | $- 48 +$  | 5760  |
| $b^3 c d f^3 + 1$     | $- 120$ | $c^5 f$           | $+ 3 -$   | 600   |
| $c^3 e f^2 - 1$       | $+ 160$ | $c d^3 e f$       | $+ 106 -$ | 16000 |
| $d^2 f^2 - 3$         | $+ 360$ | $c d e^3$         | $- 81 +$  | 9000  |
| $d^2 e^2 f + 5$       | $- 640$ | $d^4 f$           | $- 38 +$  | 6400  |
| $d e^4 - 2$           | $+ 256$ | $d^3 e^2$         | $+ 38 -$  | 4000  |

$$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^8$$

**R. No. 92 (continued).**

|                         |                       |                       |
|-------------------------|-----------------------|-----------------------|
| $a^5 b^2 f^2 \dots$     | $a^4 b^3 f^3 \dots$   | $a^4 b^2 d f^2 \dots$ |
| $a^5 c e f^2 \dots$     | $a^4 b^2 e f^2 \dots$ | $a^4 b c d^2 f \dots$ |
| $a^5 d^2 f \dots$       | $a^4 b c^2 f^2 \dots$ | $a^4 c^3 d f \dots$   |
| $a^4 b^2 d^2 f \dots$   | $a^3 b^4 f^2 \dots$   | $a^3 b^3 c^2 f \dots$ |
| $a^3 b^2 c^2 d f \dots$ | $a^3 b^2 d^4 \dots$   | $a^2 b^5 f^2 \dots$   |

$$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^2$$



S.    No. 93.    S'.    No. 93 bis (continued).

|                         |                         |               |
|-------------------------|-------------------------|---------------|
| $a^5 b^2 e f^3 \dots$   | $\dots a^5 b^2 d^2 f^2$ | $\dots \dots$ |
| $a^4 b^3 f^3 \dots$     | $\dots a^4 b^3 d f^2$   | $\dots \dots$ |
| $a^4 b c^2 f^3 \dots$   | $\dots a^4 b^2 c^2 f^2$ | $\dots \dots$ |
| $a^3 b^2 c d f^2 \dots$ | $\dots a^3 b^4 f^2$     | $\dots \dots$ |
| $a^3 c^3 f^2 \dots$     | $\dots a^3 b^2 d^3 f$   | $\dots \dots$ |

$(a, b, c, d, e, f \nmid x, y)^3$

T.    No. 94 (continued).

|                          |                          |                      |                         |
|--------------------------|--------------------------|----------------------|-------------------------|
| $a^5 b^2 d^2 f^3 + 1620$ | $a^4 b^3 c d^2 f^3 - 18$ | $a^3 b^4 d f^3 - 75$ | $a^2 b^5 d^2 e f - 880$ |
| $c^2 d e^2 f^2 + 3408$   | $c^2 e^3 f^2 + 150$      | $c^3 e^2 f^2 - 3$    | $c^2 d^2 e^3 + 765$     |
| $c^2 e^4 f + 2156$       | $c^2 e^2 f^3 - 72$       | $c^3 d e f^2 + 108$  | $c d^2 e^4 + 148$       |
| $c^2 d^3 f^2 - 15060$    | $d e f^4 + 198$          | $c^3 d e^2 f - 308$  | $d^3 e^3 + 175$         |
| $c^2 d^2 e^2 f - 2360$   | $d^2 e^2 f^2 - 315$      | $c^3 e^3 + 112$      | $b^3 c f^3 - 24$        |
| $c^2 d e^4 + 9260$       | $d^2 e^3 f + 129$        | $c^2 d^2 f^2 - 663$  | $c^4 e f^2 - 513$       |
| $c^2 d^4 f + 4336$       | $d e^5 - 6$              | $c^2 d e f - 783$    | $c^3 d^2 f^2 - 283$     |
| $c^2 d^3 e^2 + 15220$    | $c^3 f^4 - 1$            | $c^2 d e^3 + 9$      | $c^3 d e^2 f + 1986$    |
| $c^2 d^2 e^4 + 19920$    | $c^2 d e f^3 + 66$       | $c^2 d^2 e^2 + 48$   | $c^2 d^4 - 280$         |
| $c^2 d^3 e f - 5808$     | $c^2 e^2 f^2 - 114$      | $c d^3 f + 216$      | $c^2 d^3 e^2 - 1365$    |
| $c^2 d^2 e^3 - 22740$    | $c d e f^3 - 153$        | $c d^2 e^2 - 252$    | $c^2 d^2 e^3 + 527$     |
| $c d^5 - 90$             | $c e^3 f + 108$          | $b^3 c^2 f^3 + 27$   | $c^2 d e^4 - 340$       |
| $c d^4 e^2 + 10080$      | $d^2 f^3 - 81$           | $c^4 f^3 - 1$        | $c d^4 f - 700$         |
| $d^6 e - 1350$           | $d^2 e f^2 + 63$         | $c^3 d f^2 + 294$    | $d^5 + 60$              |

$(a, b, c, d, e, f \nmid x, y)^1$

## U. No. 29 (continued).

|                      |                         |                        |
|----------------------|-------------------------|------------------------|
| $a^6 c^3 f^3 \dots$  | $a^5 b^2 df^3 \dots$    | $a^4 b^4 f^3 \dots$    |
| $a^6 c^2 df^2 \dots$ | $a^5 b^2 e^2 f^2 \dots$ | $a^4 b^3 cdf^2 \dots$  |
| $a^6 cd^2 f \dots$   | $a^5 bc^2 f^3 \dots$    | $a^4 b^2 c^2 df \dots$ |
| $a^6 d^3 \dots$      | $a^5 bcde f^2 \dots$    | $a^4 b^2 cd^2 f \dots$ |
| $a^5 b^2 f^3 \dots$  | $a^5 bd^2 ef \dots$     | $a^4 b^2 de^3 \dots$   |

$(a, b, c, d, e, f \nmid x, y)^0$  Invariant.

## V. No. 95 (continued).

|                          |                      |                           |
|--------------------------|----------------------|---------------------------|
| $a^5 b^2 d^2 f^3 + 1620$ | $a^4 b^4 df^3 - 276$ | $a^5 b^2 c^2 de^4 + 7100$ |
| $c^4 de^2 f^2 + 3408$    | $d^2 e^3 f^3 + 908$  | $c^2 d^5 ef + 12440$      |
| $c^4 e^3 f + 2156$       | $d^2 e^4 f^2 - 810$  | $c^2 d^4 e^2 f - 5200$    |
| $c^3 d^2 e f^2 - 15060$  | $d^6 \dots$          | $c^2 d^3 e^3 - 5340$      |
| $c^3 d^2 e^2 f - 2360$   | $b^3 c^3 f^4 - 12$   | $c^2 d^2 e^4 - 11900$     |
| $c^3 de^4 + 9260$        | $c^5 def^3 + 832$    | $c^3 d^3 e^2 + 10800$     |
| $c^2 d^3 f^2 + 4336$     | $c^5 e^2 f^2 + 1244$ | $c^3 d^2 e^3 - 2250$      |
| $c^2 d^2 e^2 f + 15220$  | $c^4 d^2 f^3 + 1112$ | $b^3 c^3 ef^2 + 486$      |
| $c^2 de^4 + 19920$       | $c^4 de^2 f^2 - 168$ | $c^4 d^2 f^2 + 810$       |
| $c^2 d^3 ef - 5808$      | $c^4 d^2 ef - 3510$  | $c^4 def^2 + 3330$        |
| $c^2 d^2 e^3 - 22740$    | $c^4 de^3 - 1350$    | $c^4 e^3 - 750$           |
| $cd^4 f - 90$            | $cd^5 f^2 + 1172$    | $c^3 d^2 ef - 8160$       |
| $cd^3 e^2 + 10080$       | $d^6 ef - 2060$      | $c^5 d^2 e^2 + 5400$      |
| $d^5 e - 1350$           | $d^4 e^3 + 450$      | $c^5 de^3 + 3100$         |

$(a, b, c, d, e, f \nmid x, y)^1$

W. No. 29A (continued).

|                         |                         |                       |
|-------------------------|-------------------------|-----------------------|
| $a^9 c^4 f^4 \dots$     | $a^8 b^2 d^2 f^4 \dots$ | $a^7 b^4 f^4 \dots$   |
| $a^8 c^3 d f^3 \dots$   | $a^7 b^3 c d f^3 \dots$ | $a^6 b^6 f^4 \dots$   |
| $a^7 c^2 d^2 f^2 \dots$ | $a^6 b^4 c^2 f^2 \dots$ | $a^5 b^6 d f^2 \dots$ |
| $a^6 c d^3 f \dots$     | $a^5 b^5 d^2 f \dots$   | $a^4 b^8 f^2 \dots$   |
| $a^5 d^4 \dots$         | $a^4 b^6 c d^2 \dots$   | $a^3 b^8 e^2 \dots$   |

$(a, b, c, d, e, f \text{ } \S \text{ } x, y)^0$  Invariant.

---

The tables of the covariants M to W of the binary quintic are thus completed. The entire system of 23 covariants (including the quintic itself and the four invariants) has been given; and the numerical verifications have been furnished in all cases where they are requisite. The present reprint has been arranged in a form which facilitates reference; and it is hoped that the results will be found useful in future investigations.

### Final Note on the Covariants of the Binary Quintic.

The present paper completes the series of investigations on the covariants of the binary quintic, which have been published in my several Memoirs on Quantics. The results are collected in a form which admits of easy reference; and it is believed that they are substantially correct.

The system of 23 covariants of the binary quintic is connected by a network of syzygies, which have been investigated in the Tenth Memoir on Quantics. These syzygies establish relations between the covariants of the same degree and order; and they enable us to express any covariant as a linear combination of the irreducible covariants. The principal syzygies are those which connect the covariants of degrees 2, 3, 4, and 5; and they have been fully set forth in the Memoir referred to.

The numerical tables supplementary to the Second Memoir, 142, give the number of literal terms of each degree and weight for the cubic, quartic, quintic, sextic, septic, and octavic functions. These tables have been carried to a sufficient extent to include the highest invariants of each function; and they serve as a verification of the completeness of the covariant tables.

The theory of the covariants of the binary quintic is a necessary preliminary to the theory of the equations of the fifth degree. The invariants  $G, Q, U, W$  are the fundamental invariants of the quintic; and the covariants  $A, B, C, \dots$  furnish a complete system, by means of which any covariant can be expressed. The syzygies enable us to reduce any expression to a linear combination of the fundamental covariants; and the numerical tables afford a ready means of ascertaining the number of terms in any covariant of a given degree and order.

I have endeavoured to present the results in a clear and systematic manner; and I have added references to the Memoirs in which the several covariants were first obtained. The notation has been made uniform throughout; and the numerical coefficients have been carefully verified. It is hoped that the present reprint will contribute to the advancement of this important branch of algebra.

A. CAYLEY.

## A THIRD MEMOIR UPON QUANTICS.

[Philosophical Transactions, vol. CXLVII (for the year 1857), pp. 415–427.

Presented December 18, 1856.]

1. *The object of the present Memoir is to continue the investigation of the covariants and invariants of binary quantics, which was commenced in the Second Memoir, Phil. Trans. vol. CXLVI (1856), pp. 101; and to apply the general theory to the sextic, the septic, and the octavic functions.*

2. I have shown in the Second Memoir that for a binary quantic of the order  $n$ , the number of aszygetic covariants of the degree  $m$  is equal to the coefficient of  $x^m$  in the development of

$$\frac{1}{(1-x)(1-x^2)\dots(1-x^n)}$$

when  $m$  is odd; and to the coefficient of  $x^m$  in the development of

$$\frac{1}{(1-x^2)(1-x^3)\dots(1-x^n)}$$

when  $m$  is even. I have also shown that the generating function for the covariants is

$$\frac{1}{(1-x)(1-xz)(1-xz^2)\dots(1-xz^n)};$$

and that the number of invariants of the degree  $m$  is equal to the number of partitions of  $m$  into the parts 2, 3, ...,  $n$ .

3. In the present Memoir I propose to develop these formulæ for the particular cases of the sextic ( $n = 6$ ), the septic ( $n = 7$ ), and the octavic ( $n = 8$ ); and to determine the complete systems of irreducible covariants and invariants for each of these functions. The investigation will be conducted on the same principles as those which were employed in the Second Memoir for the cubic, quartic, and quintic; but the increased complexity of the functions will render the results more extensive and the calculations more laborious.

4. The sextic function presents the first case in which the complete system of irreducible covariants contains more than four invariants. In fact, the sextic has five irreducible invariants, of the degrees 2, 4, 6, 8, 10; and the complete system of covariants is of considerable extent. The septic has still more irreducible invariants; and the octavic has a very extensive system, which has not yet been completely determined.

5. I shall begin with the sextic, and shall first determine the number and character of the irreducible covariants by means of the generating function. I shall then proceed to the actual determination of the covariants by the method of the differential equations; and I shall conclude with the numerical verification of the results. The same method will then be applied to the septic and the octavic; but the complete investigation of these functions will be reserved for future Memoirs.

6. I take this opportunity of referring to a former Memoir "On the Linear Transformations of the Quantics," Phil. Mag. vol. VI (1853), p. 1, in which I considered the general theory of the linear transformation of a binary quantic. The results there obtained are intimately connected with the theory of covariants; and they furnish a direct method of obtaining the invariants of a quantic by elimination. The connexion between the two theories will be pointed out in the course of the present investigation.

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[To be continued.]

143.

TABLES OF THE COVARIANTS M  
TO W OF THE BINARY QUINTIC.

[These tables continue from the preceding paper, giving the  
covariants M to W of the binary quintic  $(a, b, c, d, e, f \mid x, y)^5$ .]

V. No. 95 (CONTINUED).  $y$  coefficient.

| $a^5$         |      | $a^4$          |       | $a^3$          |        | $a^2$           |        |
|---------------|------|----------------|-------|----------------|--------|-----------------|--------|
| $b^5df^5$     | ...  | $b^5c^2def^4$  | -116  | $b^4c^4f^5$    | ...    | $b^4c^5d^2e^2f$ | +7312  |
| $e^5f^5$      | ...  | $c^5ef^5$      | +80   | $b^2c^5f^4$    | -20    | $c^5de^5$       | +2344  |
| $a^4 b^4cf^6$ | ...  | $cd^2f^4$      | +240  | $c^6df^3$      | -280   | $c^4d^3f^2$     | -124   |
| $d^2ef^5$     | ...  | $cde^2f^3$     | -160  | $c^5e^2f^2$    | +80    | $c^4d^2e^2f$    | -8020  |
| $e^4f^4$      | ...  | $cde^3f^2$     | -120  | $c^4e^3f$      | +300   | $c^4de^4$       | -10100 |
| $b^3c^2ef^5$  | ...  | $ce^4f$        | +76   | $cd^3ef^3$     | +160   | $c^3d^3ef$      | +3792  |
| $cd^2f^5$     | -2   | $d^3ef^3$      | -120  | $c^3d^2e^2f^2$ | ...    | $c^3d^2e^3$     | +14648 |
| $cde^2f^4$    | +4   | $d^2e^2f^2$    | -80   | $cde^4f$       | -192   | $c^3de^4$       | +4     |
| $ce^3f^3$     | -2   | $d^2e^3f$      | +368  | $ce^5$         | -108   | $c^2d^4f$       | -702   |
| $d^3ef^4$     | +6   | $de^4$         | -180  | $c^3d^3e^2$    | -10296 | $c^2d^3e^2$     | -2     |
| $d^2e^2f^3$   | -16  | $d^3e^2f^2$    | -56   | $c^2d^2e^3$    | +3564  | $c^2d^2e^2f$    | +368   |
| $de^4f$       | +14  | $b^2c^4ef^4$   | +24   | $d^3e^3f$      | +940   | $c^2de^4$       | -180   |
| $e^5$         | -4   | $d^2e^4$       | -1580 | $d^4e^2$       | -486   | $c^2e^5$        | -56    |
| $a^3 b^3f^6$  | ...  | $c^3de^2f^3$   | +320  | $d^5e$         | +756   | $cd^4e$         | +3564  |
| $b^2ce^2f^5$  | ...  | $c^3e^3f^2$    | -280  | $b^2c^4df^4$   | +360   | $d^3e^2f^2$     | -16    |
| $d^2f^6$      | +2   | $c^2d^2ef^3$   | -560  | $c^4e^2f^2$    | -420   | $b^2c^5f^3$     | +24    |
| $de^2f^5$     | -4   | $c^2de^2f^2$   | +1280 | $c^4def^3$     | +1440  | $d^3ef^3$       | +940   |
| $e^3f^4$      | +2   | $cd^3f^3$      | +332  | $c^3d^2f^3$    | -2160  | $d^2e^2f^2$     | -486   |
| $b^2c^2df^5$  | +10  | $c^2de^3f$     | -688  | $cde^4f$       | -496   | $e^6$           | +14    |
| $c^2e^2f^4$   | -10  | $c^2e^4$       | +984  | $e^5f^2$       | +2     | $b^2c^4def^3$   | -160   |
| $cd^2ef^4$    | -26  | $c^2d^3f^2$    | +288  | $c^4d^2ef$     | +184   | $d^3e^2f$       | -1580  |
| $cde^2f^3$    | +32  | $c^2d^2e^2f^2$ | -480  | $c^3e^3$       | +216   | $a^2 b^2c^4f^4$ | +6     |
| $ce^3f^2$     | -6   | $d^3ef^2$      | +304  | $b^2c^3d^2f^3$ | +10    | $e^2f^4$        | -4     |
| $d^3f^5$      | -30  | $cde^4$        | ...   | $cd^2f^3$      | +288   | $b^2c^3d^2f^3$  | +6     |
| $d^2e^2f^4$   | +84  | $c^3d^2ef^2$   | -10   | $c^3d^2e^2f^2$ | -480   | $c^3d^2ef^2$    | ...    |
| $de^3f^3$     | -50  | $cd^2ef^2$     | -240  | $c^2de^3f$     | +304   | $b^2c^3def^3$   | +72    |
| $e^4f^2$      | +22  | $c^2de^2f^2$   | -1320 | $c^2e^4$       | ...    | $c^3de^2f^2$    | -420   |
| $b^2c^3f^6$   | +6   | $d^2e^3f$      | -312  | $cde^3f^2$     | -26    | $b^2c^3e^2f^2$  | +44    |
| $c^3def^4$    | +32  | $c^2d^2e^2f$   | -480  | $c^2d^2e^2f$   | -480   | $d^3f^4$        | +2     |
| $c^2d^2f^5$   | -8   | $c^2de^3$      | +264  | $c^2de^2f^2$   | +2040  | $c^2d^2ef^2$    | -10    |
| $c^2def^4$    | +4   | $c^3d^3$       | -732  | $de^5$         | +32    | $c^2de^2f^2$    | +1280  |
| $c^2e^2f^3$   | +124 | $ce^4$         | -6    | $cd^2e^2$      | +264   | $c^2de^3f$      | -1320  |



Tabelle 2 — continued

|                          |        |              |       |                 |       |                |       |
|--------------------------|--------|--------------|-------|-----------------|-------|----------------|-------|
| $cd^3f^4$                | +36    | $d^2e^3$     | -144  | $c^3d^3$        | -732  | $c^2e^4$       | -312  |
| $cd^2ef^3$               | -36    | $cd^2e^2f$   | -768  | $e^6$           | -6    | $cd^3ef^2$     | -26   |
| $cd^2e^2f^2$             | -36    | $d^3f^2$     | -320  | $d^3e^3$        | -144  | $cd^2e^2f^2$   | -480  |
| $de^4f^2$                | +72    | $d^2e^2f$    | -30   | $cd^2e^3$       | -768  | $c^2d^2e^2f$   | +2040 |
| $e^5f$                   | -60    | $d^2e^2f$    | +336  | $d^3e^2f$       | +336  | $cde^4f$       | +32   |
| $d^4f^3$                 | -36    | $cd^3f^2$    | +2640 | $d^2e^4$        | +860  | $c^2de^3$      | +264  |
| $c^3d^2e^2f$             | -168   | $d^2e^3$     | +84   | $b^2c^2d^2ef^3$ | +84   | $c^2d^3$       | -732  |
| $c^3de^3f$               | +1668  | $c^2d^3f$    | -144  | $c^2d^2f^2$     | -144  | $e^6$          | +2    |
| $c^2d^2e^2f$             | +4800  | $c^2d^2ef$   | -1440 | $c^3d^2e^2f$    | -960  | $c^2d^2e^2f$   | -6    |
| $d^3e^3f^2$              | +72    | $c^2de^3f$   | +420  | $d^3e^2f^2$     | -50   | $d^3e^2f$      | -144  |
| $d^2e^3f$                | -576   | $c^2d^3$     | -220  | $b^2c^2df^4$    | +24   | $c^2d^2ef$     | -768  |
| $d^2e^4$                 | -6420  | $c^2e^3$     | +540  | $d^4f^2$        | +504  | $d^3f^3$       | -320  |
| $c^3e^3$                 | +2520  | $c^2d^2e^2$  | -30   | $c^3d^3f^2$     | +1740 | $d^2e^2f$      | -30   |
| $d^3e^3$                 | -60    | $d^2e^3f$    | +60   | $d^3ef$         | -1296 | $c^2d^2f^2$    | +336  |
| $d^2e^4$                 | +36    | $d^2e^2f$    | -1296 | $d^2e^3$        | -2160 | $cd^3f^2$      | +2640 |
| $cd^2e^2f$               | -576   | $e^5$        | +18   | $c^2d^2e^2f$    | +280  | $d^2e^3$       | +860  |
| $d^3e^2f$                | +8160  | $c^2d^2ef$   | +280  | $d^2e^4$        | +648  | $c^2d^2e^2$    | +84   |
| $d^2e^3$                 | +18    | $c^2de^3$    | +420  | $cd^3f^2$       | +420  | $c^2d^3$       | -144  |
| $c^2d^2e^2$              | +2988  | $b^2c^2f^5$  | -6    | $b^2c^2def^3$   | -440  | $c^2d^2ef$     | -1440 |
| $c^2de^3f$               | -13360 | $c^2d^2f^3$  | -108  | $c^2d^2ef^2$    | +400  | $c^3d^2e^2f$   | -960  |
| $a^2 \quad b^2c^2e^2f^3$ | ...    | $cd^2e^2f$   | -2640 | $c^2d^2e^2f$    | +2910 | $d^3e^2f^2$    | -50   |
| $e^3f^3$                 | -6     | $c^2de^2f$   | +32   | $c^2d^3$        | -8    | $b^2c^2df^4$   | +24   |
| $d^3f^3$                 | +162   | $c^2e^2f^2$  | +400  | $c^2d^3f^2$     | -140  | $d^4f^2$       | +504  |
| $c^2d^2e^2$              | -6000  | $c^2def^2$   | -1296 | $d^3e^2f^2$     | +2344 | $c^3d^2f^2$    | +1740 |
| $b^2cdf^5$               | -16    | $cc^2e^2f$   | +2910 | $cde^4$         | +540  | $d^3ef$        | -1296 |
| $b^2c^2f^5$              | -10    | $c^2d^3$     | -8    | $b^2c^2d^2f^3$  | +4    | $d^2e^3$       | -2160 |
| $c^4e^2f^2$              | -594   | $c^2d^2f^2$  | -140  | $c^3d^3$        | +420  | $e^5$          | +18   |
| $cd^3f^3$                | -2288  | $d^2e^2f^2$  | +2344 | $d^3e^2f^2$     | -700  | $c^2d^2e^2f$   | +280  |
| $ce^3f^2$                | +16    | $cde^4$      | +540  | $c^2d^3e^2f$    | -104  | $c^2d^2e^3$    | +648  |
| $cde^2f^2$               | +180   | $b^2cdf^4$   | +6    | $c^2d^2e^3$     | +40   | $c^2de^3f$     | -440  |
| $b^2c^3ef^3$             | +432   | $cd^2e^2f$   | -6000 | $c^3d^2ef^2$    | +600  | $c^2d^2e^2f^2$ | +2910 |
| $cd^2e^2f$               | -1312  | $c^2e^3$     | -36   | $d^3e^2f$       | +1840 | $c^2d^3$       | -8    |
| $d^3ef^2$                | +8     | $c^2d^2e^2$  | +2988 | $c^3d^2f^2$     | +90   | $c^2d^2f^2$    | -140  |
| $c^4f^2$                 | -160   | $c^2d^2e^2f$ | -36   | $c^2d^2e^2$     | -368  | $d^3e^2f^2$    | +2344 |
| $c^3de^2f$               | -144   | $cd^2e^2f$   | +36   | $c^3de^3$       | -3420 | $cde^4$        | +540  |
| $cd^2e^3$                | +9360  | $d^2e^3f$    | +72   | $d^3e^2f^2$     | -1530 | $b^2cdf^4$     | +6    |
| $d^3e^2f$                | ...    | $d^2e^4$     | -60   | $c^2e^4$        | -26   | $c^2de^2f$     | +400  |
| $d^2e^2f^2$              | -130   | $d^2e^2f$    | +36   | $c^2d^2e^2f$    | +108  | $c^2def^2$     | -1296 |
| $d^2e^3f$                | -1656  | $c^2d^2e^2f$ | -576  | $c^3e^3$        | -1172 | $cc^2e^2f$     | +2910 |
| $d^2ef^2$                | +1824  | $c^2e^3f$    | +3360 | $b^2c^2f^5$     | +96   | $c^2d^3$       | -8    |
| $d^2e^3$                 | -8     | $d^3e^2f^2$  | -36   | $cd^2e^2f$      | +96   | $c^2d^2f^2$    | +420  |
| $d^2e^2f$                | +60    | $cd^2e^3$    | -168  | $c^2d^2e^2f$    | -40   | $d^3e^2f^2$    | -700  |
| $c^3d^2e^2$              | -1516  | $c^2d^2e^2f$ | +1668 | $c^2d^2ef^2$    | +900  | $c^2d^3e^2f$   | -104  |
| $d^4f^2$                 | -2880  | $c^2de^3f$   | +4800 | $c^2de^3$       | +80   | $c^2d^2e^3$    | +40   |

Tabelle 2 — continued

|                |      |             |       |               |       |              |       |
|----------------|------|-------------|-------|---------------|-------|--------------|-------|
| $b^2c^2f^4$    | +12  | $d^2e^3f$   | +72   | $cd^2e^3f$    | -160  | $c^3d^2ef^2$ | +600  |
| $de^2f^2$      | +60  | $c^2d^2e^2$ | -576  | $c^2d^2e^2f$  | +376  | $d^3e^2f$    | +1840 |
| $b^2c^2df^4$   | +912 | $d^3e^3$    | -6420 | $b^2c^2def^3$ | -1280 | $c^3d^2f^2$  | +90   |
| $b^2c^2e^2f^3$ | -72  | $c^2e^4$    | +2520 | $c^3e^3f^2$   | -3240 | $c^2d^2e^2$  | -368  |

For the Numerical Verifications see p. 308. C. II. 38

V. NO. 95 (CONCLUDED).  $y$  coefficient.

| $a^4$ |                 |        | $a^3$           |        | $a^2$          |        | $a^1$          |      |
|-------|-----------------|--------|-----------------|--------|----------------|--------|----------------|------|
| $a^4$ | $b^4df^5$       | +1620  | $b^4cde^2f^3$   | +3408  | $b^3c^3f^5$    | +276   | $b^3c^4def^2$  | +71  |
|       | $c^4ef^5$       | +908   | $c^5e^2f^3$     | +12440 | $c^6e^2f^2$    | +2156  | $c^5d^2ef$     | +34  |
| $a^3$ | $b^4c^2de^2f^2$ | -810   | $c^4d^2e^2f^2$  | -15060 | $c^5de^3$      | -5200  | $c^4d^3e^2$    | -53  |
|       | $c^4d^2e^3f$    | -2360  | $c^3d^4e^2$     | -11900 | $b^2c^4df^4$   | -12    | $c^4d^2e^2f$   | -92  |
|       | $b^2c^4f^6$     | -12    | $c^3d^3e^3$     | +10800 | $c^4d^2ef^3$   | +832   | $c^4d^3e^3$    | +108 |
|       | $c^4d^3e^2f$    | +4336  | $c^3d^2e^4$     | +1244  | $c^4d^2e^2f^2$ | -2250  | $b^2c^3d^2f^4$ | +152 |
|       | $c^3d^5f$       | +1112  | $b^2c^3de^2f^3$ | +486   | $c^3d^3ef^2$   | +19920 | $c^4d^2e^2f$   | +8   |
|       | $c^3d^4e^2$     | -5808  | $c^4d^2e^3f$    | -3510  | $c^3d^2e^3f$   | +3330  | $c^3d^3e^2f^2$ | -1   |
|       | $c^2d^6$        | -22740 | $c^3d^3e^3$     | -1350  | $c^2d^5f$      | -750   | $c^3d^2e^3$    | +8   |
|       | $c^2d^4e^2$     | -1350  | $cd^6$          | +90    | $c^3d^4e^2$    | -1350  | $c^2d^4ef$     | +11  |
|       | $c^2d^3e^3$     | +1350  | $c^3d^4f^2$     | -864   | $c^2d^4e^2$    | +1350  | $c^2d^3e^3$    | +31  |
|       | $b^2c^3d^2f^5$  | -864   | $c^3d^3e^2f$    | +1800  | $c^2d^3e^3$    | -1008  | $c^2d^2e^4$    | -20  |
|       | $c^3d^3e^2f$    | +1172  | $c^3d^2e^4$     | -5344  | $c^2d^2e^4$    | +450   | $c^2d^5e$      | -91  |
|       | $b^2c^3e^2f^4$  | -240   | $c^3d^2e^3f$    | +1800  | $b^2c^3df^5$   | -240   | $c^2d^3e^2f^2$ | +66  |
|       | $b^2c^3d^2f^5$  | -162   | $c^2d^4ef$      | -1620  | $b^2c^3e^2f^4$ | -162   | $c^2d^3e^3$    | +4   |
|       | $c^4d^2f^3$     | +1720  | $c^2d^3e^2f^2$  | -810   | $c^4df^4$      | +1720  | $c^2d^2e^4$    | +18  |
|       | $c^3d^3ef^2$    | -1620  | $c^2d^2e^3f$    | -1912  | $c^3d^2ef^3$   | -1620  | $c^2d^2e^3f$   | -53  |
|       | $c^2d^4e^2f$    | -810   | $c^2d^2e^4$     | +1620  | $c^2d^3e^2f^2$ | -3480  | $b^2c^2d^2f^5$ | -2   |
|       | $c^2d^3e^3$     | +3712  | $c^2d^2e^4$     | +1500  | $c^2d^2e^3f$   | +1620  | $b^2c^2e^2f^4$ | -1   |
|       | $c^2d^2e^4$     | -430   | $c^2d^5e$       | +4920  | $c^2de^5$      | +4920  | $c^3d^2f^4$    | +17  |
|       | $c^2d^4e^2$     | +1500  | $c^2d^3e^3$     | +2800  | $c^2d^5e$      | -4768  | $c^2d^2ef^3$   | -16  |
|       | $c^2d^3e^3$     | -3150  | $c^2d^2e^4$     | -600   | $c^2d^4e^2$    | +3840  | $c^2de^2f^2$   | -8   |
|       | $c^2d^2e^4$     | -16520 | $cd^5e^2$       | -3150  | $c^2d^3e^3$    | -3150  | $c^2d^3e^3$    | +37  |
|       | $cd^6f$         | +7200  | $b^2c^2d^2ef^4$ | +1620  | $cd^6e$        | +7200  | $c^2d^2e^4$    | -4   |
|       | $c^2d^5e$       | +2000  | $c^2d^3e^3$     | +3240  | $b^2c^2f^6$    | +162   | $c^2d^4e^2$    | +15  |
|       | $c^2d^4e^2f$    | +1920  | $c^2d^2e^3f$    | -918   | $b^2c^2df^5$   | -918   | $c^2d^3e^3$    | +49  |
|       | $c^2d^3e^3$     | -2540  | $d^5ef$         | -1290  | $c^2d^2f^5$    | -1290  | $c^2d^2e^4$    | +28  |
|       | $c^2d^2e^4$     | -400   | $c^2d^4e^2$     | +1956  | $c^2d^2e^2f^3$ | +1956  | $c^2d^2e^4$    | -6   |
|       | $c^2de^5$       | +19440 | $d^4e^2f$       | +900   | $c^2de^4f$     | +900   | $c^2d^2e^4$    | -165 |
|       | $c^2d^4e^2$     | +1180  | $c^2d^3e^2f$    | +240   | $c^2d^4e^2$    | +876   | $c^2d^3e^3$    | +72  |
|       | $c^2d^3e^3$     | -9540  | $b^2c^2d^2f^4$  | +876   | $c^2d^3e^2f$   | -2952  | $cd^5ef$       | +20  |
|       | $c^2d^2e^4$     | -10300 | $c^2d^2e^3f$    | -3440  | $c^2d^2e^4$    | +1224  | $c^2d^4e^2f$   | +19  |
|       | $cd^5e^2$       | +1620  | $c^2d^3e^3$     | +2052  | $c^2d^3e^3$    | -3440  | $c^2d^3e^3$    | -25  |
|       | $c^2d^3e^3$     | +3240  | $c^2d^2e^3f$    | +2608  | $c^2d^2e^3f$   | +2052  | $c^2d^2e^4$    | -4   |
|       | $b^2c^2f^6$     | +162   | $c^2de^4f$      | +2800  | $c^2de^4f$     | +2608  | $c^2de^5$      | +194 |
|       | $c^2d^2e^4$     | +600   | $c^2d^2e^4$     | +8760  | $c^2d^2e^4$    | +2800  | $c^2d^3e^3$    | +11  |
|       | $d^4e^2f^2$     | -918   | $c^2d^2e^4$     | -1900  | $c^2d^2e^4$    | +8760  | $c^2d^2e^4$    | -95  |
|       | $d^3e^3f$       | -1290  | $c^2d^3e^2f$    | -796   | $d^4e^3$       | -1900  | $c^2d^2e^4$    | -103 |
|       | $c^2d^3e^3$     | +2100  | $c^2d^4e^2$     | -9160  | $c^2d^4e^2$    | -796   | $c^2d^3e^3$    | +16  |
|       | $c^2d^2e^4$     | -9160  | $c^2d^3e^3$     | -8540  | $c^2d^3e^3$    | +2100  | $c^2d^2e^4$    | +32  |

Tabelle 3 — continued

|                   |        |                   |        |                |        |                   |      |
|-------------------|--------|-------------------|--------|----------------|--------|-------------------|------|
| $c^2de^5$         | +4260  | $c^2d^2e^4$       | +4260  | $c^2d^2e^4$    | -9160  | $b^2c^2f^6$       | +1   |
| $c^2d^2e^4$       | -10100 | $c^2d^2e^4$       | -720   | $c^2de^5$      | -8540  | $c^2d^2e^4$       | +6   |
| $c^2d^2e^4$       | -720   | $c^2d^2e^4$       | -6240  | $c^2d^2e^4$    | +4260  | $d^4e^2f^2$       | -9   |
| $c^2de^5$         | -6240  | $a^2$ $b^2c^2f^5$ | +2     | $c^2d^2e^4$    | -10100 | $d^3e^3f$         | -12  |
| $a^2$ $b^2c^2f^5$ | +2     | $c^2d^3e^3$       | +23300 | $c^2d^2e^4$    | -720   | $c^2d^3e^3$       | +21  |
| $c^2d^2e^4f$      | +34    | $b^2c^2ef^4$      | +3640  | $c^2de^5$      | -6240  | $c^2d^2e^4$       | -91  |
| $c^2d^2e^4$       | -54    | $c^3e^2f^3$       | -54    | $b^2c^2f^5$    | +2     | $c^2de^5$         | +42  |
| $c^2d^3e^3$       | -8800  | $de^4f^2$         | +252   | $c^2d^2e^4$    | +34    | $c^2d^2e^4$       | -101 |
| $d^3e^4$          | -750   | $c^2d^2e^3$       | -216   | $c^2de^5$      | -54    | $c^2d^2e^4$       | -7   |
| $d^2e^4f$         | +450   | $b^2c^2d^2f^4$    | +38    | $d^3e^4$       | -750   | $c^2de^5$         | -62  |
| $b^2c^2df^5$      | +38    | $b^2c^2ef^3$      | -162   | $d^2e^4f$      | +450   | $a^2$ $b^2c^2f^5$ | .    |
| $b^2c^2e^2f^4$    | -162   | $c^2d^2ef^3$      | -404   | $b^2c^2df^5$   | +38    | $c^2d^2e^4f$      | +    |
| $c^2d^2ef^3$      | -404   | $c^2de^2f^2$      | -2304  | $b^2c^2e^2f^4$ | -162   | $c^2d^2e^4$       | -    |
| $c^2de^2f^2$      | -2304  | $cd^2e^2f^2$      | -376   | $c^2d^2ef^3$   | -404   | $c^2d^3e^3$       | -88  |
| $cd^2e^2f^2$      | -376   | $c^2d^2e^3f$      | -1680  | $c^2de^2f^2$   | -2304  | $d^3e^4$          | -7   |
| $c^2d^2e^3f$      | -1680  | $cde^3f^2$        | -216   | $cd^2e^2f^2$   | -376   | $d^2e^4f$         | +4   |
| $cde^3f^2$        | -216   | $c^2d^3f^2$       | -1560  | $c^2d^2e^3f$   | -1680  | $b^2c^2df^5$      | +    |
| $c^2d^3f^2$       | -1560  | $c^2e^2f^2$       | +1080  | $cde^3f^2$     | -216   | $b^2c^2e^2f^4$    | -1   |
| $c^2e^2f^2$       | +1080  | $c^2d^2e^2f$      | ...    | $c^2d^3f^2$    | -1560  | $c^2d^2ef^3$      | -4   |

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| $a^6$                 | $a^5$                  | $a^4$                   | $a^3$                     |
|-----------------------|------------------------|-------------------------|---------------------------|
| $b^6 f^9$ ...         | $b^4 c^2 e f^7$ -15    | $b^3 c^2 d e^2 f^5$ -90 | $b^2 c^3 d^2 e^2 f$ +1320 |
| $a^5$ $b^5 e f^8$ ... | $d^2 e^2 f^6$ +10      | $c^4 e^2 f^5$ +30       | $c^4 d e^5$ -260          |
| $b^4 c d f^8$ ...     | $d^2 e^3 f^4$ -35      | $c d^3 e f^5$ -210      | $c^3 d^2 f^4$ +60         |
| $c e^2 f^7$ ...       | $d^2 e^4 f^2$ +40      | $c d^2 e^2 f^4$ +120    | $c^3 d e^2 f^3$ -500      |
| $d^2 e f^7$ ...       | $d e^5 f$ -10          | $c d^2 e^3 f^2$ -420    | $c^3 d^2 e^2 f^2$ -1995   |
| $d e^3 f^6$ ...       | $e^7$ +5               | $c^2 d e^4 f$ -10       | $d^5 e^2 f^2$ +360        |
| $e^5 f^5$ ...         | $c e^5 f^2$ +130       | $c^2 d^2 e^2 f^3$ +360  | $c^2 d^3 e^3 f$ +2235     |
| $b^3 c^2 e f^7$ ...   | $c^2 d^2 e^3 f$ +370   | $d^4 e^2 f^2$ -10       | $d^4 e^3 f$ -1995         |
| $c^2 d^2 f^7$ ...     | $b^3 c^2 f^7$ -1       | $c d e^4 f^2$ -420      | $e^9$ +5                  |
| $c^2 d e^2 f^6$ ...   | $d^4 f^4$ -5           | $c^2 d^3 e f^2$ +360    | $c e^5 f^2$ +130          |
| $c^2 e^3 f^5$ ...     | $c^2 d e^2 f^4$ +15    | $d^4 e^2 f^2$ +195      | $c^2 d^2 e^3 f$ +370      |
| $c d e^3 f^5$ ...     | $d^3 e f^3$ -35        | $c d^3 e^2 f^2$ -315    | $a^3$ $b^3 c^2 d f^5$ -1  |
| $c d^2 e^2 f^5$ ...   | $c^2 e^3 f^2$ -10      | $c^2 d^2 e^2 f^2$ +40   | $b^3 c^2 e f^5$ -5        |
| $c d^2 e^3 f^4$ ...   | $d^2 e^3 f^2$ +40      | $c d^3 e^2 f$ -210      | $c^2 d^2 e^2 f^3$ +360    |
| $c e^4 f^4$ ...       | $c^2 d^2 e^2 f^2$ -90  | $d^3 e^2 f^2$ +165      | $c d e f^4$ ...           |
| $c^2 d^2 e f^5$ ...   | $d^2 e^2 f^2$ +120     | $d^3 e^3 f$ -81         | $c^2 d^2 e^2 f^2$ +195    |
| $c^2 d e^2 f^4$ ...   | $d e^5$ -75            | $d^2 e^4$ +270          | $c d^3 e^2 f^2$ -315      |
| $c^2 e^3 f^3$ ...     | $d^2 e^3 f$ +40        | $b^2 c^2 e^2 f^4$ -40   | $c^2 d^2 e^2 f^2$ +40     |
| $c d^3 f^6$ -15       | $b^2 c^2 e^2 f^4$ -10  | $d^2 e^3 f^2$ -225      | $d^3 e^2 f^2$ -210        |
| $c d^2 e^2 f^5$ +60   | $c^2 d^2 f^3$ +60      | $d e^5 f$ -60           | $d^3 e^3 f$ +165          |
| $c d^2 e^3 f^4$ -90   | $c^2 d e^2 f^2$ +30    | $c^2 d^2 e^2 f^2$ +30   | $d^2 e^4$ -81             |
| $c d e^4 f^3$ +60     | $c^2 d e p$ +210       | $c^2 d e^3 f^2$ +120    | $d e^5$ +270              |
| $d^4 f^6$ ...         | $a^4$ $b^4 e f^6$ -180 | $c^2 d^2 e^2 f$ -180    | $d^2 e^3 f$ +120          |
| $d^3 e f^5$ +1        | $c^2 e^3 f$ -110       | $c^2 d^2 e^3$ +120      | $d^2 e^2 f^2$ -75         |
| $d^3 e^2 f^4$ -5      | $b^3 c d f^5$ -15      | $c^2 d e^3 f$ ...       | $e^6$ +45                 |
| $d^2 e^3 f^3$ +10     | $c^2 d^2 e^2 f^2$ -120 | $c^2 d^2 e^3$ -20       | $b^2 c^2 d^2 e^2 f^2$ +30 |
| $d^2 e^4 f^2$ -10     | $c^2 d^2 e^3$ -360     | $d e^5$ +60             | $c^2 d^2 e^3 f$ -180      |
| $d e^5 f$ +5          | $d e f^5$ -15          | $c^2 d^2 e^3$ +210      | $c^2 d^2 e^2 f$ +120      |
| $e^6$ -1              | $d^2 e^2 f^3$ +60      | $c^2 d^2 e^2 f^2$ -240  | $c^2 d e^3 f$ -180        |
| $a^4$ $b^4 c f^7$ ... | $c d e^3 f^2$ +240     | $d^2 e^3 f$ +265        | $c^2 d^2 e^3$ -210        |
| $d e f^7$ ...         | $c^2 d^2 e^2 f^2$ +30  | $b^2 c^2 d^2 f^3$ +30   | $c d e^4$ +1140           |
| $e^2 f^6$ ...         | $c d^2 e^2 f^2$ -200   | $c^2 d^2 e^2 f^2$ -200  | $c d^2 e^3 f$ -210        |
| $b^3 c^2 e f^6$ ...   | $c^2 d e^3 f^2$ -210   | $c^2 d e^3 f^2$ -210    | $c^2 d^2 e^3$ -180        |
| $c^2 d^2 f^6$ ...     | $c^2 d^2 e^2 f$ +65    | $c^2 d^2 e^2 f$ +90     | $d e^5$ +60               |
| $c^2 d e^2 f^5$ ...   | $c^2 d e^3$ -10        | $b^2 c d^2 f^4$ -10     | $d^2 e^3 f^2$ -100        |
| $c^2 e^3 f^4$ ...     | $d^2 e^3 f^2$ +1140    | $c^2 d^2 e^3$ -10       | $c^2 d^2 e^2 f^2$ +130    |
| $c d^3 f^6$ ...       | $c d^2 e^3 f$ -120     | $d^2 e^3 f^2$ +1140     | $c^2 d^2 e^3 f$ +310      |
| $c d^2 e f^5$ ...     | $d^2 e^4$ -5           | $c d^2 e^3 f$ -120      | $c^2 d^3$ -10             |
| $c d e^3 f^4$ ...     | $d^3 e^2 f$ +45        | $d^2 e^4$ -5            | $c^2 d^2 e^2 f$ -240      |
| $c d^2 e^2 f^4$ ...   | $c^2 d e^2 f^2$ -870   | $c^2 d e^2 f^2$ +45     | $d^2 e^3 f^2$ +310        |

Tabelle 4 — continued

|                |     |                |       |                |       |                |       |
|----------------|-----|----------------|-------|----------------|-------|----------------|-------|
| $ce^4f^3$      | ... | $cd^2e^3f$     | +90   | $c^2def^2$     | -870  | $c^2d^2e^3$    | -50   |
| $d^3e^2f^4$    | ... | $c^2e^3f$      | +10   | $cd^2e^3f$     | +90   | $a^3b^3c^2f^6$ | -10   |
| $d^2e^3f^3$    | ... | $d^2e^3f^2$    | -100  | $c^2e^3f$      | +10   | $b^3ce^2f^5$   | +81   |
| $d^2e^4f^2$    | ... | $c^2d^2e^2$    | +130  | $d^2e^3f^2$    | -100  | $c^2d^2e^2f^2$ | -5    |
| $d^2e^4f$      | ... | $cde^3f$       | +310  | $c^2d^2e^2$    | +130  | $d^2e^4$       | +45   |
| $de^5f^2$      | ... | $c^2d^3$       | -10   | $cde^3f$       | +310  | $c^2de^2f^2$   | -870  |
| $e^6f$         | ... | $d^2e^3f$      | +280  | $c^2d^3$       | -10   | $cd^2e^3f$     | +90   |
| $a^3b^3cf^7$   | ... | $c^2d^2e^2f$   | +30   | $d^2e^3f$      | +280  | $c^2e^3f$      | +10   |
| $b^3c^2f^7$    | ... | $de^4f^2$      | -180  | $c^2d^2e^2f$   | +30   | $d^2e^3f^2$    | -100  |
| $b^3ce^2f^6$   | ... | $e^4f^2$       | -90   | $de^4f^2$      | -180  | $c^2d^2e^2$    | +130  |
| $c^2df^6$      | ... | $b^2c^2e^2f^4$ | -180  | $e^4f^2$       | -90   | $cde^3f$       | +310  |
| $c^2e^2f^5$    | ... | $de^4f^2$      | ...   | $b^2c^2e^2f^4$ | -180  | $c^2d^3$       | -10   |
| $cd^2f^6$      | ... | $d^2e^3f^2$    | +300  | $de^4f^2$      | ...   | $d^2e^3f$      | +280  |
| $cd^2e^2f^5$   | ... | $b^2c^2d^2f^4$ | +35   | $d^2e^3f^2$    | +300  | $c^2d^2e^2f$   | +30   |
| $cde^3f^5$     | ... | $cd^2f^3$      | -30   | $b^2c^2d^2f^4$ | +35   | $de^4f^2$      | -180  |
| $ce^4f^4$      | ... | $c^2e^2f^3$    | -30   | $cd^2f^3$      | -30   | $e^4f^2$       | -90   |
| $d^3f^6$       | ... | $d^2e^2f^2$    | +50   | $c^2e^2f^3$    | -30   | $b^2c^2e^2f^4$ | -180  |
| $d^2ef^5$      | ... | $d^2e^3f$      | +15   | $d^2e^2f^2$    | +50   | $de^4f^2$      | ...   |
| $d^2e^2f^4$    | ... | $c^2d^2e^2f^2$ | -60   | $d^2e^3f$      | +15   | $d^2e^3f^2$    | +300  |
| $d^2e^3f^3$    | ... | $c^2de^2f^2$   | +360  | $c^2d^2e^2f^2$ | -60   | $b^2c^2d^2f^4$ | +35   |
| $de^4f^3$      | ... | $e^4f^2$       | +60   | $c^2de^2f^2$   | +360  | $cd^2f^3$      | -30   |
| $e^5f^2$       | ... | $c^2d^2e^2f^2$ | -270  | $e^4f^2$       | +60   | $c^2e^2f^3$    | -30   |
| $b^2c^2df^6$   | ... | $c^2d^2e^2f^2$ | -210  | $c^2d^2e^2f^2$ | -270  | $d^2e^2f^2$    | +50   |
| $b^2c^2e^2f^5$ | ... | $b^2c^2f^5$    | +10   | $c^2d^2e^2f^2$ | -210  | $d^2e^3f$      | +15   |
| $b^2c^2def^5$  | ... | $c^2d^2e^2f^2$ | +575  | $b^2c^2f^5$    | +10   | $c^2d^2e^2f^2$ | -60   |
| $b^2c^2e^2f^4$ | ... | $c^2d^2e^3f$   | -40   | $c^2d^2e^2f^2$ | +575  | $c^2de^2f^2$   | +360  |
| $c^2d^2e^2f^4$ | ... | $c^2d^2e^2f^2$ | +225  | $c^2d^2e^3f$   | -40   | $e^4f^2$       | +60   |
| $c^2de^3f^4$   | ... | $cd^2e^3f^2$   | -1700 | $c^2d^2e^2f^2$ | +225  | $c^2d^2e^2f^2$ | -270  |
| $c^2d^2e^2f^3$ | ... | $c^2d^3$       | +10   | $cd^2e^3f^2$   | -1700 | $c^2d^2e^2f^2$ | -210  |
| $c^2d^2e^3f^2$ | ... | $c^2d^2e^2f^2$ | +615  | $c^2d^3$       | +10   | $b^2c^2f^5$    | +10   |
| $c^2de^4f^2$   | ... | $cd^2e^2f^2$   | +480  | $c^2d^2e^2f^2$ | +615  | $c^2d^2e^2f^2$ | +575  |
| $cd^2e^3f^3$   | ... | $c^2d^2e^3f$   | -660  | $cd^2e^2f^2$   | +480  | $c^2d^2e^3f$   | -40   |
| $cd^2e^4f^2$   | ... | $cd^2f^3$      | +670  | $c^2d^2e^3f$   | -660  | $c^2d^2e^2f^2$ | +225  |
| $cde^5f$       | ... | $cd^2e^3f$     | -15   | $cd^2f^3$      | +670  | $cd^2e^3f^2$   | -1700 |
| $d^3e^2f^3$    | ... | $c^2de^2f^2$   | -60   | $cd^2e^3f$     | -15   | $c^2d^3$       | +10   |
| $d^2e^3f^3$    | ... | $d^2e^2f^2$    | +45   | $c^2de^2f^2$   | -60   | $c^2d^2e^2f^2$ | +615  |
| $d^2e^4f^2$    | ... | $ce^5f$        | -315  | $d^2e^2f^2$    | +45   | $cd^2e^2f^2$   | +480  |
| $d^2e^5f$      | ... | $ce^4f^2$      | +60   | $ce^5f$        | -315  | $c^2d^2e^3f$   | -660  |
| $de^6$         | ... | $c^2d^2e^2f^2$ | +40   | $ce^4f^2$      | +60   | $cd^2f^3$      | +670  |

For the Numerical Verifications see p. 309. 38—2

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| $a^4$          |        | $a^4$ cont.       |        | $a^3$             |        | $a^3$ cont.       |        |
|----------------|--------|-------------------|--------|-------------------|--------|-------------------|--------|
| $b^4c^2de^3f$  | -2080  | $b^3c^2d^2e^2f^3$ | +600   | $b^3d^2f^6$       | -196   | $b^3c^2d^2e^2f^2$ | -3880  |
| $d^4e^3$       | +705   | $d^6ef^2$         | -945   | $d^4e^2f^4$       | -660   | $b^2c^4ef^4$      | +300   |
| $b^2c^3df^6$   | +110   | $d^5e^2f$         | +900   | $d^3e^3f^3$       | +1840  | $c^5d^2f^3$       | +500   |
| $c^5e^2f^4$    | -195   | $d^4e^3$          | +45    | $c^4d^2f^4$       | +600   | $c^4d^2ef^3$      | +3810  |
| $c^4d^2ef^4$   | +210   | $b^2c^3ef^5$      | +180   | $c^4de^3f^3$      | -195   | $c^4de^2f^2$      | +210   |
| $c^4de^2f^3$   | +575   | $c^4d^2f^4$       | -60    | $d^4e^2f^2$       | +45    | $b^2c^3d^2e^2f^2$ | +180   |
| $c^4e^3f^2$    | +660   | $c^3de^2f^3$      | -1420  | $d^3e^3f$         | -180   | $d^4e^2f^2$       | +1840  |
| $c^3d^3f^3$    | -225   | $b^2c^3e^2f^4$    | -300   | $c^3e^4f^2$       | -3710  | $c^3d^2e^2f^2$    | -180   |
| $c^3d^2e^2f^2$ | +1350  | $c^3d^3ef^2$      | -265   | $c^3d^2e^2f^2$    | -575   | $c^3de^3f$        | -3710  |
| $c^3de^3f^2$   | -1440  | $c^3d^2e^3f$      | +16120 | $e^7$             | +216   | $c^3d^2e^2f^2$    | -1420  |
| $c^3d^2e^3f$   | -225   | $c^3d^2e^2f^2$    | -225   | $c^3d^3e^2f$      | -14040 | $c^3d^2e^3f$      | +16120 |
| $c^3e^4f$      | +25    | $c^3e^4f$         | +315   | $c^3d^2e^3f$      | +660   | $c^3d^2e^2f^2$    | +780   |
| $c^2d^4f^2$    | -540   | $c^2d^4ef$        | -540   | $b^2c^3d^2e^2f^3$ | -265   | $c^3de^3f$        | +180   |
| $c^2d^3e^2f$   | +1350  | $c^2d^3e^2f^2$    | +1350  | $c^3e^4f^2$       | -225   | $d^4e^2f$         | +600   |
| $c^2d^2e^3f^2$ | +780   | $c^2d^2e^3f$      | -2945  | $c^2d^4e^2$       | -540   | $c^3d^2e^2f^2$    | +5760  |
| $c^2d^2e^4f$   | +180   | $c^2de^4$         | -1840  | $c^2d^3e^3$       | +1700  | $c^3de^3f^2$      | +1700  |
| $c^2de^5$      | +600   | $c^2d^3e^2f$      | +5760  | $c^2d^2e^4$       | +7020  | $c^2d^3e^3$       | +7020  |
| $c^2d^3e^3$    | +5760  | $c^2d^2e^3f$      | -1440  | $c^2d^2e^3f$      | -2945  | $c^2d^2e^4$       | -1440  |
| $c^2d^2e^4$    | -1440  | $c^2de^4f^2$      | -2945  | $c^2de^5f$        | -1840  | $c^2d^2e^3f$      | -2945  |
| $c^2d^2e^4$    | -2945  | $c^2d^2e^4$       | -1840  | $c^2d^2e^4$       | ...    | $c^2de^5$         | -1840  |
| $c^2de^5f$     | -75    | $c^2d^2e^4$       | +1390  | $c^2e^6$          | -75    | $c^2d^2e^4$       | +1390  |
| $c^2d^2e^4$    | +1390  | $c^2d^3e^2f$      | -615   | $c^2d^3e^3$       | +1390  | $c^2d^2e^4$       | -615   |
| $c^2d^3e^3$    | -615   | $c^2d^2e^3f^2$    | -1950  | $c^2d^2e^3f$      | -615   | $c^2d^2e^3f^2$    | -1950  |
| $c^2d^2e^3f^2$ | -1950  | $c^2d^2e^4f$      | -1965  | $c^2d^2e^4$       | -1950  | $c^2d^2e^3f$      | -1965  |
| $cd^5ef$       | ...    | $c^2d^3e^3$       | ...    | $cd^5e^2f$        | ...    | $c^2d^2e^4$       | ...    |
| $c^2d^3e^3$    | ...    | $c^2d^2e^3f^2$    | -1350  | $c^2d^3e^3$       | ...    | $c^2d^2e^3f^2$    | -1350  |
| $c^2d^2e^3f^2$ | -1350  | $c^2d^2e^3f^2$    | -17670 | $c^2d^2e^3f^2$    | -1350  | $c^2d^2e^4f$      | -17670 |
| $c^2d^2e^4f$   | -17670 | $c^2d^3e^2f^2$    | +6000  | $c^2d^2e^4f$      | -17670 | $c^2d^3e^2f^2$    | +6000  |
| $c^2d^3e^2f^2$ | +6000  | $c^2d^3e^3f$      | -7020  | $c^2d^3e^2f^2$    | +6000  | $c^2d^3e^3f$      | -7020  |
| $c^2d^3e^3f$   | -7020  | $c^2d^2e^4$       | ...    | $c^2d^3e^3f$      | -7020  | $c^2d^2e^4$       | ...    |
| $c^2d^2e^4$    | +4170  | $c^2d^2e^3f^2$    | -180   | $c^2d^2e^4$       | +4170  | $c^2d^2e^3f^2$    | -180   |
| $c^2d^2e^3f^2$ | -180   | $c^2d^2e^3f^2$    | +1560  | $c^2d^2e^3f^2$    | -180   | $c^2d^2e^3f^2$    | +1560  |
| $c^2d^2e^3f^2$ | +1560  | $c^2d^2e^3f^2$    | +480   | $c^2d^2e^3f^2$    | +1560  | $c^2d^2e^3f^2$    | +480   |
| $c^2d^2e^3f^2$ | +480   | $c^2d^2e^3f^2$    | +3000  | $c^2d^2e^3f^2$    | +480   | $c^2d^2e^3f^2$    | +3000  |
| $c^2de^4f$     | -1275  | $c^2d^2e^4$       | +135   | $c^2de^4f$        | -1275  | $c^2d^2e^4$       | +135   |
| $c^2d^2e^4$    | +135   | $c^2d^3e^3$       | -31040 | $c^2d^2e^4$       | +135   | $c^2d^3e^3$       | -31040 |
| $c^2d^3e^3$    | -31040 | $c^2d^2e^4$       | +265   | $c^2d^3e^3$       | -31040 | $c^2d^2e^4$       | +265   |
| $c^2d^2e^4$    | +265   | $c^2de^5$         | -1110  | $c^2d^2e^4$       | +265   | $c^2de^5$         | -1110  |
| $cd^4e^2f$     | +2210  | $c^2d^2e^4$       | +45180 | $cd^4e^2f$        | +2210  | $c^2d^2e^4$       | +45180 |
| $c^2d^3e^3$    | +1420  | $c^2d^2e^4f$      | +3120  | $c^2d^3e^3$       | +1420  | $c^2d^2e^4f$      | +3120  |

Tabelle 5 — continued

|                |        |                |        |                |        |                |        |
|----------------|--------|----------------|--------|----------------|--------|----------------|--------|
| $c^2d^2e^4$    | +3120  | $c^2d^3e^3$    | -4100  | $c^2d^2e^4$    | +3120  | $c^2d^3e^3$    | -4100  |
| $c^2d^2e^4$    | -4100  | $c^2d^2e^4f$   | -3160  | $c^2d^2e^4$    | -4100  | $c^2d^2e^4f$   | -3160  |
| $c^2d^2e^4f$   | -3160  | $c^2d^2e^3f^2$ | -3810  | $c^2d^2e^4f$   | -3160  | $c^2d^2e^3f^2$ | -3810  |
| $c^2d^2e^3f^2$ | -3810  | $c^2d^2e^3f^2$ | +3900  | $c^2d^2e^3f^2$ | -3810  | $c^2d^2e^3f^2$ | +3900  |
| $c^2d^2e^3f^2$ | +3900  | $c^2d^2e^4f$   | +6000  | $c^2d^2e^3f^2$ | +3900  | $c^2d^2e^4f$   | +6000  |
| $c^2d^2e^4f$   | +6000  | $c^2d^2e^4$    | -140   | $c^2d^2e^4f$   | +6000  | $c^2d^2e^4$    | -140   |
| $c^2d^2e^4$    | -140   | $c^2d^2e^3f^2$ | +2310  | $c^2d^2e^4$    | -140   | $c^2d^2e^3f^2$ | +2310  |
| $c^2d^2e^3f^2$ | +2310  | $c^2d^3e^3f$   | +1240  | $c^2d^2e^3f^2$ | +2310  | $c^2d^3e^3f$   | +1240  |
| $c^2d^3e^3f$   | +1240  | $c^2d^2e^4f$   | -4880  | $c^2d^3e^3f$   | +1240  | $c^2d^2e^4f$   | -4880  |
| $c^2d^2e^4f$   | -4880  | $c^2d^2e^4$    | +18000 | $c^2d^2e^4f$   | -4880  | $c^2d^2e^4$    | +18000 |
| $c^2d^2e^4$    | +18000 | $c^2d^2e^3f^2$ | +1795  | $c^2d^2e^4$    | +18000 | $c^2d^2e^3f^2$ | +1795  |
| $c^2d^2e^3f^2$ | +1795  | $c^2d^3e^4$    | +3155  | $c^2d^2e^3f^2$ | +1795  | $c^2d^3e^4$    | +3155  |
| $c^2d^3e^4$    | +3155  | $c^2d^2e^4$    | +990   | $c^2d^3e^4$    | +3155  | $c^2d^2e^4$    | +990   |
| $c^2d^2e^4$    | +990   | $c^2d^2e^3f^2$ | -12180 | $c^2d^2e^4$    | +990   | $c^2d^2e^3f^2$ | -12180 |



## W, 29 A (CONTINUED).

| $a^3$                 | $a^3$ cont.           | $a^2$                 | $a^2$ cont.            |
|-----------------------|-----------------------|-----------------------|------------------------|
| $b^4c^2d^2e^2f$ -8820 | $b^3cd^2e^2f^4$ +1440 | $b^4cd^3e^2$ -18750   | $b^2c^3d^2e^4$ +243000 |
| $c^4d^2e^4$ -34620    | $cd^4e^2f^3$ -1560    | $d^5e^2f^2$ +14115    | $c^4d^{10}ef$ +2340    |
| $cd^{11}ef$ +1080     | $cde^5f^2$ ...        | $d^5e^3f$ -23790      | $c^3d^9e^2$ -89550     |
| $cd^{10}e^3$ +12060   | $ce^8f$ ...           | $d^4e^5$ +8175        | $c^2d^8e^3$ +12060     |
| $d^{12}f$ ...         | $d^5ef^4$ +960        | $c^3d^2f^5$ +1320     | $cd^7e^4$ +270         |
| $d^{11}e^2$ -1620     | $d^4e^2f^3$ -1340     | $b^3c^4df^5$ -30      | $d^6e^5$ -810          |
| $b^3c^{10}f^5$ +81    | $d^3e^3f^2$ -2440     | $c^4de^2f^4$ +540     | $b^4c^{10}ef^4$ +945   |
| $c^4def^4$ -990       | $c^4d^2e^2f^3$ +4320  | $c^3d^3e^2f^3$ -990   | $c^4d^8f^2$ -675       |
| $c^3e^2f^4$ +980      | $c^4de^3f^2$ -7240    | $c^3d^2e^4f^2$ -20390 | $c^4d^7e^2f$ +980      |
| $c^3d^2f^5$ +515      | $c^3d^4e^2f^2$ -675   | $c^3d^2e^5f$ -1620    | $d^6e^4$ -20390        |
| $c^3d^2e^2f^3$ +140   | $c^3d^3e^3f$ -14115   | $d^7e^2f^2$ +7200     | $d^6e^5$ +515          |
| $c^3de^4f^2$ +390     | $c^3d^2e^4f$ +31040   | $d^6e^3f$ -12860      | $c^4d^6e^3$ -3900      |
| $c^3e^5f$ -195        | $c^3de^5$ -195        | $c^3d^4e^3$ +140      | $c^3d^7e^2$ -14115     |
| $c^2d^4ef^2$ -1515    | $c^2d^5e^2f$ +32370   | $c^3d^3e^4$ +390      | $c^2d^8e^2$ +8220      |
| $c^2d^3e^2f^2$ -5575  | $c^2d^4e^3f$ +38820   | $c^3d^2e^5$ -1515     | $c^2d^7e^3$ -5575      |
| $c^2d^3e^3f$ +980     | $c^2d^3e^4$ -49680    | $c^2d^5e^3f$ +980     | $c^2d^6e^4$ -12860     |
| $c^2d^2e^4f^2$ -49680 | $c^2d^2e^5f$ -480     | $c^2d^4e^4f$ -49680   | $c^2d^5e^5$ -195       |
| $c^2d^2e^5f$ +22600   | $c^2d^2e^6$ +26700    | $c^2d^3e^5f$ +22600   | $c^2d^4e^6$ +31040     |
| $c^2de^6$ +2440       | $c^2de^7$ -15435      | $c^2d^2e^6$ +2440     | $cd^8e^3$ -7240        |
| $c^2d^3e^4$ -480      | $c^2d^4e^5$ +91260    | $c^2d^4e^5$ -480      | $cd^7e^4$ +150         |
| $c^2d^2e^5f$ +480     | $c^2d^3e^6$ -50550    | $c^2d^3e^5f$ +480     | $cd^6e^5$ +6000        |
| $c^2d^2e^5$ +3000     | $c^2d^2e^7$ -6660     | $c^2d^2e^6$ +3000     | $d^7e^4$ +26700        |
| $c^2de^5f$ -1275      | $c^2d^3e^6$ -1260     | $c^2de^6f$ -1275      | $d^6e^5$ -15435        |
| $c^2d^2e^5$ +135      | $c^2d^4e^5f^2$ ...    | $c^2d^2e^6$ +135      | $d^5e^6$ -480          |
| $cd^5e^3$ -800        | $c^2d^3e^5f^2$ +15240 | $cd^5e^4f$ -800       | $c^3d^6e^4$ -800       |
| $c^2d^3e^4f$ -8220    | $c^2d^2e^6f$ -10625   | $c^2d^3e^5f$ -8220    | $c^2d^5e^5$ -8220      |
| $cd^4e^4$ +480        | $c^2d^2e^7$ -940      | $cd^4e^6$ +480        | $c^2d^4e^6$ -10625     |
| $cd^3e^5$ -480        | $de^8$ +30            | $cd^3e^7$ -480        | $c^2d^3e^7$ -940       |
| $c^2d^3e^5$ -6000     | $c^2d^3e^5f$ -4300    | $c^2d^3e^5$ -6000     | $c^2d^2e^8$ +30        |
| $c^2d^2e^6$ +9875     | $c^2d^2e^6$ +9875     | $c^2d^2e^6$ +9875     | $cd^6e^5$ -4300        |
| $c^2de^6f$ -4300      | $c^2d^2e^6$ +615      | $c^2de^6f$ -4300      | $cd^5e^6$ +615         |
| $c^2d^2e^6$ +615      | $c^2d^3e^5f$ +675     | $c^2d^2e^6$ +615      | $cd^4e^7$ +675         |
| $c^2d^2e^6$ -2845     | $d^5e^5f^2$ +1285     | $c^2d^2e^6$ -2845     | $c^2d^5e^5$ +1285      |
| $d^6e^3f^2$ +1890     | $c^2d^4e^5$ +360      | $d^6e^4f$ +1890       | $c^2d^4e^6$ +360       |
| $d^5e^4f$ -45         | $c^2d^3e^6$ +11420    | $d^5e^5f$ -45         | $c^2d^3e^7$ +11420     |
| $c^2d^4e^4f$ +510     | $c^2d^2e^7$ +12860    | $c^2d^4e^5f$ +510     | $c^2d^2e^8$ +12860     |
| $b^2c^4e^2f^4$ +100   | $cd^5e^5$ -980        | $b^2c^4e^3f^3$ +100   | $c^2de^8$ -980         |
| $d^5e^4$ -540         | $c^2d^3e^5f^2$ +32000 | $d^5e^6$ -540         | $c^2d^5e^5$ +32000     |
| $c^2d^4e^4$ -310      | $c^2d^2e^6f$ +3150    | $c^2d^4e^6$ -310      | $c^2d^4e^6$ +3150      |
| $c^2d^3e^5$ +2940     | $c^2d^3e^6$ +46160    | $c^2d^3e^7$ +2940     | $c^2d^3e^7$ +46160     |

Tabelle 6 — continued

|                   |       |                |       |                   |       |                |       |
|-------------------|-------|----------------|-------|-------------------|-------|----------------|-------|
| $b^2c^3d^2e^2f^3$ | +240  | $c^2d^2e^7f$   | -3600 | $b^2c^3d^2e^3f^2$ | +240  | $c^2d^2e^8f$   | -3600 |
| $d^4e^5$          | -1710 | $de^9$         | ...   | $d^4e^7$          | -1710 | $de^{10}$      | ...   |
| $c^2d^2e^5f^2$    | +685  | $cd^4e^6f$     | ...   | $c^2d^2e^7f$      | +685  | $cd^4e^8$      | ...   |
| $c^2d^2e^6f$      | -540  | $cd^3e^7$      | ...   | $c^2d^2e^8$       | -540  | $cd^3e^9$      | ...   |
| $b^2c^3e^2f^4$    | -360  | $c^2d^2e^6f^2$ | ...   | $b^2c^3e^3f^3$    | -360  | $c^2d^2e^8f$   | ...   |
| $c^2d^2e^5f$      | +120  | $b^2c^3e^3f^3$ | ...   | $c^2d^2e^7$       | +120  | $b^2c^3e^4f^2$ | ...   |
| $cd^4e^4f$        | -135  | $c^2d^2e^5f^2$ | ...   | $cd^4e^6$         | -135  | $c^2d^2e^7f$   | ...   |
| $c^2d^3e^4f$      | +220  | $c^2d^2e^6f$   | ...   | $c^2d^3e^7$       | +220  | $c^2d^2e^8$    | ...   |
| $c^2d^2e^5f^2$    | -240  | $c^2d^2e^5f$   | ...   | $c^2d^2e^7f^2$    | -240  | $c^2d^2e^7f$   | ...   |
| $c^2d^2e^6f$      | +1965 | $cd^4e^5f$     | ...   | $c^2d^2e^8f$      | +1965 | $cd^4e^7f$     | ...   |
| $c^2d^2e^5f$      | -990  | $cd^3e^6f$     | ...   | $c^2d^2e^7f$      | -990  | $cd^3e^8f$     | ...   |
| $c^2d^2e^6$       | -6000 | $c^2d^2e^5f$   | ...   | $c^2d^2e^8$       | -6000 | $c^2d^2e^7f$   | ...   |
| $c^2d^2e^6f$      | +5840 | $c^2d^2e^6$    | ...   | $c^2d^2e^8f$      | +5840 | $c^2d^2e^8$    | ...   |

W, 29 A (CONTINUED).

| $a^2$             |        |  | $a^1$             |         | $a^1$ cont.          | $a^0$   |                      |         |
|-------------------|--------|--|-------------------|---------|----------------------|---------|----------------------|---------|
| $b^4 c^2 d^{12}$  | -1620  |  | $b^4 d^3 f^5$     | +5575   | $b^4 c d^3 e^3$      | +41250  | $b^4 c^{11} e^2 f^2$ | +5580   |
| $a^1 b^{10} f^7$  | +1     |  | $d^7 e^2 f^2$     | -5030   | $d^{12} f$           | +5445   | $c^{10} d^3 f^3$     | -9540   |
| $b^9 c e f^6$     | +10    |  | $d^6 e^3 f$       | -4255   | $d^{11} e^2$         | -6525   | $c^9 d^5 e^2 f^2$    | -2340   |
| $d^7 f^6$         | +20    |  | $d^5 e^4$         | +2175   | $b^3 c^5 e f^4$      | -900    | $c^8 d^5 e^3 f$      | +20610  |
| $d e^2 f^6$       | -130   |  | $b^3 c^4 d f^5$   | -1110   | $c^8 d^2 f^4$        | +510    | $c^7 d^7 e^4$        | -4350   |
| $e^4 f^5$         | +90    |  | $c^8 e^2 f^4$     | +870    | $c^7 d e^2 f^3$      | +120    | $c^6 d^9 e^5$        | +45130  |
| $b^3 c^3 d f^6$   | -65    |  | $c^7 d^2 e f^4$   | -5550   | $c^6 e^2 f^4$        | +23790  | $c^5 d^{11} e^6$     | -92200  |
| $c^7 e^2 f^5$     | -165   |  | $c^6 d e^2 f^3$   | +24240  | $c^5 d^2 e^2 f^3$    | -32000  | $c^4 d^{13} e^7$     | -25050  |
| $c d^6 e f^5$     | +870   |  | $c^5 e^2 f^3$     | +16194  | $c^4 d e^3 f^2$      | +58080  | $c^3 d^{15} e^8$     | +19020  |
| $c d e^2 f^5$     | -670   |  | $c^4 d^2 f^4$     | -1240   | $c^3 e^4 f^2$        | -15440  | $c^2 d^{17} e^9$     | +46050  |
| $c e^2 f^5$       | +180   |  | $c^3 d^2 e^2 f^3$ | +45180  | $c^2 d^5 e^2 f^2$    | -48360  | $c d^{19} e^{10}$    | -138750 |
| $d^4 f^6$         | -45    |  | $c^2 d^2 e^3 f^2$ | +12180  | $c^2 d^4 e^3 f$      | +41360  | $d^{21} e^{11}$      | -103950 |
| $d^3 e^2 f^5$     | +75    |  | $c^2 d e^4 f$     | -66650  | $c d^7 e^2 f^2$      | -181600 | $b^3 c^{12} d f^4$   | ...     |
| $d^2 e^3 f^4$     | -135   |  | $c^2 e^5$         | -8550   | $c^2 d^6 e^3 f$      | +18400  | $c^{11} d^3 e^2 f^3$ | ...     |
| $d e^5 f^3$       | ...    |  | $c^3 d^3 e^2 f^2$ | +17520  | $c^2 d^5 e^4$        | ...     | $c^{10} d^5 e^3 f^2$ | +1215   |
| $e^7 f^2$         | ...    |  | $c^2 d^2 e^3 f^2$ | -91260  | $c d^8 e^3 f$        | ...     | $c^9 d^7 e^4 f$      | -270    |
| $b^3 c^4 f^7$     | +30    |  | $c^2 d^2 e^4 f$   | +62100  | $c^2 d^7 e^4$        | ...     | $c^8 d^9 e^5$        | -5445   |
| $c^6 d e f^5$     | -280   |  | $c^2 d^2 e^5$     | +22600  | $d^9 e^4 f$          | ...     | $c^7 d^{11} e^6$     | +5580   |
| $c^5 e^2 f^5$     | +2080  |  | $c^2 d^2 e^4 f^2$ | -85800  | $c^2 d^6 e^5$        | ...     | $c^6 d^{13} e^7$     | -17520  |
| $c^4 d^2 f^6$     | -1320  |  | $c^2 d e^5 f^2$   | +148890 | $c^3 d^5 e^5 f$      | ...     | $c^5 d^{15} e^8$     | -30250  |
| $c^3 d^2 e^2 f^5$ | +3000  |  | $c^2 d^2 e^5 f$   | +1850   | $c d^{10} e^5$       | ...     | $c^4 d^{17} e^9$     | -8700   |
| $c^3 d e^2 f^5$   | +4880  |  | $c^2 d e^6$       | +150    | $c^2 d^5 e^6$        | ...     | $c^3 d^{19} e^{10}$  | +17520  |
| $c^2 d^4 f^6$     | -4320  |  | $c^2 d^4 e^3 f$   | -15440  | $d^{11} e^6$         | ...     | $c^2 d^{21} e^{11}$  | +8700   |
| $c^2 d^3 e^2 f^5$ | -2760  |  | $c^2 d^3 e^4 f^2$ | +10350  | $b^2 c^{11} d f^4$   | +135    | $c d^{23} e^{12}$    | -2481   |
| $c^2 d^2 e^3 f^5$ | +6960  |  | $d^9 e^3 f^2$     | -8256   | $c^{10} d^2 e^2 f^3$ | +540    | $b^2 c^{13} f^5$     | ...     |
| $c^2 d e^4 f^5$   | -1820  |  | $d^8 e^4 f$       | +12210  | $c^9 d^2 e^3 f^2$    | -8820   | $b^2 c^{12} e f^4$   | ...     |
| $c^2 e^5 f^4$     | +555   |  | $d^7 e^5$         | -7050   | $c^8 d^2 e^4 f$      | +41640  | $c^{13} d^2 e^2 f^2$ | ...     |
| $c^2 d^5 f^5$     | +40    |  | $b^3 c^5 f^6$     | -225    | $c^7 d^2 e^5$        | -12210  | $c^{12} d^2 e^3 f$   | ...     |
| $c^2 d^4 e f^4$   | -5760  |  | $c^7 d e f^4$     | +3600   | $c^6 d^3 e^5$        | +30265  | $c^{11} d^2 e^4$     | ...     |
| $c^2 d^3 e^2 f^4$ | +49680 |  | $c^6 e^2 f^4$     | -8100   | $c^5 d^3 e^6$        | -16170  | $a^0 b^{11} e f^5$   | -5      |
| $c^2 d^3 e^3 f^3$ | +14040 |  | $c^5 d^2 e^2 f^3$ | +940    | $c^4 d^4 e^6$        | -62025  | $b^{10} c d f^5$     | +10     |
| $c^2 d^2 e^4 f^3$ | -30    |  | $c^4 d^2 e^3 f^2$ | -12180  | $c^3 d^4 e^7$        | +44225  | $c^{10} e^2 f^4$     | -75     |
| $c^2 d^2 e^5 f^2$ | -16120 |  | $c^3 d^2 e^4 f$   | +38950  | $c^2 d^5 e^7$        | -141240 | $d^9 e f^5$          | +130    |
| $c^2 d^2 e^6 f$   | ...    |  | $c^2 d^2 e^5 f^2$ | +42330  | $c d^6 e^8$          | +87000  | $d^8 e^2 f^4$        | -315    |
| $c^2 d e^6 f^2$   | ...    |  | $c^2 d^2 e^6 f$   | -181980 | $d^7 e^8$            | -129000 | $d^7 e^3 f^3$        | +216    |
| $c^2 d^2 e^7$     | ...    |  | $c^2 d^2 e^7$     | ...     | $c^4 d^5 e^6$        | -46050  | $e^{11} f^3$         | -5      |
| $c^2 d^2 e^6 f$   | ...    |  | $c^2 d e^8$       | ...     | $c^3 d^5 e^7$        | +122800 | $b^9 c^2 f^6$        | -5      |
| $d^7 e^3 f^2$     | ...    |  | $b^3 c^4 e^2 f^5$ | -30     | $c^2 d^6 e^7$        | -105950 | $c^8 d e f^5$        | +30     |
| $d^6 e^4 f$       | ...    |  | $c^6 d^2 e^2 f^3$ | +1995   | $c d^7 e^8$          | +88125  | $c^7 e^2 f^5$        | -705    |
| $d^5 e^5$         | ...    |  | $c^5 d^2 e^3 f^2$ | -1795   | $d^8 e^8$            | -27300  | $c^6 d^2 f^6$        | +260    |

Tabelle 7 — continued

|                |        |                |        |                   |        |                |       |
|----------------|--------|----------------|--------|-------------------|--------|----------------|-------|
| $b^3c^5ef^5$   | -30    | $c^4d^2e^4f$   | -9875  | $c^3d^6e^7$       | +3375  | $c^5d^2e^2f^5$ | -265  |
| $c^6d^2e^2f^3$ | +1995  | $c^3d^2e^5f^2$ | +3220  | $b^2c^{12}f^4$    | ...    | $c^4d^2e^3f^4$ | +990  |
| $c^5d^2e^3f^2$ | -1795  | $c^2d^2e^6f$   | -5240  | $c^{11}d^2e^2f^2$ | ...    | $c^3d^2e^4f^3$ | -1620 |
| $c^4d^2e^4f$   | -9875  | $c^2d^2e^7$    | +4180  | $c^{10}d^2e^3f$   | +1215  | $c^2d^2e^5f^2$ | +555  |
| $c^3d^2e^5f^2$ | +3220  | $c^2de^8$      | +12600 | $c^9d^2e^4$       | -270   | $cd^2e^6f$     | -1620 |
| $c^2d^2e^6f$   | -5240  | $c^2d^3e^4f$   | -600   | $c^8d^3e^4$       | -5445  | $d^3e^7$       | +1215 |
| $c^2d^2e^7$    | +4180  | $c^2d^2e^5f^2$ | +1275  | $c^7d^3e^5$       | +5580  | $b^7c^4f^6$    | -30   |
| $c^2de^8$      | +12600 | $c^2d^2e^6f$   | -17670 | $c^6d^3e^6$       | -17520 | $c^5d^4f^5$    | +360  |
| $c^2d^3e^4f$   | -600   | $c^2d^2e^7$    | +36510 | $c^5d^3e^7$       | -30250 | $c^4d^4e^2f^4$ | -600  |
| $c^2d^2e^5f^2$ | +1275  | $c^2de^8$      | ...    | $c^4d^4e^7$       | -8700  | $c^3d^4e^3f^3$ | +420  |
| $c^2d^2e^6f$   | -17670 | $c^2d^2e^6$    | ...    | $c^3d^4e^8$       | +17520 | $c^2d^4e^4f^2$ | -360  |
| $c^2d^2e^7$    | +36510 | $cd^4e^5f$     | ...    | $c^2d^5e^8$       | +8700  | $cd^4e^5f$     | +300  |
| $c^2de^8$      | ...    | $cd^3e^6f$     | ...    | $cd^6e^9$         | -2481  | $d^4e^6$       | -45   |

W, 29 A (CONCLUDED).

|                   | $a^0$  |                   | $a^0$ cont. |                   | $a^0$ cont. |                      | $a^0$ cont. |
|-------------------|--------|-------------------|-------------|-------------------|-------------|----------------------|-------------|
| $b^6 e^9$         | ...    | $b^4 c^2 d^2 f^5$ | +34340      | $b^4 c^4 d^2 e f$ | -138750     | $b^4 c^{10} d^2 e^3$ | -17875      |
| $b^5 c^4 e f^5$   | +30    | $c^3 d^5 e^2 f^3$ | -153480     | $c^4 d^7 e^3$     | -1250       | $c^9 d f$            | -6600       |
| $c^3 d^2 f^5$     | -370   | $c^2 d^6 e^3 f^2$ | +220125     | $c^3 d^9 f$       | +31150      | $c^8 d^3 e^2$        | +4125       |
| $c^3 d e^2 f^4$   | +1800  | $c^2 d^5 e^4 f$   | ...         | $c^3 d^8 e^2$     | +4125       | $c^7 d^5 e^3$        | ...         |
| $c^3 e^3 f^3$     | +2845  | $c^2 d^4 e^5$     | +6660       | $c^2 d^{10} f^2$  | +1800       | $c^6 d^7 e^4$        | +2845       |
| $c^2 d^3 e f^4$   | +570   | $c d^7 e^2 f^2$   | +18400      | $c^2 d^9 e^2 f$   | -18750      | $b^6 c^{15} f^3$     | +72         |
| $c^2 d^2 e^2 f^3$ | +1640  | $c d^6 e^3 f$     | +2250       | $c^2 d^8 e^4$     | +570        |                      |             |
| $c^2 d e^3 f^2$   | -4185  | $b^3 c^5 e f^4$   | +72         | $c d^{11} f$      | -40         |                      |             |
| $c^2 e^4 f$       | -4050  | $c^7 d^2 e^2 f^3$ | -120        | $c d^{10} e^2$    | -1640       |                      |             |
| $c d^4 f^5$       | +1110  | $c^6 d^2 e^3 f^2$ | -5840       | $c^4 d^5 e^2 f^2$ | -120        |                      |             |
| $c d^3 e f^4$     | -4170  | $c^5 d^2 e^4 f$   | +6560       | $c^4 d^4 e^3 f$   | +180        |                      |             |
| $c d^2 e^2 f^3$   | ...    | $c^4 d^2 e^5$     | -32280      | $c^3 d^6 e^2 f^2$ | +6560       |                      |             |
| $c d^2 e^3 f^2$   | +6075  | $c^4 d^2 e^4 f$   | -360        | $c^3 d^5 e^3 f$   | -32280      |                      |             |
| $c d e^5$         | ...    | $c^3 d^3 e^4 f^2$ | +7240       | $c^3 d^4 e^4$     | -360        |                      |             |
| $d^5 e^2 f^3$     | +3620  | $c^2 d^4 e^4 f$   | -46880      | $c^2 d^8 e^2 f$   | +7240       |                      |             |
| $d^4 e^3 f^2$     | -9310  | $c^2 d^3 e^5$     | +5550       | $c^2 d^7 e^3$     | +24240      |                      |             |
| $d^3 e^4 f$       | +8550  | $c^2 d^3 e^4 f^2$ | -30040      | $c^2 d^6 e^4$     | +990        |                      |             |
| $d^3 e^5$         | -3375  | $c^2 d^2 e^6 f$   | +1285       | $c d^9 e^2 f$     | +3150       |                      |             |
| $d^2 e^5$         | +210   | $c^2 d^2 e^5 f^2$ | -360        | $c d^8 e^3$       | -980        |                      |             |
| $b^3 c^5 f^6$     | -45    | $c^2 d e^7$       | ...         | $d^{10} e^2 f^2$  | -360        |                      |             |
| $c^4 d^2 e f^4$   | +615   | $c d^5 e^4 f^2$   | +32000      | $d^9 e^3 f$       | +3150       |                      |             |
| $c^3 d^2 e^2 f^3$ | -1285  | $c d^4 e^5 f$     | +46160      | $c^3 d^5 e^4$     | ...         |                      |             |
| $c^3 d e^3 f^2$   | -1140  | $c d^3 e^6$       | -3600       | $c^2 d^7 e^4 f$   | +8100       |                      |             |
| $c^2 d^3 f^4$     | -3765  | $d^7 e^4 f$       | ...         | $c^2 d^6 e^5 f$   | -2700       |                      |             |
| $c^2 d^2 e^2 f^3$ | +3160  | $d^6 e^5$         | ...         |                   |             |                      |             |
| $c^2 d e^3 f^2$   | -9850  | $b^2 c^6 e^2 f^4$ | +16         |                   |             |                      |             |
| $c^2 e^4 f$       | +19800 |                   |             |                   |             |                      |             |
| $c^2 d^2 e^3 f^2$ | +3375  |                   |             |                   |             |                      |             |
| $c^2 d^2 e^4 f$   | -13500 |                   |             |                   |             |                      |             |
| $c^2 d^2 e^5 f^2$ | +50550 |                   |             |                   |             |                      |             |
| $c^2 d^2 e^6 f$   | -62100 |                   |             |                   |             |                      |             |
| $c^2 d^2 e^7$     | ...    |                   |             |                   |             |                      |             |
| $c d^4 e^3 f^2$   | -7240  |                   |             |                   |             |                      |             |
| $c d^3 e^4 f$     | +28710 |                   |             |                   |             |                      |             |
| $c d^3 e^5$       | -38950 |                   |             |                   |             |                      |             |
| $c d^2 e^5$       | +25875 |                   |             |                   |             |                      |             |
| $d^5 e^3 f^2$     | -6155  |                   |             |                   |             |                      |             |
| $d^4 e^4 f$       | +12500 |                   |             |                   |             |                      |             |
| $d^3 e^5$         | -7375  |                   |             |                   |             |                      |             |
| $b^2 c^5 f^6$     | -45    |                   |             |                   |             |                      |             |



For the lower covariants the numerical verifications are given for the entire coefficient, but for the higher ones where the number of terms in a coefficient is considerable they are given separately for the different powers of  $a$ ; and it is also interesting to consider them for the separate combinations of  $a$  and  $b$ .

I recall that the positive and negative numerical coefficients are summed separately, so that  $(\pm \text{a number})$  means that the sum of the positive numerical coefficients is equal to the sum of the negative numerical coefficients and thus that the whole sum is  $= 0$ .

It is to be observed that for the lower covariants the sums of the numerical coefficients do not vanish for the separate powers of  $a$ : thus in the invariant  $G$ , 141, the sums of the numerical coefficients for the terms in  $a^2, a^1, a^0$  are  $= 1, -2, 1$  respectively.

As regards the invariants  $Q$  and  $Q'$ ; for the first of these,  $Q$ , the sums of the numerical coefficients for the terms in  $a^4, a^3, a^2, a^1, a^0$  are each of them  $= 0$ , but this is not the case as regards  $Q'$ ; in fact  $Q'$  is  $= G^2 + \text{a multiple of } Q$ ; hence the sums for  $Q$  are the same as those for  $G^2$ , viz. they are  $= 1, -4, +6, -4, +1$  respectively. Like results present themselves in other cases, and they might probably be accounted for in a similar manner; we have a series of sums not each  $= 0$ , but which are equal to a set of binomial coefficients taken with the signs  $+$  and  $-$  alternately and thus the sum of these sums is  $= 0$ .

For  $R, S$  and  $S'$ , I have given the sums for the different powers of  $a$ ; and in regard to  $S$  I give here the following paragraphs from the Tenth Memoir on Quantics:—

I remark that I calculated the first two coefficients  $S_0, S_1$ , and deduced the other two,  $S_2$  from  $S_1$ , and  $S_3$  from  $S_0$ , by reversing the order of the letters (or which is the same thing, interchanging  $a$  and  $f, b$  and  $e, c$  and  $d$ ) and reversing also the signs of the numerical coefficients. This process for  $S_2, S_3$  is to a very great extent a verification of the values of  $S_0, S_1$ . For, as presently mentioned, the terms of  $S_0$  form subdivisions such that in each subdivision the sum of the numerical coefficients is  $= 0$ : in passing by the reversal process to the value of  $S_3$ , the terms are distributed into an entirely new set of subdivisions, and then in each of these subdivisions the sum of the numerical coefficients is found to be  $= 0$ ; and the like as regards  $S_1$  and  $S_3$ .

If in the expressions for  $S_0, S_1, S_2, S_3$  we first write  $d = e = f = 1$ , thus in effect combining the numerical coefficients

for the terms which contain the same powers in  $a, b, c$ , we find

$$\begin{aligned}
 S_0 = & a^3(-2c^3 + 6c^2 - 6c + 2) \\
 & + a^2\{b^2(6c^2 - 12c - 6) + b(-15c^3 + 33c^2 - 21c + 3) \\
 & \quad + b^0(42c^4 - 147c^3 + 195c^2 - 117c + 27)\} \\
 & + a\{b^4 \cdot 0 + b^3(30c^2 - 36c + 6) + b^2(-117c^3 + 249c^2 - 183c + 51) \\
 & \quad + b(9c^5 + 148c^4 - 378c^3 + 330c^2 - 99c) + b^0(-63c^6 + 165c^5 - 147c^4 \\
 & \quad + a^0\{b^6 \cdot 2 + b^5(-15c + 3) + b^4(75c^2 - 69c + 24) + b^3(-9c^4 - 167c^3 + 225c^2 - 87c \\
 & \quad + b^2(72c^5 + 48c^4 - 186c^3 + 96c^2) + b(-126c^6 + 201c^5 - 87c^4) \\
 & \quad + b^0(27c^8 - 45c^7 + 20c^6)\}
 \end{aligned}$$

which for  $c = 1$  becomes

$$= 2b^6 - 12b^5 + 30b^4 - 40b^3 + 30b^2 - 12b + 2, \text{ that is } 2(b-1)^6,$$

and for  $b = 1$ , becomes  $= 0$ .

$$\begin{aligned}
 S_2 = & a^3(0c^2 + 0c + 0) \\
 & + a^2\{b^2(0c + 0) + b(3c^3 - 9c^2 + 9c - 3) + b^0(24c^4 - 99c^3 + 153c^2 - 105c + 27)\} \\
 & + a\{b^4 \cdot 0 + b^3(-6c^2 + 12c - 6) + b^2(-24c^3 + 90c^2 - 108c + 42) \\
 & \quad + b(33c^4 - 90c^3 + 54c^2 + 30c - 27) + b^0(-27c^6 + 78c^5 - 66c^4 + 6c^3 \\
 & + a^0\{b^5(3c - 3) + b^4(-15c + 15) + b^3(6c^3 - 12c^2 + 36c - 30) \\
 & \quad + b^2(9c^5 - 42c^4 + 84c^3 - 108c^2 + 57c) + b(9c^6 - 54c^5 + 96c^4 - 51c^3 \\
 & \quad + b^0(9c^7 - 9c^6)\}
 \end{aligned}$$

which for  $c = 1$  becomes  $= 0$ .

$$\begin{aligned}
 S_3 = & a^3(0c + 0) \\
 & + a^2\{b^2 \cdot 0 + b(0c^2 + 0c + 0) + b^0(18c^4 - 72c^3 + 108c^2 - 72c + 18)\} \\
 & + a\{b^3(0c + 0) + b^2(-33c^3 + 99c^2 - 99c + 33) + b(57c^4 - 162c^3 + 144c^2 - 30c \\
 & \quad + b^0(-60c^5 + 207c^4 - 261c^3 + 141c^2 - 27c)\} \\
 & + a^0\{b^5 \cdot 0 + b^4(15c^2 - 30c + 15) + b^3(-54c^3 + 102c^2 - 42c - 6) \\
 & \quad + b^2(123c^4 - 297c^3 + 243c^2 - 87c + 18) + b(-27c^5 + 102c^4 - 96c^3 + \\
 & \quad + b^0(27c^7 - 60c^6 + 51c^5 - 12c^4)\}
 \end{aligned}$$



which for  $c = 1$  becomes  $= 0$ .

$$\begin{aligned}
 S_4 = & a^3 \cdot 0 \\
 & + a^2 \{b(0c + 0) + b^0(0c^2 + 0c + 0)\} \\
 & + a \{b^3 \cdot 0 + b^2(0c^2 + 0c + 0) + b(-9c^4 + 36c^3 - 54c^2 + 36c - 9) \\
 & \quad + b^0(36c^5 - 171c^4 + 324c^3 - 306c^2 + 144c - 27)\} \\
 & + a^0 \{b^4(0c + 0) + b^3(7c^3 - 21c^2 + 21c - 7) + b^2(-39c^4 + 135c^3 - 171c^2 + 93c - 9) \\
 & \quad + b(66c^5 - 243c^4 + 333c^3 - 201c^2 + 45c) \\
 & \quad + b^0(-27c^7 + 101c^6 - 141c^5 + 87c^4 - 20c^3)\}
 \end{aligned}$$

which for  $c = 1$  becomes  $= 0$ .

It follows that for  $c = d = e = f = 1$ , the value of the covariant  $S$  is  $= 2(b - 1)^6 x^2$ , which might be easily verified. C. II. 39

For  $T$ ,  $U$ ,  $V$  and  $W$ , I look at the sums for the different combinations of  $a$  and  $b$ .

Thus for  $T$  we have

| $x$ coefficient. |       |       |         | $y$ coefficient. |       |          |       |
|------------------|-------|-------|---------|------------------|-------|----------|-------|
| $a^4$            | $b^0$ | 26    |         | $a^4$            | $b^0$ | $\pm 12$ |       |
|                  |       | $\pm$ | 26      |                  |       | $\pm$    | 12    |
| $a^3$            | $b^2$ | $\pm$ | 14      | $a^3$            | $b^2$ | $\pm 2$  |       |
|                  | $b^4$ | 141   |         |                  | $b^1$ | 112      |       |
|                  | $b^0$ | 281   |         |                  | $b^0$ | 281      |       |
|                  |       | $\pm$ | 436     |                  |       | $\pm$    | 395   |
| $a^2$            | $b^4$ | $\pm$ | 1       | $a^2$            | $b^3$ | $\pm 42$ |       |
|                  | $b^3$ | 106   |         |                  | $b^2$ | 546      |       |
|                  | $b^2$ | 186   |         |                  | $b^1$ | 696      |       |
|                  | $b^1$ | 1173  |         |                  | $b^0$ | 366      |       |
|                  | $b^0$ | 2272  |         |                  |       | $\pm$    | 1650  |
|                  |       | $\pm$ | 3738    | $a^1$            | $b^5$ | $\pm 5$  |       |
| $a^1$            | $b^6$ | $\pm$ | 16      |                  | $b^4$ | 179      |       |
|                  | $b^5$ | 359   |         |                  | $b^3$ | 821      |       |
|                  | $b^4$ | 1411  |         |                  | $b^2$ | 2097     |       |
|                  | $b^3$ | 3103  |         |                  | $b^1$ | 2147     |       |
|                  | $b^2$ | 3030  |         |                  | $b^0$ | 1262     |       |
|                  | $b^1$ | 1197  |         |                  |       | $\pm$    | 6511  |
|                  |       | $\pm$ | 9116    | $a^0$            | $b^7$ | $\pm 28$ |       |
| $a^0$            | $b^7$ | $-2$  |         |                  | $b^6$ | 342      |       |
|                  | $b^6$ | 92    | $-78$   |                  | $b^5$ | 1790     |       |
|                  | $b^5$ | 307   | $-349$  |                  | $b^4$ | 3496     |       |
|                  | $b^4$ | 1073  | $-1003$ |                  | $b^3$ | 3445     |       |
|                  | $b^3$ | 2040  | $-2110$ |                  | $b^2$ | 2064     |       |
|                  | $b^2$ | 1930  | $-1880$ |                  | $b^1$ | 463      |       |
|                  | $b^1$ | 1207  | $-1221$ |                  |       | $\pm$    | 11628 |
|                  | $b^0$ | 231   | $-239$  |                  |       | $\pm$    | 20196 |
|                  |       | $\pm$ | 6880    |                  |       |          |       |
|                  |       | $\pm$ | 20196   |                  |       |          |       |

Observe here that in the  $x$ -coefficient for the terms in  $a^0$  the successive sums are  $-2, +14, -42, +70, -70, +42, -14, +2$ , which are the coefficients of  $-2(b-1)^7$ .

For  $U$  we have

|       |       |          |       |
|-------|-------|----------|-------|
| $a^4$ | $b^0$ | $\pm 36$ | 36    |
|       |       | $\pm$    |       |
| $a^3$ | $b^2$ | $\pm 24$ | 464   |
|       | $b^1$ | 198      |       |
|       | $b^0$ | 242      |       |
|       |       | $\pm$    |       |
| $a^2$ | $b^4$ | $\pm 2$  | 2608  |
|       | $b^3$ | 208      |       |
|       | $b^2$ | 286      |       |
|       | $b^1$ | 866      |       |
|       | $b^0$ | 1246     |       |
|       |       | $\pm$    |       |
| $a^1$ | $b^5$ | $\pm 64$ | 7278  |
|       | $b^4$ | 328      |       |
|       | $b^3$ | 1258     |       |
|       | $b^2$ | 2586     |       |
|       | $b^1$ | 2186     |       |
|       | $b^0$ | 856      |       |
|       |       | $\pm$    |       |
| $a^0$ | $b^7$ | $\pm 4$  | 6878  |
|       | $b^6$ | 70       |       |
|       | $b^5$ | 448      |       |
|       | $b^4$ | 1488     |       |
|       | $b^3$ | 2140     |       |
|       | $b^2$ | 1678     |       |
|       | $b^1$ | 884      |       |
|       | $b^0$ | 166      |       |
|       |       | $\pm$    |       |
|       |       | $\pm$    | 17264 |

For  $V$  we have

| $x$ coefficient. |       |          |          | $y$ coefficient. |       |          |              |
|------------------|-------|----------|----------|------------------|-------|----------|--------------|
| $a^5$            | $b^0$ | $\pm 36$ |          | $a^5$            | $b^0$ | $\pm 24$ |              |
|                  |       | $\pm$    | 36       |                  |       | $\pm$    | 24           |
| $a^4$            | $b^2$ | $\pm 20$ |          | $a^4$            | $b^2$ | $\pm 4$  |              |
|                  | $b^1$ | 284      |          |                  | $b^1$ | 144      |              |
|                  | $b^0$ | 1094     |          |                  | $b^0$ | 436      |              |
|                  |       | $\pm$    | 1398     |                  |       | $\pm$    | 584          |
| $a^3$            | $b^4$ | 2        |          | $a^3$            | $b^3$ | $\pm 24$ |              |
|                  | $b^3$ | 184      |          |                  | $b^2$ | 776      |              |
|                  | $b^2$ | 1656     |          |                  | $b^1$ | 2696     |              |
|                  | $b^1$ | 3624     |          |                  | $b^0$ | 1264     |              |
|                  | $b^0$ | 4898     |          |                  |       | $\pm$    | 4760         |
|                  |       | $\pm$    | 10364    | $a^2$            | $b^5$ | $\pm 6$  |              |
| $a^2$            | $b^5$ | $\pm 14$ |          |                  | $b^4$ | 300      |              |
|                  | $b^4$ | 666      |          |                  | $b^3$ | 2236     |              |
|                  | $b^3$ | 6608     |          |                  | $b^2$ | 8616     |              |
|                  | $b^2$ | 10512    |          |                  | $b^1$ | 15442    |              |
|                  | $b^1$ | 22042    |          |                  | $b^0$ | 33044    |              |
|                  | $b^0$ | 9162     |          |                  |       | $\pm$    | 59644        |
|                  |       | $\pm$    | 49004    | $a^1$            | $b^6$ | $\pm 78$ |              |
| $a^1$            | $b^7$ | $-4$     |          |                  | $b^5$ | 852      |              |
|                  | $b^6$ | 76       | $-48$    |                  | $b^4$ | 8310     |              |
|                  | $b^5$ | 2956     | $-3040$  |                  | $b^3$ | 30200    |              |
|                  | $b^4$ | 11946    | $-11806$ |                  | $b^2$ | 56740    |              |
|                  | $b^3$ | 23924    | $-24064$ |                  | $b^1$ | 39956    |              |
|                  | $b^2$ | 25110    | $-25026$ |                  | $b^0$ | 17986    |              |
|                  | $b^1$ | 25524    | $-25552$ |                  |       | $\pm$    | 154122       |
|                  | $b^0$ | 8822     | $-8812$  |                  | $a^0$ | $b^8$    | $-2$         |
|                  |       | $\pm$    | 98358    |                  |       | $b^7$    | 286          |
| $a^0$            | $b^8$ | 18       |          |                  |       | $b^6$    | 2026         |
|                  | $b^7$ | 184      | $-324$   |                  |       | $b^5$    | 9360         |
|                  | $b^6$ | 4098     | $-3622$  |                  |       | $b^4$    | 19760        |
|                  | $b^5$ | 19350    | $-20274$ |                  |       | $b^3$    | 36442        |
|                  | $b^4$ | 42398    | $-41278$ |                  |       | $b^2$    | 30340        |
|                  | $b^3$ | 51872    | $-52740$ |                  |       | $b^1$    | 23426        |
|                  | $b^2$ | 44320    | $-43900$ |                  |       | $b^0$    | 5120         |
|                  | $b^1$ | 20624    | $-20740$ |                  |       |          | $\pm 126760$ |
|                  | $b^0$ | 3870     | $-3856$  |                  |       |          | $\pm 345894$ |
|                  |       | $\pm$    | 186734   |                  |       |          |              |
|                  |       | $\pm$    | 345894   |                  |       |          |              |

Here in the  $x$ -coefficient for  $a^1$  the successive sums are  $-4, +28, -84, +140, -140, +84, -28, +4$ , which are the coefficients of

$-4(b-1)^7$ ; and for  $a^0$  the successive sums are 18,  $-140$ ,  $+476$ ,  $-924$ ,  $+1120$ ,  $-868$ ,  $+420$ ,  $-116$ ,  $+14$ , which are the coefficients of  $18(b-1)^8 + 4(b-1)^7$ . In the  $y$ -coefficient the successive sums are  $-2$ ,  $+16$ ,  $-56$ ,  $+112$ ,  $-140$ ,  $+112$ ,  $-56$ ,  $+16$ ,  $-2$ , which are the coefficients of  $-2(b-1)^8$ .

Finally for  $W$  we have

|       |       |               |        |          |       |          |               |               |  |  |
|-------|-------|---------------|--------|----------|-------|----------|---------------|---------------|--|--|
| $a^7$ | $b^0$ | $\pm 16$      |        |          |       |          |               |               |  |  |
|       |       |               |        | 16       | $a^1$ | $b^{10}$ | + 1           | $\pm 2972759$ |  |  |
| $a^6$ | $b^1$ | $\pm 175$     |        |          |       | $b^9$    | 120           | - 130         |  |  |
|       | $b^0$ | 806           |        |          |       | $b^8$    | 1125          | - 1080        |  |  |
|       |       | $\pm$         |        | 981      |       | $b^7$    | 30350         | - 30470       |  |  |
| $a^5$ | $b^3$ | $\pm 80$      |        |          |       | $b^6$    | 122400        | - 122190      |  |  |
|       | $b^2$ | 1175          |        |          |       | $b^5$    | 332494        | - 332746      |  |  |
|       | $b^1$ | 2760          |        |          |       | $b^4$    | 729150        | - 728940      |  |  |
|       | $b^0$ | 6871          |        |          |       | $b^3$    | 880750        | - 880870      |  |  |
|       |       | $\pm$         |        | 10886    |       | $b^2$    | 466935        | - 466890      |  |  |
| $a^4$ | $b^4$ | $\pm 570$     |        |          |       | $b^1$    | 363670        | - 363680      |  |  |
|       | $b^3$ | 5200          |        |          |       | $b^0$    | 76116         | - 76115       |  |  |
|       | $b^2$ | 18005         |        |          |       |          | $\pm 3003111$ |               |  |  |
|       | $b^1$ | 44720         |        |          |       |          |               |               |  |  |
|       | $b^0$ | 23810         | $a^0$  | $b^{11}$ |       |          | + - 5         |               |  |  |
|       |       | $\pm$         | 92305  |          |       | $b^{10}$ | 400           | - 346         |  |  |
| $a^3$ | $b^6$ | $\pm 90$      |        |          |       | $b^9$    | 3500          | - 3765        |  |  |
|       | $b^5$ | 2386          |        |          |       | $b^8$    | 26240         | - 25460       |  |  |
|       | $b^4$ | 26675         |        |          |       | $b^7$    | 154030        | - 155560      |  |  |
|       | $b^3$ | 84680         |        |          |       | $b^6$    | 409700        | - 407600      |  |  |
|       | $b^2$ | 107730        |        |          |       | $b^5$    | 747985        | - 750043      |  |  |
|       | $b^1$ | 199160        |        |          |       | $b^4$    | 745920        | - 744480      |  |  |
|       | $b^0$ | 240499        |        |          |       | $b^3$    | 613100        | - 613805      |  |  |
|       |       | $\pm$         | 661220 |          |       | $b^2$    | 311790        | - 311560      |  |  |
| $a^2$ | $b^8$ | $\pm 15$      |        |          |       | $b^1$    | 89215         | - 89260       |  |  |
|       | $b^7$ | 640           |        |          |       | $b^0$    | 9999          | - 9995        |  |  |
|       | $b^6$ | 8260          |        |          |       |          | $\pm 3111879$ |               |  |  |
|       | $b^5$ | 59135         |        |          |       |          |               |               |  |  |
|       | $b^4$ | 182055        |        |          |       |          | $\pm 9087749$ |               |  |  |
|       | $b^3$ | 341470        |        |          |       |          |               |               |  |  |
|       | $b^2$ | 699260        |        |          |       |          |               |               |  |  |
|       | $b^1$ | 612015        |        |          |       |          |               |               |  |  |
|       | $b^0$ | 304501        |        |          |       |          |               |               |  |  |
|       |       | $\pm 2207351$ |        |          |       |          |               |               |  |  |

Here for the terms in  $a^1$  the successive sums are

1, -10, +45, -120, +210, -252, +210, -120, +45, -10, +1,

which are the coefficients of  $(b-1)^{10}$ ; and for the terms in  $a^0$  the successive sums are

$$-5, +54, -265, +780, -1530, +2100, -2058, +1440, -705, +230, -45, +4,$$

which are the coefficients of  $-5(b-1)^{11} - (b-1)^{10}$ .

## 144.

## A THIRD MEMOIR UPON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. cxlvi. for the year 1856, pp. 627—647. Received March 13,—Read April 10, 1856.]

My object in the present memoir is chiefly to collect together and put upon record various results useful in the theories of the particular quantics to which they relate. The tables at the commencement relate to binary quantics, and are a direct sequel to the tables in my Second Memoir upon Quantics, vol. cxlvi. (1856), [141]. The definitions and explanations in the next part of the present memoir are given here for the sake of convenience, the further development of the subjects to which they relate being reserved for another occasion. The remainder of the memoir consists of tables and explanations relating to ternary quadrics and cubics.

COVARIANT AND OTHER TABLES, NOS. 27 TO 50 (NOS. 1 TO 50  
BINARY QUANTICS).<sup>8</sup>

Nos. 27 to 29 are a continuation of the tables relating to the  
quintic

$$(a, b, c, d, e, f \wr x, y)^5.$$

No. 27 gives the values of the different determinants of the  
matrix

$$\begin{vmatrix} a, & 4b, & 6c, & 4d, & e \\ a, & 4b, & 6c, & 4d, & e \\ & b, & 4c, & 6d, & 4e, & f \\ & b, & 4c, & 6d, & 4e, & f \end{vmatrix}$$

determinants which are represented by 1234, 1235, &c., where the numbers refer to the different columns of the matrix. No. 28 gives

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<sup>8</sup>The Tables 49 and 50 were inserted October 6, 1856.—A. C.

the values of certain linear functions of these determinants, viz.

$$\begin{aligned}
 L &= 1256 + 2345 - 2 \cdot 1346, \\
 L' &= 3 \cdot 1256 - 1346, \\
 8M &= -1345 + 2 \cdot 1246, \\
 8M' &= -2346 + 2 \cdot 1356, \\
 8N &= -1245 + 3 \cdot 1236, \\
 8N' &= -2356 + 3 \cdot 1456, \\
 80P &= L' - 3L = 5 \cdot 1346 - 3 \cdot 2345, \\
 16P' &= -5L' - L = -16 \cdot 1256 - 3 \cdot 1346 - 2356.
 \end{aligned}$$

At the end of the two tables there are given certain relations which exist between the terms of Tables 14, 16, 25, 26, 27 and 28.



We have also the following relations between  $L, L',$  &c. and  $a, b, c, d, e, f,$  viz.

$$\begin{aligned} aP - bM + cN &= 0, \\ aM' + bP' - 2cM + 3dN &= 0, \\ aN' + 2bM' - cL' + 3eN &= 0, \\ 3bN' - dL' + 2eM + fN &= 0, \\ 3cN' - 2dM' + eP' + fM &= 0, \\ dN' - eM' + fP &= 0. \end{aligned}$$

The quartinvariant No. 19  $[G]$  is equal to  $-AC + B^2$ , i.e. it is in fact equal to  $-4$  into the discriminant of the quintic No. 14,  $[A]$ .

The octinvariant No. 25  $[Q]$  is expressible in terms of the coefficients of Nos. 14 and 16, viz.  $A, B, C,$  as before, and  $\alpha, \beta, \gamma, \delta$  the coefficients of No. 16,  $[D]$ , i.e.

$$\begin{aligned} \alpha &= 3(ace - ad^2 - b^2e + 2bcd - c^3), \\ \beta &= acf - ade - b^2f + bd^2 + bce - cd^2, \\ \gamma &= adf - ae^2 - bcf + bde + c^2e - cd^2, \\ \delta &= 3(bdf - be^2 + 2cde - c^2f - d^3), \end{aligned}$$

then No. 25 is equal to

$$\begin{vmatrix} A, & B, & C \\ \alpha, & \beta, & \gamma \\ \beta, & \gamma, & \delta \end{vmatrix}.$$

The value of the discriminant No. 26,  $[Q']$ , is

$$(\text{No. 19})^2 - 128 \text{ No. 25},$$

$$[\text{that is } Q' = G^2 - 128Q.]$$

We have also an expression for the discriminant in terms of  $L, L',$  &c., viz. three times the discriminant No. 26 is equal to

$$[\text{or say } 3Q' =] LL' + 64MM' - 64NN',$$

a remarkable formula, the discovery of which is due to Mr Salmon.

It may be noticed, that in the particular case in which the quintic has two square factors, if we write

$$(a, b, c, d, e, f \wr x, y)^5 = \{(p, q, r \wr x, y)^2\}^2 \cdot (\lambda, \mu \wr x, y),$$

The table No. 29 is the invariant of the twelfth degree of the quintic, given in its simplest form, i.e. in a form not containing any power higher than the fourth of the leading coefficient  $a$ : this invariant was first calculated by M. Faa de Bruno.

NO. 29. [See U. No. 29, p. 294.]

The tables Nos. 30 to 35 relate to a **sextic**. No. 30 is the sextic itself; No. 31 the quadrinvariant; Nos. 32 and 33 the quadricovariants (the latter of them the Hessian); No. 34 is the quartinvariant or catalecticant; and No. 35 is the sextinvariant in its best form, i.e. a form not containing any power higher than the second of the leading coefficient  $a$ .

NO. 30.

$$(a, b, c, d, e, f, g \wr x, y)^6.$$

NO. 34.

$$\begin{aligned} & \pm 12a^2d^2g^2 + 1 \cdot ace^2g - 42 \cdot bcdefg + 60 \cdot a^2defg - 6 \cdot ace^3f + 12 \cdot bce^2f^2 - 30 \cdot ab \\ & + 4 \cdot ad^4g - 20 \cdot bcd^2eg + 24 \cdot a^2e^2g - 3 \cdot ad^3e^2 - 8 \cdot bc^2d^2f + 66 \cdot abcdfg - 6 \cdot abc \\ & + 4 \cdot b^2c^2ef - 84 \cdot abc^2eg + 18 \cdot abce^2f + 24 \cdot b^2cde^2 - 12 \cdot b^3e^2g + 8 \cdot c^4dg + 12 \cdot a \\ & - 27 \cdot c^3e^3 + 30 \cdot c^3d^2g - 8 \cdot a^3e^3 + 6 \cdot b^2ce^2f - 24 \cdot c^2e^2f + 66 \cdot ac^2d^2e - 4 \cdot b^3c^2g \\ & - 24 \cdot b^2c^2d^2 - 15 \cdot a^2c^2f^2 + 60 \cdot b^2c^2ef - 12 \cdot c^4e^2 - 8 \cdot ac^3df + 36 \cdot a^2cd^2f + 54 \cdot \\ & + 6 \cdot b^2c^3f \pm 565 \end{aligned}$$

The sextinvariant may be thus represented by means of a determinant of the sixth order and of the quadrinvariant and quartinvariant.

$$5 \times \text{No. 35} = \begin{vmatrix} a, & 2b, & 3c, & 4d, & e \\ b, & 2c, & 3d, & 4e, & f \\ c, & 2d, & 3e, & 4f, & g \\ a, & 4b, & 3c, & 2d, & e \\ b, & 4c, & 3d, & 2e, & f \\ c, & 4d, & 3e, & 2f, & g \end{vmatrix} + 4(ag - 6bf + 15ce - 10d^2) \begin{vmatrix} a, & b, & c, & d \\ b, & c, & d, & e \\ c, & d, & e, & f \\ d, & e, & f, & g \end{vmatrix}.$$

**Definitions.**

The quantic or quantics to which it belongs, and with indeterminate coefficients  $(\xi, \eta, \dots)$ . The invariants of a quantic or quantics, and of an adjoint linear form, may be considered as quantics having  $(\xi, \eta, \dots)$  for facients, and of which the coefficients are of course functions of the coefficients of the given quantic or quantics. An invariant of the class in question is termed a *contravariant* of the quantic or quantics. The idea of a contravariant is due to Mr Sylvester.

In the theory of binary quantics, it is hardly necessary to consider the contra-variants; for any contravariant is at once turned into an invariant by writing  $(y, -x)$  for  $(\xi, \eta)$ .

**57.** If we imagine, as before, a system of quantics of the form

$$(\bullet \wr x, y, \dots)^n,$$

where the number of qua

The coefficients of this quantic are termed *Emanants*, viz., excluding the first coefficient, which is the quantic itself (but which might be termed the 0-th emanant), the other coefficients are the first, second, and last or ultimate emanants.

The ultimate emanant is, it is clear, nothing else than the quantic itself, with  $(X, Y, \dots)$  instead of  $(x, y, \dots)$  for facients: the penultimate emanant is, in like manner, obtained from the first emanant by interchanging  $(x, y, \dots)$  with  $(X, Y, \dots)$ , and similarly for the other emanants. The facients  $(X, Y, \dots)$  may be termed the facients of emanation, or simply the new facients.

Such covariants are called *Combinants*, and they possess the property of being invariantive, quoad the system, i.e. the covariant remains unaltered to a factor près, when each quantic is replaced by a linear function of all the quantics. This extremely important theory is due to Mr Sylvester.

COVARIANT AND OTHER TABLES, NOS. 51 TO 56 (A TERNARY QUADRIC).

NO. 51. The quadric is represented by  $(a, b, c, f, g, h \wr x, y, z)^2$ , which means

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy.$$

NO. 52. The first derived functions (omitting the factor 2) are

$$(a, h, g \wr x, y, z), \quad (h, b, f \wr x, y, z), \quad (g, f, c \wr x, y, z).$$

NO. 53. The operators which reduce a covariant to zero are

$$\begin{aligned} &(j, 3f, c, 2i, g, 2h \wr \partial_x, \partial_y, \partial_z)^3 - y \partial_z, \\ &(a, k, 3f, 2i, 2j, h \wr \partial_x, \partial_y, \partial_z)^3 - z \partial_x, \\ &(3h, b, i, f, 2i, 2k \wr \partial_x, \partial_y, \partial_z)^3 - x \partial_y, \\ &\&c. \end{aligned}$$

NO. 55. The discriminant is

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix}$$

which is equal to

$$abc - af^2 - bg^2 - ch^2 + 2fgh.$$

NO. 56. The reciprocant is

$$\begin{vmatrix} \xi, & \eta, & \zeta \\ \xi, & h, & g \\ \eta, & b, & f \\ \zeta, & f, & c \end{vmatrix}^2$$

which is equal to

$$(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch \wr \xi, \eta, \zeta)^2.$$

The discriminant is, it will be noticed, the same function as the Hessian. The reciprocant is the evectant of the discriminant. The covariants are the quadric itself and the discriminant; the reciprocant is the only contravariant.

COVARIANT AND OTHER TABLES, NOS. 57 TO 70 (A TERNARY CUBIC).

NO. 57. The cubic is  $U = (a, b, c, f, g, h, i, j, k, l \wr x, y, z)^3$ , which means

$$ax^3 + by^3 + cz^3 + 3fyz^2 + 3gz^2x + 3hx^2y + 3iy^2z + 3jzx^2 + 3kxy^2 + 6lxyz.$$

NO. 58. The first derived functions (omitting the factor 3) are

$$\begin{aligned} &(a, h, j, l \wr x, y, z)^2, \\ &(l, b, f, i \wr x, y, z)^2, \\ &(g, l, c, f \wr x, y, z)^2, \\ &(h, k, l, g \wr x, y, z)^2. \end{aligned}$$

NO. 59. The operators which reduce a covariant to zero are

$$\begin{aligned} &(j, 3f, c, 2i, g, 2l \wr \partial_x, \partial_y, \partial_z)^3 - y \partial_z, \\ &(a, k, 3h, 2l, 2j, g \wr \partial_x, \partial_y, \partial_z)^3 - z \partial_x, \\ &(3h, b, i, f, 2l, 2k \wr \partial_x, \partial_y, \partial_z)^3 - x \partial_y, \\ &(3j, f, i, c, 2g, 2l \wr \partial_x, \partial_y, \partial_z)^3 - z \partial_x, \\ &(a, l, g, 2j, h, 2l \wr \partial_x, \partial_y, \partial_z)^3 - x \partial_z, \\ &(2l, b, 3i, f, k, 2h \wr \partial_x, \partial_y, \partial_z)^3 - y \partial_x. \end{aligned}$$

NO. 60. The evector is

$$(3a, 3b, 3c, 3f, 3g, 3h, 3i, 3j, 3k, 3l \wr \xi, \eta, \zeta)^3.$$

NO. 61. The Hessian is  $HU =$

$$\begin{vmatrix} (a, h, j, l \wr x, y, z), & (h, k, l, g \wr x, y, z), & (j, l, g, i \wr x, y, z) \\ (h, k, l, g \wr x, y, z), & (k, b, f, i \wr x, y, z), & (l, f, i, c \wr x, y, z) \\ (j, l, g, i \wr x, y, z), & (l, f, i, c \wr x, y, z), & (g, i, c, f \wr x, y, z) \end{vmatrix}$$

or in the developed form

$$a^2bcd + \dots$$

$$\begin{aligned} \Theta_3 U = & -2(1 + 3l^2)(yz^2 + z^2x + x^2y) + (1 - 9l^2)(x^3 + y^3 + z^3) \\ & + (3l - 9l^2 + 96l^3)(x^2 + y^2 + z^2)xyz + \dots \end{aligned}$$

No. 62. The quartinvariant is  $\Omega =$

$$abcl - 1 \cdot fgh + 1 \cdot abgi + 1 \cdot f^2hk + 3 \cdot c^2fjk + 1 \cdot fgjk - 1 \cdot a^2i^2 - 1 \cdot h^2ij - 1 \cdot a \\ - 2 \cdot a^2k^2 + 1 \cdot c^2h^2k + 1 \cdot bchj + 1 \cdot ghki - 1 \cdot bf^2h - 2 \cdot g^2h^2 - 2 \cdot bgji + 1 \cdot h^3l + \\ - 1 \cdot irj^2 - 1 \cdot cf^2i + 3 \cdot c^2h^2k + 1 \cdot l^4 + 1 \cdot cjk^2 \pm 16$$

No. 63. The sextinvariant is  $\Sigma =$

$$a^2c^2i^2 + 1 \cdot acf^4m - 24 \cdot bcf^2hl - 12 \cdot cf^2h^2l^2 - 12 \cdot fg^2ik^2l^2 - 12 \cdot a^2b^2cfi - 6 \cdot a \\ - 12 \cdot bc^2gh^2l^2 + 18 \cdot bcgh^2jk + 60 \cdot cf^2hj^2kl + 36 \cdot fh^2ij^2 - 12 \cdot a^2b^2c^2fi + 4 \cdot ac^2f \\ - 12 \cdot bc^2gh^2jk + 36 \cdot bch^3j^2 - 12 \cdot cf^2h^3l^2 + 24 \cdot c^2gh^3m + \dots \pm 54$$

No. 64. The cubicontravariant is  $\Psi U =$

$$bcl \cdot \xi^3 - 1 \cdot acl \cdot \eta^3 + 1 \cdot abl \cdot \zeta^3 + 1 \cdot ack \cdot \xi^2\eta + 1 \cdot abi \cdot \xi^2\zeta + 1 \cdot bcj \cdot \xi\eta^2 \\ + 1 \cdot abg \cdot \xi\zeta^2 + 1 \cdot bch \cdot \eta^3 + 1 \cdot acf \cdot \eta^2\zeta + 1 \cdot ajc \cdot \zeta^2\xi - 1 \cdot bgi \cdot \xi\eta\zeta + 1 \cdot agi \cdot \xi^2\zeta \\ + 1 \cdot afk \cdot \xi^2\eta + af^2 \cdot \xi\eta^2 - 2 \cdot ai^2 \cdot \eta^2\zeta - 1 \cdot b^2j \cdot \eta^3 + 1 \cdot ai^2 \cdot \xi\zeta^2 + 1 \cdot hgl \cdot \xi\eta\zeta \\ - 1 \cdot af^2 \cdot \xi^2\eta + 1 \cdot ah^2 \cdot \eta^3 + 1 \cdot cfk \cdot \eta^2\zeta + 1 \cdot chj \cdot \xi\eta^2 + 1 \cdot bhj \cdot \eta^3 + 1 \cdot aij \cdot \xi^2\zeta \\ + 1 \cdot bgh \cdot \xi\eta\zeta - 2 \cdot c^2h \cdot \eta^2\zeta + aik \cdot \xi^2\eta - 2 \cdot g^2i \cdot \xi\zeta^2 + bgj \cdot \xi\eta^2 + \dots$$

No. 65. The quintic contravariant is  $\Phi U =$

$$ab^2c^2 \cdot \xi^5 + 1 \cdot a^2b^2c^2 \cdot \xi^3\eta^2 + 1 \cdot a^2b^3c \cdot \xi^3\zeta^2 + 1 \cdot a^2bc^3i \cdot \xi^4\zeta - 3 \cdot ab^3c^2j \cdot \xi^3\eta^2 \\ - 3 \cdot abc^3k \cdot \xi^4\zeta - 3 \cdot a^2bc^2l \cdot \xi^4\eta - 3 \cdot ab^2c^2g \cdot \xi^3\eta\zeta - 3 \cdot a^2b^2c^2h \cdot \xi^3\eta\zeta \\ - 3 \cdot abc^2fj \cdot \xi^3\eta\zeta + 6 \cdot abc^2fi \cdot \xi^3\eta\zeta - 6 \cdot a^2c^2fi \cdot \xi^4\zeta + 3 \cdot a^2bf^2i \cdot \xi^3\zeta^2 \\ + 6 \cdot ab^2c^2g \cdot \xi^3\eta\zeta + 6 \cdot abc^2gk \cdot \xi^3\eta\zeta + 6 \cdot a^2bcl \cdot \xi^4\eta + 6 \cdot ab^2chl \cdot \xi^3\eta^2 \\ + 6 \cdot abf^2g^2i \cdot \xi^3\eta\zeta + 9 \cdot ab^2c^2i^2 \cdot \xi^3\zeta^2 - 6 \cdot abc^2l^2 \cdot \xi^3\eta^2 - 30 \cdot a^2b^2c^2l \cdot \xi^3\eta\zeta \\ + 4 \cdot a^2c^2g^2 \cdot \xi^4\zeta - 3 \cdot ab^2c^2g \cdot \xi^3\eta\zeta + 4 \cdot ab^2c^2h \cdot \xi^3\eta^2 + 3 \cdot abeh^2k \cdot \xi^3\eta\zeta \\ - 6 \cdot abc^2hl \cdot \xi^3\eta\zeta + 6 \cdot abf^2ij \cdot \xi^3\eta\zeta + 9 \cdot abf^2gi \cdot \xi^2\eta\zeta^2 + \dots$$

No. 66. The reciprocant is  $FU = (\Psi \wr \xi, \eta, \zeta)^2,$

$$\Phi' = b^2c^2l^2 + 1 \cdot bcf^2i - 6 \cdot b^2c^2h^2 + 4 \cdot c^2f^4 + 4 \cdot f^2h^2k^2 - 3 \cdot a^2i^4 + 1 \cdot ac^2gj \\ - 6 \cdot a^2c^2k^2 + 4 \cdot c^2g^4 + 4 \cdot g^2i^2j^2 - 3 \cdot a^2j^4 + 1 \cdot ab^2hk - 6 \cdot a^2b^2i^2 + 4 \cdot b^2h^4 \\ + 4 \cdot h^4k^2i^2 - 3 \cdot b^2k^4 + 6 \cdot a^2cfi + 6 \cdot b^2c^2gk + 6 \cdot c^2abhl + 6 \cdot abc^2fj \\ + 12 \cdot abc^2gl - 12 \cdot ab^2ch^2 - 12 \cdot ac^2bi^2 - 12 \cdot bc^2aj^2 + \dots \pm 9$$

No. 67. The following table shows the number of covariants and contravariants:

|         | $\xi^4$ | $\eta^4$ | $\zeta^4$ | $\eta^2\zeta^2$ | $\zeta^2\xi^2$ | $\xi^2\eta^2$ | $\xi^2\eta\zeta$ | $\eta^2\zeta\xi$ | $\zeta^2\xi\eta$ |
|---------|---------|----------|-----------|-----------------|----------------|---------------|------------------|------------------|------------------|
| $a^2cf$ | +6      |          |           |                 |                |               |                  |                  |                  |
| $ab^2e$ |         | +6       |           |                 |                |               |                  |                  |                  |
| $ac^2k$ |         |          | +6        |                 |                |               |                  |                  |                  |
| $a^2bc$ | -2      | -2       |           |                 |                |               |                  | +6               |                  |
| $ab^2c$ |         | -2       |           |                 |                |               | +6               |                  |                  |
| $abc^2$ |         |          | -2        |                 |                | +6            |                  |                  |                  |

We have, for the covariants and contravariants of the cubic following Table No. 69:

$$\begin{aligned}
 \Theta(\alpha U + 6\beta HU) &= \Theta\alpha PU + \Theta\beta QU, \\
 H(\Theta\alpha PU + \Theta\beta QU) &= (-2T, 48S^2, 18TS, T^2 + 16S^3 \wr \alpha, \beta)^2 PU \\
 &\quad + (8S, T, -24S^2, -TS \wr \alpha, \beta)^2 QU, \\
 P(\Theta\alpha PU + \Theta\beta QU) &= (32S^2, 12TS, T^2 + 32S^3, 4TS^2 \wr \alpha, \beta)^3 U \\
 &\quad + (4T, 8S^2, TS, T^2 + 16S^3 \wr \alpha, \beta)^3 HU. \\
 S(\Theta\alpha PU + \Theta\beta QU) &= \alpha^{12} + 192\beta S^3 + 128T\beta^2 S^2 + 18T^2\beta^3 S + 384S^4\beta^4 \\
 &\quad + \alpha^{10}\beta T + 64\alpha\beta^2 TS^2 + 8T^2\beta^3 S - 64S^4\beta^4, \\
 T(\Theta\alpha PU + \Theta\beta QU) &= 8T^7 + 4608T\beta S^6 + 1920T^2\beta^2 S^5 + 73728S^8\beta^3 \\
 &\quad + 3840T^3\beta^3 S^4 + 20T^4\beta^2 + 8960T^2\beta^4 S^3 \\
 &\quad + 840T^3\beta^5 S^2 + 7680T\beta^6 S^5 + 36T^5\beta^3 S \\
 &\quad + 384T^2\beta^7 S^4 + 24576S^8\beta^8 + T^8\beta^4 - 40T^4\beta^8 S^2 + 2560T\beta^9 S
 \end{aligned}$$

The tables for the ternary cubic become much more simple if we suppose that the cubic is expressed in Hesse's canonical form; we have then the

$$\begin{aligned}
 \Theta_3 U &= -2(1 + 3l^2)(yz^2 + z^2x + x^2y) + (1 - 9l^2)(x^3 + y^3 + z^3) \\
 &\quad + (3l - 180l^2 + 96l^3)(x^2 + y^2 + z^2)xyz + (3l^2 - 624l^3 + 432l^4)xyz + \dots
 \end{aligned}$$

145.

## A MEMOIR UPON CAUSTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. cxlvii. for the year 1857, pp. 273—312. Received May 1,—Read May 8, 1856.]

The following memoir contains little or nothing that can be considered new in principle; the object of it is to collect together the principal results relating to caustics in plano, the reflecting or refracting curve being a right line or a circle, and to discuss, with more care than appears to have been hitherto bestowed upon the subject, some of the more remarkable cases.

The memoir contains in particular researches relating to the caustic by refraction of a circle for parallel rays, the caustic by reflexion of a circle for rays proceeding from a point, and the caustic by refraction of a circle for rays proceeding from a point; the result in the last case is not worked out, but it is shown how the equation in rectangular coordinates is to be obtained by equating to zero the discriminant of a rational and integral function of the sixth degree.

The memoir treats also of the secondary caustic, or orthogonal trajectory of the reflected or refracted rays, in the general case of a reflecting or refracting circle and rays proceeding from a point; the curve in question, or rather a secondary caustic, is, as is well known, the Oval of Descartes or 'Cartesian': the equation is discussed by a method which gives rise to some forms of the curve which appear to have escaped the notice of geometers.

## I.

Consider a ray of light reflected or refracted at a curve, and suppose that  $\xi, \eta$  are the coordinates of a point Q on the incident ray,  $\alpha, \beta$  the coordinates of the point G of incidence upon the reflecting or refracting curve,  $a, b$  the coordinates of a point N upon the normal at the point of incidence,  $x, y$  the coordinates of a point  $q$  on the reflected or refracted ray. Write for shortness,

$$(b - \beta)(\xi - \alpha) - (a - \alpha)(\eta - \beta) = \nabla QGN,$$

$$(a - \alpha)(\xi - \alpha) + (b - \beta)(\eta - \beta) = \Delta QGN,$$

then  $\nabla QGN$  is equal to twice the area of the triangle  $QGN$ , and if  $\xi, \eta$  instead of being the coordinates of a point Q on the incident



ray were current coordinates, the equation  $\nabla QGN = 0$  would be the equation of the line through the points G and N, i.e. of the normal at the point of incidence; and in like manner the equation  $\Delta QGN = 0$  would be the equation of the line through G perpendicular to the line through the points G and N, i.e. of the tangent at the point of incidence. We have identically,

$$NG = (a - \alpha)^2 + (b - \beta)^2,$$

$$NG \cdot QG = \nabla QGN + \Delta QGN.$$

Suppose for a moment that  $\phi$  is the angle of incidence and  $\phi'$  the angle of reflexion or refraction; and let  $\mu$  be the index of refraction, then

$$\sin \phi' = \mu \sin \phi.$$

## II.

But instead of investigating the nature of the caustic itself, we may begin by finding the secondary caustic or orthogonal trajectory of the refracted rays, i.e. a curve having the caustic for its evolute; suppose that the incident rays are all of them normal to a certain curve, and let Q be a point upon this curve, and considering the ray through the point Q, let G be the point of incidence upon the refracting curve; then if the point G be made the centre of a circle the radius of which is  $\mu^{-1} \cdot GQ$ , the envelope of the circles will be the secondary caustic.

## IV.

If rays proceeding from a point Q are reflected at a line, the reflected rays will proceed from a point  $q$  situate on the perpendicular let fall from Q, and at an equal distance on the other side of the reflecting line. The point  $q$  may be spoken of as the image of Q; it is clear that if Q be considered as a variable point, then the locus of the image  $q$  will be a curve equal and similar but oppositely situated to the curve, the locus of Q, and which may be spoken of as the image of such curve. Hence it at once follows, that if the incidental rays are tangent, or normal, or indeed in any other manner related to a curve, then the reflected rays will be tangent, or normal, or related in a corresponding manner to a curve the image of the first-mentioned curve.

### V.

The incident rays are here parallel rays, and the refracting curve a right line. We have  $\sin \phi' = k \sin \phi$ , and the equation  $y - x \tan \phi' = \tan \phi$  of the refracted ray becomes, putting for  $\phi'$  its value,

$$y - \frac{k \sin \phi}{\sqrt{1 - k^2 \sin^2 \phi}} x - \tan \phi = 0.$$

Differentiating with respect to the variable parameter and combining the two equations, we obtain, after a simple reduction,

$$kx = -\frac{k^2 y - k'^2 \sin^2 \phi}{\cos^2 \phi},$$

where  $k' = \sqrt{1 - k^2}$ ; hence eliminating,

$$(kx)^2 - (k'y)^2 = k^2 k'^2,$$

which is the equation of the caustic. When the refraction takes place into a denser medium  $k$  is less than 1, and  $k'^2$  is positive, the caustic is therefore the evolute of a hyperbola (see fig. 1); but when the refraction takes place in a rarer medium  $k$  is greater than 1, and  $k'^2$  is negative, the caustic is therefore the evolute of an ellipse (see fig. 2). These results appear to have been first obtained by Gergonne.

The conic (hyperbola or ellipse) is the secondary caustic, and as such may be obtained as follows.

### VI.

The equation of the variable circle is

$$x^2 + (y - \tan \phi)^2 - k^2 \sec^2 \phi = 0;$$

or reducing, the equation is

$$x^2 + y^2 - 2y \tan \phi + k'^2 \tan^2 \phi - k^2 = 0 :$$

whence, considering  $\tan \phi$  as the variable parameter, the equation of the envelope is

$$k'^2(x^2 + y^2 - k^2) - y^2 = 0,$$

that is,

$$k'^2 x^2 - k^2 y^2 - k^2 k'^2 = 0,$$

is the equation of the secondary caustic, or conic having the caustic for its evolute. The radiant point, it is clear, is a focus of the conic.

## VII.

Let the equation of the refracted ray be represented by

$$Xx + Yy + Z = 0,$$

we have

$$X : Y : Z = -\frac{k^2 \sin \phi}{\sqrt{1 - k^2 \sin^2 \phi}} : 1 : -\tan \phi,$$

from which we obtain

$$\frac{\xi^2}{X^2} + \frac{\eta^2}{Y^2} = \frac{1}{k^2}$$

for the tangential equation of the caustic; or if we represent the equation of the refracted ray by

$$Xx + Yy - k = 0,$$

then we have

$$\frac{\xi^2}{X^2} + \frac{\eta^2}{Y^2} = \frac{1}{k^2}$$

for the tangential equation of the caustic.

## VIII.

If a ray be reflected at a circle; we may take  $a, b$  as the coordinates of the centre of the circle, and supposing as before that  $\xi, \eta$  are the coordinates of a point  $Q$  in the incident ray,  $\alpha, \beta$  the coordinates of the point  $G$  of incidence, and  $x, y$  the coordinates of a point  $q$  in the reflected ray, the equation of the reflected ray, treating  $x, y$  as current coordinates, is

$$\{(b - \beta)(x - \alpha) - (a - \alpha)(y - \beta)\}\{(a - \alpha)(\xi - \alpha) + (b - \beta)(\eta - \beta)\} \\ + \{(a - \alpha)(x - \alpha) + (b - \beta)(y - \beta)\}\{(b - \beta)(\xi - \alpha) - (a - \alpha)(\eta - \beta)\} = 0.$$

Write for shortness,

$$\nabla_{q,0} = (b - \beta)(x - \alpha) - (a - \alpha)(y - \beta), \\ \Delta_{q,0} = (a - \alpha)(x - \alpha) + (b - \beta)(y - \beta),$$

and similarly for  $\nabla'_{q,0}$  &c.; the equation of the reflected ray is

$$\nabla_{q,0'} \Delta_{Q,G} + \Delta_{q,0} \nabla_{Q,G} = 0.$$

Suppose that the reflected ray meets the circle again in  $G'$  and undergoes a second reflexion, and let  $x', y'$  be the coordinates of a point  $q$  in the ray thus twice reflected. We see first ( $G'$  being a point in the first reflected ray) that

$$\nabla_{q',G'} \Delta_{Q,G} + \Delta_{q',G'} \nabla_{Q,G} = 0.$$

## IX.

Consider a ray reflected any number of times at a circle; and let  $\theta_0 G$  be the ray incident at  $G$ , and  $\theta G'$  the last reflected ray, the point at which the reflexion takes place or last point of incidence being  $G$ . Take the centre of the circle for the origin, and any two lines  $Ox, Oy$  through the centre and at right angles to each other for axes, and let  $Ox$  meet the circle in the point  $A$ . Write

$$\angle AOG = \theta_0, \quad \angle xG_0G = \psi_0, \quad \angle AOG = \theta, \quad \angle xGG' = \psi, \quad \angle G_0GG' = \phi;$$

then the radius of the circle being taken as the centre of the circle, the equation of the reflected ray is

$$y - \sin \theta = \tan \psi'(x - \cos \theta);$$

and if there have been  $n$  reflexions, then

$$\psi' = \psi_0 + n(\pi - 2\phi) = \psi_0 + n\pi - 2n\phi,$$

$$\psi = \psi_0 - 2n\phi,$$

and therefore the equation of the reflected ray is

$$y \cos(\psi_0 - 2n\phi) - x \sin(\psi_0 - 2n\phi) + (-1)^n \sin(\psi_0 - \theta_0) = 0.$$

### X.

If a pencil of parallel rays is reflected any number of times at a circle, then taking A for the direction of the incident rays, we may write  $\theta_0 = \phi$ ,  $\psi_0 = \pi$ , and the equation of a reflected ray is

$$x \sin 2n\phi + y \cos 2n\phi = (-1)^n \sin \phi;$$

differentiating with respect to the variable parameter, we find

$$x \cos 2n\phi - y \sin 2n\phi = (-1)^n \frac{1}{2} \cos \phi;$$

and these equations give

$$x = \frac{1}{2} \{ (2n+1) \cos(2n-1)\phi - (2n-1) \cos(2n+1)\phi \},$$

$$y = (-1)^n \{ (2n+1) \sin(2n-1)\phi + (2n-1) \sin(2n+1)\phi \},$$

which may be taken for the equation of the caustic; the caustic is therefore an epicycloid: this is a well-known result.

### XI.

If rays proceeding from a point upon the circumference are reflected any number of times at a circle, then taking the point A for the radiant point, we have  $\theta_0 = 0$ ,  $\psi_0 = \pi - \phi$ , and the equation of a reflected ray is

$$x \sin(2n+1)\phi + y \cos(2n+1)\phi = (-1)^n \sin \phi;$$

differentiating with respect to the variable parameter, we find

$$x \cos(2n+1)\phi - y \sin(2n+1)\phi = (-1)^n \frac{1}{2} \sin \phi;$$

and these equations give

$$x = \frac{1}{2} \{ (n+1) \cos 2n\phi - n \cos(2n+2)\phi \},$$

$$y = \frac{1}{2} \{ -(n+1) \sin 2n\phi + n \sin(2n+2)\phi \},$$

which may be taken as the equation of the caustic; the caustic is therefore in this case also an epicycloid: this is a well-known result.

## XII.

Consider a pencil of parallel rays refracted at a circle; take the radius of the circle as unity, and let the incident rays be parallel to the axis of  $x$ , then if  $\phi, \phi'$  be the angles of incidence and refraction, and  $\mu$  or  $k$  be the index of refraction, so that  $\sin \phi' = k \sin \phi$ , the coordinates of the point of incidence are  $\cos \phi, \sin \phi$ , and the equation of the refracted ray is

$$y - \sin \phi = \tan(\phi - \phi')(x - \cos \phi),$$

or

$$y \cos(\phi - \phi') - x \sin(\phi - \phi') = \sin \phi',$$

which may also be written

$$(y \cos \phi - x \sin \phi) \cos \phi' + (y \sin \phi + x \cos \phi - 1) \sin \phi' = 0.$$

Writing  $k \sin \phi, \sqrt{1 - k^2 \sin^2 \phi}$  instead of  $\sin \phi', \cos \phi'$ , and putting for shortness

$$Y = y \cos \phi - x \sin \phi, \quad X = y \sin \phi + x \cos \phi, \quad \omega = \frac{k \sin \phi}{\sqrt{1 - k^2 \sin^2 \phi}},$$

the equation of the refracted ray becomes

$$Y + \omega(X - 1) = 0;$$

and differentiating with respect to the variable parameter  $\phi$ ,  
observing that

$$\frac{dY}{d\phi} = y, \quad \frac{dX}{d\phi} = x, \quad \frac{d\omega}{d\phi} = \frac{k \cos \phi}{(1 - k^2 \sin^2 \phi)^{3/2}} = \frac{\omega \cot \phi}{1 - k^2 \sin^2 \phi},$$

we have

$$y + \omega \cdot x + \frac{\omega \cot \phi}{1 - k^2 \sin^2 \phi}(X - 1) = 0,$$

and the combination of the two equations gives

$$Y = \frac{\omega(1 - k^2 \sin^2 \phi)}{\omega \cot \phi - 1}, \quad X = \frac{\omega \cot \phi - k^2 \sin^2 \phi}{\omega \cot \phi - 1};$$

and we have therefore

$$w = Y \cos \phi + X \sin \phi = \frac{\omega \sin \phi (1 - k^2 \sin^2 \phi) + \omega \cot \phi \sin \phi - k^2 \sin^3 \phi}{\omega \cot \phi - 1} = k \sin^3 \phi,$$

$$\begin{aligned} x &= X \cos \phi - Y \sin \phi \\ &= \frac{\omega \cot \phi \cos \phi - k^2 \sin^2 \phi \cos \phi - \omega \sin^2 \phi \sqrt{1 - k^2 \sin^2 \phi} + \omega \sin \phi \cos \phi}{\omega \cot \phi - 1} \\ &= \frac{k(1 - k^2 \sin^2 \phi) - k^2 \sin^2 \phi \cos \phi}{k \cos \phi - \sin \phi \sqrt{1 - k^2 \sin^2 \phi}}. \end{aligned}$$

### XIII.

Or multiplying the numerator and denominator by  $(1 - k^2 \sin^2 \phi)^{1/2}(k \cos \phi + \sin \phi \sqrt{1 - k^2 \sin^2 \phi})$ , the numerator becomes

$$\begin{aligned} &(1 - k^2 \sin^2 \phi)\{k \cos \phi(1 - k^2 \sin^2 \phi) - k^2 \sin^2 \phi \cos \phi \sqrt{1 - k^2 \sin^2 \phi} \\ &\quad + k \sin \phi \sqrt{1 - k^2 \sin^2 \phi}(\sin \phi(1 - k^2 \sin^2 \phi) - k^2 \sin^2 \phi \cos \phi)\} \\ &\quad = k^3 \sin^6 \phi \cos \phi \{(1 - k^2 \sin^2 \phi) - \sin^2 \phi(1 - k^2 \sin^2 \phi)\} \\ &+ k \sin^2 \phi(1 - k^2 \sin^2 \phi)^{3/2} = k^3 \sin^6 \phi \cos^3 \phi + k \sin^2 \phi(1 - k^2 \sin^2 \phi)^{3/2}, \end{aligned}$$

and the denominator becomes

$$k^2 \sin^2 \phi \cos^2 \phi - (1 - k^2 \sin^2 \phi) \sin^2 \phi = -k'^2 \sin^2 \phi,$$

$$\text{if } k'^2 = 1 - k^2.$$

Hence we have for the coordinates of the point of the caustic,

$$k'^2 x = -k'^2 \cos^3 \phi - \frac{k}{k'}(1 - k^2 \sin^2 \phi)^{3/2}, \quad y = k \sin^3 \phi;$$

and eliminating  $\phi$ , we obtain for the equation of the caustic,

$$k'^2 x = -k'^2(1 - y^2/k^2)^{3/2} - \frac{k}{k'}(1 - y^2)^{3/2};$$

or writing  $-\mu$  instead of  $k$ , we find

$$(1 - \mu^2)^{3/2} x = (1 - \mu^2 y^2/k^2)^{3/2} + \mu(1 - y^2)^{3/2}$$

for the equation of the caustic by refraction of the circle, for parallel rays. The equation was first obtained by St Laurent.

The discussion of the preceding equation presents considerable interest. In the first place to obtain the rational form write

$$\alpha^2 = \beta^2 + \gamma^2,$$

this gives

$$\alpha^4 - 2\alpha^2(\beta^2 + \gamma^2) + (\beta^2 - \gamma^2)^2 = 0,$$

and we have

$$\gamma^2 = \mu^2 - 3\mu^2\beta^2 + 3\mu^4\beta^2 - \mu^6\beta^6,$$

and consequently

$$\alpha^4 - \beta^4 = (1 - \mu^2\beta^2)\{1 - 3\mu^2\beta^2 + (1 + \mu^2)\beta^4\}.$$

Hence dividing out by the factor  $(1 - \mu^2\beta^2)$ , the equation becomes

$$(1 - \mu^2)^2x^2 - 2(1 + \mu^2)x^2\{1 - 3\mu^2y^2 + (1 + \mu^2)y^4\} \\ + 2(1 + \mu^2)x^2y^2 + \{1 - 3\mu^2y^2 + (1 + \mu^2)y^4\}^2 = 0,$$

or reducing and arranging,

$$(1 - \mu^2)^2x^4 - 2(1 + \mu^2)x^2 + 2(1 + \mu^2)x^2y^2 + 1 + 2(1 + \mu^2)y^2 + (1 + \mu^2)^2y^4 \\ + (1 - 2\mu^2y^2 + 9\mu^4y^4)2x^2 - \{6\mu^2(1 + \mu^2)x^2 + 6\mu^2 + 6\mu^2(1 + \mu^2)y^2\}y^2 = 0,$$

which is of the form

$$A + 3\mu^2By^2 - 6\mu^2Cy^4 = 0;$$

and the rationalized equation is

$$A^2 + 27\mu^4B^2y^4 - 216\mu^4Cy^4 + 54\mu^2ABCy^2 = 0,$$

where the values of  $A$ ,  $B$ ,  $C$  may be written

$$B = 4x^2 + 3y^2;$$

the caustic is therefore a curve of the 12th order.

To find where the axis of  $x$  meets the curve, we have  $y = 0$ ,

$$A^2 = 0, \text{ where}$$

$$A_0 = (1 - \mu^2)x^2 - 2(1 + \mu^2)x + 1,$$



i.e.

$$x = \frac{1 + \mu \pm 2\sqrt{\mu}}{1 - \mu^2} = \frac{(1 \pm \sqrt{\mu})^2}{(1 - \mu)(1 + \mu)},$$

or there are in all four points, each of them a point of triple intersection.

To find where the line at infinity meets the curve, we have  $x = 0$ ,

$$A'^2 = 0,$$

where

$$A' = (x^2 + y^2)\{(1 - \mu^2)x^2 + (1 + \mu^2)y^2\},$$

i.e.

$$x = \pm iy, \quad x = \pm \sqrt{\frac{1 + \mu^2}{1 - \mu^2}} iy.$$

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## A MEMOIR UPON CAUSTICS.

XXVII. (*Secondary Caustic by Refraction at a Circle is the Cartesian.*)

Suppose that rays proceeding from a point are refracted at a circle, and let it be required to find the equation of the secondary caustic: take the centre of the circle as origin, let  $c$  be the radius,  $\xi$ ,  $\eta$  the coordinates of the radiant point,  $\alpha$ ,  $\beta$  the coordinates of a point upon the circle,  $\mu$  the index of refraction; the secondary caustic will be the envelope of the circle,

$$\mu^2\{(x - \alpha)^2 + (y - \beta)^2\} - \{(\xi - \alpha)^2 + (\eta - \beta)^2\} = 0,$$

where  $\alpha$ ,  $\beta$  are variable parameters connected by the equation  $\alpha^2 + \beta^2 - c^2 = 0$ ; the equation of the circle may be written in the form

$$\mu^2(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2) - 2(\mu^2\xi - x)\alpha - 2(\mu^2\eta - y)\beta = 0.$$

But in general the envelope of  $A\alpha + B\beta + C = 0$ , where  $\alpha$ ,  $\beta$  are connected by the equation  $\alpha^2 + \beta^2 - c^2 = 0$ , is  $c^2(A^2 + B^2) - C^2 = 0$ , and hence in the present case the equation of the envelope is

$$\{\mu^2(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2)\}^2 = 4c^2[(\mu^2\xi - x)^2 + (\mu^2\eta - y)^2],$$

which may also be written

$$\{\mu^2(x^2 + y^2 - c^2) - (\xi^2 + \eta^2 - c^2)\}^2 = 4c^2\mu^2[(\xi - x)^2 + (\eta - y)^2].$$

If the axis of  $x$  be taken through the radiant point, then  $\eta = 0$ , and writing also  $\xi = a$ , the equation becomes

$$\{\mu^2(x^2 + y^2 - c^2) - a^2 + c^2\}^2 = 4c^2\mu^2[(a - x)^2 + y^2];$$

or taking the square root of each side,

$$\mu^2(x^2 + y^2 - c^2) - a^2 + c^2 = 2c\mu\sqrt{(x - a)^2 + y^2};$$

whence multiplying by  $\frac{1}{\mu}$  and adding on each side

$$\frac{c}{\mu}\left(\mu - \frac{1}{\mu}\right) + (x - a)^2 + y^2, \text{ we have}$$

$$\frac{1}{\mu}\sqrt{(x - a)^2 + y^2} = \sqrt{(x - a)^2 + y^2 + c^2\left(\mu - \frac{1}{\mu}\right)^2},$$

which shows that the secondary caustic is the Oval of Descartes, or as it will be convenient to call it, the **Cartesian**.

**XXVIII.** (*Equivalent Forms of the Equation of the Cartesian; Six Systems giving the same Caustic.*)

Reverting to the original form of the equation of the secondary caustic, multiplying by  $-\frac{3}{4}(1-\mu^2)$  and adding on each side

$-\frac{1}{\mu}\left(\mu - \frac{1}{\mu}\right) + \frac{1}{\mu^2}\{(x-a)^2 + y^2\}$ , the equation becomes

$$\left(\frac{1}{\mu}\sqrt{(x-a)^2 + y^2}\right)^2 - \left(\sqrt{(x-a)^2 + y^2 + c^2\left(\mu - \frac{1}{\mu}\right)^2}\right)^2 = \frac{c^2}{4}\left(\mu - \frac{1}{\mu}\right)^2,$$

or extracting the square root,

$$\frac{1}{\mu}\sqrt{(x-a)^2 + y^2} - \sqrt{(x-a)^2 + y^2 + c^2\left(\mu - \frac{1}{\mu}\right)^2} = \pm \frac{c}{2}\left(\mu - \frac{1}{\mu}\right).$$

Combining this with the former result, we see that the equation may be expressed indifferently in any one of the four forms.

It follows, that if we write successively

$$\begin{aligned} \alpha' &= \alpha, & c' &= c, & \mu' &= \mu; & (1) \\ \alpha' &= \frac{a\alpha}{c}, & c' &= c, & \mu' &= \frac{a}{c\mu}; & (\alpha) \\ \alpha' &= -\frac{a\alpha}{c}, & c' &= c, & \mu' &= -\frac{a}{c\mu}; & (\beta) \\ \alpha' &= \frac{a\alpha}{c^2\mu^2}, & c' &= c, & \mu' &= \frac{c}{a\mu}; & (\gamma) \\ \alpha' &= \frac{a\alpha}{c^2\mu^2}, & c' &= c, & \mu' &= -\frac{c}{a\mu}; & (\delta) \\ \alpha' &= a, & c' &= c\mu, & \mu' &= \frac{1}{\mu}; & (\epsilon) \end{aligned}$$

or what is the same thing, writing the formulae with  $a'$ ,  $c'$ ,  $\mu'$  on the left-hand side,

$$a' = a, \quad c' = c, \quad \mu' = \mu; \quad (1)$$

$$a' = -\frac{a\alpha}{a'}, \quad c' = \frac{a^2}{c'}, \quad \mu' = \frac{c'}{a'\mu'}; \quad (\alpha)$$

$$a' = a, \quad c' = \frac{a^2}{c'}, \quad \mu' = -\frac{c'}{a'\mu'}; \quad (\beta)$$

$$a' = \frac{a\alpha'}{c'\mu'}, \quad c' = c', \quad \mu' = \frac{a}{a'\mu'}; \quad (\gamma)$$

$$a' = \frac{a\alpha'}{c'\mu'}, \quad c' = c', \quad \mu' = -\frac{a}{a'\mu'}; \quad (\delta)$$

$$a' = a, \quad c' = c'\mu', \quad \mu' = \frac{c'}{a\mu'}; \quad (\epsilon),$$

we have in each case identically the same secondary caustic, and therefore also identically the same caustic; in other words, the same caustic is produced by six different systems of a radiant point and refracting circle.

It is proper to remark that if we represent the six systems of equations by  $(\alpha', c', \mu') = (\alpha, c, \mu)$ ,  $(\alpha', c', \mu') = \alpha(\alpha, c, \mu)$ , then,  $\alpha, \beta, \gamma, \delta, \epsilon$  will be functional symbols satisfying the conditions

$$1 = \alpha\beta = \beta\alpha = \gamma^2 = \delta^2 = \epsilon^2,$$

$$\alpha = \beta\gamma = \delta\gamma = \epsilon\delta = \gamma\epsilon,$$

$$\beta = \alpha\gamma = \gamma\delta = \delta\epsilon = \epsilon\gamma,$$

$$\gamma = \delta\alpha = \alpha\epsilon = \epsilon\beta = \beta\delta,$$

$$\delta = \epsilon\alpha = \alpha\gamma = \gamma\beta = \beta\epsilon,$$

$$\epsilon = \gamma\alpha = \alpha\delta = \delta\beta = \beta\gamma.$$

### XXIX. (*Particular Cases: Theorems of St. Laurent.*)

The preceding formulae, which were first given by me in the *Philosophical Magazine*, December 1853, [124], include as particular cases a preceding theorem with respect to the caustic by refraction of parallel rays, and also two theorems of St. Laurent, *Gergonne*, t. XVII. [1827, pp. 1–19], if we suppose first that  $a = c$ , that the radiant point is in the circumference of the refracting circle, then the system  $(\alpha)$  shows that the same caustic would be obtained by writing  $c, \frac{a}{c\mu}, 1$  (or what is the same thing  $-1$ ) in the place of  $c, \alpha, \mu$ , and we have

**Theorem.** *The caustic by refraction for a circle when the radiant point is in the circumference is also the caustic by reflexion for the*

same radiant point, and for a reflecting circle concentric with the refracting circle, but having its radius equal to the quotient of the radius of the refracting circle by the index of refraction.

Next, if we write  $a = c\mu$ , then the refracted rays all of them pass through a point which is a double point of the secondary caustic, the entire curve being in this case the orthogonal trajectory, not of the refracted rays, but of the false refracted rays; the formula ( $\delta$ ) shows that the same caustic is obtained by writing  $\frac{a\alpha}{c^2\mu^2}$ ,  $c$ , 1 (or what is the same thing  $-1$ ) in the place of  $a$ ,  $c$ ,  $\mu (= \frac{a}{c})$ , and we have

**Theorem.** *The caustic by refraction for a circle when the distance of the radiant point from the centre is to the radius of the circle in the ratio of the index of refraction to unity, is also the caustic by reflexion for the same circle considered as a reflecting circle, and for a radiant point the image of the former radiant point.*

**XXX.** (*Description of the Forms of the Cartesian.*)

The curve is most easily traced by means of the preceding construction; thus if we take the radiant point outside the refracting circle, and consider  $\mu$  as varying from a small to a large value (positive or negative values of  $\mu$  give the same curve), we see that when  $\mu$  is small the curve consists of two ovals, one of them within and the other without the refracting circle (see fig. 14).

As  $\mu$  increases the exterior oval continually increases, but undergoes modifications in its form; the interior oval in the first instance diminishes until we arrive at a curve, in which the interior oval is reduced to a conjugate point (see fig. 15); then as  $\mu$  continues to increase the interior oval reappears (see fig. 16), and at last connects itself with the exterior oval, so as to form a curve with a double point (see fig. 17); and as  $\mu$  increases still further the curve again breaks up into an exterior and an interior oval (see fig. 18); and henceforward as  $\mu$  goes on increasing consists always of two ovals; the shape of the exterior oval is best perceived from the figures. An examination of the figures will also show how the same curves may originate from a different refracting circle and radiant point.

**XXXI.** (*Deduction of a Theorem.*)

The theorem, "If a variable circle have its centre upon a circle  $S$ , and its radius proportional to the tangential distance of the centre

from a circle  $C$ , the envelope is a Cartesian," is at once deducible from the theorem "If a variable circle have its centre upon a circle  $S$  and its radius proportional to the distance of the centre from a point  $C$ , the locus is a Cartesian," which last theorem was in effect given in discussing the theory of the secondary caustic.

In fact, the locus of a point  $P$  such that its tangential distances from the circles  $C, C'$  are in a constant ratio, is a circle  $S$ .

Conversely, if there be a circle  $C$ , and the locus of  $P$  be a circle  $S$ , then the circle  $C'$  may be found such that the tangential distances of  $P$  from the two circles are in a constant ratio, and the circle  $C'$  may be taken to be a point, if there be a circle  $C$  and the locus of

$P$  be a circle  $S$ , then a point  $C$  may be found such that the tangential distance of  $P$  from the circle  $C$  is in a constant ratio to the distance from the point  $C$ . Hence treating  $P$  as the centre of the variable circle, it is clear that the variable circle is determined in the two cases by equivalent constructions, and the envelope is therefore the same in both cases.

### XXXII. (*Standard Form of the Equation of the Cartesian.*)

The equation of the secondary caustic developed and reduced is

$$\mu^4(x^2 + y^2)^2 - 2\mu^2\{a^2 + (\mu^2 + 1)c^2\}(x^2 + y^2) + 8c^2\mu^2\alpha^2 + a^4 - 2a^2c^2(\mu^2 + 1) + (\mu^2 - 1)^2c^4 = 0,$$

or, what is the same thing,

$$\{\mu^2(x^2 + y^2) - [a^2 + (\mu^2 + 1)c^2]\}^2 + 8c^2\mu^2\alpha^2 - 4c^2[c^2\mu^2 + (\mu^2 + 1)a^2] = 0,$$

which may also be written

$$\{\mu^2(x^2 + y^2) - [a^2 + (\mu^2 + 1)c^2]\}^2 = 4c^2\mu^2\{(a - c)^2 + (\mu^2 - 1)c^2\},$$

which is of the form

$$(x^2 + y^2 + Dx + F)^2 = A(x - \alpha)(x - \beta),$$

and the values of the coefficients are

### XXXIII.

The equation just obtained should, I think, be taken as the standard form of the equation of the Cartesian, and the form of the equation shows that the Cartesian may be defined as the locus of a

point, such that the fourth power of its tangential distance from a given circle is in a constant ratio to its distance from a given line.

**XXXIV.** (*The Cartesian as a Quartic Curve: Dirigent Circle and Directrix.*)

The Cartesian is a curve of the fourth order, symmetrical about a certain line which it intersects in four arbitrary points, and these points determine the curve. Taking the line in question (which may be called the **axis**) as the axis of  $x$ , and a line at right angles to it as the axis of  $y$ , let  $a, b, c, d$  be the values of  $x$  corresponding to the points of intersection with the axis, then the equation of the curve is

$$\begin{aligned} y^2 + x^4 - (a + b + c + d)x^3 \\ + \frac{1}{2}(a^2 + b^2 + c^2 + d^2 - 2ab - 2ac - 2ad - 2bc - 2bd - 2cd)x^2 \\ + (abc + abd + acd + bcd)x \\ - \frac{1}{4}(a^2b^2 + a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 + c^2d^2 - 2abcd) = 0, \end{aligned}$$

which is of the form

$$(x^2 + y^2 - \alpha)^2 + 16A(x - m) = 0,$$

and, as already remarked, signifies that the fourth power of the tangential distance of a point in the curve from a given circle, is proportional to the distance of the same point from a given line.

The circle in question (which may be called the **dirigent circle**) has for its equation

$$x^2 + y^2 + \frac{1}{2}(ab + ac + ad + bc + bd + cd) = 0;$$

the line in question, which may be called the **directrix**, has for its equation

$$x = \frac{a^2b^2 + a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 + c^2d^2 - 2abcd}{4(abc + abd + acd + bcd)};$$

the multiplier of the distance from the directrix is  
 $abc + abd + acd + bcd$ .

It may be remarked that  $a, b, c, d$  being real, the dirigent circle is real; the equation may, in fact, be written

$$x^2 + y^2 = \frac{1}{8}[(a+b)^2 + (a+c)^2 + (a+d)^2 + (b+c)^2 + (b+d)^2 + (c+d)^2].$$

**XXXV.** (*Discussion of the Equation: Conditions for Double Points and Cusps.*)

Considering the equation of the Cartesian under the form

$$(x^2 + y^2 - \alpha)^2 + 16A(x - m) = 0,$$

the centre of the dirigent circle  $x^2 + y^2 - \alpha = 0$  must be considered as a real point, but  $\alpha$  may be positive or negative, the radius may be either a real or a pure imaginary distance: the coefficients  $A$ ,  $m$  must be real, the directrix is therefore a real line.

The equation shows that for all points of the curve  $x - m$  is always negative or always positive, according as  $A$  is positive or negative, that the curve lies wholly on one side of the directrix, on the same side with the centre of the dirigent circle if  $A$  is positive, but on the contrary side if  $A$  is negative. In the former case the curve may be said to be an “inside” curve, in the latter an “outside” curve. If  $m = 0$ , or the directrix passes through the centre of the dirigent circle, then the distinction between an inside curve and an outside curve no longer exists.

It is clear that the curve touches the directrix in the points of intersection of this line and the dirigent circle, and that the points in question are the only points of intersection of the curve with the directrix or the dirigent circle; hence if the directrix and dirigent circle do not intersect, the curve does not meet either the directrix or the dirigent circle.

To discuss the equation

$$(x^2 + y^2 - \alpha)^2 + 16A(x - m) = 0,$$

I write first  $y = 0$ , which gives

$$x^4 - 2\alpha x^2 + 16Ax + \alpha^2 - 16Am = 0$$

for the points of intersection with the axis of  $x$ . If this equation has equal roots, there will be a double point on the axis of  $x$ , and it is important to find the condition that this may be the case.

The equation may be written in the form

$$(1, 0, -\alpha, 12A, \alpha^2 - 48Am \mid x, 1)^2 = 0,$$

the condition for a pair of equal roots is then at once seen to be

$$-(\alpha^2 - 12Am)^3 + (\alpha^3 - 18Am\alpha + 54A^2)^2 = 0;$$



or reducing and throwing out the factor  $A^2$ , this is

$$27A^2 + 2m(8m^2 - 9\alpha)A - \alpha^2(m^2 - \alpha) = 0.$$

This equation will give two equal values for  $A$  if

$$m^2(8m^2 - 9\alpha)^2 + 27\alpha^2(m^2 - \alpha) = 0,$$

an equation which reduces itself to

$$(4m^2 - 3\alpha)^3 = 0.$$

Hence, if  $4m^2 - 3\alpha$  be negative, if  $\alpha > \frac{4}{3}m^2$ , the values of  $A$  will be imaginary, but if  $4m^2 - 3\alpha$  be positive, or  $\alpha < \frac{4}{3}m^2$ , the values of  $A$  will be real. If  $\alpha = \frac{4}{3}m^2$ , then there will be two equal values of  $A$ , which in fact corresponds to a cusp upon the axis of  $x$ .

Whenever the curve is real there will be at least two real points on the axis of  $x$ ; and when  $\alpha < \frac{4}{3}m^2$ , but not otherwise, then for properly selected values of  $A$  there will be four real points on the axis of  $x$ .

Differentiating the equation of the curve, we have

$$\{(x^2 + y^2 - \alpha)x + 4A\} dx + (x^2 + y^2 - \alpha)y dy = 0;$$

and if in this equation we put  $dx = 0$ , we find  $y = 0$ , or  $x^2 + y^2 - \alpha = 0$ , that the points on the axis of  $x$ , and the points of intersection with the circle  $x^2 + y^2 = \alpha$ , are the only points at which the curve is perpendicular to the axis of  $x$ .

To find the points at which the curve is parallel to the axis of  $x$ , we must write  $dy = 0$ , this gives

$$(x^2 + y^2 - \alpha)x + 4A = 0,$$

and thence

$$x^2 + y^2 - \alpha = -\frac{4A}{x};$$

this equation will have three real roots if  $A^2 < \frac{4}{27}\alpha^3$ , and only a single real root if  $A^2 > \frac{4}{27}\alpha^3$ ; for  $A^2 = \frac{4}{27}\alpha^3$ , the equation in question will have a pair of equal roots. It is easy to see that there is always a single real root of the equation which gives rise to a real value of  $y$ , to a real point upon the curve; but, when the equation

has three real roots, two of the roots may or may not give rise to real points upon the curve.

### XXXVI. (*Tracing of the Curve.*)

It is now easy to trace the curve.

First, when  $m = 0$ , or the directrix passes through the centre of the dirigent circle, the curve is here an oval bent in so as to have double contact with the directrix, and lying on the one or the other side of the directrix according to the sign of  $A$ . See fig. *a*.

Next, when the directrix does not pass through the centre of the dirigent circle, it will be convenient to suppose always that  $m$  is positive, and to consider  $A$  as passing first from 0 to  $\infty$  and then from 0 to  $-\infty$ , to consider first the different inside curves, and then the different outside curves.

Suppose  $\alpha > \frac{4}{3}m^2$ , the inside curve is at first an oval, as in fig. *b*, where (attending to one side only of the axis) it will be noticed that there are three tangents parallel to the axis, one for the convexity of the oval, and two for the concavity. For  $A^2 = \frac{4}{27}\alpha^3$  the two tangents for the concavity come together, and give rise to a stationary tangent (a tangent at an inflection) parallel to the axis, and for  $A^2 > \frac{4}{27}\alpha^3$  the two tangents for the concavity disappear. The outside curve is an oval (of course on the opposite side of, and) bent in so as to have double contact with the directrix.

Next, if  $\alpha = \frac{4}{3}m^2$ , the inside curve is at first an oval, as in fig. *c*, and there are, as before, three tangents parallel to the axis: for  $A^2 = \frac{4}{27}\alpha^3$ , the tangents for the concavity of the oval come to coincide with the axis, and are tangents at a cusp, and for  $A^2 > \frac{4}{27}\alpha^3$  the cusp disappears, and there are not for the concavity of the oval any tangents parallel to the axis. The outside curve is an oval as before, but smaller and more compressed.

Next,  $\alpha < \frac{4}{3}m^2 > m^2$ , then the inside curve is at first an oval, as in fig. *d*, and there are, as before, three tangents parallel to the axis; when  $A$  attains a certain value which is less than  $\frac{4}{27}\alpha^3$ , the curve acquires a double point; and as  $A$  further increases, the curve breaks up into two separate ovals, and there are then only two tangents parallel to the axis, one for the exterior oval and one for the interior oval. As  $A$  continues to increase, the interior oval decreases; and when  $A$  attains a certain value which is less than  $\frac{4}{27}\alpha^3$ , the interior oval reduces itself to a conjugate point, and it afterwards disappears altogether. The outside curve is an oval as

before, but smaller and more compressed.

Next, if the directrix touch the dirigent circle, if  $\alpha = m^2$ . Then the inside curve is at first composed of an exterior oval which touches the dirigent circle, and of an interior oval which lies wholly within the dirigent circle. As  $A$  increases the interior oval decreases, reduces itself to a conjugate point, and then disappears. The outside curve is an oval which always touches the dirigent circle.

Next, when the directrix does not meet the dirigent circle, if  $\alpha < m^2$ . The inside curve consists at first of two ovals, an exterior oval lying without the dirigent circle, and an interior oval lying within the dirigent circle. As  $A$  increases the interior oval decreases, reduces itself to a conjugate point and disappears. The outside curve is at first imaginary, but when  $A$  attains a sufficiently large negative value, it makes its appearance as a conjugate point, and afterwards becomes an oval which gradually increases.

Next, when the dirigent circle reduces itself to a point, if  $\alpha = 0$ .

The inside curve makes its appearance as a conjugate point (corresponding to  $A = 0$ ), and as  $A$  increases it becomes an oval and continually increases. The outside curve comports itself as in the last preceding case.

Finally, when the dirigent circle becomes imaginary, or has for its radius a pure imaginary distance, if  $\alpha$  is negative. The inside curve is at first imaginary, but when  $A$  attains a certain value it makes its appearance as a conjugate point, and as  $A$  increases becomes an oval and continually increases. The outside curve, as in the preceding two cases, comports itself in a similar manner.

The discussion, in the present section, of the different forms of the curve is not a very full one, and a large number of figures would be necessary in order to show completely the transition from one form to another. The forms delineated in the four figures were selected as forms corresponding to imaginary values of the parameters by means of which the equation of the curve is usually represented, e.g. the equations in Section XXVIII.

### XXXVII. (*Tangential Equation of the Caustic: the Caustic is of Class 6.*)

It has been shown that for rays proceeding from a point and refracted at a circle, the secondary caustic is the Cartesian; the caustic itself is therefore the evolute of the Cartesian; this affords a means of finding the tangential equation of the caustic.

In fact, the equations which give the normal to the secondary caustic at the point  $(x, y)$  are

$$X = \partial_x Z, \quad Y = \partial_y Z,$$

where  $Z$  is the value obtained from the equation of the secondary caustic by writing it in the form  $Z = 0$ ; and then writing the equation of the normal in the form  $Xx + Yy + Z = 0$ , the envelope of this line is the caustic.

The equation of the secondary caustic being

$$(x^2 + y^2 - \alpha)^2 + 16A(x - m) = 0,$$

we have

$$X = 4(x^2 + y^2 - \alpha)x + 16A,$$

$$Y = 4(x^2 + y^2 - \alpha)y,$$

and thence, writing for shortness  $x^2 + y^2 - \alpha = P$ ,

$$X = 4Px + 16A,$$

$$Y = 4Py,$$

and from these equations we obtain

$$\begin{aligned} X^2 + Y^2 &= 16P^2(x^2 + y^2) + 128APx + 256A^2, \\ &= 16P^2(P + \alpha) + 128APx + 256A^2, \\ &= 16P^3 + 16\alpha P^2 + 128APx + 256A^2. \end{aligned}$$

Also  $Xx + Yy = 4P(x^2 + y^2) + 16Ax = 4P(P + \alpha) + 16Ax$ , and therefore

$$Z = -\frac{1}{4}(X^2 + Y^2) + \alpha P^2 + 4Ax(3P + 2\alpha) + 16A^2.$$

The equation  $Xx + Yy + Z = 0$  becomes therefore

$$\begin{aligned} Z^6 + 2Z^4X(mZ^2 - AX^2) + (Z^2 + Y^2)(mZ^2 - AX^2)^2 \\ - Z^2Y^2(\alpha Z^2 + 4AX^2) = 0, \end{aligned}$$

where, if as before,  $c$  denotes the radius of the refracting circle,  $a$  the distance of the radiant point from the centre, and  $\mu$  the index of refraction, we have

$$\alpha = -a^2 + (1 + \mu^2)c^2, \quad A = \frac{a^2}{c^2\mu^2}\{a^2 + (1 + \mu^2)c^2\}, \quad m = \frac{1}{2c^2\mu^2}\{1 + (1 + \mu^2)a^2\}.$$

The above equation is the condition in order that the line  $Xx + Yy + Z = 0$  may be a normal to the secondary caustic  $(x^2 + y^2 - \alpha)^2 + 16A(x - m) = 0$ , or it is the tangential equation of the caustic, which is therefore a curve of the class 6 only. The equation may be written in the more convenient form

$$Z^6 + 2Z^4X(mZ^2 - AX^2) + (Z^2 + Y^2)(mZ^2 - AX^2)^2 - Z^2Y^2(\alpha Z^2 + 4AX^2) = 0.$$

**XXXVIII.** (*Comparison with the result for Caustic by Reflexion.*)

To compare the last result with that previously obtained for the caustic by reflexion, I write  $\mu = -1$ , and putting also  $c = 1$  and  $Z = a$  (for the equation of the reflected ray was assumed to be  $Xx + Yy + a = 0$ ), we have

$$\alpha = a^2 + 2, \quad A = \frac{1}{4}a, \quad m = \frac{1}{4}(1 + 2a^2).$$

and the equation becomes, after a slight reduction,

$$4a^4 + 4a^2Z(2a^2 + 1 - Z^2) + (X^2 + Y^2)(2a^2 + 1 - Z^2)^2 - 4a^2Y^2(a^2 + 2 + 2Z) = 0,$$

which may be written

$$(2a^2 + Z(2a^2 + 1 - Z^2))^2 + Y^2(-4a^4 + 1 - 8a^2Z - 2(2a^2 + 1)Z^2 + Z^4) = 0;$$

this divides out by the factor  $(Z + 1)^2$ , and the equation then becomes,

$$(Z^2 - Z - 2a^2)^2 + Y^2((Z - 1)^2 - 4a^2) = 0,$$

which agrees with the result before obtained.

**XXXIX.** (*Comparison with the result for Parallel Rays refracted at a Circle.*)

Again, to compare the general equation with that previously obtained for parallel rays refracted at a circle, we must write  $\mu = \infty$ ,  $c = 1$ ,  $a = \infty$ ,  $Z = k$  (for the equation of the refracted ray was taken to be  $Xx + Yy + k = 0$ ); we have then

$$\alpha = 1 + \mu^{-2} + k^2a^2, \quad A = \frac{1}{4}k^2a^2, \quad m = \frac{1}{2}(1 + (1 + k^2)a^2),$$

and, after the substitution,  $a = \infty$ .

The equation becomes in the first instance

$$k^4 + 2k^2 Z \left\{ \frac{1}{2}(1 + (1 + k^2)a^2) - \frac{1}{4}k^2 a^2 \right\} + (X^2 + Y^2) \left\{ \frac{1}{2}(1 + (1 + k^2)a^2) - \frac{1}{4}k^2 a^2 \right\}^2 - k^2 Y^2 \{1 + k^{-2} + k^2 a^2 + 2k^2 a Z\} = 0;$$

and then putting  $a = \infty$ , or, what is the same thing, attending only to the terms which involve  $a^2$ , and throwing out the constant factor  $k^4$ , we obtain

$$(X^2 + Y^2)(X^2 - 1 - k^2) - k^2 Y^2 = 0,$$

or

$$X^2 \{X^2 - (1 + k^2)\} + Y^2 \{X^2 + 1 + k^2\} (X^2 - 1 - k^2) (X^2 + 1 - k^2) = 0,$$

which agrees with the former result.

**XL.** (*Secondary Caustic for Parallel Rays: Equation of 8th order.*)

It was remarked that the ordinary construction for the secondary caustic could not be applied to the case of parallel rays (the entire curve would in fact pass off to an infinite distance), and that the simplest course was to measure the distance  $GQ$  from a line through the centre of the refracting circle perpendicular to the direction of the rays.

To find the equation of the resulting curve, take the centre of the circle as the origin and the direction of the incident rays for the axis of  $x$ ; let the radius of the circle be taken equal to unity, and let  $\mu$  denote, as before, the index of refraction. Then if  $\alpha, \beta$  are the coordinates of the point of incidence of a ray, we have  $\alpha^2 + \beta^2 = 1$ , and considering  $\alpha, \beta$  as variable parameters connected by this equation, the required curve is the envelope of the circle,

$$\mu^2 \{(x - \alpha)^2 + (y - \beta)^2\} - \alpha^2 = 0.$$

Write now  $\alpha = \cos \theta$ ,  $\beta = \sin \theta$ , then multiplying the equation by  $-2$ , and writing  $1 + \cos 2\theta$  instead of  $2 \cos^2 \theta$ , the equation becomes

$$1 + \cos 2\theta - 2\mu^2(x^2 + y^2 - 2x \cos \theta - 2y \sin \theta + 1) = 0,$$

which is of the form

$$A \cos 2\theta + B \sin 2\theta + C \cos \theta + D \sin \theta + E = 0,$$

and the values of the coefficients are

$$A = 1, \quad B = 0, \quad C = 4\mu^2x, \quad D = 4\mu^2y, \quad E = -2\mu^2(x^2+y^2)-2\mu^2+1.$$

Substituting these values in the equation

$$\{12(A^2 + B^2) - 3(C^2 + D^2) + 4E^2\}^2 - \{11A(C^2 - D^2) + 5BCD - (7A^2(C^2 + D^2) + 9(C^4 + D^4))E + 8E^3\}^2 = 0,$$

the equation of the envelope is found to be

$$16\{(1 - \mu^2)^2 + 4\mu^4\} - (\mu^2 + \mu^4)(x^2 + y^2) + 16\mu^4(x^2 + y^2)^2 - (6\mu^2 + 3\mu^4 + 6\mu^6)(x^2 + y^2) - 27\mu^4(x^2 - y^2) + 4\mu^6(x^2 + y^2)^2 = 0,$$

which is readily seen to be only of the 8th order.

But to simplify the result, write first  $(x^2 + y^2 - 1) + 1$ , and  $2x^2 - 1 = (x^2 + y^2 - 1) + (x^2 - y^2)$  in the place of  $x^2 + y^2$  and  $x^2 - y^2$  respectively, the equation becomes

$$4\{(1 - \mu^2)^2 - \mu^2(1 - \mu^2)(x^2 + y^2 - 1) + \mu^4(x^2 + y^2 - 1)^2\} - 16\mu^2(1 - \mu^2)^2(x^2 + y^2 - 1) - 27\mu^4x^2 + 4\mu^4(x^2 + y^2 - 1)^2 = 0.$$

Write for a moment  $1 - \mu^2 = q$ ,  $\mu^2(x^2 + y^2 - 1) = p$ , the equation becomes

$$4(q^2 - qp + p^2)^2 - 16(q^2 - 3q^2p - 3qp^2 + 2p^2)^2 + 54(q^2 - 3q^2p - 3qp^2 + 2p^2)p^2x^2 - 72p^3x^2 = 0.$$

or developing,

$$4(q^2 - qp + p^2)^2 - (22q^2 - 8q^2p - 3qp^2 + 2p^2)^2 + 54(q^2 - 3q^2p - 3qp^2 + 2p^2)p^2x^2 - 72p^3x^2 = 0,$$

and reducing and dividing out by 27, this gives

$$\{p^2(p - q)^2 + 2(p + q)(2p - q)(p - 2q)p^2x^2 - 27p^3x^4\} = 0,$$

$p = 0$ , or

$$p^2(p - q)^2 + 2(p + q)(2p - q)(p - 2q)x^2 - 27px^4 = 0.$$

**XLI.** (*Final Equation; Orthogonal Trajectory of Refracted Rays.*)

Whence replacing  $q, p$  by their values, the required equation is

$$\begin{aligned} & (1 - \mu^2)^2(x^2 + y^2 - 1)^2\{\mu^2(x^2 + y^2) - 1\}^2 \\ & + 2(\mu^2(x^2 + y^2) - 2\mu^2 + 1)(2\mu^2(x^2 + y^2) - \mu^4 - 1) \\ & \times (\mu^2(x^2 + y^2) - 2 + \mu^2)x^2 - 27\mu^2x^4 = 0, \end{aligned}$$

which is the equation of an orthogonal trajectory of the refracted rays. In the case of reflexion,  $\mu = -1$ , and the equation becomes

$$4(x^2 + y^2 - 1)^3 - 27x^2 = 0.$$

Comparing this with the equation of the caustic, it is easy to see,

**Theorem.** *In the case of parallel rays and a reflecting circle, there is a secondary caustic which is a curve similar to and double the magnitude of the caustic, the position of the two curves differing by a right angle.*

**XLI.** (*System of Orthogonal Trajectories.*)

The entire system of the orthogonal trajectories of the refracted rays might in like manner be determined by finding the envelope of the circle (where, as before,  $\alpha, \beta$  are variable parameters connected by the equation  $\alpha^2 + \beta^2 = 1$ ),

$$\begin{aligned} & (3 - 12[\mu^2 + 2\mu^2\xi x + \mu^2(x^2 + y^2)] + [1 - 2\mu^2 + 2\mu^4 - 2\mu^2(x^2 + y^2)]^2)^2 \\ & - ([1 - 2\mu^2 + 2\mu^4 - 2\mu^2(x^2 + y^2)][9 + 18\mu^2 + 36\mu^2x + 18\mu^2(x^2 + y^2)] \\ & - 54[\mu^2 + 2\mu^2\xi x + \mu^2(x^2 - 3y^2)] - [1 - 2\mu^2 + 2\mu^4 - 2\mu^2(x^2 + y^2)]^2)^2 = 0, \end{aligned}$$

which, it is easy to see, is an equation of the order 8 only.

*Added Sept. 12.—A. C.*



[146]

## A MEMOIR ON CURVES OF THE THIRD ORDER.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLVII. for the year 1857, pp. 415–446. Received October 30,—Read December 11, 1856.]

A CURVE of the third order, or cubic curve, is the locus represented by an equation such as  $U = (\cap x, y, z)^3 = 0$ ; and it appears by my “Third Memoir on Quantics,” [144], that it is proper to consider, in connexion with the curve of the third order  $U = 0$ , and its Hessian  $HU = 0$  (which is also a curve of the third order), two curves of the third class, the curves represented by the equations  $PU = 0$  and  $QU = 0$ .

These equations, I say, represent curves of the third class; in fact,  $PU$  and  $QU$  are contravariants of  $U$ , and therefore, when the variables  $x, y, z$  of  $U$  are considered as point coordinates, the variables  $\xi, \eta, \zeta$  of  $PU$  and  $QU$  must be considered as line coordinates, and the curves will be curves of the third class.

I propose (in analogy with the form of the word Hessian) to call the two curves in question the **Pippian** and **Quippian** respectively.

[The curve  $PU = 0$  is now usually called the **Cayleyan**.]

A geometrical definition of the Pippian was readily found; the curve is in fact Steiner’s curve mentioned in the memoir “Allgemeine Eigenschaften der algebraischen Curven,” *Crelle*, t. XLVII. [1854] pp. 1–6, in the particular case of a basis-curve of the third order; and I also found that the Pippian might be considered as occurring implicitly in my “Mémoire sur les courbes du troisième ordre,” *Liouville*, t. IX. [1844] pp. 285–291 [26] and “Nouvelles remarques sur les courbes du troisième ordre,” *Liouville*, t. X. [1845] pp. 102–109 [27].

As regards the Quippian, I have not succeeded in obtaining a satisfactory geometrical definition; but the search after it led to a variety of theorems, relating chiefly to the first-mentioned curve, and the results of the investigation are contained in the present memoir. Some of these results are due to Mr Salmon, with whom I was in correspondence on the subject. The character of the results makes it difficult to develop them in a systematic order; but the

results are given in such connexion one with another as I have been able to present them.

### Article No. 1. Definitions, &c.

1. It may be convenient to premise as follows:—Considering, in connexion with a curve of the third order or cubic, a point, we have

- (a) The first or conic polar of the point.
- (b) The second or line polar of the point.

The meaning of these terms is well known, and they require no explanation.

Next, considering, in connexion with the cubic, a line—

- (c) The first or conic polars of each point of the line meet in four points, which are the four poles of the line.
- (d) The second or line polars of each point of the line envelope a conic, which is the lineo-polar envelope of the line.

And reciprocally considering, in connexion with a curve of the third class, a line, we have

- (e) The first or conic pole of the line.
- (f) The second or point-pole of the line.

And considering, in connexion with the curve of the third class, a point—

- (g) The first or conic poles of each line through the point touch four lines, which are the four polars of the point.
- (h) The second or point poles of each line through the point generate a conic which is the point-pole locus of the point.

But I shall not have occasion in the present memoir to speak of these reciprocal figures, except indeed the first or conic pole of the line.

The term **conjugate poles** of a cubic is used to denote two points, such that the first or conic polar of either of them, with respect to the cubic, is a pair of lines passing through the other of them.

Reciprocally, the term **conjugate polars** of a curve of the third class denotes two lines, such that the first or conic pole of either of them, with respect to the curve of the third class, is a pair of points lying in the other of them.

2. The expression, a **syzygetic cubic**, used in reference to two cubics, denotes a curve of the third order passing through the points

of intersection of the two cubics; but in the present memoir the expression is in general used in reference to a single cubic, to denote a curve of the third order passing through the points of intersection of the cubic and its Hessian.

As regards curves of the third class, I use in the memoir the full expression, a curve of the third class syzygetically connected with two given curves of the third class.

It is a well-known theorem, that if at the points of intersection of a given line with a given cubic tangents are drawn to the cubic, these tangents again meet the cubic in three points which lie in a line; such line is in the present memoir termed the **satellite line** of the given line, and the point of intersection of the two lines is termed the **satellite point** of the given line; the given line in reference to its satellite line or point is termed the **primary line**.

In particular, if the primary line be a tangent of the cubic, the satellite line coincides with the primary line, and the satellite point is the point of simple intersection of the primary line and the cubic.

### **Article No. 2. Group of Theorems relating to the Conjugate Poles of a Cubic.**

3. The theorems which I have first to mention relate to or originate out of the theory of the conjugate poles of a cubic, and may be conveniently connected together and explained by means of the accompanying figure.

The point  $E$  is a point of the Hessian; this being so, its first or conic polar, with respect to the cubic, will be a pair of lines passing through a point  $F$  of the Hessian; and not only so, but the first or conic polar of the point  $F$ , with respect to the cubic will be a pair of lines passing through  $E$ .

The pair of lines through  $F$  are represented in the figure by  $FBA$ ,  $FDC$ , and the pair of lines through  $E$  are represented by  $EGA$ ,  $EDB$ , and the lines of the one pair meet the lines of the other pair in the points  $A$ ,  $B$ ,  $C$ ,  $D$ .

The point  $O$ , which is the intersection of the lines  $AD$ ,  $BC$ , is a point of the Hessian, and joining  $EO$ ,  $FO$ , these lines are tangents to the Hessian at the points  $E$ ,  $F$ , that is, the points  $E$ ,  $F$  are corresponding points of the Hessian, in the sense that the tangents to the Hessian at these points meet in a point of the Hessian.

The two points  $E$ ,  $F$  are, according to a preceding definition, conjugate poles of the cubic.

The line  $EF$  meets the Hessian in a third point  $G$ , and the points  $O, G$  are conjugate poles of the cubic.

The first or conic polar of  $G$ , with respect to the cubic, is the pair of lines  $AOD, BOC$  meeting in  $O$ .

The first or conic polar of  $O$ , with respect to the cubic, is the pair of lines  $GEF$  and  $Ge'e'$  meeting in  $G$ .

The four poles of the line  $EO$ , with respect to the cubic, are the points of intersection of the first or conic polars of the two points  $E$  and  $O$ , that is, the four poles in question are the points  $F, f', e, e'$ .

Similarly, the four poles of the line  $FO$ , with respect to the cubic, are the points  $E, E', f, f'$ .

The line  $EF$ , that is, any line joining two conjugate poles of the cubic, is a tangent to the Pippian, and the point of contact  $T$  is the harmonic with respect to the points  $E, F$  (which are points on the Hessian) of  $G$ , the third point of intersection with the Hessian.

Conversely, any tangent of the Pippian meets the Hessian in three points, two of which are conjugate poles of the cubic, and the point of contact is the harmonic, with respect to these two points, of the third point of intersection with the Hessian.

The line  $GO$  in the figure is of course also a tangent of the Pippian, and moreover the lines  $FBA, FDC$  (that is, the pair of lines which are the first or conic polar of  $E$ ) and the lines  $EGA, EDB$  (that is, the pair of lines which are the first or conic polar of  $F$ ) touch the Pippian, the points of contact being the intersections of these lines with the satellite lines of the lines  $EF, EG$  respectively.

It will be presently seen that the analytical theory leads to the consideration of a line  $IJ$  (not represented in the figure): the line in question is the polar of  $E$  (or  $F$ ) with respect to the conic which is the first or conic polar of  $F$  (or  $E$ ) with respect to any syzygetic cubic.

The line  $IJ$  is a tangent of the Pippian, and moreover the lines  $EF$  and  $IJ$  are conjugate polars of a curve of the third class syzygetically connected with the Pippian and Quippian, and which is moreover such that its Hessian is the Pippian.

### **Article Nos. 3 to 19. Analytical investigations, comprising the proof of the theorems, Article No. 2.**

4. The analytical theory possesses considerable interest. Take as

the equation of the cubic,

$$U = x^3 + y^3 + z^3 + 6lxyz = 0;$$

then the equation of the Hessian is

$$HU = l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0;$$

and the equation of the Pippian in line coordinates (that is, the equation which expresses that  $\xi x + \eta y + \zeta z = 0$  is a tangent of the curve) is

$$PU = -l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0.$$

The equation of the Quippian in line coordinates is

$$QU = (1 - 10l^3)(\xi^3 + \eta^3 + \zeta^3) - 6l^2(9 + 4l^3)\xi\eta\zeta = 0;$$

and the values of the two invariants of the cubic form are

$$S = -l + l^4, \quad T = 1 - 20l^3 - 8l^6,$$

values which give identically,

$$T^2 - 64S^3 = (1 + 8l^3)^3,$$

the last-mentioned function being in fact the discriminant.

5. Suppose that the cubic  $U = 0$  is represented in the canonical form

$$U = x^3 + y^3 + z^3 + 6lxyz = 0.$$

The Hessian is

$$HU = l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0.$$

A syzygetic cubic is

$$\Theta U + \Phi HU = 0,$$

where  $\Theta, \Phi$  are arbitrary parameters; or substituting,

$$(\Theta l^2 + \Phi)(x^3 + y^3 + z^3) + (6\Theta l - 2\Phi l^3)xyz = 0.$$

The condition that this may be a pair of lines is that the discriminant may vanish, that

$$(\Theta l^2 + \Phi)^3 - (6\Theta l - 2\Phi l^3)^3 = 0,$$

or, what is the same thing,

$$(\Theta l^2 + \Phi) - (6\Theta l - 2\Phi l^3) = 0,$$

or

$$\Theta(l^2 - 6l) + \Phi(1 + 2l^3) = 0.$$

6. Let  $E = (x_1, y_1, z_1)$  be a point of the Hessian; then

$$l^2(x_1^3 + y_1^3 + z_1^3) - (1 + 2l^3)x_1y_1z_1 = 0.$$

The first or conic polar of  $E$  is

$$x_1(x^2 + 2lyz) + y_1(y^2 + 2lzx) + z_1(z^2 + 2lxy) = 0.$$

This is a pair of lines meeting in the point  $F = (x_2, y_2, z_2)$ , where

$$\begin{aligned} x_2 &= y_1z_1 - lx_1^2, \\ y_2 &= z_1x_1 - ly_1^2, \\ z_2 &= x_1y_1 - lz_1^2. \end{aligned}$$

And since  $E$  is a point of the Hessian, it follows that  $F$  is also a point of the Hessian, and that the first or conic polar of  $F$  is a pair of lines meeting in  $E$ ; that is,  $E$  and  $F$  are conjugate poles of the cubic.

7. The coordinates of the point  $F$  being as above, we have

$$\begin{aligned} x_1 &= y_2z_2 - lx_2^2, \\ y_1 &= z_2x_2 - ly_2^2, \\ z_1 &= x_2y_2 - lz_2^2, \end{aligned}$$

and the line  $EF$  meets the Hessian in a third point  $G$ , the coordinates of which are

$$x_3 : y_3 : z_3 = x_1y_1z_2 - x_2y_2z_1 : y_1z_1x_2 - y_2z_2x_1 : z_1x_1y_2 - z_2x_2y_1.$$

The first or conic polar of  $G$  is the pair of lines  $AOD$ ,  $BOC$  meeting in the point  $O$ , where  $O$  is the intersection of the lines  $AD$ ,  $BC$ .

8. The point  $O$  is the pole of the line  $EF$  with respect to the conic which is the first or conic polar of  $E$  (or  $F$ ) with respect to any syzygetic cubic; and the line  $GO$  is a tangent to the Pippian.

The line  $IJ$ , which is the polar of  $E$  (or  $F$ ) with respect to the conic which is the first or conic polar of  $F$  (or  $E$ ) with respect to any syzygetic cubic, is also a tangent to the Pippian.

9. The points  $E$ ,  $F$  being conjugate poles of the cubic, the line  $EF$  is a tangent to the Pippian. The point of contact  $T$  is given by the harmonic condition

$$(E, F; T, G) = -1,$$

where  $G$  is the third point in which the line  $EF$  meets the Hessian.

Conversely, any tangent of the Pippian meets the Hessian in three points  $E$ ,  $F$ ,  $G$ ; two of these, say  $E$ ,  $F$ , are conjugate poles of the cubic, and the point of contact  $T$  is the harmonic of  $G$  with respect to  $E$ ,  $F$ .

10. The coordinates of the point of contact  $T$  of the tangent  $EF$  of the Pippian are

$$\begin{aligned} x_t &= x_1(y_2z_3 + y_3z_2) + x_2(y_3z_1 + y_1z_3) + x_3(y_1z_2 + y_2z_1), \\ y_t &= y_1(z_2x_3 + z_3x_2) + y_2(z_3x_1 + z_1x_3) + y_3(z_1x_2 + z_2x_1), \\ z_t &= z_1(x_2y_3 + x_3y_2) + z_2(x_3y_1 + x_1y_3) + z_3(x_1y_2 + x_2y_1). \end{aligned}$$

These may be written in the more convenient form

$$\begin{aligned} x_t &= \partial_\xi PU, \\ y_t &= \partial_\eta PU, \\ z_t &= \partial_\zeta PU, \end{aligned}$$

where  $PU = 0$  is the equation of the Pippian in line coordinates, and  $\xi$ ,  $\eta$ ,  $\zeta$  are the coordinates of the line  $EF$ .

11. The Pippian may be defined geometrically as the envelope of the line  $EF$ , where  $E$ ,  $F$  are conjugate poles of the cubic; or, what is the same thing, as the locus of the point  $T$ , the harmonic of  $G$  with respect to  $E$ ,  $F$ .

12. The equation of the Pippian in point coordinates is obtained by eliminating  $\xi$ ,  $\eta$ ,  $\zeta$  between the equations

$$\xi x + \eta y + \zeta z = 0, \quad PU = 0, \quad \partial_\xi PU = 0, \quad \partial_\eta PU = 0, \quad \partial_\zeta PU = 0.$$

This gives

$$(x^3 + y^3 + z^3)^2 + (2 + 16l^3)(x^3 + y^3 + z^3)xyz \\ + (1 - 20l^3 - 8l^6)x^2y^2z^2 = 0,$$

which is a curve of the sixth order.

13. The Pippian has nine cusps, which are the nine points of inflection of the cubic. The cuspidal tangent at each cusp is the harmonic polar of the corresponding point of inflection.

14. The Pippian has three double points, which are the points of the Hessian corresponding to the three points of the cubic at which the tangent meets the cubic in three consecutive points ( the sextactic points).

15. The Quippian is the locus of a point  $P$  such that the point-polar locus of  $P$  (with respect to the curve of the third class  $PU = 0$ ) is a conic; or, what is the same thing, it is the envelope of the line which is the first or conic polar of the point-polar locus of a variable point.

16. The equation of the Quippian in line coordinates is

$$QU = (1 - 10l^3)(\xi^3 + \eta^3 + \zeta^3) - 6l^2(9 + 4l^3)\xi\eta\zeta = 0,$$

and in point coordinates it is a curve of the sixth order.

17. The Quippian has nine cusps and three double points, similar to the Pippian, but the positions of these singular points are different.

18. The Hessian of the cubic is the locus of a point such that its first or conic polar is a pair of intersecting lines. The Steinerian of the cubic is the locus of a point such that its first or conic polar is a pair of coincident lines. The Steinerian is the reciprocal of the Pippian.

19. The relations between the different curves are as follows:



- The **Hessian** is a curve of the 3rd order; it is the locus of a point whose first or conic polar is a pair of intersecting lines.
- The **Steinerian** is a curve of the 3rd order; it is the locus of a point whose first or conic polar is a pair of coincident lines.
- The **Pippian** is a curve of the 3rd class; it is the envelope of the line joining two conjugate poles.
- The **Quippian** is a curve of the 3rd order; it is the locus of the first or conic polar of the point-polar locus of a variable point.
- The **Cayleyan** is identical with the Pippian.

#### Article Nos. 20 to 28. Further Developments.

20. The theorems of Hesse relating to the Hessian have their analogues in relation to the Pippian. Thus, Hesse has shown that the Hessian of a cubic passes through the nine points of inflection of the cubic; and in like manner, the Pippian touches the nine harmonic polars of the nine points of inflection.

21. Hesse has shown that the first or conic polar of any point of the Hessian is a pair of lines; and in like manner, the first or conic pole of any tangent of the Pippian is a pair of points.

22. Hesse has shown that the second or line polar of any point of the Hessian is a line touching the Hessian; and in like manner, the second or line pole of any tangent of the Pippian is a point of the Pippian.

23. Hesse has shown that the Hessian is the locus of a point whose first or conic polar is a pair of lines meeting in a point of the Hessian; and in like manner, the Pippian is the envelope of a line whose first or conic pole is a pair of points lying on a tangent of the Pippian.

24. Hesse has shown that the Hessian of a cubic and the cubic itself have the same points of inflection; and in like manner, the Pippian of a cubic and the cubic itself have the same satellite lines.

25. The theory of the Pippian may be extended to curves of higher order. For a curve of the  $n$ th order, the Pippian is a curve of the  $3(n - 2)$ th class; and for a curve of the  $n$ th class, the Pippian is a curve of the  $3(n - 2)$ th order.

26. The Pippian of a curve of the fourth order is a curve of the sixth class; it has 24 cusps and 12 double points. The Pippian of a curve of the fourth class is a curve of the sixth order; it has 24 cusps and 12 double points.

27. The theory of the Pippian is intimately connected with the theory of covariants and contravariants. The Pippian is in fact a contravariant of the cubic; and the theory of the Pippian may be regarded as a chapter in the general theory of covariants and contravariants.

28. The results of the present memoir may be summarized as follows:

1. The Pippian is a curve of the third class, which may be defined geometrically as the envelope of the line joining two conjugate poles of the cubic.
2. The Quippian is a curve of the third order, which may be defined geometrically as the locus of the first or conic polar of the point-pole locus of a variable point.
3. The Pippian and Quippian are contravariants of the cubic; their equations in the canonical form are

$$PU = -l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0,$$

and

$$QU = (1 - 10l^3)(\xi^3 + \eta^3 + \zeta^3) - 6l^2(9 + 4l^3)\xi\eta\zeta = 0.$$

4. The Pippian has nine cusps (the points of inflection of the cubic) and three double points (the sextactic points).
5. The Quippian has nine cusps and three double points, but these are in general distinct from those of the Pippian.
6. The Hessian and Steinerian are related to the Pippian in the same way that the Hessian is related to the cubic itself.

#### **Article Nos. 29 to 31. Formulae for the intersection of a cubic curve and a line.**

29. If the cubic  $U = x^3 + y^3 + z^3 + 6lxyz = 0$  intersects the line  $\xi x + \eta y + \zeta z = 0$  in the three points  $(x_1, y_1, z_1)$ ,  $(x_2, y_2, z_2)$ ,

$(x_3, y_3, z_3)$ , then the coordinates of these points are given by the equations

$$\begin{aligned}x_1 + x_2 + x_3 &= 0, \\y_1 + y_2 + y_3 &= 0, \\z_1 + z_2 + z_3 &= 0, \\x_1y_1z_1 + x_2y_2z_2 + x_3y_3z_3 &= 0.\end{aligned}$$

30. The equation of the three tangents at the points of intersection is

$$\begin{aligned}\Pi &= [(x_1^2 + 2ly_1z_1)x + (y_1^2 + 2lz_1x_1)y + (z_1^2 + 2lx_1y_1)z] \\&\quad \times [(x_2^2 + 2ly_2z_2)x + (y_2^2 + 2lz_2x_2)y + (z_2^2 + 2lx_2y_2)z] \\&\quad \times [(x_3^2 + 2ly_3z_3)x + (y_3^2 + 2lz_3x_3)y + (z_3^2 + 2lx_3y_3)z] = 0.\end{aligned}$$

31. Now considering the equation

$$F \cdot U - \Pi = (\xi'x + \eta'y + \zeta'z)(\xi''x + \eta''y + \zeta''z),$$

in order that this may be identically true, we must have

$$\begin{aligned}\xi'\xi'' &= F \cdot 1 - (x_1^2 + 2ly_1z_1)(x_2^2 + 2ly_2z_2)(x_3^2 + 2ly_3z_3), \\ \eta'\eta'' &= F \cdot 1 - (y_1^2 + 2lz_1x_1)(y_2^2 + 2lz_2x_2)(y_3^2 + 2lz_3x_3), \\ \zeta'\zeta'' &= F \cdot 1 - (z_1^2 + 2lx_1y_1)(z_2^2 + 2lx_2y_2)(z_3^2 + 2lx_3y_3),\end{aligned}$$

and similarly for the other coefficients.

The coefficient of  $x^3$  in  $F \cdot U - \Pi$  is simply  $F$ , which must therefore be equal to the product of the coefficients of  $x$  in the three tangent lines; that is,

$$F = (x_1^2 + 2ly_1z_1)(x_2^2 + 2ly_2z_2)(x_3^2 + 2ly_3z_3).$$

#### **Article Nos. 32 to 40. The satellite line and satellite point.**

32. The satellite line of a given line with respect to the cubic is the line which passes through the three points in which the tangents at the points of intersection of the given line with the cubic again meet the cubic.

33. The satellite point of a given line is the point of intersection of the given line and its satellite line.

34. If the given line is  $\xi x + \eta y + \zeta z = 0$ , then the satellite line is given by the equation

$$\partial_{\xi} PU \cdot x + \partial_{\eta} PU \cdot y + \partial_{\zeta} PU \cdot z = 0,$$

where  $PU = 0$  is the equation of the Pippian.

35. The satellite point of the line  $\xi x + \eta y + \zeta z = 0$  is the point whose coordinates are

$$x : y : z = \partial_{\xi} PU : \partial_{\eta} PU : \partial_{\zeta} PU.$$

36. The primary line and its satellite line are conjugate polars of a curve of the third class syzygetically connected with the Pippian and Quippian; and the satellite point is the pole of the primary line with respect to the conic which is the first or conic polar of the satellite line with respect to any syzygetic cubic.

37. If the primary line is a tangent of the cubic, the satellite line coincides with the primary line, and the satellite point is the point of simple intersection of the primary line and the cubic.

38. The envelope of the satellite line of a variable line is the Pippian; and the locus of the satellite point of a variable line is the Quippian.

39. The satellite line of the line joining two conjugate poles of the cubic is the tangent to the Pippian at the point of contact of the line joining the two poles.

40. The satellite point of the line joining two conjugate poles is the point of contact of the line with the Pippian.

#### **Article No. 41. Recapitulation of geometrical definitions of the Pippian.**

In conclusion, I will recapitulate the different modes of generation of the Pippian.

- (1) The Pippian is the envelope of the first or conic pole of the lineo-polar envelope of a variable line.
- (2) The Pippian is the locus of the first or conic pole of the lineo-polar envelope of a variable line.
- (3) The Pippian is the envelope of the line which is the first or conic pole of the lineo-polar envelope of a variable line.

- (4) The Pippian is the locus of the line which is the first or conic pole of the lineo-polar envelope of a variable line.
- (5) The Pippian is the envelope of the second or line polar of the first or conic pole of a variable conic.
- (6) The Pippian is the locus of the second or line polar of the first or conic pole of a variable conic.
- (7) The Pippian is the envelope of the line which is the second or line polar of the first or conic pole of a variable conic.
- (8) The Pippian is the locus of the line which is the second or line polar of the first or conic pole of a variable conic.
- (9) The Pippian is the envelope of the line which is the lineo-polar envelope of a variable point.

These nine definitions are all equivalent to each other, and each of them leads to the same curve, the Pippian of the given cubic.

147.

# A MEMOIR ON THE SYMMETRIC FUNCTIONS OF THE ROOTS OF AN EQUATION.

[From the Philosophical Transactions of the Royal Society of London,  
vol. CXLVII. for the year 1857, pp. 489—499.  
Received December 18, 1856,—Read January 8, 1857.]

There are contained in a work, which is not, I think, so generally known as it deserves to be, the “Algebra” of Meyer Hirsch<sup>1</sup> some very useful tables of the symmetric functions up to the tenth degree of the roots of an equation of any order. It seems desirable to join to these a set of tables, giving reciprocally the expressions of the powers and products of the coefficients in terms of the symmetric functions of the roots. The present memoir contains the two sets of tables, viz. the new tables distinguished by the letter (*a*), and the tables of Meyer Hirsch distinguished by the letter (*b*); the memoir contains also some remarks as to the mode of calculation of the new tables, and also as to a peculiar symmetry of the numbers in the tables of each set, a symmetry which, so far as I am aware, has not hitherto been observed, and the existence of which appears to constitute an important theorem in the subject. The theorem in question might, I think, be deduced from a very elegant formula of M. Borchardt (referred to in the sequel), which gives the generating function of any symmetric function of the roots, and contains potentially a method for the calculation of the Tables (*b*), but which, from the example I have given, would not appear to be a very convenient one for actual calculation.

Suppose in general

$$(1, b, c \dots \wr x)^n = (1 - \theta x)(1 - \beta x)(1 - \gamma x) \dots,$$

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<sup>1</sup>The work referred to is entitled *Sammlung von Beispielen Formeln und Aufgaben aus der Buchstabenrechnung und Aufgaben aus der Buchstabenrechnung und Algebra*, 8vo. Berlin, 1804 (8 ed. 1853), English translation by Ross, 8vo. London, 1827.

so that

$$\begin{aligned} -b &= \Sigma\alpha, & +c &= \Sigma\alpha\beta, \\ -d &= \Sigma\alpha\beta\gamma, & \&c., \end{aligned}$$

and if in general

$$(pqr\dots) = \Sigma\alpha^p\beta^q\gamma^r\dots,$$

where as usual the summation extends only to the distinct terms, so that e.g.  $(p^2)$  contains only half as many terms as  $(pq)$ , and so in all similar cases, then we have

$$-b = (1), \quad +c = (1^2), \quad -d = (1^3), \quad \&c.;$$

and the two problems which arise are, first to express any combination  $b^pc^qd^r\dots$  in terms of the symmetric functions  $(1^\lambda 2^\mu 3^\nu\dots)$ , and secondly, or conversely, to express any symmetric function  $(1^\lambda 2^\mu 3^\nu\dots)$  in terms of the combinations  $b^pc^qd^r\dots$ .

It will conduce materially to brevity if  $1^p 2^q\dots$  be termed the *partition* belonging to the combination  $b^pc^q\dots$ ; and in like manner if  $1^\lambda 2^\mu 3^\nu\dots$  be termed the *partition* belonging to the symmetric function  $(1^\lambda 2^\mu 3^\nu\dots)$ , and if the sum of the component numbers of the partition is termed the *weight*.

Consider now a line of combinations corresponding to a given weight, e.g. the weight 4, this will be

|        |     |      |       |         |       |
|--------|-----|------|-------|---------|-------|
|        | $e$ | $bd$ | $c^2$ | $b^2c$  | $b^4$ |
| (line) | 4   | 13   | $2^2$ | $1^2 2$ | $1^4$ |

where I have written under each combination the partition which belongs to it, and in like manner a column of symmetric functions of the same weight, viz.

|          |          |
|----------|----------|
| $(4)$    |          |
| $(31)$   |          |
| $(2^2)$  | (column) |
| $(21^2)$ |          |
| $(1^4)$  |          |

where, as the partitions are obtained by simply omitting the  $(\quad)$ , I have not separately written down the partitions. It is at once

obvious that the different combinations of the line will be made up of numerical multiples of the symmetric functions of the column; and conversely, that the symmetric functions of the column will be made up of numerical multiples of the combinations of the line; but this implies the existence of two square tables, viz.

- (a) A table giving the expressions of the combinations of the line in terms of the symmetric functions of the column.
- (b) A table giving reciprocally the expressions of the symmetric functions of the column in terms of the combinations of the line.

The Tables (a) are the new tables above referred to; the Tables (b) are those of Meyer Hirsch.

The mode of calculation of the Tables (a) is as follows. Suppose the table for any weight  $w$  has been calculated, and that it is required to calculate the table for the next superior weight  $w + 1$ . If the last column of the new table is known, then all the rest is known. For any combination  $b^p c^q \dots$  is the product of  $b$  into the combination  $b^{p-1} c^q \dots$  of the next inferior weight; the expression of the latter combination in terms of the symmetric functions of the corresponding column is given by the table for the weight  $w$ , and multiplying by  $b$ , i.e. by  $-(1)$ , we have the expression of the combination  $b^p c^q \dots$  in terms of the symmetric functions of the column for the weight  $w + 1$ . It therefore only remains to show how the last column is to be calculated. Let the combination belonging to the last place be  $b^p$ , and let the partition belonging to it be  $1^p = (11 \dots \text{to } p \text{ units})$ ; let the partitions of the same weight ( $p$ ), arranged in the order of the column, be  $\lambda, \lambda', \lambda'', \dots$ , and write

$$\begin{array}{cccc} \lambda, & \lambda', & \lambda'', & \dots \\ \alpha, & \alpha', & \alpha'', & \dots \end{array}$$

where  $\alpha, \alpha', \alpha'', \dots$  are arbitrary; then we have

$$\begin{aligned} \alpha\lambda + \alpha'\lambda' + \alpha''\lambda'' + \dots \\ = b^p = (-1)^p (1)^p \\ = (-1)^p \{(p) + A(1)(p-1) + A'(1^2)(p-2) + A''(1^3)(p-3) + \dots\}, \end{aligned}$$

where  $A, A', A''$  are the coefficients of the equation

$$z^p - pz^{p-1} + Ap(p-1)z^{p-2} - A'p(p-1)(p-2)z^{p-3} + \dots = 0,$$



i.e. the equation

$$(z - 1)^p - 1 = 0,$$

so that  $A, A', A''$  are given explicitly by

$$\begin{aligned} A &= \frac{1}{2}, \\ A' &= \frac{1}{2 \cdot 3}, \\ A'' &= \frac{1}{2 \cdot 3 \cdot 4}, \end{aligned}$$

&c.; and we have

$$\alpha = (-1)^p, \quad \alpha' = (-1)^p A, \quad \alpha'' = (-1)^p A', \quad \&c.;$$

or the terms of the last column are obtained from the binomial coefficients with the signs alternately positive and negative.

The following example will serve to render the process intelligible. Suppose the table for the weight 6 has been calculated, and that it is required to calculate the table for the weight 7. Write

|  |          |         |          |                 |            |         |     |
|--|----------|---------|----------|-----------------|------------|---------|-----|
|  | 7        | 61      | 52       | 51 <sup>2</sup> | 43         | 421     | ... |
|  | $\alpha$ | $\beta$ | $\gamma$ | $\delta$        | $\epsilon$ | $\zeta$ | ... |

then the first six lines of the new table are obtained from the table for the weight 6, by multiplying in succession the combinations  $c^2$ ,  $bcd$ ,  $bd^2$ ,  $b^2c$ ,  $b^3$ ,  $b^2c$  (the six combinations of the line for the weight 6) by  $b$ , i.e. by  $-(1)$ ; the last line is obtained from the equation

$$\alpha(7) + \beta(61) + \gamma(52) + \delta(51^2) + \epsilon(43) + \zeta(421) + \dots = b^7 = -(1)^7,$$

which gives

$$\alpha = -1, \quad \beta = \frac{7}{2}, \quad \gamma = -\frac{7 \cdot 6}{2 \cdot 3}, \quad \delta = \frac{7 \cdot 6 \cdot 5}{2 \cdot 3 \cdot 4}, \quad \&c.;$$

so that the line in question is

$$-1(7) + 7(61) - 21(52) + 35(51^2) - 35(43) + 21(421) - \dots;$$

and we thus obtain the complete table for the weight 7.

The theorem as to the symmetry of the numbers of the Tables (a) and (b) respectively is as follows:

*In either table, if the numbers be read in the reverse order, i.e. in reading each line from the last column towards the first, and the lines in order from the last to the first, then (the signs being disregarded) the numbers will be the same as before; i.e. there is the same number in the first line and first column as in the last line and last column, and so for the other terms.*

If in the Tables (a) we attend to the signs, then the rule is, that the number in the first line and first column is equal to that in the last line and last column, but that for any other two corresponding numbers, those in the places which are symmetrical with respect to the centre of the table are to each other in the ratio of the numbers of the partitions to which they respectively belong; or, what is the same thing, the number in any place, divided by the number of terms of the corresponding symmetric function, is equal to the number in the reverse place, divided by the number of terms of the corresponding combination.

I was led to the foregoing theorem by the consideration of a formula of M. Borchardt (Crelle, tom. LIII. p. 193, 1855), for the generating function of any symmetric function of the roots. Writing the equation under the form

$$1 - bz + cz^2 - dz^3 + \dots = (1 - \theta z)(1 - \beta z)(1 - \gamma z) \dots,$$

and putting

$$\{1 - bz + cz^2 - dz^3 + \dots\}^m = 1 - Bz + Cz^2 - Dz^3 + \dots,$$

then if  $\lambda, \mu, \nu \dots$  denote the partitions of any number  $w$  into not more than  $m$  parts, none of which exceed  $n$ , the coefficient of

$$(-1)^{\lambda+\mu+\nu+\dots} (\lambda) (\mu) (\nu) \dots$$

in  $B^\alpha C^\beta D^\gamma \dots$  is given explicitly as a function of  $\lambda, \mu, \nu \dots$  and of  $m$ . In fact, if  $u$  be the number of numbers in the complex  $(\lambda, \mu, \nu \dots)$  which are equal to  $r$ , then the coefficient in question is

$$\frac{m(m+1) \dots (m + \Sigma u - 1)}{\prod_r \{u! (n - r + 1)^{u_r}\}}.$$

This formula contains potentially a method for the calculation of the Tables (b); but I proceed to explain what the formula becomes for the first few cases.

And in general it is easy to see that the left-hand side of  
M. Borchardt's formula is equal to

$$[0] + [1] (1) (1)' + [2] (2) (2)' + [1^2] (1^2) (1^2)' + \&c.,$$

where  $(1)$ ,  $(2)$ ,  $(1^2)$ , &c. are the symmetric functions of the roots  $(\alpha, \beta, \dots \kappa)$ ,  $(1)'$ ,  $(2)'$ ,  $(1^2)'$ , &c. are the corresponding symmetric functions of  $(x, y, \dots u)$ , and  $[0]$ ,  $[1]$ ,  $[2]$ ,  $[1^2]$ , &c. are mere numerical coefficients; viz.  $[0]$  is equal to  $1 \cdot 2 \cdot 3 \dots n$ , and  $[1]$ ,  $[2]$ ,  $[1^2]$ , &c. are such that the product of one of these factors into the number of terms in the corresponding symmetric function  $(1)$ ,  $(2)$ ,  $(1^2)$ , &c. may be equal to  $1 \cdot 2 \cdot 3 \dots n$ . The right-hand side of M. Borchardt's formula is therefore, as in the particular case, the generating function of the symmetric functions of the roots  $(\alpha, \beta, \dots \kappa)$ , and if a convenient expression of such right-hand side could be obtained, we might by means of it express all the symmetric functions of the roots in terms of the coefficients.

*Tables relating to the Symmetric Functions of the Roots of an Equation.*

The outside line of letters contains the combinations (powers and products) of the coefficients, the coefficients being all with the positive sign, and the coefficient of the highest power being unity; thus in the case of a cubic equation the equation is

$$x^3 + bx^2 + cx + d = (x - \alpha)(x - \beta)(x - \gamma) = 0.$$

The outside line of numbers is obtained from that of letters merely by writing 1, 2, 3... for  $b$ ,  $c$ ,  $d$ ..., and may be considered simply as a different notation for the combinations. The outside column contains the different symmetric functions in the notation above explained, viz.  $(1)$  denotes  $\Sigma\alpha$ ,  $(2)$  denotes  $\Sigma\alpha^2$ ,  $(1^2)$  denotes  $\Sigma\alpha\beta$ , and so on. The Tables (*a*) are to be read according to the columns; thus Table II (*a*) means  $b^2 = 1(2) + 2(1^2)$ ,  $c = (1^2)$ . The Tables (*b*) are to be read according to the lines; thus Table II (*b*) means

$$(2) = -2c + 1b^2, (1^2) = +1c.$$

**Table**                      **I (a).**                      **Weight**                      **1.**

|     | $b$ |
|-----|-----|
| (1) | -1  |

**Table**                      **I (b).**                      **Weight**                      **1.**

|     | $b$ |
|-----|-----|
| (1) | -1  |

**Table**                      **II (a).**                      **Weight**                      **2.**

|                   | $c$ | $b^2$ |
|-------------------|-----|-------|
| (2)               |     | +1    |
| (1 <sup>2</sup> ) | +1  | +2    |

**Table**                      **II (b).**                      **Weight**                      **2.**

|                   | $c$ | $b^2$ |
|-------------------|-----|-------|
| (2)               | -2  | +1    |
| (1 <sup>2</sup> ) | +1  |       |

**Table**                      **III (a).**                      **Weight**                      **3.**

|                   | $d$ | $bc$ | $b^3$ |
|-------------------|-----|------|-------|
| (3)               |     |      | -1    |
| (21)              |     | -1   | -3    |
| (1 <sup>3</sup> ) | -1  | -3   | -6    |

**Table**                      **III (b).**                      **Weight**                      **3.**

|                   | $d$ | $bc$ | $b^3$ |
|-------------------|-----|------|-------|
| (3)               | -3  | +3   | -1    |
| (21)              | +3  | -1   |       |
| (1 <sup>3</sup> ) | -1  |      |       |

Table

IV (a).

Weight

4.

|                    | $e$ | $bd$ | $c^2$ | $b^2c$ | $b^4$ |
|--------------------|-----|------|-------|--------|-------|
| (4)                |     |      |       |        | +1    |
| (31)               |     |      |       | +1     | +4    |
| (2 <sup>2</sup> )  |     |      | +1    | +2     | +6    |
| (21 <sup>2</sup> ) |     | +1   | +2    | +7     | +12   |
| (1 <sup>4</sup> )  | +1  | +4   | +6    | +12    | +24   |

Table

IV (b).

Weight

4.

|                    | $e$ | $bd$ | $c^2$ | $b^2c$ | $b^4$ |
|--------------------|-----|------|-------|--------|-------|
| (4)                | -4  | +4   | -2    | +1     |       |
| (31)               | +4  | -3   | +1    |        |       |
| (2 <sup>2</sup> )  | -2  | +1   |       |        |       |
| (21 <sup>2</sup> ) | +1  |      |       |        |       |
| (1 <sup>4</sup> )  |     |      |       |        |       |

Note: The complete Table IV (b) entries are:  
(4) =  $-4e + 4bd - 2c^2 + 1b^2c$ , (31) =  $+4e - 3bd + 1c^2$ ,  
(2<sup>2</sup>) =  $-2e + 1bd$ , (21<sup>2</sup>) =  $+1e$ .

Table

V (a).

Weight

5.

|                    | $f$ | $be$ | $cd$ | $b^2d$ | $bc^2$ | $b^3c$ | $b^5$ |
|--------------------|-----|------|------|--------|--------|--------|-------|
| (5)                |     |      |      |        |        |        | -1    |
| (41)               |     |      |      |        |        | -1     | -5    |
| (32)               |     |      |      |        | -1     | -2     | -10   |
| (31 <sup>2</sup> ) |     |      |      | -1     | -2     | -7     | -20   |
| (2 <sup>2</sup> 1) |     |      | -1   | -2     | -5     | -12    | -30   |
| (21 <sup>3</sup> ) |     | -1   | -3   | -7     | -12    | -27    | -60   |
| (1 <sup>5</sup> )  | -1  | -5   | -10  | -20    | -30    | -60    | -120  |

Table

 $V(b).$ 

Weight

5.

|                    | $f$ | $be$ | $cd$ | $b^2d$ | $bc^2$ | $b^3c$ | $b^5$ |
|--------------------|-----|------|------|--------|--------|--------|-------|
| (5)                | -5  | +5   | -5   | +5     | -5     | +1     |       |
| (41)               | +5  | -5   | +3   | -2     | +1     |        |       |
| (32)               | -5  | +3   | -1   | +1     |        |        |       |
| (31 <sup>2</sup> ) | +5  | -3   | +1   |        |        |        |       |
| (2 <sup>2</sup> 1) | -5  | +2   | -1   |        |        |        |       |
| (21 <sup>3</sup> ) | +1  | -1   |      |        |        |        |       |
| (1 <sup>5</sup> )  | -1  |      |      |        |        |        |       |



## Tables

## VII-X.

**Table VII (a) and (b). Weight 7.** The combinations are:  $h, bg, cf, de, b^2f, bce, b^2d, c^2d, b^3e, bc^2, b^2cd, b^4d, c^3, b^2c^2, b^3c, b^7$  (15 combinations). The symmetric functions are: (7), (61), (52), (51<sup>2</sup>), (43), (421), (41<sup>3</sup>), (3<sup>2</sup>1), (32<sup>2</sup>), (321<sup>2</sup>), (31<sup>4</sup>), (2<sup>3</sup>1), (2<sup>2</sup>1<sup>3</sup>), (21<sup>5</sup>), (1<sup>7</sup>) (15 partitions).

**Table VIII (a) and (b). Weight 8.** 22 partitions and combinations. The combinations are:  $i, bh, cg, df, b^2g, cef, b^2f, d^2e, bcd, b^3f, c^2e, b^2ce, b^2d^2, bc^2d, b^4e, c^4, b^2c^3, b^3c^2, b^5c, b^8$ , and others.

The symmetric functions are: (8), (71), (62), (61<sup>2</sup>), (53), (521), (51<sup>3</sup>), (4<sup>2</sup>), (431), (42<sup>2</sup>), (421<sup>2</sup>), (41<sup>4</sup>), (3<sup>2</sup>2), (3<sup>2</sup>1<sup>2</sup>), (32<sup>2</sup>1), (321<sup>3</sup>), (31<sup>5</sup>), (2<sup>4</sup>), (2<sup>3</sup>1<sup>2</sup>), (2<sup>2</sup>1<sup>4</sup>), (21<sup>6</sup>), (1<sup>8</sup>).

**Table IX (a) and (b). Weight 9.** 30 partitions and combinations.

**Table X (a) and (b). Weight 10.** 42 partitions and combinations.

*[The complete numerical tables for Weights 7-10 are extensive, filling many pages of the original. The tables exhibit the symmetry property described in the text: the coefficient of symmetric function (P) in combination (Q) equals the coefficient of symmetric function (Q) in combination (P) (for Tables b), and with the appropriate ratio of terms (for Tables a).]*



148.

## MEMOIR ON THE RESULTANT OF A SYSTEM OF TWO EQUATIONS.

[From the Philosophical Transactions of the Royal Society of London,  
vol. CXLVII. for the year 1857, pp. 703—715.  
Received December 18, 1856,—Read January 8, 1857.]

The Resultant of two equations such as

$$(a, b, \dots \wr x, y)^m = 0, \quad (p, q, \dots \wr x, y)^n = 0,$$

is, it is well known, a function homogeneous in regard to the coefficients of each equation separately, viz. of the degree  $n$  in regard to the coefficients  $(a, b, \dots)$  of the first equation, and of the degree  $m$  in regard to the coefficients  $(p, q, \dots)$  of the second equation; and it is natural to develop the resultant in the form

$$kAP + k'A'P' + \&c.,$$

where  $A, A', \&c.$  are the combinations (powers and products) of the degree  $n$  in the coefficients  $(a, b, \dots)$ ,  $P, P', \&c.$  are the combinations of the degree  $m$  in the coefficients  $(p, q, \dots)$ , and  $k, k', \&c.$  are mere numerical coefficients. The object of the present memoir is to show how this may be conveniently effected, either by the method of symmetric functions, or from the known expression of the Resultant in the form of a determinant, and to exhibit the developed expressions for the resultant of two equations, the degrees of which do not exceed 4. With respect to the first method, the formula in its best form, or nearly so, is given in the Algebra of Meyer Hirsch, and the application of it is very easy when the necessary tables are calculated: as to this, see my “Memoir on the Symmetric Functions of the Roots of an Equation”<sup>2</sup>. But when the expression for the Resultant of two equations is to be calculated without the assistance of such tables, it is I think by far the most simple process to develop the determinant according to the second of the two methods.

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<sup>2</sup>Philosophical Transactions, 1857, pp. 489—497, [147].

Consider first the method of symmetric functions, and to fix the ideas, let the two equations be

$$(a, b, c, d \wr x, y)^3 = 0, \quad (p, q, r \wr x, y)^2 = 0.$$

Then writing

$$(a, b, c, d \wr 1, z)^3 = a(1 - \alpha z)(1 - \beta z)(1 - \gamma z),$$

so that

$$-\frac{b}{a} = \alpha + \beta + \gamma = (1), \quad \frac{c}{a} = \alpha\beta + \alpha\gamma + \beta\gamma = (1^2), \quad -\frac{d}{a} = \alpha\beta\gamma = (1^3);$$

the Resultant is

$$(p, q, r \wr \alpha, 1)^2 \cdot (p, q, r \wr \beta, 1)^2 \cdot (p, q, r \wr \gamma, 1)^2,$$

which is equal to

$$r^6 + qr^5(\alpha + \beta + \gamma) + pr^4(\alpha^2 + \beta^2 + \gamma^2) + pqr(\alpha^2\beta + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + \gamma^2\alpha + \gamma\alpha^2) + \&c.;$$

or adopting the notation for symmetric functions used in the memoir above referred to, this is

$$\left\{ \begin{array}{c} +q^2 \\ +pr \end{array} \right\} (1) \quad \left\{ \begin{array}{c} +q^2 \\ +pr \end{array} \right\} (2) \quad \left\{ \begin{array}{c} +q^2 \\ +pr \end{array} \right\} (1^2) \quad \left\{ \begin{array}{c} +pqr \\ +p^3 \end{array} \right\} (21) \\ \left\{ \begin{array}{c} +p^2q \\ +p^3 \end{array} \right\} (2^2) \quad \left\{ \begin{array}{c} +p^2q \\ +p^3 \end{array} \right\} (21^2) \quad \left\{ \begin{array}{c} +p^2q \\ +p^3 \end{array} \right\} (1^4) \quad \left\{ \begin{array}{c} +p^3 \end{array} \right\} (2^3)$$

the law of which is best seen by dividing by  $r^6$  and then writing

$$\frac{q}{r} = [1], \quad \frac{p}{r} = [2],$$

and similarly,  $\frac{pq}{r^2} = [21]$ ,  $\frac{p^2}{r^2} = [2^2]$ , &c.; the expression would then become

$$1 + [1] (1) + [2] (2) + [1^2] (1^2) + [21] (21) + [1^3] (1^3) + [2^2] (2^2) + [21^2] (21^2) + [2^3] (2^3) + \dots,$$

where the terms within the [ ] and ( ) are simply all the partitions of the numbers 1, 2, 3, 4, 5, 6, the greatest part being 2, and the greatest number of parts being 3. And in like manner in the general case we have all the partitions of the numbers 1, 2, 3, ...  $mn$ , the greatest part being  $n$ , and the greatest number of parts being  $m$ .

The symmetric functions (1), (2),  $(1^2)$ , &c. are given in the Tables (*b*) of the Memoir on Symmetric Functions, but it is necessary to remark that in the Tables the first coefficient  $a$  is put equal to unity, and consequently that there is a power of the coefficient  $a$  to be restored as a factor: this is at once effected by the condition of homogeneity. And it is not by any means necessary to write down (as for clearness of explanation has been done in the foregoing example) the terms in  $q^2$ ,  $pr$ , &c.; the terms in  $[1]$ ,  $[2]$ ,  $[1^2]$ , &c. being given, those in  $q^2$ ,  $pr$ , &c. are known by the simple rule, that the number of parts expresses the degree in  $p$ , and the greatest part the degree in  $q$ .

**Method of the Determinant.** The resultant may be expressed as the determinant

$$\begin{vmatrix} a & b & c & d & \dots & & \\ & a & b & c & d & \dots & \\ & & a & b & c & d & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ p & q & r & \dots & \dots & \dots & \\ & p & q & r & \dots & & \\ & & p & q & r & \dots & \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{vmatrix}$$

of the order  $m + n$ , the first  $n$  lines being formed with the coefficients of the first equation, and the last  $m$  lines with the coefficients of the second equation, the places corresponding to terms which do not present themselves being left blank or filled with zeros.

The development of this determinant gives at once the resultant in the form

$$kAP + k'A'P' + \&c.,$$

and the numerical coefficients  $k, k', \&c.$  are obtained by inspection from the determinant. For example, if the equations are

$$(a, b, c \wr x, y)^2 = 0, \quad (p, q, r \wr x, y)^2 = 0,$$

the resultant is

$$\begin{vmatrix} a & b & c \\ & a & b & c \\ p & q & r \\ & p & q & r \end{vmatrix} = a^2r^2 - 2abqr + b^2q^2 + acq^2 - 2acpr + 2bp^2c - b^2pr + aq^2c - 2bqcp + c^2p^2,$$

which may be arranged in the form

|       |       |      |       |
|-------|-------|------|-------|
|       | $r^2$ | $qr$ | $q^2$ |
| $a^2$ | +1    |      |       |
| $ar$  |       | -2   | +1    |
| $ac$  |       |      | +1    |
| $b^2$ |       |      | +1    |
| $bc$  |       |      |       |
| $c^2$ | +1    |      |       |

*[The memoir continues with detailed formulae and tables for the resultants of two equations of degrees (2, 2), (3, 2), (3, 3), (4, 2), and (4, 3), with complete numerical coefficients for each case.]*

149.

## ON THE SYMMETRIC FUNCTIONS OF THE ROOTS OF CERTAIN SYSTEMS OF TWO EQUATIONS.

[From the Philosophical Transactions of the Royal Society of London,  
vol. CXLVII. for the year 1857, pp. 717—726.]

Received December 18, 1856,—Read January 8, 1857.]

Suppose in general that  $\phi = 0$ ,  $\psi = 0$ , &c. denote a system of  $(n - 1)$  equations between the  $n$  variables  $(x, y, z, \dots)$ , where the functions  $\phi$ ,  $\psi$ , &c. are quantics (i.e. rational and integral homogeneous functions) of the variables. Any values  $(x_1, y_1, z_1, \dots)$  satisfying the equations, are said to constitute a *set* of roots of the system; the roots of the same set are, it is clear, only determinate to a common factor pres, i.e. only the ratios inter se and not the absolute magnitudes of the roots of a set are determinate. The number of sets, or the degree of the system, is equal to the product of the degrees of the component equations.

Imagine a function of the roots which remains unaltered when any two sets  $(x_1, y_1, z_1, \dots)$  and  $(x_2, y_2, z_2, \dots)$  are interchanged (that is, when  $x_1$  and  $x_2$ ,  $y_1$  and  $y_2$ , &c. are simultaneously interchanged), and which is besides homogeneous of the same degree as regards each entire set of roots, although not of necessity homogeneous as regards the different roots of the same set; thus, for example, if the sets are  $(x_1, y_1)$ ,  $(x_2, y_2)$ , then the functions  $x_1x_2$ ,  $x_1y_2 + x_2y_1$ ,  $y_1y_2$  are each of them of the form in question; but the first and third of these functions, although homogeneous of the first degree in regard to each entire set, are not homogeneous as regards the two variables of each set. A function of the above-mentioned form may, for shortness, be termed a *symmetric function of the roots*; such function (disregarding an arbitrary factor depending on the common factors which enter implicitly into the different sets of roots) will be a rational and integral function of the coefficients of the equations, i.e. any symmetric function of the roots may be considered as a rational and integral function of the coefficients.

The general process for the investigation of such expression for a symmetric function of the roots is as follows. Consider in the first place a single set of roots  $(x_1, y_1, z_1, \dots)$ ; it will be convenient to

deal with the ratios rather than with the actual values of the roots, and I write accordingly  $(f_1, g_1, h_1, \dots)$  in place of  $(x_1, y_1, z_1, \dots)$ , or,

to avoid subscripts,  $(f, g, h, \dots)$ ; the actual values are  $x_1 = \theta f_1$ ,  $y_1 = \theta g_1$ , &c., where  $\theta$  is indeterminate. The symmetric function to be considered will, as regards each set of roots, be a function only of

the ratios  $(f, g, h, \dots)$ ; or, what is the same thing, it will be a homogeneous function, of any degree, as regards each set of roots. Let the degree as regards each set be  $k$ , and let  $n$  be the number of variables (which is one more than the number of equations); then

we have

$$(\Sigma f)^k = \Sigma f^k + A \Sigma f^{k-1}g + B \Sigma f^{k-2}g^2 + C \Sigma f^{k-2}gh + \dots,$$

where  $A, B, C$ , &c. are the coefficients of the expansion of  $(f + g + h + \dots)^k$ , divided respectively by the numbers of terms of the corresponding symmetric functions. The several functions  $\Sigma f^k$ ,  $\Sigma f^{k-1}g$ , &c. are termed *elementary symmetric functions*, and the number of such functions is equal to the number of partitions of  $k$  into not more than  $n$  parts. And any symmetric function of the roots can be expressed as a linear function of these elementary symmetric functions, with coefficients which are rational and integral functions of the coefficients of the given equations.

The process in detail is as follows. Let the equations be, for instance,

$$\phi = (a, b, c \wr x, y, z)^2 = 0, \quad \psi = (p, q, r \wr x, y, z)^1 = 0,$$

so that the degrees are 2 and 1 respectively, and the number of variables is 3. The degree of the system (number of sets of roots) is  $2 \times 1 = 2$ . Let the symmetric function to be considered be  $\Sigma x_1 x_2$  (which is of the first degree in each set). We have  $x_1 x_2 = f_1 f_2 \cdot \theta \theta'$ , where  $\theta, \theta'$  are the indeterminate factors; but it is more convenient to consider the function as homogeneous of the degree 2 in each set

by writing  $x_1^2 x_2^2$  in place of  $x_1 x_2$  (or rather by considering the symmetric function  $x_1^2 x_2^2 + x_1^2 y_2^2 + \dots$ ), and then restoring the proper degree at the end of the calculation by dividing by the proper power of the resultant. The process then consists in forming the product

$$(x_1^2 + y_1^2 + z_1^2)(x_2^2 + y_2^2 + z_2^2),$$

and expressing this in terms of the coefficients by means of the equations  $\phi = 0, \psi = 0$ .

*[The memoir continues with detailed examples and general formulae for symmetric functions of the roots of systems of two equations. Tables are given showing the developed expressions for the elementary symmetric functions in terms of the coefficients, for various degrees of the equations.]*

150.

## A MEMOIR ON THE CONDITIONS FOR THE EXISTENCE OF GIVEN SYSTEMS OF EQUALITIES AMONG THE ROOTS OF AN EQUATION.

[From the Philosophical Transactions of the Royal Society of London,  
vol. CXLVII. for the year 1857, pp. 727—731.  
Received December 18, 1856,—Read January 8, 1857.]

It is well known that there is a symmetric function of the roots of an equation, viz. the product of the squares of the differences of the roots, which vanishes when any two roots are put equal to each other, and that consequently such function expressed in terms of the

coefficients and equated to zero, gives the condition for the existence of a pair of equal roots. And it was remarked long ago by

Professor Sylvester, in some of his earlier papers in the Philosophical Magazine, that the like method could be applied to finding the conditions for the existence of other systems of equalities among the roots, viz. that it was possible to form symmetric functions, each of them a sum of terms containing the product of a certain number of the differences of the roots, and such that the entire function might vanish for the particular system of equalities in question and that such functions expressed in terms of the coefficients and equated to zero would give the required conditions.

The object of the present memoir is to extend this theory and render it exhaustive, by showing how to form a series of types of all the different functions which vanish for one or more systems of equalities among the roots; and in particular to obtain by the method distinctive conditions for all the different systems of equalities between the roots of a quartic or a quintic equation, viz. for each system conditions which are satisfied for the particular system, and are not satisfied for any other systems, except, of course, the more special systems included in the particular system.

The question of finding the conditions for any particular system of equalities is essentially an indeterminate one, for given any set of functions which vanish, a function syzygetically connected with



these will also vanish; the discussion of the nature of the syzygetic relations between the different functions which vanish for any particular system of equalities, and of the order of the system composed of the several conditions for the particular system of equalities, does not enter into the plan of the present memoir. I have referred here to the indeterminateness of the question for the sake of the remark that I have availed myself thereof, to express by means of invariants or covariants the different systems of conditions obtained in the sequel.

1. Let the roots of an equation be  $\alpha, \beta, \gamma, \delta, \epsilon$ , &c.; and let it be required to find the condition in order that there may be a given system of equal roots. Suppose, for instance, that the roots are  $\alpha, \beta, \gamma, \delta, \epsilon$ , the condition to be found is that which expresses that  $\alpha = \beta$ . A function which vanishes for  $\alpha = \beta$  is evidently  $(\alpha - \beta)$ , but this not being a symmetric function, we must, in order to obtain a symmetric function, take the product of all the like functions; the function so obtained will vanish if any two roots are equal, and it is therefore the discriminant of the equation. Or, what is the same thing, instead of  $(\alpha - \beta)$ , we may take  $(\alpha - \beta)^2$ , which gives at once the discriminant without any superfluous factor.

2. Suppose, in the case of a quartic equation, that the condition to be found is that which expresses that there are two pairs of equal roots, i.e.  $\alpha = \beta, \gamma = \delta$ . The function  $(\alpha - \beta)(\gamma - \delta)$  is not symmetrical, but the product of the three functions

$$(\alpha - \beta)(\gamma - \delta), \quad (\alpha - \gamma)(\beta - \delta), \quad (\alpha - \delta)(\beta - \gamma),$$

i.e. the square root of the product of these functions by their several like functions obtained by permutations of the letters, is a symmetric function; and the square of this symmetric function is the discriminant. But instead of forming the product of the three functions, we may take the sum of their squares, or more simply, we may consider the symmetric function

$$(\alpha - \beta)^2(\gamma - \delta)^2 + (\alpha - \gamma)^2(\beta - \delta)^2 + (\alpha - \delta)^2(\beta - \gamma)^2,$$

which, as will appear, contains the discriminant as a factor, and gives, when equated to zero, the condition for two pairs of equal roots.

3. Suppose, for a quintic equation, that the condition is that there may be a single pair of equal roots, i.e.  $\alpha = \beta$ . The function

$\prod(\alpha - \beta)^2$ , where the product extends to all pairs of roots, is the discriminant, which vanishes when any two roots are equal. This of course gives, as the function to be equated to zero, the discriminant of the quartic.

4. Suppose, for a quintic equation, that the condition is that there may be two pairs of equal roots, i.e.  $\alpha = \beta, \gamma = \delta$ . The type of the system of roots is 221. The simplest type to be considered is the product of all functions of the form  $(\alpha - \beta)^2(\gamma - \delta)^2$ , where  $\alpha, \beta$  are any two of the equal roots, and  $\gamma, \delta$  are any other two. But this is not a symmetric function, and the symmetric function formed from it is too complicated. The simplest symmetric function which vanishes for this system is

$$\sum(\alpha - \beta)^2(\gamma - \delta)^2,$$

where the summation extends to all distinct terms.

5. Suppose, for a quintic equation, that the condition is that there may be three equal roots, i.e.  $\alpha = \beta = \gamma$ . The type is 311. The simplest symmetric function which vanishes for this system is

$$\sum(\alpha - \beta)^2(\alpha - \gamma)^2(\beta - \gamma)^2,$$

which is in fact the discriminant of the cubic formed by equating to zero the first derived function of the quintic.

6. In order that the quintic may have two pairs of equal roots, or that the system of roots may be 221, the simplest type to be considered is the product of all functions of the form  $(\alpha - \beta)^2(\gamma - \delta)^2(\epsilon - \zeta)^2$ , where  $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$  are the roots of the equation. But the function in its most simple form is the symmetric function

$$\sum(\alpha - \beta)^2(\gamma - \delta)^2,$$

which vanishes for the system 221, and gives when equated to zero the required condition.

*[The memoir continues with the complete system of conditions for all different systems of equalities between the roots of a quartic and a quintic equation. The conditions are expressed as symmetric functions, which when expressed in terms of the coefficients by means of the Tables of Symmetric Functions, give the required equations of condition. Tables are given showing the different types of conditions and the corresponding symmetric functions for the quartic and quintic equations.]*

### Summary of Conditions for Equalities among Roots.

**Quartic Equation.** The different systems of equalities are:

- (i) One pair of equal roots: type 211. Condition: discriminant = 0.
- (ii) Two pairs of equal roots: type 22. Condition:  $\sum(\alpha - \beta)^2(\gamma - \delta)^2 = 0$ .
- (iii) Three equal roots: type 31. Condition:  $\sum(\alpha - \beta)^2(\alpha - \gamma)^2 = 0$ .
- (iv) Four equal roots: type 4. Condition:  $b^2 - 4c = 0$ , etc.

**Quintic Equation.** The different systems of equalities are:

- (i) One pair of equal roots: type 2111. Condition: discriminant = 0.
- (ii) Two pairs of equal roots: type 221. Condition:  $\sum(\alpha - \beta)^2(\gamma - \delta)^2 = 0$ .
- (iii) Three equal roots: type 311. Condition:  $\sum(\alpha - \beta)^2(\alpha - \gamma)^2(\beta - \gamma)^2 = 0$ .
- (iv) Three equal roots and a pair: type 32. Condition:  $\sum(\alpha - \beta)^2(\alpha - \gamma)^2(\beta - \gamma)^2(\delta - \epsilon)^2 = 0$ .
- (v) Four equal roots: type 41. Condition:  $\sum(\alpha - \beta)^2(\alpha - \gamma)^2(\alpha - \delta)^2(\beta - \gamma)^2(\beta - \delta)^2(\gamma - \delta)^2 = 0$ .
- (vi) Five equal roots: type 5. Condition:  $b^2 - 4c = 0$ , etc.

*The memoir concludes with remarks on the syzygetic relations between the different conditions, and on the expression of the conditions by means of invariants and covariants.*

# 150. On the Conditions for the Existence of Given Systems of Equalities among the Roots of an Equation.

[From the *Philosophical Transactions* of the Royal Society of London, vol. CXLVII for the year 1857, pp. 727–731. Received December 18, 1856,—Read January 8, 1857.]

10. In order that the quintic may have two pairs of equal roots, or that the system of roots may be 221, the simplest type to be considered is

$$14.15.24.25.34.35.45;$$

a type which gives the function

$$\Sigma(\alpha-\beta)(\alpha-\delta)(\beta-\delta)(\beta-\epsilon)(\gamma-\delta)(\gamma-\epsilon)(\delta-\epsilon)^2(x-\alpha y)^2(x-\beta y)^2(x-\gamma y)^2.$$

This is a covariant of the degree 5 in the coefficients and of the degree 9 in the variables; but it appears from the memoir above referred to, that there is not any irreducible covariant of the form in

question; such covariant must be a sum of the products (No. 13)(No. 20), (No. 13)(No. 14)<sup>2</sup>, (No. 15)(No. 16) (the numbers refer to the Covariant Tables given in the memoir), each multiplied by a merely numerical coefficient. These numerical coefficients may be determined by the consideration that there being two pairs of equal roots, we may by a linear transformation make these roots 0,

0,  $\infty$ ,  $\infty$ , or what is the same thing, we may write  $\alpha = \beta = \delta = \epsilon = 0$ , the covariant must then vanish identically. The coefficients are thus found to be 1, -4, 50, and we have for a covariant vanishing in the case of two pairs of equal roots,

$$1 \cdot (\text{No. 13})(\text{No. 20}) - 4 \cdot (\text{No. 13})(\text{No. 14})^2 + 50 \cdot (\text{No. 15})(\text{No. 16})$$

[or in the new notation  $AH - 4AR + 50CD$ ].

In fact, writing  $\alpha = \beta = \delta = \epsilon = 0$ , and rejecting, where it occurs, a factor  $xy$ , the several covariants become functions of  $cx$ ,  $dy$ ; and putting, for shortness,  $x$ ,  $y$  instead of  $cx$ ,  $dy$ , the equation to be verified is

$$\begin{aligned} &1 \cdot 10(x+y)(6x^4 + 8x^3y + 15x^2y^2 + 8xy^3 + 6y^4) \\ &\quad - 4 \cdot 10(x+y)(3x^2 - 12xy + 3y^2)^2 \\ &\quad + 50 \cdot (6x^2 + 8xy + 6y^2)(x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4) = 0; \end{aligned}$$

and dividing out by  $(x + y)$  and reducing, the equation is at once seen to be identically true.

- 11.** In order that the quintic may have three equal roots, or that the system of roots may be of the form 311, the simplest type to be considered is

$$12.13.23.45;$$

this gives the function

$$\Sigma(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2(\delta - \epsilon)^2,$$

which being an invariant, and being of the fourth degree in the coefficients, must be the quartinvariant of the quintic [that is No. 19, =  $G$ ]. The same type gives also the function

$$\Sigma(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2(\delta - \epsilon)^2(x - \delta y)^2(x - \epsilon y)^2,$$

which is a covariant of the degree 4 in the coefficients and the degree 4 in the variables; and it must vanish when  $\alpha = \beta = \gamma = 0$ , this can only be the covariant

$$3(\text{No. } 20) - 2(\text{No. } 14)^2, \quad [= 3H - 2S^2],$$

which it is clear vanishes as required.

- 12.** In order that the quintic may have three equal roots and two equal roots, or that the system of roots may be of the form 32, the simplest type to be considered is

$$12.13.14.15,$$

which gives the function

$$\Sigma(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\alpha - \epsilon)(x - \beta y)^2(x - \gamma y)^2(x - \delta y)^2(x - \epsilon y)^2,$$

a covariant of the degree 4 in the coefficients, and the degree 12 in the variables; and it must vanish when  $\alpha = \beta = \gamma = 0$ ,  $\delta = \epsilon \neq 0$ ; this can only be the covariant

$$3(\text{No. } 13)^2(\text{No. } 14) - 25(\text{No. } 15)^2, \quad [= 3A^2B - 25C^2],$$

which it is clear vanishes as required.

**13.** In order that the quintic may have four equal roots, or that the system may be of the form 41, the simplest type to be considered is

$$12.34,$$

which gives the function

$$\Sigma(\alpha - \beta)^2(\gamma - \delta)^2(x - \epsilon y)^2,$$

a covariant of the degree 2 in the coefficients, and of the same degree in the variables; this can only be the covariant (No. 14), [=  $B$ ].

**14.** Finally, in order that all the roots may be equal, or that the system of roots may be of the form 5, the type to be considered is

$$12;$$

and this gives the function

$$\Sigma(\alpha - \beta)^2(x - \gamma y)^2(x - \delta y)^2(x - \epsilon y)^2,$$

a covariant of the degree 2 in the coefficients, and the degree 6 in the variables, and this can only be the Hessian (No. 15), [=  $C$ ].

It will be observed that all the preceding conditions are distinctive; for instance, the covariant which vanishes when the system of roots is of the form 311, does not vanish when the system is of the form 221, or of any other form not included in the form 311.

# 151. Tables of the Sturmi- an Functions for Equations of the Second, Third, Fourth, and Fifth Degrees.

[From the *Philosophical Transactions* of the Royal Society of London, vol. CXLVII for the year 1857, pp. 733—736. Received December 18, 1856,—Read January 8, 1857.]

The general expressions for the Sturmi-  
an functions in the form of  
determinants are at once deducible from the researches of Professor  
Sylvester in his early papers on the subject in the *Philosophical  
Magazine*, and in giving these expressions in the Memoir ‘Nouvelles  
Recherches sur les Fonctions de M. Sturm,’ Liouville, t. xiii. p. 269  
(1848), [65], I was wrong in claiming for them any novelty. The  
expressions in the last-mentioned memoir admit of a modification by  
which their form is rendered somewhat more elegant; I propose on  
the present occasion merely to give this modified form of the general  
expression, and to give the developed expressions of the functions in  
question for equations of the degrees two, three, four, and five.

Consider in general the equation

$$U = (a, b, \dots j, k | x, 1)^n,$$

and write

$$P = (a, b, \dots j | x, 1)^{n-1}, \quad Q = (b, \dots j, k | x, 1)^{n-1},$$

then supposing as usual that the first coefficient  $a$  is positive, and  
taking for shortness  $n_1, n_2, \&c.$  to represent the binomial  
coefficients  $\frac{n-1}{1}, \frac{(n-1)(n-2)}{1 \cdot 2}, \&c.$  corresponding to the index  $(n-1)$ ,  
the Sturmi-  
an functions, each with its proper sign, are as follows, viz.

**Degree 2.**  $P, Q$ :

$$\begin{array}{rcl} a, & b & \dot{x}P, P, \dot{x}Q, Q \\ & & a, \cdot, b, \cdot \\ b, & c & n_1c, b, n_1c, \cdot \end{array}$$

**Degree 3.**  $xP, P, Q$ :

$$\begin{array}{rcl} & & x^2P, xP, P, x^2Q, xQ, Q \\ a, & b, & c & a, \cdot, \cdot, b, \cdot, \cdot \\ b, & c, & d & n_1c, b, \cdot, n_1d, c, \cdot \\ & & & n_2d, n_1c, b, n_2e, n_1d, c \end{array}$$

where the terms containing the powers of  $x$ , which exceed the degrees of the several functions respectively, vanish identically (as is in fact obvious from the form of the expressions), but these terms may of course be omitted *ab initio*.

The following are the results which I have obtained; it is well known that the last or constant function is in each case equal to the discriminant, and as the expressions for the discriminant of equations of the fourth and fifth degrees are given, Tables No. 12 and No. 26 [ $Q'$ , see 143] in my 'Second Memoir upon Quantics'<sup>3</sup>, I have thought it sufficient to refer to these values without repeating them at length.

### Table for the degree 2.

The Sturmian functions for the quadric  $(a, b, c \mid x, 1)^2$  are

$$\frac{a+1}{ac-1} \mid \frac{b+2}{b^2+1} \mid c+1 \quad \begin{pmatrix} (x,1)^2, \\ (a+1 & b+1) \\ (b+1 & c+1) \\ ac-b^2 \end{pmatrix} (x,1),$$

### Table for the degree 3.

The Sturmian functions for the cubic  $(a, b, c, d \mid x, 1)^3$  are

| $a+1$ |       | $b+3$  |         | $c+3$      | $d+1$ |            |
|-------|-------|--------|---------|------------|-------|------------|
|       |       |        |         | $a^2d+1$   |       | $(x,1)^3,$ |
|       |       |        |         | $abcd+6$   |       | $(x,1)^2,$ |
| $a+1$ | $b+2$ | $ac-2$ | $b^2+2$ | $ac^3-4$   |       | $(x,1),$   |
| $b+2$ | $c+1$ | $ad-1$ | $bc+1$  | $bd^3-4$   |       | Discr.     |
|       |       |        |         | $b^2c^2-3$ |       |            |

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<sup>3</sup>Philosophical Transaction, t. cxlvi. p. 101 (1856), [141].



**Table for the degree 4.**

The Sturmian functions for the quartic  $(a, b, c, d, e \mid x, 1)^4$  are

$$\left\{ \begin{array}{ccccc} a+1 & b+4 & c+6 & d+4 & e+1 \end{array} \mid x, 1 \right\}^4,$$

$$\begin{pmatrix} a+1 & b+3 \\ b+3 & c+2 \end{pmatrix} \begin{pmatrix} ac-3 & b^2+3 \\ ad-3 & bc+3 \\ ae-1 & bd+1 \end{pmatrix} (x, 1)^3, \\ \begin{pmatrix} c+2 & d+3 \\ d+3 & e+1 \end{pmatrix}$$

$$3 \left( \begin{array}{cccccc} a^2ce-1, & a^2de+1, & a^2d^2+3, & abce-4, & ab^2e+1, \\ abd^2-1, & abcd-14, & ac^2d+3, & ac^3+9, & b^3e+3, \\ b^2d^2+8, & b^2cd-2, & b^2c^2-6 \end{array} \right) (x, 1)^2, \quad \text{Discr. Tab. No. 12.}$$

**Table for the degree 5.**

The Sturmian functions for the quintic  $(a, b, c, d, e, f \mid x, 1)^5$  are

$$\left\{ \begin{array}{cccccc} a+1 & b+5 & c+10 & d+10 & e+5 & f+1 \end{array} \mid x, 1 \right\}^5,$$

$$\begin{pmatrix} a+1 & b+4 \\ b+4 & c+3 \end{pmatrix} \begin{pmatrix} ac-4 & b^2+4 \\ ad-6 & bc+6 \\ ae-4 & bd+4 \\ af-1 & be+1 \end{pmatrix} (x, 1)^4, \\ \begin{pmatrix} c+3 & d+4 \\ d+4 & e+3 \end{pmatrix} \begin{pmatrix} ce-4 & cd+6 \\ cf-1 & de+4 \\ . & ef-1 \end{pmatrix} \\ . \quad \begin{pmatrix} e+3 & f+4 \\ f+4 & . \end{pmatrix}$$

$$\begin{aligned}
& 2 \left( a^2 ce - 8, \quad a^2 cf - 2, \quad a^2 df + 3, \quad a^2 e^2 + 18, \quad a^2 de + 12, \right. \\
& \quad a^2 ef - 11, \quad ab^2 f + 8, \quad ab^2 e + 2, \quad abde - 3, \quad abcd - 76, \\
& \quad ac^2 e - 42, \quad ac^2 f + 8, \quad ac^2 d + 48, \quad abd^2 - 12, \quad b^3 f + 8, \\
& \quad b^2 d^2 + 40, \quad ac^2 d + 32, \quad b^2 ce - 5, \quad b^2 e^2 - 30, \quad b^3 e + 30, \\
& \quad \left. b^2 cd - 20 \right) (x, 1)^3,
\end{aligned}$$

$$\begin{aligned}
& 2 \left( a^2 cd^2 - 2, \quad a^2 d^3 + 3, \quad a^3 c^2 f + 24, \quad a^2 c^2 f - 8, \quad a^2 c^3 - 32, \right. \\
& a^2 bcf^2 - 11, \quad a^2 b^2 f^2 + 2, \quad a^2 bdef + 58, \quad a^2 b^2 e^2 + 264, \quad a^2 b^2 df + 8, \\
& a^2 bcef - 52, \quad a^2 be^2 f + 104, \quad a^2 bd^2 f - 96, \quad a^2 b^2 ef - 156, \quad a^2 cd^2 f + 64, \\
& a^2 cde^2 - 96, \quad a^2 c^4 + 352, \quad a^2 b^2 de + 108, \quad a^2 cd^2 e - 938, \quad ab^4 f - 8, \\
& a^2 d^4 + 432, \quad ab^2 cef - 266, \quad ab^3 ef + 28, \quad ab^3 df - 8, \quad ab^2 d^2 - 970, \\
& ab^2 e^3 + 35, \quad ab^2 cde + 120, \quad abc^2 d^2 f + 584, \quad abc^2 de + 2480, \quad abc^2 f + 120, \\
& ab^2 cdf + 264, \quad abc^2 de - 360, \quad ab^2 c^2 d - 1440, \quad ac^3 f - 288, \quad abc^2 f - 192, \\
& ac^3 d^2 + 160, \quad ac^4 e - 960, \quad b^4 f + 120, \quad ac^3 d + 640, \quad b^3 cdf - 320, \\
& b^3 df^2 - 160, \quad b^3 ce^2 - 75, \quad b^3 e^3 + 450, \quad b^3 de + 200, \quad b^2 cdef - 1400, \\
& b^3 ef + 180, \quad b^3 d^2 + 800, \quad b^2 cde - 100, \quad b^2 c^2 f + 120, \quad b^3 ce + 600, \\
& \quad \left. b^2 cd^2 - 400 \right) (x, 1)^2, \quad \text{etc. Discr. Tab. No. 26, } [Q'].
\end{aligned}$$

## 152. A Memoir on the Theory of Matrices.

[From the *Philosophical Transactions* of the Royal Society of London, vol. CXLVIII for the year 1858, pp. 17—37. Received December 10, 1857,—Read January 14, 1858.]

The term matrix might be used in a more general sense, but in the present memoir I consider only square and rectangular matrices, and the term matrix used without qualification is to be understood as meaning a square matrix; in this restricted sense, a set of quantities arranged in the form of a square, e.g.

$$\begin{pmatrix} a, & b, & c \\ b', & b, & c' \\ b'', & c'', & c'' \end{pmatrix}$$

is said to be a matrix.

The notion of such a matrix arises naturally from an abbreviated notation for a set of linear equations, viz. the equations

$$X = ax + by + cz, \quad Y = a'x + b'y + c'z, \quad Z = a''x + b''y + c''z,$$

may be more simply represented by

$$(X, Y, Z) = \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} (x, y, z),$$

and the consideration of such a system of equations leads to most of the fundamental notions in the theory of matrices.

It will be seen that matrices (attending only to those of the same order) comport themselves as single quantities; they may be added, multiplied or compounded together, &c.: the law of the addition of matrices is precisely similar to that for the addition of ordinary algebraical quantities; as regards their multiplication (or composition), there is the peculiarity that matrices are not in general convertible; it is nevertheless possible to form the powers (positive or negative, integral or fractional) of a matrix, and thence to arrive at the notion of a rational and integral function, or generally of any algebraical function, of a matrix. I obtain the remarkable theorem that any matrix whatever satisfies an algebraical equation of its own order, the coefficient of the highest

power being unity, and those of the other powers functions of the terms of the matrix, the last coefficient being of course the determinant; the rule for the formation of this equation may be stated in the following condensed form, which is a **generalization** of a known theorem for the formation of the equation producing the squared differences of the roots of a given equation. Write down the

$$\begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix},$$

as a matrix, and form the determinant

$$\begin{vmatrix} a-M, & b, & c \\ a', & b'-M, & c' \\ a'', & b'', & c''-M \end{vmatrix};$$

the developed expression of this determinant is

$$M^2 - AM^2 + BM - C;$$

the values of  $M^2$ ,  $M^1$ ,  $M^0$  are respectively

$$\begin{pmatrix} a^2 + bb' + cc'', & ab + b(b' + c'') + cc'', & ac + bc' + c(c'' + b'') \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}, \quad \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix},$$

and substituting these values the determinant becomes equal to the matrix zero, viz. we have

$$\begin{vmatrix} a-M, & b, & c \\ a', & b'-M, & c' \\ a'', & b'', & c''-M \end{vmatrix} = 0,$$

or expanding,

$$M^3 - (a+b'+c'')M^2 + (b'c'' - c'^2 + ca'' - c^2 + ab' - b'^2)M - \det \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} = 0,$$

which is the general theorem.

The theorem shows that every rational and integral function, or indeed every rational function of a matrix, can be expressed as a

rational and integral function of an order at most equal to that of the matrix, less unity; and it has been already seen that the order of the function may be still further restricted. The theorem leads also to a general expression for the inverse or reciprocal matrix, and to many other important results.

1. I use the term matrix in the sense explained in the introductory remarks; a matrix is not a determinant, but an abbreviated notation for an array of quantities, by means of which a system of equations can be expressed. I use the notation

$$\begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix}$$

to denote the matrix; if the matrix be multiplied by a single quantity, the product is the matrix each of the terms of which is multiplied by such quantity; if the matrix be multiplied by a single quantity, considered as involving the matrix unity, this is equivalent to the multiplication of each term of the matrix by such quantity.

The matrix unity is

$$\begin{pmatrix} 1, & 0, & 0 \\ 0, & 1, & 0 \\ 0, & 0, & 1 \end{pmatrix}.$$

2. The addition of matrices is effected by adding the corresponding terms; thus

$$\begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} + \begin{pmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{pmatrix} = \begin{pmatrix} a + \alpha, & b + \beta, & c + \gamma \\ a' + \alpha', & b' + \beta', & c' + \gamma' \\ a'' + \alpha'', & b'' + \beta'', & c'' + \gamma'' \end{pmatrix}.$$

The zero matrix is that in which each term is zero; a matrix is not altered by the addition of the zero matrix. Matrices of different orders cannot be added together.

3. The multiplication or composition of matrices is effected by

means of the composition of linear forms; thus if

$$(X, Y, Z) = \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} (x, y, z),$$

$$(x, y, z) = \begin{pmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{pmatrix} (\xi, \eta, \zeta),$$

then

$$(X, Y, Z) = \begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} (\xi, \eta, \zeta),$$

where

$$\begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} = \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} \begin{pmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{pmatrix}$$

is the product or compound matrix.

4. The rule for the multiplication is that the compound matrix is formed by combining each line of the first component matrix successively with each column of the second component matrix; or more explicitly: the term in any line and any column of the compound matrix is obtained by combining the corresponding line of the first component matrix with the corresponding column of the second component matrix.

5. It may be noticed that if the terms of the first component matrix are  $a_{11}$ ,  $a_{12}$ ,  $a_{13}$ , &c., and those of the second are  $b_{11}$ ,  $b_{12}$ ,  $b_{13}$ , &c., then the terms of the compound matrix are  $c_{11}$ ,  $c_{12}$ ,  $c_{13}$ , &c., where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots.$$

6. A matrix may be compounded, either as first or second component matrix, with the matrix zero; the result is in each case the matrix zero.

7. A matrix is not altered by its composition, either as first or second component matrix, with the matrix unity.

8. A matrix may be multiplied by a single quantity, by which is meant that each term of the matrix is multiplied by such quantity; this is equivalent to compounding the matrix with the single quantity considered as involving the matrix unity, and the same result is obtained whether the matrix is considered as first or second component matrix.

9. The notion of a positive and integer power of a matrix follows at once from the definition of multiplication; the different powers of the same matrix are convertible, and we have  $L^p \cdot L^q = L^{p+q}$ .

10. The notion of the inverse or reciprocal matrix follows from the equation  $L \cdot L^{-1} = 1$ ; if

$$(X, Y, Z) = \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} (x, y, z),$$

then

$$(x, y, z) = \begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} (X, Y, Z),$$

where  $(A, B, C; A', B', C'; A'', B'', C'')$  is the inverse or reciprocal matrix. If  $\nabla$  denote the determinant, then

$$A = \frac{1}{\nabla} \begin{vmatrix} 1, & 0, & 0 \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix}, \quad B = \frac{1}{\nabla} \begin{vmatrix} 0, & 1, & 0 \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix}, \quad \&c.,$$

or what is the same thing,

$$\begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix}^{-1} = \frac{1}{\nabla} \begin{pmatrix} \partial_a \nabla, & \partial_{a'} \nabla, & \partial_{a''} \nabla \\ \partial_b \nabla, & \partial_{b'} \nabla, & \partial_{b''} \nabla \\ \partial_c \nabla, & \partial_{c'} \nabla, & \partial_{c''} \nabla \end{pmatrix},$$

where of course the differentiations must in every case be performed as if the terms  $a, b, \&c.$  were all of them independent arbitrary quantities.

### 11. The equations

$$(X, Y, Z) = \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} (x, y, z),$$

$$(x, y, z) = \begin{pmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{pmatrix} (\xi, \eta, \zeta),$$

give

$$(X, Y, Z) = \begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} (\xi, \eta, \zeta),$$

and thence, substituting for the matrix  
( $A, B, C; A', B', C'; A'', B'', C''$ ) its value, we obtain

$$\begin{aligned} & \left( (a, b, c \$ \alpha, \alpha', \alpha''), (a, b, c \$ \beta, \beta', \beta''), (a, b, c \$ \gamma, \gamma', \gamma'') \right) \\ &= \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} \begin{pmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{pmatrix} \end{aligned}$$

as the rule for the multiplication or composition of two matrices. It is to be observed, that the operation is not a commutative one; the component matrices may be distinguished as the first or further component matrix, and the second or nearer component matrix, and the rule of composition is as follows, viz. any line of the compound matrix is obtained by combining the corresponding line of the first or further component matrix successively with the several columns of the second or nearer compound matrix.

[We may conveniently write

$$\begin{pmatrix} (a, b, c \$ \alpha, \alpha', \alpha''), & (a, b, c \$ \beta, \beta', \beta''), & (a, b, c \$ \gamma, \gamma', \gamma'') \\ (a', b', c' \$ \alpha, \alpha', \alpha''), & (a', b', c' \$ \beta, \beta', \beta''), & (a', b', c' \$ \gamma, \gamma', \gamma'') \\ (a'', b'', c'' \$ \alpha, \alpha', \alpha''), & (a'', b'', c'' \$ \beta, \beta', \beta''), & (a'', b'', c'' \$ \gamma, \gamma', \gamma'') \end{pmatrix}$$

to denote the left-hand side of the last preceding equation.]

**12.** A matrix compounded, either as first or second component matrix, with the matrix zero, gives the matrix zero. The case where



any of the terms of the given matrix are infinite is of course excluded.

**13.** A matrix is not altered by its composition, either as first or second component matrix, with the matrix unity. It is compounded either as first or second component matrix, with the single quantity  $m$  considered as involving the matrix unity, by multiplication of all its terms by the quantity  $m$ : this is in fact the before-mentioned rule for the multiplication of a matrix by a single quantity, which rule is thus seen to be a particular case of that for the multiplication of two matrices.

**14.** We may in like manner multiply or compound together three or more matrices: the order of arrangement of the factors is of course material, and we may distinguish them as the first or furthest, second, third, &c., and last or nearest component matrices: any two consecutive factors may be compounded together and replaced by a single matrix, and so on until all the matrices are compounded together, the result being independent of the particular mode in which the composition is effected; that is, we have  $L \cdot MN = LM \cdot N = LMN$ ,  $LM \cdot NP = L \cdot MN \cdot P$ , &c., or the operation of multiplication, although, as already remarked, not commutative, is associative.

**15.** We thus arrive at the notion of a positive and integer power  $L^p$  of a matrix  $L$ , and it is to be observed that the different powers of the same matrix are convertible. It is clear also that  $p$  and  $q$  being positive integers, we have  $L^p \cdot L^q = L^{p+q}$ , which is the theorem of indices for positive integer powers of a matrix.

**16.** The last-mentioned equation,  $L^p \cdot L^q = L^{p+q}$ , assumed to be true for all values whatever of the indices  $p$  and  $q$ , leads to the notion of the powers of a matrix for any form whatever of the index. In particular,  $L^p \cdot L^0 = L^p$  or  $L^0 = 1$ , that is, the 0th power of a matrix is the matrix unity. And then putting  $p = 1$ ,  $q = -1$ , or  $p = -1$ ,  $q = 1$ , we have  $L \cdot L^{-1} = L^{-1} \cdot L = 1$ ; that is,  $L^{-1}$ , or as it may be termed the inverse or reciprocal matrix, is a matrix which, compounded either as first or second component matrix with the original matrix, gives the matrix unity.

**17.** We may arrive at the notion of the inverse or reciprocal matrix,

directly from the equation

$$(X, Y, Z) = \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} (x, y, z),$$

in fact this equation gives

$$(x, y, z) = \begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} (X, Y, Z),$$

and we have, for the determination of the coefficients of the inverse or reciprocal matrix, the equations

$$\begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} = \begin{pmatrix} 1, & 0, & 0 \\ 0, & 1, & 0 \\ 0, & 0, & 1 \end{pmatrix},$$

$$\begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} \begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} = \begin{pmatrix} 1, & 0, & 0 \\ 0, & 1, & 0 \\ 0, & 0, & 1 \end{pmatrix},$$

which are equivalent to each other, and either of them is by itself sufficient for the complete determination of the inverse or reciprocal matrix. It is well known that if  $\nabla$  denote the determinant, that is, if

$$\nabla = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix},$$

then the terms of the inverse or reciprocal matrix are given by the equations

$$A = \frac{1}{\nabla} \begin{vmatrix} 1, & 0, & 0 \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix}, \quad B = \frac{1}{\nabla} \begin{vmatrix} 0, & 1, & 0 \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix}, \quad \&c.,$$

or what is the same thing, the inverse or reciprocal matrix is given by the equation

$$\begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix}^{-1} = \frac{1}{\nabla} \begin{pmatrix} \partial_a \nabla, & \partial_{a'} \nabla, & \partial_{a''} \nabla \\ \partial_b \nabla, & \partial_{b'} \nabla, & \partial_{b''} \nabla \\ \partial_c \nabla, & \partial_{c'} \nabla, & \partial_{c''} \nabla \end{pmatrix},$$

where of course the differentiations must in every case be performed as if the terms  $a$ ,  $b$ , &c. were all of them independent arbitrary quantities.

**18.** The formula shows, what is indeed clear *a priori*, that the notion of the inverse or reciprocal matrix fails altogether when the determinant vanishes: the matrix is in this case said to be indeterminate, and it must be understood that in the absence of express mention, the particular case in question is frequently excluded from consideration. It may be added that the matrix zero is indeterminate; and that the product of two matrices may be zero, without either of the factors being zero, if only the matrices are one or both of them indeterminate.

**19.** The notion of the inverse or reciprocal matrix once established, the other negative integer powers of the original matrix are positive integer powers of the inverse or reciprocal matrix, and the theory of such negative integer powers may be taken to be known. The theory of the fractional powers of a matrix will be further discussed in the sequel.

**20.** The positive integer power  $L^m$  of the matrix  $L$  may of course be multiplied by any matrix of the same degree: such multiplier, however, is not in general convertible with  $L$ ; and to preserve as far as possible the analogy with ordinary algebraical functions, we may restrict the attention to the case where the multiplier is a single quantity, and such convertibility consequently exists. We have in this manner a matrix  $cL^m$ , and by the addition of any number of such terms we obtain a rational and integral function of the matrix  $L$ .

**21.** The general theorem before referred to will be best understood by a complete development of a particular case. Imagine a matrix

$$M = \begin{pmatrix} a, & b \\ c, & d \end{pmatrix}$$

and form the determinant

$$\begin{vmatrix} a - M, & b \\ c, & d - M \end{vmatrix};$$

the developed expression of this determinant is

$$M^2 - (a + d)M + (ad - bc);$$

the values of  $M^2$ ,  $M^1$ ,  $M^0$  are

$$\begin{pmatrix} a^2 + bc, & b(a + d) \\ c(a + d), & d^2 + bc \end{pmatrix}, \quad \begin{pmatrix} a, & b \\ c, & d \end{pmatrix}, \quad \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix},$$

and substituting these values the determinant becomes equal to the matrix zero, viz. we have

$$\begin{vmatrix} a-M, & b \\ c, & d-M \end{vmatrix} = \begin{pmatrix} a^2+bc, & b(a+d) \\ c(a+d), & d^2+bc \end{pmatrix} - (a+d) \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \\ + (ad-bc) \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix} = \begin{pmatrix} 0, & 0 \\ 0, & 0 \end{pmatrix},$$

that is

$$\begin{vmatrix} a-M, & b \\ c, & d-M \end{vmatrix} = 0,$$

where the matrix of the determinant is

$$\begin{pmatrix} a, & b \\ c, & d \end{pmatrix} - M \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix},$$

that is, it is the original matrix, diminished by the same matrix considered as a single quantity involving the matrix unity. And this is the general theorem, viz. the determinant, having for its matrix a given matrix less the same matrix considered as a single quantity involving the matrix unity, is equal to zero.

**22.** The following symbolical representation of the theorem is, I think, worth noticing: let the matrix  $M$ , considered as a single quantity, be represented by  $\tilde{M}$ , then writing 1 to denote the matrix unity,  $M \cdot 1$  will represent the matrix  $M$ , considered as a single quantity involving the matrix unity. Upon the like principles of notation,  $1 \cdot M$  will represent, or may be considered as representing, simply the matrix  $M$ , and the theorem is

$$\text{Det. } (1 \cdot M - \tilde{M} \cdot 1) = 0.$$

**23.** I have verified the theorem, in the next simplest case of a matrix of the order 3, viz. if  $M$  be such a matrix, suppose

$$M = \begin{pmatrix} a, & b, & c \\ d, & e, & f \\ g, & h, & i \end{pmatrix},$$

then the derived determinant vanishes, or we have

$$\begin{vmatrix} a-M, & b, & c \\ d, & e-M, & f \\ g, & h, & i-M \end{vmatrix} = 0,$$

or expanding

$$M^3 - (a + e + i)M^2 + (ei + ia + ae - fh - cg - bd)M \\ - (aei + bfg + cdh - afh - bdi - ceg) = 0;$$

but I have not thought it necessary to undertake the labour of a formal proof of the theorem in the general case of a matrix of any degree.

**24.** If we attend only to the general form of the result, we see that any matrix whatever satisfies an algebraical equation of its own order, which is in many cases the material part of the theorem.

**25.** It follows at once that every rational and integral function, or indeed every rational function of a matrix, can be expressed as a rational and integral function of an order at most equal to that of the matrix, less unity. But it is important to consider how far or in what sense the like theorem is true with respect to irrational functions of a matrix. If we had only the equation satisfied by the matrix itself, such extension could not be made; but we have besides the equation of the same order satisfied by the irrational function of the matrix, and by means of these two equations, and the equation by which the irrational function of the matrix is determined, we may express the irrational function as a rational and integral function of the matrix, of an order equal at most to that of the matrix, less unity.

**26.** It is to be observed, that an equation such as  $X^2 = L$  (where  $L$  is a given matrix of the order 2, and  $X$  the matrix to be determined), does not of itself completely determine  $X$ : in fact, writing

$$L = \begin{pmatrix} a, & b \\ c, & d \end{pmatrix}, \quad X = \begin{pmatrix} x, & y \\ z, & w \end{pmatrix},$$

the equation gives

$$x^2 + yz = a, \quad y(x + w) = b, \quad z(x + w) = c, \quad w^2 + yz = d;$$

and we have hence only three equations for the determination of the four quantities  $x, y, z, w$ . The fourth equation is supplied by the equation of the same order satisfied by  $X$ , viz.

$$X^2 - (x + w)X + (xw - yz) = 0,$$

or say

$$L - (x + w)X + (xw - yz) = 0.$$

**27.** The equation satisfied by the matrix is in many cases the only condition which must be satisfied; thus, for instance, if we have  $X^2 = 1$ , the equation satisfied by  $X$  is  $X^2 - 1 = 0$ , and the general solution is

$$X = \begin{pmatrix} x, & y \\ z, & -x \end{pmatrix},$$

where  $x^2 + yz = 1$ .

**28.** If we have  $X^2 = L$ , where  $L$  is a given matrix, and if the equation satisfied by  $X$  contains as a factor the equation satisfied by the matrix, then the assumed equation will be satisfied identically, and the condition is sufficient as well as necessary: in the other case, where the assumed equation is of an order inferior to that of the matrix, the condition is necessary, but it is not sufficient.

**29.** The equation satisfied by the matrix may be of the form  $M^n = 1$ ; the matrix is in this case said to be periodic of the  $n$ th order. The preceding considerations apply to the theory of periodic matrices; thus, for instance, suppose it is required to find a matrix of the order 2, which is periodic of the second order. Writing

$$M = \begin{pmatrix} a, & b \\ c, & d \end{pmatrix},$$

we have

$$M^2 - (a + d)M + ad - bc = 0,$$

and the assumed equation is  $M^2 - 1 = 0$ . These equations will be identical if  $a + d = 0$ ,  $ad - bc = -1$ , that is, these conditions being satisfied, the equation  $M^2 - 1 = 0$  required to be satisfied, will be identical with the equation which is always satisfied, and will therefore itself be satisfied. And in like manner the matrix  $M$  of the order 2 will satisfy the condition  $M^3 - 1 = 0$ , or will be periodic of the third order, if only  $M^3 - 1$  contains as a factor  $M^2 - (a + d)M + ad - bc$ , and so on.

**30.** The equation satisfied by the matrix may be of the form  $M^n = 1$ ; the matrix is in this case said to be periodic of the  $n$ th order. The preceding considerations apply to the theory of periodic

matrices; thus, for instance, suppose it is required to find a matrix of the order 2, which is periodic of the second order. Writing

$$M = \begin{pmatrix} a, & b \\ c, & d \end{pmatrix},$$

we have

$$M^2 - (a + d)M + ad - bc = 0,$$

and the assumed equation is  $M^2 - 1 = 0$ . These equations will be identical if  $a + d = 0$ ,  $ad - bc = -1$ , that is, these conditions being satisfied, the equation  $M^2 - 1 = 0$  required to be satisfied, will be

identical with the equation which is always satisfied, and will therefore itself be satisfied. And in like manner the matrix  $M$  of the order 2 will satisfy the condition  $M^3 - 1 = 0$ , or will be periodic of the third order, if only  $M^3 - 1$  contains as a factor

$$M^2 - (a + d)M + ad - bc, \text{ and so on.}$$

**31.** But suppose it is required to find a matrix of the order 3,

$$M = \begin{pmatrix} a, & b, & c \\ d, & e, & f \\ g, & h, & i \end{pmatrix},$$

which shall be periodic of the second order. Writing for shortness

$$\begin{vmatrix} a - M, & b, & c \\ d, & e - M, & f \\ g, & h, & i - M \end{vmatrix} = -(M^3 - AM^2 + BM - C),$$

the matrix here satisfies  $M^3 - AM^2 + BM - C = 0$ , and, as before,

the assumed equation is  $M^2 - 1 = 0$ . Here, if we have  $1 + B = 0$ ,  $A + C = 0$ , the left-hand side will contain the factor  $(M^2 - 1)$ , and

the equation will take the form  $(M^2 - 1)(M + C) = 0$ , and we

should have then  $M^2 - 1 = 0$ , provided  $M + C$  were not an indeterminate matrix. But  $M + C$  denotes the matrix

$$\begin{pmatrix} a + C, & b, & c \\ d, & e + C, & f \\ g, & h, & i + C \end{pmatrix},$$

the determinant of which is  $C + AC + BC + C^3$ , which is equal to zero in virtue of the equations  $1 + B = 0$ ,  $A + C = 0$ , and we



cannot, therefore, from the equation  $(M^2 - 1)(M + C) = 0$ , deduce the equation  $M^2 - 1 = 0$ . This is as it should be, for the two conditions are not sufficient, in fact the equation

$$M^2 = \begin{pmatrix} a^2 + bd + cg, & ab + be + ch, & ac + bf + ci \\ da + ed + fg, & db + e^2 + fh, & dc + ef + fi \\ ga + hd + ig, & gb + he + ih, & gc + hf + i^2 \end{pmatrix} = 1$$

gives nine equations, which are however satisfied by the following values, involving in reality four arbitrary coefficients; viz. the value of the matrix is

$$\frac{1}{\alpha + \beta + \gamma} \begin{pmatrix} -(\beta + \gamma)\alpha^{-1} & \beta & \gamma \\ \alpha & -(\gamma + \alpha)\beta^{-1} & \gamma \\ \alpha & \beta & -(\alpha + \beta)\gamma^{-1} \end{pmatrix},$$

so that there are in all five relations (and not only two) between the coefficients of the matrix.

**32.** Instead of the equation  $M^n - 1 = 0$ , which belongs to a periodic matrix, it is in many cases more convenient, and it is much the same thing to consider an equation  $M^n - k = 0$ , where  $k$  is a single quantity. The matrix may in this case be said to be periodic to a factor pres.

**33.** Two matrices  $L, M$  are convertible when  $LM = ML$ . If the matrix  $M$  is given, this equality affords a set of linear equations between the coefficients of  $L$  equal in number to these coefficients, but these equations cannot be all independent, for it is clear that if  $L$  be any rational and integral function of  $M$  (the coefficients being single quantities), then  $L$  will be convertible with  $M$ ; or what is apparently (but only apparently) more general, if  $L$  be any algebraical function whatever of  $M$  (the coefficients being always single quantities), then  $L$  will be convertible with  $M$ . But whatever the form of the function is, it may be reduced to a rational and integral function of an order equal to that of  $M$ , less unity, and we have thus the general expression for the matrices convertible with a given matrix, viz. any such matrix is a rational and integral function (the coefficients being single quantities) of the given matrix, the order being that of the given matrix, less unity. In particular, the general form of the matrix  $L$  convertible with a given matrix  $M$  of the order 2, is  $L = \alpha M + \beta$ , or what is the same thing, the matrices

$$\begin{pmatrix} a, & b \\ c, & d \end{pmatrix}, \quad \begin{pmatrix} a', & b' \\ c', & d' \end{pmatrix}$$

will be convertible if  $a' - d' : b' : c' = a - d : b : c$ .

**34.** Two matrices  $L, M$  are skew convertible when  $LM = -ML$ ; this is a relation much less important than ordinary convertibility, for it is to be noticed that we cannot in general find a matrix  $L$  skew convertible with a given matrix  $M$ . In fact, considering  $M$  as given, the equality affords a set of linear equations between the coefficients of  $L$  equal in number to these coefficients; and in this case the equations are independent, and we may eliminate all the coefficients of  $L$ , and we thus arrive at a relation which must be satisfied by the coefficients of the given matrix  $M$ . Thus, suppose the matrices

$$\begin{pmatrix} a, & b \\ c, & d \end{pmatrix}, \quad \begin{pmatrix} a', & b' \\ c', & d' \end{pmatrix}$$

are skew convertible, we have

$$\begin{aligned} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} a', & b' \\ c', & d' \end{pmatrix} &= \begin{pmatrix} aa' + bc', & ab' + bd' \\ ca' + dc', & cb' + dd' \end{pmatrix}, \\ \begin{pmatrix} a', & b' \\ c', & d' \end{pmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} &= \begin{pmatrix} aa' + b'c, & a'b + b'd \\ c'a + d'c, & c'b + d'd \end{pmatrix}, \end{aligned}$$

and the conditions of skew convertibility are

$$\begin{aligned} 2aa' + bc' + b'c &= 0, \\ b'(a + d) + b(a' + d') &= 0, \\ c'(a + d) + c(a' + d') &= 0, \\ 2dd' + bc' + b'c &= 0. \end{aligned}$$

Eliminating  $a', b', c', d'$ , the relation between  $a, b, c, d$  is

$$\begin{vmatrix} 2a, & c, & b, & \cdot \\ b, & a + d, & \cdot, & b \\ c, & \cdot, & a + d, & c \\ \cdot, & b, & c, & 2d \end{vmatrix} = 0,$$

which is

$$(a + d)^2(ad - bc) = 0.$$

Excluding from consideration the case  $ad - bc = 0$ , which would imply that the matrix was indeterminate, we have  $a + d = 0$ .

The resulting system of conditions then is

$$a + d = 0, \quad a' + d' = 0, \quad aa' + bc' + b'c + dd' = 0,$$

the first two of which imply that the matrices are respectively periodic of the second order to a factor pres.

**35.** It may be noticed that if the compound matrices  $LM$  and  $ML$  are similar, they are either equal or else opposite; that is, the matrices  $L, M$  are either convertible or skew convertible.

**36.** Two matrices such as

$$\begin{pmatrix} a, & b \\ c, & d \end{pmatrix}, \quad \begin{pmatrix} a, & c \\ b, & d \end{pmatrix}$$

are said to be formed one from the other by transposition, and this may be denoted by the symbol  $\text{tr.}$ ; thus we may write

$$\begin{pmatrix} a, & c \\ b, & d \end{pmatrix} = \text{tr.} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix}.$$

The effect of two successive transpositions is of course to reproduce the original matrix.

**37.** It is easy to see that if  $M$  be any matrix, then

$$(\text{tr. } M)^p = \text{tr. } (M^p),$$

and in particular,

$$(\text{tr. } M)^{-1} = \text{tr. } (M^{-1}).$$

**38.** If  $L, M$  be any two matrices,

$$\text{tr.}(LM) = \text{tr. } M \cdot \text{tr. } L,$$

and similarly for three or more matrices,  $L, M, N$ , &c.,

$$\text{tr.}(LMN) = \text{tr. } N \cdot \text{tr. } M \cdot \text{tr. } L, \quad \&c.$$

**39.** (No text in original.)

**40.** A matrix such as

$$\begin{pmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{pmatrix}$$

which is not altered by transposition, is said to be symmetrical.

**41.** A matrix such as

$$\begin{pmatrix} 0, & \nu, & -\mu \\ -\nu, & 0, & \lambda \\ \mu, & -\lambda, & 0 \end{pmatrix}$$

which by transposition is changed into its opposite, is said to be skew symmetrical.

**42.** It is easy to see that any matrix whatever may be expressed as the sum of a symmetrical matrix, and a skew symmetrical matrix; thus the form

$$\begin{pmatrix} a, & h + \nu, & g - \mu \\ h - \nu, & b, & f + \lambda \\ g + \mu, & f - \lambda, & c \end{pmatrix}$$

which may obviously represent any matrix whatever of the order 3, is the sum of the two matrices last before mentioned.

**43.** The following formulae, although little more than examples of the composition of transposed matrices, may be noticed, viz.

$$\begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} a, & c \\ b, & d \end{pmatrix} = \begin{pmatrix} a^2 + b^2, & ac + bd \\ ac + bd, & c^2 + d^2 \end{pmatrix}$$

which shows that a matrix compounded with the transposed matrix gives rise to a symmetrical matrix. It does not however follow, nor is it the fact, that the matrix and transposed matrix are convertible.

And also

$$\begin{pmatrix} a, & c \\ b, & d \end{pmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} a, & c \\ b, & d \end{pmatrix} = \begin{pmatrix} a^3 + bcd + a(b^2 + c^2), & c^3 + abd + c(a^2 + d^2) \\ b^3 + acd + b(a^2 + d^2), & d^3 + abc + d(b^2 + c^2) \end{pmatrix}$$

which is a remarkably symmetrical form. It is needless to proceed further, since it is clear that

$$\begin{pmatrix} a, & c \\ b, & d \end{pmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} a, & c \\ b, & d \end{pmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} = \left( \begin{pmatrix} a, & c \\ b, & d \end{pmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \right)^2.$$

44. In all that precedes, the matrix of the order 2 has frequently been considered, but chiefly by way of illustration of the general theory; but it is worth while to develop more particularly the theory of such matrix. I call to mind the fundamental properties which have been obtained, viz. it was shown that the matrix satisfies the equation

$$M^2 - (a + d)M + ad - bc = 0,$$

and that the two matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$$

will be convertible if

$$a' - d' : b' : c' = a - d : b : c,$$

and that they will be skew convertible if

$$a + d = 0, \quad a' + d' = 0, \quad aa' + bc' + b'c + dd' = 0,$$

the first two of these equations being the conditions in order that the two matrices may be respectively periodic of the second order to a factor pres.

45. It may be noticed in passing, that if  $L, M$  are skew convertible matrices of the order 2, and if these matrices are also such that  $L^2 = -1, M^2 = -1$ , then putting  $N = LM = -ML$ , we obtain

$$L^2 = -1, \quad M^2 = -1, \quad N^2 = -1, \quad L = MN = -NM, \quad M = NL = -LN, \quad N =$$

which is a system of relations precisely similar to that in the theory of quaternions.

46. The integer powers of the matrix

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

are obtained with great facility from the quadratic equation; thus we have, attending first to the positive powers,

$$M^2 = (a + d)M - (ad - bc),$$

$$M^3 = [(a + d)^2 - (ad - bc)]M - (a + d)(ad - bc),$$

&c., whence also the corresponding negative powers.

47. The square root of the matrix, or the matrix  $X$  such that  $X^2 = M$ , may be found as follows: the equation satisfied by  $X$  is  $X^2 - (x + w)X + (xw - yz) = 0$ , and combining this with  $X^2 = M$ , we have

$$(x + w)X = M + (xw - yz),$$

and thence, writing for shortness  $x + w = \theta$ ,  $xw - yz = \phi$ ,

$$X = \frac{1}{\theta}(M + \phi),$$

where  $\theta$ ,  $\phi$  are to be determined from the equation

$$\frac{1}{\theta^2}(M + \phi)^2 = M.$$

But

$$(M + \phi)^2 = (a + d + 2\phi)M + (ad - bc) + \phi(a + d) + \phi^2,$$

and this is to be equal to  $\theta^2 M$ , that is, we must have

$$\theta^2 = a + d + 2\phi, \quad (ad - bc) + \phi(a + d) + \phi^2 = 0.$$

These equations give

$$\theta^2 = (a + d + 2\phi), \quad \phi = \frac{1}{2}[-(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}],$$

and thence

$$X = \frac{1}{\theta} \begin{pmatrix} a + \phi, & b \\ c, & d + \phi \end{pmatrix},$$

where

$$\theta = \pm \sqrt{(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}}.$$

48. The equation for a matrix of the order 2, corresponding to the ordinary quadratic equation, is as follows. Let  $X$  be the matrix to be determined, and let the given equation be  $X^2 + pX + q = 0$ , where  $p$  and  $q$  are single quantities. The equation satisfied by  $X$  is

$X^2 - (x + w)X + (xw - yz) = 0$ , and the two equations will be identical if

$$x + w = -p, \quad xw - yz = q.$$

These conditions may be satisfied in a variety of ways; in fact, taking  $y$  and  $z$  arbitrarily, we have  $xw = q + yz$ , and thence  $x$  and  $w$  from the equation  $x + w = -p$ .

**49.** To complete the theory, it is necessary to consider the rational and integral functions of a matrix of the order 2. Any such function may be reduced to the form  $\alpha M + \beta$ , that is, a linear function of the matrix.

**50.** The general expression for the matrices convertible with a given matrix  $M$  of the order 2 is  $L = \alpha M + \beta$ , where  $\alpha$  and  $\beta$  are single quantities.

**51.** The general expression for the matrices skew convertible with a given matrix  $M$  of the order 2 is obtained as follows. The conditions are

$$a + d = 0, \quad a' + d' = 0, \quad aa' + bc' + b'c + dd' = 0,$$

and we have

$$L = \begin{pmatrix} a' & b' \\ c' & -a' \end{pmatrix},$$

where  $b'$ ,  $c'$  are arbitrary, and  $a'$  is given by the equation

$$aa' + bc' + b'c - da' = 0,$$

or, since  $d = -a$ ,

$$2aa' + bc' + b'c = 0.$$

**52.** The theory of the symmetrical and skew symmetrical matrices of the order 2 is very simple. A symmetrical matrix is of the form

$$\begin{pmatrix} a & h \\ h & b \end{pmatrix},$$

and a skew symmetrical matrix is of the form

$$\begin{pmatrix} 0 & \nu \\ -\nu & 0 \end{pmatrix}.$$



- 53.** Any matrix of the order 2 may be expressed as the sum of a symmetrical matrix and a skew symmetrical matrix:

$$\begin{pmatrix} a, & h + \nu \\ h - \nu, & b \end{pmatrix} = \begin{pmatrix} a, & h \\ h, & b \end{pmatrix} + \begin{pmatrix} 0, & \nu \\ -\nu, & 0 \end{pmatrix}.$$

**54.** The notion of composition applies to rectangular matrices, but it is necessary that the number of lines in the second or nearer component matrix should be equal to the number of columns in the first or further component matrix; the compound matrix will then have as many lines as the first or further component matrix, and as many columns as the second or nearer component matrix.

**55.** As examples of the composition of rectangular matrices, we have

$$\begin{pmatrix} a, & b, & c \\ d, & e, & f \end{pmatrix} \begin{pmatrix} a', & b', & c', & d' \\ e', & f', & g', & h' \\ i', & j', & k', & l' \end{pmatrix} = \begin{pmatrix} (a, b, c \$ a', e', i'), & (a, b, c \$ b', f', j'), & (a, b, c \$ c', g', k'), & (a, b, c \$ d', h', l') \\ (d, e, f \$ a', e', i'), & (d, e, f \$ b', f', j'), & (d, e, f \$ c', g', k'), & (d, e, f \$ d', h', l') \end{pmatrix}$$

and

$$\begin{pmatrix} a, & d \\ b, & e \\ c, & f \end{pmatrix} \begin{pmatrix} a', & b', & c', & d' \\ e', & f', & g', & h' \end{pmatrix} = \begin{pmatrix} (a, d \$ a', e'), & (a, d \$ b', f'), & (a, d \$ c', g'), & (a, d \$ d', h') \\ (b, e \$ a', e'), & (b, e \$ b', f'), & (b, e \$ c', g'), & (b, e \$ d', h') \\ (c, f \$ a', e'), & (c, f \$ b', f'), & (c, f \$ c', g'), & (c, f \$ d', h') \end{pmatrix}$$

**56.** In the particular case where the lines and columns of the one component matrix are respectively equal in number to the columns and lines of the other component matrix, the compound matrix is square, thus we have

$$\begin{pmatrix} a, & b, & c \\ d, & e, & f \end{pmatrix} \begin{pmatrix} a', & d' \\ b', & e' \\ c', & f' \end{pmatrix} = \begin{pmatrix} (a, b, c \$ a', b', c'), & (a, b, c \$ d', e', f') \\ (d, e, f \$ a', b', c'), & (d, e, f \$ d', e', f') \end{pmatrix}$$

and

$$\begin{pmatrix} a', & d' \\ b', & e' \\ c', & f' \end{pmatrix} \begin{pmatrix} a, & b, & c \\ d, & e, & f \end{pmatrix} = \begin{pmatrix} (a', d' \$ a, d), & (a', d' \$ b, e), & (a', d' \$ c, f) \\ (b', e' \$ a, d), & (b', e' \$ b, e), & (b', e' \$ c, f) \\ (c', f' \$ a, d), & (c', f' \$ b, e), & (c', f' \$ c, f) \end{pmatrix}$$

The two matrices in the case last considered admit of composition in the two different orders of arrangement, but as the resulting square matrices are not of the same order, the notion of the convertibility of two matrices does not apply even to the case in question.

**57.** Since a rectangular matrix cannot be compounded with itself, the notions of the inverse or reciprocal matrix and of the powers of

the matrix and the whole resulting theory of the functions of a matrix, do not apply to rectangular matrices.

**58.** The notion of transposition and the symbol  $\text{tr.}$  apply to rectangular matrices, the effect of a transposition being to convert a broad matrix into a deep one and reciprocally. It may be noticed that the symbol  $\text{tr.}$  may be used for the purpose of expressing the law of composition of square or rectangular matrices. Thus treating  $(a, b, c)$  as a rectangular matrix, or representing it by  $(a, b, c)$ , we have

$$\text{tr.} \begin{pmatrix} a' \\ b' \\ c' \end{pmatrix} = (a', \quad b', \quad c'),$$

and thence

$$(a, b, c) \text{ tr.} \begin{pmatrix} a' \\ b' \\ c' \end{pmatrix} = (a, b, c \$ a', b', c'),$$

so that the symbol  $(a, b, c \$ a', b', c')$  would upon principle be replaced by  $(a, b, c) \text{ tr.} (a', b', c')$ ; it is however more convenient to retain the symbol  $(a, b, c \$ a', b', c')$ . Hence introducing the symbol  $\text{tr.}$  only on the left-hand sides, we have

$$\begin{pmatrix} a, & b, & c \\ d, & e, & f \end{pmatrix} \text{ tr.} \begin{pmatrix} a', & b', & c' \\ d', & e', & f' \end{pmatrix} = \begin{pmatrix} (a, b, c \$ a', b', c'), & (a, b, c \$ d', e', f') \\ (d, e, f \$ a', b', c'), & (d, e, f \$ d', e', f') \end{pmatrix}$$

or to take an example involving square matrices,

$$\begin{pmatrix} a, & b \\ d, & e \end{pmatrix} \text{ tr.} \begin{pmatrix} a', & b' \\ d', & e' \end{pmatrix} = \begin{pmatrix} (a, b \$ a', b'), & (a, b \$ d', e') \\ (d, e \$ a', b'), & (d, e \$ d', e') \end{pmatrix}$$

it thus appears that in the composition of matrices (square or rectangular), when the second or nearer component matrix is expressed as a matrix preceded by the symbol  $\text{tr.}$ , any line of the compound matrix is obtained by compounding the corresponding line of the first or further component matrix successively with the several lines of the matrix which preceded by  $\text{tr.}$  gives the second or nearer component matrix. It is clear that the terms 'symmetrical' and 'skew symmetrical' do not apply to rectangular matrices.

## 153. A Memoir on the Automorphic Linear Transformation of a Bipartite Quadric Function.

[From the *Philosophical Transactions* of the Royal Society of London, vol. CXLVIII for the year 1858, pp. 39—46. Received December 10, 1857,—Read January 14, 1858.]

The question of the automorphic linear transformation of the function  $x^2 + y^2 + z^2$ , that is the transformation by linear substitutions, of this function into a function  $x_1^2 + y_1^2 + z_1^2$  of the same form, is in effect solved by some formulae of Euler's for the transformation of coordinates, and it was by these formulae that I was led to the solution in the case of the sum of  $n$  squares, given in my paper "Sur quelques propriétés des déterminants gauches"<sup>4</sup>.

A solution grounded upon an *a priori* investigation and for the case of any quadric function of  $n$  variables, was first obtained by M. Hermite in the memoir "Remarques sur une Mémoire de M. Cayley relatif aux déterminants gauches"<sup>5</sup>.

This solution is in my Memoir "Sur la transformation d'une fonction quadratique en elle-même par des substitutions linéaires"<sup>6</sup>, presented under a somewhat different form involving the notation of matrices. I have since found that there is a like transformation of a bipartite quadric function, that is a lineo-linear function of two distinct sets, each of the same number of variables, and the development of the transformation is the subject of the present memoir.

1. For convenience, the number of variables is in the analytical formulae taken to be 3, but it will be at once obvious that the formulae apply to any number of variables whatever. Consider the bipartite quadric

$$\begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} (x, y, z \text{ \& } x, y, z),$$

<sup>4</sup>Crelle, t. XXXII. (1846) pp. 119—123, [52].

<sup>5</sup>Cambridge and Dublin Mathematical Journal, t. ix. (1854) pp. 63—67.

<sup>6</sup>Crelle, t. I. (1855) pp. 288—299, [136].

which stands for

$$(ax + by + cz)x + (a'x + b'y + c'z)y + (a''x + b''y + c''z)z,$$

and in which  $(x, y, z)$  are said to be the nearer variables, and  $(x, y, z)$  the further variables of the bipartite.

**2.** It is clear that we have

$$\begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} (x, y, z \text{ \$} x, y, z) = \begin{pmatrix} a, & a', & a'' \\ b, & b', & b'' \\ c, & c', & c'' \end{pmatrix} (x, y, z \text{ \$} x, y, z),$$

and the new form on the right-hand side of the equation may also be written

$$\text{tr.} \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} (x, y, z \text{ \$} x, y, z),$$

that is, the two sets of variables may be interchanged, provided that the matrix is transposed.

**3.** Each set of variables may be linearly transformed: suppose that the substitutions are

$$(x, y, z) = \begin{pmatrix} l, & m, & n \\ l', & m', & n' \\ l'', & m'', & n'' \end{pmatrix} (x_1, y_1, z_1)$$

and

$$(x, y, z) = \begin{pmatrix} l, & l', & l'' \\ m, & m', & m'' \\ n, & n', & n'' \end{pmatrix} (x_1, y_1, z_1).$$

Then first substituting for  $(x, y, z)$  their values in terms of  $(x_1, y_1, z_1)$ , the bipartite becomes

$$\begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} (x_1, y_1, z_1 \text{ \$} x, y, z),$$

where

$$\begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} = \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix} \begin{pmatrix} l, & m, & n \\ l', & m', & n' \\ l'', & m'', & n'' \end{pmatrix}.$$

4. Suppose now that it is required to find the automorphic linear transformation of the bipartite

$$(\Omega)(x, y, z, x, y, z),$$

or as it will henceforward for shortness be written,

$$\Omega(x, y, z, x, y, z);$$

this may be effected by a method precisely similar to that employed by M. Hermite for an ordinary quadric. For this purpose write

$$\begin{aligned} x + x_1 &= 2\xi, & y + y_1 &= 2\eta, & z + z_1 &= 2\zeta, \\ x + x_1 &= 2\Xi, & y + y_1 &= 2H, & z + z_1 &= 2Z, \end{aligned}$$

or, as these equations may be represented,

$$\begin{aligned} (x + x_1, y + y_1, z + z_1) &= 2(\xi, \eta, \zeta), \\ (x + x_1, y + y_1, z + z_1) &= 2(\Xi, H, Z); \end{aligned}$$

then we ought to have

$$\begin{aligned} \Omega(2\xi - x, 2\eta - y, 2\zeta - z, 2\Xi - x, 2H - y, 2Z - z) \\ = \Omega(x, y, z, x, y, z). \end{aligned}$$

5. The left-hand side is

$$\begin{aligned} 4\Omega(\xi, \eta, \zeta, \Xi, H, Z) - 2\Omega(x, y, z, \Xi, H, Z) \\ - 2\Omega(\xi, \eta, \zeta, x, y, z) + \Omega(x, y, z, x, y, z), \end{aligned}$$

and the equation becomes

$$\begin{aligned} 2\Omega(\xi, \eta, \zeta, \Xi, H, Z) - \Omega(x, y, z, \Xi, H, Z) \\ - \Omega(\xi, \eta, \zeta, x, y, z) = 0, \end{aligned}$$

or as it may be written,

$$\begin{aligned} \Omega(\xi, \eta, \zeta, \Xi, H, Z) - \Omega(x, y, z, \Xi, H, Z) \\ + \Omega(\xi, \eta, \zeta, \Xi, H, Z) - \Omega(\xi, \eta, \zeta, x, y, z) = 0, \end{aligned}$$

or again,

$$\begin{aligned} \Omega(\xi - x, \eta - y, \zeta - z \, \S \Xi, H, Z) \\ + \Omega(\xi, \eta, \zeta \, \S \Xi - x, H - y, Z - z) = 0, \end{aligned}$$

or what is the same thing,

$$\begin{aligned} \Omega(\xi - x, \eta - y, \zeta - z \, \S \Xi, H, Z) \\ + \text{tr. } \Omega(\Xi - x, H - y, Z - z \, \S \xi, \eta, \zeta) = 0, \end{aligned}$$

and it is easy to see that the equation will be satisfied by writing

$$\begin{aligned} \Omega(\xi - x, \eta - y, \zeta - z) &= T(\xi, \eta, \zeta), \\ \text{tr. } \Omega(\Xi - x, H - y, Z - z) &= -\text{tr. } T(\Xi, H, Z), \end{aligned}$$

where  $T$  is any arbitrary matrix. In fact we have then

$$\begin{aligned} \Omega(\xi - x, \eta - y, \zeta - z \, \S \Xi, H, Z) &= T(\xi, \eta, \zeta \, \S \Xi, H, Z), \\ \text{tr. } \Omega(\Xi - x, H - y, Z - z \, \S \xi, \eta, \zeta) &= -\text{tr. } T(\Xi, H, Z \, \S \xi, \eta, \zeta) \\ &= -T(\xi, \eta, \zeta \, \S \Xi, H, Z), \end{aligned}$$

and the sum of the two terms consequently vanishes.

**6.** The equation

$$\Omega(\xi - x, \eta - y, \zeta - z) = T(\xi, \eta, \zeta)$$

gives

$$\Omega(\xi, \eta, \zeta) - \Omega(x, y, z) = T(\xi, \eta, \zeta),$$

and we then have

$$2\Omega(\xi, \eta, \zeta) = \Omega(x + x_1, y + y_1, z + z_1),$$

$$2(\xi, \eta, \zeta) = (x + x_1, y + y_1, z + z_1),$$

which is the equation assumed as the definition of  $(\xi, \eta, \zeta)$ ; and conversely, this equation, combined with either of the two equations, gives the other of them.

**7.** We have consequently

$$(x, y, z) = (\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1),$$

$$(x, y, z) = \Omega^{-1}(\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1).$$

**8.** But in like manner the equation

$$\text{tr. } \Omega(\Xi - x, H - y, Z - z) = -\text{tr. } T(\Xi, H, Z)$$

gives

$$\text{tr. } (\Omega + T)(\Xi, H, Z) = \text{tr. } \Omega(x, y, z),$$

and we then obtain

$$\text{tr. } (\Omega + T)(\Xi, H, Z) = \text{tr. } \Omega(x_1, y_1, z_1).$$

**9.** In fact these equations give

$$\text{tr. } 2\Omega(\Xi, H, Z) = \text{tr. } \Omega(x + x_1, y + y_1, z + z_1),$$

or

$$2(\Xi, H, Z) = (x + x_1, y + y_1, z + z_1),$$

which is the equation assumed as the definition of  $(\Xi, H, Z)$ .



**10.** We have thus the complete solution of the problem of the automorphic linear transformation of the bipartite quadric function.

The result is that if  $T$  is any arbitrary skew symmetrical matrix, then writing

$$\Omega(\xi, \eta, \zeta) - \Omega(x, y, z) = T(\xi, \eta, \zeta),$$

the equations

$$(x, y, z) = (\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1),$$

$$(x, y, z) = \Omega^{-1}(\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1)$$

give the required transformation.

**11.** In the case where the two transformations are identical, we have a simplification. The equations then become

$$\begin{aligned}(x, y, z) &= (\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1), \\ (x, y, z) &= \Omega^{-1}(\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1),\end{aligned}$$

and the condition for this is that  $\Omega$  is a symmetrical matrix.

**12.** When  $\Omega$  is a symmetrical matrix, the automorphic transformation is obtained by means of an arbitrary skew symmetrical matrix  $T$ , and we have

$$(x, y, z) = (\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1).$$

**13.** When  $\Omega$  is a skew symmetrical matrix, the automorphic transformation is obtained by means of an arbitrary symmetrical matrix  $T$ .

**14.** In the particular case where  $\Omega$  is the matrix unity, the formula becomes

$$(x, y, z) = (1 - T)(1 + T)^{-1}(x_1, y_1, z_1),$$

which is the formula for the automorphic transformation of the ordinary quadric  $x^2 + y^2 + z^2$ .

**15.** The case where  $\Omega$  is the matrix unity corresponds to the ordinary quadric function  $x^2 + y^2 + z^2$ , and the transformation is

$$(x, y, z) = (1 - T)(1 + T)^{-1}(x_1, y_1, z_1),$$

where  $T$  is an arbitrary skew symmetrical matrix; this agrees with the formulae obtained in the memoir above referred to.

**16.** The mode in which this happens will be best seen by an example. Suppose

$$\Omega^{-1} = \begin{pmatrix} A, & H + \nu \\ H - \nu, & B \end{pmatrix}, \quad \text{tr. } \Omega^{-1} = \begin{pmatrix} A, & H - \nu \\ H + \nu, & B \end{pmatrix},$$

and write

$$T_1 = \begin{pmatrix} 0, & \theta \\ -\theta, & 0 \end{pmatrix},$$

then we have

$$T = -(\Omega^{-1} + \text{tr. } \Omega^{-1})^{-1} \begin{pmatrix} 0, & \nu \\ -\nu, & 0 \end{pmatrix} \begin{pmatrix} 0, & \theta \\ -\theta, & 0 \end{pmatrix} + \begin{pmatrix} 0, & \theta \\ -\theta, & 0 \end{pmatrix}.$$

**17.** Hence, finally, we have the following Theorem for the automorphic linear transformation of the bipartite quadric,

$$\Omega(x, y, z \text{ \$ } x, y, z),$$

when the two transformations are identical, viz. if  $T_1$  be a skew symmetrical matrix, and if

$$T = (\Omega^{-1} + \text{tr. } \Omega^{-1})^{-1}(\Omega^{-1} - \text{tr. } \Omega^{-1})T_1,$$

then if we have

$$\begin{aligned} (x, y, z) &= (\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1), \\ (x, y, z) &= \Omega^{-1}(\Omega - T)(\Omega + T)^{-1}\Omega(x_1, y_1, z_1); \end{aligned}$$

then

$$\Omega(x, y, z \text{ \$ } x, y, z) = \Omega(x_1, y_1, z_1 \text{ \$ } x_1, y_1, z_1);$$

and in particular,

- If  $\Omega$  is a symmetrical matrix, then  $T$  is an arbitrary skew symmetrical matrix.
- If  $\Omega$  is a skew symmetrical matrix, then  $T$  is an arbitrary symmetrical matrix.

## 154. Supplementary Researches on the Partition of Numbers.

[From the *Philosophical Transactions* of the Royal Society of London, vol. CXLVIII for the year 1858, pp. 47—52. Received March 19,—Read June 18, 1857.]

The general formula given at the conclusion of my memoir, “Researches on the Partition of Numbers”<sup>7</sup> is somewhat different from the corresponding formula of Professor Sylvester’s, and leads more directly to the actual expression for the number of partitions, in the form made use of in my memoir; to complete my former researches, I propose to explain the mode of obtaining from the formula the expression for the number of partitions.

The formula referred to is as follows, viz. if  $\frac{\phi x}{fx}$  be a rational fraction, the denominator of which is made up of factors (the same or different) of the form  $1 - x^m$ , and if  $a$  is a divisor of one or more of the indices  $m$ , and  $k$  is the number of indices of which it is a divisor, then

$$\left[ \frac{\phi x}{fx} \right]_{[1-x^a]} = \Sigma \frac{1}{\Pi(s-1)!} \left( \frac{d}{dt} \right)^{s-1} \frac{\phi(\rho e^{-t})}{\psi(\rho e^{-t})} \cdot \frac{1}{(1 - \rho e^{-t})^s},$$

where

$$\psi x = \frac{fx}{[1-x^a]},$$

in which formula  $[1-x^a]$  denotes the irreducible factor of  $1-x^a$ , that is, the factor which equated to zero gives the prime roots, and  $\rho$  is a root of the equation  $[1-x^a] = 0$ ; the summation of course extends to all the roots of the equation.

The index  $s$  extends from  $s = 1$  to  $s = k$ ; and we have then the portion of the fraction depending on the denominator  $[1-x^a]$ .

1. In the partition of numbers, we have  $\phi x = 1$ , and the formula becomes therefore

$$\left[ \frac{1}{fx} \right]_{[1-x^a]} = \Sigma \frac{1}{\Pi(s-1)!} \left( \frac{d}{dt} \right)^{s-1} \frac{1}{\psi(\rho e^{-t})} \cdot \frac{1}{(1 - \rho e^{-t})^s},$$

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<sup>7</sup>Philosophical Transactions, t. cxlvi. (1856) p. 127, [140].

where

$$\psi x = \frac{fx}{[1 - x^a]}.$$

**2.** We may write

$$fx = \Pi(1 - x^m),$$

where  $m$  has a given series of values the same or different. The indices not divisible by  $a$  may be represented by  $\varpi$ , the other indices by  $a\varpi'$ ; we have then

$$fx = \Pi(1 - x^\varpi) \cdot \Pi(1 - x^{a\varpi'}),$$

where the number of indices  $a\varpi'$  is equal to  $k$ .  
Hence

$$f(\rho e^{-t}) = \Pi(1 - \rho^\varpi e^{-\varpi t}) \cdot \Pi(1 - \rho^{a\varpi'} e^{-a\varpi' t});$$

or since  $\rho$  is a root of  $[1 - x^a] = 0$ , and therefore  $\rho^a = 1$ , we have

$$f(\rho e^{-t}) = \Pi(1 - \rho^\varpi e^{-\varpi t}) \cdot \Pi(1 - e^{-a\varpi' t});$$

and it may be remarked that if  $n \equiv \nu \pmod{a}$ , where  $\nu < a$ , then instead of  $\rho^n$  we may write  $\rho^\nu$ , a change which may be made at once, or at the end of the process of development.

We have consequently to find

$$\chi\rho = \text{coeff. } \frac{1}{t} \text{ in } \frac{e^{-t}}{\Pi(1 - \rho^\varpi e^{-\varpi t}) \cdot \Pi(1 - e^{-a\varpi' t})}.$$

3. The development of a factor  $\frac{1}{1-\rho^{\varpi}e^{-\varpi t}}$  is at once deduced from that of  $\frac{1}{1-ce^t}$ , and is a series of positive powers of  $t$ . The development of a factor  $\frac{1}{1-e^{-a\varpi't}}$  is deduced from that of  $\frac{1}{1-e^{-t}}$ , and contains a term involving  $\frac{1}{t}$ .

4. Hence we have

$$\chi\rho = \text{coeff. } \frac{1}{t} \text{ in } \frac{1}{\Pi(1-\rho^{\varpi})} \cdot \frac{1}{\Pi(a\varpi't)},$$

and thence

$$\chi\rho = \frac{1}{\Pi(1-\rho^{\varpi})} \cdot \frac{1}{a^k \cdot \Pi(\varpi')},$$

5. The actual development, when  $k$  is small (for instance  $k = 1$  or  $k = 2$ ), is most readily obtained by developing each factor separately and taking the product. To do this we have

$$\frac{1}{1-ce^t} = \frac{1}{1-c} + \frac{ct}{(1-c)^2} + \frac{c(c+1)t^2}{2(1-c)^3} + \dots,$$

where by a general theorem for the expansion of any function of  $e^t$ , the coefficient of  $t^s$  is  $= (z)^s \frac{1}{\Pi(1-c)}$ , and

$$\frac{1}{1-e^{-t}} = \frac{1}{t} + \frac{1}{2} + \frac{t}{12} - \frac{t^3}{720} + \frac{t^5}{30240} - \dots,$$

where, except the constant term, the series contains odd powers only and the coefficient of  $t^{2n-1}$  is  $\frac{(-1)^{n-1}B_n}{(2n)!}$ ;  $B_1, B_2, B_3, \dots$  denoting the series  $\frac{1}{6}, \frac{1}{30}, \frac{1}{42}, \dots$  of Bernoulli's numbers.

6. But when  $k$  is larger, it is convenient to obtain the development of the fraction from that of the logarithm, the logarithm of the fraction being equal to the sum of the logarithms of the simple factors, and these being found by means of the formula:

$$\log \frac{1}{1-ce^t} = \log \frac{1}{1-c} + \frac{ct}{1-c} + \frac{ct^2}{2(1-c)^2} + \frac{c(c+1)t^3}{6(1-c)^3} + \dots$$

The fraction is thus expressed in the form

$$\frac{1}{\Pi(1-\rho^{\varpi}) \cdot \Pi(a\varpi')} \cdot \frac{1}{t^k} e^{M_1 t + M_2 t^2 + \dots},$$

and by developing the exponential we obtain, as before, the series commencing with  $\frac{1}{t^k}$ .

7. Resuming now the formula

$$\frac{1}{[1-x^a]} = \frac{\chi\rho}{\rho-x},$$

which gives  $\chi\rho$  as a function of  $\rho$ , we have

$$[1-x^a] = (x-\rho)(x-\rho^{\alpha_1})\cdots(x-\rho^{\alpha_n}),$$

if  $1, \alpha_1, \dots, \alpha_n$  are the integers less than  $a$  and prime to it ( $a$  is of course the degree of  $[1-x^a]$ ).

Hence

$$\frac{1}{[1-x^a]} = \frac{1}{(\rho-x)(\rho-\rho^{\alpha_1})\cdots(\rho-\rho^{\alpha_n})},$$

and we have

$$\chi\rho = \frac{1}{(\rho-\rho^{\alpha_1})\cdots(\rho-\rho^{\alpha_n})},$$

and therefore

$$\theta\rho = -\rho^{\frac{a-1}{2}} \Pi(1-\rho^{\alpha_i-1})\chi\rho;$$

or putting for  $\chi\rho$  its value

$$\theta\rho = -\rho^{\frac{a}{2}} \Pi(1-\rho^{\alpha_i-1}) \cdot \chi\rho,$$

where  $a$  is the degree of  $[1-x^a]$  and  $\alpha_i$  denotes in succession the integers (exclusive of unity) less than  $a$  and prime to it.

The function on the right hand, by means of the equation  $[1-\rho^a] = 0$ , may be reduced to an integral function of  $\rho$  of the degree  $a-1$ , and then by simply changing  $\rho$  into  $x$  we have the required function  $\theta x$ .

8. The fraction  $\frac{\theta x}{1-x^a}$  can then by multiplication of the terms by the proper factor be reduced to a fraction with the denominator  $1-x^a$ , and the coefficients of the numerator of this fraction are the coefficients of the corresponding prime circulator ( ) per  $a_q$ .

9. Thus, let it be required to find the terms depending on the denominator  $[1-x^3]$  in

$$\frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)(1-x^5)(1-x^6)},$$

these are

$$\frac{1}{[1-x^3]} = \frac{\chi\rho}{\rho-x},$$

where

$$\chi\rho = \text{coeff. } \frac{1}{t} \text{ in } \frac{e^{-t}}{f(\rho e^{-t})}$$

and

$$f(\rho e^{-t}) = (1 - \rho e^{-t})(1 - \rho^2 e^{-2t})(1 - \rho^4 e^{-4t})(1 - \rho^5 e^{-5t}) \\ \times (1 - e^{-3t})(1 - e^{-6t})$$

$$= A_{-2}t^{-2} + A_{-1}t^{-1} + \cdots,$$

where it is easy to see that

$$A_{-2} = \frac{1}{18(1-\rho)(1-\rho^2)(1-\rho^4)(1-\rho^5)},$$

$$A_{-1} = -\frac{1}{18(1-\rho)(1-\rho^2)(1-\rho^4)(1-\rho^5)} \left\{ \frac{1}{1-\rho} + \frac{2}{1-\rho^2} + \frac{4}{1-\rho^4} + \frac{5}{1-\rho^5} \right\},$$

and we have

$$\theta\rho = -\rho^{\frac{3}{2}}(1-\rho)A_{-1}.$$

But  $[1-\rho^3] = 1 + \rho + \rho^2 = 0$ . Hence  $\rho^3 = 1$ , and therefore

$$(1-\rho)(1-\rho^2)(1-\rho^4)(1-\rho^5) = (1-\rho)^2(1-\rho^2)^2 = 9.$$

Hence

$$\theta\rho = \frac{1}{18}(2+\rho),$$

whence

$$\frac{1}{[1-x^3]} = \frac{1}{162} \cdot \frac{2+x}{1+x+x^2},$$

which is

$$\frac{1}{162} \cdot \frac{1-x-x^2}{1-x^3},$$

and gives rise to the prime circulator  $\frac{1}{162}(2, -1, -1)$  per  $3_1$ .



10. The reduction of  $\theta_2$  is somewhat less simple; we have

$$\begin{aligned}\theta_2 &= \frac{1}{9}\rho^2(1-\rho) \left\{ \frac{1}{(1-\rho)^2} + \frac{5}{(1-\rho^2)^2} + \frac{7}{(1-\rho^4)^2} \right\} \\ &= \frac{1}{9}(1-\rho^2)(61-10\rho-14\rho^2) \\ &= \frac{1}{9}(61+4\rho-65\rho^2); \end{aligned}$$

hence finally

$$\frac{1}{[1-x^3]} = \frac{1}{324} \cdot \frac{42+23x}{1+x+x^2},$$

which is

$$\frac{1}{324} \cdot \frac{3 \cdot 42 - 19x - 23x^2}{1-x^3},$$

and gives rise to the prime circulator  $\frac{1}{324}(42, -19, -23)$  per  $3_2$ .

11. The part depending on the denominator  $1-x$  is

$$\begin{aligned} & \frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)(1-x^5)(1-x^6)} \\ &= \frac{1}{720t^6} \left( 1 + \frac{21}{2}t + \frac{91}{8}t^2 + \frac{105}{16}t^3 + \dots \right), \end{aligned}$$

where

$$\begin{aligned} (1-e^{-t})(1-e^{-2t})(1-e^{-3t})(1-e^{-4t})(1-e^{-5t})(1-e^{-6t}) \\ = 720t^6 + \dots \end{aligned}$$

We have here

$$\frac{1}{720t^6} \left( 1 + \frac{21}{2}t + \frac{91}{8}t^2 + \dots \right),$$

and thence the fraction is

$$\frac{1}{720t^6} + \frac{21}{1440t^5} + \frac{91}{5760t^4} + \dots$$

which is equal to

$$\begin{aligned} \frac{1}{720} \cdot \frac{1}{t^6} + \frac{21}{1440} \cdot \frac{1}{t^5} + \frac{441}{11520} \cdot \frac{1}{t^4} + \frac{3087}{69120} \cdot \frac{1}{t^3} + \frac{64827}{829440} \cdot \frac{1}{t^2} \\ + \frac{1361367}{9953280} \cdot \frac{1}{t} + \dots \end{aligned}$$

$$= \frac{1}{720} \cdot \frac{1}{t^6} + \frac{7}{480} \cdot \frac{1}{t^5} + \frac{77}{1920} \cdot \frac{1}{t^4} + \frac{77}{5120} \cdot \frac{1}{t^3} + \frac{231}{20480} \cdot \frac{1}{t^2} + \frac{1463}{245760} \cdot \frac{1}{t} + \dots$$

and consequently the partial fractions are

$$\begin{aligned} \frac{1}{720} \cdot \frac{1}{1-x} + \frac{7}{480} \cdot \frac{1}{(1-x)^2} + \frac{77}{1920} \cdot \frac{1}{(1-x)^3} + \frac{77}{5120} \cdot \frac{1}{(1-x)^4} \\ + \frac{231}{20480} \cdot \frac{1}{(1-x)^5} + \frac{1463}{245760} \cdot \frac{1}{(1-x)^6}, \end{aligned}$$

from which the non-circulating part is at once obtained.

**12.** The complete expression for the number of partitions is

$$\begin{aligned}
 P(1, 2, 3, 4, 5, 6)q = & \\
 & \frac{1}{1036800} \left( \begin{array}{l} 11q^5 + 63q^4 + 270q^3 + 441q^2 \\ + 110250q + 439810q^2 + 598731 \end{array} \right) \\
 & + \frac{1}{768} (q^4 + 12q^3 + 581)(1, 1) \text{ per } 2_1 \\
 & \quad + \frac{1}{162} (2, -1, -1) \text{ per } 3_1 \\
 & + \frac{1}{324} (42, -19, -23) \text{ per } 3_2 \\
 & \quad + \frac{1}{32} (1, 1, -1, -1) \text{ per } 4_1 \\
 & \quad + \frac{1}{5} (2, 1, 0, -1, -2) \text{ per } 5_1 \\
 & \quad \quad + \frac{1}{6} (2, 1, -1, -2, -1, 1) \text{ per } 6_1.
 \end{aligned}$$

## 155. A Fourth Memoir upon Quantics.

[From the *Philosophical Transactions* of the Royal Society of London, vol. CXLVIII for the year 1858, pp. 415—427. Received February 11,—Read March 18, 1858.]

The object of the present memoir is the further development of the theory of binary quantics; it should therefore have preceded so much of my third memoir, t. 147 (1857), p. 627, [144], as relates to ternary quadrics and cubics. The paragraphs are numbered continuously with those of the former memoirs. The first three paragraphs, Nos. 62 to 64, relate to quantics of the general form  $(*|x, y, \dots)^m$ , and they are intended to complete the series of definitions and explanations given in Nos. 54 to 61 of my third memoir; Nos. 68 to 71, although introduced in reference to binary quantics, relate or may be considered as relating to quantics of the like general form. But with these exceptions the memoir relates to binary quantics of any order whatever: viz. Nos. 65 to 80 relate to the covariants and invariants of the degrees 2, 3 and 4; Nos. 81 and 82 relate to the quintic.

### 62. (Definitions relating to quantics in general.)

**63.** If  $U, V$  are any two quantics of the same order, then the coefficients  $(A, B, \dots B', A')$  of the quantic  $(A, B, \dots B', A' | \lambda, \mu)^n$ , where  $n$  is the degree of the quantic in the facients  $(\lambda, \mu)$ , will be covariants of the quantics  $U, V$ ; viz.  $A$  will be a covariant of the quantic  $U$  alone;  $A'$  will be the same covariant of the quantic  $V$  alone, and the other coefficients (which in reference to  $A, A'$  may be termed the Connectives) will be covariants of the two quantics; and any coefficient may be obtained from the one which precedes it by operating on such preceding coefficient with the combinative operator  $a'\partial_a + b'\partial_b + \dots$ , or from the one which succeeds it by operating on such succeeding coefficient with the combinative operator  $a\partial_{a'} + b\partial_{b'} + \dots$ , the result being divided by a numerical coefficient which is greater by unity than the index of  $\mu$  or (as the case may be)  $\lambda$  in the term corresponding to the coefficient operated upon. It may be added, that any invariant in regard to the facients  $(\lambda, \mu)$  of the quantic  $(A, B, \dots B', A' | \lambda, \mu)^n$  is not only a covariant, but it is also a combinant of the two quantics  $U, V$ .

As an example, suppose the quantics  $U, V$  are the quadrics

$(a, b, c \mid x, y)^2$  and  $(a', b', c' \mid x, y)^2$ ; then the quadrinvariant of  $\lambda U + \mu V$  is  $(\lambda a + \mu a')(\lambda c + \mu c') - (\lambda b + \mu b')^2$ , which is equal to

$$(ac - b^2, ac' - 2bb' + ca', a'c' - b'^2 \mid \lambda, \mu)^2,$$

and  $ac' - 2bb' + ca'$  is the connective of the two discriminants  $ac - b^2$  and  $a'c' - b'^2$ .

**65.** The law of reciprocity for the number of the invariants of a binary quantic leads at once to the theorems in regard to the number of the quadrimvariants, cubinvariants and quartinvariants of a binary quantic of a given degree, first obtained by the method in the second part of my original memoir<sup>8</sup>. Thus a quadric has only a single invariant, which is of the degree 2; hence, by the law of reciprocity, the number of quadrimvariants of a quantic of the order  $m$  is equal to the number of ways in which  $m$  can be made up with the part 2, which is of course unity or zero, according as  $m$  is even or odd. And we conclude that

*The quadrimvariant exists only for quantics of an even order, and for each such quantic there is one, and only one, quadrimvariant.*

**66.** Again, a cubic has only one invariant, which is of the degree 4, and the number of cubinvariants of a quantic of the degree  $m$  is equal to the number of ways in which  $m$  can be made up with the part 4. Hence

*A cubinvariant only exists for quantics of an evenly even order, and for each such quantic there is one, and only one, cubinvariant.*

**67.** But a quartic has two invariants, which are of the degrees 2 and 3 respectively, and the number of quartinvariants of a quantic of the degree  $m$  is equal to the number of ways in which  $m$  can be made up with the parts 2 and 3. When  $m$  is even, there is of course a quartinvariant which is the square of the quadrimvariant, and which, if we attend only to the irreducible quartinvariants, must be excluded from consideration. The preceding number must therefore, when  $m$  is even, be diminished by unity.

The result is easily found to be

*Quartinvariants exist for a quantic of any order, even or odd, whatever, the quadric and the quartic alone excepted; and according as the order of the quantic is*

$$6g, \quad 6g + 1, \quad 6g + 2, \quad 6g + 3, \quad 6g + 4, \quad 6g + 5,$$

*the number of quartinvariants is*

$$g, \quad g, \quad g, \quad g + 1, \quad g, \quad g + 1.$$

In particular, for the orders 2, 3, 4, 5; 6, 7, 8, 9, 10, 11; 12, &c., the numbers are

$$0, \quad 1, \quad 0, \quad 1; \quad 1, \quad 1, \quad 1, \quad 2, \quad 1, \quad 2; \quad 2, \quad \&c.$$

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<sup>8</sup>Introductory Memoir, [189], No. 20. Ibid. Nos. 10—17.

Thus the ninthic is the lowest quantic which has more than one quartinvariant.

**68.** But the whole theory of the invariants or covariants of the degrees 2, 3, 4 is most easily treated by the method above alluded to, contained in the second part of my original memoir; and indeed the method appears to be the appropriate one for the treatment of the theory of the invariants or covariants of any given degree whatever, although the application of it becomes difficult when the degree exceeds 4. I remark, in regard to this method, that it leads naturally, and in the first instance, to a special class of the covariants of a system of quantics, viz. these covariants are linear functions of the derived functions of any quantic of the system. (It is hardly necessary to remark that the derived functions referred to are the derived functions of any order of the quantic with regard to the facients.) Such covariants may be termed tantipartite covariants; but when there are only two quantics, I use in general the term lineo-linear.

The tantipartite covariants, while the system remains general, are a special class of covariants, but by particularizing the system we obtain all the covariants of the particularized system. The ordinary case is when all the quantics of the system reduce themselves to one and the same quantic, and the method then gives all the covariants of such single quantic. And while the order of the quantic remains indefinite, the method gives covariants (not invariants); but by particularizing the order of the quantic in such manner that the derived functions become simply the coefficients of the quantic, the covariants become invariants: the like applies of course to a system of two or more quantics.

**69.** To take the simplest example, in seeking for the covariants of a single quantic  $U$ , we in fact have to consider two quantics  $U, V$ . An expression such as  $12UV$  is a lineo-linear covariant of the two quantics; its developed expression is

$$\partial_x U \cdot \partial_y V - \partial_y U \cdot \partial_x V,$$

which is the Jacobian. In the particular case of two linear functions  $(a, b \mid x, y)$  and  $(a', b' \mid x, y)$ , the lineo-linear covariant becomes the lineo-linear invariant  $ab' - a'b$ , which is the Jacobian of the two linear functions.

In the example we cannot descend from the two quantics  $U, V$  to the single quantic  $U$  (for putting  $V = U$  the covariant vanishes); but this is merely accidental, as appears by considering a different lineo-linear covariant  $12^2 UV$ , the developed expression of which is

$$\partial_x^2 U \cdot \partial_y^2 V - 2\partial_x \partial_y U \cdot \partial_x \partial_y V + \partial_y^2 U \cdot \partial_x^2 V.$$

In the particular case of two quadrics  $(a, b, c \mid x, y)^2$ ,  $(a', b', c' \mid x, y)^2$ , the lineo-linear covariant becomes the lineo-linear invariant  $ac' - 2bb' + ca'$ . If we have  $V = U$ , then the lineo-linear covariant gives the quadricovariant of the single quantic  $U$  (such quadricovariant is in fact the Hessian); and if in the last-mentioned formula we put for  $U$  the quadric  $(a, b, c \mid x, y)^2$  or what is the same thing, if in the expression of the lineo-linear invariant  $ac' - 2bb' + ca'$ , we put the two quadrics equal to each other, we have the quadrinvariant  $ac - b^2$  of the single quadric.

**70.** The lineo-linear invariant  $ab' - a'b$  of two linear functions may be considered as giving the lineo-linear covariant

$\partial_x U \cdot \partial_y V - \partial_y U \cdot \partial_x V$  of the two quantics  $U$  and  $V$ , and in like manner the lineo-linear invariant  $ac' - 2bb' + ca'$  may be considered as giving the lineo-linear covariant

$\partial_x^2 U \cdot \partial_y^2 V - 2\partial_x \partial_y U \cdot \partial_x \partial_y V + \partial_y^2 U \cdot \partial_x^2 V$  of the quantics  $U, V$ . And generally, any invariant whatever of a quantic or quantics of a given order or orders leads to a covariant of a quantic or quantics of any higher order or orders: viz. the coefficients of the original quantic or quantics are to be replaced by the derived functions of the quantic or quantics of a higher order or orders.

**71.** The same thing may be seen by means of the theory of Emanants. In fact, consider any emanants whatever of a quantic or



quantics; then, attending only to the facients of emanation, the emanants will constitute a system of quantics the coefficients of which are derived functions of the given quantic or quantics; the invariants of the system of emanants will be functions of the derived functions of the given quantic or quantics, and they will be covariants of such quantic or quantics; and we thus pass from the invariants of a quantic or quantics to the covariants of a quantic or quantics of a higher order or orders.

**72.** It may be observed also, that in the case where a tantipartite invariant, when the several quantics are put equal to each other, does not become equal to zero, we may pass back from the invariant of the single quantic to the tantipartite invariant of the system; thus the lineo-linear invariant  $ac' - 2bb' + ca'$  of two quadrics leads to the quadrinvariant  $ac - b^2$  of a single quantic; and conversely, from the quadrinvariant  $ac - b^2$  of a single quadric, we obtain by an obvious process of derivation the expression  $ac' - 2bb' + ca'$  of the lineo-linear invariant of two quadrics. This is in fact included in the more general theory explained, No. 64.

**73.** Reverting now to binary quantics, two quantics of the same order, even or odd, have a lineo-linear invariant. Thus the two quadrics  $(a, b, c \mid x, y)^2$ ,  $(a', b', c' \mid x, y)^2$  have (it has been seen) the lineo-linear invariant  $ac' - 2bb' + ca'$ ; and in like manner the two cubics  $(a, b, c, d \mid x, y)^3$ ,  $(a', b', c', d' \mid x, y)^3$  have the lineo-linear invariant  $ad' - 3bc' + 3cb' - da'$ , which examples are sufficient to show the law.

**74.** The lineo-linear invariant of two quantics of the same odd order is a combinant, but this is not the case with the lineo-linear invariant of two quantics of the same even order. Thus the last-mentioned invariant is reduced to zero by each of the operations

$$a\partial_{a'} + b\partial_{b'} + c\partial_{c'} + d\partial_{d'}$$

and

$$a'\partial_a + b'\partial_b + c'\partial_c + d'\partial_d;$$

but the invariant  $ac' - 2bb' + ca'$  is by the operations

$$a\partial_{a'} + b\partial_{b'} + c\partial_{c'}$$

and

$$a'\partial_a + b'\partial_b + c'\partial_c$$

reduced respectively to

$$2(ac - b^2) \quad \text{and} \quad 2(a'c' - b'^2).$$

**75.** For two quantics of the same odd order, when the quantics are put equal to each other, the lineo-linear invariant vanishes; but for two quantics of the same even order, when these are put equal to each other, we obtain the quadrinvariant of the single quantic. Thus the quadrinvariant of the quadric  $(a, b, c \mid x, y)^2$  is  $ac - b^2$ ; and in like manner the quadrinvariant of the quartic  $(a, b, c, d, e \mid x, y)^4$  is  $ae - 4bd + 3c^2$ .

**76.** When the two quantics are the first derived functions of the same quantic of any odd order, the lineo-linear invariant does not vanish, but it is not an invariant of the single quantic. Thus the lineo-linear invariant of  $(a, b, c \mid x, y)^2$  and  $(b, c, d \mid x, y)^2$  is

$(ad - 2bc + cb =) ad - bc$ , which is not an invariant of the cubic  $(a, b, c, d \mid x, y)^3$ . But for two quantics which are the first derived functions of the same quantic of any even order, the lineo-linear invariant is the quadrinvariant of the single quantic. Thus the lineo-linear invariant of  $(a, b, c, d \mid x, y)^3$  and  $(b, c, d, e \mid x, y)^3$  is  $(ae - 3bd + 3c^2 - db =) ae - 4bd + 3c^2$  which is the quadrinvariant of the quartic  $(a, b, c, d, e \mid x, y)^4$ .

**77.** I do not stop to consider the theory of the lineo-linear covariants of two quantics, but I derive the quadricovariants of a single quantic directly from the quadrinvariant. Imagine a quantic of any order even or odd. Its successive even emanants will be in regard to the facients of emanation quantics of an even order, and they will each of them have a quadrinvariant, which will be a quadricovariant of the given quantic.

The emanants in question, beginning with the second emanant, are (in regard to the facients of the given quantic assumed to be of the order  $m$ ) of the orders  $m - 2, m - 4, \dots$  down to 1 or 0, according as  $m$  is odd or even, or writing successively  $2p + 1$  and  $2p$  in the place of  $m$ , and taking the emanants in a reverse order, the emanants for a quantic of any odd order  $2p + 1$  are of the orders 1, 3, 5  $\dots$   $2p - 1$ , and for a quantic of any even order  $2p$ , they are of the orders 0, 2, 4  $\dots$   $2p - 2$ .

The quadricovariants of a quantic of an odd order  $2p + 1$ , are consequently of the orders 2, 6, 10  $\dots$   $4p - 2$ , and the quadricovariants of a quantic of an even order  $2p$ , are of the orders 0, 4, 8  $\dots$   $4p - 4$ .

We might in each case carry the series one step further, and consider a quadricovariant of the order  $4p + 2$ , or (as the case may be)  $4p$ , which arises from the 0th emanant of the given quantic; such quadricovariant is, however, only the square of the given quantic.

**78.** In the case of a quantic of an evenly even order (but in no other case) we have a quadricovariant of the same order with the quantic itself. We may in this case form the lineo-linear invariant of the quantic and the quadricovariant of the same order: such lineo-linear invariant is an invariant of the given quantic, and it is of the degree 3 in the coefficients, that is, it is a cubinvariant. This agrees with the before-mentioned theorem for the number of cubinvariants.

**79.** In the case of the quartic  $(a, b, c, d, e \mid x, y)^4$ , the cubinvariant is, by the preceding mode of generation, obtained in the form

$$e(ac - b^2) - 4d(ad - bc) + 6c^2(ae - 4bd + 3c^2) - 4b^2(be - cd) + a(ce - d^2),$$

which is in fact equal to

$$3(ace - ad^2 - b^2e + 2bcd - c^3);$$

and omitting the numerical factor 3, we have the cubinvariant of the quartic.

**80.** In the case of a quantic of any order even or odd, the quadrinvariants of the quadricovariants are quartinvariants of the quantic. But these quartinvariants are not all of them independent, and there is no obvious method grounded on the preceding mode of generation for obtaining the number of the independent (asyzygetic) quartinvariants, and thence the number of the irreducible quartinvariants of a quantic of a given order.

**81.** I take the opportunity of giving some additional developments in relation to the discriminant of a quantic  $(a, b, \dots b', a' | x, y)^m$ . To render the signification perfectly definite, it should be remarked that the discriminant contains the term  $a^{m-1}a'^{m-1}$ , and that the coefficient of this term may be taken to be

**82.** (End of the memoir.)

[155]

## A FOURTH MEMOIR UPON QUANTICS.

## 0.0.81

It was noticed in the Introductory Memoir, that, by Joachimsthal's theorem, the discriminant, on putting  $a = 0$ , becomes divisible by  $b^2$ , and that throwing out this factor it is to a numerical factor *près* the discriminant of the quantic of the order  $(m - 1)$  obtained by putting  $a = 0$  and throwing out the factor  $x$ ; and it was also remarked, that

this theorem, combined with the general property of invariants, afforded a convenient method for the calculation of the discriminant of a quantic when that of the order immediately preceding is known.

Thus let it be proposed to find the discriminant of the cubic

$$(a, b, c, d \setminus x, y)^3.$$

Imagine the discriminant expanded in powers of the leading coefficient  $a$  in the form

$$Aa^2 + Ba + C,$$

then this function *quà* invariant must be reduced to zero by the operation  $3b\partial_a + 2c\partial_b + d\partial_c$ ; or putting for shortness

$\nabla = 2c\partial_b + d\partial_c$ , the operation is  $\nabla + 3b\partial_a$ , and we have

$$\left. \begin{aligned} a^2\nabla A + a\nabla B + \nabla C \\ + a6bA + 3bB \end{aligned} \right\} = 0,$$

and consequently

$$B = -\frac{1}{3b}\nabla C, \quad A = -\frac{1}{6b}\nabla B, \quad \nabla A = 0.$$

But  $C$  is equal to  $b^2$  into the discriminant of  $(3b, 3c, d \setminus x, y)^2$ , that is, its value is  $b^2(12bd - 9c^2)$ , or throwing out the factor 3, we may write

$$C = 4b^2d - 3b^2c^2;$$

this gives

$$B = -\frac{1}{3b}(-6b^2cd + 24b^2cd - 12bc^3),$$

or reducing

$$B = -6bcd + 4c^3;$$

and thence

$$A = -\frac{1}{6b}(-6bd^2 + 12c^2d - 12c^2d),$$

or reducing

$$A = d^2,$$

which verifies the equation  $\nabla A = 0$ , and the discriminant is, as we know,

$$a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2.$$

## 0.0.82

If we consider the quantic  $(a, b, \dots \setminus x, 1)^m$  as expressed in terms of the roots in the form  $a(x - \alpha y)(x - \beta y) \dots$ , then the discriminant ( $= a^{2m-2}\alpha^{m-1} + \&c.$  as above) is to a factor *près* equal to the product of the squares of the differences of the roots, and the factor may be determined as follows: viz. denoting by  $\zeta(\alpha, \beta, \dots)$  the product of the squares of the differences of the roots, we may write

$$a^{2m-2}\zeta(\alpha, \beta, \dots) = N(a^{m-1}\alpha^{m-1} + \&c.),$$

where  $N$  is a number; and then considering the equation  $a^{m-1} = 0$ , we have to determine  $N$  the equation

$$\zeta(\alpha, \beta, \dots) = (-)^{m-1}N.$$

But in general

$$\zeta(\alpha, \beta, \dots) = (-)^{\frac{1}{2}m(m-1)}(\alpha - \beta)(\alpha - \gamma) \dots (\beta - \alpha)(\beta - \gamma) \dots,$$

and if

$$\phi x = (x - \alpha)(x - \beta) \dots,$$

then

$$(\alpha - \beta)(\alpha - \gamma) \dots = \phi' \alpha, \quad \&c.,$$



or

$$\zeta(\alpha, \beta, \dots) = (-)^{\frac{1}{2}m(m-1)} \phi' \alpha \phi' \beta \dots;$$

here

$$\phi' \alpha \phi' \beta \dots = m^2 a^{m-2} (\alpha \beta \gamma \dots)^{m-1},$$

but

$$(-)^m a \alpha \beta \gamma \dots = -1,$$

or

$$\alpha \beta \gamma \dots = (-)^{m-1} \frac{1}{a},$$

and

$$\phi' \alpha \phi' \beta \dots = (-)^{m(m-1)} \frac{1}{a^{m-2}} m^m = (-)^{m-1} \frac{m^m}{a^{m-2}};$$

whence

$$\zeta(\alpha, \beta, \dots) = (-)^{\frac{1}{2}m(m-1)+m(m-1)} \frac{m^m}{a^{m-2}} = (-)^{\frac{1}{2}m(m-1)} \frac{m^m}{a^{m-2}},$$

or

$$a^{m-2} \zeta(\alpha, \beta, \dots) = (-)^{\frac{1}{2}m(m-1)} m^m,$$

and consequently

$$a^{2m-2} \zeta(\alpha, \beta, \dots) = (-)^{\frac{1}{2}m(m-1)} m^m a^{m-1};$$

so that the discriminant of the quantic is equal to the product of the squares of the differences of the roots multiplied by  $(-)^{\frac{1}{2}m(m-1)} m^m a^{m-1}$ .

It would have been allowable to define the discriminant so as that the leading term should be  $(-)^{\frac{1}{2}m(m-1)} m^m a^{m-1}$ , in which case the discriminant would have constantly the same sign as the product of the squared differences; but I have upon the whole thought it better to make the leading term of the discriminant always positive.

### 0.0.83

A quantic of an even order  $2p$  has an invariant of peculiar simplicity, viz. the determinant the terms of which are the coefficients of the  $p$ th differential coefficients, or derived functions of the quantic with respect to the facients; such invariant may also be considered as a multipartite invariant of the  $p$ th emanants. Thus the sextic

$$(a, b, c, d, e, f, g \setminus x, y)^6$$

has for one of its invariants, the determinant

$$\begin{vmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \\ d & e & f & g \end{vmatrix}.$$

The invariant in question is termed by Professor Sylvester the **Catalecticant**.

### 0.0.84

Professor Sylvester also remarked, that we may from the catalecticant form a function containing an indeterminate quantity  $\lambda$ , such that the coefficients of the different powers of  $\lambda$  are invariants of the quantic; thus for the sextic, the function in question is

$$\begin{vmatrix} a & b & c & d - \lambda \\ b & c & d + \lambda & e \\ c & d - \lambda & e & f \\ d + \lambda & e & f & g \end{vmatrix},$$

where the law of formation is manifest; the terms in the sinister diagonal are modified by annexing to their numerical submultiples of  $\lambda$  with the signs  $+$  and  $-$  alternately, and in which the multipliers are the reciprocals of the binomial coefficients. The function so obtained is termed the **Lambdaic**.

### 0.0.85

If we consider a quantic of an odd order, and form the catalecticant of the penultimate emanant, we have the covariant termed the

**Canonisant.** Thus in the case of the quintic

$$(a, b, c, d, e, f \setminus x, y)^5,$$

the canonisant is

$$\begin{vmatrix} ax + by & bx + cy & cx + dy \\ bx + cy & cx + dy & dx + ey \\ cx + dy & dx + ey & ex + fy \end{vmatrix},$$

which is equivalent to

$$\begin{vmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \end{vmatrix}$$

and a like transformation exists with respect to the canonisant of a quantic of any odd order whatever. The canonisant and the lambdaic (which includes of course the catalecticant) form the basis of Professor Sylvester's theory of the Canonical Forms of quantics of an odd and an even order respectively.

### 0.0.86

There is another family of covariants which remains to be noticed.

Consider any two quantics of the same order,

$$(a, b, \dots \setminus x, y)^m, \quad (a', b', \dots \setminus x, y)^m,$$

and join to these a quantic of the next inferior order,

$$(u, v, \dots \setminus \eta, -\xi)^{m-1},$$

where the coefficients  $(u, v, \dots)$  are considered as indeterminate, and which may be spoken of as the *adjoint quantic*.

Take the odd lineo-linear covariants (viz. those which arise from the odd emanants) of the two quantics; the term arising from the  $(2i + 1)$ th emanants is of the form

$$(A, B, \dots \setminus x, y)^{m-1-2i}(\xi, \eta)^{2i},$$

where  $(A, B, \dots)$  are lineo-linear functions of the coefficients of the two quantics.

Take also the quadricovariants of the adjoint quantic; the term arising from the  $(2i - 1)$ th emanant is of the form

$$(U, V, \dots \setminus \eta, -\xi)^{m-1-2i}(\xi, \eta)^{2i},$$

where  $(U, V, \dots)$  are quadric functions of the indeterminate coefficients  $(u, v, \dots)$ . We may then form the quadrinvariant of the two quantics of the order  $\frac{1}{2}(m - 1 - 2i)$ : this will be an invariant of the two quantics and the adjoint quantic, lineo-linear in the coefficients of the two quantics and of the degree 2 in regard to the coefficients  $(u, v, \dots)$  of the adjoint quantic; or treating the last-mentioned coefficients as facients, the result is a lineo-linear  $m$ -ary quadric of the form

$$(\mathfrak{A}, \mathfrak{B}, \dots \setminus u, v, \dots)^2,$$

viz. in this expression the coefficients  $\mathfrak{A}, \mathfrak{B}, \dots$  are lineo-linear functions of the coefficients of the two quantics. And giving to  $i$  the different admissible values, viz. from  $i = 0$  to  $i = \frac{1}{2}(m - 1)$  or  $\frac{1}{2}(m - 1) - 1$ , according as  $m$  is even or odd, the number of the functions obtained by the preceding process is  $\frac{1}{2}m$  or  $\frac{1}{2}(m - 1)$ , according as  $m$  is even or odd. The functions in question, the theory of which is altogether due to Professor Sylvester, are termed by him

**Cobezoutiants**; we may therefore say that a cobezoutiant is an invariant of two quantics of the same order  $m$ , and of an adjoint quantic of the next preceding order  $m - 1$ , viz. treating the coefficients of the adjoint quantic as the facients of the cobezoutiant, the cobezoutiant is an  $m$ -ary quadric, the coefficients of which are lineo-linear functions of the coefficients of the two quantics, and the number of the cobezoutiants is  $\frac{1}{2}m$  or  $\frac{1}{2}(m - 1)$ , according as  $m$  is even or odd.

### 0.0.87

If the two quantics are the differential coefficients or first derived functions (with respect to the facients) of a single quantic

$$(a, b, \dots \setminus x, y)^m,$$

then we have what are termed the **Cobezoutoids** of the single quantic, viz. the cobezoutoid is an invariant of the single quantic of the order  $m$ , and of an adjoint quantic of the order  $(m - 2)$ ; and

treating the coefficients of the adjoint quantic as facients, the cobezoutoid is an  $(m-1)$ -ary quadric, the coefficients of which are quadric functions of the coefficients of the given quantic. The number of the cobezoutoids is  $\frac{1}{2}(m-1)$  or  $\frac{1}{2}(m-2)$ , according as  $m$  is odd or even.

### 0.0.88

Consider any two quantics of the same order,

$$(a, \dots \setminus x, y)^m, \quad (a', \dots \setminus x, y)^m,$$

and introducing the new facients  $(X, Y)$ , form the quotient of determinants,

$$\frac{(a, \dots \setminus x, y)^m, \quad (a, \dots \setminus x', y')^m}{(x, y)^m, \quad (x', y')^m, \quad X, Y},$$

which is obviously an integral function of the order  $(m-1)$  in each set of facients separately, and lineo-linear in the coefficients of the two quantics; for instance, if the two quantics are

$$(a, b, c, d \setminus x, y)^3, \quad (a', b', c', d' \setminus x, y)^3,$$

the quotient in question may be written

$$\frac{\frac{1}{3}(ab' - a'b), \quad \frac{1}{3}(ac' - a'c), \quad ad' - a'd}{\frac{1}{3}(ac' - a'c), \quad ad' - a'd + \frac{1}{3}(bc' - b'c), \quad \frac{1}{3}(bd' - b'd) \setminus (x, y)^2(X, Y)^1.}$$

$$ad' - a'd, \quad \frac{1}{3}(bd' - b'd), \quad \frac{1}{3}(cd' - c'd)$$

The function so obtained may be termed the **Bezoutic Emanant** of the two quantics.

### 0.0.89

The notion of such function was in fact suggested to me by Bezout's abbreviated process of elimination, viz. the two quantics of the order  $m$  being put equal to zero, the process leads to  $(m-1)$  equations each of the order  $(m-1)$ : these equations are nothing else than the equations obtained by equating to zero the coefficients of the different terms of the series  $(X, Y)^{m-1}$  in the Bezoutic emanant, and the result of the elimination is consequently obtained by equating to zero the determinant formed with the matrix which

enters into the expression of the Bezoutic emanant. In other words, this determinant is the **Resultant** of the two quantics. Thus the resultant of the last-mentioned two cubics is the determinant

$$\begin{vmatrix} \frac{1}{3}(ab' - a'b) & \frac{1}{3}(ac' - a'c) & ad' - a'd \\ \frac{1}{3}(ac' - a'c) & ad' - a'd + \frac{1}{3}(bc' - b'c) & \frac{1}{3}(bd' - b'd) \\ ad' - a'd & \frac{1}{3}(bd' - b'd) & \frac{1}{3}(cd' - c'd) \end{vmatrix}.$$

### 0.0.90

If the two quantics are the differential coefficients or first derived functions (with respect to the facients) of a single quantic of the order  $m$ , then we have in like manner the **Bezoutoidal Emanant** of the single quantic; this is a function of the order  $(m - 2)$  in each set of facients, and the coefficients whereof are quadric functions of the coefficients of the single quantic. Thus the Bezoutoidal emanant of the quartic

$$(a, b, c, d, e \setminus x, y)^4$$

is

$$\begin{vmatrix} \frac{1}{3}(ac - b^2), & \frac{1}{3}(ad - bc), & ae - bd \\ \frac{1}{3}(ad - bc), & ae + \frac{1}{3}bd - \frac{9}{3}c^2, & \frac{1}{3}(be - cd) \\ ae - bd, & \frac{1}{3}(be - cd), & \frac{1}{3}(ce - d^2) \end{vmatrix} \setminus (x, y)^2(X, Y)^1;$$

and of course the determinant formed with the matrix which enters into the expression of the Bezoutoidal Emanant, is the discriminant of the single quantic.

### 0.0.91

Professor Sylvester forms with the matrix of the Bezoutic emanant and a set of  $m$  facients  $(u, v, \dots)$  an  $m$ -ary quadric function, which he terms the **Bezoutiant**. Thus the Bezoutiant of the before-mentioned two cubics is

$$\begin{pmatrix} \frac{1}{3}(ab' - a'b) & \frac{1}{3}(ac' - a'c) & ad' - a'd \\ \frac{1}{3}(ac' - a'c) & ad' - a'd + \frac{1}{3}(bc' - b'c) & \frac{1}{3}(bd' - b'd) \\ ad' - a'd & \frac{1}{3}(bd' - b'd) & \frac{1}{3}(cd' - c'd) \end{pmatrix} \setminus (u, v, w)^2;$$

and in like manner with the Bezoutoidal emanant of the single quantic of the order  $m$  and a set of  $(m - 1)$  new facients  $(u, v, \dots)$ ,

an  $(m-1)$ -ary quadric function, which he terms the **Bezoutoid**.

Thus the Bezoutoid of the before-mentioned quartic is

$$\begin{pmatrix} \frac{1}{3}(ac-b^2) & \frac{1}{3}(ad-bc) & ae-bd \\ \frac{1}{3}(ad-bc) & ae+\frac{8}{3}bd-3c^2 & \frac{1}{3}(be-cd) \\ ae-bd & \frac{1}{3}(be-cd) & \frac{1}{3}(ce-d^2) \end{pmatrix} \setminus (u, v, w)^2.$$

To him also is due the important theorem, that the Bezoutiant is an invariant of the two quantics of the order  $m$  and of the adjoint quantic  $(u, v, \dots \setminus \eta, -\xi)^{m-1}$ , being in fact a linear function with mere numerical coefficients of the invariants called Cobezoutiants, and in like manner that the Bezoutoid is an invariant of the single quantic of the order  $m$  and of the adjoint quantic  $(u, v, \dots \setminus \eta, -\xi)^{m-2}$ , being a linear function with mere numerical coefficients of the invariants called Cobezoutoids.

The modes of generation of a covariant are infinite in number, and it is to be anticipated that, as new theories arise, there will be frequent occasion to consider new processes of derivation, and to single out and to define and give names to new covariants. But I have now, I think, established the greater part by far of the definitions which are for the present necessary.

[156]

## A FIFTH MEMOIR UPON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*,  
vol. cxLviii. for the year 1858, pp. 429—460. Received February  
11,—Read March 18, 1858.]

The present memoir was originally intended to contain a development of the theories of the covariants of certain binary quantics, viz. the quadric, the cubic, and the quartic; but as regards the theories of the cubic and the quartic, it was found necessary to consider the case of two or more quadrics, and I have therefore comprised such systems of two or more quadrics, and the resulting theories of the harmonic relation and of involution, in the subject of the memoir; and although the theory of homography or of the anharmonic relation belongs rather to the subject of bipartite binary quadrics, yet from its connexion with the theories just referred to, it is also considered in the memoir. The paragraphs are numbered continuously with those of my former memoirs on the subject: Nos. 92 to 95 relate to a single quadric: Nos. 96 to 114 to two or more quadrics, and the theories above referred to; Nos. 115 to 127 to the cubic, and Nos. 128 to 145 to the quartic. The several quantics are considered as expressed not only in terms of the coefficients, but also in terms of the roots,—and I consider the question of the determination of their linear factors, a question, in effect, identical with that of the solution of a quadric, cubic, or biquadratic equation. The expression for the linear factor of a quadric is deduced from a well-known formula; those for the linear factors of a cubic and a quartic were first given in my “Note sur les Covariants d’une fonction quadratique, cubique ou biquadratique à deux indéterminées,” Crelle, vol. L. (1855), pp. 285—287, [135]. It is remarkable that they are in one point of view more simple than the expression for the linear factor of a quadric.

## 0.0.92

In the case of a quadric the expressions considered are

$$(a, b, c \setminus x, y)^2, \quad (1)$$

$$ac - b^2, \quad (2)$$



where (1) is the quadric, and (2) is the discriminant, which is also the quadrinvariant, catalecticant, and Hessian.

And where it is convenient to do so, I write

$$\begin{aligned}(1) &= U, \\ (2) &= D.\end{aligned}$$

### 0.0.93

We have

$$(a, b, c \setminus x, y)^2 = D U,$$

which expresses that the evectant of the discriminant is equal to the quadric; and

$$(a, b, c \setminus \partial_x, \partial_y)^2 \cdot U = D,$$

which expresses that the provectant of the quadric is equal to the discriminant; and

$$(a, b, c \setminus bx + cy, -ax - by)^2 = D U,$$

which expresses that a transmutant of the quadric is equal to the product of the quadric and the discriminant.

### 0.0.94

When the quadric is expressed in terms of the roots, we have

$$\begin{aligned}a^{-1}U &= (x - \alpha y)(x - \beta y), \\ a^{-2}D &= -\frac{1}{4}(\alpha - \beta)^2;\end{aligned}$$

and in the case of a pair of equal roots,

$$\begin{aligned}a^{-1}U &= (x - \alpha y)^2, \\ D &= 0.\end{aligned}$$

**0.0.95**

The problem of the solution of a quadratic equation is that of finding a linear factor of the quadric. To obtain such linear factor in a symmetrical form, it is necessary to introduce arbitrary quantities which do not really enter into the solution, and the form obtained is thus in some sort more complicated than in the like problem for a cubic or a quartic. The solution depends on the linear transformation of the quadric, viz. if we write

$$(a, b, c \setminus \lambda x + \mu y, \nu x + \rho y)^2 = (a', b', c' \setminus x, y)^2,$$

so that

$$\begin{aligned} a' &= (a, b, c \setminus \lambda, \nu)^2, \\ b' &= (a, b, c \setminus \lambda, \nu \setminus \mu, \rho), \\ c' &= (a, b, c \setminus \mu, \rho)^2, \end{aligned}$$

then

$$a'c' - b'^2 = (ac - b^2)(\lambda\rho - \mu\nu)^2,$$

an equation which in a different notation is

$$(a, b, c \setminus x, y)^2 \cdot (a, b, c \setminus X, Y)^2 - \{(a, b, c \setminus x, y \setminus X, Y)\}^2 = D(Yx - Xy)^2,$$

in which form it is a theorem relating to the quadric and its first and second emanants. The equation shows that

$$(a, b, c \setminus x, y \setminus X, Y) + \sqrt{D}(Yx - Xy),$$

where  $(X, Y)$  are treated as supernumerary arbitrary constants, is a linear factor of  $(a, b, c \setminus x, y)^2$ , and this is the required solution.

**0.0.96**

In the case of two quadrics, the expressions considered are

$$(a, b, c \setminus x, y)^2, \quad (1)$$

$$(a', b', c' \setminus x, y)^2, \quad (2)$$

$$ac - b^2, \quad (3)$$

$$ac' - 2bb' + ca', \quad (4)$$

$$a'c' - b'^2, \quad (5)$$

$$\begin{vmatrix} a & 2b & c \\ a' & 2b' & c' \end{vmatrix}, \quad (6)$$

$$(ab' - a'b, ac' - a'c, bc' - b'c \setminus x, y)^1, \quad (7)$$

$$(\lambda a + \mu a', \lambda b + \mu b', \lambda c + \mu c' \setminus x, y)^2, \quad (8)$$

$$(ac - b^2, ac' - 2bb' + ca', a'c' - b'^2 \setminus \lambda, \mu)^2, \quad (9)$$

(1) and (2) are the quadrics, (3) and (5) are the discriminants, and (4) is the lineo-linear invariant, or connective of the discriminants; (6) is the resultant of the two quadrics, (7) is the Jacobian, (8) an intermediate, and (9) is the discriminant of the intermediate. And where it is convenient to do so, I write

$$\begin{aligned} (1) &= U, & (2) &= U', \\ (3) &= D, & (4) &= Q, & (5) &= D', \\ (6) &= R, & (7) &= J, & (8) &= F, & (9) &= \Delta. \end{aligned}$$

**0.0.97**

The Jacobian (7) may also be written in the form

$$\begin{vmatrix} y^2 & -yx & x^2 \\ a & b & c \\ a' & b' & c' \end{vmatrix}.$$

The Resultant (6) may be written in the form

$$\begin{vmatrix} a & 2b & c \\ a' & 2b' & c' \end{vmatrix},$$

and also, taken negatively, in the form

$$4(ab' - a'b)(bc' - b'c) - (ac' - a'c)^2,$$

which is the discriminant of the Jacobian; and in the form

$$4(ac - b^2)(a'c' - b'^2) - (ac' - 2bb' + ca')^2,$$

which is the discriminant of the Intermediate.

### 0.0.98

We have the following relations

$$\begin{aligned} (a, b, c \setminus b'x + c'y, -a'x - b'y)^2 &= -(a'c' - b'^2)(a, b, c \setminus x, y)^2 \\ &\quad + (ac' - 2bb' + ca')(a', b', c' \setminus x, y)^2, \\ (a', b', c' \setminus bx + cy, -ax - by)^2 &= +(ac' - 2bb' + ca')(a, b, c \setminus x, y)^2 \\ &\quad - (ac - b^2)(a', b', c' \setminus x, y)^2, \end{aligned}$$

and moreover

$$(ac - b^2, ac' - 2bb' + ca', a'c' - b'^2 \setminus bx + cy, -ax - by)^2 = -\{(ab' - a'b, ac' - a'c, bc' - b'c \setminus x, y)^2\}$$

an equation, the interpretation of which will be considered in the sequel.

### 0.0.99

The most important relations which may exist between the two quadrics are:

First, when the connective vanishes, or

$$ac' - 2bb' + ca' = 0,$$

in which case the two quadrics are said to be *harmonically related*:

the nature of this relation will be further considered.

Secondly, when  $R = 0$ , the two quadrics have in this case a common root, which is given by any of the equations,

$$\begin{aligned} x^2 : 2xy : y^2 &= \partial_a R : \partial_b R : \partial_c R \\ &= \partial_{a'} R : \partial_{b'} R : \partial_{c'} R \\ &= bc' - b'c : ca' - c'a : ab' - a'b. \end{aligned}$$

The last set of values express that the Jacobian is a perfect square, and that the two roots are each equal to the common root of the two quadrics.

The preceding values of the ratios  $x^2 : 2xy : y^2$  are consistent with each other in virtue of the assumed relation  $R = 0$ , hence in general the functions

$$\partial_a R \cdot \partial_c R - (\partial_b R)^2, \quad \partial_a R \cdot \partial_b R - \partial_c R \cdot \partial_a R, \quad \&c.,$$

all of them contain the Resultant  $R$  as a factor.

It is easy to see that the Jacobian is harmonically related to each of the quadrics; in fact we have identically

$$\begin{aligned} a(bc' - b'c) + b(ca' - c'a) + c(ab' - a'b) &= 0, \\ a'(bc' - b'c) + b'(ca' - c'a) + c'(ab' - a'b) &= 0, \end{aligned}$$

which contain the theorem in question.

### 0.0.100

When the quadrics are expressed in terms of the roots, we have

$$\begin{aligned} a^{-1}U &= (x - \alpha y)(x - \beta y), \\ a'^{-1}U' &= (x - \alpha' y)(x - \beta' y), \\ 4a^{-2}D &= -(\alpha - \beta)^2, \\ 2(aa')^{-1}Q &= 2\alpha\beta + 2\alpha'\beta' - (\alpha + \beta)(\alpha' + \beta'), \\ 4a'^{-2}D' &= -(\alpha' - \beta')^2, \\ (aa')^{-1}R &= (\alpha - \alpha')(\alpha - \beta')(\beta - \alpha')(\beta - \beta'), \\ (aa')^{-1}J &= \begin{vmatrix} \xi & 2\eta & \zeta \\ 1 & \alpha + \beta & \alpha\beta \\ 1 & \alpha' + \beta' & \alpha'\beta' \end{vmatrix}. \end{aligned}$$

### 0.0.101

The comparison of the last-mentioned value of  $R$  with the expression in terms of the roots obtained from the equation

$$-\frac{1}{4}\Delta = 4DD' - Q^2,$$

gives the identical equation

$$(\alpha - \beta)^2(\alpha' - \beta')^2 - \{2\alpha\beta + 2\alpha'\beta' - (\alpha + \beta)(\alpha' + \beta')\}^2 = -4(\alpha - \alpha')(\alpha - \beta')(\beta - \alpha')(\beta - \beta'),$$

which may be easily verified.

**0.0.102**

We have identically

$$\begin{aligned}
 & 2\alpha\beta + 2\alpha'\beta' - (\alpha + \beta)(\alpha' + \beta') \\
 = & 2(\alpha - \alpha')(\alpha - \beta') - (\alpha - \beta)(-2\beta - \alpha' - \beta') \\
 = & 2(\beta - \alpha')(\beta - \beta') - (\beta - \alpha)(2\beta - \alpha' - \beta') \\
 = & 2(\alpha' - \alpha)(\alpha' - \beta) - (\alpha' - \beta')(2\alpha' - \alpha - \beta) \\
 = & 2(\beta' - \alpha)(\beta' - \beta) - (\beta' - \alpha')(2\beta' - \alpha - \beta);
 \end{aligned}$$

and the equation  $Q = ac' - 2bb' + ca' = 0$  may consequently be written in the several forms

$$\begin{aligned}
 \frac{2}{\alpha - \alpha'} &= \frac{1}{\alpha - \beta} + \frac{1}{\alpha - \beta'}, \\
 \frac{2}{\beta - \alpha'} &= \frac{1}{\beta - \alpha} + \frac{1}{\beta - \beta'}, \\
 \frac{2}{\alpha' - \alpha} &= \frac{1}{\alpha' - \beta} + \frac{1}{\alpha' - \beta'}, \\
 \frac{2}{\beta' - \alpha} &= \frac{1}{\beta' - \alpha'} + \frac{1}{\beta' - \beta},
 \end{aligned}$$

so that the roots  $(\alpha, \beta)$ ,  $(\alpha', \beta')$  are *harmonically related* to each other, and hence the notion of the harmonic relation of the two quadrics.

**0.0.103**

In the case where the two quadrics have a common root  $\alpha = \alpha'$ ,

$$\begin{aligned}
 a^{-1}U &= (x - \alpha y)(x - \beta y), \\
 a'^{-1}U' &= (x - \alpha y)(x - \beta' y), \\
 4a^{-2}D &= -(\alpha - \beta)^2, \\
 2(aa')^{-1}Q &= (\alpha - \beta)(\alpha - \beta'), \\
 4a'^{-2}D' &= -(\alpha - \beta')^2, \\
 R &= 0, \\
 (aa')^{-1}J &= (\beta - \beta')(x - \alpha y)^2.
 \end{aligned}$$

**0.0.104**

In the case of three quadrics, of the expressions which are or might be considered, it will be sufficient to mention

$$(a, b, c \setminus x, y)^2, \quad (1)$$

$$(a', b', c' \setminus x, y)^2, \quad (2)$$

$$(a'', b'', c'' \setminus x, y)^2, \quad (3)$$

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix}, \quad (4)$$

where (1), (2), (3) are the quadrics themselves, and (4) is an invariant, linear in the coefficients of each quadric. And where it is convenient to do so, I write

$$(1) = U, \quad (2) = U', \quad (3) = U'',$$

$$(4) = \Theta.$$

**0.0.105**

The invariant  $\Theta$  may be expressed in terms of the roots in the form

$$\Theta = (aa'a'')^{-1}(\alpha - \alpha')(\alpha - \alpha'')(\alpha' - \alpha'')(\beta - \beta')(\beta - \beta'')(\beta' - \beta''),$$

or more symmetrically

$$\Theta = (aa'a'')^{-1} \begin{vmatrix} 1 & \alpha + \beta & \alpha\beta \\ 1 & \alpha' + \beta' & \alpha'\beta' \\ 1 & \alpha'' + \beta'' & \alpha''\beta'' \end{vmatrix}.$$

**0.0.106**

When  $\Theta = 0$ , the three quadrics have a common root. This is the condition that the three quadrics may be simultaneously divisible by a linear factor.

**0.0.107**

The theory of involution is closely connected with the theory of two or more quadrics. Two pairs of roots  $(\alpha, \beta)$  and  $(\alpha', \beta')$  are said to be in involution when the connective  $Q = 0$ . This is equivalent to the condition that the Jacobian of the two quadrics is a perfect square.

**0.0.108**

When three pairs of roots are in involution, any two of the corresponding quadrics are harmonically related. The condition for this is that the invariant  $\Theta = 0$ .

**0.0.109**

The theory of homography, or the anharmonic relation, belongs to the subject of bipartite binary quadrics. Two pairs of roots  $(\alpha, \beta)$  and  $(\alpha', \beta')$  are said to be homographically related when the anharmonic ratio of the four roots is the same for both pairs.

**0.0.110**

The anharmonic ratio of four roots  $\alpha, \beta, \gamma, \delta$  is defined as

$$\frac{(\alpha - \gamma)(\beta - \delta)}{(\alpha - \delta)(\beta - \gamma)}.$$

When two pairs of roots have the same anharmonic ratio, they are said to be homographically related.

**0.0.111**

The condition that two pairs of roots  $(\alpha, \beta)$  and  $(\alpha', \beta')$  may be homographically related may be expressed in terms of the invariants of the corresponding quadrics. The condition is that the resultant of the Jacobian and the connective should vanish.

**0.0.112**

In the case of a cubic, the expressions considered are

$$(a, b, c, d \setminus x, y)^3, \quad (1)$$

$$(ac - b^2, ad - bc, bd - c^2 \setminus x, y)^2, \quad (2)$$

$$\left\{ \begin{array}{l} -a^2d + 3abc - 2b^3, \\ -abd + 2ac^2 - b^2c, \\ +acd - 2b^2d + bc^2, \\ +ad^2 - 3bcd + 2c^3 \end{array} \right\} (x, y)^3, \quad (3)$$

$$a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2, \quad (4)$$



where (1) is the cubic, (2) is the quadricovariant or Hessian, (3) is the cubicovariant, and (4) is the quartinvariant or discriminant.

And where it is convenient to do so, I write

$$\begin{aligned}(1) &= U, \\ (2) &= H, \\ (3) &= \Phi, \\ (4) &= \square.\end{aligned}$$

so that we have

$$\Phi^2 - \square U^2 + 4H^3 = 0.$$

### 0.0.113

The Hessian may be written under the form

$$(ax + by)(cx + dy) - (bx + cy)^2,$$

(which, indeed, is the form under which *quâ* Hessian it is originally given), and under the form

$$\begin{vmatrix} y^2 & -yx & x^2 \\ a & b & c \\ b & c & d \end{vmatrix}.$$

The cubicovariant may be written under the form

$$\begin{aligned}&\{2(ac - b^2)x + (ad - bc)y\}(bx^2 + 2cxy + dy^2) \\ &- \{(ad - bc)x + 2(bd - c^2)y\}(ax^2 + 2bxy + cy^2),\end{aligned}$$

that is, as the Jacobian of the cubic and Hessian; and under the form

$$\frac{1}{2}(\partial_a, \partial_b, \partial_c, \partial_d \setminus y, -x)^3 \square,$$

that is, as the evectant of the discriminant.

The discriminant, taken negatively, may be written under the form

$$+4(ac - b^2)(bd - c^2) - (ad - bc)^2,$$

that is, as the discriminant of the Hessian.

**0.0.114**

We have

$$(a, b, c, d \setminus bx^2 + 2cxy + dy^2, -ax^2 - 2bxy - cy^2)^3 = U\Phi,$$

which expresses that a transmutant of the cubic is the product of the cubic and the cubicovariant. The equation

$$\{(\partial_a, \partial_b, \partial_c, \partial_d \setminus y, -x)^3\}^2 \square = 2U^2$$

expresses that the second evectant of the discriminant is the square of the cubic.

The equation

$$\begin{vmatrix} d^2 & -3cd & -3bd + 6c^2 & -3bc + 2ad \\ -3cd & -3c^2 + 12bd & -3ad - 6bc & -3ac + 6b^2 \\ -3bd + 6c^2 & -3ad - 6bc & -3b^2 + 12ac & -3ab \\ -3bc - 12ad & -3ac + 6b^2 & 3ab & a^2 \end{vmatrix} = 27\square^2$$

expresses that the determinant formed with the second differential coefficients of the discriminant gives the square of the discriminant.

The covariants of the intermediate  $\alpha U + \beta \Phi$  are as follows, viz.

**0.0.115**

For the Hessian, we have

$$\begin{aligned} \tilde{H}(\alpha U + \beta \Phi) &= (1, 0, -\square \setminus \alpha, \beta)^2 H \\ &= (\alpha^2 - \beta^2 \square) H; \end{aligned}$$

for the cubicovariant,

$$\begin{aligned} \tilde{\Phi}(\alpha U + \beta \Phi) &= (0, \square, 0, -\square^2 \setminus \alpha, \beta)^3 U \\ &\quad + (1, 0, -\square, 0 \setminus \alpha, \beta)^3 \Phi \\ &= (\alpha^2 - \beta^2 \square)(\alpha \Phi + \beta \square U); \end{aligned}$$

and for the discriminant,

$$\begin{aligned} \tilde{\square}(\alpha U + \beta \Phi) &= (1, 0, -2\square, 0, \square^2 \setminus \alpha, \beta)^4 \square \\ &= (\alpha^2 - \beta^2 \square)^2 \square, \end{aligned}$$

where on the left-hand sides I have, for greater distinctness, written  $\tilde{H}$ , &c. to denote the functional operation of taking the Hessian, &c. of the operand  $\alpha U + \beta \Phi$ .

**0.0.116**

When the cubic is expressed in terms of the roots, we have

$$\begin{aligned} a^{-1}U &= (x - \alpha y)(x - \beta y)(x - \gamma y), \\ a^{-2}H &= -\frac{1}{4}\{(\alpha - \beta)^2(x - \gamma y)^2 + (\beta - \gamma)^2(x - \alpha y)^2 + (\gamma - \alpha)^2(x - \beta y)^2\}, \\ a^{-3}\Phi &= -(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)(x - \alpha y)(x - \beta y)(x - \gamma y), \\ a^{-4}\square &= -(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2. \end{aligned}$$

**0.0.117**

In the case of a pair of equal roots, we have

$$\begin{aligned} a^{-1}U &= (x - \alpha y)^2(x - \gamma y), \\ a^{-2}H &= -\frac{1}{4}(\alpha - \gamma)^2(x - \alpha y)^2, \\ a^{-3}\Phi &= 0, \\ \square &= 0. \end{aligned}$$

And in the case of all the roots equal, we have

$$\begin{aligned} U &= (x - \alpha y)^3, \\ H &= 0, \\ \Phi &= 0, \\ \square &= 0. \end{aligned}$$

**0.0.118**

In the solution of a biquadratic equation, the problem is that of finding a linear factor of the quartic. The linear factor may be found by means of the cubic resolvent, which is obtained by equating to zero the cubicovariant of the quartic.

**0.0.119**

The linear factors of the quartic  $(a, b, c, d, e \setminus x, y)^4$  are given by the formula

$$(a, b, c, d, e \setminus x, y)^4 = a(x - \alpha y)(x - \beta y)(x - \gamma y)(x - \delta y),$$

and the expression for a linear factor may be obtained by means of the invariants  $I$  and  $J$ .

**0.0.120**

The expression for a linear factor of the quartic in terms of the invariants is more complicated than the corresponding expression for the cubic. It involves the solution of a cubic equation, and the expression of the roots of this cubic equation in terms of the invariants  $I$  and  $J$ .

**0.0.121**

The theory of the cubic and quartic quantics is thus complete. The covariants and invariants of these quantics have been completely determined, and their properties have been fully investigated.

**0.0.122**

The memoir concludes with some remarks on the bearing of the theory of covariants and invariants on the theory of equations and on geometry.

**0.0.123**

The theory of invariants is of great importance in the theory of equations. It affords a method for determining the nature of the roots of an equation without actually solving it.

**0.0.124**

The discriminant of a quantic vanishes when the quantic has equal roots. The invariants of a quantic give information about the mutual relations of the roots.

**0.0.125**

The theory of covariants is of great importance in geometry. The covariants of a quantic correspond to geometric loci, and their properties give information about the geometric configuration.

**0.0.126**

The Hessian of a quantic corresponds to the locus of points of inflection of the curve represented by the quantic. The cubicovariant

corresponds to the locus of points of contact of tangents from a point on the curve.

**0.0.127**

The theory of invariants and covariants is thus seen to be a fundamental part of the theory of algebra and geometry. It affords a method for the investigation of the properties of quantics and their geometric representations.

**0.0.128**

The memoir concludes with a brief summary of the results obtained, and with some indications of the directions in which the theory may be further extended.

[157]

## ON THE TANGENTIAL OF A CUBIC.

[From the *Philosophical Transactions of the Royal Society of London*, vol. xLviii. for the year 1858, pp. 461—463. Received February 11,—Read March 18, 1858.]

In my “Memoir on Curves of the Third Order”<sup>9</sup>, I had occasion to consider a derivative which may be termed the “tangential” of a cubic, viz. the tangent at the point  $(x, y, z)$  of the cubic curve  $(a, b, c, f, g, h, i, j, k, l \setminus x, y, z)^3 = 0$  meets the curve in a point  $(\xi, \eta, \zeta)$ , which is the tangential of the first-mentioned point; and I showed that when the cubic is represented in the canonical form  $x^3 + y^3 + z^3 + 6lxyz = 0$ , the coordinates of the tangential may be taken to be  $x(y^3 - z^3) : y(z^3 - x^3) : z(x^3 - y^3)$ . The method given for obtaining the tangential may be applied to the general form  $(a, b, c, f, g, h, i, j, k, l \setminus x, y, z)^3$ : it seems desirable, in reference to the theory of cubic forms, to give the expression of the tangential for the general form<sup>10</sup>; and this is what I propose to do, merely indicating the steps of the calculation, which was performed for me by Mr Creedy.  
The cubic form is

$$(a, b, c, f, g, h, i, j, k, l \setminus x, y, z)^3,$$

which means

$$ax^3 + by^3 + cz^3 + 3fy^2z + 3gz^2x + 3hx^2y + 3iyz^2 + 3jzx^2 + 3kxy^2 + 6lxyz;$$

and the expression for  $\xi$  is obtained from the equation

$$x^2\xi = (b, f, i, c \setminus j, f, c, i, g, l \setminus x, y, z)^2, \quad -(h, b, i, f, l, k \setminus x, y, z)^2)^3 \\ - (a, b, c, f, g, h, i, j, k, l \setminus x, y, z)^3(\mathfrak{C}x + \mathfrak{D}),$$

<sup>9</sup>Philosophical Transactions, vol. cxLvii. (1857), [146].

<sup>10</sup>At the time when the present paper was written, I was not aware of Mr Salmon's theorem (*Higher Plane Curves*, p. 156), that the tangential of a point of the cubic is the intersection of the tangent of the cubic with the first or line polar of the point with respect to the Hessian; a theorem, which at the same time that it affords the easiest mode of calculation, renders the actual calculation of the coordinates of the tangential less important. Added 7th October, 1858.—A.C.

where the second line is in fact equal to zero, on account of the first factor, which vanishes. And  $\mathfrak{C}$ ,  $\mathfrak{D}$  denote respectively quadric and cubic functions of  $(y, z)$ , which are to be determined so as to make the right-hand side divisible by  $x^2$ ; the resulting value of  $\xi$  may be modified by the adjunction of the evanescent term

$$(ax + hy + jz)(a, b, c, f, g, h, i, j, k, l \setminus x, y, z)^3,$$

where  $a, h, j$  are arbitrary coefficients; but as it is not obvious how these coefficients should be determined in order to present the result in the most simple form, I have given the result in the form in which it was obtained without the adjunction of any such term.

Write for shortness,

$$\begin{aligned} P &= (k, l \setminus y, z), \\ Q &= (b, f, i \setminus y, z)^2, \\ R &= (l, g \setminus y, z), \\ S &= (f, i, c \setminus y, z)^2, \\ B &= (h, j \setminus y, z), \\ C &= (k, l, g \setminus y, z)^2, \\ D &= (b, f, i, c \setminus y, z)^3, \end{aligned}$$

so that

$$\begin{aligned} (h, b, i, f, l, k \setminus x, y, z)^3 &= (h, P, Q \setminus x, 1)^2, \\ (j, f, c, i, g, l \setminus x, y, z)^3 &= (j, R, S \setminus x, 1)^2, \\ (a, b, c, f, g, h, i, j, k, l \setminus x, y, z)^3 &= (a, B, C, D \setminus x, 1)^3, \\ \mathfrak{C}x + \mathfrak{D} &= (\mathfrak{C}, \mathfrak{D} \setminus x, 1), \end{aligned}$$

and then for greater convenience writing  $(h, 2P, Q \setminus x, 1)^2$ , &c. for  $(h, P, Q \setminus x, 1)^2$ , &c., and omitting the  $(x, 1)^2$ , &c. and the arrow-heads, or representing the functions simply by  $(h, 2P, Q)$ , &c., we have

$$\begin{aligned} x^2\xi &= b(j, 2R, S)^3 \\ &\quad - 3f(j, 2R, S)^2 \cdot (h, 2P, Q) \\ &\quad + 3i(j, 2R, S) \cdot (h, 2P, Q)^2 \\ &\quad - c \cdot (h, 2P, Q)^3 \\ &\quad - (a, 3B, 3C, D) \cdot (\mathfrak{C}, \mathfrak{D}), \end{aligned}$$

which can be developed in terms of the quantities which enter into it. The conditions, in order that the coefficients of  $x, x^2$  may vanish, are thus seen to be

$$D\mathfrak{D} = bS^3 - 3fS^2Q + 3iSQ^2 - cQ^3,$$

$$D\mathfrak{C} - 3C\mathfrak{D} = b(6RS^2) - 3f(2S^2P + 4RSQ) + 3i(2RQ^2 + 4SPQ) - c(6PQ^2),$$

and from these we obtain

$$\mathfrak{C} = \begin{pmatrix} bck - 3 & big + 6 & bck + 3 \\ bil + 6 & cfk - 6 & cfl - 6 \\ fik + 3 & f^2g - 6 & fgi - 3 \\ f^2l - 6 & i^2k + 6 & i^2l + 6 \end{pmatrix} (y, z)^2$$

and

$$\mathfrak{D} = \begin{pmatrix} b^2c - 1 & bcf - 3 & bei + 3 & bc^2 + 1 \\ bfi + 3 & bi^2 + 6 & cf^2 - 6 & cfi - 3 \\ f^3 - 2 & f^2i - 3 & fi^2 + 3 & i^3 + 2 \end{pmatrix} (y, z)^3$$

and substituting these values, the right-hand side of the equation divides by  $x^2$ , and throwing out this factor we have the value of  $\xi$ ; and the values of  $\eta, \zeta$  may be thence deduced by a mere interchange of letters. The value for  $\xi$  is [given by the following table]:

| $x^4$       | $x^3y$      | $x^3z$      | $x^2y^2$     | $x^2yz$      | $x^2z^2$     | $xy^3$       | $xyz$      |
|-------------|-------------|-------------|--------------|--------------|--------------|--------------|------------|
| $bj^2 + 1$  | $bj^2l + 6$ | $bgj^2 + 6$ | $abck + 3$   | $abgi - 6$   | $abcg - 3$   | $ab^2c + 1$  | $abck + 3$ |
| $ch^3 - 1$  | $ch^2k - 6$ | $ch^2l - 6$ | $abil - 6$   | $acfh + 6$   | $acfl + 6$   | $abfi - 3$   | $abck + 3$ |
| $fhy^3 - 3$ | $fhlj - 12$ | $fghj - 12$ | $af^2l + 6$  | $af^2g + 6$  | $afgi + 3$   | $af^3 + 2$   | $abck + 3$ |
| $h^3j + 3$  | $fj^2k - 6$ | $fj^2l - 6$ | $afik - 3$   | $ai^2k - 6$  | $ai^2l - 6$  | $bchk - 3$   | $abck + 3$ |
|             | $h^2il + 6$ | $gh^2i + 6$ | $bch^2 - 3$  | $bgjl + 24$  | $bcj^2 + 3$  | $bhil - 6$   | $abck + 3$ |
|             | $hijk + 12$ | $hijl + 12$ | $bhij + 6$   | $bj^2i + 6$  | $bg^2j + 12$ | $bijk + 12$  | $abck + 3$ |
|             |             |             | $bjl^2 + 12$ | $cfh^2 - 6$  | $cfhj - 6$   | $bf^3 + 8$   | $abck + 3$ |
|             |             |             | $chk^2 - 12$ | $chkl - 24$  | $fj^3 - 24$  | $ck^3 - 8$   | $abck + 3$ |
|             |             |             | $f^2hj - 6$  | $f^2j^2 - 6$ | $fy^2h - 12$ | $f^2hl + 6$  | $abck + 3$ |
|             |             |             | $fh^2i - 3$  | $fghl - 24$  | $fgjl - 24$  | $f^2jk - 12$ | $abck + 3$ |
|             |             |             | $fhl^2 - 12$ | $fgjk - 24$  | $fi^3 - 3$   | $fhik + 3$   | $abck + 3$ |
|             |             |             | $fjkl - 24$  | $fj^3 - 24$  | $ghil + 24$  | $fk^3 - 24$  | $abck + 3$ |
|             |             |             | $hikl + 24$  | $ghik + 24$  | $hi^2j + 6$  | $ik^2l + 24$ | $abck + 3$ |
|             |             |             | $ijl^2 + 12$ | $hi^3 + 24$  | $ij^3 + 12$  |              | $abck + 3$ |
|             |             |             |              | $hij^2 + 24$ |              |              | $abck + 3$ |
|             |             |             |              | $ijkl + 24$  |              |              | $abck + 3$ |



| $xyz^2$      | $xz^3$       | $y^4$       | $y^3z$        | $y^2z^2$      | $yz^3$       | $z^4$       |
|--------------|--------------|-------------|---------------|---------------|--------------|-------------|
| $abci - 3$   | $abc^2 - 1$  | $b^2ij + 3$ | $b^2cj + 3$   | $bcf^2 + 9$   | $bc^2h - 3$  | $bcg^2 + 3$ |
| $acf^2 + 6$  | $acfi + 3$   | $bck^2 - 3$ | $bcfh - 3$    | $bchi - 9$    | $bcgl + 6$   | $c^2fh - 3$ |
| $afi^2 - 3$  | $ai^3 - 2$   | $bf^2j - 3$ | $bckl - 3$    | $bgil + 18$   | $bcij + 3$   | $cfgl - 6$  |
| $begh - 9$   | $bcgj + 3$   | $bphi - 3$  | $bfi j + 3$   | $cfkl - 18$   | $bg^2i + 6$  | $cfij + 3$  |
| $bcjl + 12$  | $bg^3 + 8$   | $bikl + 6$  | $bgik + 3$    | $f^2gl - 18$  | $cf^2j + 6$  | $chi^2 + 3$ |
| $bg^2l + 24$ | $cfgh - 12$  | $f^3h + 3$  | $bhi^2 - 6$   | $f^2ij - 9$   | $cfgk - 6$   | $fg^2i - 3$ |
| $bgij + 6$   | $cfjl + 3$   | $f^2kl - 6$ | $bil^2 + 12$  | $fhi^2 + 9$   | $cfhi - 3$   | $g^3l + 6$  |
| $cfhl - 18$  | $ci^3 - 8$   | $fik^2 + 3$ | $cfk^2 - 6$   | $i^3kl + 18$  | $cf^3 - 12$  | $i^3j - 3$  |
| $cfjk + 6$   | $fg^3l - 24$ |             | $f^2j - 6$    | $f^2g^3 - 6$  | $f^2g^2 - 6$ |             |
| $ck^2l - 24$ | $fgij - 3$   |             | $f^2gk - 6$   | $fgil - 6$    | $fi^2j - 9$  |             |
| $f^2gj - 6$  | $ghi^2 + 12$ |             | $f^2hi + 9$   | $gi^2k + 6$   |              |             |
| $fg^3k - 24$ | $gi^3 + 24$  |             | $f^2l^2 - 12$ | $hi^3 + 6$    |              |             |
| $fghi + 9$   | $i^3jl - 6$  |             | $fikl + 6$    | $i^2l^2 + 12$ |              |             |
| $fgl^2 - 48$ |              |             | $i^3k^2 + 6$  |               |              |             |
| $fijl - 12$  |              |             |               |               |              |             |
| $gikl + 48$  |              |             |               |               |              |             |
| $hi^2l + 18$ |              |             |               |               |              |             |
| $i^2jk - 6$  |              |             |               |               |              |             |
| $il^3 + 24$  |              |             |               |               |              |             |

and it is not necessary to write down the corresponding values for  $\eta$ ,  
 $\zeta$ .

[158]

## A SIXTH MEMOIR UPON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*,  
vol. cxLix. for the year 1859, pp. 61—90. Received November 18,  
1858,—Read January 6, 1859.]

I PROPOSE in the present memoir to consider the geometrical theory: I have alluded to this part of the subject in the articles Nos. 3 and 4 of the Introductory Memoir, [139]. The present memoir relates to the geometry of one dimension and the geometry of two dimensions, corresponding respectively to the analytical theories of binary and ternary quantics. But the theory of binary quantics is considered for its own sake; the geometry of one dimension is so immediate an interpretation of the theory of binary quantics, that for its own sake there is no necessity to consider it at all; it is considered with a view to the geometry of two dimensions. A chief object of the present memoir is the establishment, upon purely descriptive principles, of the notion of distance. I had intended in this introductory paragraph to give an outline of the theory, but I find that in order to be intelligible it would be necessary for me to repeat a great part of the contents of the memoir in relation to this subject, and I therefore abstain from entering upon it. The paragraphs of the memoir are numbered consecutively with those of my former Memoirs on Quantics.

## 0.0.147

It will be seen that in the present memoir, the geometry of one dimension is treated of as a geometry of points in a line, and the geometry of two dimensions as a geometry of points and lines in a plane. It is, however, to be throughout borne in mind, that, in accordance with the remarks No. 4 of the Introductory Memoir, the terms employed are not (unless this is done expressly or by the context) restricted to their ordinary significations. In using the geometry of one dimension in reference to geometry of two dimensions considered as a geometry of points and lines in a plane, it is necessary to consider,—1°, that the word point may mean *point* and the word line mean *line*; 2°, that the word point may mean *line*

and the word line mean *point*; and so for the other terms employed in the theory.

### 0.0.148

The geometry of one dimension is the geometry of points on a line. A system of points on a line is represented analytically by a quantic, the facients of which are the coordinates of a point on the line. The roots of the quantic correspond to the points of the system, and the coefficients of the quantic correspond to the symmetric functions of the points.

### 0.0.149

The distance between two points on a line is defined as follows: let the points be represented by the coordinates  $x_1$  and  $x_2$ ; then the distance between the points is proportional to  $x_1 - x_2$ . The constant of proportionality depends on the absolute scale of measurement, which is arbitrary.

### 0.0.150

The anharmonic ratio of four points on a line is defined as follows: let the points be represented by the coordinates  $x_1, x_2, x_3, x_4$ ; then the anharmonic ratio is

$$\frac{(x_1 - x_3)(x_2 - x_4)}{(x_1 - x_4)(x_2 - x_3)}.$$

This ratio is independent of the choice of coordinates, and is thus a projective invariant.

### 0.0.151

The geometry of two dimensions is the geometry of points and lines in a plane. A point is represented analytically by its coordinates  $(x, y, z)$ , and a line is represented by its equation  $ax + by + cz = 0$ .

### 0.0.152

The distance between two points in a plane may be defined by means of the absolute conic. The absolute conic is a conic which is

fixed in the plane, and the distance between two points is defined in terms of their relation to this conic.

### **0.0.153**

In ordinary Euclidean geometry, the absolute conic is the imaginary circle at infinity. The distance between two points is then given by the ordinary formula of Euclidean geometry.

### **0.0.154**

In non-Euclidean geometry, the absolute conic is a real conic. The distance between two points is then given by a formula involving the logarithm of the anharmonic ratio of the four points formed by the two given points and the two points of intersection of the line joining them with the absolute conic.

### **0.0.155**

The notion of distance thus established is purely descriptive. It depends only on the projective properties of the figures, and not on any metrical assumptions. The geometry founded on this notion of distance includes Euclidean geometry as a particular case.

### **0.0.156**

The geometry of one dimension is a particular case of the geometry of two dimensions. A line in the plane is a one-dimensional space, and the points on it form a system which may be treated by the methods of the geometry of one dimension.

### **0.0.157**

The binary quantic corresponds to a system of points on a line. The invariants of the quantic correspond to projective properties of the system of points, and the covariants correspond to derived systems of points.

### **0.0.158**

The discriminant of a binary quantic vanishes when two of the points of the system coincide. The Hessian corresponds to the

system of points which are the harmonic conjugates of the points of the original system with respect to the pairs of points formed by the remaining points.

### 0.0.159

The theory of binary quantics is thus seen to be identical with the projective geometry of systems of points on a line. The invariants and covariants of the quantic have simple geometric interpretations.

### 0.0.160

The ternary quantic corresponds to a curve in the plane. The invariants of the quantic correspond to projective properties of the curve, and the covariants correspond to derived curves.

### 0.0.161

The discriminant of a ternary quantic vanishes when the curve has a singular point. The Hessian corresponds to the locus of points of inflection of the curve.

### 0.0.162

The theory of ternary quantics is thus seen to be identical with the projective geometry of curves in the plane. The invariants and covariants of the quantic have simple geometric interpretations.

### 0.0.163

The theory of distance established in the present memoir is applicable to the geometry of quantics. The distance between two points on a curve may be defined in terms of the absolute conic, and this definition leads to interesting results.

### 0.0.164

In particular, the distance between two points on a conic may be expressed in terms of the anharmonic ratio of the four points formed by the two given points and the two points of intersection of the line joining them with the conic. This leads to a generalization of the ordinary formula for the distance between two points on a circle.

**0.0.165**

The theory of distance may also be applied to the geometry of surfaces. The absolute quadric is a quadric surface which is fixed in space, and the distance between two points is defined in terms of their relation to this quadric.

**0.0.166**

The geometry of three dimensions is the geometry of points, lines, and planes in space. The quaternary quantic corresponds to a surface in space, and the theory of quaternary quantics is identical with the projective geometry of surfaces.

**0.0.167**

The memoir concludes with some remarks on the bearing of the theory on the philosophy of geometry. The purely descriptive character of the theory shows that the metrical properties of figures are consequences of their projective properties, and that the distinction between Euclidean and non-Euclidean geometry is merely a matter of the choice of the absolute conic.

**0.0.168**

The coordinates of the point of intersection of the lines

$$\begin{aligned}\alpha x + \beta y + \gamma z &= 0, \\ \alpha' x + \beta' y + \gamma' z &= 0,\end{aligned}$$

are given by the equations

$$x : y : z = \beta\gamma' - \beta'\gamma : \gamma\alpha' - \gamma'\alpha : \alpha\beta' - \alpha'\beta;$$

and if in these equations we consider  $\alpha', \beta', \gamma'$  as indeterminate, we have the coordinates of a point subjected to the single condition of lying in the line  $\alpha x + \beta y + \gamma z = 0$ ; the result, as in the last case, contains in appearance two arbitrary parameters, but these really reduce themselves to a single one.

**0.0.169**

The condition in order that the points  $(a, b, c)$ ,  $(a', b', c')$ ,  $(a'', b'', c'')$  may lie in a line is

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix} = 0,$$

which may also be expressed by the equations

$$a'', b'', c'' = \lambda a + \mu a', \quad \lambda b + \mu b', \quad \lambda c + \mu c',$$

where  $\lambda, \mu$  are arbitrary multipliers; these equations give therefore the coordinates of an indeterminate point in the line joining the points  $(a, b, c)$  and  $(a', b', c')$ .

**0.0.170**

The condition that the lines

$$\begin{aligned} \alpha x + \beta y + \gamma z &= 0, \\ \alpha' x + \beta' y + \gamma' z &= 0, \\ \alpha'' x + \beta'' y + \gamma'' z &= 0 \end{aligned}$$

may meet in a point is

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{vmatrix} = 0,$$

a relation which may also be expressed by the equations

$$\alpha'', \beta'', \gamma'' = l\alpha + m\alpha', \quad l\beta + m\beta', \quad l\gamma + m\gamma',$$

where  $l, m$  are arbitrary multipliers; and substituting these values in the equation  $\alpha'' x + \beta'' y + \gamma'' z = 0$ , we have for the equation of a line subjected to the single condition of passing through the point of intersection of the lines  $\alpha x + \beta y + \gamma z = 0$ ,  $\alpha' x + \beta' y + \gamma' z = 0$ , the equation

$$l(\alpha x + \beta y + \gamma z) + m(\alpha' x + \beta' y + \gamma' z) = 0,$$

which contains in appearance the two arbitrary parameters  $l$  and  $m$ , but which, as the equation is homogeneous in regard to these parameters, really contains only the single parameter  $l : m$ .

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## [End of Paper 156. A Fifth Memoir upon Quantics]

Mr Salmon has also performed the laborious calculation of Hermite's 18-thic invariant of a binary quintic, and has kindly permitted me to publish the result, which is given in the following Table:

### No. 29<sub>a</sub>.

[This is the Table No. 29 A given pp. 299—308, the form being slightly altered as appears p. 282.]

Mr Salmon has also remarked to me, that in the Table No. 50 of the cubin-variant of a binary dodecadic, the coefficients are altogether erroneous. There was, in fact, a fundamental error in the original calculation; instead of repeating it, I have, with a view to the deduction therefrom of the cubinvariant (see Fourth Memoir, No. 78), first calculated the dodecadic quadricovariant, the value of which is given in the following Table:

### No. 49<sub>a</sub>.

[For this Table see p. 319.]

It is now very easy to obtain the cubinvariant, which is

### No. 50 bis.

[This is the Table No. 50, p. 319, the original No. 50 with coefficients which were altogether erroneous having been omitted.]

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*Third Memoir, Philosophical Transactions*, t. cxlvi. (1856) p. 635.

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## 157. On the Tangential of a Cubic.

[From the *Philosophical Transactions of the Royal Society of London*, vol. XLVIII. for the year 1858, pp. 461—463. Received February 11,—Read March 18, 1858.]

In my “Memoir on Curves of the Third Order”<sup>11</sup>, I had occasion to consider a derivative which may be termed the “tangential” of a cubic, viz. the tangent at the point  $(x, y, z)$  of the cubic curve  $(\ast)x, y, z)^3 = 0$  meets the curve in a point  $(\xi, \eta, \zeta)$  which is the tangential of the first-mentioned point; and I showed that when the cubic is represented in the canonical form  $x^3 + y^3 + z^3 + 6lxyz = 0$ , the coordinates of the tangential may be taken to be  $x(y^2 - z^2) : y(z^2 - x^2) : z(x^2 - y^2)$ . The method given for obtaining the tangential may be applied to the general form  $(a, b, c, f, g, h, i, j, k, \lambda x, y, z)^3 = 0$ ; it seems desirable, in reference to the theory of cubic forms, to give the expression of the tangential for the general form; and this is what I propose to do, merely indicating the steps of the calculation, which was performed for me by Mr Greedy.

The cubic form is

$$(a, b, c, f, g, h, i, j, k, \lambda x, y, z)^3,$$

which means

$$ax^3 + by^3 + cz^3 + 3fyz^2 + 3gz^2x + 3hz^2y + 3izx^2 + 3jx^2y + 3kxy^2 + 3ly^2z;$$

and the expression for  $\xi$  is obtained from the equation

$$ax^2\xi = (j, f, i, c \setminus x, y, z)^3 - (h, b, i, f, l, k \setminus x, y, z)^3 - (a, b, c, f, g, h, i, j, k, \lambda x, y, z)^3 (\xi x + \eta y + \zeta z),$$

where the second line is in fact equal to zero, on account of the first factor, which vanishes. And  $\mathfrak{C}$ ,  $\mathfrak{D}$  denote respectively quadric and cubic functions of  $(y, z)$ , which are to be determined so as to make the right-hand side divisible by  $x^2$ ; the resulting value of  $\xi$  may be modified by the adjunction of the evanescent term

$$(\alpha x + \beta y + \gamma z)(a, b, c, f, g, h, i, j, k, \lambda x, y, z)^3,$$

where  $\alpha, \beta, \gamma$  are arbitrary coefficients; but as it is not obvious how these coefficients should be determined in order to present the result

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<sup>11</sup>Philosophical Transactions, vol. cxlvii. (1857), [146].

in the most simple form, I have given the result in the form in which it was obtained without the adjunction of any such term.

Write for shortness,

$$\begin{aligned} P &= (k, \lambda y, z), & Q &= (b, f, \lambda y, z)^2, \\ R &= (l, g \lambda y, z), & S &= (f, i, c \lambda y, z)^2, \\ \mathfrak{B} &= (h, j \lambda y, z), & \mathfrak{C} &= (k, l, g \lambda y, z)^2, \\ \mathfrak{D} &= (h, f, i, c \lambda y, z)^3, \end{aligned}$$

so that

$$\begin{aligned} (h, b, i, f, l, k \lambda x, y, z)^3 &= (h, P, Q \lambda x, 1)^3, \\ (j, f, i, c, g, \lambda x, y, z)^3 &= (j, R, S \lambda x, 1)^3, \\ (a, b, c, f, g, h, i, j, k, \lambda x, y, z)^3 &= (a, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \lambda x, 1)^3, \\ \xi x + \eta y + \zeta z &= (\mathfrak{A}, \mathfrak{B} \lambda x, 1), \end{aligned}$$

and then for greater convenience writing  $(h, 2P, Q \lambda x, 1)$ , &c. for  $(h, P, Q \lambda x, 1)$ , &c., and omitting the  $(x, 1)^3$ , &c. and the arrow-heads, or representing the functions simply by  $(h, 2P, Q)$ , &c., we have

$$\begin{aligned} ax^2\xi &= c(j, 2R, S)^3 - 3f(j, 2R, S)^2 \cdot (h, 2P, Q) \\ &+ 3i(j, 2R, S) \cdot (h, 2P, Q)^2 - c \cdot (h, 2P, Q)^3 - (a, 3\mathfrak{B}, 3\mathfrak{C}, \mathfrak{D}) \cdot (\mathfrak{A}, \mathfrak{B}), \end{aligned}$$

which can be developed in terms of the quantities which enter into it. The conditions, in order that the coefficients of  $x$ ,  $x^0$  may vanish, are thus seen to be

$$\begin{aligned} \mathfrak{D}\mathfrak{A} &= cS^3 - 3fS^2Q + 3iSQ^2 - cQ^3, \\ \mathfrak{D}\mathfrak{B} - 3\mathfrak{C}\mathfrak{A} &= b(6RS^2) - 3f(2S^2P + 4RSQ) + 3i(2RQ^2 + 4SPQ) - c(6P^2Q), \end{aligned}$$

and from these we obtain

$$\begin{aligned} \mathfrak{A} &= \{ch^2 - 3fig + cgi + 3fhj - 2ci^2\}(y, z)^4, \\ \mathfrak{B} &= \{-3bfj + bgi + 3bch - 2bk^2\}(y, z)^3, \end{aligned}$$

and substituting these values, the right-hand side of the equation divides by  $x^2$ , and throwing out this factor we have the value of  $\xi$ ; and the values of  $\eta$ ,  $\zeta$  may be thence deduced by a mere interchange of letters. The value for  $\xi$  is

$$\begin{aligned}\xi = & a^2 cl \cdot x^4 + a^2(-3cj + bk) \cdot x^3 y + a^2(3bh - 2ci) \cdot x^3 z + a^2(-3bj + 3bf + 3ch - al) \cdot x^2 y^2 \\ & + a^2(6bg - 6cf - 3ah) \cdot x^2 yz + a^2(3ci - 3ag + ac) \cdot x^2 z^2 \\ & + a(-3bj^2 + 6bfj - 3bch + bkl - 3c^2g + 3ach) \cdot xy^3 \\ & + a(6bgj - 6bh^2 - 12cfj + 6c^2h + 6chi - 3bik - 3a^2g + 6aci) \cdot xy^2z \\ & + a(-6cfi + 6cgi - 6chi + 6ag^2 - 6a^2f + 3ack + 3a^2l) \cdot xyz^2 \\ & + a(3cgi - 3ag^2 - 2ci^2 - 3a^2f) \cdot xz^3 \\ & + (bj^3 - 3bfj^2 + 3bchj - bk^2l + c^2hj - c^2fk + c^2il - 3cghj + 3cg^2i + 3cfhj \\ & - 3cf^2i - ch^2i + chkl - bg^3 + 3bfjg^2 - 3bf^2g + bf^2l + 3bghk - bh^2j + bh^2l \\ & - 2bhik + 2bi^2k + bjkl - bk^2i - c^2gh + c^2fi + cghi - cgi^2 + cfh^2 - cfhi \\ & - 3cfhk + 3cfij + 3c^2g^2 - 6cfgi + 3cf^2h - 3cg^2l + 3cghi - 3cfhl - c^2j^2 \\ & + 2c^2jk - c^2k^2 - 3afg^2 + 3af^2g - af^2l - 3aghk + ah^2j - ah^2l + 2ahik \\ & - ai^2k - ajkl + ak^2i + ag^3)y^4 + \dots\end{aligned}$$

and it is not necessary to write down the corresponding values for  $\eta$ ,  $\zeta$ .

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*At the time when the present paper was written, I was not aware of Mr Salmon's theorem (Higher Plane Curves, p. 156), that the tangential of a point of the cubic is the intersection of the tangent of the cubic with the first or line polar of the point with respect to the Hessian; a theorem, which at the same time that it affords the easiest mode of calculation, renders the actual calculation of the coordinates of the tangential less important. Added 7th October, 1858. A. C.*

## 158. A Sixth Memoir upon Quantics.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLIX. for the year 1859, pp. 61—90. Received November 18, 1858,—Read January 6, 1859.]

I PROPOSE in the present memoir to consider the geometrical theory: I have alluded to this part of the subject in the articles Nos. 3 and 4 of the Introductory Memoir, [139]. The present memoir relates to the geometry of one dimension and the geometry of two dimensions, corresponding respectively to the analytical theories of binary and ternary quantics. But the theory of binary quantics is considered for its own sake; the geometry of one dimension is so immediate an interpretation of the theory of binary quantics, that for its own sake there is no necessity to consider it at all; it is considered with a view to the geometry of two dimensions. A chief object of the present memoir is the establishment, upon purely descriptive principles, of the notion of distance. I had intended in this introductory part to make a reference to Mr W. R. Hamilton's memoir, "On Symbolical Geometry," *Camb. and Dubl. Math. Jour.* t. iv. (1849), pp. 124—137, but I am unable to do so: in fact, without at all under- taking to understand the principle or philosophy of the author's symbolical method, I find that the notion of distance enters into his definitions of the symbolical imaginaries  $i$ ,  $j$ ,  $k$  (see p. 128). The memoir contains (p. 135) a "construction for a fourth pro- portional to three given lines in the same plane," which it is proper to notice is the construction already given by M. Chasles for the product of two lines, *Comptes Rendus*, t. xxxv. (1852), p. 292. See also the memoir by the same author, "Sur la construction de la notion de distances dans la géométrie descriptive" (see in particular No. 2, p. 757, Sur les Multiplications, &c.), *Comptes Rendus*, t. xliii. (1856), pp. 753—759; and the memoir by M. Cayley (the author of the present memoir), "Sur la Géométrie de la distance générale," *Comptes Rendus*, t. xlvii. (1858), pp. 489—492, the references in which last- mentioned memoir require to be completed as follows:—The theory of the anharmonic ratio, including the theorem that a pencil of four lines and the transversal have the same anharmonic ratio, is due to M. Chasles (*Comptes Rendus*, t. xxi. (1845), p. 383),— the theorem that an involution of points gives rise to a pencil in involution, and the theory of poles and

polars, including the theorem of the anharmonic property of the four points formed by a variable point on a conic with the four foci, and including the definition of a circle by means of the two imaginary foci, are due to M. Chasles (*Comptes Rendus*, t. xxx. (1850), p. 326, and t. xxxii. (1851), p. 560); the theorem of the anharmonic property of the foci of a conic was in fact demonstrated by Mr J. T. Graves in the year 1850 (see *Philosophical Magazine*, February, 1850); the theorem of the pencil in involu- tion is due to Mr Kirkman, *Philosophical Transactions*, 1849, p. 83; the definition of a circle by means of the two circular points at infinity is to be found in Poncelet's *Traité des Propriétés Projectives*, [1822], p. 371.

The foregoing extracts show, and it is besides obvious, that the theory of distance generalizes the ordinary theory of similar figures, by substituting for the Absolute, viz. the two circular points at infinity, a conic (the Absolute) lying in the plane or other space of two dimensions: the generalized notion of distance is applicable not only to ordinary plane geometry, but to descriptive geometry, and to geometry in a superficies. The leading idea is as old as Desargues:

thus in the theorem that six points in a conic have a single anharmonic ratio, Desargues considered the conic as a conic of the figure; the notion is very explicitly contained in a posthumous memoir of Lagrange, "Solution de quelques problèmes d'astronomie sphérique par le moyen des analogies," *OEuvres de Lagrange*, t. iii. pp. 331—354, see p. 351, Nos. 27—30: in particular, considering on a sphere a triangle, the sides whereof are arcs of great circles, and substituting for the angles at the centre (these being the distances) the trigonometrical functions (the ratios of the coordinates to the distance from the centre), or, as we should now say, projecting on the osculating plane of the point which is the pole of the hemisphere, the sides of the spherical triangle become lines in a plane; and instead of the spherical triangle we have the projection thereof on the osculating plane; and Lagrange expressly remarks that the formulæ so obtained are in fact those of plane trigonometry.

The memoir just referred to may be considered as containing the germ at least of the notion of distance.

### On Geometry of One Dimension, Nos. 149 to 168.

**149.** In geometry of one dimension we have the line as a space or locus in quo of the points of the figure; a point is determined by means of its coordinate ratio ( $x : y$ ), and a figure whatever consists

of a system of points, in general distinct from each other. The coordinates  $(x, y)$  may be considered as denoting two given points of the line, or as the multiples of the distances of the variable point from these points; but it is not necessary to do so; the coordinate ratio is all that it is necessary to attend to. Any function whatever of the coordinates, equated to zero, is an equation of the figure; but only those equations which are homogeneous in regard to the coordinates are equations of any figure; or if the equation is not homogeneous, then it must be capable of being made so by the adjunction of a proper constant. The order of a figure is the order of its equation in regard to the coordinates.

**150.** A linear equation  $(a, b \setminus x, y)^1 = 0$  denotes a single point; the single point may be considered as a point-pair, the two points of which coincide together.

**151.** A quadric equation  $(a, b, c \setminus x, y)^2 = 0$  denotes a pair of points. The point  $(a, b \setminus x, y)^1 = 0$  is in a certain sense the harmonic of either of the points of the point-pair in respect to the other; but as the harmonic of a point in respect to itself and another point is altogether indefinite, it is proper to speak of the point as the centre or mid-point of the point-pair, and to denote it as such by the centre of the point-pair; the notion of the mid-point has reference to the theory of distance which will be explained in the sequel.

**152.** A system of points, the number of which is equal to the order, is a harmonic system, if the several points are the centres of the point-pairs formed by the remaining points; thus a point-pair is always a harmonic system of two points; a system of three points cannot be a harmonic system; and a system of four points is a harmonic system, when each point of it is the centre of the remaining point-pair.

**153.** In geometry of one dimension the Absolute is a pair of points; any point whatever and the Absolute determine a point-pair, and the centre of this point-pair is the harmonic of the point in respect to the Absolute; or more simply, the harmonic of the point in respect to the Absolute is the point which is the harmonic of the first point in respect to the two points of the Absolute. The harmonic of a point in respect to the Absolute may also be termed the opposite point. The Absolute may be real, or it may consist of a pair of conjugate imaginary points. The equation of the Absolute may always be taken to be  $x^2 + y^2 = 0$ , or, what is the same thing,  $(1, 0, 1 \setminus x, y)^2 = 0$ .

**154.** Taking the equation of the Absolute to be  $x^2 + y^2 = 0$ , the harmonic of the point  $(x, y)$  in respect to the Absolute is the point  $(x, -y)$ . Any two points  $(x, y)$  and  $(x', y')$  determine a point-pair, the centre of which is the point  $xx' + yy'$ ,  $xy' - x'y$ ; and if the coordinates of the first point are  $(x, y)$  and those of the second point are  $(x', y')$ , then the distance between the two points is

$$\sin^{-1} \frac{xy' - x'y}{\sqrt{x^2 + y^2} \sqrt{x'^2 + y'^2}};$$

or what is the same thing, the distance is

$$\cos^{-1} \frac{xx' + yy'}{\sqrt{x^2 + y^2} \sqrt{x'^2 + y'^2}}.$$

**155.** It follows at once, from the sine formula, that the distance of a point from itself is always  $= 0$ ; and from the cosine formula, that the distance of the point  $(x, y)$  from the point  $(x, -y)$  is  $= \pi$ ; and from the sine formula, that the distance of either of the points of the Absolute from any point whatever is  $= 0$ ; and from the cosine formula, that the distance of either of the points of the Absolute from any point whatever is  $= \frac{\pi}{2}$ .

**156.** The quadrant or distance of either of the points of the Absolute from any other point, is the unit of distance; but since either of the points of the Absolute is at a distance  $= 0$  from itself, there is here a certain apparent inconsistency. It may be observed, that in the case where the Absolute is a pair of imaginary points, the unit of distance is imaginary.

**157.** A lineo-linear equation

$$(a, b, c \backslash x, y)^2 (\alpha, \beta, \gamma \backslash x', y')^2 - (ac - b^2)(xy' - x'y)^2 = 0$$

denotes that the distance of the points  $(x, y)$  and  $(x', y')$  is a given constant; for this equation is equivalent to

$$\cos^{-1} \frac{xx' + yy'}{\sqrt{x^2 + y^2} \sqrt{x'^2 + y'^2}} = \text{const.}$$

The curve (the only curve in geometry of one dimension) represented by the above equation may be called a circle; the point  $(\alpha, \beta, \gamma \backslash x', y')^2 = 0$  is the centre of the circle; and the sine of the radius is  $= \frac{\sqrt{ac-b^2}}{\sqrt{a}}$ , where  $a$  means  $(a, b, c \backslash \alpha, \beta)^2$ .

**158.** The equation of the Absolute being  $x^2 + y^2 = 0$ , the distance of the points  $(x : y)$  and  $(x' : y')$  is in fact

$$\log \frac{xy' - x'y + \sqrt{x^2 + y^2} \sqrt{x'^2 + y'^2}}{xy' - x'y - \sqrt{x^2 + y^2} \sqrt{x'^2 + y'^2}};$$

or, as this expression may also be written,

$$\log \frac{(x + iy)(x' - iy')}{(x - iy)(x' + iy')},$$

which, putting  $\frac{x+iy}{x-iy} = u$  and  $\frac{x'+iy'}{x'-iy'} = u'$ , becomes  $\log \frac{u}{u'}$ ; and the theorem that the distance of the first and second of three points is equal to the sum of the distances of the first and second and second and third respectively, is at once verified: we have only to write

$$\log \frac{u}{u'} + \log \frac{u'}{u''} = \log \frac{u}{u''}.$$

**159.** What precedes is not to be understood as precluding the existence of a relation between the spaces which are the loci in quibus of the coordinates  $(x, y)$  and  $(x', y')$  respectively: not only may these be spaces of the same kind, but they may be one and the same space or line; and the points of the two systems may then be points of the same kind; and further, the coordinates  $(x, y)$  and  $(x', y')$  may belong to the same system of coordinates, that is, the equations  $(x : y) = (\alpha : 1)$  and  $(x' : y') = (\alpha : 1)$  may denote one and the same point.

**160.** If the two point-systems are systems of the same kind, and are in one and the same line, then there are in general two points of the first system which coincide each of them with the corresponding point of the second system; such two points may be said to be the *sibiconjugate points* of the homography. In particular, the two sibiconjugate points of the homography may coincide together.

**161.** A system in involution affords an example of two homographic systems in the same line; in fact, taking arbitrarily a point out of each pair, the points so obtained form a system which is homographic with the system formed with the other points of the several pairs; and in this case the sibiconjugate points of the involution are also the sibiconjugate points of the homography. Thus if  $A$  and  $A'$ ,  $B$  and  $B'$ ,  $C$  and  $C'$ ,  $D$  and  $D'$  are pairs of the system in involution, then  $(A, B, C, D)$  and  $(A', B', C', D')$  will be homographic point-systems; and, as a particular case,  $(A, B, C, C')$  and  $(A', B', C, C')$  will be homographic point-systems. It is proper



to notice that if  $F$  is a sibiconjugate point of the involution, then  $(A, B, F, F)$  and  $(A', B', F, F)$  are not (what at first sight they appear to be) homographic point-systems.

**162.** Imagine an involution of points; take on the line which is the locus in quo of the point-system a point  $O$ , and consider the point-system formed by the harmonics of  $O$  in respect to the several pairs of the involution; and in like manner take on the line any other point  $O'$ , and consider the point-system formed by the harmonics of  $O'$  in respect to the several pairs of the involution; these two point-systems are homographically related to each other. See Fifth Memoir, No. 111.

**163.** Two involutions may be homographically related to each other; in fact, take on the line which is the locus in quo of the first involution a point  $O$ , and consider the point-system formed by the harmonics of  $O$  in relation to the several pairs of the involution; take in like manner on the line which is the locus in quo of the second involution a point  $O'$ , and consider the point-system formed by the harmonics of  $O'$  with respect to the several pairs of the involution; then if the two point-systems are homographically related, the two involutions are said to be themselves homographically related: the last preceding article shows that the nature of the relation does not in anywise depend on the choice of the points  $O$  and  $O'$ . And it is not necessary that, as regards the two involutions respectively, the words line and point should have the same significations. See Fifth Memoir, No. 111.

**164.** Four or more tetrads of points in a line may be homographically related to the same number of tetrads in another line. This is the case when the anharmonic ratios of the tetrads of the first system are homographically related to the anharmonic ratios of the tetrads of the second system. And it is not material which of the three anharmonic ratios of a tetrad of either system is selected, provided that the same selection is made for each of the other tetrads of the same system. The order of the points of a tetrad must be attended to, but there are in all four admissible permutations of the points of a tetrad, viz. if  $A, B, C, D$  are the points of a tetrad, then  $(A, B, C, D)$ ,  $(B, A, D, C)$ ,  $(C, D, A, B)$ ,  $(D, C, B, A)$  may be considered as one and the same tetrad. Any three tetrads whatever in the second system may correspond to any three tetrads of the first system; and then given a fourth tetrad of the first system, and three out of the four points of the

corresponding tetrad of the second system, the remaining point of the tetrad may be determined. The words line and point need not, as regards the two systems of tetrads respectively, be understood in the same significations. See Fifth Memoir, No. 112.

**165.** The foregoing theory of the harmonic relation shows that if we have a point-pair

$$(a, b, c \backslash x, y)^2 = 0,$$

the equation of any other point-pair whatever can be expressed, and that in two different ways, in the form

$$(a, b, c \backslash x, y)^2 + (lx + my)^2 = 0;$$

the points  $(lx + my = 0)$  corresponding to the two admissible values of the linear function being in fact the harmonics of the point-pair in respect to the given point-pair  $(a, b, c \backslash x, y)^2 = 0$ , or what is the same thing, the sibiconjugate points of the involution formed by the two point-pairs (see Fifth Memoir, No. 105). The point-pair represented by the equation in question does not in itself stand in any peculiar relation to the given point-pair  $(a, b, c \backslash x, y)^2 = 0$ ; but when thus represented it is said to be *inscribed* in the given point-pair, and the point  $lx + my = 0$  is said to be the *axis of inscription*. And the harmonic of this point with respect to the given point-pair (that is, the other sibiconjugate point of the involution of the two point-pairs) is said to be the *centre of inscription*.<sup>12</sup>

**166.** We may, if we please,  $(x', y')$  and  $\theta$  being constants, exhibit the equation of the inscribed point-pair in the form

$$(a, b, c \backslash x, y)^2 (a, b, c \backslash x', y')^2 \sin^2 \theta - (ac - b^2)(xy' - x'y)^2 = 0,$$

where we have for the axis of inscription and centre of inscription respectively, the equations

$$xy' - x'y = 0, \quad (a, b, c \backslash x, y)^2 (a, b, c \backslash x', y')^2 = 0;$$

the axis and centre are in fact the points  $xy' - x'y = 0$  and  $xx' + yy' = 0$  respectively.

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<sup>12</sup>The words inscribed, inscription, are used not in opposition to, but as identical with, the words circumscribed, circumscription; and in like manner the words centre and axis of inscription.

**167.** The condition that the point-pair  $(a', b', c \wedge x, y)^2 = 0$  may be inscribed in the point-pair  $(a, b, c \wedge x, y)^2 = 0$  is

$$(ac - b^2)(a'c' - b'^2) - (ac' + ca' - 2bb')^2 = 0.$$

**168.** The point-pair  $(a, b, c \wedge x, y)^2 = 0$  is harmonically inscribed in the point-pair  $(a', b', c' \wedge x, y)^2 = 0$ , if the first point-pair is inscribed in the second, and if at the same time the second is inscribed in the first. The condition is

$$aa' + 2bb' + cc' = 0.$$

The condition that two point-pairs may be harmonically inscribed each of them in the other, may also be expressed by saying that the centre of either of them is the axis of the other; and the two centres or two axes form a point-pair which is the harmonic of either of the given point-pairs in respect to the other.

### On the Geometry of Two Dimensions, Nos. 169 to 231.

**169.** In geometry of two dimensions we have the plane as a space or locus in quo of the points and lines of the figure. A point is determined by means of the coordinate ratios  $(x : y : z)$ , and a line by means of the coordinate ratios  $(\xi : \eta : \zeta)$ ; the equation

$$\xi x + \eta y + \zeta z = 0$$

expresses that the point  $(x, y, z)$  and the line  $(\xi, \eta, \zeta)$  are corresponding point and line; and in like manner the equation

$$\xi x + \eta y + \zeta z = 0$$

expresses that the point  $(x, y, z)$  and the line  $(\xi, \eta, \zeta)$  are corresponding point and line. It is convenient to use the word *ineunt* (from in eo) to denote a point considered as belonging to a curve; thus a point on a curve is an ineunt of the curve; and in like manner a line touching a curve is a tangent of the curve. A line may of course be considered as a curve of the first class, and a point as a curve of the first order.

**170.** The point-coordinates  $(x, y, z)$  and the line-coordinates  $(\xi, \eta, \zeta)$  are absolutely independent of each other; the point  $(x, y, z)$  and the line  $(\xi, \eta, \zeta)$  are a corresponding point and line, when they satisfy the equation  $\xi x + \eta y + \zeta z = 0$ . The coordinates of any point

or line may be replaced by any constant multiples of their values; but it is in general not allowable to do this simultaneously for the point and line coordinates: when it is allowable, the equations of the figure are said to be *reduced*. The unreduced form of the equation of a line in point-coordinates is

$$\xi x + \eta y + \zeta z = 0,$$

and the unreduced form of the equation of a point in line-coordinates is

$$X\xi + Y\eta + Z\zeta = 0.$$

**171.** A linear equation  $(a, b, c \backslash x, y, z)^1 = 0$  denotes a line, and a linear equation  $(A, B, C \backslash \xi, \eta, \zeta)^1 = 0$  denotes a point.

**172.** A quadric equation  $(a, b, c, f, g, h \backslash x, y, z)^2 = 0$  denotes a conic; the conic may be considered as a proper conic, or as a line-pair or a point-pair. The line-pair may be two real lines, or two conjugate imaginary lines; the point-pair may be two real points, or two conjugate imaginary points.

**173.** But the same equation, considering  $(x, y, z)$  as constant coefficients, and  $(\xi, \eta, \zeta)$  as line-coordinates, is the equation of a point, viz. the point which is the pole of the line  $(\xi, \eta, \zeta)$  in respect to the conic  $(a, b, c, f, g, h \backslash x, y, z)^2 = 0$ ; and the same equation, considering  $(\xi, \eta, \zeta)$  as constant coefficients and  $(x, y, z)$  as point-coordinates, is the equation of a line, viz. the polar of the point  $(x, y, z)$  in respect to the conic.

**174.** The equation of the two lines through the point  $(x', y', z')$  to the conic  $(a, b, c, f, g, h \backslash x, y, z)^2 = 0$  is

$$\{(a, b, c, f, g, h \backslash x, y, z)^2 (a, b, c, f, g, h \backslash x', y', z')^2 - [(a, b, c, f, g, h \backslash x, y, z \backslash x', y', z')^1]^2\} =$$

or as this may also be written,

$$\{[(a, b, c, f, g, h \backslash x', y', z')^1]^2 - (a, b, c, f, g, h \backslash x', y', z')^2 (a, b, c, f, g, h \backslash x, y, z)^2\} \times \\ \{(x \partial_x + y \partial_y + z \partial_z) (x' \partial_{x'} + y' \partial_{y'} + z' \partial_{z'}) (a, b, c, f, g, h \backslash x, y, z)^2\}^2 = 0.$$

**175.** In the same equation, writing  $z = 0$ , we have the equation of the two lines through the point  $(x', y', 0)$  to the conic; and if in the equation of the conic we write  $z = 0$ , we have the equation of

the pair of points in which the conic is met by the line  $z = 0$ . These are respectively,

$$(a, b, h \setminus x, y)^2 (a, b, h \setminus x', y')^2 - [(a, b, h \setminus x, y \setminus x', y')^1]^2 = 0,$$

$$(a, b, h \setminus x, y)^2 = 0.$$

**176.** The coordinates of the point of intersection of the lines

$$(a, b, c \setminus x, y, z)^1 = 0, \quad (a', b', c' \setminus x, y, z)^1 = 0$$

are

$$(bc' - b'c, ca' - c'a, ab' - a'b);$$

and the coordinates of the line through the points

$$(A, B, C \setminus \xi, \eta, \zeta)^1 = 0, \quad (A', B', C' \setminus \xi, \eta, \zeta)^1 = 0$$

are

$$(BC' - B'C, CA' - C'A, AB' - A'B).$$

**177.** The line through the point  $(x, y, z)$  and the point of intersection of the lines  $(\xi, \eta, \zeta)$  and  $(\xi', \eta', \zeta')$  has for its equation

$$(\eta\zeta' - \eta'\zeta)X + (\zeta\xi' - \zeta'\xi)Y + (\xi\eta' - \xi'\eta)Z = 0,$$

where  $(X : Y : Z)$  are current coordinates; or what is the same thing, the line in question is

$$(x(\eta\zeta' - \eta'\zeta) + y(\zeta\xi' - \zeta'\xi) + z(\xi\eta' - \xi'\eta) = 0).$$

**178.** The coordinates of the point of intersection of the lines

$$(a, b, c \setminus x, y, z)^1 = 0, \quad (a', b', c' \setminus x, y, z)^1 = 0$$

are

$$(bc' - b'c, ca' - c'a, ab' - a'b);$$

which is, in fact, at once obtained by the consideration that the values of  $(x, y, z)$  must be such as to satisfy simultaneously the two equations.

- 179.** The condition that the line  $(\xi, \eta, \zeta)$  may pass through the point  $(x, y, z)$  is

$$\xi x + \eta y + \zeta z = 0.$$

- 180.** The condition that the two lines  $(\xi, \eta, \zeta)$  and  $(\xi', \eta', \zeta')$  may intersect on the line  $(\xi'', \eta'', \zeta'')$  is

$$\begin{vmatrix} \xi & \eta & \zeta \\ \xi' & \eta' & \zeta' \\ \xi'' & \eta'' & \zeta'' \end{vmatrix} = 0.$$

- 181.** The condition that the three lines

$$(a, b, c \setminus x, y, z)^1 = 0, \quad (a', b', c' \setminus x, y, z)^1 = 0, \quad (a'', b'', c'' \setminus x, y, z)^1 = 0$$

may meet in a point is

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix} = 0.$$

- 182.** The condition that the three points

$$(A, B, C \setminus \xi, \eta, \zeta)^1 = 0, \quad (A', B', C' \setminus \xi, \eta, \zeta)^1 = 0, \quad (A'', B'', C'' \setminus \xi, \eta, \zeta)^1 = 0$$

may lie in a line is

$$\begin{vmatrix} A & B & C \\ A' & B' & C' \\ A'' & B'' & C'' \end{vmatrix} = 0.$$

- 183.** The equation of the line joining the two points  $(x, y, z)$  and  $(x', y', z')$  is

$$\begin{vmatrix} X & Y & Z \\ x & y & z \\ x' & y' & z' \end{vmatrix} = 0,$$

or in the equivalent form,

$$(yz' - y'z)X + (zx' - z'x)Y + (xy' - x'y)Z = 0.$$

**184.** The equation of the point of intersection of the two lines  $(\xi, \eta, \zeta)$  and  $(\xi', \eta', \zeta')$  is

$$\begin{vmatrix} \Xi & H & Z \\ \xi & \eta & \zeta \\ \xi' & \eta' & \zeta' \end{vmatrix} = 0,$$

or in the equivalent form,

$$(\eta\zeta' - \eta'\zeta)\Xi + (\zeta\xi' - \zeta'\xi)H + (\xi\eta' - \xi'\eta)Z = 0.$$

**185.** The equation of the tangent of the conic  $(a, b, c, f, g, h \backslash x, y, z)^2 = 0$  at the point  $(x', y', z')$  is

$$(a, b, c, f, g, h \backslash x', y', z' \backslash x, y, z)^1 = 0,$$

or in the developed form,

$$(ax' + hy' + gz')x + (hx' + by' + fz')y + (gx' + fy' + cz')z = 0.$$

**186.** The equation of the point of contact of the tangent  $(\xi', \eta', \zeta')$  to the conic  $(A, B, C, F, G, H \backslash \xi, \eta, \zeta)^2 = 0$  is

$$(A, B, C, F, G, H \backslash \xi', \eta', \zeta' \backslash \xi, \eta, \zeta)^1 = 0.$$

**187.** The line-equation of the conic  $(a, b, c, f, g, h \backslash x, y, z)^2 = 0$  is

$$\begin{vmatrix} a & h & g & \xi \\ h & b & f & \eta \\ g & f & c & \zeta \\ \xi & \eta & \zeta & 0 \end{vmatrix} = 0.$$

**188.** The point-equation of the conic  $(A, B, C, F, G, H \backslash \xi, \eta, \zeta)^2 = 0$  is

$$\begin{vmatrix} A & H & G & x \\ H & B & F & y \\ G & F & C & z \\ x & y & z & 0 \end{vmatrix} = 0.$$

**189.** If the point-equation of a conic be

$$(a, b, c, f, g, h \backslash x, y, z)^2 = 0,$$

then its line-equation is

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} (a, b, c, f, g, h \backslash \xi, \eta, \zeta)^2 = 0,$$

or writing

$$\begin{aligned} \mathfrak{A} &= bc - f^2, & \mathfrak{B} &= ca - g^2, & \mathfrak{C} &= ab - h^2, \\ \mathfrak{F} &= gh - af, & \mathfrak{G} &= hf - bg, & \mathfrak{H} &= fg - ch, \end{aligned}$$

and, to complete the system,

$$\mathfrak{K} = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

then the line-equation of the conic is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2 = 0.$$

**190.** The quantities  $\mathfrak{A}$ , &c. satisfy the relations

$$\mathfrak{K}^2 = \mathfrak{A}\mathfrak{B}\mathfrak{C} - \mathfrak{A}\mathfrak{F}^2 - \mathfrak{B}\mathfrak{G}^2 - \mathfrak{C}\mathfrak{H}^2 + 2\mathfrak{F}\mathfrak{G}\mathfrak{H},$$

$$\mathfrak{A}a + \mathfrak{H}h + \mathfrak{G}g = \mathfrak{K},$$

$$\mathfrak{H}a + \mathfrak{B}h + \mathfrak{F}g = 0, \quad \mathfrak{G}a + \mathfrak{C}h + \mathfrak{F}g = 0,$$

$$\mathfrak{A}h + \mathfrak{H}b + \mathfrak{G}f = 0,$$

$$\mathfrak{H}h + \mathfrak{B}b + \mathfrak{F}f = \mathfrak{K}, \quad \mathfrak{G}h + \mathfrak{C}b + \mathfrak{F}f = 0,$$

$$\mathfrak{A}g + \mathfrak{H}f + \mathfrak{G}c = 0,$$

$$\mathfrak{H}g + \mathfrak{B}f + \mathfrak{F}c = 0, \quad \mathfrak{G}g + \mathfrak{C}f + \mathfrak{F}c = 0,$$

and moreover

$$\mathfrak{K}(a, b, c, f, g, h \backslash x, y, z)^2 = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash ax+hy+gz, \quad hx+by+fz, \quad gx+fy+cz)^2.$$



**191.** A system of points in a line is said to be a *range*, and a system of lines through a point is said to be a *pencil*. The theories of ranges and pencils, considered irrespectively of each other, are in fact a single theory, constituting the geometry of one dimension. It has been seen how in geometry of one dimension a range of points and a pencil of lines, although considered (as they must be considered) as existing in distinct spaces, may nevertheless stand in certain relations to each other. In geometry of two dimensions, the range and pencil may of course coexist in one and the same plane as their common locus in quo; and such coexistence occurs in fact very frequently: thus if we have a line and a point, and if lines are drawn joining the point with the several points of the line, these lines constitute a pencil, and the points of the line constitute a range, and such pencil and range are homographically related.

**192.** The theory of homography in geometry of two dimensions may be made to depend upon the corresponding theory in geometry of one dimension, or what is the same thing, upon the homography of ranges or pencils. For consider two figures existing in distinct planes or spaces of two dimensions, any four points (not in a line) in the second figure may correspond to any four points (not in a line) in the first figure; and when this is so, we may, by the process about to be explained, given any other point of the first figure, construct the corresponding point of the second figure; and the two figures are then, by definition, homographically related. Suppose that the points  $D', B', C', E'$  of the second figure correspond respectively to the points  $A, B, C, D$  of the first figure, and let  $E$  be any other point of the first figure; suppose that  $E'$  is the corresponding point of the second figure; the pencils  $A(B, C, D, E)$  and  $A'(B', C', D', E')$  should be homographic to each other, that is,  $E'$  must lie on a given line through  $A'$ ; and in like manner the pencils  $B(A, C, D, E)$  and  $B'(A', C', D', E')$  should be homographic to each other, that is,  $E'$  must lie on a given line through  $B'$ . And then, as a theorem,  $C(A, B, D, E)$  and  $C'(A', B', D', E')$ , or  $D(A, B, C, E)$  and  $D'(A', B', C', E')$  will be homographic pencils, that is, the construction will be a determinate one whichever two of the four points are selected for the points  $A$  and  $B$ . The foregoing construction leads to an analytical relation, which I think constitutes a better foundation of the theory. Consider the first plane as the locus in quo of the coordinates  $(x, y, z)$ , and the second plane as the locus in quo of the coordinates  $(X, Y, Z)$ , these two

coordinate systems being absolutely independent of each other.

Consider any point of the first plane and a corresponding point of the second plane such that its coordinates  $(X, Y, Z)$  are given linear functions of the coordinates  $(x, y, z)$  of the point in the first plane.

Any figure whatever in the first plane gives rise to a corresponding figure in the second plane, and the two figures are said to be

homographic to each other. To a point of the first figure there corresponds in the second figure a point, to a line a line, to a range of points or pencil of lines, a homographic range of points or pencil of lines; the line or point which is the locus in quo of the range or pencil in the one figure corresponding with the line or point which is

the locus in quo of the range or pencil in the other figure. And generally, to any curve of any order and class in the first figure, and to its ineunts and tangents, there correspond in the second figure a curve of the same order and class, and the ineunts and tangents of such curve.

**193.** It is to be remarked, that it is not by any means necessary that the word plane, or the words point and line, or consequently the words order and class, should have the same significations as regards the two figures respectively. The theory of homography, as above explained, in fact comprises what is commonly termed the theory of homography and also the theory of reciprocity.

**194.** Let the word plane have the ordinary signification as regards the two figures respectively; and suppose, first, that the words point and line, and therefore order and class, have also the ordinary significations as regards the two figures respectively; we have here the ordinary theory of homography, in which, to any range of points or pencil of lines in the first figure, there corresponds a homographic range of points or pencil of lines in the second figure, and to a curve of any order and class in the first figure there corresponds a curve of the same order and class in the second figure.

**195.** We may, as a specialization giving rise to further developments, assume that the two figures exist in one and the same plane. There is here in general a triangle, each of whose angles or sides, considered as a point or line in the first figure, corresponds to itself, considered as a point or line in the second figure: such triangle may be called the *sibiconjugate triangle*. Any one point of the plane, considered as belonging to the first figure, may correspond to any other point of the plane, considered as belonging to the second figure, and the second figure is then completely

determined. The equations

$$(x : y : z) = (\alpha X + \beta Y + \gamma Z : \alpha' X + \beta' Y + \gamma' Z : \alpha'' X + \beta'' Y + \gamma'' Z)$$

show that the points

$$(\alpha : \alpha' : \alpha''), \quad (\beta : \beta' : \beta''), \quad (\gamma : \gamma' : \gamma'')$$

are the angles of the sibiconjugate triangle; for to a line through any one of these points in the second figure there corresponds in the first figure a line through the same point; or, what is the same thing, the point considered as belonging to either figure is a point corresponding to itself.

**196.** It is proper to notice that the sibiconjugate triangle may be a real triangle, or one or two of its angles may be imaginary. In particular, the sibiconjugate triangle may be one of the infinity of triangles the angles of which are each of them one of the two circular points at infinity; these are the triangles which are similar each to each, and in this case the homography gives rise to the theory of similar figures.

**197.** Let the word plane have the ordinary signification as regards the two figures respectively; and suppose, secondly, that the words point and line, order and class, have interchanged significations, that is, to a point in the first figure there corresponds a line in the second figure, and to a line in the first figure there corresponds a point in the second figure; the locus in quo of the line corresponding to a variable point on a curve of the first figure is a curve, the ineunts of which are the lines corresponding to the tangents of the curve in the first figure; or the ineunts of the curve in the first figure correspond to the tangents of the curve in the second figure. The two curves are said to be reciprocal to each other; the order of the one is equal to the class of the other, and vice versâ. In particular, the two figures may exist in one and the same plane; the sibiconjugate triangle has here a different signification.

**198.** The equation of the Absolute, or say the point-equation of the Absolute, is

$$(a, b, c, f, g, h \backslash x, y, z)^2 = 0;$$

and the line-equation of the Absolute is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2 = 0,$$

where  $\mathfrak{A}$ , &c. have the significations explained in No. 189.

**199.** The point  $(x, y, z)$  and the line  $(\xi, \eta, \zeta)$  are *quadrantal* to each other when they satisfy the equation of the Absolute, that is, when

$$\xi x + \eta y + \zeta z = 0;$$

and in particular, two points  $(x, y, z)$  and  $(x', y', z')$  are quadrantal to each other when

$$(a, b, c, f, g, h \backslash x, y, z \backslash x', y', z')^1 = 0,$$

and two lines  $(\xi, \eta, \zeta)$  and  $(\xi', \eta', \zeta')$  are quadrantal to each other when

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta \backslash \xi', \eta', \zeta')^1 = 0.$$

**200.** The point  $(x, y, z)$  and the line  $(\xi, \eta, \zeta)$  are *normal* to each other when the line passes through the pole of the point in respect to the Absolute; that is, when the line  $(\xi, \eta, \zeta)$  is identical with the line

$$(\mathfrak{A}x + \mathfrak{H}y + \mathfrak{G}z, \mathfrak{H}x + \mathfrak{B}y + \mathfrak{F}z, \mathfrak{G}x + \mathfrak{F}y + \mathfrak{C}z \backslash X, Y, Z)^1 = 0.$$

The relation between the two is not a reciprocal one; but we may, if we please, speak of the point as the *normal point* of the line, and of the line as the *normal line* of the point. In particular, two lines are *normal* to each other when each of them passes through the pole of the other in respect to the Absolute; and two points are *normal* to each other when each of them lies on the polar of the other in respect to the Absolute.

**201.** The polar of a point in respect to a conic is the normal line of the point; and the pole of a line in respect to a conic is the normal point of the line. If the conic is the Absolute, then the polar and the normal line coincide together.

**202.** The pole of the line  $(\xi, \eta, \zeta)$  in respect to the Absolute is the point

$$(\mathfrak{A}\xi + \mathfrak{H}\eta + \mathfrak{G}\zeta, \mathfrak{H}\xi + \mathfrak{B}\eta + \mathfrak{F}\zeta, \mathfrak{G}\xi + \mathfrak{F}\eta + \mathfrak{C}\zeta);$$

and the polar of the point  $(x, y, z)$  in respect to the Absolute is the line

$$(\mathfrak{A}x + \mathfrak{H}y + \mathfrak{G}z, \mathfrak{H}x + \mathfrak{B}y + \mathfrak{F}z, \mathfrak{G}x + \mathfrak{F}y + \mathfrak{C}z).$$

**203.** The theory of the inscription of a conic in a triangle may be considered as arising out of the theory of the homography of two figures in the same plane. Thus, if the conic is  $(\sqrt{x} + \sqrt{y} + \sqrt{z} = 0)$ ,

and the triangle is that formed by the coordinate axes, then the equation of the conic shows that the conic is a conic inscribed in the triangle; that is, it touches the sides of the triangle. The equation

may be written in the rationalized form

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0, \text{ or}$$

$$(1, 1, 1, -1, -1, -1 \backslash x, y, z)^2 = 0.$$

**204.** The equation of the pair of tangents to the conic  $(a, b, c, f, g, h \backslash x, y, z)^2 = 0$  through the point  $(x', y', z')$  has been given in No. 174. The condition in order that the line  $(\xi, \eta, \zeta)$  may touch the conic is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2 = 0.$$

**205.** The equation of the polar of the point  $(x', y', z')$  in respect to the conic  $(a, b, c, f, g, h \backslash x, y, z)^2 = 0$  is

$$(a, b, c, f, g, h \backslash x', y', z' \backslash x, y, z)^1 = 0.$$

The polar passes through the points of contact of the conic with the tangents through the given point.

**206.** If  $U = (a, b, c, f, g, h \backslash x, y, z)^2 = 0$  is the equation of a conic, then

$$\Theta U = (A, B, C, F, G, H \backslash x, y, z)^2 = 0$$

is the equation of a conic having double contact with the conic at its points of intersection with the conic  $(A, B, C, F, G, H \backslash x, y, z)^2 = 0$ , that is, the conic  $\Theta U = 0$  has double contact with the conic  $U = 0$  at the points where the conic  $U = 0$  is met by the conic  $\Theta U = 0$ .

But this is a theorem which requires to be more precisely stated.

**207.** The theory of distance in geometry of two dimensions is founded upon the following definitions. The distance of two points is equal to the angle which the normal lines of the points subtend at any point of the Absolute; the distance of two lines is equal to the angle subtended at the centre of the Absolute by the two normal points of the lines; the distance of a point and a line is equal to the complement of the angle subtended at the point by the two points of intersection of the line with the Absolute. The quadrant is the

distance of two quadrantal lines, or of two quadrantal points; it is the distance of a point from any line through its normal point, or of a line from any point on its normal line. The quadrant is the unit of distance.

**208.** Write for shortness

$$\begin{aligned}\Pi &= (a, b, c, f, g, h \setminus x, y, z)^2, & \Pi' &= (a, b, c, f, g, h \setminus x', y', z')^2, \\ \nabla &= (a, b, c, f, g, h \setminus x, y, z \setminus x', y', z')^1, & \Omega &= (xy' - y'z, zx' - z'x, xy' - x'y).\end{aligned}$$

Then the distance of the points  $(x, y, z)$  and  $(x', y', z')$  is

$$\sin^{-1} \frac{\sqrt{(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \setminus \Omega)^2}}{\sqrt{\Pi} \sqrt{\Pi'}};$$

or as it may also be written,

$$\cos^{-1} \frac{\nabla}{\sqrt{\Pi} \sqrt{\Pi'}};$$

and the distance of the lines  $(\xi, \eta, \zeta)$  and  $(\xi', \eta', \zeta')$  is

$$\sin^{-1} \frac{\sqrt{(a, b, c, f, g, h \setminus \xi \eta' - \xi' \eta, \eta \zeta' - \eta' \zeta, \zeta \xi' - \zeta' \xi)^2}}{\sqrt{\Sigma} \sqrt{\Sigma'}};$$

or as it may also be written,

$$\cos^{-1} \frac{(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \setminus \xi, \eta, \zeta \setminus \xi', \eta', \zeta')^1}{\sqrt{\Sigma} \sqrt{\Sigma'}},$$

where  $\Sigma$  and  $\Sigma'$  denote respectively  $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \setminus \xi, \eta, \zeta)^2$  and  $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \setminus \xi', \eta', \zeta')^2$ .

**209.** It follows from the sine formula, that the distance of a point from itself, or of a line from itself, is always = 0; and from the cosine formula, that the distance of two quadrantal points, or of two quadrantal lines, is =  $\frac{\pi}{2}$ ; and from the sine formula, that the distance of a point from any point on the Absolute, or of a line from any tangent of the Absolute, is = 0; and from the cosine formula, that the distance of a point from its normal line is =  $\frac{\pi}{2}$ .

**210.** It follows from the cosine formula that, if  $P, P', P''$  be three points on the same line, then

$$\text{Dist.}(P, P') + \text{Dist.}(P', P'') = \text{Dist.}(P, P'').$$

And the like theorem holds for three lines through the same point.

**211.** The condition that the distance of the points  $(x, y, z)$  and  $(x', y', z')$  may be a given constant is

$$\Pi \Pi' \sin^2 \theta - (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \Omega)^2 = 0.$$

This is the equation of a conic, the point-equation of which, considering  $(x', y', z')$  as given, is as just written down; and the line-equation of which, considering  $(\xi, \eta, \zeta)$  as the line-coordinates, is

$$\Pi \Pi' \sin^2 \theta - \nabla^2 = 0.$$

**212.** The conic represented by either of these equations may be called a *circle*; the point  $(x', y', z')$  is the *centre* of the circle; and the normal line of the centre is the *axis* of the circle. The sine of the radius is  $= \sin \theta$ . The point-equation of the circle shows that all the ineunts of the circle are at the same distance from the centre; and the line-equation shows that all the tangents of the circle are at the same distance from the axis. In the particular case where the Absolute is a pair of points, the circle is any conic through the two points of the Absolute, and the centre is the pole of the line joining the two points.

**213.** The equation of a circle, centre  $(x', y', z')$  and radius  $\theta$ , may be written in the form

$$(a, b, c, f, g, h \backslash x, y, z)^2 (a, b, c, f, g, h \backslash x', y', z')^2 \sin^2 \theta - [(a, b, c, f, g, h \backslash x, y, z \backslash x', y', z')^1]^2 = 0.$$

**214.** The line-equation of the same circle, the line-coordinates of the axis being  $(\xi', \eta', \zeta')$ , is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2 (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi', \eta', \zeta')^2 \sin^2 \theta - [(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta \backslash \xi', \eta', \zeta')^1]^2 = 0.$$

**215.** The properties of the circle just referred to lead at once to the following theorem. If through the centre of a circle we draw any two lines, and if from the points where these lines meet the circle we draw lines parallel respectively to the two lines first drawn, these lines will meet in a point on the axis of the circle. And again, if we draw through the centre two other lines, and from the points where these meet the circle we draw lines parallel respectively to the lines last drawn, these lines will meet in a point on the axis of the circle;

and so on. The theorem is, in fact, equivalent to Euclid's definition of similar figures.

**216.** These properties of the circle lead immediately to the analytical expressions for the distances of points or lines in terms of the coordinates. In fact, take

$$(a, b, c, f, g, h \backslash x, y, z)^2 = 0$$

for the point-equation of the Absolute; its line-equation will be

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2 = 0.$$

The point-equation of the circle having the point  $(x', y', z')$  for its centre, is

$$(a, b, c, f, g, h \backslash x, y, z)^2 (a, b, c, f, g, h \backslash x', y', z')^2 \cos^2 \theta - [(a, b, c, f, g, h \backslash x, y, z \backslash x', y', z')^2]$$

or

$$(a, b, c, f, g, h \backslash x, y, z)^2 (a, b, c, f, g, h \backslash x', y', z')^2 \sin^2 \theta - [(a, b, c, f, g, h \backslash yz' - y'z, zx' - z'x, xy' - x'y)^2] = 0,$$

from which (by the same reasoning as for the case of geometry of one dimension) it follows that the distance of the points  $(x, y, z)$ ,  $(x', y', z')$  is

$$\sin^{-1} \frac{(a, b, c, f, g, h \backslash x, y, z \backslash x', y', z')^1}{\sqrt{(a, b, c, f, g, h \backslash x, y, z)^2} \sqrt{(a, b, c, f, g, h \backslash x', y', z')^2}},$$

or what is the same thing,

$$\sin^{-1} \frac{(a, b, c, f, g, h \backslash yz' - y'z, zx' - z'x, xy' - x'y)^1}{\sqrt{(a, b, c, f, g, h \backslash x, y, z)^2} \sqrt{(a, b, c, f, g, h \backslash x', y', z')^2}};$$

and it appears from the cosine formula (see ante, No. 208), that if  $P, P', P''$  be points on the same line, then we have, as we ought to have,

$$\text{Dist.}(P, P') + \text{Dist.}(P', P'') = \text{Dist.}(P, P'').$$

**217.** In like manner, the line-equation of the same circle, the line-coordinates of the axis being  $(\xi', \eta', \zeta')$ , is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2 (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi', \eta', \zeta')^2 \sin^2 \theta - [(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta \backslash \xi', \eta', \zeta')^2] = 0,$$



from which it follows that the distance of the lines  $(\xi, \eta, \zeta)$  and  $(\xi', \eta', \zeta')$  is

$$\sin^{-1} \frac{(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta \backslash \xi', \eta', \zeta')^1}{\sqrt{(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2} \sqrt{(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi', \eta', \zeta')^2}},$$

or what is the same thing

$$\sin^{-1} \frac{(a, b, c, f, g, h \backslash \eta \zeta' - \eta' \zeta, \zeta \xi' - \zeta' \xi, \xi \eta' - \xi' \eta)^1}{\sqrt{(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2} \sqrt{(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi', \eta', \zeta')^2}}.$$

**218.** And we may from the first formula of either set, deduce for the distance of the point  $(x, y, z)$  and the line  $(\xi, \eta, \zeta)$ , the expression

$$\sin^{-1} \frac{\xi x + \eta y + \zeta z}{\sqrt{(a, b, c, f, g, h \backslash x, y, z)^2} \sqrt{(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \backslash \xi, \eta, \zeta)^2}},$$

as may be easily seen by writing  $\xi \xi' + \eta \eta' + \zeta \zeta'$  for  $x', y', z'$ , or  $ax + hy + gz$ , &c. for  $\xi, \eta, \zeta$ , and putting  $\sin^{-1}$  for  $\cos^{-1}$ .

**219.** It may be noticed that there are certain lines, viz. the tangents of the Absolute, in regard to which, considered as a space of one dimension, the Absolute is a pair of coincident points; and in like manner certain points, viz. the ineunts of the Absolute, in regard to which, considered as a space of one dimension, the Absolute is a pair of coincident lines.

**220.** We may, in particular, suppose that the Absolute, instead of being a proper conic, is a pair of points. The line through the two points may be called the *Absolute line*; such line is to be considered as a pair of coincident lines. Any point whatever determines with the Absolute, two lines, viz. the lines joining the point with the two points of the Absolute; this line-pair is the Absolute for the point considered as a space of one dimension or locus in quo of a pencil of lines, and the theory of the distances of lines through a point is therefore precisely the same as in the general case. But any line whatever determines with the Absolute (meets the Absolute line in) a pair of coincident points, which pair of coincident points is the Absolute in regard to such line considered as a space of one dimension or locus in quo of a range of points, and the theory of the distance of points on a line is therefore the theory before explained for this special case. But we cannot, in the same way as before, compare the distances of points upon different lines, since we have

not in the present case the quadrant as a unit of distance. The comparison must be made by means of the circle, viz. in the present case any conic passing through the two points of the Absolute is termed a circle, and the point of intersection of the tangents to the circle at the two points of the Absolute (or what is the same thing, the pole of the Absolute line in respect to the circle) is the centre of the circle. The Absolute line itself may, if it is necessary to do so, be considered as the axis of the circle. It is assumed that the points of the circle are all of them equidistant from the centre, and by this assumption we are enabled to compare distances upon different lines. In fact we may, by a construction precisely similar to that of Euclid, Book I. Prop. II., from a given point draw a finite line equal to a given finite line  $BC$ , and thence also upon a given line through  $A$ , determine the finite line  $AD$  equal to the given finite line  $BC$ . Since the unit of distance for points on a line is arbitrary, we cannot of course compare the distances of points with the distances of lines.

The distance of a point from a line does, however, admit of comparison with the distance of two points; we have only to assume as a definition that the distance of a point from a line is the distance of the point from the point of intersection of the line with the quadrantal line through the point.

**221.** As regards the analytical theory, suppose that the point-coordinates of the two points of the Absolute are  $(p, q, r)$ ,  $(p_0, q_0, r_0)$ , then the line-equation of the Absolute is

$$(pp_0, qq_0, rr_0, qr_0 + q_0r, rp_0 + r_0p, pq_0 + p_0q, \xi, \eta, \zeta)^2 = 0,$$

or, as this may be written,

$$(\xi p + \eta q + \zeta r)(\xi p_0 + \eta q_0 + \zeta r_0) = 0;$$

so that the line-equation of the Absolute is the equation of the two points, or, what is the same thing, the equation of the Absolute line regarded as a pair of coincident lines. And in like manner, the point-equation of the Absolute is the equation of the Absolute line regarded as a pair of coincident lines.

**222.** In the foregoing case of a pair of points as the Absolute, the formulæ for the distance of two points and of two lines continue to subsist; and the formula for the distance of a point and a line also continues to subsist. The line-equation of the circle, centre

$(x', y', z')$  and radius  $\theta$ , is

$$(\xi x' + \eta y' + \zeta z')^2 (pp_0, qq_0, rr_0, qr_0 + q_0 r, rp_0 + r_0 p, pq_0 + p_0 q \backslash \xi, \eta, \zeta)^2 \sin^2 \theta \\ - [(pp_0, qq_0, rr_0, qr_0 + q_0 r, rp_0 + r_0 p, pq_0 + p_0 q \backslash \xi, \eta, \zeta \backslash x', y', z')^1]^2 = 0,$$

where  $\sin^2 \theta$  is arbitrary. The point-equation of the same circle is

$$(pp_0, qq_0, rr_0, qr_0 + q_0 r, rp_0 + r_0 p, pq_0 + p_0 q \backslash x, y, z)^2 (\xi x' + \eta y' + \zeta z')^2 \sin^2 \theta \\ - [(pp_0, qq_0, rr_0, qr_0 + q_0 r, rp_0 + r_0 p, pq_0 + p_0 q \backslash x, y, z \backslash x', y', z')^1]^2 = 0.$$

**223.** In particular, if the two points of the Absolute are the circular points at infinity, the theory of distance becomes the ordinary theory of distance in plane geometry. The Absolute line is the line at infinity. The distance of two points is the ordinary distance; the distance of two lines is the angle between them; and the distance of a point from a line is the perpendicular distance.

The circle is the ordinary circle. The quadrant is  $= \frac{\pi}{2}$ .

**224.** It may be remarked, that the condition in order that the distance of two points may be  $= 0$  is that the line joining the two points may pass through one of the points of the Absolute; and in particular, if the Absolute is the two circular points at infinity, that the line joining the two points may pass through one of the circular points, that is, that the two points may be in one and the same direction from each other. The condition in order that two lines may be at the distance  $= 0$  from each other is that the two lines may meet at a point of the Absolute; and in particular, if the Absolute is the two circular points at infinity, that the two lines may meet at a circular point, that is, that the two lines may be parallel to each other.

**225.** The condition that two points may be at a quadrantal distance from each other is that the line joining them may be a tangent of the Absolute; and the condition that two lines may be at a quadrantal distance from each other is that their point of intersection may lie on the Absolute.

**226.** The normal of a point is the line joining the point with the pole of the Absolute line; that is, the normal of a point is the line through the point and the centre of any circle; and the normal of a line is the line through the intersection of the line with the Absolute line, and the pole of the line in respect to any circle.

**227.** If the Absolute is a pair of points, the harmonic of a point with respect to the Absolute is the point such that the two points of the Absolute, the given point and its harmonic, form an involution; that is, the point-pair consisting of the two points of the Absolute, and the point-pair consisting of the given point and its harmonic, form an involution. The harmonic of a line with respect to the Absolute is the line through the pole of the given line in respect to any circle; that is, the line through the centre of any circle and the intersection of the given line with the Absolute line.

**228.** If the Absolute is a proper conic, the harmonic of a point is the pole of the line joining the point with its normal point; that is, the point of intersection of the tangents to the Absolute from the given point. And the harmonic of a line is the polar of the point of intersection of the line with its normal line; that is, the line joining the points of contact of the Absolute with the tangents from the point of intersection of the line and its normal line.

**229.** In ordinary plane geometry, the Absolute degenerates into a pair of points, viz. the two circular points at infinity. The general formulæ for distance, &c., thus become the ordinary formulæ of plane geometry. It is hardly necessary to remark that in this case the notion of the quadrantal relation of two points has altogether disappeared, and that the unit of distance has become the right angle.

**230.** In the geometry of similar figures, the Absolute is a proper conic; but the homography is such that the two circular points at infinity are the sibiconjugate points of the homography; that is, the homography is a similarity. The theory of distance applies to similar figures; but the distances in different figures are not comparable with each other, except in the case of similar figures, where the ratio of the distances is the ratio of similarity.

**231.** In spherical geometry, the Absolute is a proper conic; the plane of the Absolute is the plane at infinity; and the points and lines of the figure are the points and great circles of a sphere. The theory of distance applies to spherical geometry; and the distances upon different great circles are comparable with each other, since the Absolute is the same for all the great circles.

## Notes and References.

**101.** No. V. of this paper gives a correction of a formula (18) in the paper 8, *Philosophical Transactions*, 1854. I may notice that the denominator of the second term on the right-hand side of the equation (18) is there written by an error  $(1 + \alpha\beta'\gamma'')$  instead of  $(1 - \alpha\beta'\gamma'')$ , which error is not of much consequence, since it affects only the sign of the term in question.

**119.** I attach some value to the process here explained: the most simple case of it is the elimination of  $x$  from the two equations  $ax^2 + 2bx + c = 0$  and  $a'x^2 + 2b'x + c' = 0$ , which can be effected by putting the two equations under the forms

$$(ax + b)^2 = b^2 - ac, \quad (a'x + b')^2 = b'^2 - a'c',$$

and then by the elimination of  $x$  from the linear equations

$$ax + b \pm \sqrt{b^2 - ac} = 0, \quad a'x + b' \pm \sqrt{b'^2 - a'c'} = 0.$$

**137.** The theory developed in the Sixth Memoir upon Quantics is, I think, one of the most important contributions which I have made to Mathematics. I find that the first communication of it to the Mathematical Society was made on November 9, 1858; the memoir was received by the Royal Society November 18, 1858. Prof. Klein has called my attention to a reference (p. 132 of his second paper) to the theory developed in Staudt's *Geometrie der Lage*, 1847 (more fully in the *Beiträge zur Geometrie der Lage, Zweites Heft*, 1857). The logarithm of the formula is  $\log(A, B, P, Q)$ , and, according to Staudt's theory  $(A, B, P, Q)$ , the anharmonic ratio of any four points, has independently of any notion of distance the fundamental properties of a numerical magnitude, viz. any two such ratios have a sum and also a product, such sum and product being each of them a like ratio of four points determinable by purely descriptive constructions. The proof is easiest for the product: say the ratios are  $(A, B, P, Q)$  and  $(A', B', P', Q')$ ; then considering these as given points we can construct  $R$  such that  $(A', B', P', Q') = (A, B, Q, R)$ : the two ratios are thus  $(A, B, P, Q)$  and  $(A, B, Q, R)$ , and we say that their product is  $(A, B, P, R)$  (observe as to this that introducing the notion of distance, the two factors are  $\frac{PQ}{BQ}$  and  $\frac{QR}{BR}$ , and then their product  $= \frac{PR}{BR}$ , which is  $(A, B, P, R)$ , which is the foundation of the definition). Next for the sum, we construct  $Q_1$  such that  $(A', B', P', Q') = (A, B, P, Q_1)$ ; the

sum then is  $(A, B, P, Q) + (A, B, P, Q_1)$ ; and if we then construct  $S$  such that  $(A, R), (Q, Q_1), (B, S)$  are an involution, we say that  $(A, B, P, Q) + (A, B, P, Q_1) = (A, B, P, S)$ . (Observe as to this that again introducing the notion of distance the last mentioned equation is  $\frac{PQ}{BQ} + \frac{PQ_1}{BQ_1} = \frac{PS}{BS}$ , that is  $\frac{q}{b-q} + \frac{q_1}{b-q_1} = \frac{s}{b-s}$ , which expresses that

$S$  is determined as above; in fact the equation  $\begin{vmatrix} 1, b+s & bs \\ 1, q+q_1 & qq_1 \\ 1, 2a & a^2 \end{vmatrix} = 0$  is readily seen to be equivalent to  $\frac{q}{b-q} + \frac{q_1}{b-q_1} = \frac{s}{b-s}$ .)

It must however be admitted that, in applying this theory of Staudt's to the theory of distance, there is at least the appearance of arguing in a circle, since the construction for the product of the two ratios, is in effect the assumption of the relation,

$$\text{dist. } PQ + \text{dist. } QR = \text{dist. } PR.$$

**138.** In regard to the "Academy problem,"—I may refer to Boole's memoir, *Philosophical Transactions*, 1862, pp. 225—252; and to Wilbraham's memoir, *Phil. Mag.* July, 1854. Wilbraham's example, p. 422, is as follows: "If an event can only happen by one or both of two causes  $A$  and  $B$  acting, and the probability of  $A$  acting is  $a$ , of  $B$  acting is  $\beta$ , of both acting is  $a\beta$ , of the event happening if  $A$  acts is  $p$ , of the event happening if  $B$  acts is  $q$ , what is the probability of the event?" Boole's answer is

$$u = \frac{a\beta(1-p)(1-q)}{a(1-p) + \beta(1-q) - \nu'},$$

where  $\nu'$  is a root of the equation

$$a(1-p)(a-\nu')(\beta-\nu') + \beta(1-q)\nu'(1-a-\beta+\nu') = 0.$$

But Prof. Klein called my attention to a reference (p. 132 of his second paper) to the theory developed in Staudt's *Geometrie der Lage*, 1847 (more fully in the *Beiträge zur Geometrie der Lage*, *Zweites Heft*, 1857).

**139.** Boole, in his solution of the "Academy problem," introduces what he calls "logical probabilities" (but what these mean, I cannot make out) viz. these are Prob.  $A = x$ , or say simply  $A = x$ ; and similarly,  $B = y$ ,  $AE = s$ ,  $BE = t$ ; then in the case  $ABE$  we have  $A, B, AE, BE$ , and the logical probability is taken to be  $xyst$ ; and

we obtain in like manner the other terms of the third column. And then taking  $\xi, \xi', \eta, \eta', \zeta, \zeta', \sigma'$  to be proportional to the terms of the third column, say  $\sqrt{\xi} = xyst$ , &c. and substituting in the six equations, we have six equations for the determination of  $x, y, s, t, V, u$ , and we thus arrive at the value of the required probability  $u$ .

**140.** To complete Boole's solution: the equations easily give

$$\frac{a'tx'y}{u} = \frac{s\sigma'xy'}{a\rho - u} = \frac{s't'}{\beta q' - u} = \frac{y}{\gamma},$$

and

$$\frac{s't'\alpha'}{1 - a\rho' - u} = \frac{s't'y'}{1 - \beta q' - u} = \frac{a\tau xy'}{a\rho + \beta q - u} = \frac{y'}{\gamma'},$$

and multiplying together the first three values, and also the second three values, we have in each case the same numerator

$ss' \cdot t\sigma' \cdot xx' \cdot yy'$ , and we thus obtain the equation

$$(u - a\rho)(u - \beta q)(1 - u) - (1 - a\rho' - u)(1 - \beta q' - u)(a\rho + \beta q - u) = 0,$$

which, the term in  $u^2$  disappearing, is a quadric equation; it is in fact

$$u^2(-1 + a\rho' + \beta q') + u\{1 + a(\rho - \rho') + \beta(q - q') - a^2\rho\rho' - \beta^2qq' + \alpha\beta(-1 + p'q')\} \\ + \{-a\rho - \beta q + a^2\rho\rho' + \beta^2qq' + \alpha\beta(1 - p'q') - (a\rho + \beta q)a\beta p'q'\} = 0;$$

or, what is more simple, if we write with Boole  $a\rho = a, \beta q = b,$   
 $1 - a\rho' = a', 1 - \beta q' = b', a\rho + \beta q = c,$

$$u^2(-1 + a' + b') + u\{1 + a - a' + b - b' - a^2\frac{pp'}{\rho\rho'} - b^2\frac{qq'}{q'q'} + ab(1 - p'q')\} + \dots = 0.$$

**147.** Upon looking at any one of the Tables, for instance VIII (a), it will be noticed (1) that the partition symbols in the outside top line and left-hand column re-appear in the same order in the left-hand top line and outside column respectively; (2) that the partitions which present themselves in the outside lines and columns are not in general all the partitions of the same weight, but only a selection of such partitions.

**149.** The table of postmultiplication given in the paper on the Calculus of Forms, No. 70, is applicable to the present theory; the only difference is that instead of deriving each product by the

operation of the sign  $\Sigma$ , the products are here obtained by mere addition.

**156.** The theory of distance in geometry of two dimensions was first developed by me in the paper “Sur la Géométrie de la distance générale,” *Comptes Rendus*, t. xlvii. (1858), pp. 489—492. The leading idea is due to Lagrange (see the reference in No. 137 above).

Prof. Klein has also developed the theory in his paper “Ueber die sogenannte Nicht-Euklidische Geometrie,” *Math. Annalen*, t. iv.

(1871), pp. 573—625, especially pp. 586—590. The theory of distance is the foundation of the so-called Non-Euclidean geometry; but the notion of distance itself depends ultimately upon that of anharmonic ratio.

**157.** The condition in order that the point-pair  $(a, b, c \setminus x, y)^2 = 0$  may be in- scribed in the point-pair  $(a', b', c' \setminus x, y)^2 = 0$  is

$$(ac - b^2)(a'c' - b'^2) - (ac' + ca' - 2bb')^2 = 0.$$

This may be written in the form

$$\begin{vmatrix} a & b & c \\ b & c' & b' \\ c & b' & a' \end{vmatrix} = 0,$$

or in the form

$$(a, b, c \setminus a', b', c')^2 + 4(ac - b^2)(a'c' - b'^2) = 0.$$

**158.** The problem of the inscription of point-pairs has been considered in the present memoir, Nos. 165—168. The theory is as

follows: If the point-pair  $(a, b, c \setminus x, y)^2 = 0$  is inscribed in the point-pair  $(a', b', c' \setminus x, y)^2 = 0$ , then the axis of the first point-pair is the centre of the second, and the centre of the first is the axis of the second.



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CAMBRIDGE  
PRINTED BY C. J. CLAY, M.A. AND SONS,  
AT THE UNIVERSITY PRESS.

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